INDIA DATA CENTRE ATLAS · COMPILED RESEARCH REPORT

Data Centres in India

Power, water, carbon, cost, policy and place: the complete research compendium

Fifteen research parts on how data centres work, what they consume, what they cost, how they are regulated, where they should be built and what India should do next, compiled with every data table, illustrated with figures from the atlas, and referenced to their sources.

October 2026

Prepared for the India Data Centre Atlas research series

Contents

1.1 Types of Data Centre 12
1.2 Caveats 18

Chapter 1 · Types of data centre: the global picture and the Indian choice 12
2.1 Anatomy of Data Centre Consumption 19
2.2 AI Data Centre vs Conventional Data Centre 23
2.3 Recommendations 26
2.4 Caveats 27

Chapter 2 · Anatomy of consumption: how a data centre uses energy, and how AI data centres differ 19
3.1 Fundamentals 28
3.2 Air cooling types 30
3.3 Chilled-water plants and heat rejection 31
3.4 Liquid cooling types 32
3.5 Natural and alternative cooling 33
3.6 Market and adoption 33
3.7 Water–energy trade-offs 34
3.8 Cost 35
3.9 The Indian perspective 35
3.10 Future trends 37
3.11 Caveats 37

Chapter 3 · Cooling: every type, how it works, what it costs and what India should build 28
4.1 Key Findings 39
4.2 Assumptions 39
4.3 Power calculations 40
4.4 Water calculations 40
4.5 Comparison with other Bengaluru and Karnataka users 42
4.6 Where Bengaluru data centres get water today 42
4.7 Will data centre water cause scarcity? 43
4.8 Policy context 43
4.9 Recommendations 44
4.10 Caveats 45

Chapter 4 · What one campus asks of a city: water and power for a Bengaluru data centre 39
5.1 Key Findings 46
5.2 SOURCE — What Generates the Electron 47
5.3 TIMING — When the Load Occurs 48
5.4 LOCATION — Which Grid, Which Margin, Which Basin 50
5.5 CARBON INTENSITY — Definitions, Trajectory and Data-Centre Metrics 51
5.6 WATER INTENSITY — The Grid's Hidden Water 52
5.7 SYNTHESIS — "Every TWh Is Not the Same": Worked Matrix 53
5.8 Recommendations 54
5.9 Caveats 54

Chapter 5 · Every terawatt-hour is not the same: source, timing and location of data-centre electricity 46
6.1 Executive summary 56
6.2 How much water? 57
6.3 Water balance 59
6.4 Water intensity metrics 60
6.5 Local context 62
6.6 Water stress 63
6.7 Hyperscaler and Indian operator water reporting: practices and gaps 64
6.8 Recommendations 65
6.9 Caveats 66

Chapter 6 · Water impact: from litres used to water impact 56
7.1 How to read the labels 67
7.2 Key Findings 68
7.3 Direct (on-site) water 68
7.4 Indirect (off-site) water 71
7.5 Water by cooling technology 73
7.6 Peak water demand 74
7.7 Hourly water and electricity profiles 75
7.8 Grid electricity options in India 76
7.9 Energy calculations 77
7.10 Worked examples 78
7.11 Recommendations 80
7.12 Data gaps and how to close them 80
7.13 Caveats 81

Chapter 7 · Water and energy calculated: parameters, formulas and worked examples 67
8.1 Key Findings 82

8.2 Definitions and Archetypes 83
8.3 Energy 83
8.4 Water 85
8.5 Does Liquid Cooling for AI Raise or Lower Water Use? 89
8.6 Side-by-Side Comparison 89
8.7 India Specifics 90
8.8 Uncertainty, Disputes and Data Gaps 91
8.9 Recommendations for the atlas and the series 91
8.10 Caveats 91

Chapter 8 · Conventional and AI data centres compared: water and energy at the scale of a site 82
9.1 Key Findings 93
9.2 CAPEX (₹) 93
9.3 OPEX (₹) 98
9.4 Unit economics 101
9.5 Recommendations 105
9.6 Caveats 105

Chapter 9 · Unit economics: what a data centre costs to build and run 93
10.1 What is a "green data centre"? 106
10.2 Metrics and standards 107
10.3 Design levers with quantified savings 108
10.4 Renewable and carbon-free energy procurement 109
10.5 Global benchmarks and the AI problem 110
10.6 Economics of going green: a 20 MW Bengaluru colocation worked example 111
10.7 India and Karnataka 112
10.8 Green Data Centre maturity model (India-calibrated) 113
10.9 Policy recommendations 114
10.10 Caveats 115

Chapter 10 · Green data centres: a decision-grade synthesis for India and Karnataka 106
11.1 The question the model answers 116
11.2 How the model works 116
11.3 How the trade-off between cost and water is traced 117
11.4 Results: every option the model found 118
11.5 The hydropower uncertainty 119
11.6 Three rules for choosing an option 120
11.7 The recommended mix 120
11.8 What the results mean for buyers and for policy 120
11.9 Limitations of the model 121

Chapter 11 · Least-water power: an hourly model for a Karnataka data centre 116
12.1 Key Findings 122
12.2 Part A — Central Framework 123
12.3 Part B — State-by-State Deep Dive 123
12.4 Part C — Master Comparison Table 129
12.5 Part D — Attractiveness Index 130
12.6 Part E — Competition Dynamics: Race to the Bottom vs Sustainability Conditions 131
12.7 Recommendations 131
12.8 Caveats 132

Chapter 12 · State data-centre policies: a state-by-state deep dive 122
13.1 Key Findings 133
13.2 Method 134
13.3 Latency table (great-circle km to Mumbai / Delhi / Chennai / Bengaluru → mean RTT) 135
13.4 Connectivity backbone (applies across states) 135
13.5 State-by-state shortlists 136
13.6 National master table (default weights) 139
13.7 Alternate weightings and rank stability 140
13.8 Recommendations — top picks by use case 141
13.9 Caveats and data gaps (and how to close them) 141

Chapter 13 · Where to build: a city-by-city shortlist for India 133
14.1 Key Findings 143
14.2 Capacity (MW IT): data, three models, comparison, backtest 144
14.3 Electricity (TWh) and share of the grid 147
14.4 Carbon trajectory 147
14.5 Costs: build cost per MW and GPU-hour prices 148
14.6 Water demand by city (MLD) 149
14.7 Where new capacity goes, city by city 150
14.8 Demand drivers (regression) 151
14.9 Integrated base-case table (2026–2035) [M] 152
14.10 Parameter table for the calculator, with Monte Carlo distributions 152
14.11 Risks and swing factors, ranked by impact on 2035 MW 153
14.12 Recommendations 153

14.13 Caveats 153

Chapter 14 · India data centres to 2035: a forecast 143

15.1 Key Findings 155

15.2 Part 1 — Design and Site Recommendations 156

15.3 Part 2 — Policy Recommendations 160

15.4 Caveats 163

Chapter 15 · Recommendations: design, siting and policy for India's next data centres 155

Appendix A · Simulator assumptions and key numbers 165

Appendix B · Bibliography 167

Executive summary 9

About this report and how to read it 8

Glossary 11

List of tables

Table 1.1 · Types by ownership and business model 13
Table 1.2 · Classification by resilience: Uptime Tiers and TIA-942 13
Table 1.3 · Classification by cooling approach 14
Table 1.4 · Global market share and installed capacity by type (2024–2026) 14
Table 1.5 · Current state of the market 15
Table 1.6 · Current state of the market 15
Table 1.7 · Policy drivers 16
Table 2.1 · Electricity breakdown by subsystem 19
Table 2.2 · Inside the IT load: CPUs, GPUs, memory, storage and networking 20
Table 2.3 · Efficiency metrics: PUE, WUE, CUE and ERE 20
Table 2.4 · Water consumption: direct and indirect 21
Table 2.5 · Global and regional electricity consumption 22
Table 2.6 · Energy per workload 22
Table 2.7 · Master comparison table 23
Table 2.8 · Compute hardware and power 24
Table 2.9 · Inference versus training data centres 26
Table 3.1 · Why cooling is needed 28
Table 3.2 · Heat transfer properties: why liquid wins 29
Table 3.3 · ASHRAE TC 9.9 thermal guidelines (5th edition, 2021) 29
Table 3.4 · ASHRAE TC 9.9 thermal guidelines (5th edition, 2021) 29
Table 3.5 · Key metrics 30
Table 3.6 · Chiller types 31
Table 3.7 · Liquid cooling comparison 33
Table 3.8 · Natural and alternative cooling 33
Table 3.9 · Market size and forecasts (2024–2031) 33
Table 3.10 · Vendor landscape and consolidation (2024–2026) 34
Table 3.11 · The core trade 35
Table 3.12 · Capex by cooling approach (indicative, 2025–26) 35
Table 3.13 · Climate challenge 35
Table 3.14 · Indian operators and liquid cooling deployments 36
Table 3.15 · Recommendations for India: what to use where 37
Table 4.1 · Assumptions 39
Table 4.2 · Power calculations 40
Table 4.3 · On-site water by case 41
Table 4.4 · Indirect water embedded in grid electricity 41
Table 4.5 · Comparison with other Bengaluru and Karnataka users 42
Table 4.6 · Where Bengaluru data centres get water today 42
Table 4.7 · Scenarios (75% utilisation; IT energy = 18,000 kWh/day per MW) 43
Table 4.8 · Phased roadmap 45
Table 5.1 · Lifecycle carbon intensity by technology 47
Table 5.2 · Water: withdrawal vs consumption by source 47
Table 5.3 · Country grids (operational, annual average) 50
Table 5.4 · Average vs marginal; attributional vs consequential 50
Table 5.5 · Hyperscaler disclosures: location-based vs market-based 52
Table 5.6 · Grid-average water intensity 52
Table 5.7 · SYNTHESIS — "Every TWh Is Not the Same": Worked Matrix 53
Table 5.8 · Five-factor scorecard (for a site or procurement decision) 53
Table 6.1 · Withdrawal vs consumption: definitions and why both matter 57
Table 6.2 · Cooling water 58
Table 6.3 · Non-cooling water 58
Table 6.4 · Freshwater vs recycled water 58
Table 6.5 · Daily and annual demand: worked examples 58
Table 6.6 · Worked balance: 50 MW Bengaluru hyperscale facility (evaporative) 59
Table 6.7 · Worked balance: 100 MW AI facility (DLC + adiabatic dry coolers) 60
Table 6.8 · Measurement and accounting standards 60
Table 6.9 · L/MW (per MW of IT capacity) 61
Table 6.10 · L/GPU-hour 61
Table 6.11 · L/inference and training 61
Table 6.12 · Which metric for which audience, pitfalls and normalisation 61
Table 6.13 · Single comparison table: Bengaluru scenarios 62
Table 6.14 · Data centre use against Bengaluru's water availability 62
Table 6.15 · Equivalence metrics (S2, 50 MW evaporative) 63
Table 6.16 · Frameworks 63
Table 6.17 · Methods 63
Table 6.18 · Applying the methods to Bengaluru scenarios (50 MW, annual) 64
Table 6.19 · Water impact scorecard 64
Table 6.20 · Hyperscaler and Indian operator water reporting: practices and gaps 64
Table 6.21 · Minimum measurement and disclosure framework (all sites ≥500 kW IT) 65
Table 6.22 · Suggested thresholds 65
Table 7.1 · Definitions and metrics 69
Table 7.2 · Components and physics 69
Table 7.3 · Published WUE and water values 70
Table 7.4 · Water per AI query 70

Table 7.5 · Water intensity of generation 71
Table 7.6 · Indian grid-average water intensity 71
Table 7.7 · Indian grid-average water intensity 72
Table 7.8 · Source WUE formula 73
Table 7.9 · Embedded water (scope 3), order of magnitude 73
Table 7.10 · Water by cooling technology 74
Table 7.11 · Water by cooling technology 74
Table 7.12 · Peak factors (calculator inputs) 74
Table 7.13 · IT load profile archetypes 75
Table 7.14 · Hot-day profile: 50 MW hyperscale, Hyderabad, adiabatic (Example B), May design day 75
Table 7.15 · Tariff time structures 76
Table 7.16 · DISCOM HT/EHT tariffs (FY2025-26 unless noted) 76
Table 7.17 · Open access and procurement options 77
Table 7.18 · Reliability and connection 77
Table 7.19 · Typical parameter values 78
Table 7.20 · Scale snapshot 78
Table 7.21 · Example A: 20 MW IT colocation, Mumbai (Navi Mumbai), water-cooled chillers + open cooling towe 78
Table 7.22 · Example A: 20 MW IT colocation, Mumbai (Navi Mumbai), water-cooled chillers + open cooling towe 79
Table 7.23 · Example B: 50 MW IT hyperscale, Hyderabad, air-side economiser / adiabatic-assisted cooling 79
Table 7.24 · Example B: 50 MW IT hyperscale, Hyderabad, air-side economiser / adiabatic-assisted cooling 79
Table 7.25 · Example C: 100 MW IT AI campus, Bengaluru (alt.: Visakhapatnam), direct-to-chip liquid cooling 80
Table 7.26 · Example C: 100 MW IT AI campus, Bengaluru (alt.: Visakhapatnam), direct-to-chip liquid cooling 80
Table 7.27 · Example C: 100 MW IT AI campus, Bengaluru (alt.: Visakhapatnam), direct-to-chip liquid cooling 80
Table 7.28 · Data gaps and how to close them 80
Table 8.1 · Definitions and Archetypes 83
Table 8.2 · PUE by archetype 83
Table 8.3 · National and global totals 84
Table 8.4 · Energy per query, token and training run 84
Table 8.5 · WUE by cooling type 85
Table 8.6 · Reported hyperscaler and operator water 86
Table 8.7 · Assumptions [A] 89
Table 8.8 · Per facility (annual) 89
Table 8.9 · Per MW of IT capacity (annual) 90
Table 8.10 · AI data-centre announcements 91
Table 9.1 · Global and Indian benchmarks 93
Table 9.2 · Land and buildings 94
Table 9.3 · Servers, storage and GPUs (IT capex) 95
Table 9.4 · Soft costs, interest during construction and contingency 96
Table 9.5 · Breakdown tables for the three cases 96
Table 9.6 · Breakdown tables for the three cases 97
Table 9.7 · Breakdown tables for the three cases 97
Table 9.8 · Electricity 98
Table 9.9 · Electricity 99
Table 9.10 · Water 99
Table 9.11 · Personnel 99
Table 9.12 · Opex tables per case (stabilised year) 100
Table 9.13 · Opex tables per case (stabilised year) 100
Table 9.14 · Opex tables per case (stabilised year) 100
Table 9.15 · ₹ per MW: capex, revenue, EBITDA, payback, IRR, valuation 101
Table 9.16 · ₹ per MW: capex, revenue, EBITDA, payback, IRR, valuation 101
Table 9.17 · ₹ per MW: capex, revenue, EBITDA, payback, IRR, valuation 102
Table 9.18 · ₹ per rack 102
Table 9.19 · ₹ per kWh: levelised cost of data centre capacity (LCOD) 102
Table 9.20 · Cost per GPU-hour: TCO in India vs market prices 103
Table 9.21 · Cost per GPU-hour: TCO in India vs market prices 103
Table 9.22 · Sensitivity analysis 104
Table 9.23 · Comparing models and markets 104
Table 9.24 · Comparing models and markets 104
Table 10.1 · The twelve dimensions 106
Table 10.2 · Green vs greenwashing: a quick test 107
Table 10.3 · Core metrics and formulas 107
Table 10.4 · Certifications 108
Table 10.5 · Indian standards and policy instruments 108
Table 10.6 · Design levers with quantified savings 108
Table 10.7 · Procurement options in India 109
Table 10.8 · Indian operator snapshot 110
Table 10.9 · Global benchmarks and the AI problem 110
Table 10.10 · Assumptions (flagged estimates) 111
Table 10.11 · Marginal abatement cost (MAC) 111
Table 10.12 · Green Data Centre maturity model (India-calibrated) 113
Table 11.1 · How the model works 116
Table 11.2 · Results: every option the model found 118
Table 11.3 · The hydropower uncertainty 119
Table 11.4 · The hydropower uncertainty 120
Table 11.5 · Three rules for choosing an option 120
Table 12.1 · Part A — Central Framework 123

Table 12.2 · Telangana — the inland AI challenger 126
Table 12.3 · Part C — Master Comparison Table 129
Table 12.4 · Part D — Attractiveness Index 130
Table 13.1 · Key Findings 133
Table 13.2 · Key Findings 134
Table 13.3 · Method 134
Table 13.4 · Method 134
Table 13.5 · Latency table (great-circle km to Mumbai / Delhi / Chennai / Bengaluru → mean RTT) 135
Table 13.6 · Maharashtra 136
Table 13.7 · Tamil Nadu 136
Table 13.8 · Karnataka 136
Table 13.9 · Telangana 136
Table 13.10 · Andhra Pradesh 137
Table 13.11 · Uttar Pradesh 137
Table 13.12 · Haryana 137
Table 13.13 · Delhi 137
Table 13.14 · Gujarat 138
Table 13.15 · West Bengal 138
Table 13.16 · Kerala 138
Table 13.17 · Odisha 138
Table 13.18 · Rajasthan 138
Table 13.19 · Madhya Pradesh 139
Table 13.20 · Secondary states 139
Table 13.21 · National master table (default weights) 139
Table 13.22 · Alternate weightings and rank stability 140
Table 14.1 · Historical series 144
Table 14.2 · Model (a): CAGR, C(t) = C_o(1+g)^t 145
Table 14.3 · Model (b): logistic S-curve, C(t) = K / (1 + e^(−r(t − t_o))) 145
Table 14.4 · Model (b): logistic S-curve, C(t) = K / (1 + e^(−r(t − t_o))) 145
Table 14.5 · Model (c): pipeline conversion, C(t+1) = C(t) + Σ P_stage(t − L) × c_stage 145
Table 14.6 · Model (c): pipeline conversion, C(t+1) = C(t) + Σ P_stage(t − L) × c_stage 146
Table 14.7 · Model (c): pipeline conversion, C(t+1) = C(t) + Σ P_stage(t − L) × c_stage 146
Table 14.8 · Integrated capacity path 146
Table 14.9 · Published forecasts compared 146
Table 14.10 · Electricity (TWh) and share of the grid 147
Table 14.11 · Electricity (TWh) and share of the grid 147
Table 14.12 · Electricity (TWh) and share of the grid 147
Table 14.13 · Carbon trajectory 148
Table 14.14 · Carbon trajectory 148
Table 14.15 · Carbon trajectory 148
Table 14.16 · Build cost 148
Table 14.17 · Build cost 149
Table 14.18 · Build cost 149
Table 14.19 · GPU-hour prices 149
Table 14.20 · GPU-hour prices 149
Table 14.21 · Water demand by city (MLD) 150
Table 14.22 · Water demand by city (MLD) 150
Table 14.23 · Where new capacity goes, city by city 151
Table 14.24 · Where new capacity goes, city by city 151
Table 14.25 · Where new capacity goes, city by city 151
Table 14.26 · Demand drivers (regression) 152
Table 14.27 · Integrated base-case table (2026–2035) [M] 152
Table 14.28 · Parameter table for the calculator, with Monte Carlo distributions 152
Table 15.1 · Site-selection framework (recommended weighting, estimated) 156
Table 15.2 · Siting logic by archetype 157
Table 15.3 · Hub-by-hub suitability (scores are our estimates; 5 = best) 157
Table 15.4 · Cooling design by climate 158
Table 15.5 · Performance targets by archetype (recommended design targets; estimated) 160
Table 15.6 · The current national framework at a glance 160
Table 15.7 · International comparison 161
Table 15.8 · Policy recommendation matrix 162
Table A.1 · Inputs behind the atlas simulator 165
Table A.2 · Key numbers used across the series 165

List of figures

Figure 1.1 · Who owns the world's capacity 12
Figure 1.2 · India by city 18
Figure 1.3 · What a Tier III rating allows 18
Figure 2.1 · Where the power goes, by facility class 20
Figure 2.2 · Where the power goes in a 20 MW Bengaluru colocation hall 21
Figure 2.3 · Global data-centre electricity 22
Figure 2.4 · Rack density and chip power 24
Figure 3.1 · Which Indian cities can cool 40 °C facility water without chillers 31
Figure 3.2 · Liquid-cooling market 32
Figure 4.1 · Scaling to Karnataka's 1 GW target 41
Figure 4.2 · A day in Bengaluru, hour by hour 41
Figure 5.1 · 1 TWh of data-centre load, seven ways 47
Figure 5.2 · A Karnataka day 49
Figure 5.3 · Annual matching hides the gap 51
Figure 6.1 · Scarcity-weighted water footprint 57
Figure 6.2 · The season matters 57
Figure 6.3 · Cooling-tower water balance for a 50 MW hyperscale facility 60
Figure 7.1 · All 8,760 hours of a year 68
Figure 7.2 · Where the water goes 69
Figure 7.3 · Water behind one AI prompt 69
Figure 7.4 · Where grid water comes from, by generating source 71
Figure 7.5 · Grid water intensity by state 72
Figure 7.6 · Peaks size the pipes 72
Figure 7.7 · Power versus water: every cooling technology 73
Figure 7.8 · Inside a cooling tower 74
Figure 7.9 · One hot day, hour by hour 76
Figure 7.10 · Hours a year that adiabatic coolers run wet, by city 76
Figure 7.11 · Cost versus water, by how you buy power 79
Figure 8.1 · Five kinds of data centre 84
Figure 8.2 · Two buildings, side by side 85
Figure 8.3 · On-site water by cooling type, and what operators report 86
Figure 8.4 · How hot is one rack? 87
Figure 8.5 · PUE: where each kind of operator sits 87
Figure 8.6 · Inside one Gemini prompt 88
Figure 8.7 · One AI prompt: energy and water 88
Figure 8.8 · The water you don't see 88
Figure 8.9 · Does liquid cooling save water? Run the trade 89
Figure 8.10 · Two speeds of growth 90
Figure 8.11 · India: from mostly conventional to mostly AI 90
Figure 9.1 · Build cost around the world 96
Figure 9.2 · Capex per MW, three cases 98
Figure 9.3 · Where colo opex goes 101
Figure 9.4 · What moves the colo's cost 103
Figure 9.5 · GPU-hour cost in India 104
Figure 10.1 · PUE thresholds around the world 111
Figure 10.2 · Abatement cost by measure 112
Figure 10.3 · Green data-centre maturity: a self-assessment against today's Indian leaders 113
Figure 11.1 · What gets built as the water cap tightens 117
Figure 11.2 · Rule 1 · Save water only while it is cheaper than buying it 117
Figure 11.3 · Rule 2 · Pick a mix that survives the hydro uncertainty 118
Figure 11.4 · Rule 3 · Don't trade water for carbon 118
Figure 11.5 · Cost against water: every option the model found 119
Figure 11.6 · A year, month by month 119
Figure 11.7 · A week, hour by hour 121
Figure 12.1 · Policy validity 123
Figure 12.2 · India at a glance 124
Figure 12.3 · State attractiveness index at the report weights 131
Figure 13.1 · How stable is the ranking? 140
Figure 14.1 · Capacity 144
Figure 14.2 · Our forecast against everyone else's 144
Figure 15.1 · Eighteen recommendations, on a five-year roadmap 156
Figure 15.2 · Cooling by climate 159
Figure 15.3 · Water: a hierarchy, sized to the peak day 160

About this report and how to read it

This report brings together, in one place, all of the research carried out for the India Data Centre Atlas between October 2025 and October 2026. The research was produced as a series of fifteen parts. Each part answered one question about data centres, and each part fed an interactive chapter of the atlas. The parts covered the types of data centre and how they compete, how a data centre consumes electricity and water, how cooling works, what a single campus asks of a city such as Bengaluru, why the carbon and water behind one unit of electricity differ so much, how water impact should be measured, what a data centre costs to build and run, what makes a data centre genuinely green, how the Indian states regulate and court the industry, where in India new capacity should be built, how much capacity India is likely to have by 2035, how water and energy interact in practice, how conventional and AI data centres differ, which mix of clean power uses the least water, and what design, siting and policy choices India should make next.

The report is written for readers who need detail and who also need to understand it quickly. Every chapter therefore opens with a short explanation of what the chapter covers, a box of key takeaways written as full sentences, and a note on how the chapter connects to the others. The body of each chapter then presents the full analysis, including every table of data that the research collected. Figures drawn from the atlas illustrate the main relationships. Each chapter ends with a list of the sources it relies on, and a consolidated bibliography with web addresses appears at the end of the report.

The report avoids repeating the same material in several places. Where two chapters discuss the same idea, the fuller treatment appears once and the other chapter points to it. Caveats that apply to a particular chapter are kept at the end of that chapter, so that a reader can judge each set of numbers where it is used.

Evidence, units and conventions

Numbers in this report come from three kinds of evidence, and the difference matters when the numbers are used. A **reported** figure has been published or measured by a named organisation, such as the International Energy Agency, the Central Electricity Authority or a company sustainability report. An **estimate** is a forecast, a third-party calculation or a figure whose basis is contested. A **model** figure has been calculated within this research series from stated assumptions, for example the cost of a 20 MW colocation hall or the water used by a 100 MW AI campus. Several chapters label figures with [R], [E] or [M] (and, in some chapters, [A] for an assumption or [D] for a design value). Where no label is given, the sentence names its source, and the chapter caveats describe the main uncertainties.

All money is in Indian rupees unless stated otherwise. One crore is ten million rupees and one lakh is one hundred thousand rupees. Energy is usually given in million units (MU), which is the Indian utility term for one million kilowatt-hours, the same as one gigawatt-hour (GWh). One terawatt-hour (TWh) is one thousand GWh. Water is given in litres, kilolitres (kL, one thousand litres), million litres (ML) or million litres per day (MLD). Capacity is given in megawatts of IT load unless the text says otherwise, which means the power that the servers themselves can draw, not the total power that the whole site draws from the grid.

How the original research parts map to the chapters

The text of each chapter sometimes refers to the original research part by its number, for example "Report 4" or "Part 12". The table below shows which chapter of this report contains each part.

Original part	Chapter	Subject
Report 1	1	Types of data centre
Report 2	2	Anatomy of consumption; AI vs conventional
Report 3	3	Cooling
Report 4	4	Bengaluru water and power
Report 5	5	Every TWh is not the same
Report 6	6	Water impact
Part 12	7	Water and energy calculated
Part 13	8	Conventional and AI compared
Report 7	9	Unit economics
Report 8	10	Green data centres
Part 14 (model)	11	Least-water power
Report 9	12	State policies
Part 10	13	Where to build
Part 11	14	Forecast to 2035
Part 15	15	Recommendations

Executive summary

Why this report exists

Data centres are the factories of the digital economy. Every payment made on a phone, every video watched, every government service delivered online and every question put to an artificial intelligence system is processed inside one. India is in the middle of a rapid build-out of these facilities. About 1.8 gigawatts of IT capacity was operating in the first half of 2026, almost half of it in Mumbai, and the Central Electricity Authority now projects about 26 gigawatts of additional demand from data centres by 2031–32. That is more than fourteen times the capacity that runs today. A build-out of this size will use large amounts of electricity, land and water, and it will be shaped by decisions that are being made now by companies, state governments and national regulators.

This report compiles fifteen pieces of research that together describe how data centres work, what they cost, what they consume, how they are regulated and where they should go. The emphasis throughout is on India, and on the cities and states where the build-out is concentrated, but the global evidence is presented wherever it explains what India should expect.

What a data centre is, and which kinds matter

Data centres differ by who owns them and how their space is sold. Enterprise facilities are owned and used by one organisation. Colocation facilities rent space, power and cooling to many customers. Hyperscale facilities are built by the large cloud and AI companies for their own use and can each draw hundreds of megawatts. Edge facilities are small sites close to users. Hyperscale facilities now hold about half of the world's capacity and are on course for about two-thirds by 2031, while enterprise capacity is flat. India's market is led by colocation and by hyperscale build-to-suit campuses, and the research concludes that India should build a core of hyperscale and colocation capacity in a few hubs, make every new large building ready for AI and liquid cooling, and add a thinner layer of sovereign and edge capacity (Chapter 1).

How data centres use energy, and why AI changes the picture

Data centres used about 415 terawatt-hours of electricity worldwide in 2024, about 1.5% of global demand, and the International Energy Agency expects this to more than double to about 945 terawatt-hours by 2030. Inside a typical facility the servers take about 60% of the power, and cooling takes anywhere from about 7% to more than 30% depending on design and climate. The ratio of total site energy to IT energy is called power usage effectiveness, or PUE. The global average PUE has hardly improved in six years and stood at 1.54 in 2025, while the largest cloud operators report between 1.08 and 1.15, and Indian operators report about 1.5 to 1.75 (Chapters 2 and 8).

AI data centres are not simply larger versions of conventional ones. A rack of current AI servers draws 120 to 142 kilowatts, compared with 5 to 15 kilowatts for a typical colocation rack, so the heat can no longer be carried away by air and must be removed by liquid flowing to the chips. AI campuses are also built at a far larger scale, from one hundred megawatts to more than one gigawatt, and their training workloads swing up and down together in a way that grid operators now treat as a reliability risk. Per megawatt of IT capacity, however, a well-designed AI campus uses about the same energy as a good colocation hall. Its impact comes from its size and density, not from poor efficiency (Chapters 2 and 8).

Cooling: the design decision that sets water use

Air cooling remains the most common method in existing buildings, but it stops being practical somewhere around 20 to 40 kilowatts per rack. New AI capacity is designed around direct-to-chip liquid cooling, and the newest server generations assume that the facility supplies water at up to 45 °C. Water that warm can be cooled by dry coolers, which use no water, for almost every hour of the year in most Indian cities. The research therefore recommends warm-water liquid cooling with dry or adiabatic heat rejection for AI halls in India, with evaporative cooling towers kept only for peak days, and air cooling with economisers for low-density halls in milder cities such as Bengaluru and Hyderabad (Chapter 3).

Water: what is used on site, what is used at the power station, and what it does to a place

A data centre uses water in two places. On site, evaporative cooling towers and adiabatic coolers evaporate water to reject heat. Off site, the coal and gas power stations that generate the data centre's electricity also evaporate water in their own cooling systems. Indian coal plants are legally allowed to use 3.0 to 3.5 litres of water per kilowatt-hour, and surveys show that many exceed this. As a result, for an Indian data centre the grid usually decides the total water footprint, while the cooling technology decides the local, on-site demand. A dry-cooled 100 megawatt AI campus on the average Indian grid still carries about 1.5 billion litres a year of water at power stations, which is more than six times the on-site water of a 20 megawatt colocation hall with cooling towers in Mumbai (Chapters 6, 7 and 8).

Volume alone does not describe impact. A modelled 50 megawatt data centre with evaporative cooling in Bengaluru would withdraw about 1.6 to 2.1 million litres a day, which is less than 0.1% of the city's piped supply. If the same water were drawn from groundwater in Bangalore East, it would equal about 4% of that area's sustainable groundwater, in a taluk where groundwater is already drawn at about three times its recharge. The research therefore argues that water should be reported as withdrawal, consumption and scarcity-weighted impact, by source and by season, and that peak-day demand matters more than the annual average for local supply (Chapters 4 and 6).

Carbon: every terawatt-hour is not the same

The same terawatt-hour of electricity can carry about 10,000 tonnes of carbon dioxide if it comes from nuclear, wind or a portfolio of clean power matched hour by hour, or about one million tonnes if it comes from marginal coal at the evening peak. For an Indian operator, where, when and from what source electricity is bought matters more to the carbon footprint than almost any efficiency gain inside the building. Annual renewable certificates can hide the fact that a data centre still draws coal power at night. Hourly matching, in which clean supply is matched to consumption hour by hour, is the more honest measure (Chapter 5).

What data centres cost

India is one of the cheapest places in the world to build a data centre. A Tier III colocation facility costs about ₹56 crore per megawatt of IT load to build, excluding land, and a hyperscale build-to-suit campus about ₹39 to 42 crore per megawatt. A liquid-cooled AI facility costs about ₹72 crore per megawatt for the building and plant, but about ₹340 crore per megawatt once the GPUs and networking are included, so the building is only about a fifth of the total. Electricity is about three-quarters of a colocation operator's operating costs, but the recovery of capital is the larger part of the full cost of about ₹30 per kilowatt-hour of IT energy. A fully loaded H100 GPU in India costs about ₹177 per hour at 80% utilisation, and utilisation matters more than any other variable in GPU economics (Chapter 9).

What makes a data centre green, and which power mix uses least water

A data centre is green only if it is measured, disclosed and independently verified on at least four dimensions: energy efficiency, hourly carbon-free power, freshwater impact and embodied carbon. For a Bengaluru operator the most bankable route is a design PUE of 1.20 or lower, at least 80% round-the-clock renewable supply through open access or group captive arrangements, and cooling that uses dry coolers or treated wastewater. On the worked numbers this package adds about 5% to capital cost, pays back in about two to three years, and reduces carbon at a negative cost, which means it saves money (Chapter 10).

An hourly optimisation of the power supply for a 50 megawatt-IT data centre in Karnataka shows that the cheapest clean-power mix can use more water than buying all power from the grid, because reservoir hydropower evaporates water from its reservoirs. The recommended mix, 161 megawatts of solar, 86 megawatts of wind and 379 megawatt-hours of four-hour batteries with grid top-up and no hydro, costs ₹5.39 per kilowatt-hour against ₹7.00 for grid power, uses about 218 million litres of water a year against 1,139 million litres for grid power, and supplies 83% of the hours from clean power. Each kilolitre of water it saves costs about ₹85, which is less than Bengaluru's municipal water price (Chapter 11).

Policy and place

No national data-centre law is in force in India. The real rulebook is about sixteen state regimes, and for a large project the deciding factors are the price and reliability of power, access to renewable energy, connectivity and policy stability rather than headline subsidies. Maharashtra, Telangana, Andhra Pradesh, Tamil Nadu and Uttar Pradesh lead on those factors. Karnataka's 2026–31 policy is the only one that pays for measured efficiency through power and water usage effectiveness, while Maharashtra's 2026 revision weakened its green-power condition from 100% to 51%, which is an example of states competing by lowering standards (Chapter 12).

A ranking of fifty candidate cities, built from hard filters for seismic, flood, cyclone, power and groundwater risk and a weighted score across power, connectivity, latency, water, climate, land and policy, places the periphery of Hyderabad first under every weighting tested, with Visakhapatnam second for AI-scale campuses and Navi Mumbai and Chennai irreplaceable for colocation that depends on interconnection (Chapter 13).

How much capacity India will have

The base case of this research puts India's operational capacity at about 4.7 gigawatts of IT load by the end of 2030 and about 10.8 gigawatts by the end of 2035. That is about 28% below the median of published forecasts for 2030, because a back-test shows that consultants have historically delivered only 58 to 78% of the capacity they forecast. By 2035 data centres would use about 83 terawatt-hours a year, about 2.5% of India's electricity, and would require about ₹8 lakh crore of cumulative facility investment (Chapter 14).

What India should do next

The final chapter sets out two siting rules and eighteen policy recommendations. Cloud and colocation facilities should stay near users and cable landings, in Mumbai, Pune, Chennai, Hyderabad, the National Capital Region and Bengaluru, and should use zero-freshwater cooling and treated wastewater. AI training campuses should go where renewable power, land, water and low hazard coincide, such as Visakhapatnam, Jamnagar and the interiors of Andhra Pradesh, Karnataka, Telangana and Rajasthan. The most urgent policy steps are to notify a national policy with mandatory, independently assured reporting of energy, water and carbon performance, to add water-use and hourly clean-power criteria to the IndiaAI and other public compute tenders, to create grid-connection rules for large loads, to ban fresh groundwater use for cooling in over-exploited areas, and to replace open-ended state subsidies with time-limited incentives paid only for verified results (Chapter 15).

What remains uncertain

Several important numbers are uncertain. India has no public national dataset of data-centre efficiency, water use or clean-energy share, so many figures are modelled or rest on a few operator disclosures. Capacity figures differ because organisations count different things, such as IT load, operational capacity, contracted capacity or announced capacity. The water evaporated by Indian reservoirs and the actual water use of Indian thermal plants are not measured publicly. GPU prices and economic lives are contested. Forecasts differ by factors of two to three. The caveats at the end of each chapter describe these limits in detail, and the reader should treat modelled figures as reasoned estimates rather than measurements.

Glossary

Adiabatic cooling
Cooling in which water is sprayed onto pads or into incoming air so that its evaporation lowers the air temperature before the air passes over a heat exchanger. It uses water only in hot hours.

AI training and inference
Training is the process of building an AI model from data, which uses large clusters of GPUs running together for weeks. Inference is the use of a trained model to answer requests, which is spread more evenly through the day and is more sensitive to latency.

ASHRAE TC 9.9
The technical committee of the American Society of Heating, Refrigerating and Air-Conditioning Engineers that publishes temperature and humidity guidelines for data centres, including the W17 to W45 classes for liquid cooling.

Availability zone
A separate data centre or group of data centres within a cloud region, with independent power, cooling and network, so that a failure in one does not affect the others.

Banking (of renewable power)
An arrangement in which surplus renewable energy fed into the grid at one time can be drawn back later, usually for a charge and within limits set by the state regulator.

BESS
Battery energy storage system: a large battery installation used to store electricity and release it later, for example to firm solar power into the evening.

Blowdown
Water deliberately drained from a cooling tower to remove the salts that build up as water evaporates.

Capacity factor
The energy a power plant actually produces in a year divided by what it would produce running at full output all year.

CDU
Coolant distribution unit: equipment that exchanges heat between the liquid loop serving the racks and the facility water loop.

CFE (carbon-free energy)
Electricity from sources that emit no carbon dioxide when generating, such as solar, wind, hydro and nuclear. Hourly CFE is the share of each hour's consumption matched by carbon-free supply in that hour.

Colocation
A business model in which an operator rents space, power, cooling and connectivity in its data centre to many customers.

Consumption (of water)
The part of water withdrawn that is not returned to its source, mainly because it has evaporated.

CRF (capital recovery factor)
The fraction of an initial investment that must be paid each year to repay it with interest over a given life: $r(1 + r)^n \div ((1 + r)^n - 1)$.

CUE (carbon usage effectiveness)
Kilograms of carbon dioxide emitted per kilowatt-hour of IT energy. For a site supplied only by the grid, it equals PUE multiplied by the grid emission factor.

Cycles of concentration (CoC)
How many times the dissolved salts in cooling-tower water are concentrated relative to the make-up water before blowdown. Higher cycles mean less blowdown.

Direct-to-chip liquid cooling (DLC)
Cooling in which liquid flows through cold plates mounted directly on processors and memory, carrying heat away far more effectively than air.

Dry cooler
A heat exchanger that rejects heat from a water loop to the outside air using fans only, without evaporating water.

Edge data centre
A small data centre located close to users to reduce latency, often in a smaller city or at a telecom site.

ERF (energy reuse factor)
The share of a data centre's energy that is reused outside it, for example as heat for buildings or industry.

GPU
Graphics processing unit: the processor used for AI training and inference, such as NVIDIA's H100, B200 and GB300.

Green open access
The right of a consumer above a threshold (100 kilowatts under the 2022 rules) to buy renewable power directly from a generator through the grid.

Group captive
An arrangement in which a consumer owns at least 26% of a power plant and consumes at least 51% of its output, which exempts it from some grid surcharges.

Hyperscale
Very large data centres, typically tens to hundreds of megawatts, built and operated by large cloud and AI companies for their own services.

LCOD
Levelised cost of data centre capacity: the yearly cost of capital plus operating costs divided by the IT energy delivered, expressed in rupees per kilowatt-hour.

Marginal emissions
The emissions of the power plant that changes its output when demand changes, as opposed to the average emissions of all plants running.

MU
Million units: one million kilowatt-hours, equal to one gigawatt-hour.

MVA
Megavolt-ampere: a unit of apparent power used to size grid connections and transformers. MVA equals megawatts divided by the power factor.

PUE (power usage effectiveness)
Total facility energy divided by IT energy. A value of 1.0 would mean that no energy is used for anything other than computing.

Rack
A standard cabinet that holds servers, storage and network equipment. Its power draw, in kilowatts, determines how it must be cooled.

RTC renewable energy
Round-the-clock renewable energy: a contract that supplies renewable power for most hours of the day, usually combining solar, wind and storage.

Scope 2 (location-based and market-based)
Emissions from purchased electricity. Location-based accounting uses the average intensity of the local grid; market-based accounting uses the intensity of contracts and certificates the buyer holds.

Tier (Uptime Institute)
A rating of designed reliability from Tier I (basic) to Tier IV (fault tolerant). Tier III allows maintenance without shutdown and about 1.6 hours of downtime a year.

TSE
Treated sewage effluent: wastewater treated at a sewage plant and reused, for example as cooling-tower make-up water.

Water stress
The ratio of water demand to renewable supply in a place. Highly stressed places have little water to spare for new users.

Wet-bulb temperature
The lowest temperature air can reach by evaporating water into it. It limits how cold evaporative cooling can make water.

Withdrawal (of water)
The total water taken from a source, whether or not it is later returned.

WUE (water usage effectiveness)
Litres of water used on site per kilowatt-hour of IT energy. It should state whether it measures withdrawal or consumption.

PART I · FOUNDATIONS

Chapter 1
Types of data centre: the global picture and the Indian choice

WHAT THIS CHAPTER COVERS

This chapter explains how data centres are classified, by who owns them, by how their space is sold, by how reliable they are designed to be and by how they are cooled. It then measures how much of the world's capacity each type holds and explains why hyperscale and colocation facilities have grown while enterprise facilities have not. The second half of the chapter turns to India: how large the market is, where it is concentrated, what policy drives it, what holds it back, and which types of data centre India should prioritise.

KEY TAKEAWAYS

1. Hyperscale facilities built by the large cloud and AI companies now hold about 48% of global data-centre capacity and are expected to reach about two-thirds by 2031, while enterprise on-premises capacity is flat in absolute terms and shrinking as a share.
2. Colocation, in which an operator rents space, power and cooling to many customers, remains the main way that every organisation other than the hyperscalers obtains capacity.
3. India had about 1.8 gigawatts of operational IT capacity in the first half of 2026, with about half in Mumbai, and a further 3.9 gigawatts was planned or under construction for delivery by 2030.
4. Reliability is classified by the Uptime Institute's Tiers. A Tier III facility, which can be maintained without switching off, allows about 1.6 hours of downtime a year and is the usual standard for colocation and cloud in India.
5. India should not copy the United States mix. It should build a hyperscale-plus-colocation core in a few hubs, make every new large building ready for AI and liquid cooling, and add a thin, policy-backed layer of sovereign and edge capacity, with water-light cooling and renewable power contracts required from the start.

How this chapter connects to the others. The physical consumption of each type is analysed in Chapter 2, and the cost of each type in Chapter 9. Where each type should be located in India is the subject of Chapters 13 and 15.

Hyperscale facilities run by cloud and AI giants are now the main type of data centre in the world. They hold roughly half of global capacity (about 48% in Synergy Research's April 2026 data) and are on track for about two-thirds by 2031. Third-party colocation is the main way everyone else gets capacity, and enterprise on-premises capacity is flat in absolute terms and shrinking as a share. India should not try to copy the US mix. It should build a "hyperscale-plus-colocation core" in a few coastal and inland metro hubs, make every new large build AI-ready and liquid-cooled, and add a thin, policy-backed layer of sovereign and edge capacity. Water-light cooling and renewable power contracts should be required from the start.

1.1 Types of Data Centre

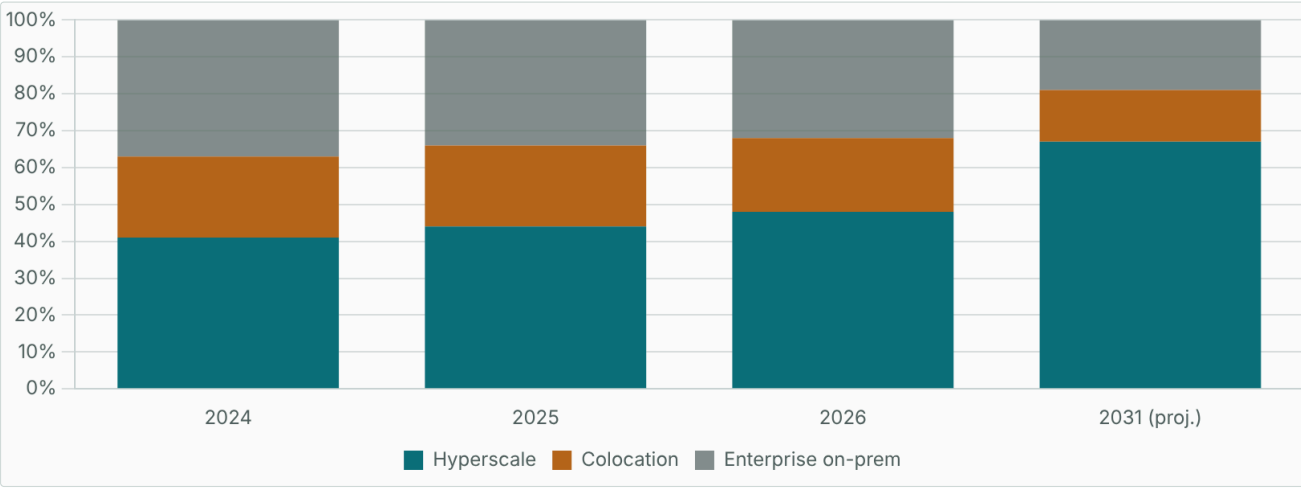


Figure 1.1 Who owns the world's capacity. Share of global data-centre capacity (Synergy Research). 2031 is a projection. *Source: India Data Centre Atlas, chapter built on this chapter's data.*

1.1.1 Deep analysis of the types: what exists, what dominates globally, and why

The classification schemes

Data centres are classified along four independent axes. A single facility can be "hyperscale + Tier III + direct-to-chip-cooled + AI factory" all at once:

1. **Ownership/business model:** enterprise on-premises, colocation (retail/wholesale), hyperscale (self-built or leased), managed services/cloud, edge/micro.

2. **Workload/purpose:** general-purpose cloud, AI/HPC "AI factories" and GPU clusters, sovereign/government.
3. **Physical form and location:** traditional built, modular/containerised/prefabricated, underground, underwater.
4. **Resilience rating:** Uptime Institute Tier I–IV and ANSI/TIA-942 Rated 1–4. **Cooling:** air, rear-door/hybrid, direct-to-chip, immersion.

Types by ownership and business model

Table 1.1 Types by ownership and business model

Type	How it works	Typical scale / density	Cost structure	Advantages	Disadvantages	Typical users
Enterprise / on-premises	Organisation owns and runs its own facility (server room up to a corporate campus)	Under 1 MW to about 20 MW; modal rack about 7.5 kW (Uptime 2025)	All capex borne by the owner; opex for power, staff and maintenance; low utilisation is common	Full control, data sovereignty, custom compliance	Poor economies of scale, older and less efficient plant, hard to upgrade for AI density	Banks, government, defence, large manufacturers, hospitals
Retail colocation	Operator leases racks, cages or suites with power, cooling and security; tenant brings servers	Facilities of 5–50 MW; customers take 1 rack to under 1 MW	Tenant pays monthly rent per rack/kW plus interconnection; operator carries building capex	Converts capex to opex, carrier-neutral interconnection, fast deployment	Higher per-kW price, less control, contract lock-in	Enterprises, BFSI, network operators, SaaS firms
Wholesale colocation / build-to-suit	Operator leases whole halls or buildings, often to a single hyperscaler on long leases	10–100+ MW per building, campuses of hundreds of MW to GW	Operator funds the build (often project-financed against an anchor lease); tenant often pays power at cost plus rent	Speed to market for hyperscalers, bankable long-term cash flows	Concentrated tenant risk, thin margins, power-procurement exposure	AWS, Microsoft, Google, Meta, Oracle, AI labs, neoclouds
Hyperscale (self-built)	Cloud/internet giants design, own and operate their own campuses	Single buildings of 30–100+ MW; campuses of 1 GW+; AI halls at 50–130+ kW/rack	Huge capex funded from balance sheet; lowest unit cost thanks to scale	Lowest cost per MW, custom hardware, best PUE (larger sites average about 1.44, per Uptime 2025)	Very concentrated power and water demand, local opposition, long grid queues	AWS, Microsoft, Google, Meta, Alibaba, Tencent, Oracle
Managed services / cloud	Provider runs the infrastructure and sells compute or storage "as a service" (IaaS/PaaS, managed hosting, GPU-as-a-Service)	Sits on top of hyperscale or colocation capacity	Customer pays per use; provider absorbs capex	Elastic, no hardware ownership, fast access to GPUs	Egress costs, vendor lock-in, data-residency questions	Start-ups, digital natives, enterprises moving to hybrid
Edge / micro	Small facilities close to users or devices (telecom central offices, towers, factory floors, cities)	50 kW to about 5 MW; micro/containerised units at single-rack scale	Low capex per site but high cost per kW; many sites to manage	Low latency, data reduction, local resilience	Hard to operate at scale, uneven demand, limited GPU density	Telcos (5G/MEC), CDNs/OTT, retail, payments, industrial IoT

Reading the table. Moving down from enterprise to hyperscale, cost per kW falls and efficiency rises, but control is lost and concentration risk grows. The same economics have driven every capacity shift of the past decade.

Newer and purpose-based categories

- **AI/HPC data centres ("AI factories") and GPU clusters.** These are facilities designed around thousands to hundreds of thousands of tightly networked accelerators, for training and increasingly for inference. Their design differs from cloud facilities in three ways: rack density, floor loading and cooling. JLL's 2026 outlook says AI training facilities demand about **10 times the power density** of traditional data centres and command **60% lease-rate premiums**. JLL estimates AI was about **a quarter of data centre workloads in 2025** and could be **half by 2030**, with inference becoming the main driver. The IEA estimates AI servers accounted for **24% of server electricity demand and 15% of total data centre energy demand in 2024**, and accelerated-server electricity use is growing about 30% per year against about 9% for conventional servers. The cost structure is dominated by IT: GPUs can outweigh the building several times over. Their main disadvantages are the risk of rapid hardware obsolescence and very large single-site power draws.
- **Sovereign/government data centres.** These are facilities, or dedicated cloud regions, where the hardware, operations, keys or data must stay under national jurisdiction. Examples include national informatics clouds, defence clouds and "sovereign AI" compute. JLL attributes EMEA's 10% CAGR partly to strong demand for sovereign AI clouds driven by privacy regulation. The advantage is legal certainty. The disadvantages are higher cost, smaller scale and slower access to frontier hardware.
- **Modular, containerised and prefabricated data centres.** Factory-built power, cooling and IT modules (from shipping-container units to full prefabricated halls) are assembled on site. They cut construction time, suit edge and remote sites, and allow phased capex. They are also increasingly how hyperscale campuses are built. AFCOM's 2025 State of the Data Center Report found 83% of respondents planned to deploy prefabricated cooling modules, up from 72% in its 2024 report, though AFCOM's 2026 report puts planned cooling-module deployment at 69% (and power modules at 77%). Downsides are design rigidity and transport and logistics limits.
- **Underground and underwater (experimental or niche).** Underground sites in mines and bunkers (common in Scandinavia) get stable temperatures and physical security. Underwater is still experimental. Microsoft's Project Natick ran 855 servers on the seabed off Scotland for just over 25 months and reported failure rates about one-eighth of the land-based control group. But Microsoft confirmed to DCD that "I'm not building subsea data centers anywhere in the world." China has gone further: Highlander deployed a 1,433-ton commercial underwater unit off Hainan, and a Shanghai Lin-gang project with a 2.3 MW first phase claims PUE below 1.15 using offshore-wind power. **Conclusion:** these designs show what passive cooling can do, but they cannot be serviced, refreshed or scaled the way AI hardware cycles require.

Classification by resilience: Uptime Tiers and TIA-942

Table 1.2 Classification by resilience: Uptime Tiers and TIA-942

Uptime Tier	Redundancy	Concurrently maintainable?	Fault tolerant?	Design-intent availability	Approx. annual downtime	Typical use
Tier I – Basic capacity	N, single path	No	No	99.671%	~28.8 hours	Small business, dev/test
Tier II – Redundant components	N+1 components, single path	No	No	99.741%	~22 hours	Non-critical enterprise
Tier III – Concurrently maintainable	N+1, multiple paths (one active)	Yes	No	99.982%	~1.6 hours	Colocation baseline, cloud, BFSI production
Tier IV – Fault tolerant	2N / 2(N+1), multiple active paths	Yes	Yes	99.995%	~26 minutes	Trading, critical health, national systems

Uptime Institute certifies its own Tiers (Roman numerals), and these focus on power and cooling topology. ANSI/TIA-942 uses **Rated 1–4** (Arabic numerals), audited by accredited third parties. It also covers telecom cabling, architecture, fire protection and physical security. TIA dropped the word "Tier" in its 2014 revision at Uptime's request. The availability percentages are customary design-intent figures, not contractual SLAs. **In practice, Tier III / Rated 3 is the commercial standard worldwide and in India.** Tier IV is used selectively, because it typically costs much more per kW. The SBI sector report (Apr 2026) puts AI or Tier IV builds in India at up to INR 75 crore/MW, against INR 50–55 crore/MW for standard Tier III air-cooled builds.

Classification by cooling approach

Table 1.3 Classification by cooling approach

Cooling approach	How it works	Practical rack density	Efficiency	Water use	Status
Air (CRAC/CRAH, hot/cold aisle, economisers)	Chilled or outside air moved through racks	Up to about 15–20 kW/rack (SBI notes air nears its limits beyond about 15 kW)	Fleet average PUE 1.54 (Uptime 2025); new sites about 1.48	High if evaporative towers are used; low with dry coolers	Still most of the installed base: about 75% use perimeter air (Uptime 2025)
Rear-door heat exchangers / hybrid	Liquid-cooled coil on the rack door removes heat from the exhaust air	About 20–50 kW	Better than room air	Moderate	Most common "first step" into liquid
Direct-to-chip (DLC)	Cold plates on CPUs/GPUs; coolant loop to a CDU; residual air cooling	30–130+ kW	PUE about 1.1–1.25 achievable	Low with closed loops and dry coolers	Becoming the standard for AI; about 22% of operators used direct liquid cooling in 2025 (Uptime)
Single-phase immersion	Servers submerged in a dielectric fluid	50–150+ kW	Very low cooling overhead	Very low	Niche; suits edge, crypto and HPC
Two-phase immersion	Fluid boils at the chip and condenses	100–250+ kW	Lowest overhead	Very low	Early stage; fluid cost and environmental concerns

The key takeaway from Uptime's 2025 survey is that the **average** facility is still air-cooled at a modal density of about **7.5 kW/rack** (6.8 kW in 2024), and few sites exceed 30 kW. AI halls, by contrast, start at 30 kW and run past 100 kW. The market has split into two classes: legacy and enterprise facilities on one side, AI-native campuses on the other. Global PUE has been stuck at about 1.5 for six years because the large air-cooled legacy fleet holds the average down, even as new large sites (over 20 MW) average about 1.44.

Global market share and installed capacity by type (2024–2026)

Table 1.4 Global market share and installed capacity by type (2024–2026)

Metric	Value	Source and date
Global data centre capacity	~103 GW (2025) → ~200 GW (2030), 14% CAGR; ~100 GW added 2026–30	JLL 2026 Global Data Center Outlook (Jan 2026)
Hyperscale share of capacity	41% (2024) → 44% (Q1 2025) → nearly half, about 48% (early 2026) → 67% projected by 2031	Synergy Research (2024; Jun 2025; Apr 2026)
Non-hyperscale colocation share	22% (2024–25) → 20% (2026); absolute capacity still growing	Synergy Research
Enterprise on-premises share	56% (2018) → 34% (Q1 2025) → 32% (2026) → 19% by 2031; absolute capacity stable and slightly boosted by on-prem GPUs	Synergy Research
Number of hyperscale data centres	1,189 (Q1 2025); about 800 more in the known pipeline (2026)	Synergy Research
Concentration	20 metro/state markets hold 60% of hyperscale capacity; Northern Virginia 11%, Beijing 6%	Synergy Research (Aug 2026)
Big three cloud providers' share of hyperscale capacity	58%	Synergy Research (Q3 2025)
Hyperscale quarterly capex	~$142 bn in Q3 2025	Synergy Research via The Register (Dec 2025)
Global occupancy	~97%; 77% of capacity under construction pre-leased	JLL (Jan 2026)
APAC capacity	32 GW → 57 GW by 2030; colocation growing ~19% CAGR; on-prem declining 6%	JLL (Jan 2026)
Liquid-cooling adoption	~22% of operators use direct liquid cooling; ~75% perimeter air	Uptime Institute 2025 survey

Electricity demand (IEA, *Energy and AI*, April 2025): - Data centres used about **415 TWh in 2024**, about 1.5% of global electricity. - Base case: about **945 TWh by 2030**, "slightly more than Japan's total electricity consumption today." That is about 15% annual growth. - Base case: about **1,200 TWh by 2035**, within a scenario range of 700–1,700 TWh. - The US and China account for nearly 80% of growth to 2030. - In the US, data centres account for nearly half of electricity-demand growth to 2030. - Globally, data centres account for only about a tenth of demand growth, less than EVs or air-conditioning. - Renewables meet about half of the growth in data centre demand, and natural gas expands by 175 TWh.

These are projections, not outcomes. The IEA's own scenario range is wide, and Sightline Climate (now Currence) estimated in February 2026 that 30–50% of the roughly 16 GW slated to come online in 2026 may be delayed, after 26% of expected capacity slipped in 2025.

Why certain types dominate

1. **Economies of scale.** Hyperscale and wholesale campuses spread fixed costs (substations, cooling plant, security, staff) over hundreds of MW. In India, cost per MW falls from INR 50–55 crore at standard Tier III to INR 35–45 crore at scale (SBI, Apr 2026). Newer and larger facilities also run at lower PUE (Uptime 2025), which compounds the cost advantage, since electricity is the largest opex line.
2. **Cloud adoption.** Synergy notes that in 2012 enterprises spent about twelve times as much on their own data centre hardware and software as on cloud services. That ratio has since reversed. Enterprises keep only the workloads that must stay on-premises, put the rest into colocation, and rent elastic compute from hyperscalers.
3. **AI workloads.** Training needs contiguous, very high-density, liquid-cooled halls with tens to hundreds of MW on one network fabric, and only hyperscalers, neoclouds and large wholesale operators can build these. Inference is spreading back toward metros and the edge, which supports colocation and edge demand.
4. **Latency.** Interconnection-rich metro colocation (for cloud on-ramps, exchanges and CDNs) and edge sites survive because physics caps how far some workloads can be from users.
5. **Regulation and data localisation.** Privacy and sovereignty rules push in-country builds and sovereign clouds. This favours domestic colocation and local hyperscale regions over offshore capacity.

6. **Energy.** Power has become the binding constraint, and it decides location more than land or tax does. Large players can sign PPAs, self-generate, or move to "power-first" frontier sites. Smaller enterprises cannot.

Leading hubs and their constraints

- **Northern Virginia (US).** It is still the world's largest market, with 4,496.5 MW of inventory in H1 2026. Vacancy is about 0.2% and all 2026 deliveries are pre-committed (CBRE, H1 2026). Atlanta has overtaken it in capacity under construction (2,882 MW). Across North America's primary markets, supply hit a record 10,903 MW while vacancy fell to 1.4%. Local approvals are tightening too: DCD reported in September 2026 that Loudoun County's board voted to pause new data centre applications.
- **Dublin (Ireland).** Dublin has about 1,150 MW in operation (mid-2025), and data centres used 22% of metered national electricity in 2024. A de facto grid-connection moratorium ran from 2021 until the CRU's new Large Energy User policy in December 2025. Under that policy, new data centres must bring on-site or nearby generation or storage able to match their demand.
- **Singapore.** A moratorium imposed in 2019, when data centres used about 7% of national electricity, was replaced by a selective allocation regime that sets strict PUE and green-energy conditions.
- **Frankfurt, London, Amsterdam, Paris.** These are capped by grid capacity. Frankfurt colocation vacancy was 3.1% in mid-2026, the tightest in the FLAP-D group against a 6.4% average, per JLL's EMEA mid-year 2026 report, which also notes Frankfurt's 24-month-plus lead times.
- **Beijing, Tokyo, Sydney, Mumbai.** These are the big APAC hubs. CBRE named Mumbai as one of the new hotspots that benefit from power constraints in older hubs.

The common lesson for India: hubs grow until power, water or local politics stop them. The winners plan grid, water and land ahead of demand.

1.1.2 Indian perspective: which types to build, why, and how

Current state of the market

India vs. global capacity

Table 1.5 Current state of the market

Metric	India	Global	Source and date
Operational capacity	~1.5 GW (MeitY/PIB, 2025); 1,123 MW (JLL, H1 2025); ~1.7 GW (CBRE, end-2025); 1,789 MW in top seven markets (C&W, H1 2026)	~103 GW (2025)	PIB 13 Mar 2026; JLL; CBRE Apr 2026; C&W Sep 2026
Growth	375 MW (2020) → ~1.5 GW (2025); 440 MW added in 2025; ~500 MW expected in 2026	14% CAGR to 2030	PIB; CBRE
2030 outlook	3,860 MW pipeline (509 MW under construction, 3,351 MW planned) to 2030 (C&W); total of ~4.5–6.5 GW (CEEW base cases), ~6.5 GW colocation (SBI), up to ~8 GW (Jefferies)	~200 GW	C&W; CEEW; SBI; JLL
Development capital needed	US$29.9 bn through 2030	Up to US$3 trn by 2030	C&W; JLL
Share of national electricity	~0.5–0.8% (2024) → ~2.6% (2030)	~1.5% (2024) → under 3% (2030)	CEEW; IEA
Penetration	~825,333 people per MW of capacity, among the lowest in APAC	n/a	C&W (Sep 2026)
Colocation vacancy	14.9% (H1 2026)	~3% (97% occupancy, per JLL)	C&W; JLL

The private estimates differ because they count different things: colocation only vs. colocation plus hyperscaler self-builds, top seven cities vs. national, and IT load vs. facility load. The direction is clear in all of them. India has roughly 1.5–1.8 GW today, and it will have 4–8 GW by 2030 if announced projects deliver. That would still be only a few per cent of global capacity, for a country of more than 1.4 billion people. India's higher vacancy (about 15%) compared with global hubs means supply is currently arriving ahead of absorption in some submarkets. That is a healthy buffer, but it is also a warning against speculative builds.

Key hubs (Cushman & Wakefield, H1 2026)

Table 1.6 Current state of the market

Hub	Operational (MW)	Share of 3,860 MW pipeline	Why it is there
Mumbai / Navi Mumbai	890 (about 50%; expected to pass 1 GW by end-2026)	45% (1,726 MW)	Most subsea cables, BFSI demand, power availability
Chennai	214	9% (368 MW)	Second subsea cable hub; Tamil Nadu policy
Delhi NCR / Noida	183	10% (387 MW)	Northern demand centre, government, UP incentives
Bengaluru	162	3% (100 MW)	IT/SaaS demand; constrained by water and land
Hyderabad	151	18% (710 MW)	Inland, low seismic risk, Telangana power policy, Microsoft/AWS regions
Pune	133	7% (258 MW)	Mumbai overflow, AWS/Microsoft presence
Kolkata	22	2%	Eastern gateway; WB policy promotes edge
Visakhapatnam (emerging)	Small today	Gigawatt-scale campuses announced	New east-coast subsea gateway, Andhra Pradesh incentives

Major players - **Colocation leaders.** STT GDC India, CtrlS, NTT, Nxtra (Airtel), Yotta and Sify together account for more than 1 GW installed and plan more than 4.5 GW more (SBI, Apr 2026). Market-intelligence vendor figures for 2025 put the portfolios at roughly: STT GDC about 400 MW, Yotta about 434 MW, NTT about 268 MW, CtrlS about 250 MW and Nxtra over 200 MW. These are vendor estimates and may count planned capacity. Other entrants named by SBI include AdaniConneX, Digital Connexion, Equinix, CapitaLand, RMZ–Colt and L&T Vyoma, plus Web Werks (now Iron Mountain). - **Hyperscaler commitments.** - **Google:** about US$15 bn over 2026–2030 for its first AI hub, a gigawatt-scale campus at Visakhapatnam with AdaniConneX and Airtel. It includes a new international subsea gateway, and Google calls it its largest AI hub investment outside the US (Oct 2025). - **Microsoft:** US$17.5 bn over four years for cloud and AI infrastructure, with a new hyperscale region due by mid-2026 (Dec 2025). - **Amazon:** more than US$35 bn across all its India businesses by 2030 (Dec 2025). This builds on AWS's US$12.7 bn cloud-infrastructure commitment for Maharashtra and Telangana. - **Domestic AI campuses.** -

Reliance: a Jamnagar AI campus targeting up to 3 GW, with an estimated US$20–30 bn cost per Bloomberg, built on Nvidia Blackwell chips and an adjacent green-energy complex. It has also announced a 1 GW twin campus in Andhra Pradesh. - **Adani:** about US$100 bn by 2035 for renewable-powered data centres, targeting 5 GW for AdaniConneX. - **Tata:** TCS's HyperVault (about ₹18,000 crore for more than 1 GW). OpenAI and Tata have announced 100 MW, scaling to 1 GW. - **Caveat:** these are announcements and targets. Many are multi-year, and some depend on power and financing that are not yet secured.

Demand drivers. - **Digital public infrastructure.** Aadhaar, UPI and ONDC generate huge transaction volumes. RBI's 2018 payments data-localisation circular requires that data to be stored in India. - **BFSI.** Banks and insurers are moving from captive data centres to colocation for compliance and cost reasons (SBI). - **OTT, e-commerce and mobile.** CBRE identifies hyperscalers as 50–55% of market activity. - **AI adoption.** This comes from enterprises, start-ups (for example Sarvam AI, which trained its models on IndiaAI compute) and Indian-language foundation models. - **Global capability centres and SaaS exporters.** - **5G.** By 2027 more than 75% of Indians are expected to have internet access, and 5G will push edge demand (SBI).

Policy drivers

- **DPDP Act 2023 and DPDP Rules (notified 14 Nov 2025).** The regime is conditional and risk-based. Cross-border transfers are allowed except to jurisdictions the government restricts. Significant Data Fiduciaries carry extra obligations, and the government can require local storage for specified data. Combined with sector rules from RBI (payments) and SEBI, this pushes regulated data into Tier III/IV domestic facilities.
- **Draft National Data Centre Policy (MeitY, Nov 2020).** It proposed infrastructure status, single-window clearances and four Data Centre Economic Zones, but **it was never notified.** MeitY has restarted consultations, and is also working on national standards for AI data centres. As of October 2026, **India still has no binding national data centre policy**, and 15 states have their own policies (CEEW).
- **Infrastructure status (2022).** Data centres (CEEW specifies those above 5 MW) were added to the Harmonised Master List of infrastructure sub-sectors. This opened access to longer-tenor infrastructure debt, ECBs and infrastructure debt funds.
- **Union Budget 2026–27 and Finance Act 2026.** These created an income-tax exemption, available until 2047, for foreign companies' income from procuring data centre services from a "specified data centre" in India and serving Indian customers through local resellers. It is aimed squarely at deepening hyperscaler commitments.
- **IndiaAI Mission (approved March 2024; ₹10,372 crore over five years; ₹4,563 crore for compute).** It uses a PPP model in which the government empanels private GPU operators and subsidises access at about ₹65/GPU-hour instead of building state data centres. **38,231 GPUs** were onboarded through 14 empanelled providers as of March 2026 (PIB). At the India AI Impact Summit in February 2026, MeitY announced about 20,000 more under "AI Mission 2.0." Press reporting suggests actual spending has lagged the headline allocation: a Daily Pioneer op-ed (2026) says the previous year's ₹2,000 crore budget "saw actual utilisation nearer Rs 800 crore," and Outlook Business reported about ₹111.85 crore in GPU subsidies disbursed; both are secondary press sources.
- **Efficiency rules.** MeitY tenders for AI cloud infrastructure require PUE below 1.35. Leading Indian operators report PUE of 1.4–1.5. Other levers are BEE/ECBC benchmarks, SEBI BRSR Core disclosures and the IGBC Green Data Centre rating (SBI).

State policies (key terms)

Table 1.7 Policy drivers

State	Policy	Key incentives	Notes
Maharashtra	IT/ITeS Policy 2023 (data centres treated as IT units, with essential-service status)	100% stamp-duty exemption; electricity-duty exemption (described as permanent in 2023, while a June 2026 notification cites 20 years); ₹1/unit power subsidy for 5 years outside Zone I; renewable open access	Hosts about 50% of capacity; green data centre parks policy reported
Tamil Nadu	Data Centre Policy 2021	Eligibility above ₹500 crore and at least 30% renewables; 100% electricity-tax subsidy for 5 years; dual grid for 50 MW+; land and stamp-duty concessions	Policy window ended March 2026; no extension notified as of April 2026
Telangana	Data Centres Policy 2016	Power at cost of generation; dual grid; exemption from power cuts; renewable open access; building-fee and lease-rental rebates	S&P Global rates it India's most attractive data centre policy package
Uttar Pradesh	DC Policy 2021 → **new policy approved 6 Jul 2026**	2021: capital and interest subsidies, 100% stamp duty, 10-year electricity-duty waiver, wheeling/transmission relief (about ₹21,343 crore of projects approved). 2026: targets 2 GW and more than ₹2 trillion, covers edge data centres, AI-compute and green incentives, 50% renewables target by 2030	Noida is the main northern hub
West Bengal	DC Policy 2021	100% stamp duty and registration; 5-year electricity-duty waiver; industrial tariff; dual grid; preferential incentives for edge data centres in IT/industrial parks	5-year term ran to about September 2026
Andhra Pradesh	DC Policy 4.0 (2024–29)	100% net SGST reimbursement or 10% capital subsidy; 100% stamp duty; industrial tariff; dual grid via PPP with VGF; custom packages for mega projects over 50 MW	Target given as 200 MW in the G.O. and "up to 1 GW" in the annexure; Vizag is the anchor
Gujarat	IT/ITeS Policy 2022–27	Capex support of 25% up to ₹150 crore for greenfield Tier III+ data centres; ₹1/unit power subsidy for 5 years; 100% electricity-duty reimbursement for 5 years	Jamnagar mega-campus

Constraints and enablers

- **Power.** This is the main long-term constraint. Data centres used about 13 TWh in 2024 (0.8% of demand), and CEEW projects about 57 TWh (2.6%) by 2030. The Ministry of Power estimates data centre electricity demand could reach **13.56 GW by 2031–32** (PIB, Mar 2026). A data centre can be built in about two years, but transmission takes longer. The main enabler is that India added a record 44.5 GW of renewable capacity in 2025 (CBRE). The difficulty is converting variable solar and wind into the round-the-clock, 99.98%-reliable supply a Tier III site needs. That requires firm-and-dispatchable contracts, storage, green open access and dual-grid feeds. Cost is driven by state tariffs, cross-subsidy surcharges and open-access charges, which is why state power concessions matter so much.
- **Water and climate.** CEEW estimates data centres used about 150 billion litres in 2025, which could more than double by 2030. The underlying estimate comes from Mordor Intelligence. A 100 MW hyperscale facility using evaporative cooling can use about 800,000 litres a day. Bengaluru, Chennai and Gurugram are water-stressed. India's high temperatures and, on the coasts, high humidity limit free-air economisation and make evaporative cooling both thirsty and less effective. This is a strong argument for closed-loop liquid cooling and dry coolers.

- **Land.** Land in Mumbai and Chennai is scarce and expensive, which is pushing builds to the periphery (Navi Mumbai, Panvel, Oragadam) and to new hubs (Vizag, Jamnagar).
- **Submarine cable landings.** India has about 17–18 international cables at 14 landing stations in five cities: Mumbai, Chennai, Kochi, Thoothukudi and Thiruvananthapuram. Carnegie India counted 18 cables across 23 distinct landings in 2025. **Fifteen of 17 cables land within about 6 km of Versova beach in Mumbai, and about 95% of international bandwidth comes through Mumbai.** That is a serious single-point-of-failure risk. Singapore, by comparison, has 26–28 cables. New east-coast landings (SEA-ME-WE 6 at Chennai in 2025, and Google's Vizag gateway) are the most important connectivity enabler for India's next phase.
- **Seismic risk.** Under India's BIS seismic zoning, Delhi NCR sits in a higher-hazard zone than the Deccan plateau. Hyderabad is among the lowest-risk major metros, which is one reason it is a natural inland disaster-recovery and hyperscale hub. Mumbai and Chennai are in moderate zones. Developers should check site-specific hazard against the current BIS map, including coastal flooding and cyclone exposure in Chennai and Vizag.
- **Skills and supply chain.** SBI flags gaps in local manufacturing of power systems, HVAC, lithium-ion storage and modular construction, which raise import dependence, costs and timelines. The workforce needs electrical, mechanical, liquid-cooling and DCIM skills at the scale of tens of thousands of technicians.

Recommendations: which types India should prioritise, and why

Priority 1: Hyperscale and wholesale colocation campuses in four or five hubs (the core). *Why:* This is where global scale economics, India's localisation rules and hyperscaler demand all point. According to SBI's April 2026 sector report, "Hyperscalers contribute 60–70% of new demand in co-location centres." Concentrating builds also lets the grid, water and fibre be planned once rather than piecemeal. *Where:* Mumbai/Navi Mumbai and Chennai for cables, Hyderabad for inland low-seismic power, Noida for the north, and Vizag as the second coastal gateway. Use paired-region siting: a coastal cable hub plus an inland availability zone about 300–700 km away (Mumbai–Pune/Hyderabad, Chennai–Hyderabad/Bengaluru, Vizag–Hyderabad).

Priority 2: Build every new large facility AI-ready. *Why:* AI is heading toward half of global workloads by 2030 (JLL). Retrofitting air-cooled halls costs far more than building them right the first time. *How:* - Design structure and electrical systems for 30–130 kW/rack in at least part of each campus: higher floor loading, taller ceilings, liquid distribution and larger busways. - Standardise on **direct-to-chip liquid cooling with closed loops and dry coolers**, with immersion reserved for specialised HPC and edge. - Require PUE of 1.35 or lower, matching the MeitY AI-tender benchmark. - Budget INR 60–75 crore/MW for AI-grade builds rather than INR 50–55 crore. - Avoid speculative GPU halls without anchor tenants. SBI itself warns that more efficient models (the "DeepSeek effect") can lower density needs.

Priority 3: Sovereign and government cloud/AI capacity, using PPP rather than state-built facilities. *Why:* DPDP, RBI and SEBI rules, defence and national-security workloads, and Indian-language AI all need assured domestic compute. The IndiaAI "empanel-and-subsidise" model reached 38,000+ GPUs without the state taking hardware-obsolescence risk. *How:* - Extend that model to sovereign cloud regions run by Indian-controlled operators in Tier III/IV facilities with MeitY empanelment. - Use NIC/MeghRaj for core government systems. - Tie subsidy to utilisation and openness, not to the number of GPUs procured.

Priority 4: Green, renewable-powered campuses, placed near renewables where latency allows. *Why:* Data centres could take about 2.6% of national demand by 2030. Carbon intensity matters to both hyperscaler buyers and SEBI BRSR disclosures, and RE-rich states (Gujarat, Rajasthan, AP, Tamil Nadu, Karnataka) have surplus green power. *How:* - Use round-the-clock hybrid PPAs (solar + wind + storage) under green open access, plus captive or group-captive plants. Jamnagar's adjacent green-energy complex and Adani's renewable-linked campuses are the template. - Site latency-tolerant AI *training* inland or near RE parks. Keep latency-sensitive *inference* and interconnection in metros. - Make state incentives conditional on renewable share. Tamil Nadu's 30% threshold and UP's 50%-by-2030 target are good models.

Priority 5: Modular edge data centres for Tier-2/3 cities. *Why:* 5G, UPI and fintech, OTT and CDN caching, gaming, and Indian-language AI inference all benefit from sub-20 ms latency to the next 500 million users. Edge is real but small in MW terms: Nxtra already runs 120+ edge facilities across 65+ cities, and the UP 2026 and West Bengal policies explicitly cover edge. *How:* - Use prefabricated 0.1–2 MW modules at telecom exchanges, towers and government IT parks, run as networks by telcos (Airtel/Nxtra, Jio, BSNL partnerships) and CDN/colocation operators. - Design for heat with sealed liquid- or immersion-cooled pods that do not depend on local water. - Do not over-invest: edge is a complement to the core, and its share of global capacity is not growing.

What not to prioritise: new large enterprise-owned data centres (move them to colocation and hybrid cloud), Tier IV everywhere (use it only for truly critical systems), and underwater or other experimental formats.

How: the implementation playbook

1. **Financing.** - Use infrastructure-status debt (long tenor, ECBs, InvITs and REITs for stabilised assets) and anchor-tenant project finance. - SBI's sector report recommends debt-service coverage covenants, margin ratchets linked to PUE, escrowed cash flows, fixed-price EPC, and anchor commitments for a significant share of initial capacity. - Sustainability-linked loans and green bonds should be tied to PUE, WUE and RE share. - Data centre EBITDA margins of 40–55% (SBI) make the sector bankable if underwriting avoids speculative builds.
2. **PPP.** Use VGF-backed dual-grid and shared substations in data centre parks, as Andhra Pradesh already offers. Use PPP sovereign cloud and GPU capacity through the IndiaAI model. Use state data centres under MeitY schemes for e-governance.
3. **Policy.** - Notify a binding National Data Centre Policy. It should create Data Centre Economic Zones with pre-cleared land, grid and water, set uniform standards for security, fire and disaster recovery, and include a national spatial plan for siting. - Harmonise state incentives around performance (PUE, WUE, RE share) rather than a race to the bottom on subsidies. Renew the lapsed Tamil Nadu and West Bengal policies. - Clarify DPDP localisation categories early so investment can be sized correctly.
4. **Power.** - Plan transmission to data centre parks 3–5 years ahead. - Standardise green open access and banking rules. - Allow behind-the-meter storage and gas or green-hydrogen backup instead of diesel. - Consider an Irish-style "bring-your-own-firming" rule for very large loads.
5. **Cooling and water.** - Make closed-loop liquid cooling plus dry or adiabatic coolers the default for new builds. - Require treated wastewater or recycled water (not potable supply) for any evaporative use, with Zero Liquid Discharge. - Mandate WUE disclosure. - Use elevated chilled-water temperatures so chiller-free hours can be extended even in hot climates.

6. **Connectivity.** Speed up permits for new cable landing stations outside Versova: Chennai, Vizag, Kochi, and possibly Odisha or Gujarat. Build diverse terrestrial routes between hubs and promote neutral interconnection exchanges in Tier-2 cities.
7. **Skills and supply chain.** - Create a data centre skills mission (through NSDC and ITIs/polytechnics) for critical-facility technicians, liquid-cooling and electrical engineers. - Use PLI-style incentives for domestic manufacture of UPS, switchgear, CDUs, chillers, Li-ion BESS and prefabricated modules.

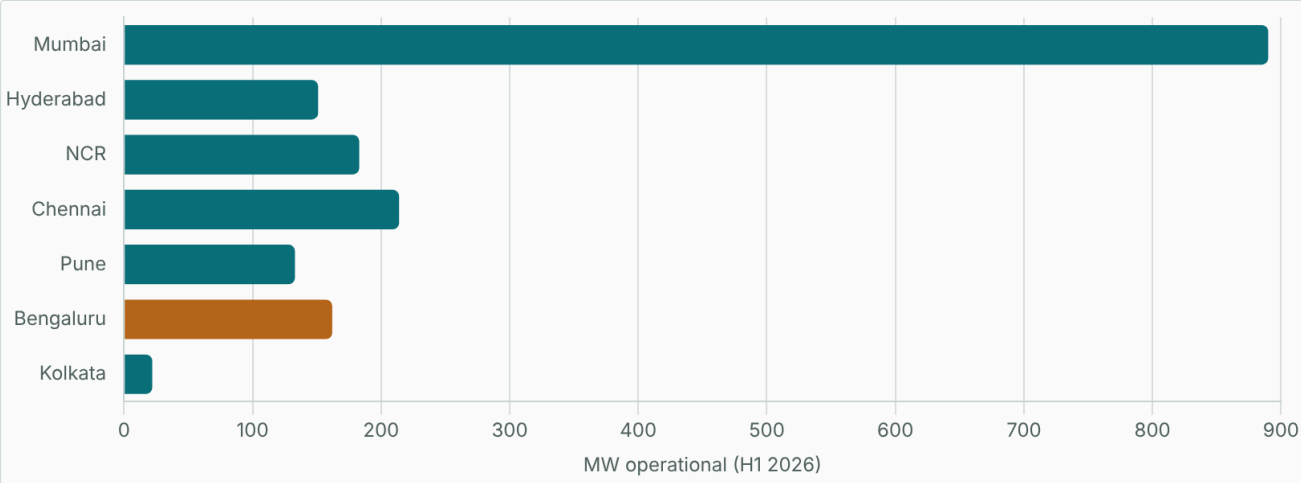


Figure 1.2 India by city. Megawatts, H1 2026 (Cushman & Wakefield). Switch between what runs today and what is coming by 2030. *Source: India Data Centre Atlas, chapter built on this chapter's data.*

1.2 Caveats

- **Capacity figures differ by methodology.** MeitY/PIB (~1.5 GW, 2025), JLL (1,123 MW, H1 2025), CBRE (~1.7 GW, end-2025) and C&W (1,789 MW, H1 2026) measure different scopes. A figure of "1.57 GW in 2026" appears in some media but could not be confirmed in an official release. Global figures (JLL's 103 GW; Synergy's shares) are also model-based estimates.
- **Projections are not outcomes.** IEA 2030 and 2035 figures, JLL's 200 GW, India's 4–10 GW range, and the hyperscaler and conglomerate pledges (Reliance 3 GW, Adani 5 GW, Google 1 GW+) are forward-looking. Sightline Climate (now Currence) estimated in February 2026 that 30–50% of the capacity slated for 2026 globally may be delayed, after 26% slipped in 2025.
- **Some figures are secondhand or vendor-sourced.** These include operator portfolio MW, India's water-use estimate (originally Mordor Intelligence), liquid-cooling market shares, and the IndiaAI disbursement claims. They should be treated as indicative.
- **State policy terms change often.** Several state policies (Tamil Nadu, West Bengal, UP 2021) expired or were replaced in 2026, and some incentive figures differ between official texts and summaries. Check against the latest gazette notification before making investment decisions.



Figure 1.3 What a Tier III rating allows. Each square is one of 120 slices of a year; red slices show allowed downtime, magnified 60 times. *Source: India Data Centre Atlas, chapter built on this chapter's data.*

Sources cited in this chapter

Organisations and publications named in this chapter's text, in order of how often they are cited. Full details are in the bibliography.

- JLL, Global and India Data Center Outlooks
- State Bank of India sector report on data centres (April 2026)
- Ministry of Electronics and Information Technology (MeitY), Data Centre Policy draft and IndiaAI
- Uptime Institute, Global Data Center Surveys and Tier Standard
- Synergy Research Group, capacity share data
- CBRE, India data centre market reports
- Microsoft sustainability reports and datacenter design posts
- Council on Energy, Environment and Water (CEEW) studies
- Google environmental reports, efficiency data and Gemini inference paper
- Amazon Web Services sustainability reporting
- AdaniConneX disclosures
- Press Information Bureau releases
- International Energy Agency (IEA), Energy and AI (2025) and updates
- Nxtra by Airtel disclosures
- Bureau of Indian Standards (IS 1893, ISO/IEC 30134 adoption)
- Meta sustainability reports and data index
- NTT DATA ESG data
- STT GDC India ESG report
- CtrlS disclosures
- Yotta Data Services disclosures
- Reliance Industries announcements
- Mordor Intelligence market estimates
- Sightline Climate (Currence) delay tracking
- DatacenterDynamics reporting
- Cushman & Wakefield, India data centre market updates (H1 2026)
- Ministry of Power, orders on ISTS waiver and green open access
- Bureau of Energy Efficiency
- NVIDIA product specifications and roadmap
- Equinix sustainability reports
- Sify Infinit Spaces sustainability report
- S&P Global (451 Research) India data centre analysis
- Ireland Commission for Regulation of Utilities, large-energy-user connection policy

PART I · FOUNDATIONS

Chapter 2
Anatomy of consumption: how a data centre uses energy, and how AI data centres differ

WHAT THIS CHAPTER COVERS

This chapter takes a data centre apart to show where its electricity goes, from the servers and their processors, memory, storage and networking to cooling, power conversion and lighting. It defines the efficiency metrics used throughout the report, describes how much electricity data centres use globally and regionally, and estimates the energy used by different kinds of workload. The second half compares AI data centres with conventional ones across purpose, hardware, rack density, cooling, power architecture, networking, campus scale, cost, utilisation, reliability, water and grid impact.

KEY TAKEAWAYS

1. Data centres used about 415 terawatt-hours of electricity in 2024, about 1.5% of global electricity demand, and the International Energy Agency's base case expects about 945 terawatt-hours by 2030.
2. In a typical facility the servers take about 60% of the electricity, and cooling takes between about 7% and more than 30%, depending on the design and the climate.
3. Power usage effectiveness (PUE) is the ratio of total facility energy to IT energy. A PUE of 1.5 means that for every unit of energy used by the servers, another half unit is used by cooling, power conversion and other loads.
4. AI racks draw ten to twenty times more power than conventional racks, liquid cooling is mandatory, power distribution is moving to 800 volts of direct current, campuses are built at gigawatt scale, and fit-out costs about three times as much per megawatt.
5. The synchronised power swings of AI training clusters are now recognised by grid operators, including the North American Electric Reliability Corporation in May 2026, as a risk to grid reliability.

How this chapter connects to the others. Chapter 3 explains how the heat described here is removed, and Chapter 8 returns to the comparison of AI and conventional data centres with a detailed treatment of water and energy at the level of a whole site.

Data centres used about 415 TWh of electricity in 2024, roughly 1.5% of global demand. Servers take about 60% of a typical facility's power and cooling takes anywhere from about 7% to over 30%. AI data centres are not simply bigger conventional facilities. They are a different class of industrial plant: racks draw 10–20 times more power, liquid cooling is mandatory, power architecture is moving to 800 VDC, campuses run at gigawatt scale, and fit-out costs about three times as much per MW. Their load behaviour is now a recognised grid-reliability risk.

2.1 Anatomy of Data Centre Consumption

2.1.1 Electricity breakdown by subsystem

The IEA's *Energy and AI* report (April 2025) gives the canonical modern split:

- **Servers:** about 60% of electricity in modern data centres on average. This rises to about 75% in large AI-optimised hyperscale sites, where chips draw 2–4 times more power than conventional ones.
- **Storage:** about 5%.
- **Networking:** up to 5%.
- **Cooling and environmental control:** about 7% in efficient hyperscale sites, rising to over 30% in less-efficient enterprise sites.
- **Remainder:** power conversion and distribution losses (UPS, transformers, PDUs), lighting and building loads.

CEEW's India study (February 2026), citing the IEA, notes that servers and cooling together take nearly 70% of electricity. The SBI report says cooling can account for up to 40% of energy use in Indian facilities.

Diagram in words: the power chain in a conventional facility Grid (e.g. 33/66/132 kV) → step-down transformer (about 1–2% loss) → switchgear → double-conversion UPS (about 3–8% loss in legacy units, about 2–4% in modern ones) → PDU/RPP (about 1–3%) → server PSU (AC to 12 V, about 4–10%) → voltage regulators on the board → chip. In parallel: chillers, CRAH/CRAC fans, pumps and cooling towers, plus lighting, security and offices.

Typical percentage splits by facility class. These are the author's synthesis, derived from PUE ranges and the IEA component shares. They are indicative, not measured.

Table 2.1 Electricity breakdown by subsystem

Subsystem	Legacy/small enterprise (PUE about 1.8–2.0)	Typical enterprise/colo (PUE about 1.5–1.6)	Hyperscale cloud (PUE about 1.1–1.2)	AI liquid-cooled (PUE about 1.1–1.3)
IT load (servers, storage, network)	50–55%	62–67%	83–91%	77–90%
Cooling (chillers, CRAH/CRAC, pumps, towers, fans)	30–40%	25–30%	5–10%	7–15% (CDUs, pumps, dry coolers, residual air)
Power conversion and distribution losses (UPS, transformers, PDUs)	8–12%	6–8%	2–5%	3–6%
Lighting and other (security, offices, humidification)	2–5%	1–3%	about 1%	about 1%

A useful cross-check comes from Google's own Gemini serving measurement. For a median prompt, TPUs used 58% of energy, host CPU and DRAM 25%, idle reserve machines 10%, and data-centre overhead (cooling and power conversion) only 8%. That is consistent with a fleet PUE of about 1.09.

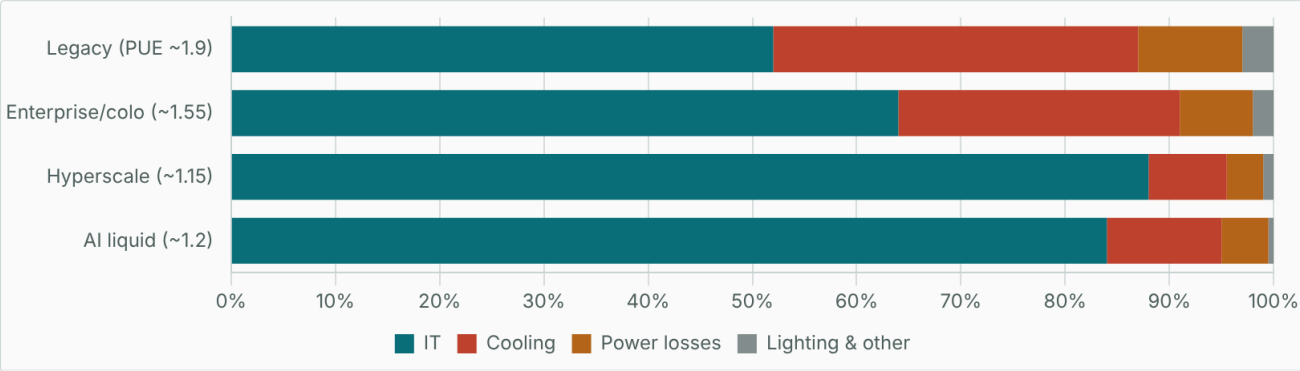


Figure 2.1 Where the power goes, by facility class. Share of facility electricity, indicative (series synthesis from PUE ranges and IEA component shares). *Source: India Data Centre Atlas, chapter built on this chapter's data.*

2.1.2 Inside the IT load: CPUs, GPUs, memory, storage and networking

Applying the IEA shares to IT load alone, a typical facility's IT power splits about 85% servers, about 7% storage and about 7% networking. Within a server, the split depends on the class:

Table 2.2 Inside the IT load: CPUs, GPUs, memory, storage and networking

Component	Conventional CPU server (indicative)	AI accelerated server/rack
CPUs	about 40–55% of server power	about 10–15% (host CPUs, e.g. Grace)
GPUs/accelerators	0% (or a small inference card)	about 60–75% (e.g. 72 × B200 at about 1.0–1.2 kW = about 72–86 kW of a 120 kW NVL72 rack)
Memory (DRAM/HBM)	about 15–25%	HBM is counted in the GPU package; host DRAM is a few %
Local storage (SSD/HDD)	about 5–10%	a few %
Fans, PSU losses, NICs	about 10–20%	Switch trays (NVLink), NICs/DPUs and PSU losses about 10–15%; fans are minimal with liquid cooling

The Gemini figures (TPUs 58%, host 25%) illustrate the accelerator-dominant profile in production.

Idle versus active power and utilisation. Conventional servers are poorly energy-proportional: many still draw roughly a third to half of peak power at idle, and enterprise fleets often run at 10–30% average CPU utilisation. That is why consolidation and virtualisation remain the cheapest efficiency lever. AI fleets differ by workload:

- **Training clusters** run near full power for weeks.
- **Inference clusters** are bursty and often lightly loaded. Epoch AI assumed about 10% utilisation for inference clusters in its ChatGPT estimate.
- **Idle reserve** is a measurable cost even at Google, where idle capacity kept for failover was 10% of per-prompt energy.

2.1.3 Efficiency metrics: PUE, WUE, CUE and ERE

Table 2.3 Efficiency metrics: PUE, WUE, CUE and ERE

Metric	Definition	Benchmarks
PUE (Power Usage Effectiveness)	Total facility energy ÷ IT energy	Uptime 2025 global weighted average 1.54 (1.58 in 2023, 1.56 in 2024). Facilities built in the last five years: 1.48. Sites over 20 MW: about 1.44. Many new high-latitude sites: 1.3 or better. Google fleet: 1.09 (2025). MeitY AI-cloud tenders in India require below 1.35.
WUE (Water Usage Effectiveness)	Site water (L) ÷ IT energy (kWh)	Microsoft FY24: 0.30 L/kWh, down 39% from 0.49 in 2021. Microsoft's regional values range from about 1.52 (Arizona) to 0.02 (Singapore), as reported secondhand.
CUE (Carbon Usage Effectiveness)	$kgCO_2e$ ÷ IT kWh	Determined mainly by the grid mix and whether emissions are counted location-based or market-based
ERE (Energy Reuse Effectiveness)	(Total energy − reused energy) ÷ IT energy	Can fall below 1.0 with district-heat export, which matters for liquid-cooled AI halls producing 40–60 °C return water

Trend. Uptime's 2025 survey (15th edition, fielded April–May 2025) found the weighted average PUE of 1.54 had "virtually stood still" for the sixth consecutive year. Legacy infrastructure and hot or humid climates hold the average up, even as new large sites improve.

Limitations of PUE. - It ignores IT efficiency: wasteful idle servers can make PUE look better. - It ignores water. Swapping evaporative cooling for chillers lowers WUE but raises PUE, as Microsoft acknowledges for its zero-water designs. - It is climate-dependent and self-reported, and measurement boundaries vary. - It says nothing about carbon or embodied impacts. - Uptime itself stresses that it was designed to track a single site over time, not to compare sites.

The rise of liquid cooling also blurs the boundary, because CDU pumps and server fans can move between the "facility" and "IT" sides of the ratio.

2.1.4 Water consumption: direct and indirect

Global (IEA). Data centres consumed about 560 billion litres of water in 2023, possibly about 1,200 billion litres by 2030. A peer-reviewed analysis of the IEA figures breaks the 2023 total down as: - about 373 billion L (two-thirds) indirect, from electricity generation; - about 140 billion L direct, on site; - about 47 billion L in hardware manufacturing.

Bloomberg, reporting on the IEA, notes that a typical 100 MW US data centre consumes about 2 million litres a day.

United States (LBNL 2024, as reported). Direct consumption was about 66 billion litres in 2023, with hyperscale direct use projected at 60–124 billion litres by 2028. Indirect use through electricity was about 800 billion litres in 2023.

Litres per kWh. Direct (WUE) runs from near 0 for dry, closed-loop or free-air sites to above 1.5 L/kWh for evaporative sites in hot, dry climates. Indirect intensity depends on the grid; one study applies about 3.4 L/kWh. The total water footprint per kWh is therefore usually dominated by the power plant, not the cooling tower.

Company disclosures. - **Google:** - 2024: about 8.1 billion gallons consumed; 4.5 billion gallons (64%) replenished. - 2025: total consumption rose 34% to 10.9 billion gallons, more than double 2021 levels, with data centres accounting for most of the increase (Axios, 30 June 2026, on Google's 2026 Environmental Report); 7.7 billion gallons (78%) replenished. - 87% of 2025 freshwater withdrawals came from low or medium-risk sources. - Target: 120% replenishment by 2030. - Google argues that water cooling cuts energy use about 10% compared with air. - **Microsoft:** - WUE 0.30 L/kWh (FY24). - All designs since August 2024 use closed-loop "zero-water evaporation" chip-level cooling, avoiding more than 125 million litres per data centre per year. - Pilots in Phoenix and Mt Pleasant from 2026, coming online from late 2027.

Per-AI-query water: a contested range.

Table 2.4 Water consumption: direct and indirect

Estimate	Water per query	Scope
UC Riverside, "Making AI Less Thirsty" (Li, Yang, Islam, Ren)	500 mL per about 10–50 medium GPT-3 responses (about 10–50 mL each); GPT-3 training about 5.4 million L total, of which 700,000 L on site	Scope 1 + 2, GPT-3-era hardware, Microsoft US data centres
Google Gemini (May 2025)	0.26 mL (0.15 mL under a narrower method)	On-site cooling only
OpenAI (Altman, June 2025)	about 0.32 mL	Unspecified, likely on-site
Mistral Le Chat LCA	about 45 mL per 400-token response	Full lifecycle

The often-quoted "a bottle per query" misreads the UC Riverside paper by a factor of 10–50. Google's 0.26 mL excludes power-plant water, which critics argue can raise the figure to roughly 1–2 mL. The paper also projects global AI water withdrawal of 4.2–6.6 billion m^3 by 2027.

2.1.5 Backup power and other resources

- **Diesel generators:** still the default backup. CEEW's 2026 India stakeholder study found DG sets "remain an indispensable asset" for uptime. They run few hours a year but have air-permit, fuel-storage and emissions implications. HVO (renewable diesel), gas gensets and BESS are emerging alternatives.
- **Batteries:** UPS batteries have shifted from valve-regulated lead-acid (VRLA) to lithium-ion. AI sites add grid-scale BESS for load smoothing, and Nvidia's 800 VDC reference places energy storage on the DC bus.
- **Refrigerants:** chillers and DX units use HFCs with high global-warming potential. The Kigali Amendment, EU F-gas rules and India's Cooling Action Plan are pushing operators toward low-GWP refrigerants. Liquid cooling with dry coolers reduces refrigerant dependence.
- **Land:** India's roughly 271 data centres occupy about 23 million m^2 (CEEW). Meta's 5 GW Hyperion campus is described as a footprint comparable to a large part of Manhattan.
- **Embodied carbon (Scope 3):** construction (concrete, steel, copper) and above all IT hardware manufacturing (chips, HBM, servers) make up a growing share of lifecycle emissions as operational power decarbonises. Google's per-prompt figure of 0.03 gCO_2e explicitly includes embodied hardware emissions. Hardware manufacturing also accounts for about 8% of the IEA's water estimate. Hyperscalers' rising Scope 3 emissions are the main reason their net-zero trajectories have slipped. Quantification is weak and methods vary, so treat any single number with caution.

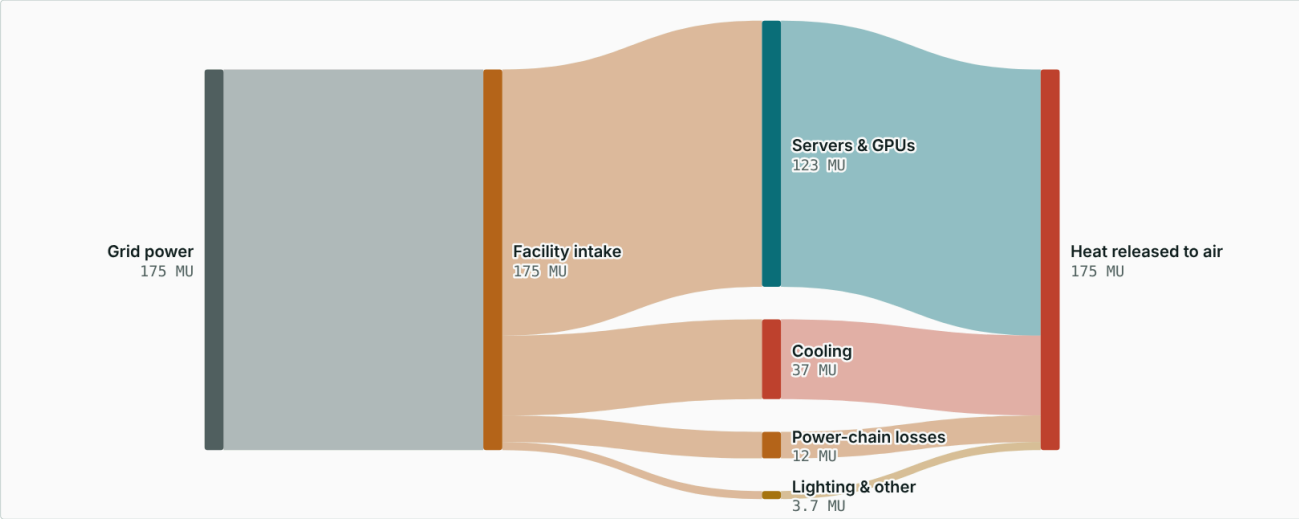


Figure 2.2 Where the power goes in a 20 MW Bengaluru colocation hall. Flow of a year's electricity from the grid through IT, cooling, losses and other loads, ending as heat. *Source: India Data Centre Atlas, chapter built on this chapter's data.*

2.1.6 Global and regional electricity consumption

IEA *Energy and AI* (April 2025), Base Case: - **2024:** about 415 TWh, about 1.5% of global electricity. The US accounts for about 45% of global data-centre consumption. - **Growth:** about 12% a year since 2019, accelerating to about 15% a year to 2030, more than four times the growth of other demand. - **2030:** about 945 TWh, just under 3% of global electricity. The "Lift-Off" case exceeds 1,260 TWh by 2030. - **2035:** about 1,200 TWh. - **AI share:** accelerated (AI) servers were about 24% of server electricity and about 15% of total data-centre demand in 2024. They grow about 30% a year, against about 9% for conventional servers, and account for almost half of the net increase to 2030; conventional servers account for about 20%. - **Regions:** - China and the US together account for nearly 80% of growth. - The US adds about 240 TWh (+130%) by 2030. - Europe adds about 45 TWh (+70%). - **Supply:** renewables meet about half of incremental demand to 2030. Natural gas expands by about 175 TWh to 2035. In 2024, US data centres drew over 40% of their power from gas, about 24% from renewables, about 20% from nuclear and about 15% from coal (Pew, citing the IEA). - **Emissions:** about 182 Mt CO_2 from data-centre electricity in 2024.

LBNL 2024 *United States Data Center Energy Usage Report* (December 2024):

Table 2.5 Global and regional electricity consumption

Year	US data-centre electricity	Share of US electricity
2014	58 TWh	—
2018	76 TWh	1.9%
2023	176 TWh	4.4%
2028 (scenarios)	325–580 TWh	6.7–12.0%

LBNL attributes the post-2017 doubling largely to AI servers. Estimates diverge widely: the IEA 4E critical review (2025) found 2030 projections ranging nearly 40-fold across more than 50 publications, and put AI data-centre energy at a plausible 200–400 TWh in 2030.

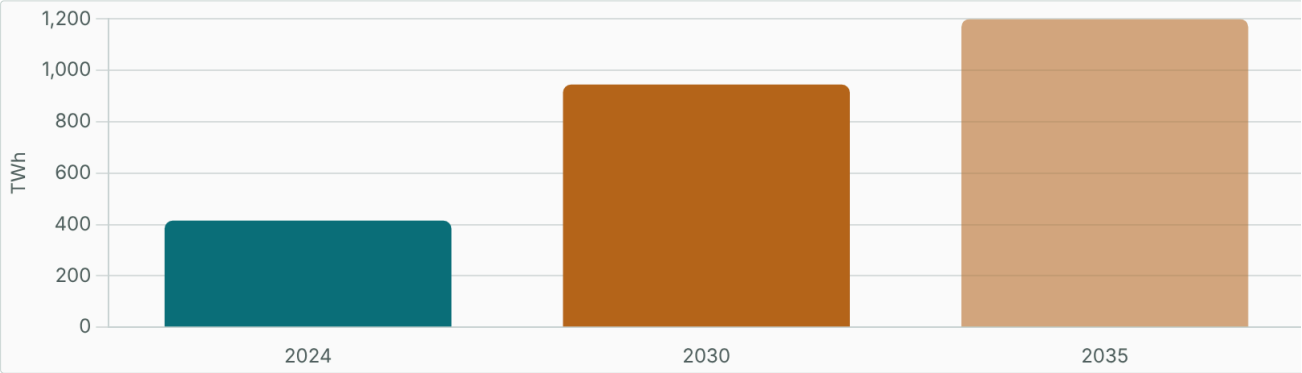


Figure 2.3 Global data-centre electricity. TWh, IEA Energy and AI (April 2025) base case. 2030 high case exceeds 1,260 TWh. *Source: India Data Centre Atlas, chapter built on this chapter's data.*

2.1.7 Energy per workload

Table 2.6 Energy per workload

Workload	Energy	Source/notes
Google web search	about 0.3 Wh	Google's oft-cited 2009 figure; likely lower today
Median Gemini Apps text prompt	0.24 Wh (0.10 Wh active-chip-only), 0.03 gCO_2e	Google, May 2025; 33× lower energy and 44× lower carbon than May 2024. Not independently verified, and prompt length is undisclosed.
ChatGPT average query	about 0.34 Wh	Sam Altman, June 2025; no methodology
GPT-4o typical query (500 output tokens)	about 0.3 Wh	Epoch AI, February 2025 (replacing the 2023 figure of about 3 Wh)
Long-input query (10k / 100k tokens)	about 2.5 Wh / about 40 Wh	Epoch AI
Reasoning/agentic queries	several to tens of Wh	Third-party benchmarks; high uncertainty

Training versus inference. - **Llama 3.1 (Meta model card, July 2024):** 39.3 million H100-80GB GPU-hours (700 W TDP) across the family. The 405B model alone took 30.84 million GPU-hours and 8,930 tCO_2e location-based (0 market-based). At full TDP that implies an IT-only upper bound of about 21.6 GWh (the author's calculation). - **GPT-4:** no official disclosure. Widely cited estimates are about 50–62 GWh, based on about 25,000 A100s running for 90–100 days; MIT Technology Review used about 50 GWh. Epoch AI says GPT-4o-class training runs drew about 20–25 MW for about three months. Epoch estimates Grok 4 training used about 310 GWh. - **Inference is now larger than training in aggregate:** at about 0.3 Wh per query and 1 billion messages a day, ChatGPT inference alone needs about 12.5 MW of continuous power (Epoch AI). Inference runs indefinitely, while a training run is a one-off. The IEA expects AI server growth to be "predominantly inference".

2.1.8 Carbon footprint and grid impacts

- **Load profile:** conventional data centres are flat 24/7 baseloads with high load factors, which suits utilities but concentrates demand. The IEA notes a 100 MW facility can consume as much electricity as 100,000 households. Clustering (Northern Virginia, Dublin, Mumbai) concentrates stress on local grids.
- **AI training swings:** thousands of GPUs alternate in lockstep between compute and communication or checkpoint phases, producing synchronised swings of tens to hundreds of MW within seconds. NERC's July 2025 Large Loads white paper documents:
- one data centre falling from about 450 MW to 40 MW in 36 seconds;

- a July 2024 Eastern Interconnection event in which a 230 kV fault caused about 1,500 MW of data-centre load to switch to backup at once, pushing frequency to about 60.047 Hz.
- **Regulatory escalation:** NERC issued a **Level 3 "Essential Actions" Alert on 4 May 2026** on computational loads, setting seven required actions with registered-entity responses due by 3 August 2026. It is developing a "Computational Load Entity" registration category for facilities of at least 20 MW aggregate, connected at 60 kV or above, with more than 1 MW of IT load.
- **Power quality:** large power-electronic loads raise concerns about harmonics, oscillations and ride-through. Mitigations include rack-level energy storage, GPU power smoothing in firmware, BESS and grid-forming inverters.
- **Carbon:** market-based figures (with PPAs and certificates) can be close to zero, as with Llama 405B's 0 market-based emissions, while location-based figures remain material. Decision-makers should ask for both.

2.1.9 The Indian angle

- **Scale:**
- Capacity rose from about 520 MW in 2020 to about 1.5 GW by mid-2025. Other estimates: JLL 1,123 MW operational (H1 2025), Cushman & Wakefield about 1.3 GW, CBRE about 1.7 GW.
- Projections: 4.5–6.5 GW by 2030; Jefferies is more bullish at about 8 GW.
- Committed investment of about USD 95 billion (2019–25).
- **Footprint:** about 0.5% of national electricity and about 150 billion litres of water in 2025 (Mordor Intelligence via CEEW). Both are expected to more than double by 2030; Mordor projects about 359 billion litres. WRI India mapping shows more than half of India's data centres sit in water-stressed regions. Small operators (under 5 MW) rely mainly on municipal water.
- **Efficiency:** the SBI report says Indian industry leaders achieve PUE below 1.5, and MeitY AI-cloud tenders since 2024 require below 1.35. Hot, humid climates (Mumbai, Chennai) limit free cooling, which is one reason Indian PUEs sit above Nordic levels. Andhra Pradesh promotes seawater cooling for coastal sites.
- **Policy gaps:** there is no binding national data-centre policy. 15 states have dedicated policies, but only 5 embed sustainability provisions, and most lack PUE or WUE standards. Renewable developers say up to about 80% renewable supply is feasible using diversified solar–wind portfolios plus storage (CEEW).
- **Implication:** India's binding constraints will be local water and grid capacity in the Mumbai and Chennai clusters, not national energy totals. Water-free liquid cooling, treated-wastewater sourcing and renewable-energy-linked siting should be conditions of state incentives.

2.2 AI Data Centre vs Conventional Data Centre

2.2.1 Master comparison table

Table 2.7 Master comparison table

Attribute	Conventional (enterprise/colo/cloud)	AI data centre (training/inference)
Primary workloads	Web, databases, ERP, storage, virtualised and cloud VMs, streaming	Large-model training (synchronous, weeks to months); inference (latency-sensitive, bursty); fine-tuning
Compute	x86/Arm CPU servers, about 300–1,000 W per server	GPUs/TPUs/ASICs: H100 700 W; B200 1,000–1,200 W; B300 about 1.4 kW; 8-GPU HGX node about 5.6–10 kW
Rack density	about 5–15 kW (Uptime: few above 30 kW; average about 7–8 kW)	about 40 kW (H100 air-cooled) → 120–132 kW (GB200 NVL72) → 135–142 kW (GB300) → about 190–230 kW (Vera Rubin, 2026) → about 600 kW to 1 MW (Kyber, from 2027)
Cooling	Air (CRAH/CRAC, hot/cold aisle containment), chillers, evaporative towers	Direct-to-chip liquid (CDUs, facility water loops), rear-door heat exchangers, immersion; residual air about 10–15%
Power distribution	400/415 V AC to rack, 12 V server PSUs, centralised UPS	415/480 V AC with 54 VDC busbars and power shelves now; 800 VDC from about 2027; BESS for smoothing; on-site gas generation and nuclear PPAs
Networking	North–south dominant; 10–100 GbE; leaf–spine	East–west dominant; NVLink scale-up (72 → 576 GPUs); InfiniBand or Ultra Ethernet scale-out at 400–800G per GPU; optics-heavy; many times more fibre
Facility	1–50 MW (the IEA treats 25 MW as conventional); raised floors	100 MW to multi-GW campuses; slab floors with rack positions over 1.3 t; taller halls for overhead busways and piping
Siting	Latency-first, close to users and internet exchanges	Power-first for training (cheap, available GW); metro or edge for inference
Capex per MW	JLL shell and core about $11.3m/MW (2026 global average). India: INR 50–55 crore/MW (SBI)	Plus up to $25m/MW tenant AI fit-out (JLL), before GPUs. India: up to INR 75 crore/MW (SBI)
Reliability	Tier III/IV, 2N power, concurrent maintainability	Training: increasingly N+1 or Tier II/III-like, relying on checkpoints. Inference: Tier III+
Load profile	Flat, diurnal and variable; low utilisation	Training: sustained near-peak with sub-second synchronised swings. Inference: bursty
PUE	1.5–1.6 average; 1.1–1.2 hyperscale	1.1–1.3 achievable with liquid cooling; IT share about 75% or more
Water	Evaporative towers can mean WUE above 1 L/kWh	Closed-loop liquid cooling can reach near-zero WUE but may raise PUE; indirect water rises with load
Obsolescence	Hardware refreshed every 5–7 years; shell lasts 15–20+ years	GPU generations every year; economic life about 3–6 years (contested); shells risk stranding on density
Grid and regulatory exposure	Moderate	High: NERC Level 3 alert, large-load tariffs, interconnection queues, local opposition

2.2.2 Purpose and workloads

- **Training:** one job spans tens of thousands to hundreds of thousands of accelerators. It is throughput-bound, tolerant of location and latency, and runs at sustained high utilisation.
- **Inference:** serving tokens to users. It is latency-sensitive, scales with adoption, is bursty, and increasingly dominates AI electricity use.
- **Conventional:** a mix of many small, independent workloads with modest density and high availability requirements. Most enterprise and colocation halls are built for this.

2.2.3 Compute hardware and power

Table 2.8 Compute hardware and power

Platform	Power	Notes
Nvidia H100 SXM	700 W	8-GPU node about 5.6 kW (GPUs) to about 10 kW (node); racks about 30–40 kW
Nvidia B200	1,000 W (HGX) to 1,200 W (NVL72)	Air-cooled HGX B200 racks about 60 kW
GB200 NVL72	about 120 kW (Nvidia); 132 kW nominal (HPE: 115 kW liquid + 17 kW air); peak electrical design power about 192 kW	72 GPUs + 36 Grace CPUs; about 1.36 t; eight 33 kW power shelves
GB300 NVL72	135 kW (Nvidia Mission Control); 142 kW (Nvidia/Vertiv reference)	B300 at about 1.4 kW per GPU
Vera Rubin NVL144 (2026)	about 190–230 kW (industry reports)	Transitional AC/54 VDC or 800 VDC input
Rubin Ultra / Kyber (2027+)	about 600 kW (GTC 2025), designed for up to 1 MW racks	Kyber redesigned at GTC 2026; SemiAnalysis reports a possible slip to 2028
Google TPU (Trillium, Ironwood); AWS Trainium	Lower per chip than Nvidia's flagships; liquid-cooled pods at scale	Google says Ironwood is 30× more efficient than its first TPU
Conventional CPU server	about 300–1,000 W	2-socket x86; 5–15 kW per rack

2.2.4 Rack power density

The trajectory runs from about 7–8 kW (industry-average rack) to about 40 kW (Hopper, 2023), about 120–132 kW (Blackwell, 2024–25), about 135–142 kW (Blackwell Ultra, 2025–26), about 200+ kW (Rubin, 2026) and about 600 kW to 1 MW (Rubin Ultra Kyber, 2027–28). That is a roughly 100-fold rise in about five years. The practical implication is that a facility built in 2023 for 30 kW racks is already stranded for frontier training. The SBI report notes air cooling "approaching physical limits" above about 15 kW per rack.

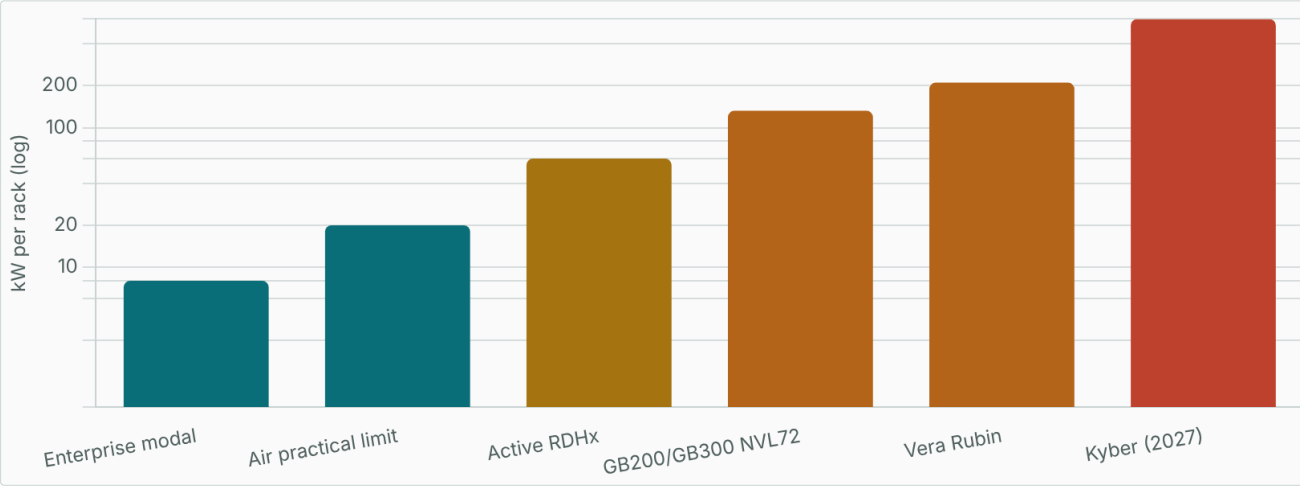


Figure 2.4 Rack density and chip power. Toggle between rack kW (log scale) and per-GPU watts. *Source: India Data Centre Atlas, chapter built on this chapter's data.*

2.2.5 Cooling

- **Air** (CRAH/CRAC, containment, economisers): practical to about 20–30 kW per rack, about 40 kW with rear-door heat exchangers.
- **Direct-to-chip (DLC):** cold plates on GPUs and CPUs. In-row or in-rack CDUs exchange heat between the technology cooling system (TCS) loop and the facility water system (FWS). Supply water can be warm (about 30–45 °C), which allows dry coolers and chiller-free operation in many climates. It is mandatory for NVL72 (about 85–90% of heat goes to liquid).
- **Immersion** (single or two-phase dielectric): the highest density and near-silent operation, but less mature, with heavier tanks and service complexity. The SBI report associates it with the INR 75 crore per MW tier.
- **Heat reuse:** warm return water makes district heating and ERE below 1.0 feasible, mostly relevant in Nordic and European markets.
- **Cost:** benchmarks cited by Archdesk put air-cooled heat rejection at about $1.8m/MW against $4.5–5.2m/MW for liquid-cooled AI halls. Turner & Townsend cites a 7–10% total premium for liquid-cooled builds. These are secondary benchmarks.

2.2.6 Power architecture

Diagram in words: Nvidia's 800 VDC target chain. Utility 13.8 kV AC → solid-state transformer or rectifier at the facility perimeter → 800 VDC busway (with BESS on the DC bus) → in-rack 800 V to 12 V DC-DC conversion (64:1) → GPU core voltage.

Nvidia says this: - removes multiple AC/DC stages; - avoids "up to 200 kg of copper busbar" per 1 MW rack, which 54 VDC would require; - frees up to 64U of rack space otherwise consumed by power shelves; - targets up to 30% lower total cost of ownership; - begins full-scale production with Kyber in 2027.

Schneider Electric argues that 400 V three-phase AC and 48 VDC become "impossible" at about 400 kW per rack.

Other changes: - busways replace whips; - UPS moves to lithium-ion and is increasingly paired with BESS; - busways must be provisioned for peak design power, about 1.5× TDP per HPE.

On-site generation and firm power: - **Gas turbines:** OpenAI's Stargate Abilene has gas turbines capable of up to 300 MW (IEEE Spectrum). xAI's Colossus was criticised for running unpermitted gas turbines. - **Nuclear PPAs:** - Microsoft–Constellation: 20-year PPA to restart Three Mile Island Unit 1 as the 835 MW Crane Clean Energy Center, targeted for 2027. - Amazon–Talen Susquehanna: originally a

960 MW co-located campus, restructured in 2025 into a reported PPA of up to 1.92 GW. - Google–Kairos Power: 500 MW of small modular reactors by about 2030.

2.2.7 Networking

AI clusters need a non-blocking east–west fabric: - **Scale-up:** NVLink inside a rack domain (72 GPUs for NVL72, up to 576 for Kyber) over copper backplanes. - **Scale-out:** InfiniBand or Ultra Ethernet/RoCE at 400–800 Gb/s per GPU, organised as rail-optimised fat trees. - **Front-end:** a separate network for storage and users.

The result is several times more optical transceivers and fibre per MW than in cloud halls, structured-cabling plants with very high fibre counts, and a coming move to co-packaged optics (Nvidia's Feynman generation, about 2028). Networking's share of power rises accordingly, though it remains small next to the GPUs.

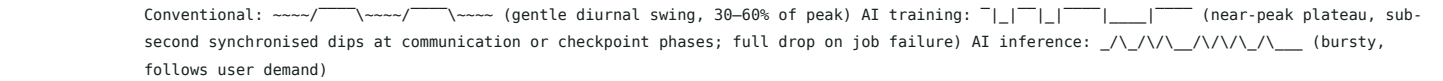
2.2.8 Facility design and campus scale

- **Structure:** an NVL72 rack weighs about 1.36 t. Liquid-filled 600 kW racks concentrate several tonnes per position, requiring slab-on-grade or reinforced floors. Halls need greater clear height for overhead busways, cable trays and piping, plus space for CDUs and secondary piping.
- **Campus examples:**
- xAI's Colossus (Memphis): built in 122 days. Colossus 2 is designed for more than 550,000 GPUs at about 1 GW+.
- OpenAI's Stargate Abilene: more than 450,000 GB200s planned, about 1.2 GW (reported). The Stargate programme overall had 5.5 GW+ across six sites as of September 2025.
- Meta: Prometheus (New Albany, Ohio), 1 GW, due in 2026; Hyperion (Louisiana), 2 GW by 2030 scaling to 5 GW.
- **Location logic:** training goes where large, cheap, fast power exists (Texas, Louisiana, the Midwest, the Gulf, Nordics). Conventional and inference capacity stays near users and exchanges.

2.2.9 Cost

- **Shell and core (JLL):** global average rose from $7.7m/MW (2020) to $10.7m (2025), with $11.3m forecast for 2026 (7% CAGR). 57% of projects were delayed by three months or more in 2025.
- **AI tenant fit-out:** up to $25m/MW on top (JLL), covering high-voltage distribution, CDUs and liquid loops, and network. GPUs and servers are extra and dominate the total: at roughly $3–4m per NVL72 rack (industry estimates), IT equipment can be several times the building cost.
- **The investment split is inverted:** in conventional data centres the building and mechanical/electrical plant are the majority of capex; in AI data centres IT is the majority, and the building is roughly 20–30% of total capex.
- **India (SBI, April 2026):** standard Tier III air-cooled INR 50–55 crore/MW; AI/immersion or Tier IV up to INR 75 crore/MW; large-scale campuses INR 35–45 crore/MW. EBITDA margins are 40–55%. India builds at roughly half the per-MW cost of US markets.
- **Opex:** electricity is about 50–70% of operating cost (CEEW cites 60–70%), so PUE and tariffs drive margins.
- **Depreciation:** hyperscalers book servers over about 5–6 years. Annual GPU cadence and falling rental prices argue for 3–4 years of economic life. This is a live, contested accounting debate.
- **Lease rates:** not independently verified here. Sources indicate AI or liquid-ready capacity commands a premium over standard colocation.

2.2.10 Utilisation and load profiles



Training is good for utilisation economics but hard on grids and on-site electrical gear. Mitigations include GPU power capping and smoothing, rack-level energy storage and site BESS (the IEA envisages about 20–25 GW of data-centre-paired storage by 2030, as reported secondhand).

2.2.11 Efficiency and obsolescence

- **PUE:** liquid-cooled AI halls can reach 1.1–1.2. Even so, Uptime's survey shows the global average is not falling.
- **Performance per watt** improves dramatically each generation. Google reports more than three times more compute per unit of energy than five years ago, and a 33× fall in per-prompt energy over a single year, driven by software and model efficiency more than by facilities.
- **Obsolescence risk:** the same rapid improvement erodes the value of last year's GPUs and of facilities sized for them. Shells should be designed with 2–3 generations of density headroom.

2.2.12 Reliability requirements

- **Training:** periodic checkpointing means a power event costs minutes to hours of recomputation, not lost data. Some operators are therefore building training halls with reduced redundancy (N+1 rather than 2N UPS, partial generator backup), trading resilience for cost and speed.
- **Inference:** customer-facing and revenue-critical, so it needs Tier III+.

This is an emerging industry practice, not a standard. In India, uptime expectations and DPDP-driven enterprise demand still favour Tier III/IV.

2.2.13 Water usage differences

AI facilities can use less water per kWh than conventional evaporative sites if they use closed-loop DLC with dry coolers; Microsoft's zero-water design is an example. The trade-off is higher PUE in hot weather. Because AI campuses draw far more electricity, their indirect (power-plant) water and absolute totals are much larger. Google's 34% jump in water consumption in 2025 shows the AI build-out outrunning efficiency gains.

2.2.14 Environmental and grid impacts, and the regulatory response

- **United States:** the NERC Level 3 Alert (May 2026) and the proposed registration category; DOE support for on-site generation and flexibility; large-load tariffs and interconnection reforms at utilities; local opposition to new data centres, with a March 2026 Gallup poll finding 71% of US adults oppose an AI data centre in their local area, including 48% strongly opposed.
- **India:** state data-centre policies (15 states); MeitY's PUE below 1.35 tender requirement; SEBI BRSR Core disclosure for the top 150 listed firms; IGBC Green Data Centre ratings; renewable-energy-linked incentives (e.g. Tamil Nadu above 30% renewables); the Budget 2026–27 tax holiday to 2047 for foreign cloud providers serving India through local entities.

2.2.15 Inference versus training data centres

Table 2.9 Inference versus training data centres

	Training DC	Inference DC
Location	Power-rich, remote, cheap land	Metro or edge, near users (latency of tens of ms)
Size	300 MW to 5 GW single campus	10–200 MW, distributed
Network	Massive internal fabric; little external	Strong external connectivity and peering
Redundancy	Lower (checkpointing)	Higher (Tier III+)
Hardware	Newest flagship GPUs	Mix, including older GPUs, inference ASICs and L40S-class cards
Trend	Fewer, larger campuses	Growth of metro or edge AI zones and retrofitted colocation

The SBI report likewise notes that training ("model grounding") stays in core data centres while inference moves to the edge. As inference becomes the majority of AI electricity, metro capacity (Mumbai, Chennai, Noida in India) becomes the competitive battleground.

2.2.16 Retrofit challenges

Converting a conventional hall to AI-ready means dealing with: 1. **Power:** a 10× density jump means the hall's MW capacity supports only a fraction of the floor area, and upstream transformers and switchgear may need replacing. 2. **Cooling:** adding a facility water loop, CDUs, leak detection and secondary piping into live, raised-floor space. 3. **Structure:** floor loading above 1.3 t per rack. 4. **Height and routing:** overhead busway and piping clearances. 5. **Network:** fibre pathways for many times more strands. 6. **Downtime and tenant disruption.**

The practical route is usually hybrid: dedicate certain halls to liquid-assisted 30–60 kW racks (rear-door or DLC with liquid-to-air CDUs) and reserve 120 kW+ for new builds. Developers should design new shells "liquid-ready" from day one.

2.2.17 The Indian angle: AI-ready builds

- **IndiaAI Mission:**
- INR 10,300 crore allocation.
- 38,231 GPUs onboarded on the common compute portal, at a subsidised rate of about ₹65 per GPU-hour (Lok Sabha reply, 25 March 2026). This was up from 34,333 in May 2025.
- An additional 20,000 GPUs were announced at the February 2026 AI Impact Summit, which would take the total above 58,000, but onboarding was unconfirmed as of March 2026.
- The pool includes about 1,050 Google Trillium TPUs.
- **Yotta Shakti Cloud:**
- NM1 (Navi Mumbai): Tier IV, 52 MW IT load, PUE below 1.5, more than 7,000 racks, with 8,192 H100s and 1,024 L40S GPUs live.
- 8,192 B200s empanelled under IndiaAI.
- In February 2026, Yotta announced 20,736 liquid-cooled Blackwell Ultra (B300) GPUs at its 60 MW D2 site in Greater Noida (more than $2 billion, targeted live by August 2026; not yet confirmed). It also has a DGX Cloud deal with Nvidia worth more than $1 billion.
- Plans for more than 80,000 GPUs by FY27–28, including at the 75 MW NM2.
- Published GPU counts vary by source.
- **Reliance Jamnagar:**
- Gigawatt-scale, green-powered AI data centres announced in August 2024. Bloomberg reported a 3 GW plan costing $20–30 billion with Nvidia chips.
- Reliance Intelligence and a Google AI-cloud partnership (August 2025).
- More than 120 MW expected online in H2 2026 (February 2026).
- A Meta built-to-suit lease announced on 10 June 2026, reported at about 168 MW.
- **Costs:** INR 50–55 crore/MW (standard) versus up to INR 75 crore/MW (AI or Tier IV), per SBI (April 2026). Other Indian advisory benchmarks put AI-ready liquid-cooled builds at INR 70–95 crore/MW, so treat SBI's 75 crore as a mid-point, not a ceiling.

2.3 Recommendations

1. **For planners and investors:** underwrite new Indian capacity as "AI-ready" (DLC loops, more than 1.5 t floors, 415 V or higher distribution, and space for future 800 VDC) even when it is leased for cloud. The roughly 35–40% capex premium (INR 75 vs 50–55 crore/MW) is cheaper than a retrofit or a stranded shell.

2. **Measure beyond PUE:** require WUE (direct and indirect), location- and market-based CUE, and IT utilisation in contracts and state incentives. Tie Indian state subsidies to PUE of 1.35 or below and WUE thresholds in water-stressed basins.
3. **Water:** in Mumbai, Chennai and Hyderabad, prefer closed-loop or dry-cooled liquid designs and treated wastewater. Accept a small PUE penalty in exchange for near-zero evaporative use.
4. **Grid:** co-plan with transmission utilities, deploy BESS and GPU power-smoothing for training loads, and design for ride-through. NERC's experience is a preview of what Indian grids will face as GW campuses arrive.
5. **Separate training from inference:** site training at power-rich, renewable-heavy locations and inference in metros. Price redundancy accordingly.

2.4 Caveats

- Subsystem splits by facility class (section 1.1) and within-server splits (section 1.2) are indicative syntheses, not measured averages.
- Per-query energy and water figures are company-reported (Google, OpenAI) or modelled (Epoch AI, UC Riverside). They are not independently audited and differ in scope by orders of magnitude.
- GPT-4 training energy is an unofficial estimate.
- Global and Indian capacity forecasts vary widely (e.g. India 2030 at 4.2–8 GW). IEA projections are scenarios, not forecasts.
- Several hardware roadmap figures (Rubin, Kyber timing and power) come from vendor keynotes and secondary reporting and have already shifted. Treat 600 kW to 1 MW racks as a direction of travel, not a delivery date.
- Some figures (Microsoft regional WUE, Uptime average rack kW, Google's 2025 water total, LBNL water volumes, Amazon–Talen capacity, IEA BESS) were obtained via secondary reporting.
- Yotta and IndiaAI GPU counts differ across sources and announcements may not reflect deployed capacity.

Sources cited in this chapter

Organisations and publications named in this chapter's text, in order of how often they are cited. Full details are in the bibliography.

- Google environmental reports, efficiency data and Gemini inference paper
- International Energy Agency (IEA), Energy and AI (2025) and updates
- NVIDIA product specifications and roadmap
- State Bank of India sector report on data centres (April 2026)
- Microsoft sustainability reports and datacenter design posts
- Epoch AI energy estimates
- Uptime Institute, Global Data Center Surveys and Tier Standard
- Ministry of Electronics and Information Technology (MeitY), Data Centre Policy draft and IndiaAI
- Council on Energy, Environment and Water (CEEW) studies
- JLL, Global and India Data Center Outlooks
- North American Electric Reliability Corporation, Large Loads alert (May 2026)
- Lawrence Berkeley National Laboratory, 2024 US Data Center Energy Usage Report
- Meta sustainability reports and data index
- Amazon Web Services sustainability reporting
- Yotta Data Services disclosures
- Reliance Industries announcements
- Mordor Intelligence market estimates
- Hewlett Packard Enterprise product data
- CBRE, India data centre market reports
- Cushman & Wakefield, India data centre market updates (H1 2026)
- Turner & Townsend, Data Centre Cost Index 2025
- World Resources Institute (Aqueduct, thermal water studies)
- Mistral AI life-cycle assessment (ADEME / Carbone 4)
- Li, Yang, Islam and Ren, "Making AI Less Thirsty"
- Kigali Amendment to the Montreal Protocol

PART I · FOUNDATIONS

Chapter 3
Cooling: every type, how it works, what it costs and what India should build

WHAT THIS CHAPTER COVERS

This chapter explains why data centres need cooling, why liquids carry heat far better than air, and how the industry's thermal guidelines define safe temperatures. It describes every family of cooling technology, from room air conditioning and economisers to chilled-water plants, cooling towers, dry coolers, direct-to-chip liquid cooling and immersion, together with their costs, their water use and the refrigerants they rely on. It reviews the market and how the large operators cool their facilities, and it ends with what works in India's hot and humid climates.

KEY TAKEAWAYS

1. Air cooling remains the most common method in existing data centres, but it stops being practical at somewhere between 20 and 40 kilowatts per rack.
2. New AI capacity is designed around single-phase direct-to-chip liquid cooling. NVIDIA's GB200 and GB300 racks are already more than 90% liquid-cooled, and its Vera Rubin generation and 600 kilowatt Kyber racks assume 100% liquid cooling with water supplied at 45 °C.
3. Warm facility water matters because a dry cooler can only cool water to a few degrees above the outside air temperature. Water at 40 to 45 °C can therefore be cooled without evaporation for almost all hours in most Indian cities.
4. For India, the best default for AI halls is warm-water direct-to-chip cooling that rejects heat through dry or adiabatic coolers, with evaporative towers kept as an option for summer peaks. Air cooling with economisers remains right for low-density enterprise and cloud halls, especially in Bengaluru and Hyderabad.
5. India's phase-down of hydrofluorocarbon refrigerants under the Kigali Amendment means that chillers bought now should use low-global-warming refrigerants.

How this chapter connects to the others. The water consequences of each cooling choice are quantified in Chapters 6 and 7, and Chapter 15 sets out cooling recommendations by Indian climate zone.

Bottom line: Air cooling is still the most common method in the installed base. But it stops being practical somewhere around 20–40 kW per rack, and new AI capacity is now designed around single-phase direct-to-chip (DLC) liquid cooling. Nvidia's GB200 and GB300 racks are already 90%+ liquid-cooled. Its Vera Rubin generation (2026) and 600 kW Kyber racks (2027) assume 100% liquid cooling with 45°C supply water. For India, the best default for AI halls is warm-water DLC rejecting heat through dry or adiabatic coolers, with evaporative towers kept as a summer peak-shaving option. Air plus economisers remains right for low-density enterprise and cloud halls, especially in Bengaluru and Hyderabad.

3.1 Fundamentals

3.1.1 Why cooling is needed

Almost all electrical energy delivered to IT equipment becomes heat. A 100 MW IT load is a 100 MW heater that must be cooled continuously. Silicon junction temperatures typically need to stay below about 85–105°C, and the problem is heat flux (W/cm²), not just total watts.

Accelerator power trend (per GPU):

Table 3.1 Why cooling is needed

GPU	Year	Max TDP	Cooling implication
Nvidia H100 SXM	2022–23	700 W	Air-coolable (HGX); DLC optional
Nvidia B200	2024–25	1,000 W (up to ~1,200 W in GB200 configurations)	Air HGX options exist; GB200 NVL72 rack is liquid
Nvidia B300 (Blackwell Ultra)	H2 2025	1,400 W (liquid bin); 1,100 W air bin	GB300 NVL72 requires DLC; DGX B300 has an air-cooled 14.5 kW variant
Nvidia Rubin (Vera Rubin NVL72/NVL144)	H2 2026	~1,800–2,300 W (industry estimates; not formally confirmed)	100% liquid; 45°C inlet water
Rubin Ultra (Kyber NVL576)	H2 2027	—	~600 kW per rack, 100% liquid, no fans, 800 VDC
Leading GPUs (Dell'Oro projection)	2029	>4,000 W	Two-phase or embedded cooling likely needed

Rack-level consequences: the GB200 NVL72 is about 120–132 kW. The GB300 NVL72 is about 130–150 kW. Vera Rubin NVL72 racks are widely estimated at 190–230 kW. Nvidia's Charlie Boyle said "GB200 & 300 are like 90 plus percent liquid-cooled… The Kyber rack design is 100 percent liquid cooling, no fans." The Kyber rack also needs a sidecar for power and cooling, so it takes two racks' worth of floor space.

3.1.2 Heat transfer properties: why liquid wins

Heat removed is $Q = \dot{m} \cdot cp \cdot \Delta T$. What matters is how much heat a given volume of fluid can carry per degree of temperature rise. The values below are standard engineering values at about 25°C.

Table 3.2 Heat transfer properties: why liquid wins

Fluid	Specific heat (kJ/kg·K)	Density (kg/m³)	Volumetric heat capacity (kJ/m³·K)	Relative to air	Notes
Air	~1.0	~1.2	~1.2	1×	Low thermal conductivity (~0.026 W/m·K)
Water	~4.18	~1,000	~4,180	~3,500×	Conductivity ~0.6 W/m·K; conductive and corrosive, so loops must be sealed and treated
PG25 (25% propylene glycol)	~3.9	~1,020	~4,000	~3,300×	Standard technology cooling system (TCS) fluid for cold plates; biocide and freeze protection
Synthetic or hydrocarbon dielectric oil (single-phase immersion)	~1.9–2.3	~800–850	~1,600–1,900	~1,400×	Higher viscosity means slower convection
Fluorocarbon or HFO dielectric (two-phase)	~1.1–1.3	~1,400–1,700	~1,700–2,000 (sensible)	—	Boiling absorbs latent heat (~90–140 kJ/kg); heat transfer coefficient 5–10× single-phase

In practice, moving 100 kW of heat with a 10 K temperature rise takes about 8 m³/s of air, but only about 2.4 L/s of water. That is why fan power, white space and air-handler size all collapse once heat is captured in liquid at the chip.

3.1.3 ASHRAE TC 9.9 thermal guidelines (5th edition, 2021)

Air-cooled classes (IT inlet air):

Table 3.3 ASHRAE TC 9.9 thermal guidelines (5th edition, 2021)

Class	Recommended	Allowable dry-bulb	Typical use
A1	18–27°C	15–32°C	Enterprise and mission-critical
A2	18–27°C	10–35°C	Volume servers
A3	18–27°C	5–40°C	Extended-range, chillerless designs
A4	18–27°C	5–45°C	Extreme economisation
H1 (high-density)	**18–22°C**	**15–25°C**	Tightly integrated, high-power air-cooled systems that cannot meet A2

H1 matters because it runs against two decades of "raise the setpoint" efficiency gains. Uptime Institute warned that H1 "challenge[s] the efficiency drive": dense air-cooled AI gear needs colder air, which reduces economiser hours. ASHRAE recommends keeping H1 equipment in separately controlled zones. One secondary source lists the H1 allowable range as 5–25°C; ASHRAE's own white paper gives 15–25°C, and this report uses that figure.

Liquid-cooling W classes (facility water supply temperature): the 5th edition renamed W1–W5 by their upper temperature limit. All classes share a 2°C lower bound.

Table 3.4 ASHRAE TC 9.9 thermal guidelines (5th edition, 2021)

Class (old name)	Facility water supply	Typical heat rejection
W17 (W1)	2–17°C	Chillers, with economiser assist
W27 (W2)	2–27°C	Chillers with large economiser share
W32 (W3)	2–32°C	Cooling towers or dry coolers; chillers for trim
W40 (new)	2–40°C	Mostly chiller-free
W45 (W4)	2–45°C	"Typically operated without chillers": dry coolers year-round in most climates
W+ (W5)	>45°C	Heat reuse and district heating

Nvidia's third-generation MGX racks (Vera Rubin) are "universally designed to operate with 45°C (113°F) warm-water inlet temperatures." At CES 2026, Jensen Huang said, "With 45 degrees C, no water chillers are necessary." That is effectively a W45 design point. In practice, chillers are still often kept for resilience and for mixed air and liquid halls.

3.1.4 The cooling chain: chip to atmosphere

Diagram in words: a modern liquid-cooled AI hall

``` [GPU/CPU die] → TIM → [cold plate] ↓ (TCS loop: PG25 / treated water, ~30–45°C supply, ~40–55°C return) [Rack manifold + quick disconnects (UQD)] ↓ [CDU: pumps, filters, plate heat exchanger, controls] ← boundary between IT loop and facility ↓ (FWS: facility water, W32–W45) [Facility pipework / headers] ↓ [Heat rejection: dry cooler | adiabatic dry cooler | cooling tower | chiller (trim) | heat-reuse HX] ↓ [Atmosphere / district heating network / seawater]

Parallel residual path (~10–30% of heat): memory, NICs, PSUs, drives → air → CRAH / fan wall / RDHx → chilled water → same plant. ```

- **TCS (technology cooling system):** the secondary loop that touches IT equipment, made up of cold plates, manifolds and hoses. It must be very clean (filtered to around 25–50 µm), chemically controlled, and built from compatible wetted materials.
- **FWS (facility water system):** the primary building loop connecting CDUs to heat rejection.
- **CDU (coolant distribution unit):** isolates the two loops, controls TCS supply temperature, flow and pressure, and handles leak and pressure alarms. It can be liquid-to-liquid (L2L, connected to FWS) or liquid-to-air (L2A, rejecting heat into room air through a coil, as in AWS's IRHX or Meta's AALC).
- **Heat rejection:** the final stage that moves heat to the atmosphere, a water body, or a heat user.

**Diagram in words: a classic air-cooled chilled-water hall**

`Server fans → hot aisle (35–45°C) → CRAH coil (chilled water 7–18°C) → cold aisle/plenum (18–27°C) → servers CRAH chilled water → chiller evaporator → compressor → condenser → cooling tower (evaporative) or air-cooled condenser → atmosphere (Economiser bypasses the compressor when outdoor conditions allow)`
```

3.1.5 Key metrics

Table 3.5 Key metrics

Metric	Definition	Typical values / notes
PUE	Total facility energy ÷ IT energy	Legacy air: 1.5–1.8; good modern air with economisers: 1.2–1.4; DLC: 1.05–1.2; immersion: ~1.03–1.08 (vendor-reported). The SBI report says cooling "can account for up to 40 percent of total energy usage" in Indian facilities.
pPUE (partial PUE)	(IT + subsystem energy) ÷ IT energy for one subsystem	A cooling pPUE of 1.05–1.15 is excellent
WUE	Annual site water (L) ÷ IT energy (kWh)	0 (dry) to more than 2 L/kWh (fully evaporative); US average 1.8 L/kWh (LBNL, Shehabi et al., 2016)
kW per rack	Cooling capacity delivered per rack	Uptime 2025: average modal density ~9 kW (up from 8.3 kW); AI racks 40–230 kW, heading to 600 kW+
ΔT	Temperature rise across a coil, server or loop	Air: 10–20 K; TCS: ~10 K. Bigger ΔT means less flow and less pump or fan energy.
Approach temperature	Gap between leaving fluid and its heat sink	Towers: 3–5 K over wet-bulb; dry coolers: 5–10 K over dry-bulb; CDU HX: 2–4 K
Chiller COP / kW per ton	Cooling output ÷ electrical input; 1 ton = 3.517 kW	Water-cooled centrifugal: ~0.5–0.6 kW/ton (COP ~6–7); air-cooled screw: ~1.0–1.2 kW/ton (COP ~3)
IPLV / NPLV	Weighted part-load efficiency (100/75/50/25% load)	Matters because data centre chillers rarely run at full load. India's ECBC data-centre guidance cites, for example, centrifugal chillers ≥1,580 kWr at COP 6.3 and IPLV 7.0.

3.2 Air cooling types

3.2.1 Room-level units

- **CRAC (computer room air conditioner):** a self-contained direct-expansion (DX) refrigeration unit with an outdoor condenser. Simple and modular, with no chilled-water plant, but less efficient at scale. Common in small Indian enterprise and edge rooms.
- **CRAH (computer room air handler):** a coil plus fans fed by a central chilled-water plant. Better efficiency at scale, and pairs with economisers. The workhorse of Indian colocation.
- **Raised floor versus slab:** raised floors deliver cold air through perforated tiles but suffer from obstructions and bypass air, and struggle beyond ~5–8 kW/rack. Modern builds favour slab floors with flooded-room or ducted supply, fan walls and overhead routing. Slab floors also handle the 1.5–2.5+ tonne weight of liquid-cooled AI racks better.

3.2.2 Airflow management

- **Hot aisle/cold aisle:** racks face each other front-to-front, so cold aisles and hot aisles alternate. This is the minimum standard.
- **Cold aisle containment (CAC):** encloses the cold aisle. Cheaper to retrofit; the rest of the room runs hot.
- **Hot aisle containment (HAC):** encloses the hot aisle and ducts it to the return plenum. Return temperatures are higher, which raises coil and economiser efficiency. Generally preferred for new builds; Nvidia's reference designs use HAC.
- **In-row coolers:** CRAH or DX units placed between racks. Good for pockets of 15–30 kW per rack.
- **Overhead cooling:** coils above the racks. Saves floor space.
- **Fan walls:** large arrays of EC fans feeding a flooded cold aisle. Common in hyperscale slab designs; scalable to ~20–30 kW/rack averages.

3.2.3 Rear-door heat exchangers (RDHx)

A water coil mounted on the rack's rear door neutralises exhaust heat before it enters the room. - **Passive RDHx:** relies on the servers' own fans. Typically ~15–30 kW per rack. - **Active RDHx:** adds fans to the door. Typically ~40–75 kW per rack. Yotta reports RDHx "manage rack densities of up to 50–60 kW." RDHx is the bridge technology. It brings water to the rack without touching servers, and it pairs with DLC to capture the 10–30% of heat that cold plates miss.

3.2.4 Economisers ("free cooling")

- **Direct airside:** filtered outside air goes straight into the hall. Unsuitable for most Indian cities because of humidity, heat, dust and pollution.
- **Indirect airside:** outside air cools hall air through a heat exchanger without mixing. Examples are the **KyotoCooling** rotary heat wheel and **indirect evaporative coolers (IEC)** from Munters, Stulz, Vertiv and Airedale. IEC is a strong option for Bengaluru, Hyderabad, Delhi and Pune during dry seasons.
- **Waterside:** when ambient conditions allow, towers or dry coolers produce chilled water directly (plate HX bypass) or a free-cooling chiller runs its integrated coils. Each degree of higher chilled-water temperature adds economiser hours.

3.2.5 Evaporative and adiabatic cooling

- **Direct evaporative:** air passes over wetted media, and the dry-bulb temperature drops toward wet-bulb.
- **Adiabatic pre-cooling:** water is sprayed or wetted onto pads in front of dry-cooler or condenser coils, only on hot hours. Both trade water for energy (see Section 7). They work well in hot-dry climates like Delhi and Hyderabad in April–May, and poorly in humid coastal monsoons, when Mumbai wet-bulb temperatures can approach 28–32°C.

3.2.6 Limits of air cooling

- Practical limits: about 15–20 kW per rack with standard raised-floor CRAH; 20–40 kW with containment, fan walls or in-row units; ~40–75 kW with active RDHx. One industry analysis puts the practical ceiling for air at ~41 kW per rack. Dell'Oro describes air as having "historically supported rack densities up to 20 kW/rack."

- The SBI sector report (April 2026) says Indian legacy air cooling is "approaching their physical limits, particularly beyond 15 kW per rack."
- Above ~30 kW, fan power inside servers and air handlers becomes a large overhead. The ASHRAE H1 class also forces cooler air, which reduces economiser hours.

3.3 Chilled-water plants and heat rejection

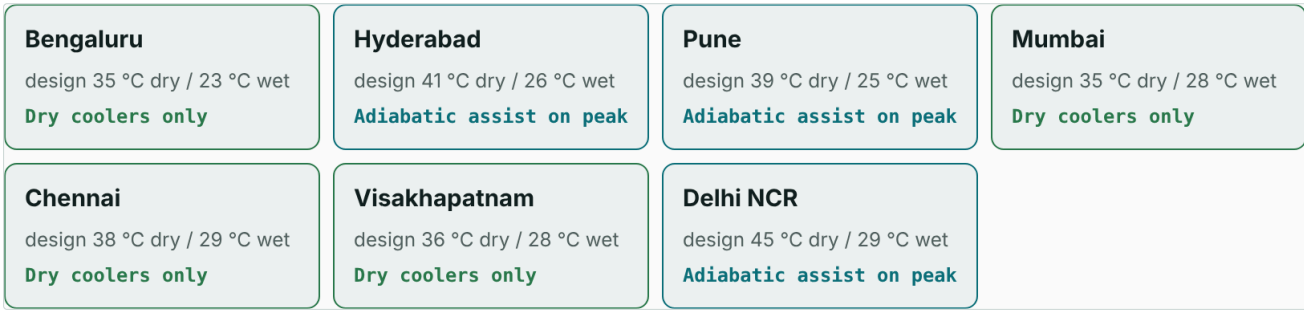


Figure 3.1 Which Indian cities can cool 40 °C facility water without chillers. Heat-rejection option that works on the design day in each city. *Source: India Data Centre Atlas, chapter built on this chapter's data.*

3.3.1 Chiller types

Table 3.6 Chiller types

Type	Compressor	Size range	Typical full-load efficiency	Notes
Air-cooled scroll	Scroll	50–500 kW	~1.1–1.3 kW/ton	Small sites, edge
Air-cooled screw	Screw	300–2,000 kW	~1.0–1.2 kW/ton	No tower water; often with integrated free-cooling coils
Water-cooled screw	Screw	500–3,500 kW	~0.6–0.7 kW/ton	Needs towers
Water-cooled centrifugal	Centrifugal	1,000–10,000+ kW	~0.5–0.6 kW/ton	Best at scale
Magnetic-bearing centrifugal (e.g., Danfoss Turbocor compressors)	Oil-free, variable speed	300–5,000 kW	Full load ~0.55; IPLV can approach ~0.3–0.35 kW/ton	Excellent part-load efficiency and low maintenance
Free-cooling chillers	Any, plus integrated dry coils	—	Compressors off when ambient is ~8–10 K below return water	Very effective at higher water temperatures

Values are typical industry ranges and vary with lift and rating conditions.

Which to choose: water-cooled plants are most efficient but consume tower water. Air-cooled plants use no water but more energy. The SBI report notes Indian operators "transitioning to dry-mode chillers, closed-loop liquid systems, and on-site tertiary treatment units."

3.3.2 Heat rejection equipment

- **Open-circuit cooling towers:** lowest approach (to wet-bulb) and highest water use from evaporation, drift and blowdown. Carries Legionella and water-treatment burden.
- **Closed-circuit fluid coolers:** the process fluid stays in a coil while spray water evaporates over it.
- **Dry coolers:** finned coils and fans. Zero water; limited on hot days.
- **Adiabatic dry coolers:** dry coolers with pads or sprays that switch on only above a set ambient temperature.
- **Hybrid coolers:** switch between wet and dry modes. A 24-month NREL trial of a thermosyphon cooler hybrid cut on-site WUE from 1.27 to 0.70 L/kWh (~50% less water) "without degrading energy performance."
- **Seawater, lake or river once-through:** discussed in Section 5.

3.3.3 Thermal energy storage (TES)

Chilled-water or ice tanks provide ride-through during chiller restarts (typically 5–15 minutes for Tier III/IV). They can also shift chiller load to cheaper or cooler night hours, which is useful under Indian time-of-day tariffs.

3.3.4 Refrigerants and the HFC phase-down

- **Kigali Amendment (2016)** to the Montreal Protocol: phases down hydrofluorocarbons by 80–85% by the late 2040s.
- **India (Article 5 Group 2; ratified 2021):** baseline is the average HFC use in 2024–2026 plus 65% of the HCFC baseline. Production and consumption **freeze on 1 January 2028**, then reduce by **10% by 2032, 20% by 2037, 30% by 2042 and 85% by 2047**. A 2026 government direction stops environmental clearances for new HFC production capacity beyond 31 December 2027. NRDC India estimates India's baseline at about 72–73 $MtCO_2e$.
- **Low-GWP alternatives in data-centre chillers:** R-1234ze (GWP <1, A2L) in centrifugal and magnetic-bearing chillers; R-513A (GWP ~630, non-flammable) as a path from R-134a; R-32 (GWP ~675) in DX units; R-1233zd in low-pressure centrifugals; and natural refrigerants (propane, ammonia, CO_2).
- **India Cooling Action Plan (ICAP, 2019):** a 20-year plan targeting a 20–25% cut in cooling demand, a 25–30% cut in refrigerant demand and a 25–40% cut in cooling energy by 2037–38. Data centres are not a named sector, but chiller and refrigerant policy affects them directly.

- **What it means:** the freeze could raise R-32 and R-134a prices after 2028. Specify low-GWP refrigerants now. Liquid cooling designs that eliminate compressors entirely (W40/W45 with dry coolers) are the most future-proof refrigerant strategy.

3.4 Liquid cooling types

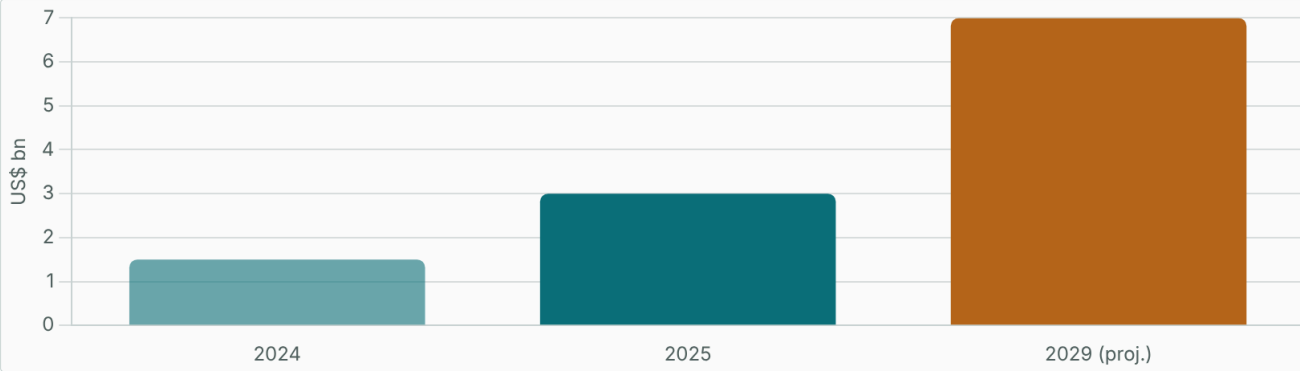


Figure 3.2 Liquid-cooling market. Manufacturer revenue, US$ bn (Dell'Oro). India: $113 m (2026) → $790 m (2031) per MarketsandMarkets. *Source: India Data Centre Atlas, chapter built on this chapter's data.*

3.4.1 Single-phase direct-to-chip (cold plate)

How it works: copper micro-channel cold plates sit on the GPUs, CPUs, switch ASICs and sometimes the memory and VRMs. Coolant (usually PG25 or treated water) absorbs heat as sensible heat (typically a ~10 K rise), returns via rack manifolds to a CDU, and the CDU passes the heat to facility water or room air.

Components: - **CDUs:** in-rack (~20–100 kW) or in-row (~500 kW–2.5 MW+). LiquidStack's CDU 2.X delivers up to 3,750 L/min at 3.5 bar, and its GigaModular platform scales to 14 MW. L2L CDUs need facility water; L2A CDUs reject heat to room air and suit retrofits. Dell'Oro counts about 40 CDU vendors, a sign of a crowded segment heading for consolidation. - **Manifolds and quick disconnects:** vertical rack manifolds and dripless blind-mate or UQD couplings (Nvidia MGX uses UQD08 manifolds). Third-generation MGX racks are "cable-free, hose-free, fanless" compute trays with liquid-cooled busbars up to 5,000 A. - **Leak detection:** rope or point sensors in trays and under racks, plus pressure-decay and flow monitoring in CDUs. Negative-pressure TCS designs draw air in rather than spilling coolant if a leak occurs. - **Share of heat captured:** about 70–80% on typical servers, about 85–90%+ on GB200/GB300 NVL72, and up to 98% claimed by Supermicro's DLC-2. Kyber is designed for 100%. The remaining heat needs air cooling.

Why it dominates: it uses standard server form factors and service models, works at W32–W45, and scales to 200 kW+ per rack. Dell'Oro says "single-phase direct liquid cooling is expected to remain the prevailing architecture for most deployments through the end of the decade." In Uptime's vendor survey, 48% rated water-cooled cold plates as currently viable, the highest of any liquid approach.

Drawbacks: two cooling systems to operate (liquid plus residual air), fluid chemistry management, leak risk, and supply-chain dependence on specific cold plates and QDs.

3.4.2 Two-phase direct-to-chip

A dielectric refrigerant boils inside the cold plate, and latent heat removes very high heat flux at near-constant temperature. Vendors include ZutaCore (HyperCool, a waterless dielectric design) and Accelsius (NeuCool). - **Pros:** handles extreme hotspots, and a leak of non-conductive fluid is benign to electronics. - **Cons:** fluid cost and regulation (fluorinated chemistries face PFAS and F-gas scrutiny), pressure-rated plumbing, and a smaller service ecosystem. - **Outlook:** Dell'Oro expects two-phase DLC "to expand gradually, with adoption accelerating once chip-level TDPs and thermal flux exceed the practical limits of single-phase systems." That is plausible with 2,000–4,000 W GPUs after 2027.

3.4.3 Single-phase immersion

Servers sit vertically in tanks of dielectric fluid (synthetic hydrocarbon, PAO oils or esters), pumped through a heat exchanger to facility water. Vendors include GRC, Submer, LiquidStack, Asperitas and Iceotope (chassis-level precision immersion). - **Pros:** captures about 100% of heat, removes server fans, PUE ~1.03–1.08 is reported, quiet, and dust- and humidity-proof. The dust and humidity protection is attractive for Indian edge sites. - **Cons:** horizontal tanks reduce floor density; tanks are heavy; servicing is messy; fluid costs; OEM warranty support is improving but not universal; and some optics and connectors are incompatible. Dell'Oro says immersion "is finding its place through selective adoption." In Uptime's vendor survey only about a third rated immersion viable, and 24% rated two-phase immersion viable. - **India example:** NTT deployed liquid immersion plus direct contact liquid cooling at Navi Mumbai NAV1 (January 2023), described as "APAC's largest… capacity of 4.8 MW spread over 13,740 square feet," and claimed it was "up to 30% more energy efficient than conventional cooling."

3.4.4 Two-phase immersion

Servers sit in sealed tanks of low-boiling fluid (~34–61°C). The fluid boils on hot components, vapour condenses on a lid coil, and drips back. - **Pros:** the highest heat-transfer coefficients and very low PUE (~1.01–1.03 claimed). - **Cons and the PFAS shock:** the market was built on 3M Novec and Fluorinert. **3M exited PFAS manufacturing at the end of 2025.** The last order date for Novec was 31 March 2025, and 3M's annual report states it "completed its exit of PFAS manufacturing at the end of 2025." Alternatives include **Chemours Opteon 2P50**,

an HFO-based fluid (GWP ≈10, boiling point ~49°C) trialled with LiquidStack and NTT. All face tightening EU PFAS restrictions and US state laws. - **Verdict:** a niche for extreme density or HPC. Not a mainstream bet in 2026. LiquidStack, the best-known two-phase immersion vendor and now owned by Trane, sells DLC CDUs as its volume product.

3.4.5 Rear-door heat exchangers as hybrid

RDHx is commonly combined with DLC to capture the residual air heat. Meta's **air-assisted liquid cooling (AALC)** reverses the idea: cold plates feed a closed loop that dumps heat into room air through a rear-door coil, driven by a reservoir and pumping unit (RPU) in an adjacent rack. That lets Meta run liquid-cooled GPUs in air-cooled halls without facility water piping.

3.4.6 Emerging: spray, jet impingement, microfluidic and embedded cooling

- **Spray or jet impingement:** jets of coolant hit the die or lid directly, giving very high local heat-transfer coefficients.
- **Microsoft microfluidics (announced 23 September 2025, with Swiss start-up Corintis):** channels etched into the back of the silicon, with AI-optimised, leaf-vein-like layouts that target hotspots. Lab tests showed heat removal "up to three times better than cold plates" (workload-dependent) and "a 65% reduction in maximum temperature rise of the silicon inside a GPU." Microsoft also claims it dissipates more than 1 kW/cm², 2–3× standard cold plates. **Caveat:** this is lab-scale, and Network World quoted analysts warning that "chip replacement [becomes] the only maintenance option."
- **Embedded and 3D-stack cooling:** necessary if 3D-stacked logic and HBM become mainstream. Expect commercial use around 2028–2030 at the earliest.

3.4.7 Liquid cooling comparison

Table 3.7 Liquid cooling comparison

Attribute	Single-phase DLC	Two-phase DLC	Single-phase immersion	Two-phase immersion	RDHx (active)
Practical rack density	50–250 kW+ (with air assist)	100–300 kW+	100–250 kW per tank	200 kW+ per tank	40–75 kW
Heat to liquid	70–98%	70–95%	~100%	~100%	100% at the door (air-side)
Facility water class	W32–W45	W32–W45	W32–W45	W32–W45	W17–W27 (needs colder water)
PUE (reported range)	1.05–1.20	1.04–1.15	1.03–1.08	1.01–1.05	1.15–1.35
Fluid	PG25 / water (cheap)	Dielectric refrigerant ($$)	Hydrocarbon or synthetic oil ($–$$)	Fluorocarbon or HFO ($$$)	Water
Serviceability	Server-like; QDs	Server-like; pressurised	Lift-out, drain	Sealed tank	Easy
Weight / floor	Heavy racks (1.5–3 t)	Heavy racks	Heavy tanks, low floor density	Heavy tanks	Moderate
OEM support / warranty	Mainstream (Nvidia reference)	Emerging	Improving; check per OEM	Limited	Mainstream
Maturity (2026)	Mainstream for AI	Early commercial	Niche but proven	Niche; PFAS risk	Mature
OCP / standards	OCP ACS cold plate, CDU, UQD specs; ORv3	OCP ACS two-phase workstream	OCP ACS immersion specs	OCP ACS immersion	OCP Door HX workstream

3.5 Natural and alternative cooling

Table 3.8 Natural and alternative cooling

Approach	Flagship examples	How it works	Relevance to India
Climate free cooling	Nordic sites (Meta Luleå, many in Finland, Norway, Iceland)	Cool, dry ambient air: indirect or direct economisers for most of the year. Cool-climate adiabatic sites "may rely on free cooling for 90% to 95% of the year."	Limited. Only hill or high-altitude sites and winter months.
Seawater cooling	Google Hamina, Finland (former paper mill using Baltic seawater)	Cold seawater through titanium HX to the facility loop	Tropical surface seawater is ~27–31°C, too warm for chilled water but usable for **heat rejection at W32–W45**. Andhra Pradesh's data-centre policy proposes seawater cooling. Indian coastal thermal plants (NTPC Simhadri, Hinduja Vizag) already use seawater for condenser cooling.
Deep lake / deep water	Enwave Deep Lake Water Cooling, Toronto (Lake Ontario at ~4°C)	Cold deep water through district cooling	No equivalent cold deep lakes near Indian metros
Groundwater / geothermal	Various European aquifer thermal energy storage sites	Aquifer as heat sink or seasonal storage	Restricted. Indian groundwater is overdrawn.
Underwater	Microsoft Project Natick (Orkney, 2018–2020); China's commercial underwater data centre off Hainan and a reported offshore-wind-linked project near Shanghai	Sealed pressure vessel cooled by surrounding seawater	Microsoft says the Natick servers "showed a failure rate of 1/8th that of our land-based control group" (vessel retrieved July 2020), but Microsoft confirmed in 2024 that the project is no longer active. China's projects are reported commercial pilots, not independently verified here. A curiosity for India.
Heat reuse / district heating	Stockholm Data Parks; Meta Odense (Denmark)	Heat pumps lift 25–45°C waste heat to 60–80°C district heating	Low heating demand in India. Better candidates are absorption chillers, desalination pre-heat, or industrial process heat.

3.6 Market and adoption

3.6.1 Market size and forecasts (2024–2031)

Table 3.9 Market size and forecasts (2024–2031)

Source (date)	Metric	Figure
Dell'Oro (Jan 2026)	Global DC liquid cooling, manufacturer revenue	~$3 billion in 2025 (≈2× 2024), ~$7 billion by 2029
Dell'Oro (Aug 2025)	Direct liquid cooling	$1.1 billion (2024) to $5.8 billion (2029); DLC grew 156% year on year in 2Q 2025
Dell'Oro (2026)	DLC 2030	~$8 billion by 2030
Dell'Oro (Aug 2025, 2026 update)	Data centre physical infrastructure (DCPI)	$63.1 billion by 2029 (15% CAGR); >$80 billion by 2030. Thermal management CAGR 19–20%.

Source (date)	Metric	Figure
MarketsandMarkets	India DC liquid cooling	$113.0 million (2026) to $789.9 million (2031), ~32% CAGR
Other analyst estimates (secondary)	Global "liquid cooling" (broader definitions)	~$5.5 billion (2025) to ~$15.8 billion (2030)

Contested figures: market estimates vary two- to three-fold depending on what is counted. Dell'Oro's manufacturer-revenue definition is the most conservative and the most useful for comparing years.

3.6.2 Adoption (Uptime Institute)

- **2024 Cooling Systems Survey (n=964):** 22% of respondents use some DLC; 61% do not but would consider it; 17% would not. Nearly half of DLC users run it in under 10% of racks.
- **2025 Cooling Systems Survey (n=1,033; July 2025 report):** "DLC adoption remains slow and steady." Perimeter air cooling is still the most popular option, used by around three-quarters of operators. High rack density is the top driver, and **lack of system standards is the top barrier.**
- **Global Data Center Survey 2025:** average modal rack density ~9 kW (up from 8.3 kW); 15% report PUE ≤1.3; PUE improvement has largely stalled.
- **What it means:** DLC adoption by count of operators is modest. By *capacity*, new AI megawatts are overwhelmingly liquid-cooled. Both statements are true.

3.6.3 AI-driven shifts

- **GB200/GB300 NVL72:** 90%+ liquid; air-cooled HGX variants remain for 8-GPU servers at lower bins.
- **Vera Rubin (in production 2026; shipping H2 2026):** 100% liquid compute and switch trays, 45°C warm-water inlet. Nvidia says 45°C cooling "unlocks up to 30% more GPUs in the same power budget."
- **Kyber / Rubin Ultra NVL576 (H2 2027):** ~600 kW per rack, 100% liquid, no fans, 800 VDC. Schneider engineers say the 1 MW rack is coming.

3.6.4 Vendor landscape and consolidation (2024–2026)

Table 3.10 Vendor landscape and consolidation (2024–2026)

Company	Position / recent moves
Vertiv	Dell'Oro names it the liquid cooling market leader. Acquired CoolTera (CDUs, UK, announced Dec 2023) and agreed to buy PurgeRite (~$1 billion, Nov 2025; fluid services). **Vertiv did not buy CoolIT.**
Ecolab to buy CoolIT	Announced 20 Mar 2026: ~$4.75 billion cash from KKR and Mubadala, about 29× next-12-month EBITDA; CoolIT expected to generate ~$550 million in sales. Closing reported in early July 2026.
Eaton to buy Boyd Thermal	$9.5 billion (Nov 2025; 22.5× 2026E EBITDA). Boyd forecast $1.7 billion of 2026 sales, $1.5 billion of it liquid cooling. Closing reported in March 2026.
Schneider Electric to buy Motivair	$850 million for a controlling stake (2024). Motivair liquid cooling factory opened in Bengaluru (~March 2026), its third globally.
Trane Technologies to buy LiquidStack	Announced 10 Feb 2026, closed 3 Mar 2026. Brings DLC CDUs and single- and two-phase immersion.
nVent, CoolIT, Boyd	Named by Dell'Oro as holding strong DLC share
Others	Asetek, Submer, GRC, Iceotope, Stulz, Rittal, Carrier, Johnson Controls (Silent-Aire), Daikin, Danfoss, Munters, Airedale, Delta, and Taiwanese cold-plate makers
Indian players	Mostly integrators and assemblers: Blue Star, Voltas and Kirloskar (chillers), plus local system integrators. Global vendors manufacture locally (Section 9.5).

Interpretation: cooling has turned into a strategic choke point. Around $16 billion of acquisitions in 2025–26 shows that the large power and HVAC companies now treat CDUs and cold plates as core products. Buyers should insist on open interfaces (OCP UQD, CDU control protocols) to avoid lock-in.

3.6.5 Hyperscaler approaches

- **Google:** liquid-cooled its TPUs since TPU v3 (2018) using its own cold plates and CDUs. For its planned 1 GW, ~$15 billion Visakhapatnam AI hub with AdaniConneX, Google has publicly said it will use air-cooled (dry) heat rejection rather than water-based cooling.
- **Microsoft:** since August 2024, all new designs use **zero-water-evaporation** closed-loop chip-level cooling: "once the system is filled during construction, it will continually circulate water between the servers and chillers." Microsoft says this avoids "more than 125 million liters of water per year per datacenter." First sites are Phoenix and Mt Pleasant (Wisconsin), from 2026. Fleet WUE was 0.30 L/kWh in 2023, down 39% from 0.49 in 2021.
- **Meta:** AALC (cold plates, RPU and rear-door coil) across its fleet by early 2025, then facility-water DLC in new AI-optimised designs. OCP Open Rack v3 supports both.
- **AWS: In-Row Heat Exchanger (IRHX)**, announced 9 July 2025 for Blackwell (P6e / GB200 NVL72). It has a water-distribution cabinet, an integrated pumping unit, and modular in-row fan-coil modules, so it is effectively an L2A CDU for existing air-cooled halls. AWS's Dave Brown said off-the-shelf options "would take up too much data center floor space or increase water usage substantially."

3.7 Water–energy trade-offs

3.7.1 The core trade

Evaporating 1 kg of water absorbs ~2,400 kJ, so towers can reject heat close to the wet-bulb temperature. Dry systems can only approach the dry-bulb temperature, which is often 8–15 K higher in hot-dry climates.

Table 3.11 The core trade

Heat rejection	On-site WUE (L/kWh IT)	Energy effect	Best climate
Open cooling tower + water-cooled chiller	~1.4–2.5 typical; 2–4 for traditional evaporative systems	Lowest chiller energy; evaporative cooling "reduces the peak cooling power demand by 25–35% in the summer compared with dry cooling"	Hot-dry with ample (ideally non-potable) water
Direct / indirect evaporative air cooling	~0.2–1.5 (seasonal)	Very low energy when humidity is low	Hot-dry (Delhi and Hyderabad pre-monsoon, Bengaluru dry season)
Adiabatic dry cooler	~0.1–0.5	Wets coils only on peak days	Most climates; India's best compromise
Hybrid / thermosyphon	~0.5–0.7 (NREL: 1.27 to 0.70)	Near-evaporative energy, about half the water	Mixed
Dry cooler + air-cooled chiller	~0	Cooling energy 5–15% higher than evaporative (more in hot climates); air-cooled chillers can use 2–3× the electricity of evaporative-cooled ones at peak	Water-scarce sites
DLC at W40–W45 + dry cooler	~0 to <0.1	Small penalty, since 45°C water can be dry-cooled even at 35–40°C ambient	**Almost anywhere, including India**
Seawater once-through	~0 freshwater	Pumping energy only; marine permitting and thermal discharge limits	Coastal

Benchmarks: the US average WUE is 1.8 L/kWh (LBNL, Shehabi et al., 2016). Across 15 US climate zones, hyperscale WUE values are 0.09–1.80 L/kWh, while traditional evaporative systems require 1.99–4.00 L/kWh; the technology effect averages 10.57× across locations (arXiv 2606.21760). Microsoft's regional WUE is 1.52 L/kWh in Arizona versus 0.02 L/kWh in Singapore (Introl, citing Microsoft data).

3.7.2 The hidden water in electricity

"Zero-water" designs move water use off-site, to power plants. India's grid is still largely coal-thermal, and MoEFCC rules cap cooling-tower thermal plants at 3.5 m³/MWh (3.5 L/kWh) for existing plants and 3.0 m³/MWh for plants installed after 1 January 2017. A dry design that raises PUE from 1.2 to 1.35 adds about 0.15 kWh per kWh of IT, or around 0.3–0.5 L/kWh of off-site water. That is still usually less than the ~1.5–2 L/kWh a tower evaporates. Academic work recommends dry cooling where its efficiency penalty is small (PUE ≤1.5) and in water-stressed basins.

What it means: warm-water liquid cooling largely ends the water-versus-energy dilemma. Where legacy air halls force a choice, use evaporative cooling only where water is non-potable or abundant, and adiabatic or hybrid cooling elsewhere.

3.8 Cost

3.8.1 Capex by cooling approach (indicative, 2025–26)

Table 3.12 Capex by cooling approach (indicative, 2025–26)

Approach	Cooling-system capex	Notes
Traditional air (CRAH + chillers + containment)	~$1.5–2 million per MW of IT	Cannot serve 100 kW+ racks
Advanced air with RDHx	~$1,800–3,200 per kW	Bridge for 30–60 kW racks
Single-phase DLC (CDU, manifolds, piping, plus heat rejection)	~$3–4 million per MW all-in cooling (some sources: $3,500–5,000 per kW). The CDU, manifold and cold-plate subset is ~$300–650 per kW.	CDUs ~$75,000–150,000 per 500 kW; cold plates and manifolds ~$5,000–10,000 per server; ~$50,000 per NVL72 rack
Single-phase immersion	~$1 million per MW for tanks and fluid systems (versus ~$650,000 for comparable DLC hardware); $2.5–3.5 million per MW all-in (vendor-adjacent estimates)	Some vendor-sponsored TCO studies claim lower total capex than air at high density; treat with caution
Retrofit an air hall to liquid	~$2–3 million per MW of cooling capacity	12–18-month lead times for raised-floor facilities

Whole-facility context: an air-cooled Tier III facility costs ~$8–14 million per MW of IT in 2026; a fully fitted DLC facility costs ~$17–25 million per MW. Most of that difference is higher-density electrical gear, heavier structures and stainless piping. **In India (SBI, April 2026, citing JM Financial):** a Tier III non-AI air-cooled facility costs **INR 50–55 crore per MW**, AI or Tier IV builds with immersion **up to INR 75 crore per MW**, and large campuses **INR 35–45 crore per MW**. Mechanical and cooling systems are typically ~15–25% of total build cost; this is an industry rule of thumb that varies by design.

Contested figures: published cooling capex figures conflict by 2–5×, because of scope, greenfield versus retrofit, and vendor sponsorship. Use immersion TCO studies only as directional evidence and get project-specific quotes.

3.8.2 Opex

- Air cooling overhead: ~0.3–0.6 kW of cooling per kW of IT in legacy halls (PUE 1.4–1.7). DLC: ~0.05–0.2.
- At ₹7–9/kWh Indian industrial tariffs, a 10 MW IT hall moving from PUE 1.5 to 1.2 saves ~26 GWh a year, or roughly ₹18–23 crore a year.
- DLC-specific opex: fluid testing and top-up, filter changes, CDU pump maintenance, and leak-response training. Immersion adds fluid replacement.
- One cost model finds DLC's 10-year TCO falls below air at ~30 kW per rack (at $0.12/kWh), consistent with industry consensus that liquid is cheaper above ~30–40 kW per rack.

3.9 The Indian perspective

3.9.1 Climate challenge

India's problem is **heat plus humidity**. High wet-bulb temperatures limit evaporative cooling, high dry-bulb temperatures limit dry coolers, and dust, pollution and monsoon humidity largely rule out direct fresh-air cooling.

Table 3.13 Climate challenge

City	Climate character	Free-cooling outlook	Indicative peak wet-bulb*
Mumbai / Navi Mumbai	Hot-humid coastal, monsoon	Poor for air-side; one estimate gives ~28% economiser hours at conventional setpoints	~28°C design; monsoon peaks reportedly 30–32°C

City	Climate character	Free-cooling outlook	Indicative peak wet-bulb*
Chennai	Hot-humid coastal, severe water stress	Poor for air-side; good for W40–W45 dry cooling	~28–29°C
Delhi NCR / Noida	Composite: very hot-dry summer (>45°C dry-bulb), humid monsoon, cool winter	Good winter economisation (Dec wet-bulb 3–8°C); summer is the worst case for dry coolers	~29°C
Bengaluru	Moderate, elevated (~900 m)	Best in India; one estimate gives ~71% free-cooling hours at a 35°C dry-cooler setpoint	~23°C
Hyderabad	Hot semi-arid	Good for indirect evaporative cooling much of the year; hot April–May	~25–26°C
Visakhapatnam	Hot-humid coastal	Seawater heat rejection possible; water-stress controversy	~28°C

*Indicative values from ASHRAE-type climatic data and engineering blogs. Verify against ASHRAE Handbook or IS 7896 / ISHRAE design tables before design. By comparison, Nordic sites achieve free cooling for most of the year.

Water stress: Bengaluru, Chennai and Gurugram are named in the SBI report as water-stressed metros, where "traditional cooling infrastructure can consume tens of millions of litres annually" in large facilities. New rules require water-management plans and prioritise treated wastewater. India's Ministry of Jal Shakti says the industry is adopting direct-to-chip and immersion cooling to reduce water demand.

3.9.2 Indian operators and liquid cooling deployments

Table 3.14 Indian operators and liquid cooling deployments

Operator	Liquid cooling status (sourced)
Sify	First in India to be certified for liquid cooling under Nvidia's DGX-Ready programme (Rabale, Navi Mumbai; Sept 2024), specified "to support 130 KW/rack." Chennai 02 (Siruseri) and Noida 02 certified for "up to 200 KW/rack" (May 2025).
Yotta	NM1 (Navi Mumbai) hosted ~16,000 H100/GH200 GPUs by mid-2025 (secondary source). Yotta says RDHx supports "50–60 kW" and DLC "up to 80 kW per rack." It has announced plans for 40,000 Vera Rubin GPUs at D4 Greater Noida (120 MW) and GB300 at NM2. These are plans, not commissioned capacity.
CtrlS	Marketing claims of AI racks up to 70 kW (RDHx, DLC, immersion) and immersion "up to 250kW per rack." Plans for more than 4 GW of AI-ready capacity. No independently verified rack-level DLC deployment details were found.
NTT DATA India	NAV1 Mahape: 4.8 MW of immersion plus direct contact liquid cooling (2023).
Nxtra (Airtel)	~300 MW capacity (March 2026); reported "high-density racks of up to 130 kW using air and liquid cooling" (secondary).
STT GDC India	Expanding AI-ready capacity; no published liquid-cooling rack density found.
Google / AdaniConneX / Nxtra (Vizag)	1 GW AI hub; Google says air-cooled (dry) heat rejection; under legal and community challenge over water.

Overall: Indian liquid cooling is at the pilot-to-first-wave stage. Leading operators report PUE of 1.4–1.5 (SBI), well above the MeitY AI-cloud tender requirement of **PUE <1.35** (in force since 2024). DLC is the most practical route to that threshold for GPU halls in Mumbai and Chennai.

3.9.3 Andhra Pradesh and seawater cooling

Andhra Pradesh's data-centre policy proposes **seawater cooling**, and the state incentivises operators to build **desalination** plants. IT Minister Nara Lokesh says one operator is already building one. AP also mandates ≥70% renewable energy for data centres. The state estimates its planned **6.5 GW** Vizag ecosystem would need about **3 TMC (~85 billion litres) of water a year** "based on current technology," supplied from the Polavaram industrial allocation. Activists and lawyers note the environmental clearance for Google's Tarluvada site does not specify water volumes or the freshwater-versus-seawater split.

Assessment: tropical seawater (~28–30°C) cannot produce chilled water without chillers. It works well as a **condenser or heat sink for W32-W45 liquid loops**, replacing evaporative towers, as coastal Indian power plants already do. It needs titanium HXs, anti-fouling measures, CRZ permits and thermal-discharge limits. Desalination for tower make-up is energy-intensive, at ~3–4 kWh/m³ for seawater reverse osmosis, and produces brine. Dry or adiabatic cooling plus DLC avoids most of that water.

3.9.4 Standards and policy

- **ECBC 2017 / BEE:** no PUE cap in the code itself. The **"User Guide for Implementing ECBC in Data Centers"** (CII-IGBC with LBNL/US DOE, May 2021) sets tiered requirements: BEE 3-star chillers minimum rising to 5-star at higher tiers, cooling-tower fan efficiency, "Measure PUE as per Green Grid," and simulation reports to show PUE for new builds. Treat this as voluntary unless notified by a state.
- **MeitY:** PUE <1.35 for AI-cloud tenders (since 2024).
- **SEBI BRSR Core:** energy and water disclosure for the top 150 listed companies.
- **IGBC Green Data Centre rating:** Platinum requires more than 80 of 100 points, and energy is the largest category.
- **TRAI:** has recommended a green data-centre certification framework.
- **Kigali / ICAP:** HFC freeze 2028 (Section 3.4).

3.9.5 Local manufacturing

- **Vertiv, Chakan (Pune):** opened January 2024 on 4.8 acres (210,000 sq ft), making "cooling solutions ranging from 200W to 2MW+" including adiabatic free-cooling chillers, AHUs and in-row and rack cooling. It is Vertiv's third Indian facility, after Ambernath and Pune.
- **Schneider Electric, Bengaluru:** a cooling factory, plus a **Motivair liquid-cooling factory** (~March 2026), its first in India and third globally. It makes CDUs, direct-to-chip hardware and RDHx.
- **Gap:** the SBI report flags an "underdeveloped domestic infrastructure ecosystem" in "power systems, HVAC, lithium-ion storage." Cold plates, QDs and specialised pumps are still largely imported.

3.9.6 Recommendations for India: what to use where

Table 3.15 Recommendations for India: what to use where

Use case	Recommended cooling	Heat rejection	Target PUE / WUE
AI training / inference (≥40 kW per rack), any city	Single-phase DLC (W32–W45), with RDHx or fan-wall for the residual 10–25%	Dry coolers with adiabatic assist on peak days; magnetic-bearing chillers only for trim and resilience	PUE 1.15–1.25; WUE <0.2
AI halls in coastal Mumbai, Chennai, Vizag (hyperscale)	DLC as above	Seawater or treated-sewage-water heat rejection where permitted; otherwise adiabatic dry coolers	PUE ~1.2; freshwater WUE ~0
Cloud / enterprise (8–20 kW per rack), Bengaluru, Hyderabad, Pune	Hot aisle containment, fan walls or CRAH	Indirect evaporative coolers or air-cooled free-cooling chillers; recycled water if evaporative	PUE 1.3–1.4; WUE 0.2–0.8
Cloud / enterprise in Mumbai and Chennai	Hot aisle containment, CRAH, higher chilled-water setpoints (18–20°C)	Water-cooled magnetic-bearing chillers on recycled water, or air-cooled free-cooling chillers in water-stressed zones	PUE 1.35–1.5
Delhi NCR	Hot aisle containment plus economiser for winter; DLC for AI	Hybrid: dry/adiabatic in winter and monsoon, evaporative trim in May–June using recycled water	PUE 1.25–1.4
Edge / telecom / dusty sites	Sealed DX (R-32 / low-GWP) or single-phase immersion micro-pods	Dry	—
Retrofits of existing Indian air halls for GPUs	L2A CDUs (AWS IRHX / Meta AALC style) or active RDHx up to ~50–70 kW; full L2L DLC needs facility water risers	Existing chilled water	—

Do not build new freshwater-evaporative designs in Chennai, Bengaluru or Gurugram. **Do not** bet on two-phase immersion until PFAS-free fluids and OEM warranties mature. **Do** design every new hall with facility-water taps and floor loading for 100–200 kW liquid racks, even if it opens air-cooled.

3.10 Future trends

1. **Racks heading to 1 MW.** GB300 is ~130–150 kW, Vera Rubin ~190–230 kW (2026), Kyber ~600 kW (2027), and 1 MW-class racks are discussed for around 2028–29.
2. **800 VDC interplay.** Moving from 54 V in-rack to 800 VDC cuts copper (Nvidia says over 150% more power through the same copper) and AC/DC losses, but brings power conversion into sidecars that themselves need liquid cooling. Power and cooling are being co-designed, which helps explain the Eaton–Boyd and Schneider–Motivair deals.
3. **Warm water (45°C+) and the decline of chillers.** W45 makes chiller-free operation feasible in most climates, including much of India. Expect chillers to be downsized to a trim and resilience role, and the dry-cooler market to grow.
4. **Two-phase and embedded cooling** once GPU TDPs pass ~2.5–4 kW (2027–2029), with PFAS-free fluids a key enabler.
5. **Heat reuse** at 45–60°C return temperatures. In India, absorption cooling, desalination pre-heat and industrial process heat are more realistic uses than district heating.
6. **AI-driven cooling optimisation.** DeepMind's 2016 work cut Google's data-centre cooling energy "by up to 40%," equivalent to "a 15% reduction in overall PUE overhead." Similar machine-learning controls are now in commercial DCIM and building-management products.
7. **Standardisation.** The OCP Advanced Cooling Solutions project covers cold-plate, CDU, UQD, door-HX, AALC and immersion specifications, and ORv3 and MGX define rack and manifold interfaces. Uptime's survey shows that missing standards are still the top barrier, so expect convergence on Nvidia/OCP interfaces.

3.11 Caveats

- **GPU TDPs for Rubin and Rubin Ultra** (1,800–2,300 W) and Vera Rubin rack power (190–230 kW) are industry estimates, not Nvidia specifications. Kyber's ~600 kW comes from Nvidia's CEO at GTC 2025, and timelines can slip.
- **Market sizes** vary widely by definition. Dell'Oro figures are manufacturer revenue only.
- **PUE, capex and TCO** figures for immersion and two-phase cooling often come from vendors or vendor-affiliated analyses. Cost figures across sources conflict by 2–5×.
- **Uptime adoption figures** measure the share of operators, not capacity. Secondary outlets report other percentages (e.g., "19% liquid adoption") that could not be verified against Uptime primary data.
- **Indian climate figures** are indicative and partly from engineering blogs. Site design must use ASHRAE / IS 7896 / ISHRAE data.
- **Indian operator rack-density claims** (CtrlS, Nxtra, Yotta) are largely company marketing or secondary reports. Yotta's Rubin and GB300 numbers are announced plans.
- **M&A closing dates** for Eaton–Boyd and Ecolab–CoolIT are from secondary and financial-press reports.

Sources cited in this chapter

Organisations and publications named in this chapter's text, in order of how often they are cited. Full details are in the bibliography.

- NVIDIA product specifications and roadmap
- Dell'Oro Group, liquid-cooling market data
- Microsoft sustainability reports and datacenter design posts
- Uptime Institute, Global Data Center Surveys and Tier Standard
- ASHRAE TC 9.9 thermal guidelines
- Google environmental reports, efficiency data and Gemini inference paper
- Meta sustainability reports and data index
- State Bank of India sector report on data centres (April 2026)
- Yotta Data Services disclosures
- Amazon Web Services sustainability reporting
- Lawrence Berkeley National Laboratory, 2024 US Data Center Energy Usage Report
- NTT DATA ESG data
- Nxtra by Airtel disclosures
- Ministry of Electronics and Information Technology (MeitY), Data Centre Policy draft and IndiaAI
- Bureau of Energy Efficiency
- CtrlS disclosures
- AdaniConneX disclosures
- Kigali Amendment to the Montreal Protocol
- National Renewable Energy Laboratory water factors

- ISHRAE climate design data
- Ministry of Environment, Forest and Climate Change, thermal plant water norms and EIA
- STT GDC India ESG report
- Sify Infinit Spaces sustainability report
- MarketsandMarkets market estimates
- The Green Grid metric definitions

PART II · RESOURCES AND IMPACTS

Chapter 4
What one campus asks of a city: water and power for a Bengaluru data centre

WHAT THIS CHAPTER COVERS

This chapter applies the physics of power and water to Bengaluru, India's technology capital and one of its most water-stressed cities. It sets out the assumptions for three worked cases (a 20 megawatt colocation hall, a 50 megawatt hyperscale facility and a 100 megawatt AI campus), calculates their electricity, grid connection, carbon and water, compares them with other users of water and power in the city and the state, describes where Bengaluru's data centres obtain water today, and asks whether data centres will cause water scarcity. It ends with recommendations for water, power, siting, transparency and national policy.

KEY TAKEAWAYS

1. Across the whole city, data-centre water use will not cause Bengaluru's water scarcity: one gigawatt of new IT load with evaporative cooling on fresh water would take about 32 million litres a day, about 1% of the city's supply of roughly 3,000 million litres a day.
2. Locally, however, a single 50 megawatt evaporative facility drawing borewells in Whitefield or Devanahalli would use about 1.4 million litres a day, the domestic water of about 10,000 people, in areas where groundwater is already over-drawn.
3. Electricity is the larger strain. A single 100 megawatt AI campus needs about 840 million units of electricity a year and about 140 megavolt-amperes of 220 kilovolt supply, which is more than 5% of all household electricity used in Karnataka.
4. The water used at power stations to generate a data centre's electricity is often larger than the water the data centre uses on site.
5. Bengaluru's data centres should use treated wastewater or dry cooling, should not draw groundwater for cooling, and should disclose their water use by source.

How this chapter connects to the others. Chapter 6 develops the method for measuring water impact that this chapter applies, and Chapter 7 extends the calculations to other Indian cities and to hour-by-hour profiles.

Citywide, data centre water use will not cause Bengaluru's water scarcity. Locally, it can make it worse. If 1 GW of new IT load used evaporative cooling on fresh water, it would take about 32 MLD, roughly 1% of the city's ~3,000 MLD supply. But one 50 MW evaporative facility drawing borewells in Whitefield or Devanahalli would use about 1.4 MLD, the domestic water of ~10,000 people. Power is the bigger strain: a single 100 MW AI campus needs about 840 MU a year and ~140 MVA of 220 kV supply, more than 5% of all household electricity used in Karnataka.

4.1 Key Findings

1. **The power is the bigger issue, not the water.** A 100 MW IT AI campus needs ~841 GWh/yr. That is ~0.9% of Karnataka's ~92,550 MU FY2024-25 energy requirement, ~2.3% of BESCOM's estimated ~37,000 MU sales, and ~5% of Karnataka's 16,089 MU household consumption (2023-24). The state's 1 GW target, built as AI capacity, would add ~8,400 MU/yr (~9% of today's state demand) and ~1.3 GW of firm peak. That is about 6.5% of the record 20,094 MW peak on 11 September 2026 (News Karnataka; Deccan Herald reports 20,093 MW, a 7.7% rise on the previous summer).
2. **On-site water depends mostly on the cooling technology.** For the same IT energy, evaporative cooling towers use about 1.8 L/kWh, adiabatic/hybrid about 0.4, and dry/closed-loop DLC about 0.05. That is a ~36× spread. Choosing the cooling technology matters more than choosing the site.
3. **Bengaluru's current data centre water use is small but poorly measured.** Estimates range from ~2.2 MLD of permitted volumes (Down To Earth's estimate of roughly 812.9 million litres a year of permissions, from 2013–2025 government meeting minutes and BWSSB data), to ~4 MLD (the Karnataka IT department's estimate of ~4,000 kL/day for the state's 32 data centres, based on informal operator inputs), to ~20 MLD (WELL Labs hydrologist Shashank Palur's estimate, which assumes ~26 million litres per MW per year). My bottom-up model (162 MW, mixed cooling) gives 2–5 MLD, which supports the lower figures. Neither BWSSB nor KSPCB tracks data centre water separately.
4. **Many comparable users are larger.** JSW Vijayanagar's water allocation (330 MLD) equals more than 200 hyperscale data centres. A 1,000-flat apartment complex uses about the same water as the 20 MW colocation site. A 10 million sq ft IT campus uses about 9× the colocation site's water for similar electricity. Data centres are unusual because they are *power-dense*, not because they are *water-dense*.
5. **Treated wastewater is abundant enough to supply all of this.** BWSSB has ~1,348.5 MLD of STP capacity. It sends 440 MLD (K&C Valley) and 210 MLD (H-N Valley) of secondary-treated water to Kolar and Chikkaballapur tanks, and reuses only ~30 MLD commercially. Even a 1 GW evaporative fleet would need ~43 MLD of feed water to RO.

4.2 Assumptions

Table 4.1 Assumptions

Parameter	Case A: Colo/enterprise	Case B: Hyperscale/cloud	Case C: AI campus
Design IT load	20 MW	50 MW	100 MW
Cooling	Water-cooled chillers + cooling towers (with water-side economiser)	B1: chillers + cooling towers; B2: adiabatic/hybrid dry coolers	DLC (~80% to liquid) + dry coolers with adiabatic assist
Annual PUE (central / range)	1.45 / 1.30–1.60	1.35 / 1.25–1.45	1.20 / 1.15–1.25

Parameter	Case A: Colo/enterprise	Case B: Hyperscale/cloud	Case C: AI campus
Average utilisation of design IT	70% / 60–85%	75% / 65–85%	80% / 70–90%
Design-day PUE (for contract demand)	1.60	1.50	1.30
On-site WUE (L/kWh IT)	1.8 / 1.5–2.0	B1: 1.6; B2: 0.4	Dry 0.05; adiabatic 0.2; (evaporative alt. 1.6)
Staff on site	~150	~300	~300
Grid water intensity	1.5 L/kWh (range 1.0–2.5)	same	same

Other constants: 8,760 h/yr. Per-capita water benchmark 135 LPCD (CPHEEO/BWSSB norm for sewered cities). Office/staff use 45 LPCD. Bengaluru household electricity 150 kWh/month = 1,800 kWh/yr. That household figure is an estimate: Gruha Jyothi caps free supply at 200 units, and Bengaluru Urban has the lowest enrolment because many homes exceed 200 units, so the city average is plausibly 150–250 units/month.

4.3 Power calculations

Formulas - IT energy (MWh/yr) = Design IT (MW) × Utilisation × 8,760 - Facility energy (MWh/yr) = IT energy × PUE (1 GWh = 1 MU) - Average grid draw (MW) = Facility energy ÷ 8,760 - Contract demand (MVA) ≈ Design IT × design-day PUE ÷ power factor (0.95), plus ~3–5% for offices and losses

Table 4.2 Power calculations

	Case A (20 MW)	Case B (50 MW)	Case C (100 MW)
IT energy	20×0.70×8,760 = **122.6 GWh**	50×0.75×8,760 = **328.5 GWh**	100×0.80×8,760 = **700.8 GWh**
Facility energy (central)	122.6×1.45 = **177.8 MU**	328.5×1.35 = **443.5 MU**	700.8×1.20 = **841.0 MU**
Sensitivity range	136.7–238.3 MU	355.9–539.8 MU	705.2–985.5 MU
Average draw	20.3 MW	50.6 MW	96.0 MW
Design peak	32 MW	75 MW	130 MW
Contract demand	~34–35 MVA	~80 MVA	~140 MVA
Supply voltage (KERC norms)	110 kV	220 kV	220 kV, dedicated substation, dual feed
Installed transformer capacity (2N)	~70 MVA	~160 MVA	~280 MVA
Bengaluru household equivalent (1,800 kWh/yr)	~98,800	~246,400	~467,200
Share of Karnataka energy (92,550 MU, FY25)	0.19%	0.48%	0.91%
Share of BESCOM sales (~37,000 MU, est.)	0.48%	1.2%	2.3%
Share of state household use (16,089 MU)	1.1%	2.8%	5.2%
Share of 2026 state peak (20,094 MW)	0.17%	0.40%	0.67%
Energy bill at HT-2a (Rs 6.70/kWh + Rs 365/kVA/month)	~Rs 134 cr/yr	~Rs 332 cr/yr	~Rs 625 cr/yr

How to read this - **Redundancy:** Tier III/IV sites use 2N utility feeds and transformers, so installed MVA is about twice contract demand. Diesel or other backup is sized for the full facility load. Redundancy doubles the network assets KPTCL/BESCOM must build, but not the energy drawn. - **Voltage thresholds:** under the KERC Conditions of Supply, 66 kV covers up to 20 MVA. 110 kV covers up to 35 MVA, raised to 55 MVA in June 2025 subject to technical feasibility and ₹10 lakh/MVA augmentation charges. Loads above 20 MVA can take 220 kV. So Case A fits at 110 kV. Cases B and C need 220 kV bays, typically fed by two independent 220 kV circuits from different 400/220 kV grid stations. Multi-phase AI campuses of 300 MW or more will push toward 400 kV. - **The load is flat.** Data centres run at a high load factor (~75–90%) all day and all year. Karnataka's demand peaks in Jan–Mar daytime and is swinging sharply: daily consumption ran 310–350 MU in August 2026 against ~200 MU in August 2025. A large flat block of data centre load raises firm-capacity needs even when the energy share looks small. - **Grid mix:** about 63% of Karnataka's power requirement is met from renewables including hydro (2024). Of contracted capacity, 32% is fossil (CEA). The state's KPCL coal stations at Raichur, Yermarus and Ballari total 5,020 MW.

4.4 Water calculations

4.4.1 Evaporation physics, checked against WUE

- Latent heat of water ≈ 2,400 kJ/kg, and 1 kWh = 3,600 kJ. So evaporating water rejects heat at **3,600 ÷ 2,400 = 1.5 L per kWh of heat**.
- Cooling towers reject IT heat plus compressor and pump work, about **1.2 × IT energy** in a chilled-water plant. That gives ~1.8 L per kWh of IT if all heat is rejected by evaporation.
- About 15% of tower heat is rejected sensibly, so evaporation E ≈ 1.8 × 0.85 ≈ **1.53 L/kWh IT**.
- **Blowdown** B = E ÷ (CoC − 1), where CoC is cycles of concentration. Drift is <0.005% of circulation and negligible. **Make-up water** M = E + B = E × CoC ÷ (CoC − 1).
- At CoC 4: M = 1.53 × 4/3 = **2.04 L/kWh**
- At CoC 5: M = **1.91 L/kWh**
- At CoC 6 (needs soft feed water such as RO-treated wastewater): M = **1.84 L/kWh**
- This matches the 1.5–2.0 L/kWh evaporative WUE range. I use **1.8** as the central value.
- Higher CoC (enabled by RO-treated feed) saves 10–15% of water. Moving away from evaporation saves 75–97%.

4.4.2 On-site water by case

Formula: annual on-site water (L) = IT energy (kWh) × WUE + staff × 45 L × 365 + landscaping. MLD = annual ML ÷ 365. Population equivalent = MLD × 10^6 ÷ 135.

Table 4.3 On-site water by case

Case / cooling	Cooling water	Domestic + landscaping	Total on-site	Range	People at 135 LPCD
A (WUE 1.8)	122.6 GWh×1.8 = 220.8 ML/yr = 0.605 MLD	~0.02 MLD	**~0.62 MLD** (226 ML/yr)	0.43–0.82 MLD	~4,600
B1 evaporative (1.6)	525.6 ML/yr = 1.44 MLD	~0.03 MLD	**~1.47 MLD**	1.1–1.9 MLD	~10,900
B2 adiabatic (0.4)	131.4 ML/yr = 0.36 MLD	~0.03 MLD	**~0.39 MLD**	0.2–0.6 MLD	~2,900
C dry/closed-loop (0.05)	35.0 ML/yr = 0.096 MLD	~0.03 MLD	**~0.13 MLD**	0.06–0.2 MLD	~930
C adiabatic assist (0.2)	140.2 ML/yr = 0.38 MLD	~0.03 MLD	**~0.41 MLD**	0.3–0.6 MLD	~3,050
C if built with cooling towers (1.6)	1,121 ML/yr = 3.07 MLD	~0.03 MLD	~3.10 MLD	2.6–3.6 MLD	~23,000

Tanker equivalent: at 6,000 L per tanker, B1 = ~245 tanker loads a day. In Whitefield, one apartment complex of 100 homes reports needing 4–5 tankers a day at Rs 1,600 each.

Climate effect: Bengaluru's design wet-bulb of ~23 °C and mild dry-bulb temperatures (rarely above ~35 °C) suit dry coolers with brief adiabatic assist. That is why Case C's adiabatic WUE can stay near 0.2. In March–May, evaporative sites use 20–30% more water than their annual average, exactly when borewells fail.

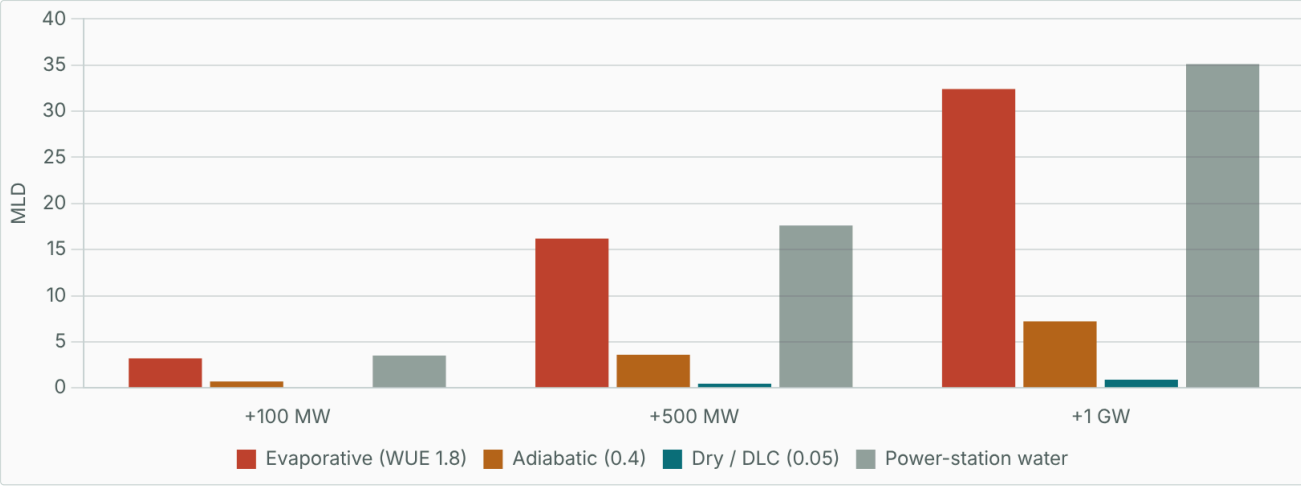


Figure 4.1 Scaling to Karnataka's 1 GW target. MLD of on-site water at 75% utilisation, by cooling choice; plus power-station water outside the city. *Source: India Data Centre Atlas, chapter built on this chapter's data.*

4.4.3 Indirect water embedded in grid electricity

- Grid water intensity ≈ Σ (generation share × water consumption per MWh):
- Coal ~38% × 3.5 m³/MWh (MoEFCC 2015 limit for existing cooling-tower plants; new plants 2.5) = 1.33
- Nuclear ~3% × 2.7 = 0.08
- Solar/wind ≈ 0.01
- Hydro: 0–0.6, depending on whether reservoir evaporation is allocated to power or to irrigation (contested)
- **Total ≈ 1.4–2.0; central 1.5 L/kWh**
- Using Raichur's reported actual of ~4.9 m³/MWh would push this towards 2.5. Raichur sits on the Krishna and has been shut during water shortages.

Table 4.4 Indirect water embedded in grid electricity

	Case A	Case B	Case C
Indirect water (1.5 L/kWh)	177.8 GWh → 267 ML/yr = **0.73 MLD**	443.5 GWh → 665 ML/yr = **1.82 MLD**	841 GWh → 1,262 ML/yr = **3.46 MLD**
Total footprint (on-site + indirect)	~1.35 MLD	B1 ~3.3 / B2 ~2.2 MLD	dry ~3.6 / adiabatic ~3.9 MLD
With 100% solar/wind supply	~0.62 MLD	B1 1.47 / B2 0.39	0.13–0.41

What this means: for DLC and adiabatic designs, ~90% of the water footprint sits at power plants in the Krishna/Tungabhadra basin, not in Bengaluru. Round-the-clock renewable supply is therefore a water policy as well as a carbon policy.

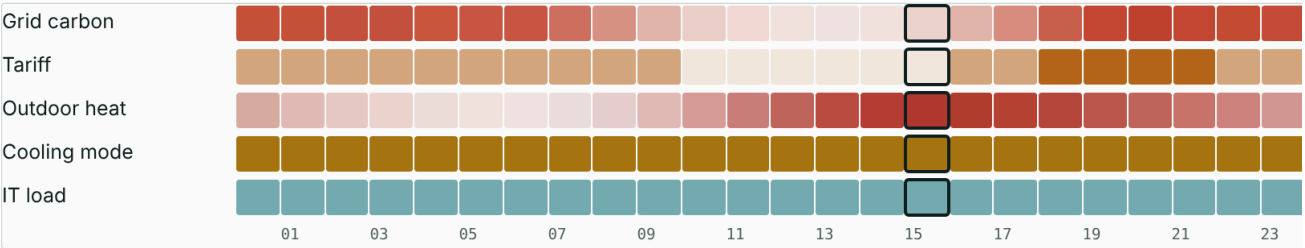


Figure 4.2 A day in Bengaluru, hour by hour. Grid carbon, tariff, outdoor heat, cooling mode and IT load for an April day. *Source: India Data Centre Atlas, chapter built on this chapter's data.*

4.5 Comparison with other Bengaluru and Karnataka users

Table 4.5 Comparison with other Bengaluru and Karnataka users

User (illustrative scale)	Water (MLD)	Per-unit norm	Power (MU/yr)	Source / status
Case A, 20 MW colo	0.62	1.8 L/kWh IT	178	This model
Case B1, 50 MW evaporative	1.47	1.6 L/kWh IT	444	This model
Case C, 100 MW DLC	0.13–0.41	0.05–0.2 L/kWh IT	841	This model
IT campus, 10 million sq ft (Manyata/ITPL scale, ~100,000 staff)	~5.5–6.5 (4.5 domestic at 45 LPCD + HVAC towers)	45 LPCD/employee	~185 (≈200 kWh/m²/yr)	Estimate from norms
Apartment complex, 1,000 flats	~0.61	135 LPCD × 4.5 persons	~18	Norm-based
500-bed hospital	~0.23	~450 L/bed/day (NBC)	~15–25	Norm-based estimate
300-room luxury hotel	~0.3	~1,000 L/room/day	~10–15	Industry-typical estimate
Large mall, ~1 million sq ft	~0.3–0.5	—	~25–35	Estimate
Brewery, 2 million hl/yr	~1.7	UBL: 2.93 kL/kL (FY24), 3.10 kL/kL (FY25)	~15–20	UBL reports; plant size illustrative
Soft-drink bottler, 1 million kL/yr	~4–6	~1.5–2.2 L/L industry-typical	~20–30	Estimate
Toyota Kirloskar, Bidadi (capacity ~342,000 vehicles)	~4.7 total (1.73 million m³/yr), ~89–90% recycled or rainwater	2.94 m³/vehicle (2016-17)	100% renewable since FY2021	TKM disclosures
BEL (Bengaluru)	~2 MLD of BWSSB treated water	—	—	Deccan Herald via BWSSB
18-hole golf course (dry season)	~1.5–2.5	~5 mm/day on ~40–50 ha	<5	Estimate
Garment factory (cut-make-trim), 2,000 workers	~0.1	Domestic-type use; wet processing 100–200 L/kg elsewhere	~5–10	Estimate
BMRCL Namma Metro (96 km)	Small (stations, depot washing)	—	~65 lakh units/month in 2018-19 (~78 MU/yr); now plausibly ~200 MU; Blue Line alone projected at 152 GWh	BMRCL (2019), IISc–BMRCL study
JSW Steel Vijayanagar (12–17.5 MTPA)	**330 MLD allocated** (Tungabhadra 32 MGD + Almatti 40 MGD)	~10 m³/t on allocation basis	Captive ~860 MW plants	JSW environmental clearance filing
Raichur TPS (1,720 MW)	~110–150 at 3.5–4.9 m³/MWh	MoEFCC norm 3.5	Generator	CPCB / reported data
Pharma/biotech (Biocon), HAL	Not publicly disaggregated	—	—	Data gap

What the comparison shows - **Per MLD, data centres are mid-sized water users.** An evaporative 50 MW site is comparable to a large brewery or a third of Toyota Bidadi. - **Per MU, they are extreme power users.** Case C uses ~4× Namma Metro's likely consumption and ~4.5× a 10 million sq ft IT campus. - **Per unit of value, the gap is widest.** A 100 MW DLC site uses less water than one large hotel cluster, but as much electricity as ~467,000 Bengaluru homes. - The best Bengaluru-region manufacturers (Toyota at ~90% recycled, BEL on treated water) show that non-potable sourcing is already standard practice for comparable industry. Data centres can meet it.

4.6 Where Bengaluru data centres get water today

Table 4.6 Where Bengaluru data centres get water today

Source	Status for data centres	Key facts
BWSSB Cauvery (Stages I–V)	Used for a few; poorly metered	Stages I–IV ~1,450 MLD; Stage V adds 775 MLD (commissioned Oct 2024), but connection uptake is slow (<1.2 lakh of 4 lakh targeted connections after 18 months). Cauvery allocation is capped. BWSSB industrial tariff Rs 99/kL. Five data centres receive ~129 kL/month in BWSSB records.
Groundwater/borewells	Likely the dominant source (WELL Labs view)	All Bengaluru Urban and Rural assessment units are over-exploited. Bangalore East is at 378.85% (Karnataka Dynamic Ground Water Resources report; 395.0% in IN-GRES 2025-26 data), Devanahalli at 169%, with "no net groundwater available for future use". The whole area is notified under the Karnataka Ground Water Act 2011, so every borewell needs a KGWA/BWSSB permit.
Private tankers	Back-up; unregulated	Tanker water is mostly borewell water from peri-urban villages, which moves the depletion elsewhere.
Treated wastewater (STP/tertiary)	Under-used; no mandate	BWSSB STP capacity ~1,348.5 MLD. Reuse ~825 MLD environmental and ~30 MLD commercial. K&C Valley sends 440 MLD and H-N Valley 210 MLD (secondary treated) to Kolar and Chikkaballapur. Treated water sells at Rs 10–15/kL. Apartment STPs may sell 50% of their output. BWSSB's chair has said there is no mandate for data centres to use recycled water.
Rainwater harvesting / on-site STP	Supplementary	Useful for domestic use and landscaping. Too small and seasonal for evaporative cooling.

Operator evidence is thin and inconsistent. - The Iron Mountain/Web Werks 4 MW Whitefield site sought just 10 kL/day from BWSSB, implying ~0.15 L/kWh, i.e. air-cooled. BWSSB says it does not supply that site. - ETL Secure Space was allotted 650 kL/day at Devanahalli, likely from panchayat borewells, with no groundwater permit specified in the order. - One major operator's 2025 ESG report, reviewed by Down To Earth, showed 216,357 kL of total use, including 9,876 kL of groundwater and 191,152 kL from third parties. - Yotta says Karnataka's policy prioritises air-cooled and closed-loop systems and requires non-potable water. Down To Earth found the policy text does not support this.

Bengaluru's share of Indian capacity. - Cushman & Wakefield (H1 2026): 162 MW operational, 8 MW under construction and 92 MW planned in Bengaluru. That is ~9% of India's 1,789 MW operational but only ~2.6% of the 3,860 MW pipeline (509 MW under construction, 3,351 MW planned). Hyderabad alone has 668 MW planned. - Other trackers give different Bengaluru figures (e.g. 78 MW in a 2025 Houlihan Lokey compilation) because they define capacity differently, so treat 162 MW as Cushman's colo-plus-hyperscale definition. - The lag reflects several factors: no submarine-cable landing (Mumbai and Chennai have them), expensive land in the core, hyperscalers' preference for states with surplus power and incentive packages (the SBI April 2026 sector report highlights Mumbai, Chennai and Noida), and growing awareness of water stress. Latency is not the constraint for Bengaluru.

4.7 Will data centre water cause scarcity?

4.7.1 City balance

- Supply: Cauvery ~2,225 MLD of capacity (less in practice while Stage V connections lag), plus groundwater of ~800 MLD (BWSSB study). Groundwater extraction in Bengaluru Urban district has been estimated as high as 1,457 MLD in 2024, a drought year.
- Freshwater demand: ~2,632 MLD (WELL Labs), of which domestic is ~1,890 MLD.
- BWSSB projects demand of ~4,100 MLD by 2051 against the Cauvery cap, a shortfall of ~2,650 MLD.
- Experts put total current availability at ~3,100 MLD. I use **3,000 MLD** as the denominator.

4.7.2 Scenarios (75% utilisation; IT energy = 18,000 kWh/day per MW)

Table 4.7 Scenarios (75% utilisation; IT energy = 18,000 kWh/day per MW)

Added IT capacity	Evaporative, WUE 1.8	Adiabatic, WUE 0.4	Dry/DLC, WUE 0.05	Indirect (PUE 1.3, 1.5 L/kWh), outside city
100 MW	3.2 MLD (0.11%)	0.7 MLD (0.02%)	0.09 MLD	3.5 MLD
500 MW	16.2 MLD (0.54%)	3.6 MLD (0.12%)	0.45 MLD	17.6 MLD
1 GW	32.4 MLD (1.1%)	7.2 MLD (0.24%)	0.9 MLD	35.1 MLD
People equivalent, 1 GW	~240,000	~53,000	~6,700	~260,000
1 GW as share of BWSSB STP capacity (1,348.5 MLD)	2.4% (≈43 MLD RO feed)	0.5%	~0	—

Percentages in brackets are shares of 3,000 MLD city supply.

Interpretation - **Citywide:** even the worst case (1 GW evaporative) is about 1% of supply. That is less than the commercial reuse already happening, and about a tenth of JSW Vijayanagar's allocation. - **Against the deficit:** 32 MLD is about 1.2% of the projected 2051 shortfall. It is also roughly the domestic supply for one or two large peri-urban wards, which is *not* trivial in a city where 80 wards are rated at risk of severe scarcity. - **Against industrial groundwater:** OpenCity reports Bengaluru industry drew ~13,700 ha-m of groundwater in 2025 (~375 MLD), about 25% of the city's groundwater availability. Today's data centres, at 2–5 MLD, are ~1% of that. A 1 GW evaporative fleet on groundwater would add ~9%. - **Locally, the risk is real.** Whitefield, Mahadevapura and Devanahalli host most data centres and are the most over-drawn aquifers. There, deep borewells (beyond 2,000 ft) are dry and residents buy tankers. A 1.4 MLD groundwater draw equals the entire domestic use of ~10,000 people concentrated in one aquifer cell. Restoring a lake elsewhere cannot offset that. - **Timing:** evaporative demand peaks in March–May, coinciding with tanker price spikes. The 2024 crisis came after the deficient 2023 monsoon. - **Equity:** data centres can pay Rs 99/kL or tanker prices. The households, dhobi ghats and villages competing for the same aquifer often cannot. - **Elsewhere in Karnataka:** a 1 GW fleet on today's grid would consume ~35 MLD at thermal plants in the Krishna basin, which already competes with irrigation. According to the CGWB's Dynamic Ground Water Resources Assessment, 88% of Karnataka's groundwater goes to irrigation, and the region already sends treated city sewage to Kolar's farm tanks.

Conclusion: data centre water will not be a primary cause of Bengaluru's scarcity. It becomes a material *local* stressor if freshwater or groundwater feeds evaporative cooling in over-exploited taluks. A treated-water and low-WUE mandate removes nearly all of the in-city risk at modest cost.

4.8 Policy context

- **Karnataka Data Centre Policy 2022-27:**
- Promised 24×7 water to investors but was silent on sustainable water use.
- Targeted Rs 10,000 crore of investment and 200 MW by 2025.
- Offered concessional industrial tariffs, electricity-duty exemption and land and stamp-duty benefits. Incentives outside Bengaluru included a 10% land subsidy and a 7% capital subsidy.
- A 2025 amendment removed the 5 MW cap on green-power tariff reimbursement, retaining Rs 0.50/unit for sites with more than 50% renewable supply.
- **Karnataka Sustainable Data Centre Policy 2026-31 (released 30 September 2026):**
- Targets 1 GW of IT load and a 10–12% share of India's AI-ready capacity.
- Requires plant-wise PUE and WUE reporting.
- Pays Rs 20 crore for sustained PUE of 1.00–1.20 over three years.
- Reimburses 100% of treated-water charges up to Rs 8 lakh/yr for five years.
- Reimburses water charges up to Rs 30 lakh/yr for a WUE of 0.7–1.3.
- Plans a power and water "resource heat map".
- Proposes parks at Hoskote, Mangaluru and Mysuru totalling ~1,000 MW. Hoskote is planned at 500 MW with Pavagada solar and 60 MLD of secondary treated water.
- **Critique of the 2026 policy:**
- At Rs 10–15/kL, Rs 8 lakh buys only 53,000–80,000 kL/yr (0.15–0.22 MLD), under a sixth of Case B1's need.
- A WUE band of 0.7–1.3 rewards designs that DLC and adiabatic systems already beat four-fold. Johor and Singapore use a 1.8 ceiling, which is lax for Bengaluru's mild climate.
- The price signal matters more than the subsidy. Fresh BWSSB water for B1 would cost ~Rs 5.2 crore/yr (525,600 kL × Rs 99), against ~Rs 0.8 crore for treated water.
- **National level:** the draft national Data Centre Policy (2020) has still not been finalised. MeitY AI-cloud tenders require PUE below 1.35. SEBI BRSR Core requires water disclosure by the top 150 listed firms. The IGBC Green Data Centre rating is voluntary. The CGWA/KGWA permit regimes exist, but enforcement and borewell metering are weak.

4.9 Recommendations

4.9.1 Water

1. **Mandate non-potable sourcing for cooling.** Every new or expanded data centre above 1 MW IT in BBMP/GBA, BMRDA and the peri-urban taluks should source 100% of process and cooling make-up from treated wastewater (BWSSB tertiary water, apartment or KIADB STPs, or an on-site STP+RO plant) or use near-zero-water cooling. Potable or Cauvery water should be limited to domestic use (~45 LPCD × staff).
2. **Ban groundwater for cooling in over-exploited taluks.** No new KGWA/BWSSB borewell NOCs for data centre cooling in Bengaluru Urban, Bengaluru Rural, Kolar or Chikkaballapur. Existing borewells should be metered with telemetry and phased out of cooling within 3 years.
3. **Cap WUE.** New builds from 2027: WUE ≤0.4 L/kWh IT annual, and ≤0.2 for liquid-cooled AI halls. Existing sites: ≤1.0 by 2029, with any evaporative cooling fed by treated water. Tie the cap to the KSPCB Consent to Establish/Operate.
4. **Issue water budgets with the permit.** A KSPCB/BWSSB consent should state an annual and peak-month allocation (kL/day) by source. Exceeding it should trigger escalating charges.
5. **Use tiered pricing.** Data centre tariff for potable water ≥Rs 150–200/kL (above the Rs 99 industrial rate; Johor applies a dedicated, higher data centre tariff). Treated water at Rs 10–20/kL with long-term take-or-pay contracts to finance tertiary and RO plants. Move incentive money from reimbursement to pipelines and plants.
6. **Drought rules.** When BWSSB declares a shortage, sites with evaporative systems must switch to dry or adiabatic mode or to stored treated water. Potable top-up should be prohibited from March to May in drought years.
7. **Accounting integrity.** Disallow offset claims such as lake restoration elsewhere against local aquifer depletion. Require reporting of withdrawal by source alongside "recycled" claims.

4.9.2 Power

8. **Cap PUE.** ≤1.30 annual for new air-cooled builds, ≤1.20 for liquid-cooled AI halls, and ≤1.45 for existing sites by 2029. Keep the Rs 20 crore PUE incentive but pay it only after metered verification.
9. **Plan grid connections.** Treat any connection above 55 MVA as a KPTCL transmission-planning item: 220 kV dual-circuit from two separate 400/220 kV sources, with the developer funding dedicated bays and substations. Publish a 5-year *hosting-capacity map* of available 220/400 kV capacity, building on the policy's resource heat map. Co-locate the 1 GW parks with 400 kV grid stations.
10. **24×7 renewable requirement.** Require ≥70% annual renewable supply, hourly-matched where possible, through green open access, group captive or Pavagada-type banking, for any campus above 50 MW. This also cuts the ~1.5 L/kWh of indirect water.
11. **Demand flexibility and storage.** Require BESS and/or dispatchable backup sized to ride through and shift load during the Jan–Mar and Sep evening peaks. Offer ToD tariffs and demand-response contracts. Replace diesel gensets with cleaner backup over time.
12. **Heat reuse and efficiency.** Require heat-reuse feasibility studies for campuses above 20 MW. Give incentives for DLC/immersion, warm-water cooling and dry coolers. Adopt BEE/ECBC data centre benchmarks as mandatory.

4.9.3 Siting

13. **Zoning.** Large campuses (>50 MW) should go to designated parks with assured treated-water supply and 400/220 kV access: Hoskote with 60 MLD treated water, and Tumakuru, Mysuru and Mangaluru. No evaporative cooling should be approved in Bangalore East, Mahadevapura or Devanahalli. Edge and enterprise sites (<10 MW, air-cooled) can stay in the city.
14. **Coastal options.** Treat desalination- or seawater-linked campuses at Mangaluru on their own merits. The 2026 policy already lists desalination plants as eligible for customised packages.

4.9.4 Transparency and enforcement

15. **Disclosure.** Require monthly metered PUE, WUE (by source), CUE and renewable share per facility, published on a KITS/KSPCB portal. This mirrors the EU Energy Efficiency Directive's database for data centres ≥500 kW.
16. **Enforcement.** Annual third-party audits, IoT flow meters on all sources, and consent suspension for repeat breaches. Coordinate KITS, KSPCB, BWSSB, KGWA, KPTCL and BESCOM through one data centre resource cell.

4.9.5 National

17. MeitY should finalise the national Data Centre Policy with water clauses (non-potable cooling, WUE reporting). CGWA should exclude data centre cooling from NOCs in over-exploited blocks. BEE should notify a data centre star-labelling scheme with PUE/WUE thresholds.

4.9.6 International precedents (background)

- **Johor (Malaysia):** stopped approving Tier 1/2 data centres, requires reclaimed water, applies a WUE threshold of 1.8 with approval still subject to local water availability, and charges a dedicated RM5.33/m³ tariff. Operators there now run their own reclamation plants (Bridge Data Centres treats ~5 MLD).
- **Singapore:** uses the same 1.8 WUE benchmark and NEWater-style reuse.
- **Others:** Ireland (Dublin grid-connection limits), Amsterdam (moratorium followed by zoning), Chile and Uruguay (projects redesigned to air cooling after drought objections), and US states such as Arizona (water-use scrutiny). These were not re-verified in this research cycle and should be checked before being cited in a formal submission.

4.9.7 Phased roadmap

Table 4.8 Phased roadmap

Phase	Timing	Actions
1	Q4 2026–2027	Operational guidelines for the 2026–31 policy. Mandatory metering and disclosure. Groundwater-for-cooling ban in over-exploited taluks. Data centre water tariff. Publish hosting-capacity map.
2	2027–2028	WUE ≤0.4 and PUE ≤1.3 written into consent conditions for new builds. Treated-water pipelines to Hoskote and Whitefield clusters. ≥70% renewable rule for >50 MW. BESS and demand-response contracts.
3	2029–2031	Existing-fleet WUE ≤1.0 and PUE ≤1.45. Hourly renewable matching. Independent audits. Review the 1 GW target against grid and water heat-map outcomes.

4.10 Caveats

- **Current-use estimates conflict.** For Bengaluru data centre water: ~2.2 MLD (Down To Earth's estimate of roughly 812.9 million litres a year of permitted use, covering Bengaluru plus one Ramanagara site), ~4 MLD (the IT department's ~4,000 kL/day for 32 data centres, from informal operator inputs) and ~20 MLD (WELL Labs hydrologist Shashank Palur's estimate, based on ~26 million litres per MW per year, which implies WUE near 3 L/kWh at full load and looks high). A widely repeated claim of "26 million litres annually" for the whole city appears to be a mis-statement of the per-MW figure.
- **Several key figures are estimates.** BESCOM sales (~37,000 MU) are inferred from KERC's FY25 true-up (Rs 2,068 crore ÷ Rs 0.56/unit). The 1,800 kWh household figure, the grid water intensity (especially hydro allocation) and the comparison-industry figures marked "estimate" are norm-based, not plant data.
- **Bengaluru capacity figures differ by source and definition** (Cushman 162 MW, others 78 MW or less).
- **Policy details** for the 2026–31 policy come from press reporting of the policy text. Operational guidelines are not yet issued.
- **Supply-voltage thresholds** come from a KERC presentation and a June 2025 amendment as reported. Specific connections depend on KPTCL feasibility studies.

Sources cited in this chapter

Organisations and publications named in this chapter's text, in order of how often they are cited. Full details are in the bibliography.

- Bangalore Water Supply and Sewerage Board
- Central Ground Water Board and state groundwater assessments
- State electricity regulatory commissions (KERC, MERC, TNERC, TGERC) tariff and open-access orders
- WELL Labs, Bengaluru water research
- Down To Earth reporting
- Cushman & Wakefield, India data centre market updates (H1 2026)
- Ministry of Electronics and Information Technology (MeitY), Data Centre Policy draft and IndiaAI
- Ministry of Environment, Forest and Climate Change, thermal plant water norms and EIA
- Central Ground Water Authority guidelines
- Bureau of Energy Efficiency
- Central Electricity Authority (CEA), CO_2 Baseline Database and grid reports
- Yotta Data Services disclosures
- State Bank of India sector report on data centres (April 2026)
- Houlihan Lokey India data centre edition (December 2025)
- CPHEEO urban water supply norms
- European Union Energy Efficiency Directive and Delegated Regulation 2024/1364

PART II · RESOURCES AND IMPACTS

Chapter 5
Every terawatt-hour is not the same: source, timing and location of data-centre electricity

WHAT THIS CHAPTER COVERS

This chapter explains why the carbon dioxide and the water behind a unit of electricity depend on which power station generated it, at what hour, and on which grid. It compares the life-cycle carbon and the water consumption of every generating technology, describes how Indian grid intensity changes through the day and the year, explains the difference between average and marginal emissions and between location-based and market-based carbon accounting, and shows how much water the Indian grid evaporates per kilowatt-hour. It ends with a worked matrix, a five-factor scorecard for procurement decisions and recommendations for operators and regulators.

KEY TAKEAWAYS

1. The same terawatt-hour of data-centre electricity can carry about 10,000 tonnes of carbon dioxide if it comes from nuclear, wind or a portfolio of clean power matched every hour, or about one million tonnes if it comes from marginal coal at the evening peak, a spread of about one hundred times.
2. The water consumed to generate that terawatt-hour ranges from almost nothing for wind and solar to more than 3.5 billion litres for coal plants with cooling towers.
3. India's grid emitted about 0.71 tonnes of carbon dioxide per megawatt-hour on average in 2024–25 according to the Central Electricity Authority, and the marginal unit at the evening peak is almost always coal.
4. Annual matching of renewable certificates can report near-zero emissions while the data centre still draws coal power at night. Hourly matching of clean supply to consumption is the more accurate measure, and Microsoft's own figures show the gap between location-based and market-based emissions growing to more than 30 times.
5. For an Indian operator, where, when and from what source power is bought matters more to the footprint than almost any efficiency gain inside the building.

How this chapter connects to the others. Chapter 7 uses these grid factors to calculate the total water of worked examples, Chapter 10 turns them into a procurement strategy, and Chapter 11 optimises an hourly clean-power portfolio.

The same 1 TWh of data-centre electricity can carry anywhere from about 10,000 tCO_2 (nuclear, wind, or a 24/7-matched clean portfolio) to roughly 1,000,000 tCO_2 (marginal Indian coal at the evening peak), and from almost zero to more than 3.5 billion litres of consumed freshwater. A ~100× spread in carbon and an even wider spread in water means that for an Indian operator, *where, when and from what* power is bought matters more to the footprint than almost any efficiency gain inside the building.

5.1 Key Findings

1. **Source is the dominant factor.** Lifecycle medians (IPCC AR5 Annex III) run from 820 gCO_2e/kWh for pulverised coal down to 11–12 for wind and nuclear. That is a 70× range before timing or location are considered.
2. **Timing matters more each year in India.** Ember finds India's fossil generation fell to around 125 GW at 1 pm in H1 2026 (from 135 GW in H1 2023) but peaked at 174 GW at 7 pm. The nearly 50 GW midday-to-evening swing, equivalent to more than 70 modern coal units, is mostly coal ramping. Keeping coal above its technical minimum forced the curtailment of around 2.1 TWh (1.3%) of renewable generation in FY2025–26, according to Ember. Moving load from 7 pm to 1 pm cuts *average*-intensity emissions, and on curtailment days it also cuts *marginal* emissions.
3. **Location is mostly about the margin, not the average.** India has one synchronous national grid, and coal sits at the margin in almost all hours. So a new Bengaluru data centre drawing grid power raises coal output somewhere in India, even though Karnataka's own mix is far cleaner than the national mix. State-level averages are useful for attributional reporting. Marginal factors are the right basis for siting and procurement decisions.
4. **India's carbon intensity is falling but remains among the world's highest.** CEA's weighted average fell from 0.774 to 0.710 tCO_2/MWh over a decade. Ember's Global Electricity Review 2026 puts India at 670 gCO_2e/kWh in 2025, down from 705 in 2024, against an Asian average of 544 and a global 458. The National Electricity Plan pathway implies about 430 g by FY2031-32.
5. **Water is the hidden half.** Indirect water from electricity often exceeds a data centre's on-site cooling water several-fold. LBNL's December 2024 US report finds the 176 TWh used by US data centres in 2023 consumed about 800 billion litres of water indirectly (≈4.5 L/kWh), against 66 billion litres consumed on site. In India, WRI's *Parched Power* found 40% of thermal power plants experience high water stress, and water shortages have shut plants at Raichur, Parli, Farakka and elsewhere.
6. **Market-based claims can hide real grid impact.** Microsoft's location-based Scope 2 rose 130% from FY20 to FY24 (4.33 to 9.96 $MtCO_2e$), while its market-based figure fell 43%. Google's 2025 Environmental Report shows 66% hourly CFE globally but 12% in Asia-Pacific, against roughly 70% in North America and 92% in Latin America. Annual REC matching is ending as a credible claim: the GHG Protocol's October 2025 Scope 2 consultation proposes hourly matching and deliverability.

5.2 SOURCE — What Generates the Electron

5.2.1 Lifecycle carbon intensity by technology

Table 5.1 Lifecycle carbon intensity by technology

Source	IPCC AR5 lifecycle median (min–max), gCO_2e/kWh	UNECE 2021 range, gCO_2e/kWh	Direct (combustion) share	India-specific note
Coal (pulverised)	820 (740–910)	751 (IGCC, US) – 1,095 (PC, China)	Direct 760 (670–870)	Indian subcritical units on high-ash coal: about 0.9–1.1 tCO_2/MWh at the busbar; UNECE notes a plant-by-plant global study at about 1,130 g
Oil	~720 (secondary compilations)	—	Mostly direct	Negligible in Indian grid
Gas CCGT	490 (410–650)	403–513 (92–220 with CCS)	Direct 370 (350–490)	Indian gas fleet runs at low PLF; mostly peaking/balancing
Biomass (dedicated)	230 (130–420)	—	Biogenic accounting contested	Bagasse co-gen in Karnataka/Maharashtra
Solar PV (utility)	48 (18–180)	8–83	0 direct	Upstream manufacturing mostly on coal-heavy Chinese grid
CSP	27 (AR5)	27–122	0	Marginal in India
Geothermal	38 (6–79)	—	Some CO_2 from geofluid	Not deployed
Hydropower	24 (1–2,200)	6–147	0	Tropical reservoirs can emit large biogenic CH_4; upper tail is real for shallow, warm reservoirs
Wind onshore / offshore	11 / 12	7.8–16 / 12–23	0	Karnataka and Tamil Nadu are major onshore wind states
Nuclear	12 (3.7–110)	5.1–6.4	0	NPCIL fleet incl. Kaiga (Karnataka)

Interpretation. - **Fossil fuels.** Combustion makes up almost all of their lifecycle emissions: 760 of 820 g for coal, 370 of 490 g for gas. Upstream methane is the main addition, mattering most for gas. - **Renewables and nuclear.** Their emissions are almost entirely embodied in manufacturing and construction, so they fall as manufacturing grids decarbonise. - **Two contested figures:** - *Coal.* UNECE argues the IPCC 820 g median rests on an optimistic efficiency assumption that "made its way to the IPCC AR5 report." Real-world coal fleets, including India's, sit nearer 1,000–1,100 g. - *Hydro.* The IPCC maximum of 2,200 g reflects reservoir methane in a few tropical cases. Indian reservoir hydro should not be assumed to sit at the 24 g median without site data.

Operational vs lifecycle. Grid factors used for Scope 2 (CEA, Ember) are *operational CO_2 only* (CEA) or *operational CO_2e* (Ember GER 2026). They count solar, wind, hydro and nuclear as zero. Lifecycle factors are right for comparing technologies and for procurement. Operational factors are what reporting regimes currently require. Mixing the two is a common error.

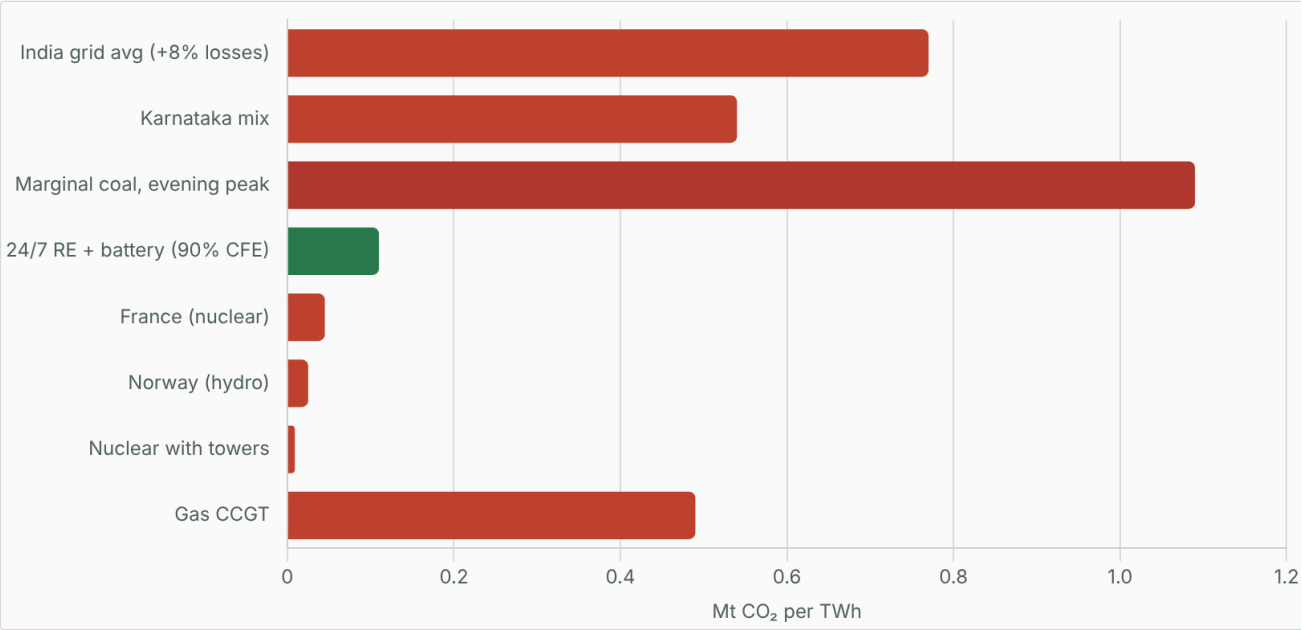


Figure 5.1 1 TWh of data-centre load, seven ways. Toggle between carbon and water. *Source: India Data Centre Atlas, chapter built on this chapter's data.*

5.2.2 Water: withdrawal vs consumption by source

- **Withdrawal** is water taken from a source, much of it returned warmer.
- **Consumption** is water evaporated or otherwise not returned.

Once-through cooling withdraws a great deal but consumes relatively little. Cooling towers withdraw less but evaporate more.

Table 5.2 Water: withdrawal vs consumption by source

Technology / cooling	Consumption, L/MWh (approx., NREL Macknick et al. 2012 medians, converted from gal/MWh)	Withdrawal, L/MWh	Comment
Coal, cooling tower	~1,800–2,600 (US); **2,800–4,000+ actual in India**	~2,500–4,000 (US); ~4,000 India	NREL notes tower-cooled plants sit near the ~1,000 gal/MWh (~3,800 L) upper bound

Technology / cooling	Consumption, L/MWh (approx., NREL Macknick et al. 2012 medians, converted from gal/MWh)	Withdrawal, L/MWh	Comment
Coal, once-through	~400–1,000	**70,000–200,000** (CSE, India)	High thermal-pollution and intake impacts
Coal/gas, dry cooling	~0–450 (India dry-cooled coal: 450–650 L/MWh)	Low	Efficiency penalty (higher CO_2/MWh)
Nuclear, cooling tower	~2,500–2,700	~4,000–5,000	Lowest carbon, among highest water
Nuclear, once-through / seawater	~1,000 (largely seawater)	~100,000–170,000	Coastal sites (Kudankulam, Tarapur) avoid freshwater stress
Gas CCGT, cooling tower	~700–1,000	~1,000	Only the steam cycle (about one-third of output) needs cooling
Gas CCGT, dry	~10	~10	
Solar PV	~20–100 (panel cleaning; higher in dusty Indian conditions)	same	Robotic dry cleaning cuts this further
Wind	~0–5	~0	
CSP (trough, wet tower)	~3,000–3,500	~3,500	Wettest renewable
Biopower (steam, tower)	~2,000 (operations; crop irrigation can add orders of magnitude)	—	
Hydro (reservoir evaporation)	**~13,000 (China avg) to ~245,000 (Mekonnen & Hoekstra 35-plant avg)**; global mean ~79,000	Not meaningful	Highly contested (see below)

Hydro allocation caveat. The definition you choose drives the result: - **Global average gross evaporation:** Mekonnen & Hoekstra report 68 m^3/GJ (≈245 m^3/MWh) for 35 plants, and 22 m^3/GJ (≈79 m^3/MWh) as a global average. - **Net and allocated approaches:** subtracting pre-dam evapotranspiration and allocating to multipurpose reservoirs gives a median of 17 m^3/GJ (≈61 m^3/MWh). - **China:** a study of 209 Chinese reservoirs finds only 3.6 m^3/GJ (≈13 m^3/MWh).

The 5,000–60,000 L/MWh range often quoted in data-centre literature therefore sits at the *low* end of the published spread. Irrigation-led Indian dams such as Krishnaraja Sagar or Almatti should not have all their evaporation charged to electricity. Run-of-river hydro is effectively zero.

5.2.3 Indian regulatory norms and actual performance

MoEFCC water norms (Environment (Protection) Amendment Rules, 7 Dec 2015; amended 28 June 2018): - Once-through plants must install cooling towers and meet **3.5 m^3/MWh**. Existing cooling-tower plants must also meet 3.5 m^3/MWh, both within two years (by Dec 2017). - New plants commissioned after 1 Jan 2017 were first held to **2.5 m^3/MWh** with zero liquid discharge. This was diluted to **3.0 m^3/MWh** in June 2018, still with ZLD. Seawater-based plants are exempt.

Actual performance: - *CSE survey.* CSE analysed self-reported data for FY2017-18 to FY2019-20 covering 154 GW. About half of non-seawater capacity (59 GW across 54 plants) was **non-compliant**: 42 GW of cooling-tower plants and 17 GW still on freshwater once-through. A typical non-compliant plant ran at about 4 m^3/MWh. Named plants included Paras (5.24), Nashik (4.09), Koradi (4.0) and NTPC Solapur (3.72). CSE warns of a "high probability of data manipulation, under-reporting." - *CEA-linked research.* Indian closed-loop plants use 2.8–3.4 m^3/MWh, "twice the world's average of 1.2–1.5 m^3/MWh." - *Raichur (Karnataka).* The plant consumed 204,889 kL/day in 2016-17, **87% above** the 3.5 m^3/MWh limit at 76% PLF. In January 2019, units were shut for lack of water, cutting output to 600 MW. Raichur sits in the Krishna basin. - *Flue-gas desulphurisation.* FGD adds about 0.3 m^3/MWh, so tightening SO_2 norms raises water use.

CEA CO_2 Baseline Database (latest): - **Version 21.0 (Dec 2025):** provisional FY2024-25 weighted average emission factor (WAEF, incl. RE) of **0.710 tCO_2/MWh**. - **Version 20.0:** FY2023-24 WAEF of **0.727**, build margin (BM) of **0.552** (first version to include RE in BM), and combined margin (CM, 50:50 OM/BM) of **0.757 tCO_2/MWh**. - **Decade trend:** WAEF fell from 0.774 ten years ago. - **Scope:** CEA factors are CO_2-only (no CH_4/N_2O), calculated on *net* generation, and follow the CDM tool ACM0002. CCTS Scope 2 calculations use the WAEF. The OM and CM are for offset baselines.

5.2.4 Other impacts (brief)

- **Air pollutants.** Coal is by far the largest emitter of SO_2, NOx, PM and mercury per TWh. India's 2015 stack norms (SO_2 100–600 mg/Nm^3 by vintage) have had their deadlines extended repeatedly. FGD retrofit status therefore matters for the health footprint of a coal-heavy TWh.
- **Land.** Solar uses roughly 1.5–2.5 ha/MW of direct footprint (~1–2 km^2 per TWh/yr in India). Wind's total spaced footprint is larger, but its direct footprint is under 5% of that. Coal's land use is dominated by open-cast mining and ash ponds. Nuclear has the smallest footprint.
- **Materials.** UNECE finds nuclear, wind and PV have low lifecycle metal and mineral intensity per kWh in absolute terms. PV and wind shift impacts toward copper, silicon, steel and concrete, and batteries toward lithium, nickel and cobalt (LFP avoids the last two).

5.3 TIMING — When the Load Occurs

5.3.1 Diurnal and seasonal patterns in India

- **Diurnal.** Ember's 2026 analysis (H1 2026 vs H1 2023):
- Fossil generation at 1 pm fell 10 GW, to about 125 GW.
- Between 5 pm and 7 am it *rose* 22 GW, to an average of about 168 GW, peaking near **174 GW at 7 pm**.
- The resulting ~50 GW intraday swing is equivalent to more than 70 modern coal units cycling.
- Keeping coal above its technical minimum forced the curtailment of around 2.1 TWh (1.3%) of renewable generation in FY2025–26, according to Ember, which says India needs around 10 GWh of battery storage immediately.
- **Trend.** A decade ago the IEA found India's hourly CO_2 intensity varied by only a factor of 1.2 between midday and the evening peak, with variable RE at 6% of generation. It projected that the spread would widen sharply as solar grows. That widening is now happening.
- **Seasonal (Karnataka/South).**
- Wind is strongest in the southwest monsoon (roughly June–September).

- Hydro peaks in and after the monsoon (Aug–Nov).
- Solar is strongest in the dry season (Feb–May), when demand also peaks from agricultural pumping and cooling.
- Monsoon months therefore have the lowest intensity in Karnataka. The pre-monsoon evening is the dirtiest window.

Chart in words — illustrative Karnataka day (dry season), state-mix average: - *00:00–06:00:* ~550–600 g. Coal and some wind. - *09:00–16:00:* falls to ~200–300 g as solar dominates. - *18:00–22:00:* climbs back to ~550–600+ g. Coal ramps, hydro is dispatched for the peak, and imports rise.

A third-party modelled dashboard (energymap.in, built on CEA v18 lifecycle factors and *modelled* state fuel mix, so treat as indicative) shows a 90-day Karnataka mean of about 410 g with a range of about 210–603 g. That is roughly a 3× daily spread.

5.3.2 Average vs marginal at different hours

The **average** intensity at 1 pm is low. The **marginal** generator, though, is usually still a coal unit backing down or ramping, because coal sets the margin in the merit order in almost all hours. In practice: - **Evening peak (18:00–23:00).** Marginal coal, sometimes gas or peaking hydro. Marginal emission factor ≈ 0.95–1.1 tCO_2/MWh, worse at part load because of heat-rate penalties. - **Solar hours on normal days.** Still marginal coal (≈0.9–1.0), because coal is backed down rather than solar curtailed. The *system* benefit of moving load to noon comes from making room for more solar build and flattening the evening ramp. - **Solar hours on curtailment days.** The marginal unit is curtailed RE, so the marginal factor approaches **~0**. Here, load shifting truly cuts emissions.

Caveat on methods. Marginal emission factors are contested and method-dependent. Short-run marginal (WattTime-style, statistical or dispatch-model) and long-run marginal (build margin, e.g. CEA BM 0.552 or NREL Cambium) can differ by 2× or more. No official hourly marginal factor exists for India.

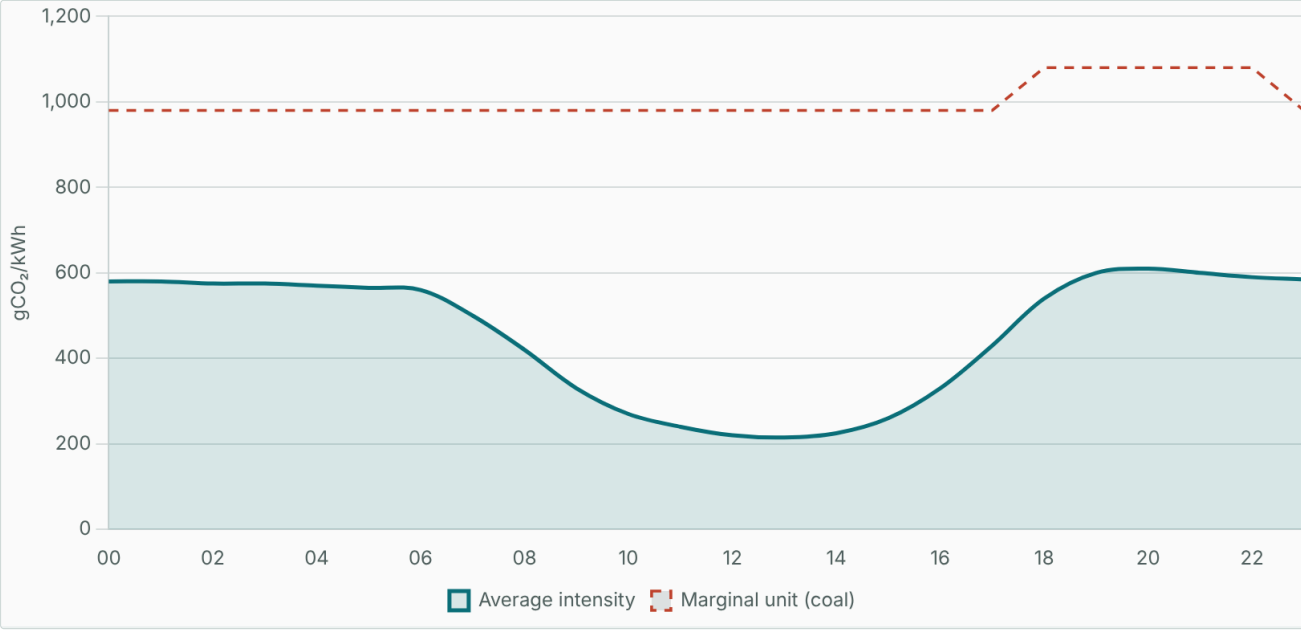


Figure 5.2 A Karnataka day. Illustrative hourly grid intensity, dry season (gCO_2/kWh). The marginal unit stays coal in most hours. *Source: India Data Centre Atlas, chapter built on this chapter's data.*

5.3.3 Data sources for hourly intensity

- **Electricity Maps and WattTime** provide hourly average and marginal signals for many grids. Indian coverage is national or regional and model-based.
- **Grid-India (formerly POSOCO)** publishes real-time and daily generation by source, from which hourly average intensity can be derived.
- **Ember** publishes monthly state-level generation and emissions for 36 states/UTs, plus periodic hourly analyses.
- **IEA** maintains a real-time CO_2 database covering average and marginal hourly intensity.

5.3.4 Implications and instruments for data centres

- **Carbon-aware computing.** Google's carbon-intelligent computing platform shifts flexible, non-urgent compute across hours and locations according to grid carbon intensity. In August 2025, Google signed its first demand-response agreements targeting **machine-learning workloads**, with **Indiana Michigan Power** (Fort Wayne data centre) and the **Tennessee Valley Authority**. Under these, ML workloads are curtailed or shifted during grid stress. Google itself cautions that flexibility "is still in the early stages" and only available at certain sites.
- **India's ToD tariff mandate.** The Electricity (Rights of Consumers) Amendment Rules, 2023 (MoP notification of 14 June 2023) make ToD tariffs mandatory for C&I consumers above 10 kW from **1 April 2024**, and for all other consumers except agriculture from 1 April 2025.
- **Peak tariff:** ≥1.20× normal for C&I.
- **Solar-hour tariff (8 hours set by the SERC): at least 20% below** normal.

- **Result:** a spread of about 40% or more between solar-hour and peak energy charges, which strongly rewards shifting load and adding batteries.
- **Implementation is uneven.** Telangana's regulator held in 2026 that solar-hour rebates require a full tariff proceeding. KERC applies ToD to HT categories through its annual combined tariff order. Operators should check the current order's slot definitions and rebates.
- **Data-centre flexibility in practice.** Training jobs, batch inference, video transcoding and backups are shiftable, with a typical 10–30% flexible share in hyperscale fleets. Latency-critical inference is not. A 100 MW Bengaluru campus with 20 MW shiftable from 19:00–23:00 into 10:00–15:00 moves about 29 GWh/yr into solar hours. At a ~0.3 t/MWh average-intensity differential, that avoids about 9,000 tCO_2/yr on an average-factor basis. On curtailment days the marginal benefit is larger.

5.3.5 Annual vs hourly matching

- **Annual matching** (buying RECs equal to annual MWh) lets 2 pm solar certificates "cover" 9 pm coal-fired consumption.
- **24/7 carbon-free energy (CFE)** matches every hour in the same grid. Google reports **66% hourly CFE** in 2024, up from 64%, with 9 of 20 grid regions at ≥80%. Its Asia-Pacific regional CFE was **12%** in 2024 (vs ~70% in North America and 92% in Latin America, per its 2025 Environmental Report), which shows how hard Asia is. The UN-backed 24/7 Carbon-Free Energy Compact promotes this standard.
- **GHG Protocol Scope 2 revision.** The October 2025 public consultation (closed 19 Dec 2025) proposes:
- hourly matching for market-based claims, with exemption thresholds for smaller reporters;
- deliverable market boundaries;
- load-profile approximations for consumers without interval data;
- a legacy clause for existing contracts.

Finalisation is expected around 2027. Firms such as EY argue hourly matching should start as optional, given data gaps. - **EU RFNBO (green hydrogen) rules** (Delegated Regulation 2023/1184) provide a working precedent. They require additionality (new RE, generally commissioned no earlier than 36 months before the electrolyser), geographic correlation (same bidding zone), and temporal correlation (monthly until end-2029, hourly from 1 January 2030).

5.4 LOCATION — Which Grid, Which Margin, Which Basin

5.4.1 Country grids (operational, annual average)

Table 5.3 Country grids (operational, annual average)

Grid	Approx. intensity, $gCO_2(e)$/kWh	Source/year	1 TWh ≈
India	670 (2025); 708 (2024, CO_2-only)	Ember GER 2026/2025	0.67–0.71 $MtCO_2$
China	~550	Ember (approx.)	~0.55 Mt
Asia average	544	Ember 2025	0.54 Mt
World	458 (2025); 473 (2024)	Ember	0.46 Mt
United States	~370	Ember (approx.)	~0.37 Mt
EU-27	~210–240	Ember (approx.)	~0.22 Mt
France	~30–60	Ember/RTE (approx.)	~0.03–0.06 Mt
Norway	~20–30	(approx.)	~0.02–0.03 Mt

Ember's GER 2026 switched from CO_2 to CO_2e, which accounts for part of India's apparent 705/708 → 670 drop. The real 2025 decline (−2.9% emissions) was driven by record solar and wind, strong hydro and milder weather.

5.4.2 Within India: state mixes on a single grid

- **Karnataka.** Ember/IEEFA's SET 2026 ranks Karnataka first on decarbonisation, with about **37% RE in its procurement mix** and low power-sector intensity. Ember's EV study likewise finds Karnataka led with 37.6% of its FY2025 mix from solar, wind and hydro clean sources. Karnataka's remaining supply is mainly state and central coal (RTPS Raichur, BTPS Ballari, Yermarus, Udupi), some hydro and Kaiga nuclear.
- **Estimate (flagged):** consumption-mix intensity is about **0.40–0.55 tCO_2/MWh**, central **~0.50**. That is about 30% below the national average.
- **Coal-heavy states** (Jharkhand, Chhattisgarh, Odisha, West Bengal, UP) generate at about 0.85–1.0 t/MWh. Hydro-rich states such as Himachal and Kerala, and wind-rich Tamil Nadu, are much lower on a generation basis.
- **Generation-based vs consumption-based state factors.** Chhattisgarh exports coal power. Karnataka imports some. A consumption-based factor (generation adjusted for net imports) is the right one for a Bengaluru load. CEA does not yet publish one.

5.4.3 Average vs marginal; attributional vs consequential

Table 5.4 Average vs marginal; attributional vs consequential

Metric	Value (India)	Use
Weighted average (WAEF), FY24-25	0.710 t/MWh	Location-based Scope 2; CCTS
Operating margin (OM), FY23-24	~0.9+ t/MWh (fossil-only)	Short-run displacement
Build margin (BM), FY23-24	0.552 t/MWh	What new capacity looks like
Combined margin (CM), FY23-24	0.757 t/MWh	Offset baselines
Short-run marginal at evening peak	~0.95–1.1 t/MWh (est.)	Consequential impact of new load

Why the margin matters for data centres. A new 100 MW Bengaluru campus is *new* load. In the short run, coal plants somewhere in the national grid ramp up to meet it, whatever Karnataka's average says. In the long run, the BM shows that new capacity is now about half RE, which is why *additional*, contracted RE matters. Attributional (average) accounting answers "what share of today's emissions is mine?"

Consequential (marginal) accounting answers "what changes because I exist?" The GHG Protocol has opened a parallel consultation on electricity-sector consequential accounting.

5.4.4 Location-based vs market-based Scope 2; instruments and their quality

- **Location-based:** grid-average factor (CEA 0.710 for India).
- **Market-based:** contractual instruments such as I-RECs, Indian RECs, green tariffs, PPAs, and open-access/group-captive supply. The remainder is covered by a residual-mix factor, which India lacks. That gap means market-based figures in India are typically computed with the full grid average, which double-counts clean energy.
- **Quality hierarchy (best to weakest):** 1. On-site or adjacent RE plus storage. 2. Hourly-matched, additional, same-grid PPA/group captive. 3. Annual-matched new-build PPA. 4. Discom green tariff (Green Energy Open Access Rules 2022). 5. Bundled RECs from existing plants. 6. Unbundled RECs/I-RECs from old hydro or wind.

The last tier rarely changes what is built. Microsoft stopped counting unbundled RECs in FY25, and its market-based Scope 2 jumped to 2.71 $MtCO_2e$ as a result. - **Deliverability.** In India, an RE plant in Rajasthan and a load in Bengaluru are on the same synchronous grid, so physical deliverability is plausible. Transmission congestion and the GHG Protocol's proposed "market boundary" definition could still narrow this.

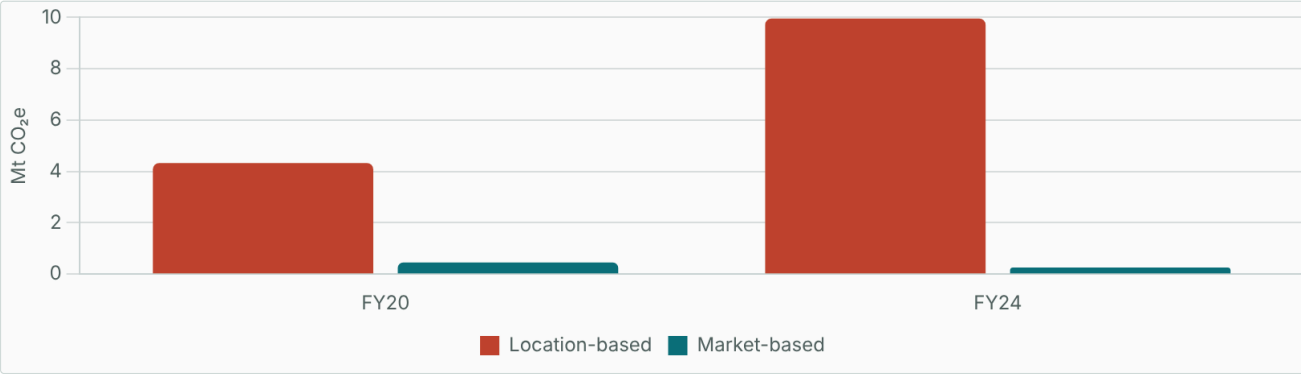


Figure 5.3 Annual matching hides the gap. Microsoft Scope 2 emissions, Mt CO_2e: location-based vs market-based. *Source: India Data Centre Atlas, chapter built on this chapter's data.*

5.4.5 Local water stress: the same litre is not the same

- WRI's *Parched Power* (2018) finds:
- 90% of India's thermal plants are freshwater-cooled, and **40% experience high water stress**.
- Plants in high-stress areas run at a **21% lower capacity factor**.
- Water shortage was the fifth-largest cause of forced outages in 2013–16.
- 14 of the top 20 utilities lost about **$1.4 billion** in potential revenue.
- Thermal freshwater consumption rose 43% between 2011 and 2016, from 1.5 to 2.1 billion m^3/yr. A 2021 ES&T study puts it at 2.39 billion m^3 in 2018.
- **Plants shut for water shortages:** Raichur (Karnataka), Parli (Maharashtra), Farakka (WB), Rihand (UP) and Ennore (TN).
- **Methods for weighting water by scarcity.** ISO 14046 (water footprint) and the **AWARE** characterisation factor, which ranges from 0.1 to 100 m^3-world-eq per m^3 (world average = 1), weight consumption by the remaining water available after human and ecosystem demand. A litre evaporated at Raichur (Krishna basin, high to extremely high stress) is far more damaging than a litre in Norway or coastal seawater.
- **Bengaluru context.** Bengaluru draws Cauvery water pumped about 100 km and up about 300 m. Its own groundwater is overdrawn. The *grid* water linked to its data centres is mostly evaporated in the Krishna and Godavari basins and in coal states further north.

5.4.6 Transmission and distribution losses

CEA factors are per unit *generated*. A consumer needs more generation than it meters:

> **Effective intensity at meter = EF_generation ÷ (1 − loss fraction)**

- **National AT&C losses:** 16.16% in FY2025 (MoP). PFC's 14th Integrated Rating puts them at 15.04%. Part of these are commercial (theft and billing) losses, not physical.
- **Physical losses for HT customers:** an HT data centre at 66/220 kV faces much lower physical losses, about 4–8% across inter-state and intra-state transmission.
- **Worked example:** 0.710 ÷ (1 − 0.08) = **0.77 tCO_2/MWh at the meter**. The same 8% gross-up applies to water: 2.0 → 2.17 L/kWh.

5.5 CARBON INTENSITY — Definitions, Trajectory and Data-Centre Metrics

5.5.1 Definitions

- **gCO_2/kWh vs gCO_2e/kWh.** CEA is CO_2-only. Ember GER 2026 and lifecycle data are CO_2e (incl. CH_4, N_2O). The difference is a few percent for coal and more for gas, because of upstream methane.

- **Generation- vs consumption-based.** Consumption-based intensity adjusts for imports and exports between states or countries. Production-based intensity flatters exporters of clean power and penalises exporters of coal power.
- **Average, marginal, residual-mix.** Each answers a different question (see §3.3).

5.5.2 Global and Indian trajectory

- **Global:** 533 g (2015) → **458 gCO₂e/kWh (2025)**, a 2.7% fall in 2025 alone, the largest in two decades. Renewables (33.8%) overtook coal in 2025.
- **India:** coal is still about 71% of generation (Ember 2025). Power-sector emissions were 1,395 $MtCO_2e$ in 2025, down 2.9%, as coal generation fell 44 TWh. Solar overtook hydro as India's largest clean source in 2025.
- **CEA WAEF:** 0.774 (a decade ago) → 0.727 (FY23-24) → **0.710 (FY24-25, provisional)**.
- **2030–32 projection:** if the NEP-14 targets and 500 GW of non-fossil capacity are delivered, clean energy reaches about 50% of generation and the grid factor falls to **~0.43 tCO_2/MWh by FY2031-32** (Ember citing CEA/NEP). This is a projection that depends on storage and transmission being built, not an outcome.

5.5.3 Data-centre carbon metrics (worked)

> **CUE (Carbon Usage Effectiveness) = Total facility CO_2 ÷ IT energy = PUE × EF_grid (for grid-only supply)**

- **Bengaluru, grid supply:**
- *National EF:* PUE 1.5 × 0.77 t/MWh (national, at meter) = **1.16 kgCO_2/kWh_IT**.
- *Karnataka mix:* 1.5 × 0.50 = **0.75 kgCO_2/kWh_IT**.
- **Same facility on 90% hourly CFE:** 1.5 × (0.9×0.04 + 0.1×0.77) ≈ **0.17 kgCO_2/kWh_IT**.
- **1 TWh of facility load:**
- *India:* ≈ 0.71–0.77 $MtCO_2$.
- *France:* ≈ 0.03–0.06 $MtCO_2$, a **~15–25× difference** for the same compute.

5.5.4 Hyperscaler disclosures: location-based vs market-based

Table 5.5 Hyperscaler disclosures: location-based vs market-based

Company	Disclosure	Location-based	Market-based	Takeaway
Microsoft (2025 report, FY24)	Scope 2, tCO_2e	4,328,916 (FY20) → **9,955,368** (FY24), **+130%**; +23% y/y	456,119 → **259,090**, **−43%**	Electricity use 10.77 → 29.83 TWh. Market-based fell while location-based more than doubled. Asia location-based Scope 2 rose from 0.91 to 2.51 Mt
Microsoft (2026 report, FY25)	Scope 2	**12,030,556**	**2,707,428**	Market-based jump reflects dropping unbundled RECs; not comparable with earlier years
Google (2025 report, CY2024)	Data-centre electricity +27% y/y	(location-based rose with load)	Data-centre energy emissions **−12%**	66% hourly CFE; ~12% in APAC; >25 clean projects (2.5 GW) came online

Amazon and Meta also report large location/market gaps driven by annual REC/PPA matching. Gaps of this kind are the main reason the GHG Protocol is moving to hourly matching.

5.6 WATER INTENSITY — The Grid's Hidden Water

5.6.1 Definitions

- **Withdrawal:** water abstracted.
- **Consumption:** water evaporated or embodied.
- **EWIF (energy water intensity factor):** litres consumed per kWh *generated*. It should be stated as consumption, with or without hydro evaporation, and freshwater-only or including seawater.

5.6.2 Grid-average water intensity

Table 5.6 Grid-average water intensity

Grid	Consumption intensity	Basis
India, all thermal + non-hydro RE (2014)	**2.09 L/kWh** (withdrawal 15.9 L/kWh)	IRENA/WRI 2018; *excludes hydro*
India 2027–2030 projections	1.57–1.94 L/kWh	IRENA/CEA scenarios
India, freshwater thermal only (2018)	~1.8 L/kWh	Derived: 2.39 bn m^3 ÷ ~1,300 TWh (estimate)
India incl. hydro evaporation	Several L/kWh, highly allocation-dependent	One multi-model study says hydro is about one-third of Indian power-sector water consumption
Karnataka mix (used in this series)	**~1.5 L/kWh** (excl. hydro)	Estimate: ~45–50% coal at ~3.3 L/kWh + small PV cleaning
US data-centre-weighted indirect (2023)	**~4.5 L/kWh** (incl. hydro evaporation)	LBNL 2024 US Data Center Energy Usage Report, as widely cited

What these numbers show. India's thermal plants are about twice as water-hungry per MWh as global peers. India's grid-average EWIF excluding hydro (~2 L/kWh) is therefore comparable to or higher than the US thermal-only figure, despite a less water-intensive gas share. As solar displaces coal, EWIF falls roughly in proportion to the coal share. This is the largest water co-benefit of decarbonisation, and it is rarely counted.

5.6.3 Combining site and source water (The Green Grid / LBNL method)

> **Total water footprint per kWh_IT = WUE_site + PUE × EWIF Scarcity-weighted footprint = Σ (water_i × CF_i)**, with CF from AWARE or Aqueduct for the basin where the water is lost

Worked example, Bengaluru campus: - *Inputs:* evaporative cooling WUE_site = 1.8 L/kWh_IT, PUE = 1.5, Karnataka EWIF = 1.5 L/kWh. - *Result:* total = 1.8 + 1.5 × 1.5 = **4.05 L/kWh_IT**. About 55% of this is *off-site*, in power plants. - *Hybrid/dry cooling:* WUE_site = 0.2, PUE = 1.35 → 0.2 + 1.35 × 1.5 = **2.2 L/kWh_IT**. Grid water is now 90% of the total. - *Same dry-cooled campus on 90% hourly RE:* 0.2 + 1.35 × (0.9 × 0.1 + 0.1 × 1.5) = **0.52 L/kWh_IT**.

The lesson: once a site goes dry-cooled, **power procurement becomes its main water lever**.

5.7 SYNTHESIS — "Every TWh Is Not the Same": Worked Matrix

Formulas (per 1 TWh = 1,000,000 MWh delivered to the data centre): - CO_2 (t) = EF (tCO_2/MWh) × 1,000,000 ÷ (1 − loss) - Water (litres) = EWIF (L/kWh) × 10^9 ÷ (1 − loss) - Scarcity-weighted water = Water × W, where W is an *illustrative* stress weight: 1 = low, 2 = low-medium, 3 = medium-high, 4 = high, 5 = extremely high (WRI Aqueduct baseline water-stress classes; not AWARE CFs)

For grid cases, losses are 8% for HT supply. For on-site/adjacent supply, losses are about 3–4% (ignored below for simplicity).

Table 5.7 SYNTHESIS — "Every TWh Is Not the Same": Worked Matrix

#	Scenario	EF used (t/MWh)	CO_2 per TWh	EWIF (L/kWh)	Water consumed per TWh	Stress weight (basin where water is lost)	Scarcity-weighted (bn L-eq)
a	**Indian grid average** (CEA WAEF 0.710, +8% losses)	0.77	**~770,000 t**	2.17	**~2.2 bn L**	4 (mixed; ~40% of thermal in high stress)	~8.7
b	**Karnataka consumption mix** (est. 0.50, +8%)	0.54	**~540,000 t** (range 430–600k)	1.6	**~1.6 bn L**	4–5 (Krishna/Raichur, Ballari)	~6.5–8
c	**Marginal coal, Indian evening peak** (1.0 at part load, +8%)	1.09	**~1,090,000 t**	3.8–4.3 (3.5–4.0 at plant)	**~3.8–4.3 bn L**	4–5	~15–21
d	**Karnataka solar hours, grid** — average basis (~0.25–0.30) / marginal basis (coal backing down, ~0.95)	0.30 / 1.03	**~300,000 t (avg)** / ~1,000,000 t (marginal); **~0 on curtailment days**	~0.8 (avg)	**~0.8 bn L** (avg)	4–5	~3–4
e	**24/7 hybrid RE + BESS via group captive/open access** (90% hourly CFE; lifecycle ~40 g for the RE part; 10% residual grid at 0.77)	~0.11 lifecycle (~0.08 operational)	**~110,000 t lifecycle**; 100% CFE → ~40,000 t	0.09 RE + 0.22 residual	**~0.3 bn L**	4–5 (residual only)	~1.2
f1	**France (nuclear-dominant grid)**	0.03–0.06	**~30,000–60,000 t**	~1–2.5 (many plants on rivers/sea)	**~1–2.5 bn L**	2–3 (Rhône/Loire summer stress)	~3–6
f2	**Nordic hydro (Norway)**	0.02–0.03	**~20,000–30,000 t**	0 (run-of-river) to >10 (gross-evaporation allocation)	**contested: ~0–10+ bn L**	1	~0–10
g	**US PJM / Northern Virginia** (~0.37–0.40)	0.38	**~380,000–420,000 t**	~2 (thermo-electric) to 4.5 (LBNL DC-weighted incl. hydro)	**~2–4.5 bn L**	2	~4–9
h	**Gas CCGT** (IPCC lifecycle 490; direct ~370)	0.49	**~490,000 t** (direct ~370,000)	0.75 (tower) / 0.01 (dry)	**~0.75 bn L** (tower)	site-specific (2–5)	~1.5–3.8
i	**Nuclear with cooling towers**	0.006–0.012	**~6,000–12,000 t**	2.5–2.7	**~2.5–2.7 bn L**	site-specific; inland Indian sites 4–5	~10–13
—	Wind only (reference)	0.011	~11,000 t	~0	~0	—	~0
—	Solar PV only (reference)	0.048	~48,000 t	0.02–0.1	~0.02–0.1 bn L	4–5	~0.1–0.5

Ranges shown by the matrix: - **Carbon:** about 1,090,000 t (marginal Indian coal) down to about 6,000–12,000 t (nuclear/wind). That is a **~100× spread**. - **Water:** about 4.3 billion litres (Indian coal) down to about 0 (wind). Scarcity-weighted, the gap is wider still: ~20 vs ~0. - **Decoupling of carbon and water:** - *Nuclear with towers* scores among the best on carbon but near coal on water (row i). - *Gas CCGT* is mid-carbon but relatively low-water (row h). - *Hydro's* water burden depends on an allocation choice (row f2).

Only **wind and dry-cleaned solar**, firmed by batteries, are low on both. - **India's middle ground.** A Bengaluru data centre on the plain grid (rows a/b) sits at 0.5–0.8 $MtCO_2$ and about 1.6–2.2 billion litres per TWh. Hourly-matched RE (row e) cuts both by about **80–90%**. No efficiency upgrade inside the building comes close to that.

Chart in words — CO_2 per TWh (log scale): marginal coal ≈ 1.1 Mt ▸ India avg 0.77 ▸ Karnataka 0.54 ▸ gas 0.49 ▸ PJM 0.40 ▸ Karnataka solar-hours avg 0.30 ▸ 24/7 RE (90%) 0.11 ▸ France 0.05 ▸ solar 0.048 ▸ Norway 0.025 ▸ wind/nuclear ~0.01.

5.7.1 Five-factor scorecard (for a site or procurement decision)

Score each factor 1 (worst) to 5 (best). Weight them by your priorities; a default is carbon 30%, water 25%, source 20%, timing 15%, location 10%.

Table 5.8 Five-factor scorecard (for a site or procurement decision)

Factor	Question	1 (poor)	3	5 (best)	Example: Bengaluru grid-only	Example: Bengaluru + 24/7 group-captive hybrid + BESS + dry cooling
Source	What physically or contractually supplies each hour?	Unbundled RECs / grid coal	Annual-matched new PPA	Additional, deliverable, hourly CFE	2	5
Timing	What share of load is hourly matched? Is flexible load in solar hours?	<30% CFE, no flexibility	50–70% CFE	>90% CFE + DR contracts	2	4–5
Location	Marginal factor of grid; consumption-based state mix; network losses	Marginal coal, high losses	Mixed grid	Low-carbon margin / on-site	2 (national coal margin)	4
Carbon intensity	Location-based and hourly market-based tCO_2/MWh	>0.7	0.3–0.5	<0.1	2 (0.54–0.77)	5 (~0.08–0.11)
Water intensity	WUE_site + PUE × EWIF, scarcity-weighted	>4 L/kWh_IT in high-stress basin	2–3 L/kWh_IT	<0.5 L/kWh_IT or non-stressed	1–2 (4.05 L/kWh_IT, high stress)	4–5 (~0.5 L/kWh_IT)
Weighted score					**≈1.8 / 5**	**≈4.6 / 5**

5.8 Recommendations

5.8.1 For data-centre operators (India, especially Bengaluru)

1. **Contract for 24/7, not annual.** Use the Green Energy Open Access Rules 2022 (100 kW threshold) and group-captive structures to build solar + wind hybrids with 2–4 h BESS. Target ≥80–90% hourly CFE. - Karnataka's wind is strongest in the monsoon, and solar is strongest in the dry season. Pairing them, plus storage for the 18:00–23:00 window, closes most of the gap. - Battery costs have fallen sharply. Ember cites a 95% fall in global installed cost, to about $140/kWh in 2025.
2. **Schedule for carbon and water.** - Move training, batch inference and backups into the eight solar hours, where ToD gives at least a 20% discount. - Avoid the 18:00–23:00 peak, which carries a ≥20% premium and is the coal margin. - Sign interruptible or flexible-load agreements with BESCOM/KPTCL modelled on Google's I&M/TVA contracts.
3. **Site for both carbon and water.** Prefer locations near RE parks or coastal and wastewater sources, and avoid extremely high-stress basins. Use dry, hybrid or liquid cooling with treated sewage water. Remember that grid water dominates once site WUE is low.
4. **Report in full.** - Location-based (CEA WAEF, grossed up for losses) and market-based Scope 2, with **hourly CFE%**. - WUE_site and source WUE (PUE × EWIF), scarcity-weighted using AWARE/Aqueduct. - BRSR Core metrics: energy and water intensity, Scope 1/2.
5. **Avoid low-quality claims.** Do not rely on unbundled RECs from old hydro or wind. Prioritise additionality (new capacity), deliverability (same grid) and temporal matching. Assume the GHG Protocol's hourly rules will apply by around 2027–2028, and build legacy-clause-eligible contracts now.

5.8.2 For CEA, MoP, KERC, MoEFCC and the Government of Karnataka

1. **Publish granular emission factors.** CEA should publish **hourly (or 15-minute) national and state consumption-based average factors and marginal factors**, alongside the annual baseline database. It should also publish a **residual-mix factor** so that market-based claims do not double-count. Grid-India data makes this feasible.
2. **Enforce and disclose plant water use.** Bring water into transparent compliance: plant-level, metered **specific water consumption disclosure** on a public CEA/CPCB portal, as WRI India has urged. Strictly enforce the 3.5/3.0 m^3/MWh norms; CSE found about half of non-seawater capacity non-compliant. Report the extra water FGD adds.
3. **Set a Karnataka 24×7 RE standard for large data centres.** Andhra Pradesh plans to require **≥70% RE** for data centres. IT Minister Nara Lokesh announced on 24–25 September 2026 that "a minimum 70 per cent of the energy used by data centres must come from renewable sources", with solar, wind, pumped storage and battery counting. Karnataka should go further by specifying **hourly** matching, with a phased path to 70% then 90% CFE. This avoids an annual-REC loophole.
4. **Strengthen price and market signals.** - Implement the full solar-hour discount (≥20%) and peak premium for HT data-centre tariffs. - Create a **green tariff with hourly attribution**. - Recognise data-centre flexibility in ancillary and demand-response markets.
5. **Site by water stress.** Require data centres above a threshold (e.g. 10 MW) to submit a **five-factor assessment**, including scarcity-weighted source and site water. Steer them away from extremely high-stress taluks and toward treated wastewater.
6. **Use carbon markets carefully.** The Carbon Credit Trading Scheme (2023) uses the CEA WAEF for Scope 2. Data centres are not yet obligated entities. Bring large operators into the offset/voluntary track only with hourly-matched, additional RE, so that recycled RECs do not generate credits.
7. **Reduce network losses.** Bringing AT&C losses down from about 15–16% toward the RDSS 12–15% target directly lowers the delivered intensity of every TWh.

5.9 Caveats

- **Contested and estimated figures** are flagged in the text. They include:
- Karnataka's consumption-mix intensity (no official consumption-based factor exists);
- hourly Karnataka profiles from a modelled third-party dashboard;
- all marginal emission factors (no official Indian series; method choice can change results 2×);
- France, Norway, EU, US and China intensities quoted approximately rather than from the latest Ember tables;
- Macknick-based water values (approximate conversions from US medians).
- **Hydro water** varies by more than two orders of magnitude depending on gross vs net evaporation and multipurpose-reservoir allocation. **Tropical reservoir methane** can push hydro carbon far above the 24 g median. Indian reservoir hydro should be treated site by site.
- **Self-reported Indian plant water data** is unverified (CSE flags likely under-reporting), so actual thermal water intensity may be higher than shown.
- **Stress weights** in the synthesis are illustrative ordinal weights from Aqueduct classes, not AWARE characterisation factors. A formal ISO 14046 study would use basin-specific AWARE CFs.
- **Policy status.** The Andhra Pradesh 70% RE rule is a stated plan, not yet a notified regulation. The GHG Protocol Scope 2 revision is a consultation draft. NEP 2031-32 intensity (~0.43 t/MWh) is a projection that depends on storage and transmission build-out.
- **Corporate disclosures** (Google, Microsoft) are self-reported and have been criticised by NGOs. Microsoft's FY25 market-based figure is not comparable with earlier years because of a methodology change.

Sources cited in this chapter

Organisations and publications named in this chapter's text, in order of how often they are cited. Full details are in the bibliography.

- Ember, Global Electricity Review and India data
- Central Electricity Authority (CEA), CO_2 Baseline Database and grid reports
- World Resources Institute (Aqueduct, thermal water studies)
- Google environmental reports, efficiency data and Gemini inference paper
- IPCC AR5 Annex III life-cycle emission factors
- Microsoft sustainability reports and datacenter design posts
- Centre for Science and Environment, thermal plant water survey

- Lawrence Berkeley National Laboratory, 2024 US Data Center Energy Usage Report
- State electricity regulatory commissions (KERC, MERC, TNERC, TGERC) tariff and open-access orders
- Ministry of Power, orders on ISTS waiver and green open access
- National Renewable Energy Laboratory water factors
- International Energy Agency (IEA), Energy and AI (2025) and updates
- Ministry of Environment, Forest and Climate Change, thermal plant water norms and EIA
- Meta sustainability reports and data index
- Amazon Web Services sustainability reporting
- The Green Grid metric definitions
- IEEFA analysis of green open access

PART II · RESOURCES AND IMPACTS

Chapter 6
Water impact: from litres used to water impact

WHAT THIS CHAPTER COVERS

This chapter explains how data centres use water and how that use should be measured. It distinguishes withdrawal from consumption, cooling water from other uses and fresh water from recycled water, builds complete water balances for an evaporatively cooled hyperscale facility and a liquid-cooled AI facility, compares every metric used to express water intensity, places data-centre use against Bengaluru's supply and other demands, and applies water-stress methods that weight each litre by how scarce water is where and when it is taken. It ends with a minimum disclosure framework and suggested thresholds.

KEY TAKEAWAYS

1. A modelled 50 megawatt evaporatively cooled data centre in Bengaluru would withdraw about 1.6 to 2.1 million litres a day, less than 0.1% of the roughly 1,935 million litres a day that the city water board supplies.
2. Drawn from groundwater in Bangalore East taluk, the same volume would equal about 4% of that taluk's sustainable groundwater, so the same litres have very different consequences depending on the source.
3. Withdrawal is the water taken in, consumption is the part that is not returned (mostly evaporated), and the difference between them can be large; treatment and higher cycles of concentration reduce withdrawal but not evaporation.
4. The power stations that supply a data centre consume more water than the site itself in most Indian cases, so litres used on site do not measure water impact.
5. India's disclosure regime currently captures neither the source, the location, the season nor the off-site electricity water, and the chapter proposes seven metrics that every site of 500 kilowatts or more should publish monthly.

How this chapter connects to the others. Chapter 4 applies these methods to Bengaluru, Chapter 7 adds hourly and peak-day calculations, and Chapter 8 compares water across conventional and AI data centres.

A modelled 50 MW evaporatively cooled data centre in Bengaluru would withdraw about 1.6–2.1 million litres a day (MLD). That is less than 0.1% of the roughly 1,935 MLD BWSSB currently supplies. Drawn from groundwater in Bangalore East taluk, though, the same volume equals about 4% of that taluk's sustainable groundwater, and its power plants consume more water than the site itself. So "litres used" does not measure "water impact". What matters is the source, the location, the season and the off-site electricity water, and India's disclosure regime currently captures none of these.

6.1 Executive summary

This report follows the series' earlier reports on consumption anatomy, cooling, Bengaluru water and power, and "Every TWh is not the same". It moves from volumes to impact:

1. **How much.** Evaporation physics sets a floor of about 1.5 L per kWh of heat rejected. Cycles of concentration (CoC) change withdrawal and discharge but barely change consumption. A 50 MW Bengaluru hyperscale facility at 80% load withdraws 1.58–2.08 MLD and consumes about 1.39 MLD at any CoC.
2. **Water balance.** Make-up = Evaporation + Blowdown + Drift, and Consumption = Withdrawal − Discharge. Balances are given at CoC 3, 5 and 8, with an RO and zero-liquid-discharge (ZLD) variant, and for a 100 MW AI facility with direct liquid cooling (DLC) and adiabatic dry coolers (about 0.13 L/kWh on site, peak-to-average about 3.5).
3. **Metrics.** WUE is necessary but not sufficient. Per-GPU-hour and per-prompt metrics swing over two orders of magnitude depending on boundary: Google's 0.26 mL per prompt (on-site, derived by applying its 2024 fleetwide WUE) against Mistral's 45 mL (life cycle).
4. **Local context.** Cauvery capacity is 2,225 MLD but about 1,935 MLD is delivered. Bengaluru Urban groundwater extraction rose from 600 MLD (2020) to 1,457 MLD (2024). Every taluk is over-exploited.
5. **Stress.** Scarcity weighting, materiality ratios and a five-dimension scorecard show that dry or DLC cooling with renewable power cuts the scarcity-weighted footprint by more than half compared with evaporative cooling on groundwater, even though dry cooling raises electricity use.

Recommendation in one line: regulate on *consumption by source, location and season*, not on annual WUE alone.

6.2 How much water?

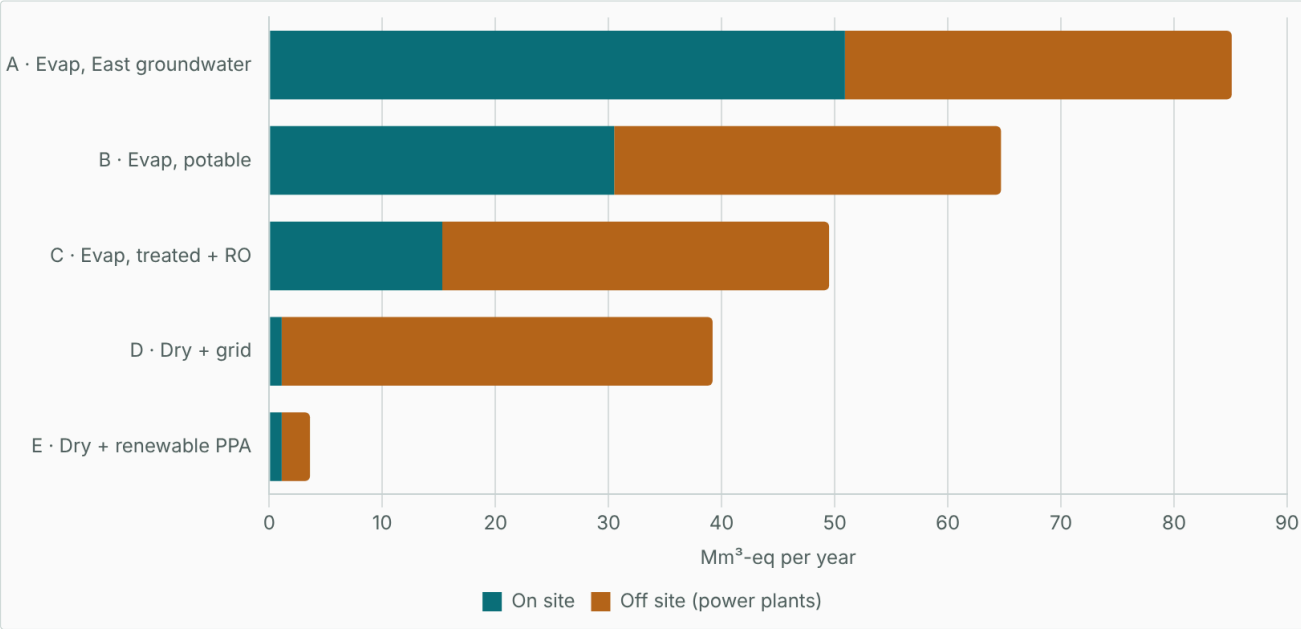


Figure 6.1 Scarcity-weighted water footprint. Mm³-equivalent a year, 50 MW, illustrative AWARE-style factors *Source: India Data Centre Atlas, chapter built on this chapter's data.*

6.2.1 Withdrawal vs consumption: definitions and why both matter

Table 6.1 Withdrawal vs consumption: definitions and why both matter

Framework	Withdrawal	Consumption	Other key terms
USGS	Water removed from ground or surface sources	Water evaporated, transpired, incorporated into products or otherwise removed from the immediate water environment	Return flow
GRI 303 (2018)	303-3: all water drawn from all sources, split freshwater (≤1,000 mg/L TDS) and "other"	303-5: **withdrawal – discharge** (± storage change)	303-4 discharge by destination and treatment; all broken out for water-stressed areas
ISO 14046 (2014)	Water use	Water not returned to the drainage basin (evaporated, integrated, transferred)	Water scarcity footprint
Water Footprint Network	—	**Blue**: surface or groundwater consumed. **Green**: soil rainwater. **Grey**: virtual dilution volume for pollutants	—

Why report both:

- **Withdrawal** measures pressure on suppliers and infrastructure: BWSSB mains, tankers, borewell drawdown.
- **Consumption** measures what is *lost to the basin*, which is the quantity used in scarcity footprints.
- **Discharge** measures quality burden: concentrated blowdown and RO reject.

Contrast:

- **Once-through cooling** (seawater or river; possible for coastal Chennai or Mumbai sites): huge withdrawal (tens of L/kWh), near-zero consumption, thermal and ecological impact.
- **Evaporative towers:** moderate withdrawal (about 1.6–2.2 L/kWh of IT energy), of which 65–90% is consumed.

Withdrawal-based WUE (Microsoft, AWS, Meta) is not directly comparable with consumption-based WUE (Google's per-prompt figure, ISO "Category 2": input minus returned).

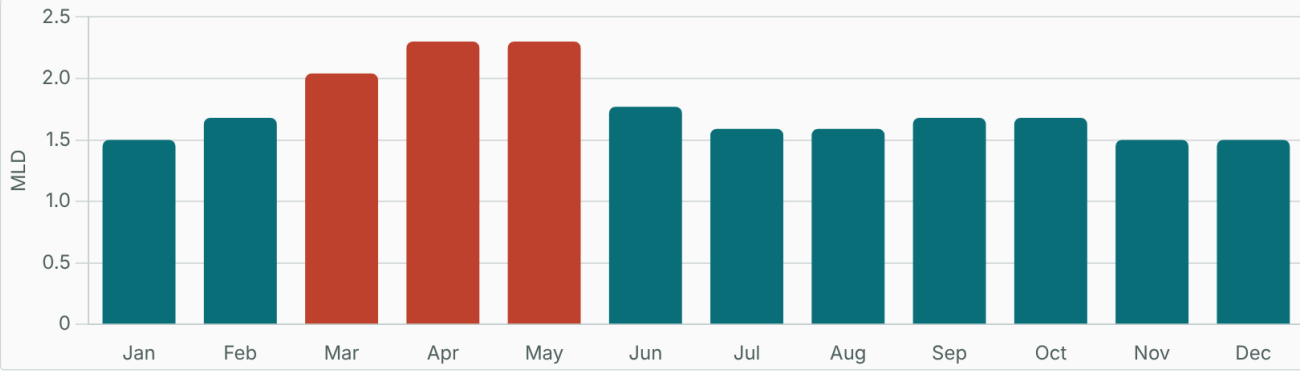


Figure 6.2 The season matters. Monthly withdrawal, MLD, for the 50 MW evaporative site. The peak lands when tankers are scarcest. *Source: India Data Centre Atlas, chapter built on this chapter's data.*

6.2.2 Cooling water

Table 6.2 Cooling water

System	Mechanism	Site water (L/kWh of IT energy)	Notes
Open cooling towers + water-cooled chillers	Latent heat (about 2,400 kJ/kg → about 1.5 L per kWh of heat)	1.4–2.5 withdrawal; 1.1–1.8 consumption	Bengaluru's moderate wet-bulb suits towers; lowest PUE
Direct / indirect evaporative air handlers	Evaporation into air	0.3–1.5	Seasonal
Adiabatic hybrid dry coolers	Dry coils; sprays above about 28–32°C	0.05–0.4	Low average, high peaks
Air-cooled chillers / dry coolers	Sensible heat only	≈0–0.05	PUE higher by about 0.1–0.2
Closed-loop DLC + dry coolers	Warm water (35–45°C) to cold plates	0.01–0.15	One-time fill about 1.5–3 L/kW (often 25% propylene glycol)
Chilled-water make-up	Closed-loop losses	<0.01	Quality matters more than volume

LBNL's 2024 *United States Data Center Energy Usage Report* puts industry-average site WUE at just over 0.36 L/kWh, ranging from near zero (air-cooled) to well over 1 L/kWh (water-cooled). Equinix reports 1.41 L/kWh at evaporative sites against 0.91 portfolio-wide (2025).

6.2.3 Non-cooling water

Table 6.3 Non-cooling water

Use	50 MW campus magnitude	Comment
Domestic / staff (about 300 people at the 45 L per head per day office norm)	≈13.5 m^3/day	About 80% returns as sewage; on-site STP reuse
Cleaning, housekeeping	3–10 m^3/day	—
Humidification	0–20 m^3/day, seasonal	Dry months (January–March, relative humidity 25–40%) need humidification; the monsoon needs dehumidification, which yields recoverable condensate
Landscaping	5–20 m^3/day	Should use STP or rainwater
Fire suppression	Hundreds of m^3 stored; 1–5 m^3/week testing	Storage, not consumption
Diesel generator cooling	Closed radiator loops; negligible	—
RO reject	20–35% of RO feed	Raises withdrawal and discharge, not consumption
Construction	Tens of thousands of m^3 over 18–36 months (estimate)	Meta reports construction water separately; Yotta's Greater Noida site had a 50 KLD STP-water connection for construction (DTE)

Rule of thumb: non-cooling uses add about 20–50 m^3/day (about 0.03–0.05 L/kWh). That is trivial for evaporative sites but most of the water at a dry site.

6.2.4 Freshwater vs recycled water

Table 6.4 Freshwater vs recycled water

Source	Bengaluru status	Cooling-tower issues	Implications
BWSSB potable (Cauvery)	Industrial tariff about ₹99/kL	Low TDS allows high CoC	Diverts drinking water; last resort
Groundwater (borewell or tanker)	All Bengaluru Urban taluks over-exploited	TDS 500–1,500 mg/L; silica 20–60 mg/L limits CoC to about 3–4	Highest local impact; KGWA permits
Tertiary treated wastewater (TTW)	Growing; K&C Valley already sends treated water to Kolar and Chikkaballapur tanks	Ammonia, phosphate, biofouling; needs UF/RO	Has downstream claimants
Rainwater	About 900 mm/yr; a 10 ha roof yields about 80,000 m^3/yr	Ideal make-up	About 13% of a 50 MW evaporative plant's need
Seawater	Not available in Bengaluru	Corrosion	Coastal sites only

Indicative recirculating-water limits (typical engineering guidance; vendor limits vary):

- **TDS:** 1,500–2,500 mg/L. Groundwater at 600 mg/L allows CoC of about 3–4; RO permeate allows CoC above 10.
- **Silica:** 150 mg/L. Make-up at 40 mg/L allows CoC of about 3.7, often the binding limit for Deccan groundwater.
- **Chlorides:** 250–500 mg/L.
- **Calcium hardness:** 600–1,000 mg/L as $CaCO_3$, with inhibitors.

The first constituent to hit its limit sets the CoC, which is why quality drives impact.

6.2.5 Daily and annual demand: worked examples

Assumptions: 80% average load, so 1 MW = 19.2 MWh/day = 7,008 MWh/yr of IT energy. WUE is withdrawal-based (L/kWh = m^3/MWh).

Table 6.5 Daily and annual demand: worked examples

IT capacity	Evaporative, CoC 5 (WUE 1.80)	Adiabatic (0.40)	Dry / DLC (0.05)	WELL Labs basis (26 ML/MW/yr)
1 MW	34.6 m^3/d · 12.6 ML/yr	7.7 m^3/d · 2.8 ML/yr	1.0 m^3/d · 0.35 ML/yr	71 m^3/d · 26 ML/yr
20 MW	0.69 MLD · 252 ML/yr	154 m^3/d · 56 ML/yr	19 m^3/d · 7 ML/yr	1.42 MLD · 520 ML/yr
50 MW	1.73 MLD · 631 ML/yr	384 m^3/d · 140 ML/yr	48 m^3/d · 17.5 ML/yr	3.56 MLD · 1,300 ML/yr
100 MW	3.46 MLD · 1,261 ML/yr	768 m^3/d · 280 ML/yr	96 m^3/d · 35 ML/yr	7.12 MLD · 2,600 ML/yr

Interpretation. WELL Labs' about 26 ML/MW/yr (similar to a Deloitte-cited 68,500 L/MW/day) implies a WUE of about 3 L/kWh at full load. That is plausible for low-CoC towers on hard groundwater or if total campus water is included, but it is about twice our central case. These figures are consistent with the series' earlier assumptions (WUE 1.8, 1.6, 0.4 and 0.05).

Peak vs average. Evaporation tracks heat load and wet-bulb temperature, which peaks in April–May.

- **Illustrative monthly multiples:** January 0.85, February 0.95, March 1.15, **April 1.30, May 1.30**, June 1.00, July–August 0.90, September–October 0.95, November–December 0.85.

- **Design peak day:** about 1.35× the average, or **2.3 MLD** for 50 MW. Afternoon hourly peaks are about 1.5–1.7× the daily average.
- **Coincidence:** the peak falls when tanker demand and borewell failures peak.

Storage. Indian Tier III and IV practice commonly provides 24–72 h of make-up storage. We could not verify a fixed Uptime Institute mandate; this is a design-basis choice. For 50 MW at peak, that is 2.3 ML (24 h), 4.7 ML (48 h) or 7.0 ML (72 h, about 3 Olympic pools). Dry or DLC sites need only domestic and fire storage.

6.3 Water balance

6.3.1 Water balance in words

1. **SOURCE.** BWSSB potable, borewell or tanker, TTW, rainwater.
2. **TREATMENT.** Softening (brine about 2–5%) and/or UF + RO (70–80% recovery, so reject is 20–30% of feed).
3. **USES:** - **Cooling towers:** evaporation (70–85% of make-up), drift (<0.005% of circulation) and blowdown (12–35%). - **Humidification:** consumed. - **Domestic:** about 80% returned as sewage. - **Cleaning and landscaping.**
4. **REUSE.** Blowdown to side-stream RO with permeate returned to make-up; sewage to STP for flushing and landscaping; ZLD (evaporator or crystalliser, salts to hazardous landfill).
5. **DISCHARGE.** To the BWSSB sewer under KSPCB consent. CPCB general standards for inland surface water include pH 5.5–9.0, BOD ≤30 mg/L, TSS ≤100 mg/L and oil and grease ≤10 mg/L; limits for public sewers are looser. Blowdown TDS and biocides are the usual issue.

6.3.2 Formulas

- **Make-up:** M = E + B + D
- **Evaporation, energy method:** E ≈ Q (kWh) × 3,600 ÷ 2,400 ≈ **1.5 L/kWh of heat** if all heat is rejected as latent heat. With 80–90% latent, use **1.25–1.35 L/kWh of heat**.
- **Evaporation, rule of thumb:** E (gpm) ≈ 0.00085 × circulation (gpm) × ΔT (°F)
- **Blowdown:** B = E ÷ (CoC − 1)
- **Drift:** D ≈ 0.001–0.005% of circulation
- **Consumption:** C = Withdrawal − Discharge = E + D + humidification + other losses
- **Heat to towers:** Q ≈ 1.15 × IT load (adds chiller compressor work)

6.3.3 Worked balance: 50 MW Bengaluru hyperscale facility (evaporative)

Inputs:

- 50 MW IT × 80% load = 40 MW average = **350,400 MWh/yr** of IT energy.
- Heat to towers 46 MW. Evaporation 1.25 × 1.15 = **1.44 L/kWh of IT energy** = 504.6 ML/yr (1.38 MLD).
- Cross-check: circulation at a 5.6 K range is about 7.06 ML/h. 0.85% evaporation = 60 m³/h = 1.44 MLD, which agrees with the energy method.
- Drift 0.002% gives **1.2 ML/yr**. Non-cooling 40 m³/d = 14.6 ML/yr (80% returned).

Annual balance (ML/yr):

Table 6.6 Worked balance: 50 MW Bengaluru hyperscale facility (evaporative)

Stream	CoC 3 (hard borewell)	CoC 5 (potable or softened)	CoC 8	CoC 8 on TTW + RO (75% recovery)	ZLD
Evaporation	504.6	504.6	504.6	504.6	504.6
Blowdown	252.3	126.2	72.1	72.1	0
Drift	1.2	1.2	1.2	1.2	1.2
Tower make-up	**758.1**	**632.0**	**577.9**	**577.9**	≈506
RO reject	—	—	—	192.6	0
Non-cooling	14.6	14.6	14.6	14.6	14.6
Withdrawal	**772.7 (2.12 MLD)**	**646.6 (1.77 MLD)**	**592.5 (1.62 MLD)**	**785.1 (2.15 MLD)**	**≈525 (1.44 MLD)**
Discharge	264.0	137.9	83.8	276.4	≈0
Consumption	**508.7 (1.39 MLD)**	**508.7**	**508.7**	**508.7**	**≈525**
Consumption ÷ withdrawal	66%	79%	86%	65%	≈100%
Site WUE, withdrawal	2.21	1.85	1.69	2.24	1.50
Site WUE, consumption	1.45	1.45	1.45	1.45	1.50

Daily at CoC 5: withdrawal 1.77 MLD, consumption 1.39 MLD, discharge 0.38 MLD; peak day about 2.4 MLD.

Insight. Raising CoC from 3 to 8 cuts withdrawal by 23% and discharge by 68%, but **consumption is unchanged**. Treated-wastewater RO raises gross withdrawal while shifting it off freshwater. ZLD removes discharge but makes everything consumption and adds 20–70 kWh/m³ of energy plus salt waste. Each option wins on a different metric, so all three flows must be reported.

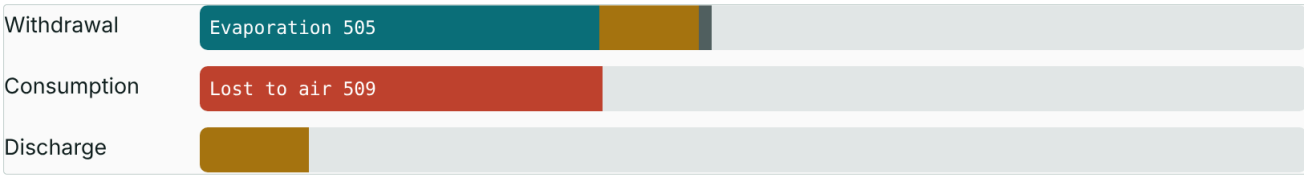


Figure 6.3 Cooling-tower water balance for a 50 MW hyperscale facility. Withdrawal, consumption and discharge at four cycles of concentration. *Source: India Data Centre Atlas, chapter built on this chapter's data.*

6.3.4 Worked balance: 100 MW AI facility (DLC + adiabatic dry coolers)

Inputs: 100 MW × 85% = **744,600 MWh/yr** of IT energy; PUE 1.15; all heat to dry coolers. Adiabatic sprays run about 1,000 h/yr (an illustrative assumption, mostly March–May afternoons) at about 1.0 L/kWh.

Table 6.7 Worked balance: 100 MW AI facility (DLC + adiabatic dry coolers)

Item	Calculation	Annual
Adiabatic water	85,000 kW × 1,000 h × 1.0 L/kWh	85 ML
Non-cooling	30 m^3/d	11 ML
Coolant top-up	—	≈1 ML
Total withdrawal	—	**≈97 ML/yr (0.27 MLD)**
Site WUE	97 ML ÷ 744,600 MWh	**≈0.13 L/kWh**
Consumption	—	≈80 ML
One-time fill	1.5–3 L/kW × 100 MW	150–300 m^3
Peak day (May)	10 h × 90 MW × 1.0 + 30 m^3	**≈0.93 MLD (peak ÷ average ≈ 3.5)**
Off-site (grid)	744,600 × 1.15 × 1.5 L/kWh	**≈1,284 ML/yr (3.5 MLD), about 13× on-site**

Interpretation. On site, DLC is about 13 times less water-intensive than evaporative cooling. But its demand is peaky and its footprint is dominated by electricity, so power sourcing is a water decision.

6.3.5 Measurement and accounting standards

Meters: M1 intake by source (borewell telemetry, tanker logs); M2 RO permeate and reject; M3 tower make-up; M4 blowdown; M5 humidifier feed; M6 domestic and STP; M7 reuse; M8 final discharge. Add conductivity for continuous CoC and 15-minute logging.

Table 6.8 Measurement and accounting standards

Standard	What it requires	Relevance
ISO 46001:2019	Water-efficiency management system (baselines, indicators, Plan-Do-Check-Act)	Management wrapper
AWS Standard 2.0	Site + catchment stewardship, certification	Best fit for "impact"
GRI 303	Withdrawal, discharge, consumption by source and stressed area	Used company-wide, not per site
CDP Water Security	Facility-level data in stressed basins	Few Indian respondents
SEBI BRSR Core	Assured withdrawal by source, consumption, intensity, discharge	Listed companies only (for example Bharti Airtel, parent of Nxtra)
EU Delegated Regulation 2024/1364	Sites ≥500 kW report total water input (W_IN) and potable input (W_IN-POT); WUE = W_IN ÷ E_IT; annually by 15 May	Template. The 2024 EU average WUE was 0.58 against the Climate Neutral Data Centre Pact target of 0.4.
ISO/IEC 30134-9:2022	WUE KPI with withdrawal-type and consumption-type categories	Like-for-like if category declared
The Green Grid WUE	Site and source WUE	Original definition

6.4 Water intensity metrics

6.4.1 L/MWh and WUE

- **Site WUE** = site water ÷ IT energy (declare withdrawal or consumption basis).
- **Source WUE** = (site water + electricity water) ÷ IT energy.
- **Total WUE ≈ site WUE + PUE × EWIF** (EWIF = energy water intensity factor, L/kWh).
- **L/MWh = WUE × 1,000.** For example, 1.80 → 1,800 L per MWh of IT energy, or 1,385 L per MWh of facility energy at PUE 1.30.

Worked example (50 MW, CoC 5, Karnataka EWIF 1.5): site 1.80 (withdrawal) or 1.45 (consumption) + off-site 1.30 × 1.5 = 1.95, giving **3.75 or 3.40 L/kWh**. If marginal power comes from Raichur Thermal Power Station (CAG norm 3.08 m^3/MWh; actual 5.1–7.9 in 2013–17), off-site becomes 4.0–10.3 L/kWh.

Benchmarks (L/kWh):

- AWS: 0.12 (2025), 0.15 (2024), withdrawal
- Meta: 0.19 (2024), withdrawal
- Microsoft: 0.30 (FY2024), down from 0.49 (2021), withdrawal
- LBNL US average: about 0.36, site
- EU average: 0.58 (2024)
- Equinix: 0.91 global; 1.41 at evaporative sites
- Industry average quoted by AWS (de Vries-Gao 2025): 0.56 consumption, 0.84 withdrawal
- LBNL US grid indirect: 4.52

6.4.2 L/MW (per MW of IT capacity)

L/MW/yr = WUE × 8,760 × load factor × 1,000.

Table 6.9 L/MW (per MW of IT capacity)

Case	WUE	Annual per MW	Daily per MW
Evaporative, CoC 5	1.80	12.6 ML	34.6 m^3
Evaporative, CoC 3	2.21	15.5 ML	42.4 m^3
WELL Labs heuristic	≈3.0 (100% load)	26 ML	71 m^3
Adiabatic	0.40	2.8 ML	7.7 m^3
Dry	0.05	0.35 ML	1.0 m^3

Capacity metrics suit allocations (for example, 650 KLD allotted to ETL Secure Space) but hide utilisation.

6.4.3 L/server-hour

L/server-hour = server power (kW) × WUE (plus P × PUE × EWIF for the total).

- **0.5 kW server:** 0.90 L (evaporative) or 0.025 L (dry), plus about 1.0–1.1 L off-site.
- **1.0 kW server:** 1.80 L or 0.05 L, plus about 2.0–2.2 L off-site.

6.4.4 L/GPU-hour

Assumptions (approximate vendor figures, not independently verified):

- **H100:** 700 W TDP; about **1.0 kWh of IT energy per GPU-hour** including host share.
- **B200:** 1,000 W; about **1.4 kWh per GPU-hour**.
- **GB200 NVL72:** about 120 kW per rack (72 GPUs).

Table 6.10 L/GPU-hour

Accelerator	On-site, evaporative (1.80)	On-site, DLC (0.13)	Off-site (PUE × 1.5)	Total (evaporative / DLC)
H100 per GPU-hour	1.8 L	0.13 L	1.95 / 1.73 L	3.75 / 1.86 L
B200 per GPU-hour	2.5 L	0.18 L	2.7 / 2.4 L	5.2 / 2.6 L
GB200 NVL72 per rack-hour	n/a (requires DLC)	15.6 L	207 L	≈223 L (≈3.1 L per GPU-hour)

A 30-day, 10,000-GPU H100 run in a Bengaluru DLC facility uses about 0.9 ML on site and 12.5 ML off site.

6.4.5 L/inference and training

Table 6.11 L/inference and training

Source	Figure	Boundary
Google (August 2025)	**0.26 mL** per median Gemini Apps text prompt (0.24 Wh); narrow approach 0.12–0.15 mL	On-site consumption; fleet-average WUE; excludes electricity water; prompt length unspecified
Sam Altman (June 2025)	≈0.32 mL; 0.34 Wh	Method undisclosed
Li, Yang, Islam & Ren (UC Riverside), *Making AI Less Thirsty*	500 mL per **10–50** GPT-3 responses. The paper's India column: **19.7 mL per request**, all off-site (on-site WUE 0, EWIF 3.445).	On-site + off-site
Mistral AI LCA (Carbone 4, ADEME)	**45 mL** per 400-token response; 1.14 gCO_2e	Full life cycle; training + inference = 91% of water
GPT-3 training	5.4 ML total incl. 0.7 ML on-site (US); India case 6.34 ML	Li et al.
Mistral Large 2	281,000 m^3 (training + 18 months of use)	LCA

Bengaluru calculation (0.3 Wh prompt):

- On-site: 0.3 ÷ 1.30 × 1.80 = **0.42 mL** (evaporative) or 0.3 ÷ 1.15 × 0.13 = **0.03 mL** (DLC).
- Off-site: 0.3 × 1.5 = **0.45 mL** (about 1.5 mL on coal).
- **Total: 0.5-0.9 mL**, up to about 2 mL on coal. That is 2–7× Google's headline but 10–50× below GPT-3-era estimates.

Image and video (indicative only):

- Image at about 2.9 Wh: **about 8 mL** (on-site + off-site).
- 5-second video at about 3.4 MJ: **about 2.7 L per clip**.

GPT-3 trained in Bengaluru (evaporative): 1,287 MWh × 1.80 = 2.3 ML on site + 2.5 ML off site = **about 4.8 ML**.

6.4.6 Which metric for which audience, pitfalls and normalisation

Table 6.12 Which metric for which audience, pitfalls and normalisation

Audience	Best metrics
Operator	Monthly site WUE, CoC, make-up per tower cell
Regulator / utility (BWSSB, KGWA, KSPCB)	Withdrawal by source (MLD, peak day), consumption, discharge quality
Customer	Source WUE, L/GPU-hour, L/inference with declared boundary
Citizen	Share of ward or taluk supply, people-equivalents, scarcity-weighted footprint
Investor	Consumption in stressed areas; replenishment vs consumption

Pitfalls:

1. Functional unit ("per prompt" without tokens).
2. Boundary: on-site vs off-site vs life cycle. Hardware manufacturing water is usually excluded.
3. Withdrawal vs consumption WUE (ratio 1.2–1.5×).

4. Utilisation effects on per-MW metrics.
5. Annual averages hide peaks. Ren: WUE "does not reflect peak water demand".
6. Location blindness.

Normalise by reporting withdrawal, consumption and discharge × source × month, then derive intensities from that.

6.4.7 Single comparison table: Bengaluru scenarios

Table 6.13 Single comparison table: Bengaluru scenarios

Metric	S1: 20 MW colocation, evaporative (PUE 1.40, 70%)	S2: 50 MW hyperscale, evaporative, CoC 5 (PUE 1.30, 80%)	S3: 50 MW dry (PUE 1.45, 80%)	S4: 100 MW AI DLC + adiabatic (PUE 1.15, 85%)
IT energy (GWh/yr)	122.6	350.4	350.4	744.6
Site withdrawal (ML/yr; MLD)	221; 0.60	647; 1.77	17.5; 0.05	97; 0.27
Peak day (MLD)	0.81	2.4	0.05	0.93
Site WUE / L per MWh	1.80 / 1,800	1.85 / 1,850	0.05 / 50	0.13 / 130
Off-site (PUE × 1.5)	2.10	1.95	2.18	1.73
Total / source WUE	**3.90**	**3.80**	**2.23**	**1.86**
ML/MW/yr (site)	11.0	12.9	0.35	0.97
L per server-hour (0.5 kW), site / total	0.90 / 1.95	0.93 / 1.90	0.025 / 1.11	0.065 / 0.93
L per H100-hour, site / total	1.8 / 3.9	1.85 / 3.8	0.05 / 2.2	0.13 / 1.9
mL per 0.3 Wh prompt, site / total	0.39 / 0.84	0.43 / 0.88	0.01 / 0.46	0.03 / 0.48

6.5 Local context

6.5.1 Data centre use against Bengaluru's water availability

Supply:

- **Cauvery.** Stages I–IV supply about 1,450 MLD. Stage V (commissioned 16 October 2024) adds 775 MLD, for 2,225 MLD of capacity. The government told the Legislative Council on 20 August 2026 that about **1,935 MLD** is actually supplied: Stage V delivers about 485 MLD, with 290 MLD to be phased in "in accordance with the demand in the 110 villages". Stage VI (500 MLD) is at DPR stage.
- **Groundwater.** Bengaluru Urban district extraction reached **1,457 MLD in 2024**, up from 600 MLD in 2020 (OpenCity, from the state's *Dynamic Ground Water Resources* report). The corporation area accounted for 547 MLD and Anekal about 295 MLD. 2024 followed a drought.
- **Total and demand.** Experts quoted by Deccan Herald estimate about **3,100 MLD** in total. WELL Labs (*How Water Flows Through Bengaluru*, October 2023) estimates total freshwater demand at about 2,632 MLD. The domestic norm alone (about 14 million people × 135 LPCD) is about 1,890 MLD before losses (historically about 30–40% unaccounted-for water). The deficit is concentrated in borewell-dependent peripheral zones (Mahadevapura, KR Puram, the 110 villages).

Groundwater (CGWB/KGWA categories):

- Karnataka: 68.44% stage of extraction (2024), 45 of 237 taluks over-exploited; 66.49% in 2025.
- All Bengaluru Urban taluks are over-exploited.
- **Bangalore East: 305% (2024)**, extracting 138.46 MLD against about 45 MLD of sustainable use.
- Devanahalli: 169%.
- **Conflict:** the 379–395% figures used earlier in this series reflect earlier assessments or boundaries. The corporation area has been assessed separately since 2022.

Where data centres sit. Whitefield and Mahadevapura host about 120 MW across seven major facilities. The other clusters (east and south-east, Devanahalli–Hoskote) are all in over-exploited, tanker-dependent zones.

Sector use: estimates disagree fivefold:

Table 6.14 Data centre use against Bengaluru's water availability

Estimate	Value	Basis
Karnataka IT department (2026)	≈4 MLD across 32 facilities statewide	Informal "industry-wide approximation"
WELL Labs (Shashank Palur)	≈20 MLD	26 ML/MW/yr × capacity (about 292 MW across 18 centres)
DTE permissions (2013–2025)	≈812.9 ML/yr (≈2.2 MLD)	Allocations, not use
BWSSB billing	129,000 L/month for five centres	Shared meters make it meaningless
This report	292 MW × 70% × 24 h × 1.8 ≈ **8.8 MLD** if all evaporative; about 4–5 MLD if half air-cooled	Cooling mix explains the gap

The 4–20 MLD range is itself the finding. KSPCB officials told reporters they could not quantify data centre water use.

S2 (50 MW evaporative, 1.77 MLD) as a share of local resources:

- BWSSB delivered supply (1,935 MLD): 0.09%
- City total (≈3,100 MLD): 0.06%
- District groundwater extraction (1,457 MLD): 0.12%
- **Bangalore East sustainable groundwater (≈45 MLD): ≈3.9%**
- Undelivered Stage V balance (290 MLD): 0.6%

6.5.2 Household, agricultural and industrial demand

- **Bengaluru Urban groundwater use (2024):** about half irrigation (peri-urban), about 30% domestic, the rest industrial. Within the corporation area: 73% domestic, 27% industrial.
- **Karnataka:** about 88% of groundwater extraction is for irrigation (CGWB); irrigation is about 85–90% of total water use.
- **Treated wastewater is already allocated.** K&C Valley and related schemes pump treated water to Kolar and Chikkaballapur tanks for aquifer recharge, so it has existing claimants.

6.5.3 Equivalence metrics (S2, 50 MW evaporative)

Table 6.15 Equivalence metrics (S2, 50 MW evaporative)

Equivalence	Result
People at 135 LPCD (1.77 MLD)	≈13,100 people
Households (675 L/d)	≈2,600 households
Paddy (≈12,500 m^3/ha per season, typical value) vs 508,700 m^3 consumed	≈41 ha
Ragi (≈4,000 m^3/ha)	≈127 ha
WELL Labs framing, 20 MLD sector-wide	"drinking water for around 1.3 lakh people"
Karnataka's 1 GW target, all evaporative at S2 intensity	≈12.9 GL/yr ≈ 35 MLD (about 1.8% of BWSSB delivered supply); about 71 MLD at WELL Labs intensity
1 GW, DLC or dry	≈1–3 MLD on site
1 GW off-site (80% load, PUE 1.3, 1.5 L/kWh)	≈37 MLD, regardless of cooling

A 50 MW evaporative site consumes about as much as a mid-sized brewery or bottling plant and far less than a coal plant (3–5 m^3/MWh). The meaningful comparison is with local renewable supply.

6.5.4 Temporal context

- **Monsoon.** Most rain falls June–October (about 900 mm/yr). Cauvery inflows depend on the Kodagu and Wayanad catchments and the KRS reservoir. A failed monsoon hits the following summer.
- **2024 crisis.** About 6,997 of 14,781 public borewells failed (Deputy Chief Minister, March 2024). BWSSB banned non-essential potable use, city groundwater extraction quadrupled, tanker prices surged and authorities capped them.
- **Peak coincidence.** Evaporative demand peaks in April–May at 1.3× average, the tanker-scarcity window.
- **2026.** DTE warned of an El Niño-weakened monsoon.
- **Implication.** Permits need peak-month numbers and drought-curtailment plans.

6.6 Water stress

6.6.1 Frameworks

Table 6.16 Frameworks

Framework	Definition	Scale	Bengaluru / India
WRI Aqueduct 4.0 baseline water stress	Withdrawal ÷ available renewable supply (sub-basin)	Low <10%, low–medium 10–20%, medium–high 20–40%, high 40–80%, extremely high >80%	WRI lists India among the 25 "extremely high" countries (2023); a 2026 preprint scores it 3.05 ("high"). **Conflict flagged.** WRI India finds more than half of India's data centres in high or extremely-high stress areas. S&P Global Sustainable1 (September 2025) projects that "60% to 80%" of India's operating data centres will face high water stress this decade, with an average data centre water stress index of 62. Sub-basin scores for Bengaluru, Cauvery, Krishna (Raichur) and Pennar were not verifiable from text sources.
AWARE (WULCA)	CF = world-average available water remaining per area ÷ regional value	0.1–100 m^3 world-eq/m^3; consumption-weighted world averages **43 (unspecified), 20 (non-agricultural), 46 (agricultural)**	India sub-national CFs exist in AWARE 1.2; AWARE 2.0 (2025) generally lowers CFs. **India and basin values not verified here.** Scenario values are used below.
WWF Water Risk Filter	Composite basin physical, regulatory and reputational risk	1–5	Portfolio screening
CGWB stage of extraction	Extraction ÷ extractable recharge	Safe <70, semi-critical 70–90, critical 90–100, over-exploited >100%	Most relevant locally: Bangalore East 305%, Devanahalli 169%
Falkenmark	Renewable m^3/person/year	<1,700 stress; <1,000 scarcity; <500 absolute	India near or below 1,700 nationally (approximate). Bengaluru's delivered water is about 80 m^3/person/year, almost all imported Cauvery water.

6.6.2 Methods

1. **Scarcity-weighted footprint (ISO 14046).** $WSF = \Sigma\ consumption_i \times CF_i$, for on-site and off-site (power-plant basin) consumption, monthly where possible.
2. **Materiality thresholds** (proposed by this report):

Table 6.17 Methods

Ratio	Disclose	Significant	Material
Consumption ÷ catchment renewable availability	>0.1%	>1%	>5%
Withdrawal ÷ ward or zone supply	>1%	>5%	>10%
Groundwater withdrawal ÷ taluk recharge	any (over-exploited)	>0.5%	>1% or any evaporative use in over-exploited taluks
Peak-month withdrawal ÷ summer deficit	>0.5%	>2%	>5%

3. **Contextual targets** (CEO Water Mandate / WRI contextual water targets; SBTN freshwater targets). Set targets by the basin's required reduction. Bangalore East (305%) needs about a two-thirds cut, so new users should target **net-zero groundwater consumption plus recharge**.

4. **Stewardship (AWS Standard).** Catchment study, stakeholder engagement (BWSSB, lake groups, Kolar farmers), and replenishment **in the same catchment and season**.

6.6.3 Applying the methods to Bengaluru scenarios (50 MW, annual)

Illustrative CFs (not verified AWARE values):

- Over-exploited aquifer: 100.
- BWSSB potable: 60.
- Reclaimed water: 30 (60 × 0.5 opportunity factor for K&C Valley users).
- Off-site grid / Krishna basin: 50.

Off-site consumption is facility energy × 1.5 L/kWh: 683 ML/yr for evaporative and 762 ML/yr for dry. A renewable PPA has an EWIF of about 0.1.

Table 6.18 Applying the methods to Bengaluru scenarios (50 MW, annual)

Scenario	On-site consumption (ML/yr)	CF	On-site WSF (Mm^3-eq)	Off-site (ML/yr)	Off-site WSF	Total WSF	Materiality
A. Evaporative, Bangalore East groundwater	508.7	100	50.9	683	34.2	**85.0**	3.9% of taluk recharge: **material**
B. Evaporative, BWSSB potable	508.7	60	30.5	683	34.2	**64.7**	0.09% of city supply, but drinking water: **significant**
C. Evaporative, TTW + RO	508.7	30	15.3	683	34.2	**49.4**	Competes with K&C Valley: disclose
D. Dry, municipal top-up	17.5	60	1.1	762	38.1	**39.2**	Immaterial on site
E. Dry + renewable PPA	17.5	60	1.1	51	2.5	**3.6**	Immaterial
F. 100 MW AI DLC on TTW	≈80	30	2.4	1,284	64.2	**66.6** (13.6 with PPA)	Off-site dominated

Findings:

1. Moving from A to D cuts the scarcity-weighted footprint by **54%**, even though off-site water rises about 12%. Adding renewables (E) cuts it by about 96%.
2. Reclaimed water helps less than claimed: consumption is unchanged and it has claimants.
3. **The off-site Krishna-basin term is the largest single component in every non-groundwater case.** Raichur units have shut in droughts for lack of Krishna water, and actual consumption there (5.1–7.9 m^3/MWh) is about double the norm. Procurement is water policy.
4. The ranking is robust for CFs from 20 to 100; absolute values must be recomputed with official AWARE factors.

6.6.4 Water impact scorecard

Scores run from 1 (low) to 5 (high impact); weights in brackets.

Table 6.19 Water impact scorecard

Dimension (weight)	A. Evaporative, groundwater	B. Evaporative, potable	C. Evaporative, TTW + RO	D. Dry	F. AI DLC on TTW	E. Dry + PPA
Volume (20%): consumption per MW	5	5	5	1	2	1
Source (25%): over-exploited groundwater 5 … none 1	5	4	3	2	3	2
Timing (15%): peak and summer coincidence	4	4	4	1	4	1
Location / stress (25%)	5	4	4	4	4	4
Off-site (15%)	3	3	3	4	3	1
Weighted score	**4.6**	**4.0**	**3.8**	**2.6**	**3.3**	**1.9**
Decision (≥4 refuse/redesign; 3–4 conditional; <3 acceptable)	**Refuse**	**Redesign**	**Conditional**	**Acceptable**	**Conditional** (peak cap, PPA)	**Preferred**

6.7 Hyperscaler and Indian operator water reporting: practices and gaps

Table 6.20 Hyperscaler and Indian operator water reporting: practices and gaps

Company	Latest disclosures	Gaps
Google	2024 consumption 8.1 billion gallons (+28%); replenished 4.5 billion gallons = **64%** (18% in 2023); original target 120% by 2030. Secondary summaries of the 2026 report show 2025 consumption of about 10.9 billion gallons (+34%) and more than 7 billion gallons replenished. The June 2026 pledge is to replenish more than it consumes at its sites by 2030 (165 projects, 97 watersheds).	Consumption growing faster than replenishment; replenishment not necessarily in the same catchment; per-prompt figure excludes electricity water
Microsoft	WUE **0.30** (FY2024, withdrawal), down from 0.49 (2021); zero-water-evaporation designs since August 2024 (pilots at Phoenix and Mt Pleasant from 2026, coming online from late 2027), which Microsoft's VP of Datacenter Infrastructure Engineering Steve Solomon says will save more than 125 million litres of water per datacenter per year	"Nominal" energy increase; claims not independently reviewed; legacy fleet still evaporative
AWS	WUE **0.15** (2024) and **0.12** (2025); **53% → 75%** toward water positive by 2030; claims water positive in India (returned 120% in 2025)	No Indian site volumes or sources; relies on replenishment accounting
Meta	WUE **0.19** (2024); consumption 3,123 ML; restores 200% (high stress) and 100% (medium stress); 1.6 billion gallons restored in 2024; construction water reported separately	US-centric

Indian operators (DTE, November 2025):

- **Nxtra:** FY2025 consumption 216,357 KL (9,876 KL groundwater; 191,152 KL third-party); 15,329 KL recycled; no prior-year data.
- **STT GDC:** global withdrawal 1,149,020 KL (2024). India was 419,204 KL in 2023 (40% of global); no 2024 India split.
- **NTT:** withdrew 413,779 KL and consumed 203,297 KL (2023); omitted in 2025.

- **Sify:** 6,132,323 KL (FY2024) fell to 306,916 KL (FY2025), unexplained.
- **CtrlS:** 409,037 KL (FY2024) against 468,330 KL; claims a 119,662 KL/yr saving from air cooling without facility counts.
- **AdaniConneX:** no standalone report.
- **Yotta:** no figures; advocates "air-cooled + liquid-cooled + closed-loop systems, not water-cooled towers".

Systemic gaps:

1. No facility-level data.
2. Withdrawal and consumption confused.
3. Inconsistent WUE.
4. Recycling claimed without the matching freshwater withdrawal.
5. Private operators fall outside BRSR. CEEW and SYSTEMIQ (February 2026) found water risk systematically underestimated, most state policies lacking PUE and WUE standards, and about 271 data centres (January 2026). Press coverage cites an industry estimate of about 150 billion litres used in 2025, rising to nearly 358 billion litres by 2030. That is about 100 ML/MW across about 1.5 GW, about four times the WELL Labs heuristic, so it likely includes off-site water. Clarify its boundary before quoting it.

6.8 Recommendations

6.8.1 Minimum measurement and disclosure framework (all sites ≥500 kW IT)

Table 6.21 Minimum measurement and disclosure framework (all sites ≥500 kW IT)

#	Metric	Granularity	Basis
1	Withdrawal by source (potable, groundwater, tanker, TTW, rainwater)	m^3, **monthly**, per site	GRI 303-3; EU 2024/1364
2	Consumption (withdrawal − discharge, cross-checked by CoC)	m^3, monthly	GRI 303-5; ISO 14046
3	Discharge volume and quality	m^3, mg/L	GRI 303-4; CPCB / KSPCB
4	Site WUE, withdrawal **and** consumption (ISO/IEC 30134-9 category declared)	L/kWh, monthly and annual	The Green Grid
5	Source WUE (location-based and market-based)	L/kWh	The Green Grid
6	Scarcity-weighted footprint (on-site + off-site)	m^3 world-eq (AWARE version declared)	ISO 14046
7	Seasonal profile	Peak-day MLD, peak-month ÷ average, storage hours	—
+	Context ratios	Share of taluk groundwater and ward supply	CGWB / BWSSB

Reporting should be site-level, assured and public. Recycled-water claims must sit alongside gross freshwater withdrawal; replenishment should be reported separately and counted only within the same catchment.

6.8.2 Regulatory adoption

- **Karnataka Sustainable Data Centre Policy 2026–31** (approved September 2026; 1 GW target; ₹1,300.45 crore outlay):
- It mandates **plant-wise PUE and WUE reporting** and "Data Centre Resource Heat Mapping" for siting. That is welcome.
- Its treated-water incentive (100% reimbursement, **capped at ₹8 lakh per year**) is too small. At ₹99/kL it buys about 22 m^3/day, about 1% of a 50 MW evaporative plant's make-up. Scale it to volume or replace it with a mandate.
- Press coverage reports no numeric WUE ceiling. We suggest:
  - New builds in Bengaluru Urban and Rural: **consumption WUE ≤0.2 and no groundwater or potable water for cooling**.
  - Elsewhere: **≤0.5**.
  - Existing sites: **≤1.0 by 2028**.
- **KGWA:** no new groundwater abstraction for cooling in over-exploited taluks; public telemetry on data centre borewells.
- **BWSSB:** dedicated metered data centre connections; summer-curtailment clauses; price TTW below potable and tanker water; allocate reclaimed water transparently against K&C Valley commitments.
- **KSPCB:** tie consents to the water balance, blowdown and RO-reject management, and drought plans. Bring campuses above about 20 MW under environmental assessment, closing the IT/ITeS exemption.
- **MeitY:** make the seven-metric disclosure mandatory nationally, regardless of listing status, with an EU-style database.
- **SEBI BRSR:** a sector annex with site-level WUE and stressed-area consumption for listed parents.

6.8.3 Suggested thresholds

Table 6.22 Suggested thresholds

Indicator	Green	Amber	Red
Site consumption WUE (L/kWh)	≤0.2	0.2–1.0	>1.0
Groundwater for cooling, over-exploited taluks	0%	—	>0%
Withdrawal ÷ taluk recharge	<0.5%	0.5–1%	>1%
Peak ÷ average without curtailment plan	<1.5	1.5–3	>3
Source WUE (L/kWh)	<1.0	1.0–2.5	>2.5
Scorecard	<3	3–4	≥4

6.9 Caveats

- **Modelled, not metered.** Balances rest on stated assumptions; real sites can differ by ±30–50%, as the twofold gap with WELL Labs shows.
- **AWARE and Aqueduct values are illustrative.** Official India and basin AWARE factors and Aqueduct sub-basin scores for Bengaluru and Raichur could not be verified from text sources. Rankings are robust; absolute m^3-eq values are not. India's national Aqueduct rating is reported inconsistently.
- **Sector totals are contested.** Bengaluru estimates (4, about 9, 20 MLD) and the national 150 GL figure use different boundaries; none is metered.
- **Groundwater figures shift with year and boundary.** Bangalore East was 305% in 2024 against 379–395% in earlier assessments; 2024 followed a drought.
- **Corporate data are self-reported.** Some 2025 Google figures come from secondary summaries. Withdrawal-based and consumption-based WUEs are not comparable.
- **AI per-unit estimates move fast.** GPU, image, video and prompt energy figures are indicative; Google reports that energy per median Gemini text prompt fell 33x between May 2024 and May 2025.
- **Equivalences and limits are typical values.** Crop water, per-capita norms, water-quality limits and CPCB limits are used for scale, not site determinations.
- **Policy details** come from press reports of the cabinet approval; the notified text may differ.

Sources cited in this chapter

Organisations and publications named in this chapter's text, in order of how often they are cited. Full details are in the bibliography.

- Bangalore Water Supply and Sewerage Board
- WELL Labs, Bengaluru water research
- World Resources Institute (Aqueduct, thermal water studies)
- Google environmental reports, efficiency data and Gemini inference paper
- Amazon Web Services sustainability reporting
- Central Ground Water Board and state groundwater assessments
- Microsoft sustainability reports and datacenter design posts
- Meta sustainability reports and data index
- Lawrence Berkeley National Laboratory, 2024 US Data Center Energy Usage Report
- Mistral AI life-cycle assessment (ADEME / Carbone 4)
- The Green Grid metric definitions
- Equinix sustainability reports
- Nxtra by Airtel disclosures
- Yotta Data Services disclosures
- Li, Yang, Islam and Ren, "Making AI Less Thirsty"
- ISO/IEC 30134 data-centre KPI standards
- European Union Energy Efficiency Directive and Delegated Regulation 2024/1364
- Uptime Institute, Global Data Center Surveys and Tier Standard
- Ministry of Electronics and Information Technology (MeitY), Data Centre Policy draft and IndiaAI
- NTT DATA ESG data
- STT GDC India ESG report
- Sify Infinit Spaces sustainability report
- CtrlS disclosures
- AdaniConneX disclosures
- Council on Energy, Environment and Water (CEEW) studies
- S&P Global (451 Research) India data centre analysis

PART II · RESOURCES AND IMPACTS

Chapter 7
Water and energy calculated: parameters, formulas and worked examples

WHAT THIS CHAPTER COVERS

This chapter gathers the parameters and formulas needed to calculate a data centre's water and energy, and applies them to three worked examples: a 20 megawatt colocation hall in Mumbai with cooling towers, a 50 megawatt hyperscale facility in Hyderabad with adiabatic cooling, and a 100 megawatt AI campus in Bengaluru with direct-to-chip liquid cooling and dry coolers. It quantifies the water intensity of the Indian grid, compares cooling technologies, sets peak factors for sizing water connections and storage, models hour-by-hour water and power on a hot day, and reviews electricity tariffs and procurement options.

KEY TAKEAWAYS

1. Cooling technology decides a site's on-site water, but the grid decides its total water footprint, because Indian power stations evaporate about 3.0 litres for every kilowatt-hour delivered to a consumer's meter.
2. A dry-cooled 100 megawatt AI campus on the average Indian grid still carries about 1.5 billion litres a year of water at power stations, more than six times the on-site water of a 20 megawatt colocation hall with cooling towers in Mumbai.
3. Adiabatic cooling uses little water on average but uses the most in April and May, exactly when municipal supply is tightest, so water connections and storage must be sized for the peak day, which can be two to five times the average.
4. Exchange power prices in India fall to about ₹1 per kilowatt-hour around noon and rise to about ₹10 late in the evening, so running adiabatic coolers dry at midday saves water at little cost.
5. Buying round-the-clock renewable or group captive power reduces total water by 70 to 85% in the worked examples and is the cheapest option in two of the three.

How this chapter connects to the others. The definitions used here come from Chapters 5 and 6. Chapter 8 compares conventional and AI sites with the same method, and Chapter 11 optimises the power supply.

For an Indian data centre, cooling technology decides the on-site (direct) water bill, but the grid decides the total water footprint. A dry-cooled 100 MW AI campus on the average Indian grid still carries about 1.5 billion litres a year of off-site water, mostly at coal plants. That is more than six times the direct water of a 20 MW evaporative colo in Mumbai. So procurement (RE-RTC, group captive) matters more than cooling choice for total water. Cooling choice matters most for local peak water demand and for securing water permits.

7.1 How to read the labels

- **[Verified: source]** means a figure from a primary or authoritative document fetched or confirmed in this research. "Verified-secondary" means it came from a secondary summary of a primary document.
- **[Proxy: basis]** means an analogue, engineering rule or other-country range, adjusted for India.
- **[Model: formula]** means a value computed in this report.

Source codes used in labels: - LBNL24 — Shehabi et al., 2024 US Data Center Energy Usage Report: https://eta-publications.lbl.gov/sites/default/files/2024-12/lbnl-2024-united-states-data-center-energy-usage-report_1.pdf - UPT25 — Uptime Institute Global Data Center Survey 2025: https://datacenter.uptimeinstitute.com/rs/711-RIA-145/images/2025.Annual.Survey.Report.pdf?version=0 - GOOG25 — Elsworth et al., Measuring the environmental impact of delivering AI at Google Scale: https://arxiv.org/pdf/2508.15734 ; Google PUE: https://datacenters.google/efficiency/ - MSFT — Microsoft datacenter efficiency: https://datacenters.microsoft.com/sustainability/efficiency/ - LI — Li, Yang, Islam & Ren, Making AI Less "Thirsty": https://dl.acm.org/doi/10.1145/3724499 - NTT — NTT DATA ESG environment data: https://www.nttdata.com/global/en/about-us/sustainability/esg-data/environment-data - DTE — Down To Earth, India's digital thirst: https://www.downtoearth.org.in/science-technology/indias-digital-thirst-what-data-centre-giants-arent-saying-about-their-water-use - CEEW — Scaling India's Data Centre Ecosystem: https://www.ceew.in/sites/default/files/ceew-data-centre-study-web-ready-final.pdf - KADC — Karnataka Sustainable Data Centre Policy 2026–31 (TNM): https://www.thenewsminute.com/karnataka/karnatakas-new-data-centre-policy-puts-water-energy-in-focus-what-does-it-say - MOEF15 — MoEFCC S.O. 3305(E), 7 Dec 2015 (CPCB): https://cpcb.nic.in/uploads/Thermal_Power_Plant_overview.pdf ; amendments: https://powerline.net.in/2026/09/04/water-stress-thermal-power-plants-take-steps-to-optimise-consumption/ - SD23 — Vulnerability assessment of thermal power plants in India under water stress: https://www.sciencedirect.com/science/article/abs/pii/S0360544223009477 - WRI — WRI, India's thermal plants in water-scarce areas: https://www.wri.org/insights/40-indias-thermal-power-plants-are-water-scarce-areas-threatening-shutdowns - NREL — Macknick et al., operational water factors: https://docs.nrel.gov/docs/fy11osti/50900.pdf - MNRE25 — MNRE Renewable Energy Statistics 2024-25: https://cdnbbsr.s3waas.gov.in/s3716e1b8c6cd17b771da77391355749f3/uploads/2025/11/202511061627678782.pdf - CREA — FY25 generation: https://www.deccanherald.com/amp/story/india%2Fthermal-generation-growth-slowest-in-fy-2024-25-since-covid-3497645 - EU24 — Delegated Regulation (EU) 2024/1364: https://www.eudca.org/documents/content/_wU9NIdhRn2O7FEPP-YZ2Qo33?download=0 ; rating scheme: https://www.philiplee.ie/eu-publishes-draft-mandatory-sustainability-rating-scheme-for-data-centres/ - NERC — NERC/FERC large loads: https://www.ferc.gov/sites/default/files/2025-04/Presentation%20NERC%20Seeks%20to%20Address%20Reliability%20Impacts%20from%C2%A0Large%20Load%20Integration_1.pdf - IEX — https://www.saurenergy.com/solar-energy-news/iex-dam-prices-drop-nearly-18-yoy-despite-rising-demand-11176360 ; FY25: https://www.business-standard.com/amp/markets/capital-market-news/iex-records-highest-ever-electricity-traded-volumes-in-fy25-

125040400233_1.html ; intraday: https://earthsync.io/resources/iex-dam-gdam-rtm-explained - SECI — FDRE-RTC: https://www.pv-magazine.com/2026/08/10/secis-1-gw-fdre-rtc-power-tender-discovers-inr-5-25-kwh-tariff/ ; FDRE-IX: https://www.pv-magazine.com/2026/08/24/seci-discovers-inr-5-99-kwh-tariff-for-4-hour-peak-renewable-power/ - ISTS — MoP ISTS waiver phase-out: https://www.tndindia.com/waiver-of-ists-charges-on-renewable-energy-to-be-phased-out/ - MERC — MSEDCL MYT press note: https://merc.gov.in/wp-content/uploads/2025/03/Press-Note_MSEDCL-MYT-Order_English.pdf ; TPC-D: https://merc.gov.in/wp-content/uploads/2025/03/Press-Note_MYT-Order_TPC_English.pdf - TGERC — TGSPDCL FY26 tariff schedule: https://tgsouthernpower.org/resources/PDF/Tariffs/63tarifffile.pdf - KERC — KERC Tariff Order 2025 schedule: https://hrecs.co.in/app/documents/TariffRates2026-27-28.pdf - TNERC — TN Tariff Order 6/2025 summary: https://www.ecloudsenergy.com/news-details.php?nid=52 - GEOA — GEOA Rules 2022 gazette: https://www.acma.in/uploads/ciculer-attachement/Electricity%20(Promoting%20Renewable%20Energy%20through%20Green%20Energy%20Open%20Access)%20Rules,2022.pdf ; FOR model: https://forumofregulators.gov.in/Data/Reports/Revised%20Model%20regulation%20GEOA.pdf - NVDA — NVIDIA DGX B200 guide: https://docs.nvidia.com/dgx/dgxb200-user-guide/introduction-to-dgxb200.html ; GB200 NVL72: https://pantheon.run/learn/nvidia-gb200-nvl72-power-and-cooling

7.2 Key Findings

1. **Direct water is a cooling-design choice, and the gap between designs is about 50×.** Year-round evaporative cooling (chillers with open towers) puts Indian sites at about 1.6–2.5 L/kWh. NTT's evaporative-only sites reported 2.93, 2.77 and 1.42 L/kWh in FY2022–24 [Verified: NTT]. Air-side economiser with adiabatic assist or IEC brings that to 0.2–0.6 L/kWh, and closed-loop liquid cooling with dry coolers to below 0.05 L/kWh. Microsoft's Asia-Pacific WUE jumped from 0.03 L/kWh (FY24) to 0.25 L/kWh (FY25), which shows how quickly regional numbers move as new sites enter the fleet [Verified: MSFT].
2. **Indirect water is 2–60× the direct water, and Indian disclosures ignore it.** LBNL estimates that the roughly 176 TWh used by US data centres in 2023 consumed "nearly 800 billion liters" (about 211 billion gallons) at power plants, or 4.52 L/kWh, against 66 billion litres direct [Verified: LBNL24]. India's coal-heavy grid (thermal was about 75% of the FY25 generation of 1,824 BU) gives a modelled 2.3–4.4 L/kWh at the consumer meter [Model].
3. **Peak water, not annual water, is the binding constraint for adiabatic designs.** A Hyderabad adiabatic hyperscale site draws about 2.1× its annual-average daily water on a May design day, and about 3.7× its average hourly rate at 14:00–15:00 [Model, Example B]. Tower-based Mumbai colos peak at only about 1.5× [Model, Example A]. Size supply connections and storage to the peak-day figure.
4. **Water peaks at solar noon, when power is cheapest; electricity is dearest in the evening.** June 2026 IEX averages bottomed at ₹1.17/kWh at 11:00 and peaked at ₹9.90/kWh at 22:30 [Verified-secondary: IEX/earthsync]. Running chillers "dry" during midday water peaks trades cheap kWh for scarce litres. This is the core co-optimisation for Indian sites.
5. **India's disclosure base is thin.** DTE found that Nxtra and CtrlS publish no WUE, and CtrlS reported only total water (409,037 kL in 2023-24) [Verified: DTE]. Karnataka's 2026–31 policy is the first to require plant-wise PUE and WUE reporting, with a water-charge reimbursement for a WUE of 0.7–1.3 L/kWh [Verified: KADC].
6. **Firm RE is now priced near or below coal.** SECI's FDRE-RTC tender cleared at ₹5.25/kWh, and 4-hour assured peak power at ₹5.99/kWh [Verified: SECI]. Coal PPAs signed in 2025 started at ₹5.38–6.30/kWh [Verified-secondary: IECC via SaurEnergy]. ISTS-charge waivers fall to zero for projects commissioned after July 2028 [Verified: ISTS]. For a data centre, landed cost depends on state surcharges more than on generation price.

7.3 Direct (on-site) water

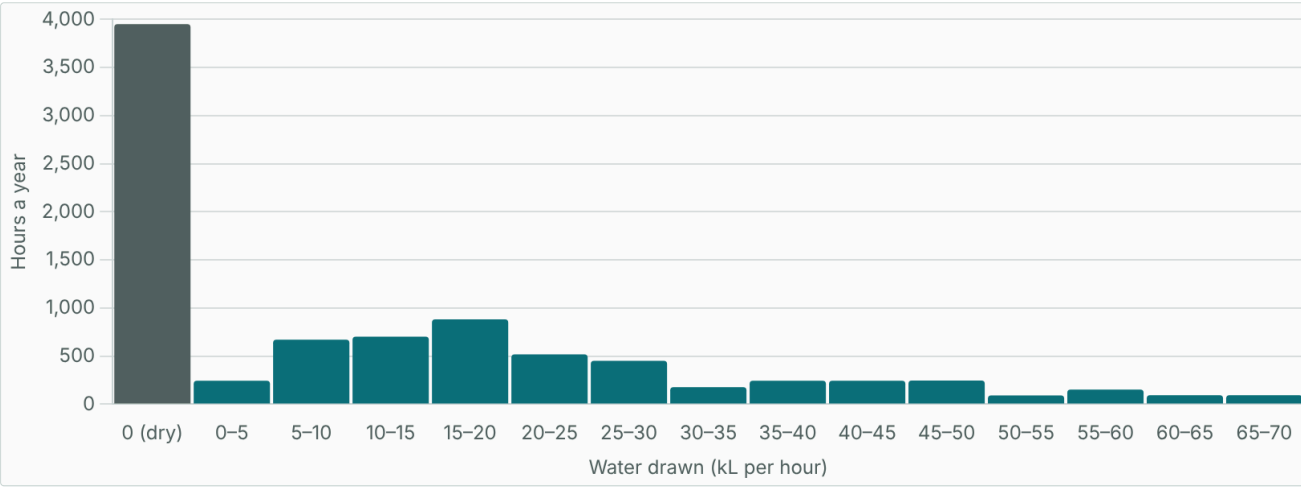


Figure 7.1 All 8,760 hours of a year. The same adiabatic site, run through a whole year of typical temperatures. The histogram counts how many hours fall at each rate of water draw; the grid shows when they happen. *Source: India Data Centre Atlas, chapter built on this chapter's data.*

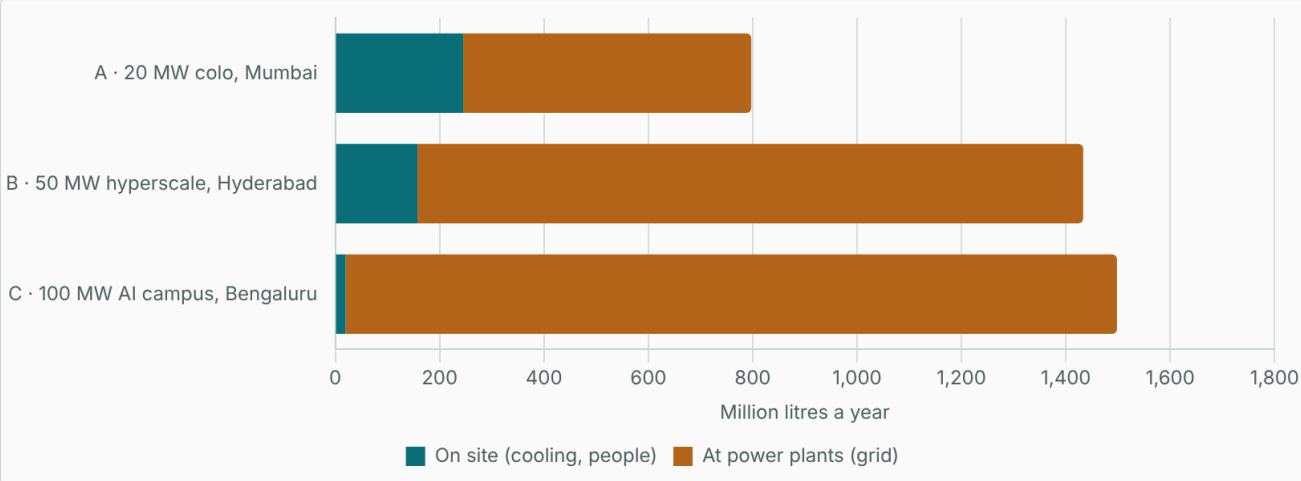


Figure 7.2 Where the water goes. Annual water for the three Report 12 worked examples: on site versus at the power plants that supply them. *Source: India Data Centre Atlas, chapter built on this chapter's data.*

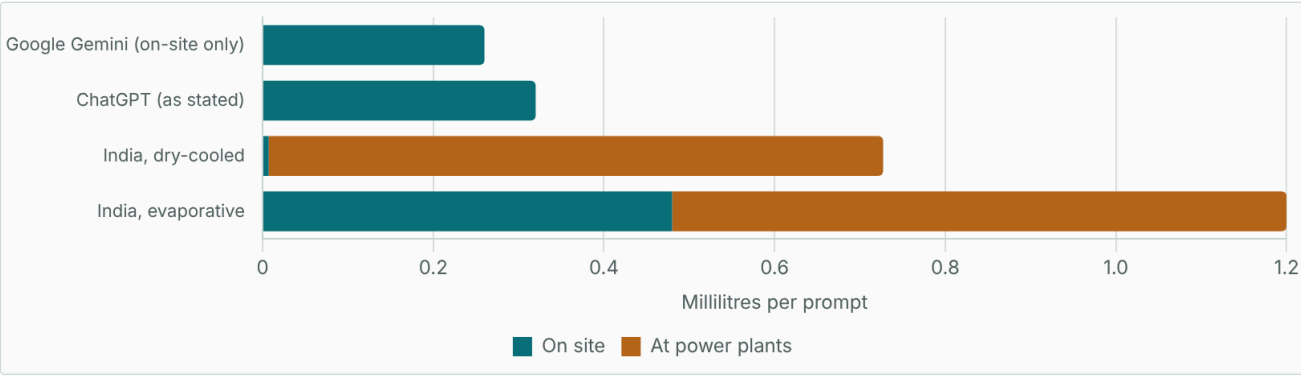


Figure 7.3 Water behind one AI prompt. Millilitres per typical text prompt. Company figures count on-site water only; in India the power plants add about three times as much. *Source: India Data Centre Atlas, chapter built on this chapter's data.*

7.3.1 Definitions and metrics

Table 7.1 Definitions and metrics

Term	Definition	Unit	Label
Withdrawal	Water taken from any source (municipal, groundwater, tanker, STP-treated)	kL, m^3	[Verified: LBNL24; WRI]
Consumption	Withdrawal minus discharge; mostly evaporation and drift	kL	[Verified: LBNL24 — "permanently removed from the immediate water cycle"]
Discharge	Blowdown, RO reject, sanitary sewage returned	kL	[Proxy: water-balance convention]
WUE (site)	Annual site water ÷ annual IT energy. Microsoft: "annual liters of water used for humidification and cooling" ÷ IT kWh	L/kWh_IT	[Verified: MSFT; LBNL24]
WUE (source)	Site WUE plus water used to generate the electricity consumed	L/kWh_IT	[Verified: LBNL24 — "source WUE accounts for the water required to generate the electricity"]
Basis caveat	NTT computes WUE as withdrawal ÷ IT energy; Microsoft's 0.30 is described as a "withdrawal WUE"	—	[Verified: NTT; MSFT blog]
ISO/IEC 30134-9	International WUE standard (site and source categories)	—	[Proxy: text not reviewed — obtain from ISO/BIS]
EU EED reporting (Delegated Reg. 2024/1364)	Sites ≥500 kW IT report total water input (W_IN, m^3) and potable water input (W_IN-POT, m^3); first returns due 15 Sep 2024, then 15 May 2025 and yearly	m^3	[Verified: EU24]
EU rating scheme (draft)	A–G energy and water labels auto-generated from 15 Aug 2027; water grade from freshwater intake ÷ IT energy	—	[Verified-secondary: EU24]

Calculator recommendation: store withdrawal, consumption and discharge separately; compute WUE_withdrawal and WUE_consumption; keep potable and non-potable apart (as the EU requires). Indian operator numbers may be on either basis, so never compare them without checking.

7.3.2 Components and physics

Table 7.2 Components and physics

Component	Formula / rule	Low	Typical	High	Unit	Label
Latent heat of vaporisation	h_fg	2,257 (100 °C)	2,430 (30 °C)	2,450 (20 °C)	kJ/kg	[Proxy: steam tables]
Pure-evaporation water per kWh of heat	3,600 ÷ h_fg	1.47	1.48	1.60	L/kWh_th	[Model]
Latent share of tower heat rejection	f_lat	0.75	0.85	0.95	—	[Proxy: tower practice]

Component	Formula / rule	Low	Typical	High	Unit	Label
Evaporation	E = Q_rej × f_lat × 3,600/h_fg	1.1	1.26	1.5	L/kWh_th	[Model]
Heat rejected per kWh IT (water-cooled chillers)	1 + compressor + pump/fan heat	1.15	1.25	1.35	kWh_th/kWh_IT	[Proxy: chiller COP 5–7]
Blowdown	B = E ÷ (CoC − 1)	E/9 (CoC 10)	E/3 (CoC 4)	E/1 (CoC 2, hard water)	L/kWh_th	[Model]
Drift	0.001–0.005% of circulation (~86 L/kWh_th at ΔT 10 K)	0.0009	0.002	0.004	L/kWh_th	[Model: 3,600/(4.19×10)]
Tower make-up	M = E + B + D	1.3	1.7	3.0	L/kWh_th	[Model]
Tower-based WUE	M × Q_rej/IT	1.5	2.1	3.5	L/kWh_IT	[Model]; NTT evaporative 1.42–2.93 [Verified: NTT]
Humidification (air-cooled halls)	—	0	0.01	0.05	L/kWh_IT	[Proxy: LBNL24 "occasional humidification"]
RO reject	feed × (1 − recovery); recovery 65–80%	0.25	0.4	0.55	L per L permeate	[Proxy]
Domestic/sanitary	45 L/person/day × staff	0.005	0.01	0.02	L/kWh_IT at >10 MW	[Proxy: NBC/CPHEEO norms]
Landscaping	2–5 L/m²/day of green area	—	—	—	—	[Proxy; use STP water]
DG radiator cooling	Closed loop	0	~0	~0	—	[Proxy]
Fire systems	One-time fill plus tests; storage per NBC/TAC	—	—	—	kL storage	[Proxy: NBC Part 4]

What this means: the "1.8 L/kWh" figure often quoted fits towers at a CoC of about 4 in a hot-humid climate. Raising CoC from 3 to 8 (softening or RO make-up) cuts make-up by about 25–35%. That is the cheapest retrofit lever for existing tower sites in Mumbai and Chennai.

7.3.3 Published WUE and water values

Table 7.3 Published WUE and water values

Entity / scope	Value	Year	Label
US all data centres, average site WUE	~0.36 L/kWh through 2023; 0.45–0.48 after 2023	2023 / 2028	[Verified: LBNL24]
US direct water consumption	66 billion L (2023); 21.2 billion L (2014); hyperscale 60–124 billion L in 2028	—	[Verified: LBNL24]
US hyperscale aggregate median WUE	0.32 L/kWh (0.40 if adiabatic runs at 0.2)	2023	[Verified: LBNL24]
Hyperscale adiabatic systems (self-reported)	0.1–0.3 L/kWh	—	[Verified: LBNL24]
Microsoft global WUE / PUE	0.30 / 1.16 (FY24); 0.27 / 1.17 (FY25)	FY24–25	[Verified: MSFT]
Microsoft Asia-Pacific WUE / PUE	0.03 / 1.25 (FY24); 0.25 / 1.28 (FY25)	FY24–25	[Verified: MSFT]
Microsoft zero-water-for-cooling design	Avoids >125 million L/yr per datacenter	from Aug 2024	[Verified: MSFT blog]
Microsoft India sites	On-site WUE 0.000 L/kWh (dry cooling) in Li et al.'s table	—	[Verified: LI]
Google fleet PUE	1.09 (2025)	2025	[Verified: GOOG25]
Google WUE used in Gemini analysis	1.15 L/kWh (fleet)	2024	[Verified-secondary: MDPI reconstruction of GOOG25]
NTT DATA WUE, whole portfolio	1.01 / 0.65 / 0.72 L/kWh	FY22 / 23 / 24	[Verified: NTT]
NTT DATA WUE, evaporative sites	2.93 / 2.77 / 1.42 L/kWh	FY22 / 23 / 24	[Verified: NTT]
NTT DATA PUE	1.40 / 1.38 / 1.38	FY22 / 23 / 24	[Verified: NTT]
Equinix global / evaporative sites WUE	0.91 / 1.41 L/kWh	2025	[Verified-secondary: Equinix 2025 report via Wikipedia]
CtrlS total water	409,037 kL (2023-24) vs 468,330 kL (2022-23); no WUE	—	[Verified: DTE]
Nxtra, CtrlS	No published WUE	2025	[Verified: DTE]
Yotta	Claims "extremely low WUE", no number	2025	[Verified: DTE]
India sector	DCs ≈ 0.5% of national electricity (2025); ~271 DCs (Jan 2026); a 100 MW hyperscale "can take ~2 million litres of water per day"	2025–26	[Verified: CEEW]
India sector share of water	~0.02–0.03% of national water demand	2025	[Verified-secondary: CEEW author in Scroll]
India siting	~75% of DCs in water-stressed regions; ~25 million L per MW per year	2026	[Verified-secondary: DTE Apr 2026 via Substack]

Interpretation: CEEW's 2 million L/day for 100 MW works out to 0.83 L/kWh at full load. That fits a hybrid/adiabatic design, not year-round towers (about 5 million L/day). DTE's 25 million L/MW/yr implies about 2.9 L/kWh, high-end tower territory. Treat both as screening values, not design values.

7.3.4 Water per AI query

Table 7.4 Water per AI query

Estimate	Energy	Water	Scope	Label
Google Gemini median text prompt (May 2025)	0.24 Wh	0.26 mL (0.15 mL narrow method); 0.03 gCO_2e	On-site only	[Verified: GOOG25]
OpenAI average ChatGPT query (Altman, Jun 2025)	0.34 Wh	~0.32 mL	Undisclosed method	[Verified-secondary: Towards Data Science]

Estimate	Energy	Water	Scope	Label
Li et al., GPT-3 medium request, India	—	19.7 mL (0 on-site + 19.7 off-site); 25.4 requests per 500 mL; PUE 1.430, EWIF 3.445 L/kWh	On-site + grid	[Verified: LI]
Gemini-like prompt, dry-cooled Indian DC	0.24 Wh	0.24×10^{-3} kWh × (0.03 + 3.0) = 0.73 mL	On-site + grid	[Model]
Same, evaporative Indian DC	0.24 Wh	0.24×10^{-3} × (2.0 + 3.0) = 1.2 mL	On-site + grid	[Model]

Interpretation: for India the grid term (~0.72 mL) is about three times Google's whole on-site figure. Per-query claims that leave out electricity understate India's footprint by 70–98%.

7.4 Indirect (off-site) water

7.4.1 Water intensity of generation

Table 7.5 Water intensity of generation

Technology	Consumption	Withdrawal	Label
Indian coal, once-through (legacy)	small	70–200 m³/MWh (CSE); 80–160 m³/MWh	[Verified: PowerLine/CSE; SD23]
Indian coal, closed-loop towers (actual)	2.8–3.4 m³/MWh, about twice the world average of 1.2–1.5	~4 m³/MWh average (CSE)	[Verified: SD23; PowerLine/CSE]
MoEFCC norm, existing CT plants	≤3.5 m³/MWh (by 7 Dec 2017)	—	[Verified: MOEF15]
MoEFCC norm, plants after 1 Jan 2017	2.5 m³/MWh with ZLD (2015), relaxed to 3.0 m³/MWh in June 2018	—	[Verified: MOEF15 amendments]
FGD additional consumption	~0.3 m³/MWh	—	[Verified: PowerLine/CSE]
Coal with air-cooled condenser	0.45–0.65 m³/MWh	—	[Verified: SD23]
Nuclear with towers (US median)	672 gal/MWh = 2.54 L/kWh	higher than coal	[Verified: NREL]
Tower-cooled thermal, upper bound	~1,000 gal/MWh ≈ 3.8 L/kWh	—	[Verified: NREL]
Gas CCGT with tower	~0.7–0.8 L/kWh	~1 L/kWh	[Proxy: NREL medians via secondary sources]
Hydro, reservoir evaporation (contested)	~17 L/kWh gross; NREL excludes hydro "due to the wide range of estimates"	—	[Verified-secondary: NREL caption + citing literature]
Solar PV (cleaning)	<0.19 L/kWh; Indian dry robots lower	—	[Verified-secondary: NREL-derived]
Wind	~0	~0	[Verified: NREL]
Coastal seawater once-through plants	Freshwater ≈ 0	Seawater 70–200 L/kWh	[Proxy: per plant]

Context: WRI (Tianyi Luo, 16 Jan 2018) found that "40 percent of the country's thermal power plants are located in areas facing high water stress" and that "70 percent of India's thermal power plants will face high water stress by 2030"; about 90% of thermal generation depends on freshwater, and "14 of India's 20 largest thermal utilities experienced at least one shutdown due to water shortages between 2013-2016, costing the companies $1.4 billion" [Verified: WRI].

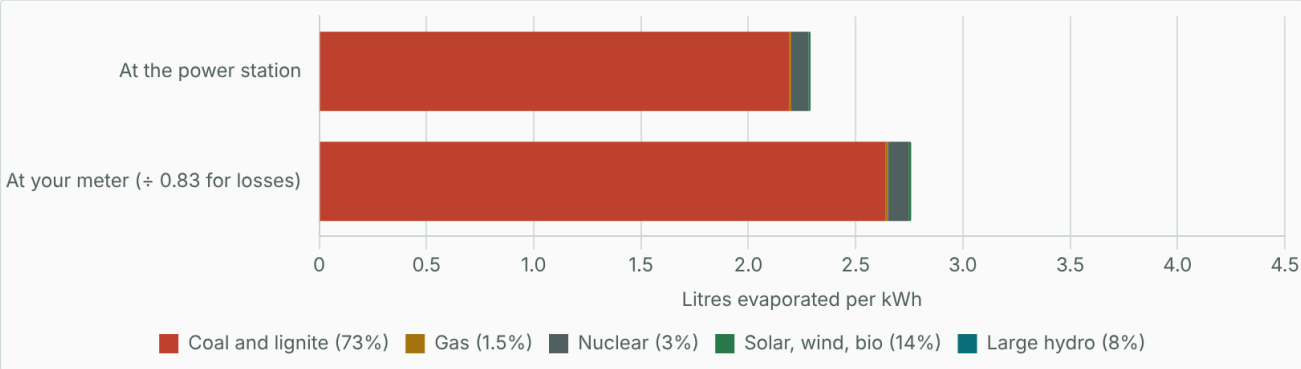


Figure 7.4 Where grid water comes from, by generating source. Litres evaporated per kWh for India's FY2025 generation mix. *Source: India Data Centre Atlas, chapter built on this chapter's data.*

7.4.2 Indian grid-average water intensity

Generation base for FY2024-25: total 1,824 BU, thermal 1,363 BU (74.7%) [Verified: CREA], and RE 403.64 BU (22.13%, which appears to include large hydro) [Verified: MNRE25]. Nuclear is the residual, about 3%; large hydro about 8% [Proxy: residual allocation].

Table 7.6 Indian grid-average water intensity

Source	Share	Consumption intensity (L/kWh)	Contribution (L/kWh generated)
Coal and lignite	73%	3.0 (2.8–3.4)	2.19
Gas	1.5%	0.7	0.01
Nuclear	3%	2.5	0.08
Solar, wind, bio	14%	0.1	0.01
Large hydro	8%	0 (operational) – 17 (gross evaporation)	0 – 1.36
Total at busbar	100%	—	**2.29 – 3.65**

Source	Share	Consumption intensity (L/kWh)	Contribution (L/kWh generated)
At consumer meter (÷ (1 − 17% losses) [Proxy])	—	—	**2.76 – 4.40**

Default for calculators: 3.0 L/kWh consumption (range 2.3–4.4). It sits inside Li et al.'s 3.445 L/kWh for India [Verified: LI] and below LBNL's US data-centre figure of 4.52 [Verified: LBNL24]. Withdrawal intensity for closed-loop coal is about 4 L/kWh.

State proxies (replace with SLDC/CEA mix and plant water data):

Table 7.7 Indian grid-average water intensity

State	Grid water proxy (L/kWh delivered)	Basis	Label
Karnataka	1.5–2.2	RE was 48.10% of state generation in 2024-25	[Verified: MNRE25 share]; [Proxy: intensity]
Tamil Nadu	1.8–2.6	High wind/solar; seawater-cooled coastal plants	[Proxy]
Andhra Pradesh	2.0–3.0	Coastal seawater plus inland coal	[Proxy]
Maharashtra	2.8–3.4	Coal-heavy MSEDCL portfolio	[Proxy]
Telangana	2.8–3.4	Coal-heavy	[Proxy]
Uttar Pradesh	3.0–3.6	Coal-heavy, inland freshwater	[Proxy]

Electrons come from the regional grid, not the state fleet. For contracted RE, use the contract mix; LBNL notes PPAs "could significantly affect water consumption" [Verified: LBNL24].

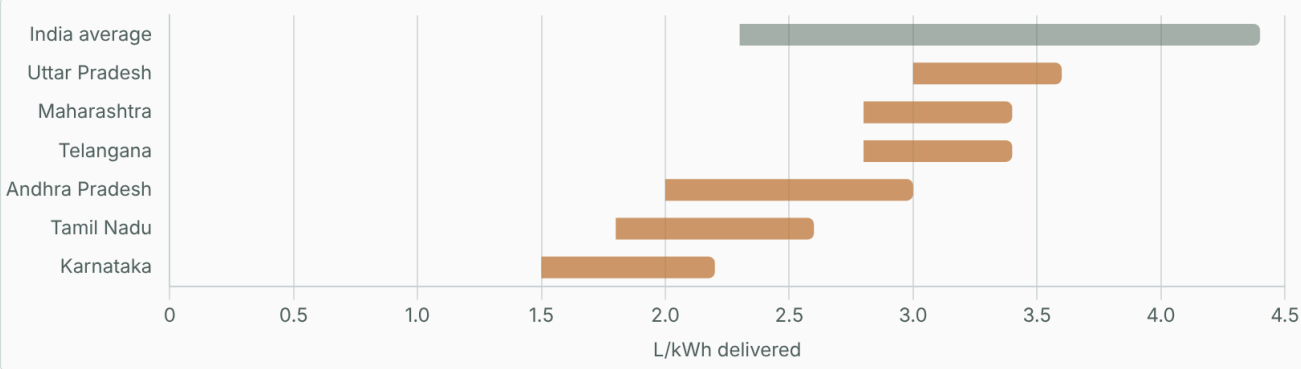


Figure 7.5 Grid water intensity by state. Range of litres evaporated per kWh delivered, by state. *Source: India Data Centre Atlas, chapter built on this chapter's data.*

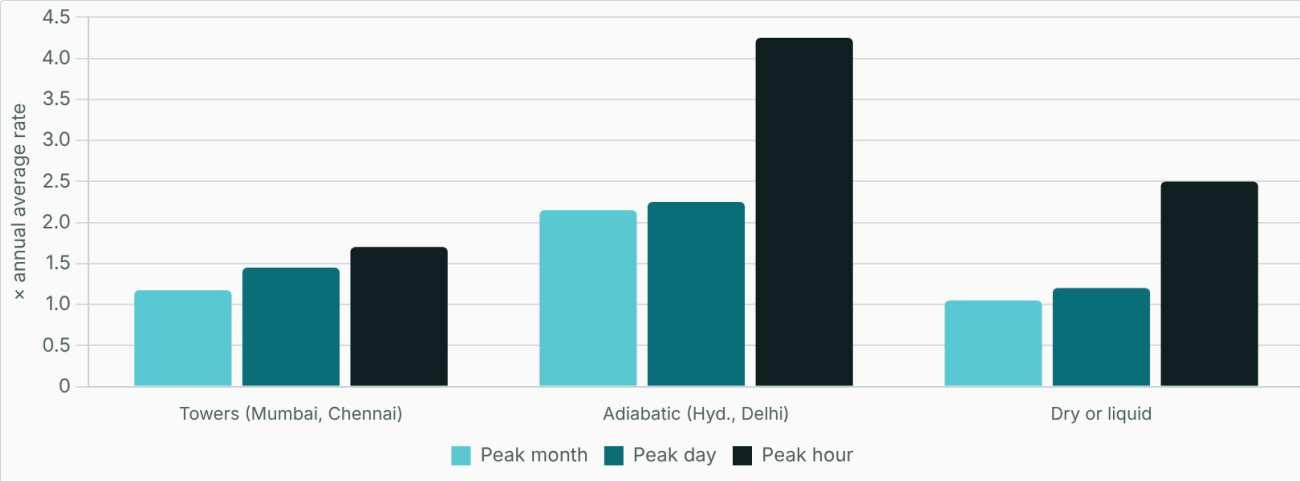


Figure 7.6 Peaks size the pipes. Peak month, peak day and peak hour as multiples of the annual average water rate, by cooling design. *Source: India Data Centre Atlas, chapter built on this chapter's data.*

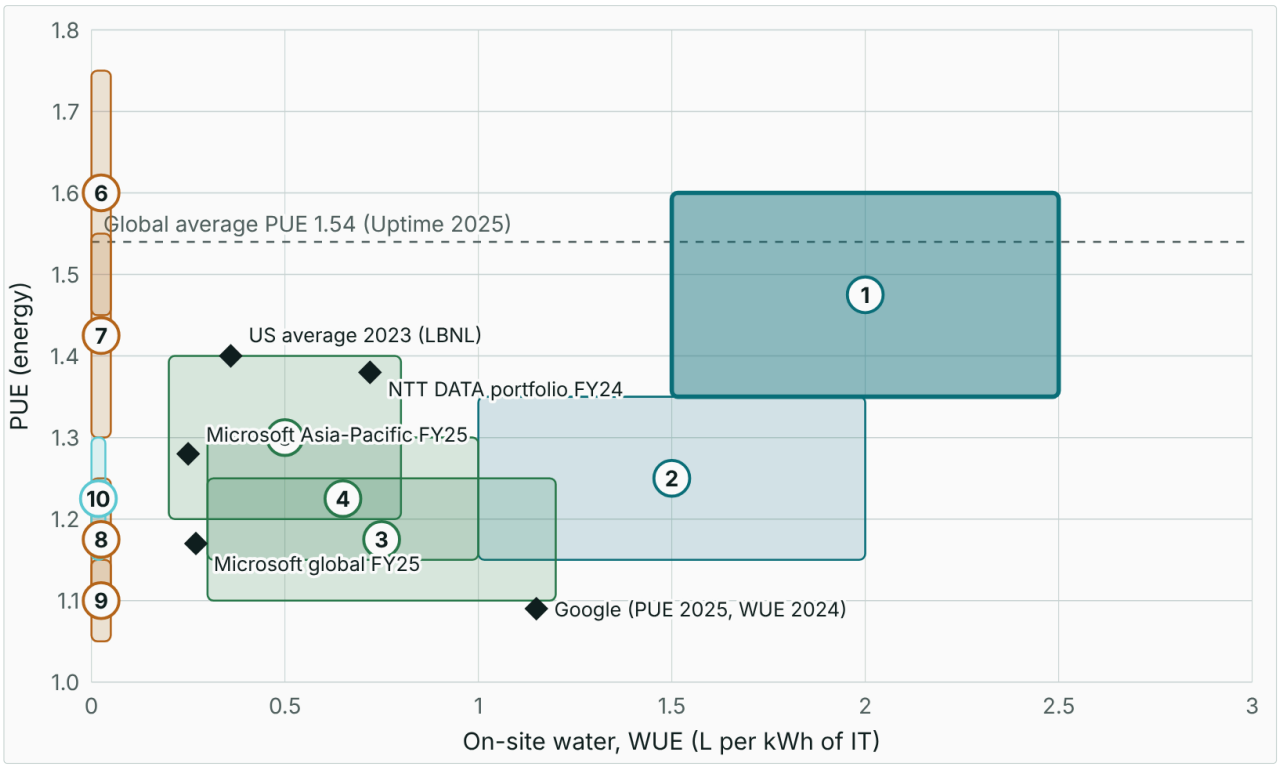


Figure 7.7 Power versus water: every cooling technology. Each box spans the typical annual PUE and on-site water (WUE) range in Indian climates. Dots are published fleet figures. Switch on the grid's water to see where each design really lands. *Source: India Data Centre Atlas, chapter built on this chapter's data.*

7.4.3 Source WUE formula

- Source WUE (L/kWh_IT) = WUE_site + PUE × EWIF_grid [Model; structure per LBNL24/LI]
- Total annual water (L) = E_IT (kWh) × Source WUE
- Contracted mix: EWIF_eff = Σ (share_i × EWIF_i); RE-RTC (solar + wind + BESS) ≈ 0.05–0.15 L/kWh [Proxy: NREL]

Table 7.8 Source WUE formula

Case	WUE_site	PUE	EWIF	Source WUE
Mumbai evaporative colo, grid	2.0	1.50	3.0	6.5
Hyderabad adiabatic hyperscale, grid	0.45	1.30	2.8	4.09
Bengaluru DLC dry-cooled AI, grid	0.03	1.25	1.8	2.28
Same, 90% RE-RTC (0.1) + 10% grid (1.8) → EWIF 0.27	0.03	1.25	0.27	0.37

7.4.4 Embedded water (scope 3), order of magnitude

Table 7.9 Embedded water (scope 3), order of magnitude

Item	Value	Label
Global DC water 2023 (IEA): 560 billion L = 373 indirect + 140 direct + 47 hardware manufacturing	—	[Verified-secondary: IEA via gpusmith]
Hardware manufacturing share	~8%	[Model: 47/560]
Concrete for a 100 MW campus	~100,000–200,000 m³ × ~150–200 L/m³ ≈ 15–40 ML, one-time	[Proxy: construction norms]
Takeaway	Embedded water ≈ 1–3 years of a dry-cooled site's direct water, but under 5% of its grid water	[Model]

7.4.5 Scarcity weighting

- Scarcity-weighted water = Σ (water_i × CF_i), with CF from AWARE (0.1–100) or the WRI Aqueduct baseline water-stress class [Proxy: method].
- Most Indian metros (Chennai, Bengaluru, Delhi, Hyderabad, Mumbai) fall in Aqueduct "high" to "extremely high" baseline stress [Proxy: pull per catchment from Aqueduct 4.0].
- CGWB's stage of groundwater extraction (>100% means over-exploited) should gate any bore-well use [Proxy: pull block-level data].

7.5 Water by cooling technology

Annual averages for Indian climates [Proxy: engineering ranges cross-checked against LBNL24's ranking and operator data where noted].

Table 7.10 Water by cooling technology

Technology	WUE site (L/kWh_IT)	Annual PUE (India)	Energy penalty	Climate driver	Capex	Maturity
Water-cooled chillers + open towers	1.5–2.5 (to 3.5 at low CoC)	1.35–1.60	baseline	Wet-bulb (coastal design WB ~28–29 °C)	baseline	Mature; highest WUE per LBNL24
Waterside economiser + towers (liquid IT)	1.0–2.0	1.15–1.35	lower	Wet-bulb	+0–10%	Mature; LBNL: "consumes substantial amounts of water"
Air-cooled chillers	~0–0.05	1.45–1.75	+8–15%	Dry-bulb; 45 °C+ Delhi peaks derate	−5 to −10%	Mature; LBNL: "use no water, they use more energy"
Dry coolers (no adiabatic) + chiller trim	~0–0.05	1.30–1.55 air IT; 1.10–1.25 liquid IT	+ air IT; ~0 W32–W45	Dry-bulb	+5–15%	Mature
Adiabatic / hybrid dry coolers	0.2–0.8	1.20–1.40	small	Wet mode above ~26–30 °C dry-bulb	+10–20%	Mature
Direct evaporative (air-side)	0.3–1.2	1.10–1.25	low	Needs low WB and clean air	+0–10%	Limited Indian fit
Indirect evaporative (IEC)	0.3–1.0	1.15–1.30	low	WB; no outside air in hall	+15–30%	Mature
Rear-door heat exchangers	per heat-rejection plant	1.20–1.40	—	Facility water temperature	+per rack	Mature to 40–70 kW/rack
Direct-to-chip single-phase + dry coolers	0–0.05	1.10–1.25	low	Dry-bulb vs W32–W45	+15–30%	Mainstream for GB200 (NVL72 ~120 kW/rack) [Verified: NVDA]
Direct-to-chip two-phase	0–0.05	1.08–1.20	low	Dry-bulb	Higher	Early commercial
Immersion single-phase	0–0.05	1.05–1.15	lowest fans	Dry-bulb	+20–40%	Growing
Immersion two-phase	0–0.05	1.03–1.10	lowest	Dry-bulb	Highest; fluorocarbon risk	Niche
Seawater / lake once-through	~0 freshwater	1.15–1.30	low	Sea 27–31 °C off India	High (CRZ, intake/outfall)	CEEW: coastal hyperscalers "leverage seawater cooling" [Verified: CEEW]
Air-side economiser (free cooling)	~0	1.15–1.30 in Bengaluru/Pune winters	—	Dry-bulb below ~24–27 °C	Low	Limited hours

Hours of adiabatic/IEC wet mode per year (above ~28 °C dry-bulb) [Proxy: climatology; replace with ISHRAE TMY]:

Table 7.11 Water by cooling technology

City	Wet-mode hours/yr	Tower design WB (0.4%)	Notes
Bengaluru	1,000–1,800	~23–24 °C	Best Indian climate for dry/adiabatic
Pune	1,800–2,800	~24–25 °C	Good
Hyderabad	3,000–4,000	~25–26 °C	Dry heat suits adiabatic; spiky April–May
Delhi NCR	3,000–3,800	~28 °C	45 °C+ peaks; humid monsoon
Mumbai	3,500–5,000	~28 °C	Humidity cuts adiabatic gain
Chennai	5,000–6,500	~28–29 °C	STP/seawater key
Kolkata	3,500–4,500	~29 °C	Hot-humid
Visakhapatnam	4,500–6,000	~28–29 °C	Seawater option

Decision rule: in hot-humid coastal cities, adiabatic gains are small because wet-bulb depression is small. Choose towers on STP/recycled water at high CoC, or liquid cooling with dry coolers. In Hyderabad, Bengaluru and Pune, adiabatic or dry designs are the right default.

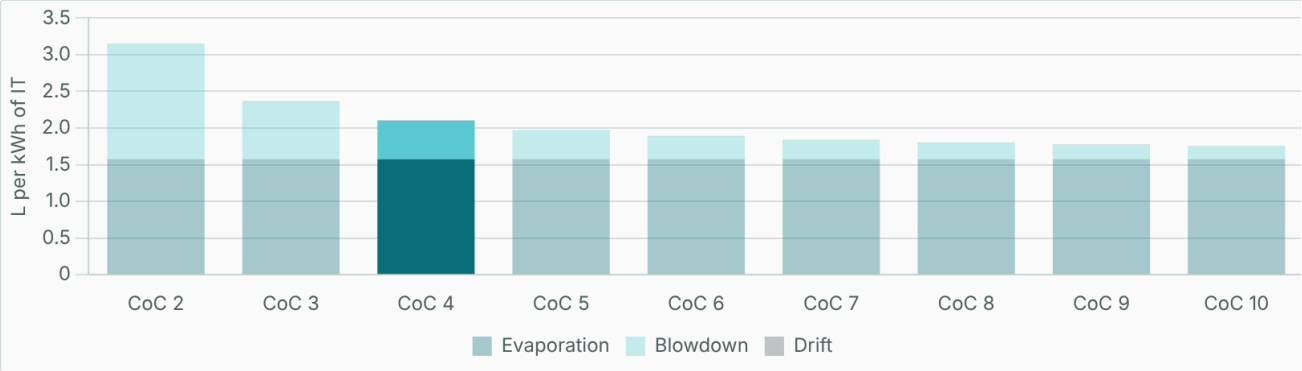


Figure 7.8 Inside a cooling tower. Where each litre of make-up water goes. Raising cycles of concentration (cleaner water, softening or RO) is the cheapest saving at existing tower sites. *Source: India Data Centre Atlas, chapter built on this chapter's data.*

7.6 Peak water demand

7.6.1 Peak factors (calculator inputs)

Table 7.12 Peak factors (calculator inputs)

System	Peak-month / avg	Peak-day / avg	Peak-hour / avg hour	Peak season	Label
Towers, Mumbai/Chennai	1.10–1.25	1.3–1.6	1.5–1.9	Apr–May; Oct	[Model: Example A]
Adiabatic/hybrid, Hyderabad/Delhi	1.8–2.5	2.0–2.5	3.5–5.0	Apr–Jun	[Model: Example B]
Adiabatic, monsoon months	0.3–0.6	—	—	Jul–Sep	[Model]

System	Peak-month / avg	Peak-day / avg	Peak-hour / avg hour	Peak season	Label
Dry/DLC (domestic only)	1.0–1.1	1.1–1.3	2–3	—	[Proxy]

7.6.2 Storage and connection sizing

- Supply connection (kL/day) = peak-day water × 1.1 [Model]
- Make-up storage (kL) = peak-hour water × storage hours [Model]
- Common practice is 12–72 h of make-up storage, with 24–48 h typical for Tier III designs [Proxy: industry practice]. Uptime Tier certification asks for on-site make-up water and fuel autonomy aligned to design, often cited as ≥12 h [Proxy: verify against Tier Standard: Topology].
- In India, add tanker contingency and STP-treated water contracts. Karnataka's policy focuses on treated water to reduce freshwater dependence [Verified: KADC].

7.6.3 Permit and allocation examples

We found no verified Indian EIA or water-allocation figures for named data-centre projects (Navi Mumbai, Chennai, Hyderabad, Visakhapatnam, Greater Noida, Hoskote). The screening values available are:

- CEEW's ~2 million L/day per 100 MW hyperscale [Verified: CEEW]
- Microsoft's >125 million L/yr avoided per zero-water datacenter, about 342 kL/day [Verified: MSFT blog; Model: ÷365]

US, UK and Singapore permit examples are left for follow-up (see Data gaps).

7.7 Hourly water and electricity profiles

7.7.1 IT load profile archetypes

Table 7.13 IT load profile archetypes

Archetype	Avg / design IT	Diurnal swing	Sub-minute behaviour	Label
Enterprise/colo	60–80% (lease fill); colo server utilisation 21% (2014) → 35% (2027)	±3–8%	Smooth	[Verified: LBNL24]; [Proxy: fill]
Hyperscale cloud	70–85%; utilisation 45% → 50%	±5–10%	Smooth	[Verified: LBNL24]; [Proxy]
AI training	80% operational time at ~70% of rated; saturated 8×H100 node 7.9 kW (~78%)	Flat	"Square-wave" drops at synchronisation	[Verified: LBNL24]
AI inference	40% operational time; power half of training	Diurnal	Bursty	[Verified: LBNL24]
Idle floor	AI ~20% of rated; conventional 36% of max (2023) → 27% (2028)	—	—	[Verified: LBNL24]

Grid risk: on 10 July 2024 a 230 kV fault led to about 1,500 MW of data-centre load disconnecting across 60 load points and 25 substations in Virginia, and a 1,800 MW load loss followed in February 2025 [Verified: NERC]. About 1,260 MW did not return for hours [Verified-secondary: POWER]. NERC's July 2025 white paper documents a site dropping from about 450 MW to 40 MW in 36 seconds [Verified-secondary: Keentel]. For India, specify UPS/BESS ride-through and ramp-rate limits in connection agreements for loads above 100 MW.

7.7.2 Hourly PUE and WUE functions

- PUE(T_db) = PUE_base + k_P × max(0, T_db − T_0); adiabatic defaults PUE_base 1.22, k_P 0.015/K, T_0 25 °C [Model]
- WUE(T_db) = 0 below 26 °C; above it, 0.1 + 0.107 × (T_db − 26), capped at 1.6 L/kWh [Model]
- Towers: WUE(t) ≈ [1 + 1/(CoC − 1)] × f_lat × 3,600/h_fg × Q_rej(t)/IT(t), nearly flat at 1.9–2.3 L/kWh [Model]
- Water(t) = IT(t) × WUE(t); electricity(t) = IT(t) × PUE(t)

7.7.3 Hot-day profile: 50 MW hyperscale, Hyderabad, adiabatic (Example B), May design day

IT is held at a flat 42 MW [Proxy]. Temperatures follow a sinusoid from 27 °C at 05:00 to 40 °C at 15:00 [Proxy: replace with ISHRAE TMY]. IEX prices are shaped to June 2026 averages: trough ₹1.17 at 11:00, peak ₹9.90 at 22:30, 08–16 h average ₹1.63, 18–23 h average ₹8.10 [Verified-secondary: earthsync; Proxy: shape]. Shown every two hours; totals come from the full 24-hour model.

Table 7.14 Hot-day profile: 50 MW hyperscale, Hyderabad, adiabatic (Example B), May design day

Hour	T_db (°C)	PUE	Facility MW	WUE (L/kWh_IT)	Water (kL/h)	IEX DAM proxy (₹/kWh)
00	31	1.31	55.0	0.64	26.9	6.0
02	29	1.28	53.8	0.42	17.6	4.0
04	27.5	1.26	52.8	0.26	10.9	3.8
06	28	1.27	53.1	0.31	13.0	4.5
08	32	1.33	55.7	0.74	31.1	2.2
10	36	1.39	58.2	1.17	49.1	1.3
12	38.5	1.42	59.7	1.44	60.5	1.2
14	40	1.45	60.7	1.60	67.2	1.5
16	39	1.43	60.1	1.49	62.6	3.0
18	35.5	1.38	57.9	1.12	47.0	6.5
20	33	1.34	56.3	0.85	35.7	9.0
22	32	1.33	55.7	0.74	31.1	9.9
Day (24 h)	—	**avg 1.347**	**1,357.7 MWh**	**avg 0.90**	**905.9 kL**	—

[Model: daily water = 42 MW × Σ WUE (21.57) = 905.9 kL; daily energy = 42 × Σ PUE (32.325) = 1,357.7 MWh]

Interpretation: about 55% of the day's water is drawn between 10:00 and 17:00, exactly when exchange power is cheapest. When water is constrained, switching adiabatic pads off and running chillers between 11:00 and 16:00 costs extra kWh at about ₹1.2–2/kWh. It avoids the most water-intensive hours and halves peak-hour demand (about 67 down to about 30 kL/h). This "dry at noon" strategy is the main co-optimisation lever for India.

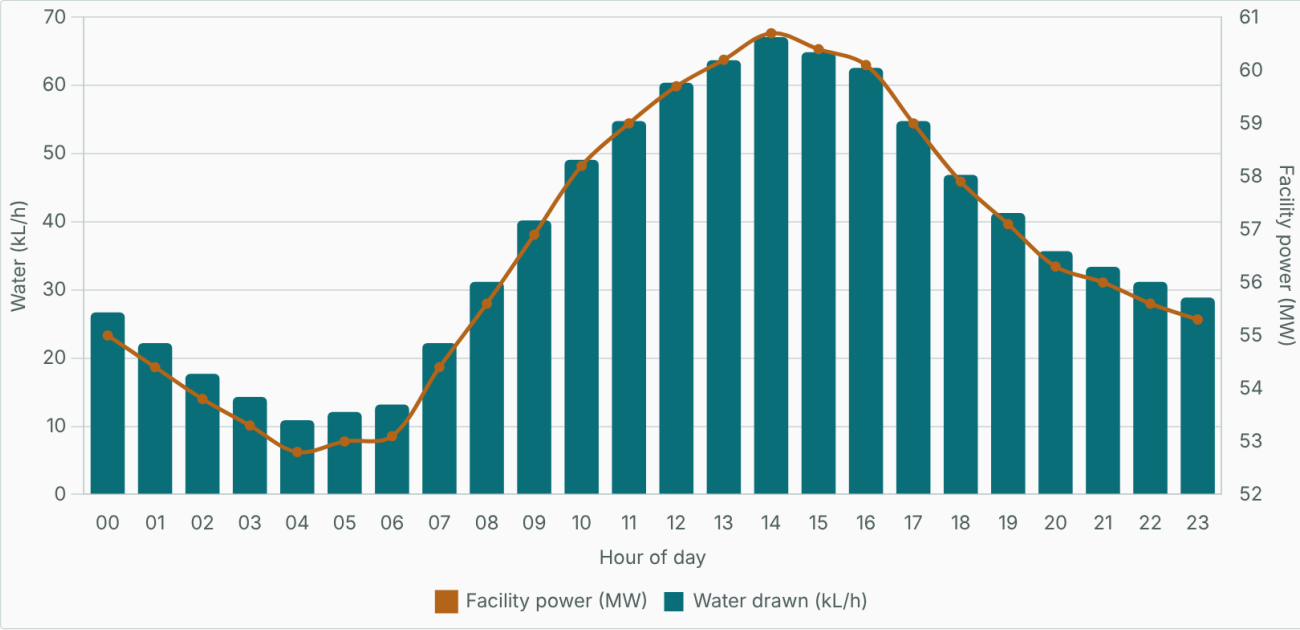


Figure 7.9 One hot day, hour by hour. A 50 MW hyperscale site in Hyderabad on a May design day, 42 MW of IT load all day. Bars are water drawn each hour; the line is the second measure you choose. *Source: India Data Centre Atlas, chapter built on this chapter's data.*

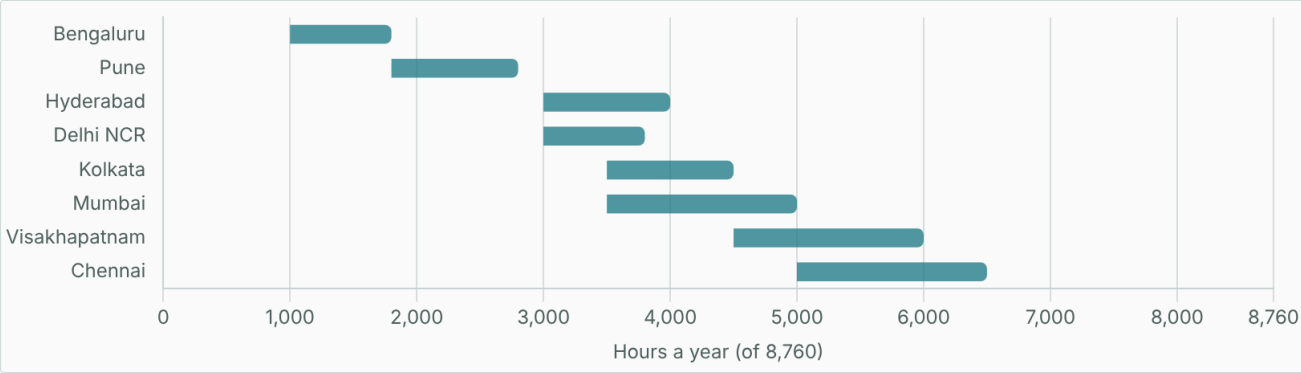


Figure 7.10 Hours a year that adiabatic coolers run wet, by city. Wet-mode hours out of 8,760. *Source: India Data Centre Atlas, chapter built on this chapter's data.*

7.7.4 Tariff time structures

Table 7.15 Tariff time structures

State	ToD rule for HT	Label
Telangana (FY26)	+₹1.00/unit 06–10 and 18–22; −₹1.50/unit 22–06	[Verified: TGERC]
Maharashtra (FY26–30)	HT industrial/commercial ToD charge 25%; solar-hour rebates from −15% (FY26)	[Verified: MERC]
Grid emission factor	Lowest in solar hours, highest in evening; India weighted average declined from 0.774 tCO_2/MWh (FY2013-14) to 0.710 tCO_2/MWh in FY2024-25 (1,234.19 MT ÷ 1,739.06 BU)	[Verified: CEA CO2 Baseline Database User Guide V21.0, Nov 2025]

7.8 Grid electricity options in India

7.8.1 DISCOM HT/EHT tariffs (FY2025-26 unless noted)

Table 7.16 DISCOM HT/EHT tariffs (FY2025-26 unless noted)

State / DISCOM	Category	Energy charge	Demand charge	Other	Label
Maharashtra MSEDCL	HT-I(A) industry	₹7.48/kVAh (FY26); then 7.27, 7.11	₹555/kVA/mo (then 560, 565)	Wheeling ~₹0.62 HT; 10% wheeling discount for DCs on OA or green tariff	[Verified: MERC]; a secondary claim of ₹8.68/kVAh after the Mar 2026 review is unverified
Maharashtra MSEDCL	HT-II commercial	₹9.26 + ₹0.62 = ₹9.88/kVAh	₹555/kVA/mo	—	[Verified: MERC]

State / DISCOM	Category	Energy charge	Demand charge	Other	Label
Mumbai TPC-D	HT-II / EHV commercial	₹5.89 + ₹0.69 wheeling / ₹5.89	₹300/kVA/mo	Green power tariff ₹0.25/kWh	[Verified: MERC TPC-D]
Telangana TGSPDCL	HT-I(A) industry	₹7.65 / 7.15 / 6.65 per kVAh (11 / 33 / ≥132 kV)	₹500/kVA/mo	Green tariff ₹0.66/unit	[Verified: TGERC]
Telangana TGSPDCL	HT-II(A) commercial	₹8.80 / 8.00 / 7.80 per kVAh	₹500/kVA/mo	—	[Verified: TGERC]
Karnataka (all ESCOMs)	HT-2(a) industry	660 paise/kWh (FY26, FY27); 650 (FY28)	₹345 / 350 / 355 per kVA/mo	P&G surcharge 36 paise (FY26), 35 (FY27); BESCOM true-up +56 paise (May 2026–Apr 2027)	[Verified: KERC; true-up Verified-secondary]
Tamil Nadu TNPDCL	HT-I (incl. IT) / HT-III commercial	₹7.50 / ₹9.40 per kWh	₹608/kVA/mo	Wheeling HT ₹1.04/kWh	[Verified-secondary: TNERC summary]

Key judgment: check whether the state DC/IT policy grants industrial tariff status. In Maharashtra, commercial versus industrial is a ₹1.78/kVAh difference [Model: 9.88 − (7.48 + 0.62)].

7.8.2 Open access and procurement options

Table 7.17 Open access and procurement options

Option	Key conditions	Generation price	Adders	Carbon / water	Label
DISCOM green tariff	Premium over normal tariff	—	+₹0.25 (TPC-D), + ₹0.66 (Telangana)	RE attribute; physical grid water unchanged unless additional RE	[Verified: MERC TPC-D; TGERC]
GEOA Rules 2022	Threshold 100 kW; CSS may not rise >50% over 12 years from grant year; additional surcharge not applicable if consumer pays fixed charges; monthly banking ≥30% of consumption	—	CSS, wheeling, transmission, losses, banking	Additional RE → EWIF ~0.1	[Verified: GEOA gazette]; 100 kW [Proxy: verify amendment]
FOR model GEOA regulations	CSS shall not exceed 20% of tariff or ABR	—	—	—	[Verified: GEOA FOR]
CSS, Telangana FY26	HT-I ₹1.82 / 1.56 / 1.46; HT-II(A) ₹2.15 / 1.86 / 1.79 per kWh (11 / 33 / 132 kV)	—	—	—	[Verified: TGERC]
CSS, Tamil Nadu	Industrial ₹1.99/kWh; commercial ₹2.57/kWh	—	—	—	[Verified-secondary: TNERC summary]
Additional surcharge, Karnataka	BESCOM proposed ₹1.65/kWh (not approved)	—	—	—	[Verified-secondary: SolarQuarter]
TN banking	8% of banked energy, in kind (TN GEOA Regs 2025)	—	—	—	[Verified-secondary: SolarQuarter]
Group captive	≥26% equity, ≥51% consumption (Electricity Rules 2005); CSS/AS exempt	₹4.0–5.25	Wheeling, transmission, losses, banking	Near-zero water if RE	[Proxy: rules]; TN exemption [Verified-secondary]
ISTS RE / FDRE via third party	ISTS charges levied at 25% for COD Jul 2025–Jun 2026, 50%, 75%, then 100% after 1 Jul 2028; waiver worth ₹0.4–1.8/unit per industry	FDRE-RTC ₹5.25; 4-h peak ₹5.99; RTC-IV ₹5.06–5.07; SJVN FDRE-3 ₹4.82–4.91	ISTS + state charges + CSS + AS	Near-zero water	[Verified: ISTS; SECI; SaurEnergy]
IEX G-DAM	FY25 weighted avg ₹4.01/unit; Jan 2026 ₹4.06; Feb 2026 ₹3.43	—	OA charges + fees	RE attribute	[Verified: IEX]
IEX DAM	Jan 2026 ₹3.86; Feb 2026 ₹3.58	—	—	Grid mix	[Verified: IEX]
On-site solar	Rooftop ≈ 0.1–0.3 MWp per MW IT; negligible for large sites	₹3–4.5/kWh	Maharashtra GSC ₹1.40/unit HT	—	[Proxy]; GSC [Verified-secondary: MERC order]
Captive gas / DG	Backup only; DG ₹25–35/kWh (diesel ~₹90/L × 0.27 L/kWh)	—	—	High CO_2	[Model]
RECs / I-RECs	Unbundled; no physical additionality	—	—	No physical water change	[Proxy]
Carbon Credit Trading Scheme	DCs not yet obligated entities (as understood)	—	—	—	[Proxy: verify with BEE]

7.8.3 Reliability and connection

Table 7.18 Reliability and connection

MW (facility)	Typical supply voltage	Label
<5 MVA	11 kV	[Proxy: state supply codes]
5–20 MVA	33 kV	[Proxy]
20–60 MVA	66/110/132 kV	[Proxy]
60–200 MVA	132/220 kV	[Proxy]
>200 MVA	220/400 kV, ISTS or dedicated substation	[Proxy]

- **Redundancy:** dual feeds from separate grid substations, N+1 transformers, 2N UPS for Tier III/IV [Proxy].
- **Timelines:** 18–36 months for ≥220 kV; transformer lead times 12–24+ months [Proxy: request from STU/CTUIL].
- **Regulation:** CEA connectivity regulations and state grid codes apply; NERC-style ride-through requirements are absent in India today [Proxy].

7.9 Energy calculations

7.9.1 Core formulas

1. IT design load (kW) = racks × kW/rack
2. Average IT (kW) = IT design × utilisation (lease fill × server load factor)

3. IT energy (kWh/yr) = average IT × 8,760; MU = kWh ÷ 10^6
4. Facility energy = IT energy × PUE_annual
5. Non-IT split ≈ cooling 60–75%, UPS/transformer losses 15–30%, lighting/other 3–8% of (PUE − 1) [Proxy]
6. Peak facility kW = IT design × PUE_peak (hot hour)
7. Contract demand (kVA) = peak facility kW ÷ PF (0.95–0.99) × diversity (0.9–1.0)
8. DG fuel (L/h) = facility kW × 0.26–0.28 L/kWh; storage (L) = L/h × hours [Proxy: genset datasheets]
9. UPS battery (kWh) = IT kW × minutes/60 ÷ (DoD × η); 5–10 min typical for Li-ion [Proxy]
10. Landed cost (₹/yr) = facility kWh × energy rate + contract kVA × demand rate × 12 + surcharges

7.9.2 Typical parameter values

Table 7.19 Typical parameter values

Parameter	Low	Typical	High	Label
Global average PUE	1.54 (2025)	1.56 (2024)	1.58 (2023)	[Verified: UPT25; Uptime blog]
New facilities / >20 MW	1.44 (>20 MW)	1.48 (last 5 yrs)	—	[Verified: UPT25 — "Facilities commissioned within five years of the survey (from around early 2020) achieved an average PUE of 1.48"]
US modelled average PUE	1.15–1.35 (2028)	1.4 (2023)	1.6 (2014)	[Verified: LBNL24]
Hyperscaler fleets	1.09 (Google 2025)	1.17 (Microsoft FY25)	1.28 (MSFT APAC FY25)	[Verified: GOOG25; MSFT]
Indian colo PUE	1.35	1.45–1.55	1.7	[Verified: Uptime Field Report 183 — Asia-Pacific (excl. China) average annual PUE 1.52 (n=79) vs global 1.54; Proxy for India range]
UPS efficiency	77–85% (edge)	80–94% (colo)	90–99% (hyperscale/AI)	[Verified: LBNL24]
Transformer loss per stage	0.5%	1.0%	1.5%	[Proxy]
Power factor	0.95	0.98	0.99	[Proxy]
Rack density	most 10–30 kW; few >30 kW	—	AI 120–142 kW	[Verified: UPT25; NVDA]
DGX H100 (8 GPU) rated	10.2 kW	—	—	[Verified: LBNL24]
DGX B200 max	14.3 kW	—	—	[Verified: NVDA]
GB200 NVL72 rack	~120 kW (NVIDIA)	132 kW (HPE nominal)	GB300: 135–142 kW	[Verified-secondary: NVDA summaries]
AI annualised power	60–80% of rated	70%	—	[Verified: LBNL24]
Energy per AI prompt	0.10 Wh (narrow)	0.24 Wh (Gemini)	0.34 Wh (ChatGPT avg)	[Verified: GOOG25; secondary for OpenAI]

Per-GPU (NVL72): 120 kW ÷ 72 = 1.67 kW IT; × PUE 1.25 = 2.08 kW; × 8,760 h = 18.2 MWh/yr at full load [Model].

7.9.3 Scale snapshot

Table 7.20 Scale snapshot

Archetype	IT design	Utilisation	IT MU/yr	PUE	Facility MU/yr	Contract MVA	DG fuel at full load (L/h)
5 MW enterprise (Pune)	5 MW	60%	26.3	1.60	42.0	9	2,300
20 MW colo (Mumbai)	20 MW	70%	122.6	1.50	184.0	35	8,600
50 MW hyperscale (Hyderabad)	50 MW	80%	350.4	1.30	455.5	76	18,900
100 MW AI campus (Bengaluru)	100 MW	75%	657.0	1.25	821.3	150	35,100

[Model. 5 MW: 5×0.6×8,760 = 26,280 MWh; ×1.6 = 42,048 MWh; peak 5×1.7 = 8.5 MW ÷ 0.95 ≈ 9 MVA; DG 8,500 kW × 0.27 ≈ 2,300 L/h.]

7.10 Worked examples

7.10.1 Example A: 20 MW IT colocation, Mumbai (Navi Mumbai), water-cooled chillers + open cooling towers

Table 7.21 Example A: 20 MW IT colocation, Mumbai (Navi Mumbai), water-cooled chillers + open cooling towers

Step	Calculation	Result	Label
Average IT	20 MW × 70%	14 MW	[Proxy: colo fill]
IT energy	14 × 8,760	122,640 MWh = 122.6 MU	[Model]
PUE	Hot-humid tower plant	1.50	[Proxy: Uptime 1.54 avg]
Facility energy	122,640 × 1.5	183,960 MWh ≈ 184 MU	[Model]
Site WUE	1.25 × 0.85 × 1.48 = 1.57 evaporation + CoC-4 blowdown 0.52 ≈ 2.1 → use 2.0	2.0 L/kWh	[Model; NTT 1.42–2.93 Verified]
Direct water	122.64 M kWh × 2.0	245,280 kL/yr = 0.67 MLD average	[Model]
Indirect water	183.96 M kWh × 3.0	551,880 kL/yr	[Model]
Total water	—	797,160 kL/yr (source WUE 6.5)	[Model]
Peak day	20 MW × 24 h × 2.2 L/kWh	1,056 kL/day (1.57× average)	[Model]
Peak hour	20,000 kWh × 2.3	46 kL/h	[Model]
Storage (48 h at peak)	46 × 48	~2,200 kL	[Model]
Contract demand	20 × PUE_peak 1.65 = 33 MW ÷ 0.95	~35 MVA (110/220 kV)	[Model]

Step	Calculation	Result	Label
DG fuel	33,000 kW × 0.26	8,580 L/h; 24 h = 206 kL	[Model]

Hot-day profile (tower): WUE stays nearly flat at 2.0–2.3 L/kWh, so water tracks IT load (≈28–46 kL/h); facility power swings only ±5% as chiller lift rises with wet-bulb [Model].

Landed electricity cost (184 MU; 35,000 kVA):

Table 7.22 Example A: 20 MW IT colocation, Mumbai (Navi Mumbai), water-cooled chillers + open cooling towers

Option	Calculation	₹ crore/yr	₹/kWh	Label
1. MSEDCL HT-I industry (if DC gets industrial status)	184 × (7.48 + 0.62) = 149.0 + demand 35,000 × 555 × 12 = 23.3	172.3	9.37	[Model on MERC rates; excl. duty, ToD]
1b. MSEDCL HT-II commercial	184 × 9.88 = 181.8 + 23.3	205.1	11.15	[Model]
2. HT-I + green tariff (₹0.25 TPC-D benchmark)	+184 × 0.25 = 4.6	176.9	9.62	[Model; MSEDCL premium not confirmed]
3. Group-captive RE-RTC (intra-state)	5.25 + wheeling 0.62 + transmission/losses ~0.6 + banking ~0.2 = 6.67 × 184 = 122.7 + 23.3	146.0	7.94	[Model; adders Proxy]
4. Third-party ISTS FDRE	5.25 + ISTS ~0.6 + 0.62 + CSS ~1.8 [Proxy] + losses 0.4 = 8.67 × 184 = 159.5 + 23.3	182.8	9.94	[Model]

Carbon and water by option: options 1 and 2 keep grid water at about 552 ML/yr. Option 3 or 4 at 90% RE cuts indirect water to about 72 ML/yr and total water from 797 to about 317 ML/yr [Model: 184 M × (0.9×0.1 + 0.1×3.0) = 71.8 ML].

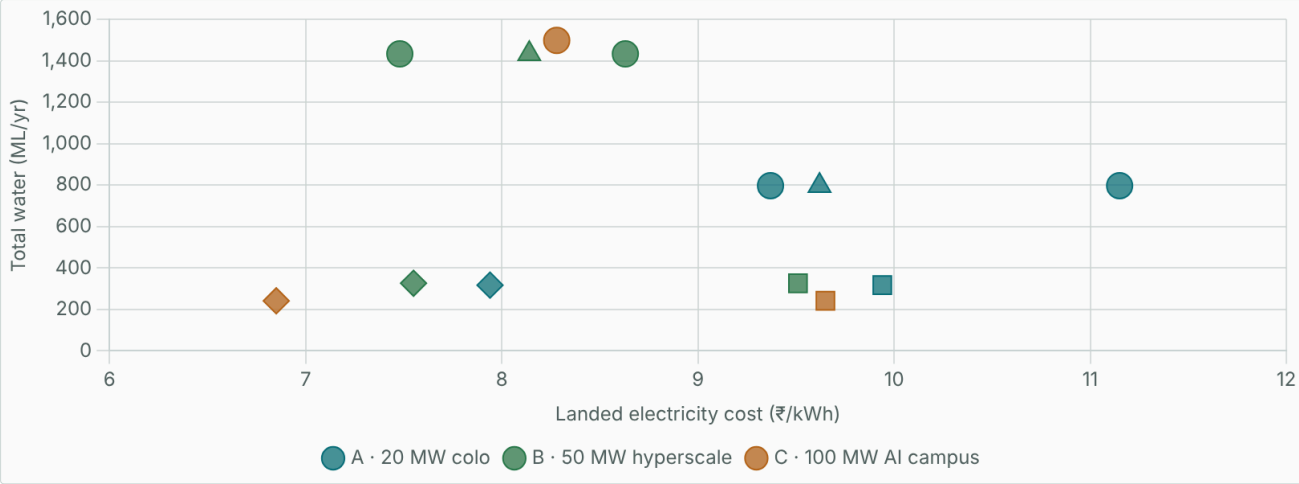


Figure 7.11 Cost versus water, by how you buy power. Each dot is one procurement option for one worked example: landed ₹ per kWh against total water per year. Down and to the left is better. *Source: India Data Centre Atlas, chapter built on this chapter's data.*

7.10.2 Example B: 50 MW IT hyperscale, Hyderabad, air-side economiser / adiabatic-assisted cooling

Table 7.23 Example B: 50 MW IT hyperscale, Hyderabad, air-side economiser / adiabatic-assisted cooling

Step	Calculation	Result	Label
Average IT	50 × 80%	40 MW	[Proxy]
IT energy	40 × 8,760	350,400 MWh	[Model]
PUE annual	—	1.30	[Proxy: MSFT APAC 1.25–1.28 Verified]
Facility energy	350,400 × 1.30	455,520 MWh	[Model]
Site WUE annual	LBNL hyperscale 0.32–0.40; MSFT APAC 0.25	0.45 L/kWh	[Proxy: ~3,000–4,000 wet-mode hours]
Direct water	350.4 M × 0.45	157,680 kL/yr (432 kL/d avg)	[Model]
Indirect water	455.52 M × 2.8	1,275,456 kL/yr	[Model]
Total water	—	1,433,136 kL/yr (source WUE 4.09)	[Model]
Peak day (Section 5.3)	42 MW × Σ WUE	906 kL/day (2.1× average)	[Model]
Peak hour	42,000 × 1.60	67.2 kL/h (3.7× the 18 kL/h average hour)	[Model]
Storage	24 h at peak-day rate	~900 kL; 48 h ≈ 1,800 kL	[Model]
Contract demand	50 × 1.45 = 72.5 MW ÷ 0.95	~76 MVA (220 kV)	[Model]
DG fuel	72,500 × 0.26	18,850 L/h	[Model]

Landed electricity cost (455.5 MU; 76,000 kVA → demand ₹45.6 cr/yr = ₹1.00/kWh):

Table 7.24 Example B: 50 MW IT hyperscale, Hyderabad, air-side economiser / adiabatic-assisted cooling

Option	Calculation	₹/kWh	Label
1. TGSPDCL HT-I(A) ≥132 kV + ToD	6.65 + 1.00 demand − 0.17 ToD net (flat load: +1.00 × 8 h, −1.50 × 8 h)	7.48 (₹340.7 cr)	[Model on TGERC rates]
1b. HT-II(A) commercial ≥132 kV	7.80 + 1.00 − 0.17	8.63	[Model]
2. HT-I + green tariff	7.48 + 0.66	8.14	[Model; TGERC Verified]
3. Group captive RE-RTC	5.25 + ISTS/STU ~0.6 + wheeling ~0.4 [Proxy] + losses 0.3 + demand 1.00 (CSS/AS exempt)	7.55	[Model]
4. Third-party OA FDRE	5.25 + 0.6 + 0.4 + CSS 1.46 (HT-I 132 kV) + AS ~0.5 [Proxy] + 0.3 + 1.00	9.51	[Model; CSS Verified]

7.10.3 Example C: 100 MW IT AI campus, Bengaluru (alt.: Visakhapatnam), direct-to-chip liquid cooling + dry coolers

Table 7.25 Example C: 100 MW IT AI campus, Bengaluru (alt.: Visakhapatnam), direct-to-chip liquid cooling + dry coolers

Step	Calculation	Result	Label
Racks	100,000 kW ÷ 120 kW (NVL72)	~833 racks ≈ 60,000 GPUs	[Model; NVDA]
Average IT	100 × 75%	75 MW	[Proxy: LBNL training 80% × 70% rated, plus inference]
IT energy	75 × 8,760	657,000 MWh	[Model]
PUE annual	Mild climate; ~80% heat to liquid at W32–W40	1.25	[Proxy]
Facility energy	657,000 × 1.25	821,250 MWh	[Model]
Site WUE	Closed loop; domestic + humidification	0.03 L/kWh	[Proxy; Microsoft zero-water design Verified]
Direct water	657 M × 0.03	19,710 kL/yr (54 kL/d)	[Model]
Indirect water	821.25 M × 1.8 (Karnataka proxy)	1,478,250 kL/yr	[Model]
Total water	—	1,497,960 kL/yr (source WUE 2.28)	[Model]
With 90% RE-RTC	821.25 M × 0.27 + 19.7 ML	~241,400 kL/yr	[Model]
Peak water day	Domestic-driven; ~1.2×	~65 kL/d (+ one-time loop fill)	[Model]
Hot-day electricity	April afternoon 34–35 °C → PUE 1.35 at 100 MW	135 MW	[Model]
Contract demand	135 ÷ 0.95 ≈ 142 → 150 MVA	220/400 kV	[Model]
Load swings	Training square-wave drops	Specify BESS/UPS smoothing and ride-through	[Verified: LBNL24 Fig 3.4; NERC]
DG fuel	135,000 × 0.26	35,100 L/h; 24 h ≈ 842 kL	[Model]

Landed electricity cost (821.25 MU; 150,000 kVA → demand ₹62.1 cr = ₹0.76/kWh):

Table 7.26 Example C: 100 MW IT AI campus, Bengaluru (alt.: Visakhapatnam), direct-to-chip liquid cooling + dry coolers

Option	Calculation	₹/kWh	Label
1. KERC HT-2(a) via BESCOM	6.60 + 0.76 demand + 0.36 P&G + 0.56 true-up	8.28 (₹679.7 cr)	[Model on KERC rates; P&G/true-up applicability to HT Proxy]
2. Intra-state group captive (solar + wind + BESS)	4.5–5.25 + wheeling/transmission 0.8–1.2 [Proxy] + banking 0.2 + demand 0.76	6.3–7.4	[Model]
3. Third-party ISTS FDRE	5.25 + ISTS 0.6–1.0 + CSS ~1.5 [Proxy] + AS 0–1.65 (BESCOM proposal; GEOA exempts if fixed charges paid) + wheeling 0.5 + 0.76	8.6–10.7	[Model]

Visakhapatnam variant: seawater heat rejection gives freshwater WUE ≈ 0, but PUE is about 1.2–1.3 at 28–30 °C sea temperatures. CRZ clearance and intake/outfall capex apply. APERC tariffs were not retrieved [Proxy].

Bottom line across the three examples:

Table 7.27 Example C: 100 MW IT AI campus, Bengaluru (alt.: Visakhapatnam), direct-to-chip liquid cooling + dry coolers

Metric	A: Mumbai colo	B: Hyderabad hyperscale	C: Bengaluru AI
Direct water (ML/yr)	245.3	157.7	19.7
Indirect water (ML/yr)	551.9	1,275.5	1,478.3
Total (ML/yr)	797.2	1,433.1	1,498.0
Direct L per facility kWh	1.33	0.35	0.02
Cheapest modelled landed ₹/kWh	7.94 (captive)	7.48 (HT-I)	6.3–7.4 (captive)

7.11 Recommendations

1. **Calculator defaults:** EWIF India 3.0 L/kWh (state overrides from Section 2.2), PUE 1.30–1.55 by design, and WUE 2.0 (towers), 0.45 (adiabatic) or 0.03 (DLC + dry). Always output direct, indirect, peak-day and peak-hour water.
2. **Siting:** in Chennai, Mumbai, Kolkata and Vizag, avoid freshwater towers; use STP-treated or seawater sources, or liquid cooling with dry coolers. In Hyderabad and Delhi, adiabatic designs need storage of at least 2× the average day and a "dry at noon" control mode.
3. **Procurement:** treat group-captive RE-RTC as the base case above 20 MW. It is the cheapest option in two of three examples and cuts total water by 70–85%. Reach COD before July 2027 to keep part of the ISTS waiver.
4. **Tariff category:** secure industrial classification under state DC/IT policies before connection; it is worth ₹1–2.5/kWh.
5. **Grid code readiness:** for AI loads above 100 MW, contract for ride-through and ramp limits now, ahead of Indian equivalents of NERC's large-load standards.

7.12 Data gaps and how to close them

Table 7.28 Data gaps and how to close them

Gap	Request from	Instrument
Site WUE (withdrawal/consumption), source split, CoC	Nxtra, CtrlS, Yotta, Sify, STT GDC, AdaniConneX, NTT India	ESG/BRSR requests; Karnataka plant-wise PUE/WUE filings [Verified: KADC]
Water allocations and consents	State PCBs (CTE/CTO), MIDC, CMWSSB, HMWSSB, BWSSB, GVMC	RTI for CTO water quantities and STP/ZLD conditions
EIA water figures	MoEFCC PARIVESH	Portal downloads
CSS, AS, wheeling (MSEDCL, KERC); MSEDCL green tariff	MERC Tables 446–447; KERC tariff/OA orders	Regulator PDFs
Hourly DC load data	Grid-India / SLDCs, DISCOM AMR	Data-sharing MoU
Hourly climate	ISHRAE TMY for eight cities	Licence

Gap	Request from	Instrument
Plant-level water intensity	CEA, plant CTO returns, WRI India	RTI; WRI satellite method
US/EU permit examples	The Dalles, Mesa, Goodyear/Quincy, Irish/Dutch permits	Follow-up research
Uptime Tier water/fuel autonomy text	Uptime Institute	Tier Standard: Topology

7.13 Caveats

- Several tariff figures (TN, BESCOM true-up, GB200 power) come from secondary summaries and are marked as such. Confirm against final tariff orders before investment decisions. A secondary blog claims MSEDCL energy charges were revised to ₹8.68/kVAh in March 2026, which conflicts with the press note; this is unresolved.
- Including hydro reservoir evaporation in grid water intensity is contested and moves the Indian figure by about 1.4 L/kWh.
- Climate-hour counts, peak factors and hourly functions are models, not measurements. Replace them with ISHRAE TMY data and operator metering.
- Operator WUEs mix withdrawal and consumption bases (NTT and Microsoft use withdrawal). Do not rank Indian operators on mixed-basis numbers.
- Per-prompt AI figures are company disclosures with undisclosed prompt lengths; Google's 0.24 Wh excludes grid water.

Sources cited in this chapter

Organisations and publications named in this chapter's text, in order of how often they are cited. Full details are in the bibliography.

- State electricity regulatory commissions (KERC, MERC, TNERC, TGERC) tariff and open-access orders
- NTT DATA ESG data
- Ministry of Power, orders on ISTS waiver and green open access
- Microsoft sustainability reports and datacenter design posts
- World Resources Institute (Aqueduct, thermal water studies)
- Indian Energy Exchange price data
- National Renewable Energy Laboratory water factors
- Google environmental reports, efficiency data and Gemini inference paper
- Council on Energy, Environment and Water (CEEW) studies
- Lawrence Berkeley National Laboratory, 2024 US Data Center Energy Usage Report
- Uptime Institute, Global Data Center Surveys and Tier Standard
- North American Electric Reliability Corporation, Large Loads alert (May 2026)
- CtrlS disclosures
- Centre for Science and Environment, thermal plant water survey
- Central Electricity Authority (CEA), CO_2 Baseline Database and grid reports
- Ministry of Environment, Forest and Climate Change, thermal plant water norms and EIA
- Solar Energy Corporation of India tenders
- Li, Yang, Islam and Ren, "Making AI Less Thirsty"
- ISHRAE climate design data
- Nxtra by Airtel disclosures
- European Union Energy Efficiency Directive and Delegated Regulation 2024/1364
- International Energy Agency (IEA), Energy and AI (2025) and updates
- NVIDIA product specifications and roadmap
- Equinix sustainability reports
- Yotta Data Services disclosures
- Central Ground Water Board and state groundwater assessments
- Bangalore Water Supply and Sewerage Board
- Bureau of Energy Efficiency
- Bureau of Indian Standards (IS 1893, ISO/IEC 30134 adoption)
- Ministry of New and Renewable Energy statistics
- STT GDC India ESG report
- Sify Infinit Spaces sustainability report
- AdaniConneX disclosures
- Down To Earth reporting
- CPHEEO urban water supply norms
- ISO/IEC 30134 data-centre KPI standards
- Hewlett Packard Enterprise product data

PART II · RESOURCES AND IMPACTS

Chapter 8
Conventional and AI data centres compared: water and energy at the scale of a site

WHAT THIS CHAPTER COVERS

This chapter compares conventional data centres (enterprise, colocation and cloud facilities with air-cooled racks of about 5 to 15 kilowatts) with AI data centres (training and inference facilities with racks of 40 to more than 140 kilowatts and liquid cooling). It sets out the archetypes, compares their efficiency, the energy and water used per AI prompt, the water used by each cooling type and reported by each operator, asks whether liquid cooling raises or lowers water use, and presents a side-by-side model of a 20 megawatt colocation hall and a 100 megawatt AI campus in India.

KEY TAKEAWAYS

1. Per megawatt of IT capacity, a 100 megawatt AI campus uses about the same energy as a 20 megawatt colocation hall, because its higher utilisation is offset by a lower PUE. Per site it uses about five times the energy and five times the water, because it is five times larger.
2. In India, water evaporated at coal power stations, assumed at about 2.6 litres per kilowatt-hour at the plant, is larger than on-site water in every case modelled and makes up between 74% and 99% of a site's total water.
3. Liquid cooling lowers water per kilowatt-hour when it rejects heat through dry coolers, but it remains water-intensive if it is paired with cooling towers, and going dry raises energy use and carbon by about 8 to 10%.
4. The water used per AI prompt is reported very differently by different organisations, from 0.26 millilitres for Google's median Gemini prompt to 45 millilitres in Mistral's life-cycle assessment, mainly because each figure counts different things.
5. Only one Indian operator, STT GDC India, publishes an independently assured water usage effectiveness (0.58 litres per kilowatt-hour), and the IndiaAI tender sets a PUE limit of 1.35 with no water requirement.

How this chapter connects to the others. This chapter builds on the anatomy in Chapter 2, the cooling options in Chapter 3 and the grid water factors in Chapters 5 to 7.

AI data centres use far more energy per site and per rack than conventional data centres, but not necessarily more water per unit of IT energy: a liquid-cooled AI campus on dry coolers can have a lower on-site WUE than an Indian colo running cooling towers, while its total water footprint — dominated in India by water evaporated at coal plants — is several times larger simply because it burns 4–5× more electricity. For India, the decisive water variable is therefore the grid, not the cooling tower; the decisive design choice at site level is whether heat is rejected by evaporation or by dry/closed-loop systems.

Label key used throughout: [R] = measured/reported by the operator or official statistics; **[E]** = estimated/modelled by researchers or analysts; **[D]** = design value or target, not an operating result; **[A]** = this report's own assumption/calculation.

8.1 Key Findings

1. **The industry average has stalled; AI builds are pulling the frontier.** Uptime Institute's 2025 survey reports a weighted-average PUE of 1.54, the sixth year of near-stagnation; facilities built in the last five years average 1.48 and sites over 20 MW about 1.44 [R, survey]. Hyperscalers report 1.08–1.15 (Google fleet 1.09 for 2025; AWS 1.15 for 2024 and 1.14 for 2025; Meta 1.08) [R]. India's operating fleet sits closer to ~1.5–1.75 (Sify 1.7554 FY2024-25 [R]; an unnamed data-centre executive quoted by The Economic Times said "the PUE of data centres in India is now about 1.5", and a Noida executive told The Secretariat that "a PUE of 1.5-1.6 was considered good in a tropical climate like India's" [E]).
2. **Rack density is the structural difference.** Uptime finds typical racks shifting into the 10–30 kW band with "few facilities" above 30 kW [R]. AI racks are an order of magnitude denser: NVIDIA GB200 NVL72 ≈120 kW (NVIDIA reference) to 132 kW (HPE: 115 kW liquid + 17 kW air); GB300 NVL72 up to 142 kW; Microsoft Fairwater Atlanta is built for ~140 kW per rack and 1,360 kW per row [R/D]. Hashrate Index estimates Vera Rubin VR200 NVL72 racks at ~190–230 kW, though DCD notes NVIDIA "has yet to disclose" Rubin rack power [E, third-party]; NVIDIA's Jensen Huang said at GTC 2025 that Rubin Ultra "Kyber" racks will reach 600 kW in 2027 [D, vendor statement].
3. **Per-query energy has fallen sharply and is small; aggregate demand is not.** Google measured 0.24 Wh, 0.03 gCO_2e and 0.26 mL water for the median Gemini Apps text prompt (May 2025), a 33× energy reduction in 12 months [R, company]. OpenAI's CEO stated 0.34 Wh per average ChatGPT query [R, unaudited statement]; Epoch AI estimates ~0.3 Wh for a GPT-4o query, rising to ~2.5 Wh at 10k input tokens and ~40 Wh at 100k [E].
4. **Water-per-query numbers differ by up to ~170× because of scope, not error.** Google's 0.26 mL counts only on-site cooling water at its fleet WUE; Mistral's 45 mL per 400-token response is a life-cycle figure including electricity-generation water and hardware manufacturing; Li et al.'s "500 mL per 10–50 responses" (≈10–50 mL each) combines on-site and scope-2 water for GPT-3-era hardware [E].
5. **In India, grid water dominates.** Indian coal plants are legally capped at 3.5 m^3/MWh (pre-2017) and 3.0 m^3/MWh (post-2017) of specific water consumption, with seawater-cooled plants exempt [R, regulation]; the CEA has noted historic consumption of 5–7 m^3/MWh [R]. At India's grid mix this implies roughly 2.6 L per kWh consumed [A] — about 2× a cooling-tower data centre's on-site WUE and ~25–50× a closed-loop AI site's.

6. **India is moving from ~1.4–2.2 GW to 8–12 GW by 2030, with AI the fastest-growing slice.** Wood Mackenzie projects operational capacity from 2.2 GW (2025) to 12 GW (2030) and AI-dedicated capacity from 275 MW to 6,546 MW; S&P Global estimates 13 TWh of data-centre electricity in 2024 (0.8% of demand) rising to 57 TWh by 2030 (2.6%) [E].
7. **Indian disclosure is thin.** Only a handful of operators publish absolute PUE/WUE; STT GDC India's 0.58 L/kWh (FY2025, DNV limited assurance) is the only audited portfolio WUE found [R]. Karnataka's 2026–31 policy and the IndiaAI PUE < 1.35 tender clause — a MeitY corrigendum issued in the final week of September 2024 requiring bidders to affirm a PUE "not greater than 1.35" — are the first policy hooks on efficiency and water.

8.2 Definitions and Archetypes

Table 8.1 Definitions and Archetypes

Archetype	Typical rack density	Typical facility size (IT)	Utilisation / load profile	Cooling norm
Enterprise / on-prem	3–10 kW	<1–5 MW	Server utilisation ~11% (2014) rising to ~20% (2027) in LBNL's model [E]; flat, low	CRAC/DX, air-cooled chillers
Retail / wholesale colocation	5–15 kW (10–30 kW growing)	5–50 MW per building	Server utilisation 21%→35% (LBNL) [E]; facility load factor often 50–70% of contracted IT [A]	Chilled water + cooling towers or air-cooled chillers
Hyperscale cloud	10–30 kW	50–300 MW campuses	Server utilisation 45%→50% (LBNL) [E]; diurnal, smooth	Air-side economiser + adiabatic, evaporative; increasingly liquid
AI training	40 kW (H100, air) to 120–142 kW (GB200/GB300, liquid); ~190–230 kW next generation	100 MW to >1 GW	LBNL assumes 80% operational time at ~70% of rated power [E]; synchronous "square-wave" swings	Direct-to-chip liquid (closed loop) + dry coolers/chillers or towers
AI inference	20–130 kW	10–300 MW, nearer users	LBNL assumes 40% operational time and half of training power [E]; spiky, demand-following	Mixed air/liquid

What "flat vs spiky" means in practice. LBNL's measurements of Llama-70B training across eight H100 nodes show a "characteristic square-wave pattern" with regular simultaneous power drops at synchronisation points [R, measured], which grid operators find taxing. Conventional colo loads change slowly and average far below nameplate. LBNL measured 8×H100 DGX nodes at an average of 7.9 kW (~78% of rated 10.2 kW) under saturating training workloads and idle at ~18% of rated power [R, measured]; it models the US data-centre fleet on 50% average capacity utilisation [E].

Server power. LBNL sets a typical 2023 dual-socket conventional server at 600 W operational, rising to 782 W by 2028, with annual average draw below 0.8 kW; an 8-GPU AI server exceeds a 5 kW annual average by 2026 [E]. Rated server power: A100-class 6.5 kW, H100-class 10.2 kW, B100-class 12.2 kW [R, vendor/Omdia]. At rack scale, a single GB200 NVL72 (~120–132 kW) draws as much as roughly 150–200 conventional servers [A].

8.3 Energy

8.3.1 PUE by archetype

Table 8.2 PUE by archetype

Type / operator	PUE	Status
Global survey average (Uptime 2025)	1.54 (1.56 in 2024; 1.58 in 2023)	[R, survey]
New builds (<5 yrs) / >20 MW sites (Uptime 2025)	1.48 / ~1.44	[R, survey]
US fleet average (LBNL)	1.6 (2014) → 1.4 (2023) → 1.15–1.35 (2028)	[E, modelled]
Google global fleet (2025)	1.09	[R]
AWS global (2024 / 2025)	1.15 / 1.14	[R]
Meta	1.08	[R]
IDC estimate for on-prem enterprise (cited by AWS)	1.63	[E]
India operator: Sify (FY2024-25)	1.7554	[R]
India operator: CtrlS	1.35	[R, unaudited claim]
India operator: Yotta NM1 / D1 / GIFT City	1.4 / 1.4 / ~1.6	[D]
India operator: STT GDC India new builds	<1.5	[D]
IndiaAI GPU tender requirement	<1.35	[D, procurement criterion]

AI-specific facilities rarely publish operating PUE. Microsoft has disclosed design features for Fairwater (closed-loop liquid cooling for >90% of capacity) but not an operating PUE; Meta has not disclosed PUE for Prometheus or Hyperion, nor OpenAI/Oracle for Stargate Abilene, nor xAI for Colossus. Treat any "PUE ~1.1–1.2" quoted for these sites as an estimate.

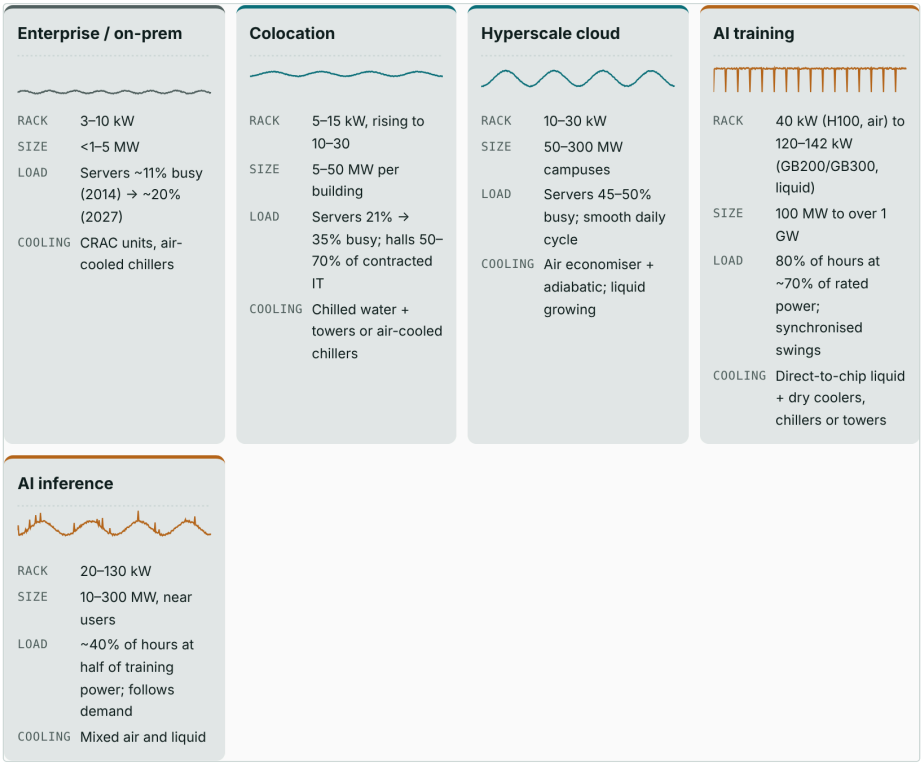


Figure 8.1 Five kinds of data centre. Rack density, size and the shape of the load. The small traces are illustrative 48-hour load profiles; the numbers come from LBNL, Uptime and vendors. *Source: India Data Centre Atlas, chapter built on this chapter's data.*

8.3.2 Where the energy goes

- **IT vs overhead:** At the Uptime average PUE of 1.54, 35% of facility energy is non-IT (cooling, power conversion, lighting); at a hyperscale PUE of 1.09–1.15 the overhead is 8–13% [A, arithmetic]. LBNL estimates US infrastructure energy fell from 40% of the total in 2014 to 30% in 2023 [E].
- **Servers vs the rest of IT:** The IEA says servers account for around 60% of electricity demand in modern data centres, with the rest split among storage, networking and cooling/overhead [E].
- **Inside an AI workload:** In Google's Gemini measurement, the AI accelerators (TPUs) accounted for only 58% of the 0.24 Wh per prompt; host CPU/DRAM, idle provisioned capacity and facility overhead made up the rest [R]. A "chip-only" methodology gives 0.10 Wh — less than half the comprehensive figure.

8.3.3 National and global totals

Table 8.3 National and global totals

Metric	Value	Status / source
Global DC electricity 2024	~415 TWh (~1.5% of world)	[E] IEA Energy and AI (2025)
Global DC electricity 2025	~485 TWh; AI-focused DCs +50% in 2025	[E] IEA Key Questions on Energy and AI update
Global DC electricity 2030	~945–950 TWh (just under 3%)	[E] IEA Base Case
Growth by server type, 2024–30	Accelerated +30%/yr; conventional +9%/yr	[E] IEA
Share of net increase	Accelerated servers ~half; conventional ~20%	[E] IEA
US DC electricity	58 TWh (2014) → 176 TWh (2023, 4.4%) → 325–580 TWh (2028, 6.7–12%)	[E] LBNL 2024
US AI vs conventional server electricity	AI servers <2 TWh (2017) → >40 TWh (2023); conventional ~30 → ~60 TWh	[E] LBNL 2024
India DC electricity	~13 TWh (2024, 0.8%) → 57 TWh (2030, 2.6%)	[E] S&P Global
India DC electricity (alt.)	10 TWh (2025) → 191 TWh (2040, ~7%)	[E] Wood Mackenzie

The IEA notes a typical AI-focused data centre consumes as much electricity as 100,000 households, with the largest under construction consuming 20 times as much [E].

8.3.4 Energy per query, token and training run

Table 8.4 Energy per query, token and training run

Measure	Value	Scope	Status
Gemini Apps median text prompt (May 2025)	0.24 Wh; 0.03 gCO_2e (market-based); 0.26 mL	Accelerators + host + idle + PUE 1.09; water at fleet WUE	[R, company technical paper]
Same, chip-only method	0.10 Wh; 0.02 g; 0.12 mL	Active TPU/GPU only	[R]
ChatGPT average query (Altman, June 2025)	0.34 Wh; ~0.000085 gal (~0.32 mL)	Undisclosed	[R, unaudited statement]
GPT-4o typical query (Epoch AI, Feb 2025)	~0.3 Wh; ~2.5 Wh (10k-token input); ~40 Wh (100k)	Modelled, H100	[E]
Older ChatGPT estimate (EPRI)	2.9 Wh vs 0.3 Wh for a web search	A100-era assumptions	[E, now considered high]
Mistral Large 2, 400-token response	1.14 gCO_2e; 45 mL water; 0.16 mg Sb eq	Full life-cycle (ADEME/Carbone 4)	[R, peer-reviewed LCA]
Mistral Large 2, training + 18 months use	20.4 $ktCO_2e$; 281,000 m^3 water; 660 kg Sb eq	Life-cycle	[R]

Measure	Value	Scope	Status
GPT-3 training	1,287 MWh	Literature estimate	[E]
GPT-4 / frontier training	Not disclosed by OpenAI	—	Data gap

Interpretation: Per-query energy has converged around 0.2–0.4 Wh for short text prompts, but reasoning models, long contexts, image and video generation can be 10–100× higher, and none of the disclosed medians include them. The efficiency gains (33× in a year at Google) are real but are being outrun by volume, which is why IEA and LBNL still project demand doubling.

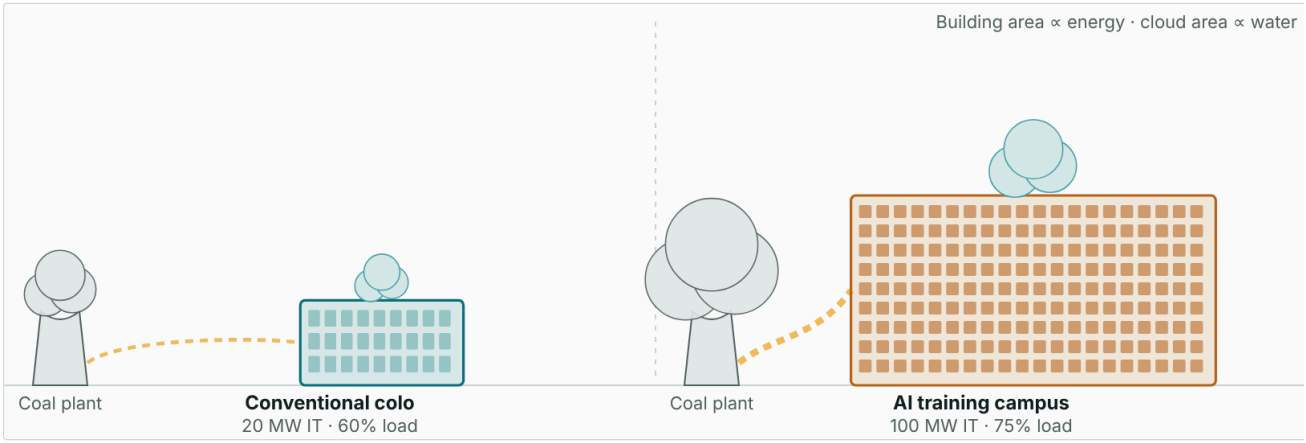


Figure 8.2 Two buildings, side by side. A 20 MW conventional colocation hall and a 100 MW AI training campus, run in India. Switch between whole-site and per-megawatt views, wet and dry cooling, and change how clean the power is. *Source: India Data Centre Atlas, chapter built on this chapter's data.*

8.4 Water

8.4.1 Withdrawal vs consumption, site vs source

- **Withdrawal** = water taken from a source; **consumption** = water not returned (mainly evaporated). A study of US data centres converts withdrawal-based WUE to consumption using a consumptive ratio of ~0.75 for data centres [E]; Shaolei Ren has said data centres typically evaporate about 80% of the water they draw [E].
- **Mixed definitions matter:** AWS's WUE is *withdrawal* per kWh of IT load; Microsoft's 0.30 L/kWh is a *withdrawal* WUE; Meta's WUE is withdrawal divided by IT electricity; LBNL's WUE is *consumption*. Comparing them directly overstates or understates by ~25%.
- **Site WUE vs source WUE:** LBNL defines site WUE as on-site water and source WUE as water used to generate the electricity, warning that "low site WUEs" are not "necessarily good" because waterless cooling raises energy use.

8.4.2 WUE by cooling type

Table 8.5 WUE by cooling type

Cooling system	Site WUE (L/kWh IT)	PUE tendency	Status / note
Air-cooled chiller / DX (CRAC)	~0 (humidification only)	Highest	[E] LBNL: "use no water, they use more energy"
Air-side economiser + occasional adiabatic (hyperscale)	Modelled very low; hyperscalers report 0.1–0.3	Low	[E/R] LBNL notes its simulations may be low
Dry cooler + adiabatic assist	Low–moderate; highest of non-tower systems	Low–moderate	[E] LBNL
Water-cooled chiller + cooling towers (no economiser)	Highest; up to ~2.5 L/kWh cited for evaporative	Low	[E]
Equinix sites using evaporative cooling (2025)	1.41	—	[R, Equinix 2025 Sustainability Report; verify against PDF]
Liquid IT cooling + waterside economiser (towers)	Lower than air-cooled towers but "consumes substantial amounts of water"	Very low	[E] LBNL
Liquid IT cooling + dry cooler (closed loop)	Near zero operationally; one-time fill	Low–moderate	[E/D]
Immersion + dry cooler	Near zero	Very low	[E]

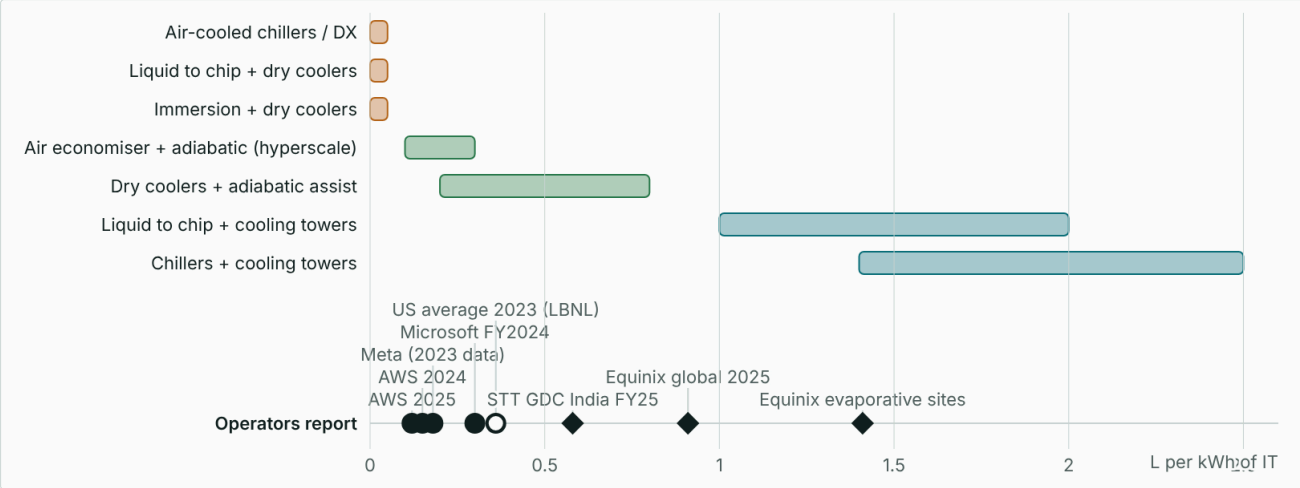


Figure 8.3 On-site water by cooling type, and what operators report. Bands are typical WUE ranges. Dots are published fleet or portfolio figures. Filled dots count water withdrawn; rings count water consumed (evaporated), about 25% less for the same site. *Source: India Data Centre Atlas, chapter built on this chapter's data.*

8.4.3 Reported hyperscaler and operator water

Table 8.6 Reported hyperscaler and operator water

Operator	Metric	Value	Status
AWS	Global WUE (withdrawal)	0.15 (2024); 0.12 (2025); 0.06 in N. Virginia 2025	[R]
Microsoft	Global WUE (withdrawal), FY2024	0.30 L/kWh (0.49 in 2021)	[R]
Meta	WUE (2023 data)	0.18 L/kWh	[R]
Meta	DC withdrawal (2023)	3,881 ML; 1,360 ML from high-stress areas	[R]
Meta	Direct DC consumption vs water embedded in purchased electricity (2024)	2,974 ML vs 72,207 ML (~24×)	[R]
Google	Total water consumption	~8.1 bn gal (2024); 10.9 bn gal (2025, +34%, "more than double 2021 levels", with data centres accounting for most of the increase, per Axios on Google's 2026 Environmental Report)	[R]
Google	Replenishment	4.5 bn gal (64%, 2024); more than 7 bn gal per Google, or 7.7 bn gal (78%) per secondary trackers (2025)	[R]
Equinix	Global WUE (2025)	0.91 L/kWh	[R, Equinix 2025 Sustainability Report; verify against PDF]
STT GDC India	Portfolio WUE FY2025	0.58 L/kWh (1.34 in FY21)	[R, DNV limited assurance]
Sify Infinit Spaces	Water withdrawal FY2024-25	296,279 kL; no WUE published	[R]
Nxtra by Airtel	Water consumption FY2025	216,357 kL; no WUE published	[R]
CtrlS	Water consumption FY2023-24	409,037 kL	[R]

US totals (LBNL): US data centres directly consumed about 66 billion litres in 2023 and indirectly ~800 billion litres via electricity [E]; hyperscale direct use could reach 60–124 billion litres by 2028 [E]. US average site WUE was ~0.36 L/kWh through 2023, rising to 0.45–0.48 L/kWh by 2028 because of the shift to hyperscale and liquid-cooled systems on evaporative heat rejection [E]. The IEA's Energy and AI report (April 2025) puts global data-centre water consumption at "around 560 billion litres per year", rising to "around 1200 billion litres per year in 2030" [E]; De Vries-Gao (Patterns, 2025) splits the 560 billion litres into ~373 billion indirect (electricity), ~140 billion direct (on-site) and ~47 billion from hardware manufacturing [E].

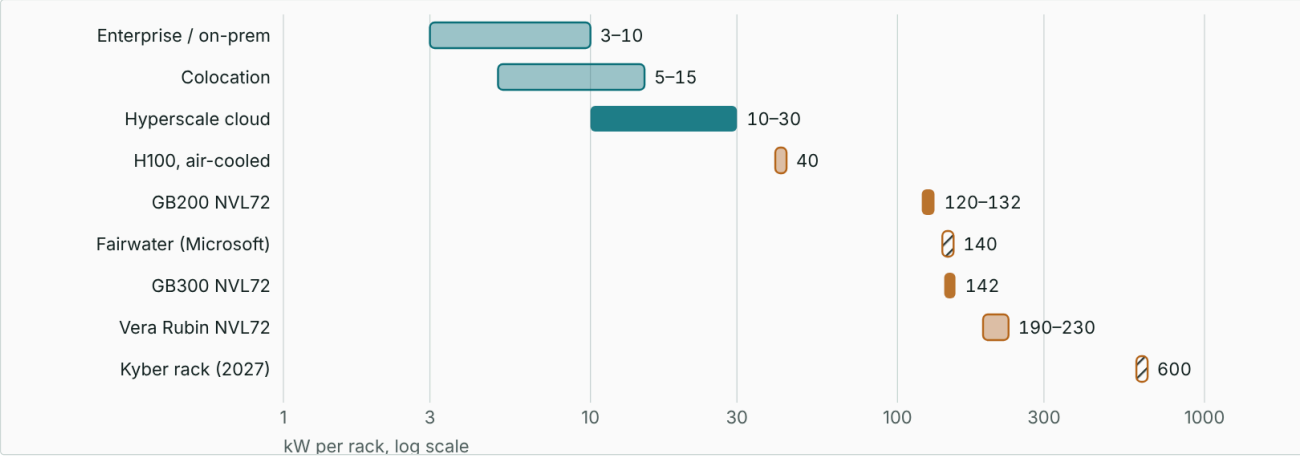


Figure 8.4 How hot is one rack?. kW per rack on a log scale. Solid bars are reported, pale bars estimates, hatched bars design targets. *Source: India Data Centre Atlas, chapter built on this chapter's data.*

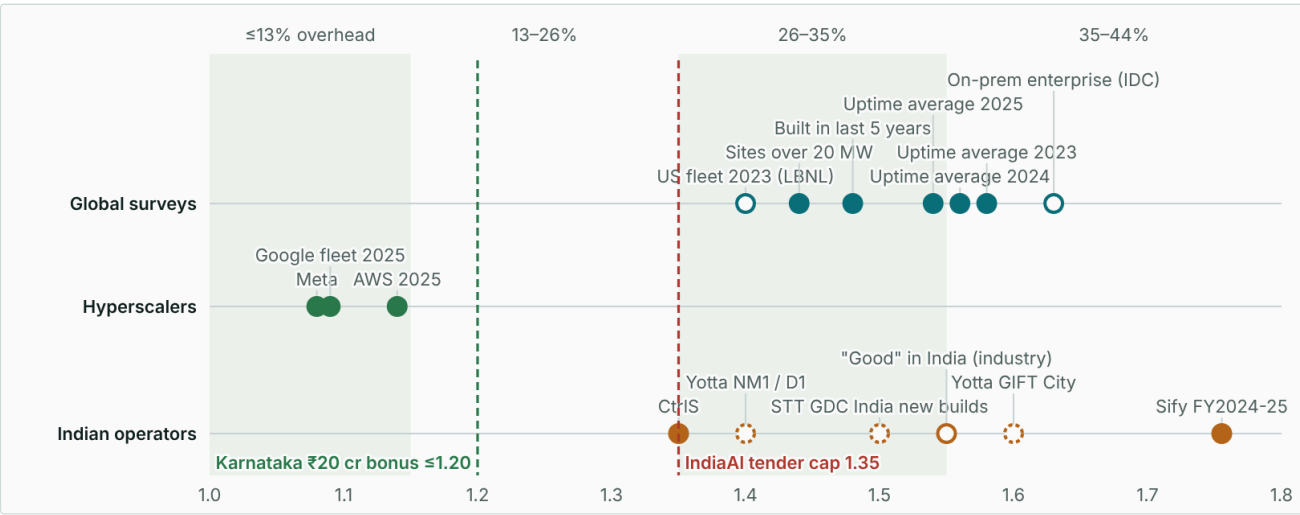


Figure 8.5 PUE: where each kind of operator sits. Lower is better. Hover or tap a dot for the figure and its status. Bands show how much of the bill is overhead rather than computing. *Source: India Data Centre Atlas, chapter built on this chapter's data.*

8.4.4 Zero-water and closed-loop AI designs

- **Microsoft:** From August 2024 all new designs use chip-level closed-loop cooling with "zero water for cooling", avoiding more than 125 million litres per data centre per year; pilots in Phoenix and Mt. Pleasant (Wisconsin) from 2026, with designs coming online from late 2027 [D]. Microsoft and trade press acknowledge that replacing evaporation with mechanical cooling slightly increases PUE.
- **Fairwater (Wisconsin/Atlanta):** >90% of capacity on a closed loop filled once; initial fill equal to ~20 homes' annual use; the remaining ~10% uses outdoor air, switching to water only on the hottest days; Wisconsin is supported by the "second largest water-cooled chiller plant on the planet" [R/D]. QTS describes ~600,000 gallons of one-time fill per building over a 50-year life [D, secondary].
- **Stargate Abilene (OpenAI/Oracle/Crusoe):** closed-loop water cooling with an initial fill of ~8 million gallons of city water, against the city's average use of 22 million gallons per day [R, local utility].
- **xAI Colossus (Memphis):** an xAI engineer said the open loop "needs 3 million gallons a day" [R]; the local utility reports a request for up to 3.7 million gallons/day for the first two sites, and an activist group says the contract caps use at 1.2 million gallons/day [R, conflicting]. A planned ~$80m, 13-million-gallon-per-day wastewater recycling plant broke ground in October 2025, but construction was paused in 2026; the Memphis mayor says xAI committed to restart in Q1 2027 [D, not operating].

8.4.5 Indirect (scope 2) water — the India-specific factor

- **Regulation:** MoEFCC's 2015 notification (amended 2018) caps specific water consumption at 3.5 m³/MWh for existing cooling-tower plants and 3.0 m³/MWh for plants installed after 1 January 2017, with zero liquid discharge; seawater-cooled plants are exempt [R]. FGD adds about 0.3 m³/MWh [E]. Air-cooled condensers cut this to 0.45–0.65 m³/MWh; NTPC's 1,980 MW North Karanpura (June 2025) is India's first supercritical plant with ACC [R].
- **Compliance is unverified:** Down To Earth reports no public plant-level compliance data, and the CEA noted actual consumption of 5–7 m³/MWh in 2012 [R].
- **Grid-average estimate:** With coal/lignite supplying roughly three-quarters of generation (CEA: 1,739 BU total, 255 BU renewable in FY2024-25) and plants at ~3–3.5 m³/MWh, this report uses **2.6 L/kWh** as India's grid-average consumptive water intensity [A]. It excludes hydro reservoir evaporation (which would raise it) and assumes norm compliance (actual may be higher). For comparison, a study of US hyperscalers' US data centres derived 3.4 L/kWh including hydro evaporation [E].

8.4.6 Water per AI query and training run

- **Li, Yang, Islam and Ren (UC Riverside), "Making AI Less Thirsty":** training GPT-3 in Microsoft's US data centres could directly evaporate ~700,000 L on-site and ~5.4 million L in total including electricity water; GPT-3 "drinks" a 500 mL bottle per roughly 10–50 medium-length responses; global AI could account for 4.2–6.6 billion m³ of withdrawal in 2027 [E].
- **Critiques:** The figures depend on assumed WUE/grid water factors and GPT-3-era hardware; Epoch AI and Google's data imply per-response energy 10× or more lower today; the 500 mL headline combines scope-1 and scope-2 water and is often misquoted as on-site use. The 2027 withdrawal projection is not a consumption figure.
- **Google 0.26 mL:** on-site consumption only, at a fleet-wide 2024 WUE, for a median text prompt — it excludes electricity-generation water, which in India would add roughly 0.6 mL at 2.6 L/kWh [A].
- **Mistral 45 mL:** full life cycle, including electricity and manufacturing, as in an LCA.

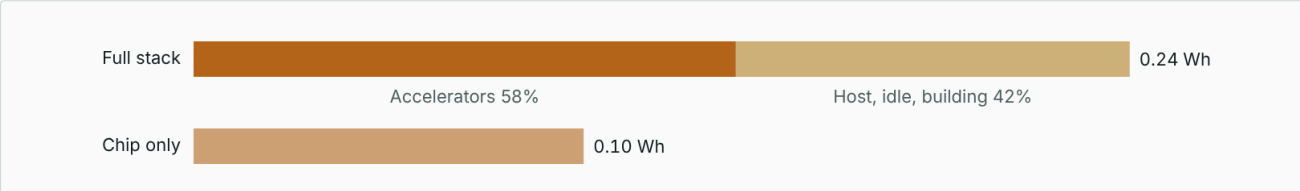


Figure 8.6 Inside one Gemini prompt. Google's 0.24 Wh, split by what draws the power, against the 0.10 Wh "chip-only" method. *Source: India Data Centre Atlas, chapter built on this chapter's data.*

8.4.7 Embedded water in chips (scope 3)

Quantitative disclosure is weak. Li et al. note "very limited data" on embodied water in chipmaking [E]. Mistral's LCA attributes 91% of life-cycle water to training and inference, implying ≤9% to manufacturing, data-centre construction and other phases, while hardware manufacturing dominates material depletion (61%) [R]. Meta separately reported 1,019 ML withdrawn for data-centre *construction* in 2024 [R]. Treat chip-fab water as a known unknown for India's atlas.

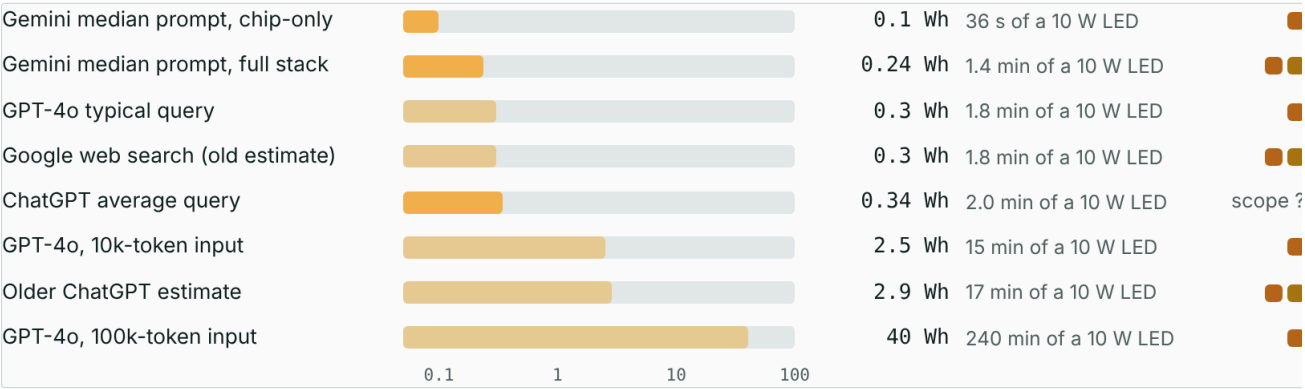


Figure 8.7 One AI prompt: energy and water. The figures disagree by up to ~170×, mostly because each counts different things. The coloured chips show what each number includes. *Source: India Data Centre Atlas, chapter built on this chapter's data.*

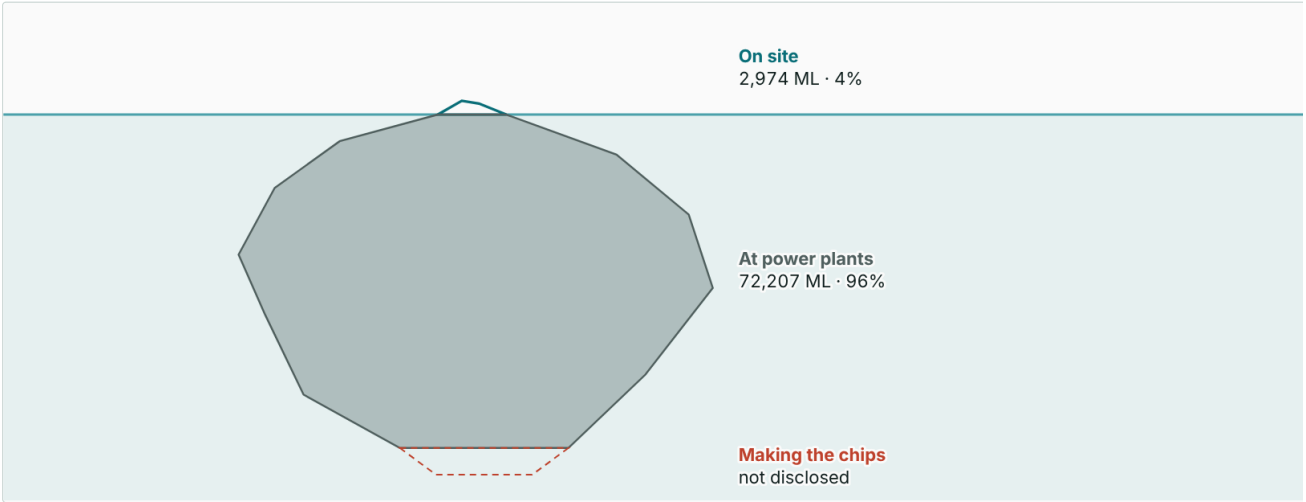


Figure 8.8 The water you don't see. Heights are to scale. Above the line is the water used on site; below is water evaporated to make the electricity, then the water used to make the chips. *Source: India Data Centre Atlas, chapter built on this chapter's data.*

8.5 Does Liquid Cooling for AI Raise or Lower Water Use?

Bottom line: it lowers water per kWh, can lower water per site if paired with dry heat rejection, and raises water per site if paired with cooling towers at higher absolute heat loads.

1. **Why liquid helps.** Direct-to-chip cooling operates at much warmer facility water (LBNL models AI liquid cooling at ASHRAE W45 class), so dry coolers can reject heat without evaporation for more hours; LBNL notes the higher operating temperature "facilitates better water and energy efficiency/performance (less chiller run hours and less adiabatic assist)" [E].
2. **Why it can still be thirsty.** LBNL's base case for liquid-cooled AI is a waterside economiser on cooling towers, which "consumes substantial amounts of water, like all evaporative cooling systems"; it projects US average site WUE *rising* to 0.45–0.48 L/kWh by 2028 partly because of liquid-cooled systems [E].
3. **Absolute heat load.** A 100 MW AI campus rejects ~5× the heat of a 20 MW colo. Even at a lower WUE, if it uses towers (≈1.0 L/kWh) it consumes ~657 ML/yr on-site, versus ~147 ML/yr for a tower-cooled colo [A].
4. **PUE–WUE trade-off.** Evaporative systems are "generally more energy efficient" than air-cooled chillers (LBNL). Microsoft acknowledges a slight PUE penalty for zero-water designs [R]. In India, each extra kWh carries ~2.6 L of grid water and 0.71 kg CO_2, so a dry design that adds 0.1 to PUE costs ~65,700 MWh, ~171 ML of grid water and ~47 kt CO_2 per year at a 100 MW AI campus — but saves ~624 ML of on-site water, a net water win of ~450 ML/yr and a carbon loss [A].
5. **Climate dependence (qualitative).** Evaporative and adiabatic cooling work best when wet-bulb temperature is low; dry coolers work best when dry-bulb is low. - **Mumbai, Chennai, Vizag (coastal, humid):** high wet-bulb most of the year limits both economiser hours and evaporative effectiveness; closed-loop liquid with chillers, or seawater/desalinated make-up, is favoured. Reliance–Meta's 168 MW Jamnagar design uses desalinated seawater [D]. - **Bengaluru (elevated plateau, milder):** the most hours of free/dry cooling among Indian hubs, so the smallest PUE penalty for going waterless — but severe groundwater stress and tanker dependence make on-site water politically costly. - **Hyderabad and NCR (hot, dry pre-monsoon):** adiabatic cooling is very effective in dry heat but peaks in water use exactly when supply is tightest; high summer dry-bulb temperatures degrade dry coolers and push chiller hours. - The Part 12 hourly adiabatic model should quantify these hours by city using typical meteorological year data, as LBNL did for 965 US weather stations.

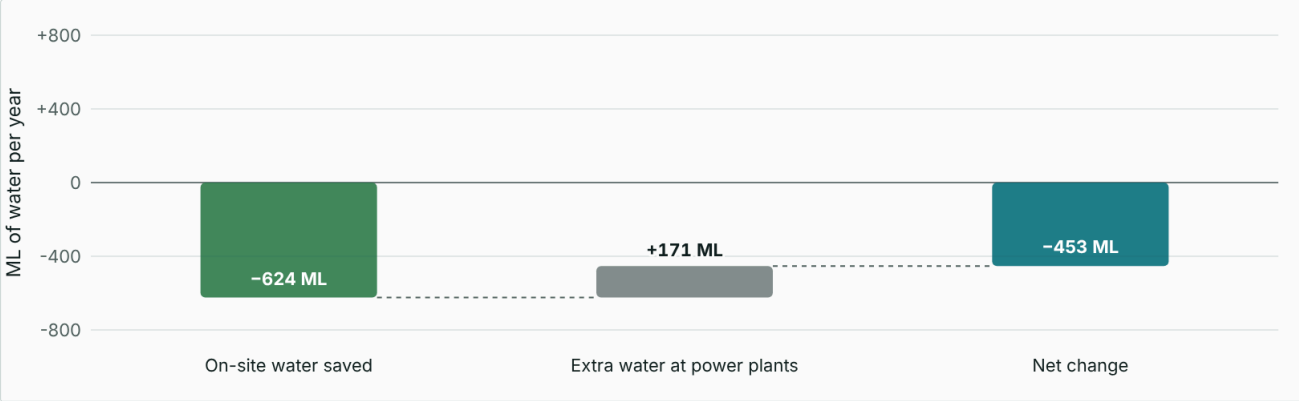


Figure 8.9 Does liquid cooling save water? Run the trade. Liquid cooling lets heat leave through dry coolers instead of evaporating it. That costs extra electricity, and in India extra electricity carries coal-plant water and CO_2. Move the sliders to see who wins. *Source: India Data Centre Atlas, chapter built on this chapter's data.*

8.6 Side-by-Side Comparison

8.6.1 Assumptions [A]

Table 8.7 Assumptions [A]

Parameter	20 MW conventional colo	100 MW AI campus
IT capacity	20 MW	100 MW
Average IT load factor	60% (12 MW)	75% (75 MW) — LBNL training: 80% time at ~70% rated power
IT energy	105,120 MWh/yr	657,000 MWh/yr
Case "wet" (cooling towers)	PUE 1.50, WUE 1.4 L/kWh	PUE 1.20, WUE 1.0 L/kWh (DLC + towers)
Case "dry" (air-cooled / closed-loop)	PUE 1.65, WUE 0.1 L/kWh	PUE 1.30, WUE 0.05 L/kWh (DLC + dry coolers)
Grid water intensity	2.6 L/kWh consumed	2.6 L/kWh consumed
Grid emission factor	0.71 tCO_2/MWh (CEA v21.0, FY2024-25)	0.71 tCO_2/MWh

8.6.2 Per facility (annual)

Table 8.8 Per facility (annual)

Metric	Colo – wet	Colo – dry	AI – wet	AI – dry
Facility energy	157.7 GWh	173.4 GWh	788.4 GWh	854.1 GWh
On-site water	147 ML (0.40 MLD)	10.5 ML (0.03 MLD)	657 ML (1.80 MLD)	33 ML (0.09 MLD)
Indirect (grid) water	410 ML (1.12 MLD)	451 ML (1.24 MLD)	2,050 ML (5.62 MLD)	2,221 ML (6.08 MLD)
Total water	557 ML	462 ML	2,707 ML	2,254 ML
Indirect share of total	74%	98%	76%	99%

Metric	Colo – wet	Colo – dry	AI – wet	AI – dry
CO_2 (location-based)	112 kt	123 kt	560 kt	606 kt

8.6.3 Per MW of IT capacity (annual)

Table 8.9 Per MW of IT capacity (annual)

Metric	Colo – wet	Colo – dry	AI – wet	AI – dry
Facility energy	7,884 MWh	8,672 MWh	7,884 MWh	8,541 MWh
On-site water	7.4 ML	0.53 ML	6.6 ML	0.33 ML
Indirect water	20.5 ML	22.5 ML	20.5 ML	22.2 ML
CO_2	5.6 kt	6.2 kt	5.6 kt	6.1 kt

What this shows. (1) Per MW of IT, AI and conventional sites are similar on energy; AI's impact comes from scale and density, not from worse efficiency. (2) Per site, the AI campus uses ~5× the energy and water. (3) Going dry cuts on-site water by 93–95% but raises CO_2 by 8–10% and grid water by ~10%; total water still falls by 17–20%. (4) In every case, India's grid water exceeds on-site water — so procurement of renewable power (solar PV and wind consume almost no operational water) cuts water more than any cooling choice.

8.7 India Specifics

8.7.1 Capacity and electricity

- **Capacity estimates conflict by definition:** 1.4 GW IT load as of Q2 2025 (S&P Global 451 Research) vs 2.2 GW operational capacity in 2025 (Wood Mackenzie) [E]. 2030 projections range from ~8.12 GW (India Capital Management) and 9–17 GW to 12 GW (Wood Mackenzie) [E]. A Ministry of Power assessment cited by activists puts upcoming data-centre demand at 5,640 MW by FY2031-32 [E, secondhand].
- **AI share:** Wood Mackenzie projects AI-dedicated capacity rising ~24× from 275 MW (2025) to 6,546 MW (2030) — over half of the 2030 total [E]. S&P estimates the IndiaAI GPUs alone add ~2 TWh/yr, ~10% of 2025 data-centre demand [E].

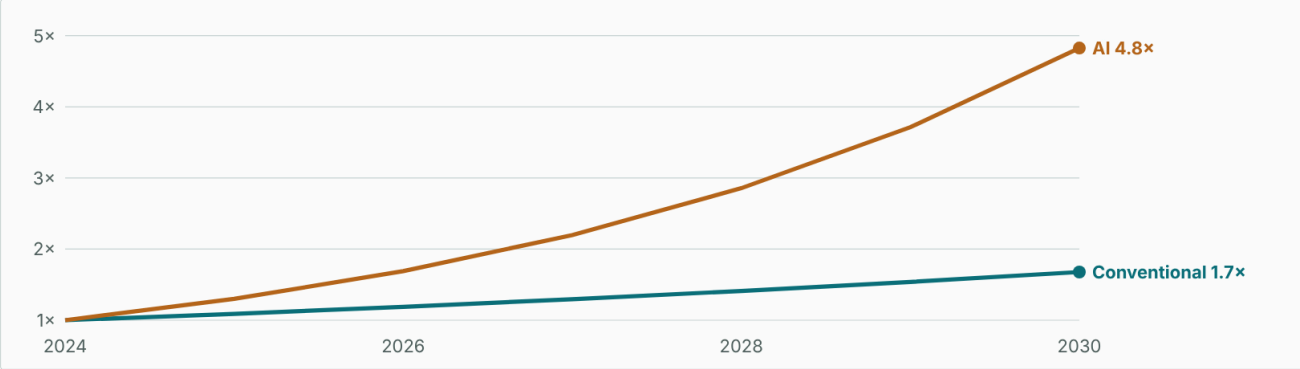


Figure 8.10 Two speeds of growth. IEA projection of electricity use, indexed to 2024 = 1. AI ("accelerated") servers grow about 30% a year, conventional servers about 9%. *Source: India Data Centre Atlas, chapter built on this chapter's data.*

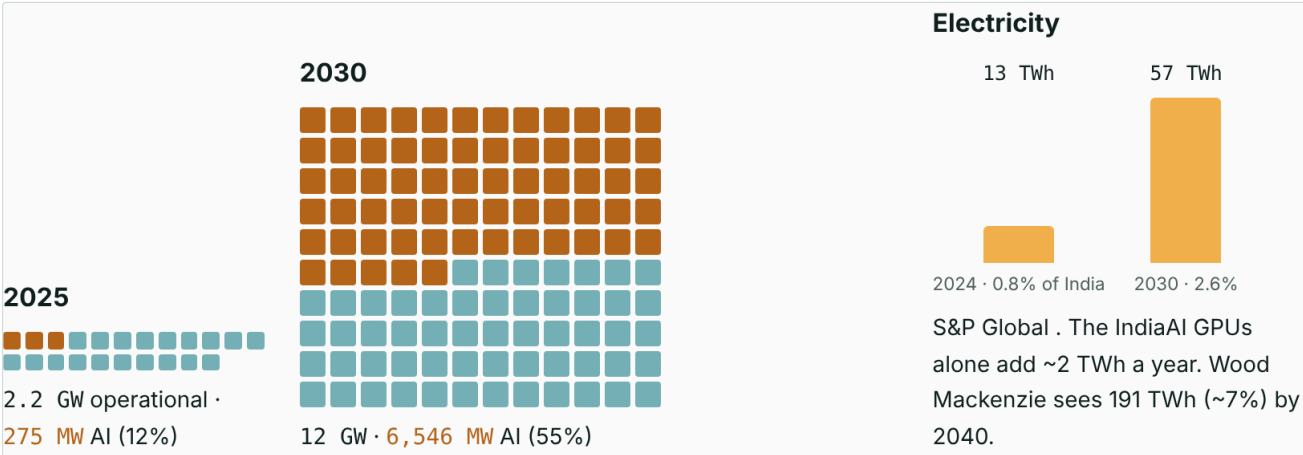


Figure 8.11 India: from mostly conventional to mostly AI. Operational capacity (Wood Mackenzie) and data-centre electricity (S&P Global). Squares are 100 MW each; amber squares are AI-dedicated capacity. *Source: India Data Centre Atlas, chapter built on this chapter's data.*

8.7.2 AI data-centre announcements

Table 8.10 AI data-centre announcements

Project	Scale	Cooling/water disclosure	Status
Google AI hub, Visakhapatnam (with AdaniConneX, Airtel/Nxtra)	~1 GW initially, "multiple gigawatts" later; US$15bn 2026–30; three campuses (Adivivaram 120 acres, Tarluvada 200, Rambilli–Achyutapuram 160)	Google cites "advanced air cooling"; state guaranteed water for 20 years; protests and court challenges over water and an adjacent wildlife sanctuary	Ground broken 28 April 2026 [R]
Reliance–Meta, Jamnagar	168 MW; Reliance Intelligence targets first 120 MW by end-2026	100% renewable; desalinated seawater for cooling; no PUE/WUE published	Announced June 2026 [D]
CoreWeave–AdaniConneX, Taloja (Navi Mumbai)	240 MW	Direct liquid cooling; non-evaporative chilled water	Announced [D]
AdaniConneX pipeline	2.6 GW	PUE-linked US$1.44bn sustainability loan; targets not public	[R/D]
IndiaAI Mission compute	Target 10,000 GPUs; cumulative ~29,000 after second round; Yotta empanelled for 8,192 B200s	Tender clause PUE < 1.35	Rs 10,372 crore mission [R]
Yotta NM1 (Navi Mumbai) / D1 (Greater Noida)	Hyperscale campuses	Design PUE 1.4; no WUE; claims to "avoid reliance on freshwater entirely"	[D]

8.7.3 Water stress and policy

- **Karnataka Sustainable Data Centre Policy 2026-31:** Rs 1,300.45 crore outlay; 1 GW cumulative IT load by 2031; 10–12% of India's AI-ready capacity; incentives linked to power and water efficiency; PUE and WUE monitoring; treated water; mapping of water and energy availability before expansion; Sustainable Data Centre Zones outside Bengaluru [R]. Neither BWSSB nor KSPCB separately tracked data-centre water use before this policy (TNM) [R]; twelve Bengaluru data centres could consume up to 1.964 million litres/day at peak [E, secondary].
- **MeitY/IndiaAI:** the PUE < 1.35 clause, added by a MeitY corrigendum to the IndiaAI GPU tender in the final week of September 2024 (bidders must affirm a PUE "not greater than 1.35"), drew industry objections that legacy sites "will not be able to handle this PUE"; an unnamed executive quoted by The Economic Times said India's data centres run at "about 1.5" [R, press]. SBI's April 2026 sector report notes that "since 2024, MeitY tenders for AI cloud infrastructure require data centres to maintain a PUE below 1.35". It is the first national efficiency requirement on AI compute, but it has no WUE counterpart.
- **Disclosure gap:** Down To Earth (November 2025) found that "Not every company discloses its WUE" and "None of these reports provide data centre-level details" [R].

8.8 Uncertainty, Disputes and Data Gaps

1. **Contested per-query numbers.** 0.24 Wh (Google), 0.34 Wh (OpenAI) and 0.3 Wh (Epoch) are not comparable: medians vs averages, different scopes, undisclosed prompt mixes; none cover reasoning, image or video. Google's per-prompt water excludes scope-2 water and uses market-based carbon.
2. **Capacity definitions.** IT load vs operational vs contracted vs announced MW explains most of the 1.4 vs 2.2 GW and 8 vs 12 GW gaps in India.
3. **WUE definitions.** Withdrawal-based (AWS, Microsoft, Meta) vs consumption-based (LBNL) differ by ~25%; few operators report site-level or seasonal figures; peak-day demand (critical for municipal supply) is almost never disclosed.
4. **Modelled vs measured.** LBNL warns simulated PUEs assume systems "operate as designed", which "is rarely the case", and that its hyperscale WUE simulations may be low.
5. **AI-specific facility data.** No operating PUE or WUE has been published for Fairwater, Stargate Abilene, Meta Prometheus/Hyperion or xAI Colossus; the xAI water figures conflict.
6. **India grid water.** Thermal plant compliance with water norms is unverified; this report's 2.6 L/kWh is an assumption.
7. **Forecast spread.** LBNL's 2028 range spans 325–580 TWh; Indian forecasts differ by factors of 2–3. Treat all forecasts as scenarios, not predictions.
8. **Scope 3.** Embodied water and energy in GPUs, HBM and construction are largely undisclosed.

8.9 Recommendations for the atlas and the series

1. **Report three water numbers per site:** on-site consumption, grid (scope-2) water at 2.6 L/kWh with a sensitivity of 2.0–3.5, and total; never show WUE alone.
2. **Use the per-MW-of-IT table (5.3) as the atlas's default normaliser,** with load factor stated, because per-site comparisons mainly reflect scale.
3. **Treat AI sites in coastal hubs as "dry-or-seawater by default"** and flag any Bengaluru, Hyderabad or NCR proposal relying on evaporative cooling or groundwater as high-risk.
4. **Score policy on WUE as well as PUE:** recommend that MeitY add a WUE or source-water requirement alongside PUE < 1.35, and that Karnataka publish facility-level PUE/WUE.
5. **Pair every water claim with a renewable-sourcing claim:** in India, shifting from coal to solar/wind power cuts more water than any cooling technology.

8.10 Caveats

All site-level comparison figures in Section 5 are this report's own modelled values from stated assumptions, not measurements of any real facility. Hyperscaler PUE/WUE values are self-reported, fleet-wide and partly unaudited. Several figures (Equinix WUE, global IEA water totals, Google 2025 water, Vera Rubin rack power, Karnataka peak water) come via secondary sources and should be checked against primary documents before publication in the atlas. Climate implications for Indian cities are qualitative and should be quantified in the Part 12 hourly model.

Sources cited in this chapter

Organisations and publications named in this chapter's text, in order of how often they are cited. Full details are in the bibliography.

- Lawrence Berkeley National Laboratory, 2024 US Data Center Energy Usage Report
- Google environmental reports, efficiency data and Gemini inference paper
- Meta sustainability reports and data index
- Ministry of Electronics and Information Technology (MeitY), Data Centre Policy draft and IndiaAI
- International Energy Agency (IEA), Energy and AI (2025) and updates
- Microsoft sustainability reports and datacenter design posts
- Amazon Web Services sustainability reporting
- Uptime Institute, Global Data Center Surveys and Tier Standard
- Equinix sustainability reports
- Mistral AI life-cycle assessment (ADEME / Carbone 4)
- Wood Mackenzie India data centre outlook
- Central Electricity Authority (CEA), CO_2 Baseline Database and grid reports
- NVIDIA product specifications and roadmap
- Epoch AI energy estimates
- STT GDC India ESG report
- Sify Infinit Spaces sustainability report
- Yotta Data Services disclosures
- AdaniConneX disclosures
- Reliance Industries announcements
- Li, Yang, Islam and Ren, "Making AI Less Thirsty"
- Nxtra by Airtel disclosures
- CtrlS disclosures
- Down To Earth reporting
- Bangalore Water Supply and Sewerage Board
- Ministry of Power, orders on ISTS waiver and green open access
- Ministry of Environment, Forest and Climate Change, thermal plant water norms and EIA
- ASHRAE TC 9.9 thermal guidelines
- State Bank of India sector report on data centres (April 2026)
- De Vries-Gao, Patterns (2025)
- Electric Power Research Institute
- Hewlett Packard Enterprise product data
- DatacenterDynamics reporting
- Omdia server market data

Web sources: the 96 web pages consulted for this chapter are listed with their full addresses in Appendix B.2.

PART III · ECONOMICS AND SUSTAINABILITY

Chapter 9
Unit economics: what a data centre costs to build and run

WHAT THIS CHAPTER COVERS

This chapter estimates what a data centre costs in India, from land, buildings, electrical and cooling plant, backup systems and networking to soft costs, interest during construction and contingency. It compares India with global benchmarks, builds operating cost models for electricity, water, maintenance, staff, networks, tax and insurance, and turns capital and operating costs into unit economics: cost per megawatt, per rack, per kilowatt-hour of IT energy and per GPU-hour. It ends with sensitivity analysis, a comparison of business models and markets, and recommendations.

KEY TAKEAWAYS

1. A Tier III colocation data centre in India costs about ₹56 crore per megawatt of IT load to build, excluding land, or about ₹61 crore including land, which is about US$6.2 million per megawatt.
2. India is among the cheapest places in the world to build: Turner & Townsend's 2025 index places Mumbai at US$6.64 per watt, 51st of 52 markets, against US$14.5 in Singapore and US$15.2 in Tokyo.
3. Electrical systems take about 39 to 46% of facility cost, cooling about 18 to 23% and land only about 4 to 8%. For an AI campus, GPUs and networking add about ₹266 crore per megawatt, about 3.7 times the cost of the facility.
4. Electricity is about 74% of a colocation operator's operating costs but only about 39% of the full cost of about ₹29.7 per kilowatt-hour of IT energy, because recovering capital is the larger part. A new mid-size colocation hall earns roughly its cost of capital at current Mumbai rents.
5. A fully loaded H100 GPU costs about ₹177 per hour at 80% utilisation. The IndiaAI rate of ₹150 per hour covers cost only at about 86 to 96% utilisation, and utilisation matters more than any other variable in GPU economics.

How this chapter connects to the others. Chapter 10 uses these costs to test the economics of going green, and Chapter 14 projects cumulative capital needs to 2035.

A Tier III colocation data centre in India costs about ₹56 crore per MW of IT load to build (≈US$6.2M/MW, excluding land) or about ₹61 crore with land. A 50 MW hyperscale build-to-suit costs about ₹39–42 crore/MW. A 100 MW AI campus costs about ₹72 crore/MW for the facility, but roughly ₹340 crore/MW once the GPUs are included. For AI, the building is only about a fifth of the bill, and electricity is under 5% of the cost of a GPU-hour.

9.1 Key Findings

1. **India's cost advantage is real but narrower than headline indices suggest.** Construction is about 45–55% cheaper than in Singapore, Tokyo or Silicon Valley. Rents are proportionally lower too: Houlihan Lokey puts India at about US$80–100/kW/month, against US$160–225 in the US and US$300–475 in Singapore. India's rent-to-capex ratio is therefore the thinnest of the three markets. India's case rests on demand growth and scale, not fat unit margins.
2. **For colo and hyperscale, the facility is a capital-recovery business.** Of every ₹100 of lifetime cost per IT-kWh in a 20 MW colo, about ₹47 is capex recovery, ₹39 electricity and ₹14 other opex. Utilisation, cost of capital and build cost per MW move returns more than the tariff, because electricity is usually passed through to tenants.
3. **For AI, the GPU is the business.** In a GB300-class campus at 80% utilisation, a ₹308 GPU-hour splits into roughly ₹242 of GPU and network capex (79%), ₹31 of facility capex (10%), ₹20 of other opex (6.5%) and ₹14 of electricity (4.6%). A ₹1/kWh change in tariff moves the GPU-hour by only about ₹3. A 20-point change in utilisation moves it by ₹95–195.
4. **Green power is now roughly at par with southern grid tariffs, not a big saving.** SECI's 1 GW round-the-clock tranche V auction (August 2026) cleared at ₹5.25–5.26/kWh across seven winners, fixed for 25 years. Delivered to a data centre, that becomes about ₹6.25–6.75 after transmission and losses, with grid demand charges still payable. That is cheaper than Maharashtra (₹7.27) or Tamil Nadu (₹7.50) energy charges. It is close to Telangana at 132 kV (₹6.65) and Karnataka (₹6.70). Third-party open access that pays cross-subsidy surcharge (CSS) is uneconomic in Tamil Nadu (CSS ₹1.99–2.06/kWh).
5. **Valuations price Indian platforms near replacement cost plus pipeline.** Nxtra's March 2026 raise valued it at US$3.1 billion post-money, about US$2.1 billion pre-money, on about 300 MW. That is roughly US$7M (₹63 crore) per operating MW pre-money, close to our ₹56–61 crore/MW replacement cost. Houlihan Lokey's Indian cap rates of 7.5–8.5% imply about 12–13× EBITDA for stabilised assets.

9.2 CAPEX (₹)

9.2.1 Global and Indian benchmarks

Turner & Townsend Data Centre Construction Cost Index 2025–26 (52 markets, traditional air-cooled cloud data centre, US$ per watt; construction only):

Table 9.1 Global and Indian benchmarks

Market	US$/W	₹ crore/MW (at ₹90/$)	Comment
Tokyo	15.2	137	Most expensive

Market	US$/W	₹ crore/MW (at ₹90/$)	Comment
Singapore	14.5	131	2nd
Zurich	14.2	128	3rd
Silicon Valley	13.3	120	Largest US premium
New Jersey	12.9	116	
Oslo	12.4	112	7th
London	12.0	108	
Frankfurt	11.6	104	
Atlanta	9.9	89	
Phoenix / Columbus	9.8	88	
Charlotte	9.5	86	
Mumbai	**6.64**	**60**	51st of 52

Turner & Townsend recorded 5.5% global cost inflation for traditional data centres in 2025, down from 9.0% in 2024; it links the slowdown to construction overall, where its 2025 market intelligence report showed average inflation of 4.2%, and 60% of its survey respondents expect a 5–15% rise in 2026. It notes that cost data for AI-specific builds remains too thin for a global benchmark. Its Northern Virginia, Chennai and Hyderabad figures were not captured in this research. Treat Chennai and Hyderabad as broadly in line with or slightly below Mumbai, which is our estimate.

Other benchmarks

- **Cushman & Wakefield, Asia Pacific Data Centre Construction Cost Guide 2026.** Total construction across India's five main cities is US$5.9–9.0M/MW (about ₹53–81 crore at ₹90/$). The guide's April 2026 release gives India at US$5.9M (low), US$7.4M (mid) and US$9.0M (high) per MW, up 3.8% year-on-year; Cushman & Wakefield's Andrew Green notes that construction cost inflation "diverges sharply, with some markets seeing increases above 15% while others remain below 5%."
- **Cushman & Wakefield, US 2026 guide.** US/Canada builds average US$17.6M/MW, and per-MW construction costs are up 21% since the previous edition. Powered land in primary US markets averages US$584,000 per MW.
- **Houlihan Lokey India edition, December 2025** (drawing on Cushman & Wakefield, CBRE and others). Development cost is US$5.5–8.5M/MW in India, US$9–12M in the US and US$9–14.5M in Singapore.
- **SBI sector report, April 2026 (prior reports).**
- Tier III air-cooled: ₹50–55 crore/MW (shell plus M&E, excluding IT).
- AI, immersion or Tier IV: up to ₹75 crore/MW.
- Large campuses: ₹35–45 crore/MW.
- **Jefferies** has cited about US$4–5M per MW for Indian capacity, which is at the low end and likely reflects large-campus or powered-shell scope.
- **Anant Raj** reports about ₹26 crore/MW for brownfield conversion of buildings it already owns. That is not a greenfield benchmark.

The indices cover different scopes. Turner & Townsend covers construction only. Cushman & Wakefield's development figures include land. SBI excludes IT. All of them land Indian Tier III facilities in the **₹50–65 crore/MW** band before land.

9.2.2 Land and buildings

Land prices per acre in Indian data centre hubs. Transaction-based figures are labelled. The rest are estimates or come from low-quality sources.

Table 9.2 Land and buildings

Hub	₹ crore/acre	Evidence
Navi Mumbai (TTC MIDC, Airoli, Mahape, Ghansoli)	15–40 greenfield; up to ~80 for in-fill with structures	STT GDC paid ₹282 crore for 14,093 m² (~3.5 acres) with an existing 6,828 m² building (May 2026), ≈₹81 crore/acre. Brookfield paid ₹600 crore for 30 acres in Ghansoli (2021), ≈₹20 crore/acre.
Thane / Palava (MMR periphery)	12–35	AWS: ~40–45 acres at Palava for ₹500–600 crore (≈₹12–14 crore/acre). Amazon: 54 acres in Thane for ₹1,870 crore (≈₹35 crore/acre).
Chennai (Ambattur, Oragadam, Siruseri)	8–15 Ambattur; 3–8 Oragadam (estimate)	A secondary blog source gives ₹8–15 crore/acre; treat as indicative.
Hyderabad	2–6	Microsoft bought 48 acres for ₹267 crore (2024), ≈₹5.6 crore/acre.
Noida / Greater Noida	3–10 (estimate, not verified)	Authority allotment rates vary; check YEIDA and GNIDA schedules.
Bengaluru Whitefield	10–25 (estimate, not verified)	
Bengaluru Devanahalli / Hoskote	4–10 (estimate, not verified)	
Pune (Hinjewadi, Chakan, Talegaon)	5–15 (estimate, not verified)	

Acres per MW. Cushman & Wakefield's Asia Pacific scenarios for a 50 MW building are:

- 20 acres for a low-rise hyperscale build: **0.4 acre/MW**
- 10 acres for a multi-level build: **0.2 acre/MW**
- 5 acres for a high-rise operator build: **0.1 acre/MW**

Mumbai-area operators usually build multi-storey at 0.1–0.25 acre/MW. Hyderabad and Chennai campuses run at 0.3–0.5 acre/MW.

Land share of capex.

- Navi Mumbai: 0.25 acre/MW × ₹20 crore = ₹5 crore/MW, about **8%** of a ₹61 crore/MW total.
- Hyderabad: 0.4 × ₹6 crore ≈ ₹2.4 crore/MW, about **4–5%**.
- Extreme Mumbai in-fill: 0.1 × ₹80 crore = ₹8 crore/MW, about **12%**.

In absolute terms, Mumbai land per MW (₹5–8 crore) is similar to US primary-market powered land (US$584k/MW ≈ ₹5.3 crore). The big India–US gap is in construction, not land.

Construction cost per sq ft (estimates).

- Indian data centres typically need about 10,000–12,000 sq ft of gross built-up area per MW of IT, more for low-density colo.
- Shell and core for data-centre-grade structures runs about **₹5,500–8,000/sq ft**. These have high floor loading (~2,500 kg/m²) and about 18 ft clear height, the specs Houlihan Lokey cites for new AI-ready builds. That works out to about **₹6–8.5 crore/MW**.
- The total fitted facility is about ₹45,000–55,000/sq ft of built-up area.

9.2.3 Servers, storage and GPUs (IT capex)

Table 9.3 Servers, storage and GPUs (IT capex)

Item	Price (USD)	India landed (₹, at ₹90/$)	Notes
CPU server, dual-socket 1U/2U cloud node	US$9–22k (estimate)	₹8–20 lakh	Not verified; varies hugely with configuration
Storage	~₹45–90 lakh per PB raw (estimate)		Not verified
H100 SXM card	US$25–40k	₹22–36 lakh	Multiple 2026 price guides
H100 HGX 8-GPU server	**US$250–320k**	**₹2.25–2.9 crore**	
H200 HGX 8-GPU server	~US$300–330k (estimate)	~₹2.7–3.0 crore	H200 cards quoted at US$30–35k
B200 HGX 8-GPU server	**US$400–500k** (DGX B200 lists ~US$515k)	**₹3.6–4.5 crore**	
GB200 NVL72 rack	**US$3.0M** (HSBC); US$3.1M bare / US$3.9M all-in (SemiAnalysis)	₹27–35 crore	72 GPUs, ~120–135 kW
GB300 NVL72 rack	**US$3.7–4.0M** (Loop Capital) vs **US$6.0–6.5M** (Tom's Hardware supply chain, March 2026); ~US$5.0M purchase / US$5.7M deployed (Data Gravity)	₹33–58 crore	Estimates disagree; Supermicro datasheet lists 132–140 kW

India landed cost.

- Servers and GPU systems generally attract **0% basic customs duty under the WTO Information Technology Agreement (ITA-1)**. Since basic customs duty is zero, the 10% social welfare surcharge, which is levied on that duty, is also zero.
- **18% IGST** is charged on the CIF value and is **fully creditable** for GST-registered buyers.
- The landed premium is therefore mainly freight, insurance and clearance (~1–2%), plus the **working-capital cost of IGST**.
- For our 100 MW AI campus, IGST paid at import is about ₹4,800 crore. Against about ₹2,100 crore a year of output GST on GPU revenue, that takes about 2+ years to recover. At about 9.5% cost of debt, the carrying cost is roughly ₹450–500 crore. Neoclouds selling to foreign customers (zero-rated exports) would have to use the refund route instead.
- Customs classification disputes do occur. Servers under HSN 8471 also need import authorisation.

IT capex per MW.

- **Cloud (CPU-heavy) halls:** about ₹90–180 crore/MW of IT capex, or **1.5–3× the facility**. This is an estimate.
- **AI halls:**
- GB300 NVL72 at ~135 kW per rack: about 530 GPUs per MW of IT.
- At about US$65–70k all-in per GPU (including the scale-out fabric), that is **₹265–310 crore/MW, about 3.5–5× the facility**. Our Case C comes out at 3.7×.
- **Back-end networking** (InfiniBand or Spectrum-X Ethernet, switches and optics) is about **10–15% of cluster cost**. We use 12%.

Refresh and depreciation.

- CPU servers are depreciated over 5–6 years by most hyperscalers. Some shortened lives for AI-era servers back to 5 years in 2025. That comes from company disclosures not re-verified in this research.
- GPUs are contested. Accounting lives run 5–6 years, but critics argue the economic life is 3–4 years, because each Nvidia generation (Hopper → Blackwell → Rubin) cuts the market rental price of the previous one.
- We model 5 years as the base and show 4 and 6 years as sensitivities. For an H100 at 80% utilisation, the cost per GPU-hour moves from ₹204 (4 years) to ₹177 (5 years) to ₹159 (6 years).

9.2.4 Electrical infrastructure

The electrical system includes:

- HV substation (66, 110 or 220 kV for colo; 220 or 400 kV for 100 MW+ AI campuses)
- Power transformers and MV/LV switchgear
- 2N UPS (increasingly lithium-ion)
- PDUs, RPPs and overhead busway

Multiple cost benchmarks put electrical at **40–50% of hard construction cost**. In our models it is 39% of the all-in facility cost and about 46% of hard cost. That is ₹22 crore/MW for Tier III colo, ₹15.5 crore/MW for hyperscale and ₹28 crore/MW for AI.

Cushman & Wakefield's US guide shows big supply-chain price rises since Q3 2021: switchgear +50%, UPS +48%, generators +45%, transformers +44%, chillers +40%. Imported or long-lead electrical gear is the main way FX feeds into Indian capex.

9.2.5 Cooling infrastructure

- **Air-cooled Tier III:** chillers (air-cooled or water-cooled with cooling towers or adiabatic coolers), CRAH or fan walls, piping and controls. Cost is about ₹8–10 crore/MW, **15–20%** of the facility.

- **Liquid-cooled AI:** coolant distribution units (CDUs), secondary loops, dry coolers and trim chillers for residual air. Cost is about ₹15–18 crore/MW, **~23%** of the facility.
- Turner & Townsend's own November 2025 release says it has, for the first time, identified a 7–10% construction cost premium between traditional and AI data centres in the US, comparing projects of similar IT capacity.
- Morgan Stanley estimates liquid-cooling content in a GB300 NVL72 rack at about US$49,860 (+20% versus GB200), as reported by Tom's Hardware. That content sits in IT capex, not the building.

9.2.6 Backup systems

- **Diesel generators** are typically N+1 for Tier III and 2N for Tier IV. Fuel storage covers 24–48 hours.
- Installed cost is about **₹1,500–2,500/kVA**, including synchronising panels, acoustic enclosures and fuel systems. This is an estimate, not verified.
- A 20 MW IT colo at PUE 1.5 needs about 42 MVA of N+1 DG. At about ₹2,000/kVA that costs about **₹84 crore (₹4.2 crore/MW)**.
- **Lithium-ion UPS batteries** cost more upfront than VRLA but last about 10–15 years against 4–5, and have a smaller footprint.
- **Battery energy storage (BESS)** is increasingly specified for AI campuses. Its job is to smooth synchronous GPU load swings and support green-power firming. We include ₹6 crore/MW for DG plus BESS in Case C.

9.2.7 Networking

- **Fibre and meet-me rooms.** Carrier-neutral meet-me rooms, diverse fibre entries and structured cabling cost about ₹0.3–1 crore/MW of facility capex. Cross-connect revenue makes this one of the highest-margin lines in retail colo.
- **AI fabrics.** InfiniBand or Ethernet back-end networking is **10–15% of cluster cost**. It is part of IT capex, not the building.
- **Subsea proximity.** Mumbai and Chennai host India's cable landing stations. This is a key reason Mumbai holds about 568 MW of the country's 1.4–1.8 GW, per Savills via Houlihan Lokey. Low-latency interconnection there supports retail colo pricing.

9.2.8 Soft costs, interest during construction and contingency

Table 9.4 Soft costs, interest during construction and contingency

Item	Typical share	Our base
Design, permits, project management (PMC), commissioning	5–8% of hard cost	5–6%
Contingency	5–10%	5–6.5%
Interest during construction (IDC)	18–24-month build; ~70% debt at ~9.5%; average 50% drawn	≈5–6% of facility cost

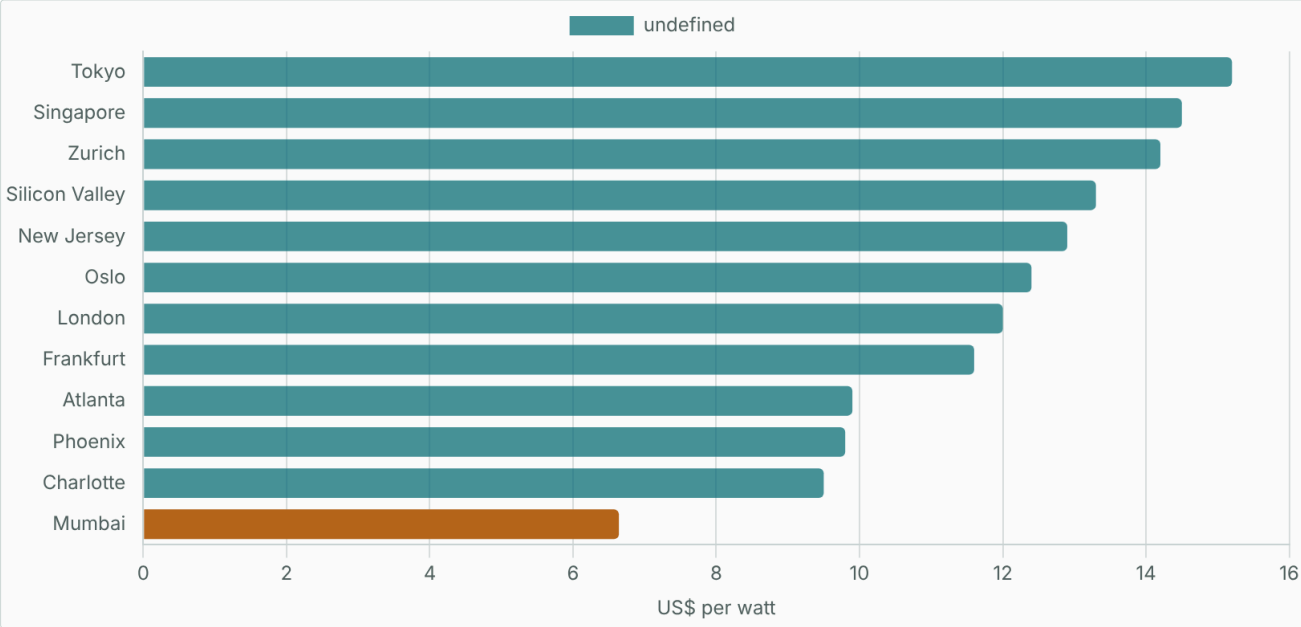


Figure 9.1 Build cost around the world. Traditional air-cooled data centre, US$ per watt, construction only *Source: India Data Centre Atlas, chapter built on this chapter's data.*

9.2.9 Breakdown tables for the three cases

Common assumptions: ₹90/US$; 2026 prices; IT MW is critical IT load.

(a) 20 MW Tier III colocation (Navi Mumbai/Chennai-type; air-cooled chilled water; 2N electrical, N+1 DG; multi-storey on 5 acres at ₹20 crore/acre)

Table 9.5 Breakdown tables for the three cases

Item	₹ crore/MW	₹ crore (20 MW)	Share of facility capex
Land (0.25 acre/MW)	5.0	100	—
Shell and core, civil/structural	8.5	170	15.2%
Electrical (substation share, transformers, switchgear, 2N UPS, PDUs, busway)	22.0	440	39.2%

Item	₹ crore/MW	₹ crore (20 MW)	Share of facility capex
Cooling (chillers, towers/adiabatic, CRAH, piping)	10.0	200	17.8%
Backup (DG N+1 ~42 MVA, fuel)	4.2	84	7.5%
Fire, BMS/DCIM, security, low-voltage	2.0	40	3.6%
Networking (meet-me rooms, fibre entry, cabling)	1.0	20	1.8%
Hard cost subtotal	**47.7**	**954**	85.0%
Design, permits, PMC (~6%)	3.0	60	5.3%
Contingency (~5%)	2.5	50	4.5%
Interest during construction	2.9	58	5.2%
Facility total (excl. land)	**56.1**	**1,122**	100%
Total incl. land	**61.1**	**1,222 (≈US$136M)**	Land = 8%

The ₹56.1 crore/MW facility cost equals ₹561/W, or US$6.2/W. That is consistent with Turner & Townsend's Mumbai US$6.64/W and the upper half of SBI's ₹50–55 crore/MW once soft costs and IDC are added.

(b) 50 MW hyperscale build-to-suit (Hyderabad/Chennai-type; developer delivers shell and M&E, tenant supplies IT; 20 acres at ₹8 crore/acre)

Table 9.6 Breakdown tables for the three cases

Item	₹ crore/MW	₹ crore (50 MW)	Share of facility capex
Land (0.4 acre/MW)	3.2	160	—
Shell and core	5.5	275	14.0%
Electrical (incl. 220 kV substation)	15.5	775	39.5%
Cooling	7.5	375	19.1%
Backup (DG)	3.3	165	8.4%
Fire, BMS, security	1.4	70	3.6%
Networking / meet-me rooms	0.3	15	0.8%
Hard cost subtotal	**33.5**	**1,675**	85.5%
Soft costs (5%)	1.7	85	4.3%
Contingency (5%)	1.7	85	4.3%
Interest during construction	2.3	115	5.9%
Facility total	**39.2**	**1,960**	100%
Total incl. land	**42.4**	**2,120 (≈US$236M)**	Land = 7.5%

A **powered-shell-only** variant would cost roughly **₹12–15 crore/MW** (estimate). That covers the building, substation and utility yard, with the hyperscaler fitting its own M&E. The returns profile is then closer to industrial real estate.

(c) 100 MW AI campus including GPUs (liquid-cooled; GB300 NVL72-class at ~135 kW/rack; Hyderabad/AP/Devanahalli-type land at ₹5 crore/acre)

Table 9.7 Breakdown tables for the three cases

Item	₹ crore/MW	₹ crore (100 MW)	Share
Land (0.3 acre/MW, 30 acres)	1.5	150	0.4% of total
Shell and core	7.0	700	9.7% of facility
Electrical (400/220 kV substation, high-ampacity busway, UPS)	28.0	2,800	38.6%
Cooling (CDUs, DLC loops, dry coolers, trim chillers)	17.0	1,700	23.4%
Backup (DG N+1 + BESS)	6.0	600	8.3%
Fire, BMS	2.0	200	2.8%
Network infrastructure (meet-me rooms, fibre)	1.0	100	1.4%
Soft costs / contingency / IDC	3.5 / 4.0 / 4.0	350 / 400 / 400	15.8%
Facility total (excl. land)	**72.5**	**7,250**	21.3% of total
GPU racks: ~590 × GB300 NVL72 at ~US$4.2M	223	22,300	
Back-end fabric (~12% of cluster)	32	3,240	
Storage, front-end, CPU, management	11	1,080	
IT total (42,480 GPUs)	**266**	**26,620**	78.2% of total
Campus total	**~340**	**~34,020 (≈US$3.8 billion)**	IT/facility = 3.7×

Per GPU, the cost is about **US$70k of IT** plus about **₹17.4 lakh (US$19k) of facility and land**. The rack price is the biggest uncertainty. At the US$6.0–6.5M Tom's Hardware figure for GB300, the campus would cost about ₹45,000 crore.

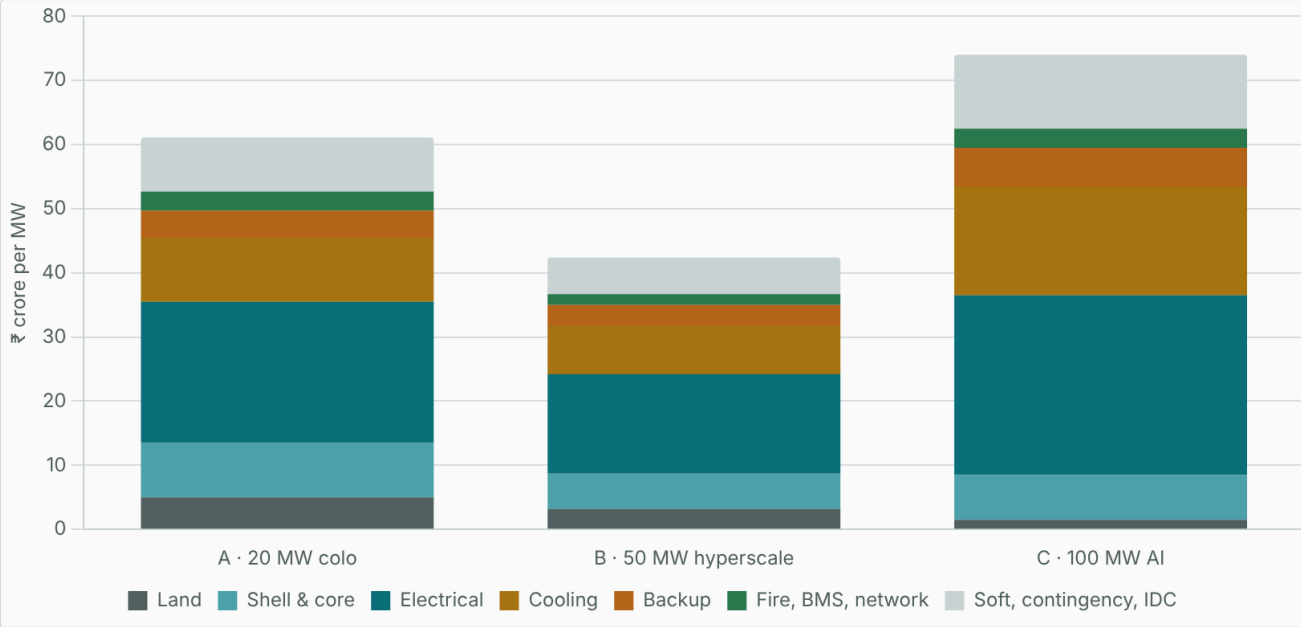


Figure 9.2 Capex per MW, three cases. ₹ crore per MW of IT load *Source: India Data Centre Atlas, chapter built on this chapter's data.*

9.2.10 India vs global: what the gap means

Picture a bar chart of US$/W. Tokyo, Singapore and Zurich stand at about 14–15. London and Frankfurt are at 11.6–12. The newer US hubs (Atlanta, Phoenix, Charlotte) are at 9.5–9.9. Mumbai's bar is less than half Singapore's, at 6.64.

The cheaper build does not translate one-for-one into better returns, because Indian rents are also low (section 3.1). Cushman & Wakefield's 2026 guide puts India's construction cost inflation at 3.8% year-on-year, below Turner & Townsend's 5.5% global figure, with some Asia Pacific markets rising above 15%, so India's cost lead is holding rather than narrowing.

9.3 OPEX (₹)

9.3.1 Electricity

State HT/EHT tariffs relevant to data centres (FY2025–26 orders; energy charge plus demand charge; excludes electricity duty, ToD and fuel adjustments):

Table 9.8 Electricity

State	Category / voltage	Energy ₹/kWh	Demand ₹/kVA/month	DC-specific relief
Karnataka	HT industrial (prior report)	~6.70	~365	See prior report
Maharashtra (MSEDCL)	HT-I(A) Industry	7.27	560	MERC: 10% wheeling-charge discount for data centres sourcing green power via open access or green tariff. State policy: ₹1/unit tariff subsidy for 5 years for certain new data centre units. MERC's March 2025 multi-year order was later subject to review; verify the current schedule.
Tamil Nadu (TNERC)	HT-I	7.50	608 (up from 589)	100% electricity tax subsidy for 5 years for data centres. Green tariff = 110% of category tariff. ToD: +25% peak, −5% night.
Telangana (TGERC)	HT-I(A) at 132 kV / 33 kV / 11 kV	6.65 / 7.15 / 7.65	500	Data centres pay 60% (wind) / 70% (solar) of conventional cross-subsidy surcharge.
Uttar Pradesh, Andhra Pradesh, Gujarat	HT industrial	Not verified in this research (typically ₹7–8)	—	Check UPERC, APERC and GERC orders. Andhra Pradesh hosts Google's planned US$6 billion, 1 GW project.

Green open access and RE PPAs.

- **Firm round-the-clock RE (government auctions):**
- SECI 1 GW round-the-clock tranche V (issued March 2026, results August 2026): **₹5.25–5.26/kWh** across seven winners, with Juniper Green taking the largest share (230 MW at ₹5.26), fixed for 25 years.
- SECI RTC-IV, adopted by CERC in November 2025: ₹5.06–5.07.
- SJVN FDRE-3: ₹4.82–4.91.
- SECI's 4-hour assured-peak tender (FDRE-IX): ₹5.99.
- **Plain solar or hybrid C&I PPAs:** about ₹3.0–4.5/kWh at the busbar.
- **Charges stacked on top:**
- **Wheeling:** about ₹0.60 (Maharashtra), ₹0.63 (Gujarat), ₹1.03 (Tamil Nadu).
- **Cross-subsidy surcharge:** Tamil Nadu ₹1.99 (FY26) and ₹2.06 (FY27); Karnataka ₹2.08.
- **Additional surcharge:** Tamil Nadu ₹0.54 for December 2024–March 2025.
- **Banking charges** and **ISTS charges**, which rise as the ISTS waiver phases out.

- **Group captive** structures (with 26% equity in the generator) avoid cross-subsidy and additional surcharge. Landed costs are then about **₹4.5–6.5/kWh**, against about ₹5.5–7.0 for third-party open access in Maharashtra.

The practical conclusion: RE PPAs save about ₹0.8–1.5/kWh against Maharashtra and Tamil Nadu grid supply. They save little against Telangana or Karnataka EHT supply. Their main value is decarbonisation plus a 25-year price lock.

Annual electricity cost at full design load (₹ crore/yr; energy plus demand charges only):

- Case A: 20 MW IT × PUE 1.5 = 30 MW, 262.8 GWh, 33 MVA contract demand
- Case B: 50 MW × PUE 1.4 = 70 MW, 613 GWh, 77 MVA
- Case C: 100 MW × PUE 1.25 = 125 MW, 1,095 GWh, 138 MVA

Table 9.9 Electricity

Tariff	Case A (20 MW colo)	Case B (50 MW hyperscale)	Case C (100 MW AI)	Effective ₹/kWh
Karnataka (6.70 + 365)	190.5	444.5	794.1	~7.25
Maharashtra (7.27 + 560)	213.3	497.5	888.8	~8.1 (before ₹1 subsidy)
Tamil Nadu (7.50 + 608)	221.2	516.1	922.0	~8.4 (≈8.9 with ToD for a flat load)
Telangana 132 kV (6.65 + 500)	194.6	454.0	811.0	~7.4
Green RTC via group captive (~6.5 + grid demand ~500)	~190.6	~444.8	~794.6	~7.25

Turner & Townsend cites a Mumbai tariff of 6.71 US cents/kWh (about ₹6.0 at its exchange rate), against 10.09 US cents in Shanghai.

Share of opex. Electricity is **74%** of total opex in our stabilised colo case. In Case B it is about 84% if the pass-through is included. That is above the 50–70% often quoted, because most other Indian opex lines (labour, land, property tax) are cheap. Houlihan Lokey puts Indian electricity at about US$0.7–0.9M per MW per year (₹6.3–8.1 crore), within total opex of about US$1.0–1.2M/MW/yr. Our Case A is ₹6.45 crore/MW of electricity within ₹8.7 crore/MW of opex.

9.3.2 Water

The prior water report gave about 0.6–1.8 MLD for evaporative cases. Tariff anchors are BWSSB industrial at about ₹99/kL and treated water at ₹10–20/kL. MIDC, CMWSSB and HMWSSB tariffs were not re-verified here.

Table 9.10 Water

Case	Assumed draw	kL/yr	Bill at ₹99/kL	Bill at ₹15/kL (treated)
A: 20 MW colo (evaporative)	0.3 MLD	109,500	₹1.08 crore	₹0.16 crore
B: 50 MW hyperscale	0.6 MLD	219,000	₹2.17 crore	₹0.33 crore
C: 100 MW AI (hybrid; DLC + dry coolers can be far lower)	1.0 MLD (up to 1.8)	365,000 (657,000)	₹3.6 (6.5) crore	₹0.55 (0.99) crore

Water is **under 2% of opex**. Its cost is about the licence to operate, not the bill.

9.3.3 Cooling opex

Cooling energy sits inside the electricity line. In Case A it accounts for about a third of non-IT power, roughly ₹20–25 crore a year.

Chiller, tower and CRAH maintenance is part of the maintenance contracts (AMC). Water treatment chemicals and blowdown add about ₹0.4–0.6 crore a year for an evaporative 20–50 MW site (estimate).

9.3.4 Maintenance

- **AMC:** about **1.5–3% of M&E capex** per year. We use 2.0% (Case A, ₹15.3 crore), 1.75% (Case B, ₹24.2 crore) and 2.0% (Case C facility, ₹106 crore).
- **DG fuel and testing:** monthly load tests and annual full-load runs cost about ₹1.5–3 crore a year.
- **Lifecycle replacement:** UPS batteries, fans, VFDs and chiller overhauls. We reserve about 1% of M&E capex a year below EBITDA.

9.3.5 Personnel

Staffing per MW (estimates).

- About 60 FTE for a 20 MW colo (3/MW), including outsourced facility management, security and housekeeping.
- About 70 for a 50 MW hyperscale site (1.4/MW).
- About 150 for a 100 MW AI campus (1.5/MW, including GPU fleet operations).

A sanity check: provident-fund (EPFO) filings show Nxtra with only about 231–263 direct employees in 2025 for about 300 MW. Facility management and security are usually outsourced. Direct headcount therefore understates on-site staffing.

Indian salary markers.

Table 9.11 Personnel

Role	Pay	Source
Data centre critical-facilities engineer	~₹20.8 lakh average; ₹14.8 lakh at 1–3 years	SalaryExpert, Aug 2026; modelled, likely high
NOC engineer	Median ₹4.18 lakh (₹3.1–6.0 lakh interquartile)	Glassdoor India, Jun 2026
NOC engineer	Median ₹4.05 lakh	PayScale, Apr 2026
Data centre manager	Median ₹8.62 lakh (75th percentile ₹20 lakh)	PayScale; large-campus heads earn far more (estimate ₹30–60 lakh)
Security guard, Mumbai/Thane	Statutory cost ₹32,315/month per head for 8-hour duty (≈₹3.9 lakh/yr)	Maharashtra Security Guards Board wage order, Jan–Jun 2026

Personnel cost. We use a loaded average of about ₹12.5 lakh per FTE. That gives ₹7.5 crore/yr for Case A, ₹8.5 crore for Case B and ₹25 crore for Case C. Personnel is about 4–5% of colo opex.

9.3.6 Network costs

- **IP transit.** TeleGeography (September 2026) puts the weighted-median 100GigE IP transit price in Mumbai at **US$0.97/Mbps/month (≈₹87)**. That is down about 35% a year compounded over three years, but still 3.7× Singapore's US$0.26. Smaller 10G commits cost more per Mbps.
- **Enterprise leased lines.** 1 Gbps internet leased lines run about ₹70,000–2,50,000 a month (₹70–250/Mbps), per vendor sources of low quality.
- **Cross-connects.** Indian fees are not published. Global norms are US$100–300 a month, and Equinix's global average is about US$254 per interconnection per month. We assume about ₹5,000–15,000 a month in India (estimate).
- **Dark fibre** is priced per route-km on IRU terms and was not researched here.
- For the operator, network opex is small (about ₹1 crore a year in Case A). In a carrier-neutral facility, carriers pay to be present.

9.3.7 Property tax, insurance, security and SG&A

- **Insurance:** about 0.15–0.25% of replacement value. That is ₹2.2 crore (A), ₹3.5 crore (B) and ₹68 crore (C, where GPUs dominate the insured value).
- **Property tax and local levies:** about ₹1.5–5 crore (estimate; varies by municipality and SEZ status).
- **Security:** included in personnel.
- **SG&A:** about 3–5% of revenue for wholesale and colo.

9.3.8 Opex tables per case (stabilised year)

Case A: 20 MW colo. 90% billed (18 MW); average IT draw 70% of billed (12.6 MW); PUE 1.5; blended energy ₹6.6/kWh (grid plus partial RE).

Table 9.12 Opex tables per case (stabilised year)

Line	₹ crore/yr
Electricity (165.6 GWh × ₹6.6 = 109.3, plus demand charges 19.8)	129.0
Water	0.6
Water treatment and chemicals	0.4
AMC (2% of ₹764 crore M&E)	15.3
DG fuel and testing	1.5
Personnel (60 FTE)	7.5
Insurance	2.2
Property tax and levies	1.5
Network	1.0
SG&A	15.0
Total opex	**174.0**
₹/kW/month (on 20 MW built)	**₹7,250 (₹1,875 excluding power)**

Case B: 50 MW hyperscale. 95% billed; IT draw 75% (35.6 MW); PUE 1.4; ₹6.2/kWh; power passed through to the tenant.

Table 9.13 Opex tables per case (stabilised year)

Line	₹ crore/yr
Electricity (437 GWh × ₹6.2 = 271, plus demand 33), passed through	304.0
Water	1.2
AMC (1.75% of ₹1,385 crore)	24.2
DG fuel and testing	2.5
Personnel (70 FTE)	8.5
Insurance / property tax / network	3.5 / 2.5 / 0.5
SG&A	10.0
PUE-overrun and SLA reserve	5.0
Chemicals	0.6
Total opex (non-power)	**58.5**
Total including pass-through power	**362.5**
₹/kW/month	**₹6,040 total; ₹975 non-power**

Case C: 100 MW AI campus. IT draw 80%; PUE 1.25; RE-RTC-led supply at ₹6.0/kWh.

Table 9.14 Opex tables per case (stabilised year)

Line	₹ crore/yr
Electricity (876 GWh × ₹6.0 = 526, plus demand 66)	592
Water	2
Facility AMC (2% of ₹5,300 crore M&E)	106
GPU/IT support and spares (1% of IT capex)	266
Personnel (150 FTE)	25
Insurance	68
Network, backbone, transit	20
Property tax	5

Line	₹ crore/yr
SG&A	100
DG	3
Total opex	**1,187**
₹/kW/month	**₹9,890 (facility-only ≈ ₹7,675)**

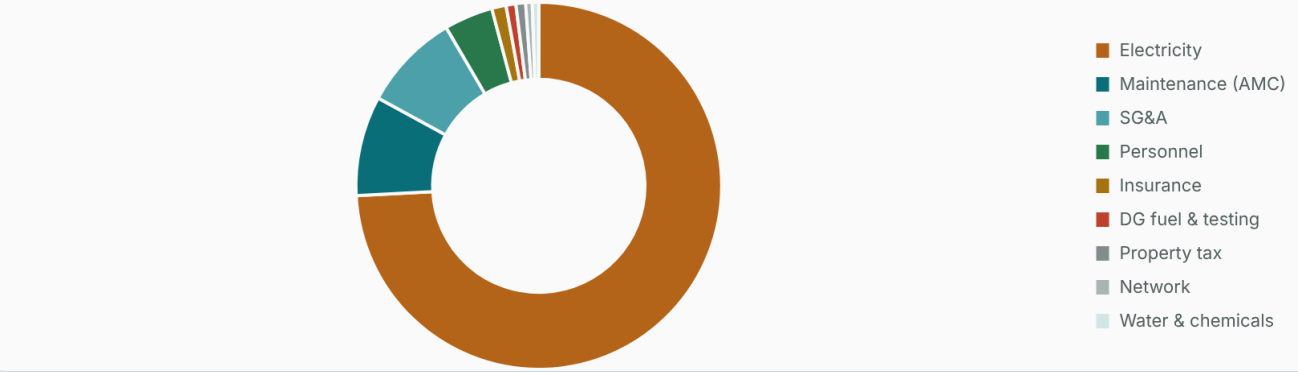


Figure 9.3 Where colo opex goes. Case A, ₹174 crore a year *Source: India Data Centre Atlas, chapter built on this chapter's data.*

9.4 Unit economics

9.4.1 ₹ per MW: capex, revenue, EBITDA, payback, IRR, valuation

Colocation pricing in India.

Table 9.15 ₹ per MW: capex, revenue, EBITDA, payback, IRR, valuation

Segment	Price	Source
Mumbai, 250 kW–1 MW	₹7,000–10,000/kW/month	Cushman & Wakefield H1 2025 (standard 5–7 kVA racks)
Mumbai, 1–5 MW	₹6,500–8,000	Cushman & Wakefield H1 2025
Mumbai, 5 MW+	₹6,000–7,500	Cushman & Wakefield H1 2025
Chennai, 1–5 MW	₹7,000–8,000	Cushman & Wakefield H1 2025
Sub-250 kW, secondary hubs	₹7,000–9,500	Cushman & Wakefield H1 2025
India average rent	US$80–100/kW/month (₹7,200–9,000)	Houlihan Lokey
North America wholesale (250–500 kW), for comparison	US$196.25/kW/month (+6.6% year-on-year), about 2× India	CBRE, H2 2025

Realised revenue per MW from company data:

- Sify Infinit Spaces: FY25 revenue ₹1,428 crore and EBITDA ₹634 crore (44.4% margin) on 188 MW built and about 127 MW sold. That is ≈₹11 crore per sold MW per year, including power.
- An industry roll-up reported by Outlook Business puts Sify at about ₹5,699/kW/month and Iron Mountain at about ₹8,445/kW/month, with about ₹11,200 crore of industry revenue on 1,520 MW.
- Anant Raj's management guides colo revenue of ₹90 lakh/MW/month (₹9,000/kW) against opex of ₹15 lakh/MW/month.

Per-MW economics by case (stabilised).

Table 9.16 ₹ per MW: capex, revenue, EBITDA, payback, IRR, valuation

Metric	A: 20 MW colo	B: 50 MW hyperscale build-to-suit	C: 100 MW AI neocloud
Capex/MW, facility only	₹56.1 crore	₹39.2 crore	₹72.5 crore
Capex/MW incl. land and IT	₹61.1 crore	₹42.4 crore	~₹340 crore
Pricing assumption	₹7,000/kW/mo rent + power at cost +10%	₹6,250/kW/mo + power at cost	₹400/GPU-hr at 80% utilisation (assumption)
Revenue (₹ crore/yr)	301 (rent 151 + power 142 + other 8)	660 (rent 356 + power 304)	11,908
Opex (₹ crore/yr)	174	362.5	1,187
EBITDA (₹ crore/yr)	127 (42% margin)	298 (45% of gross; 84% of rent)	10,721 (90%)
EBITDA per MW per year	₹6.35 crore	₹5.96 crore	₹107 crore
Yield on cost (EBITDA ÷ capex)	10.4%	14.0%	31.5% (but GPUs last ~5 years)
Simple payback	~9.6 years	~7.1 years	~3.2 years
Unlevered pre-tax IRR (method below)	**~9.5%**	**~14.7%**	**~21.5%** at 80% / ~10.5% at 60% / negative at 40%

IRR method.

- Cases A and B: 15-year hold. A ramps 40% → 75% → 100%; B ramps 70% → 100%. Lifecycle reserve of 1% of M&E. Exit at an 8.5% cap rate (A) or 7.5% (B).
- Case C: 5-year GPU life. Residual value of about ₹8,160 crore (facility net book plus 10% GPU salvage).

Interpretation. At ₹7,000/kW/month, a new ₹61 crore/MW Tier III colo earns about its cost of capital. Each ₹1,000/kW/month of extra rent adds about ₹21.6 crore of EBITDA, about 1.8 points of yield and about 2 points of IRR:

- ₹8,000/kW/month gives an IRR of about 11.5%.
- ₹9,000/kW/month gives about 13.5%.

Pre-leased hyperscale wins on capital intensity and instant lease-up, even at lower rents. These IRRs are pre-tax and before the 20-year tax holidays proposed in the draft national data centre policy (unverified secondary source). Leverage at 8–10% debt adds little when project IRRs are only 9.5–11%.

Valuations (EV/MW and EV/EBITDA).

Table 9.17 ₹ per MW: capex, revenue, EBITDA, payback, IRR, valuation

Data point	Implied value
Nxtra, March 2026: US$1 billion raise (Alpha Wave US$435M, Carlyle US$240M, Anchorage US$35M, Airtel about US$290M) at **US$3.1 billion post-money**; before the deal Airtel held 75.96% and Carlyle the rest; about 300 MW, targeting 1 GW	~US$7M/MW pre-money (≈₹63 crore/MW); ~US$10M/MW post-money
Earlier Nxtra reference points	Carlyle invested in 2020 at a US$1.2 billion valuation. April 2025 reports of Alpha Wave buying Carlyle's 24% stake implied ₹20,500 crore (US$2.4 billion) for 100%.
Houlihan Lokey cap rates: India 7.5–8.5%; US 4.5–6.5%; Singapore 5.0–6.5%	Stabilised Indian assets at about 12–13× EBITDA (Case A at 8% → about ₹1,590 crore of value vs ₹1,222 crore cost, 1.3×; Case B at 7.5% → about ₹3,970 crore vs ₹2,120 crore, 1.9×)
Sify Infinit Spaces DRHP (Oct 2025): ₹3,700 crore IPO (₹2,500 crore fresh issue); net debt ₹2,324 crore; net debt/EBITDA 3.66×	EV/EBITDA not determinable until pricing

Lumina CloudInfra (Blackstone), STT GDC India, Yotta and CtrlS valuations were not verified in this research.

9.4.2 ₹ per rack

The facility cost per rack scales with density. The wholesale-rent column uses ₹7,500/kW/month.

Table 9.18 ₹ per rack

Density	Design	Facility ₹ crore/MW	Facility capex per rack	Typical IT per rack	Implied wholesale rent/rack/month
6 kW	Air, Tier III	~58	~₹35 lakh	₹40–80 lakh of CPU servers (estimate)	~₹45,000
10 kW	Air + containment	~55	~₹55 lakh	₹0.8–1.5 crore (estimate)	~₹75,000
40 kW	Rear-door heat exchangers / DLC hybrid	~62	~₹2.5 crore	~4 × H100 HGX ≈ US$1.1M ≈ ₹10 crore	~₹3.0 lakh
130 kW	DLC (NVL72)	~72.5	~₹9.4 crore	GB300 NVL72 ₹33–58 crore + fabric	~₹9.75 lakh (₹10.4–11.7 lakh at ₹8,000–9,000/kW)

Retail rack pricing in India (vendor price lists; low-to-medium quality):

- Full 42U racks for space alone list at about ₹25,000–60,000 a month, with power at about ₹8,000–20,000 per kVA per month extra.
- Bundled offers include a 3 kW full rack at ₹50,000 a month and a 5–10 kW Tier III rack in Ludhiana at ₹50,000.
- A 5–8 kW retail rack therefore lands at roughly **₹60,000–1,50,000 a month all-in**. That is about ₹10,000–20,000 per kW-month, **1.5–2.5× wholesale**.

9.4.3 ₹ per kWh: levelised cost of data centre capacity (LCOD)

Formulas.

- Annualised capex = Capex × CRF, where **CRF = $r(1+r)^n / [(1+r)^n - 1]$**.
- Facility at r = 11%, n = 20: $(1.11)^{20} = 8.062$, so CRF = 0.11 × 8.062 / 7.062 = **0.1256**.
- Other CRFs used:
- 10%/20 years: 0.1175
- 12%/20 years: 0.1339
- 8%/20 years: 0.1019
- GPUs, 11%/5 years: **0.2706**
- GPUs, 11%/4 years: 0.3223
- GPUs, 11%/6 years: 0.2365
- **LCOD = (Annualised capex + Annual opex) ÷ IT kWh delivered.**

Table 9.19 ₹ per kWh: levelised cost of data centre capacity (LCOD)

Case	Annualised capex (₹ crore)	Opex (₹ crore)	IT energy (GWh)	LCOD (₹/IT-kWh)	Power cost (₹/IT-kWh)	Power share
A: 20 MW colo (12.6 MW average IT)	1,222 × 0.1256 = 153.5	174.0	110.4	**29.7** (~US$0.33)	11.7	39%
B: 50 MW hyperscale, owner view	2,120 × 0.1256 = 266.3	362.5	311.9	**20.2**	9.7	48%
C: AI campus, facility only	7,400 × 0.1256 = 929.4	~921	700.8	**26.4**	8.45	32%
C: AI campus incl. GPUs	929.4 + 26,620 × 0.2706 = 8,132	1,187	700.8	**133** (~US$1.48)	8.45	6%

Picture each ₹29.7 IT-kWh in Case A as a stacked bar. About ₹14 is capex recovery, ₹11.7 is electricity and ₹4 is everything else. Running the same hall at full load (175 GWh of IT) cuts LCOD to about ₹22/kWh. Filling the building is worth more than any tariff negotiation.

Self-build vs lease (hyperscaler view).

- Owning Case B costs 42.4 × 0.1256 = ₹5.33 crore/MW/yr of capex, plus ₹1.17 crore/MW/yr of non-power opex. That is **≈₹5,415/kW/month**, against leasing at ₹6,250, a saving of about 13%.
- At a hyperscaler's ~8% WACC (CRF 0.1019), the cost falls to **≈₹4,575/kW/month**, a saving of about 27%.
- This explains AWS, Microsoft and Google land purchases in MMR and Hyderabad. They self-build where they can and lease for speed.

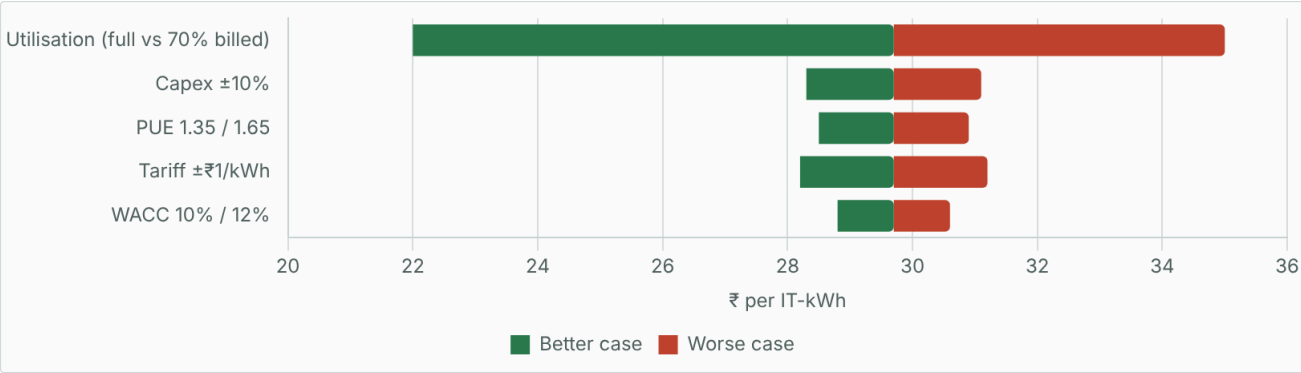


Figure 9.4 What moves the colo's cost. Case A LCOD (₹/IT-kWh), base ₹29.7. Bars show low and high cases. *Source: India Data Centre Atlas, chapter built on this chapter's data.*

9.4.4 Cost per GPU-hour: TCO in India vs market prices

Assumptions for a neocloud hosting in leased Indian colo.

- ₹90/$. 5-year life at 11% WACC (CRF 0.2706).
- Server price plus 15% for networking and 3% for storage and management.
- Hosting at ₹7,000/kW/month of IT.
- Power: average draw 80% of nameplate, PUE 1.35 (1.3 for B200), ₹7.0/kWh.
- Operations, software and insurance: ₹70,000–90,000 per GPU per year.
- Formula: **GPU-hour cost at utilisation u = (capex + hosting + ops) ÷ (8,760 × u) + power per hour**.

Table 9.20 Cost per GPU-hour: TCO in India vs market prices

₹/GPU-hour	Capex/hr (at 100%)	100%	80%	60%	40%
H100 (US$275k HGX → ₹36.5 lakh/GPU all-in; 1.4 kW)	112.8	**145**	**177** (US$1.97)	**232**	**343**
H200 (~US$315k HGX → ₹41.8 lakh; 1.4 kW)	129.1	**161**	**198**	**259**	**384**
B200 (US$450k HGX → ₹59.7 lakh; 2.0 kW)	184.5	**229**	**281** (US$3.12)	**368**	**545**
GB300 NVL72, self-built campus (Case C)	193.6 + 25 facility	**251**	**308** (US$3.42)	**404**	**598**

For a GB300 GPU-hour at 80% utilisation, the ₹308 splits into ₹242 of GPU and network capex, ₹31 of facility, ₹20 of opex and ₹14 of power.

Market prices (Oct 2026 unless stated).

Table 9.21 Cost per GPU-hour: TCO in India vs market prices

Offer	₹/GPU-hour	H100 breakeven utilisation vs our cost
IndiaAI L1 rate, high-end (H100-class); standard GPUs L1 ₹115.85. Users pay about ₹92 after a 40% subsidy, and Union Minister Ashwini Vaishnaw told the Lok Sabha that the average rate across the 34,381 GPUs onboarded from 14 providers is about ₹65/GPU-hr, with H100s at ₹92.	**150**	**~96%** (5-year life); ~86% (6-year life)
JarvisLabs H100	217.89	~64%
Global OCPI-H100 index (2 Oct 2026): US$2.79	~251	~55%
E2E Networks H100 on-demand	255.55	~54%
Yotta Shakti H100, dedicated (₹1,96,113/month)	~269	~52%
Yotta Shakti H100 on-demand	356	~39%
E2E H200 on-demand / 3-month+ commitment	379.05 / 189.20	H200: ~41% / ~84%
B200, global neocloud on-demand, US$3.70–6.00	~333–540	B200: ~67% / ~41%
Hyperscaler H100 on-demand (AWS/Azure/GCP), US$6–12	~540–1,080	Scarcity and convenience premium

What this means.

1. **IndiaAI rates sit at fully loaded cost.** Providers clear ₹150/hr only with near-full utilisation, 6-year lives, or hardware bought cheaper (at ₹83–85/$ in 2024 or at lower prices). The subsidy makes Indian sovereign compute cheap for users, but it is a thin-margin business for providers.
2. **Commercial on-demand pricing in India (₹218–356) leaves healthy margins at 50–65% utilisation.**
3. **Global prices are no longer falling uniformly.** SemiAnalysis's 1-year H100 contract index rose about 40%, from US$1.70 (October 2025) to US$2.35 (March 2026). The OCPI spot index is about US$2.5–3.0. The "steady decline" story describes 2024–25, not 2026.
4. **Why our costs run higher than some published teardowns.** Silicon Analysts estimates about US$1.45/hr all-in for an H100 at 70% utilisation, and Data Gravity about US$2.56–2.88 for GB300 at 80–90%. Those use straight-line depreciation with residual value. We use capital recovery at an 11% WACC. Indian cost of capital adds roughly US$0.3–0.5 per GPU-hour.

Case C breakeven and returns. The breakeven contracted price is about ₹308/hr at 80% utilisation, ₹404 at 60% and ₹598 at 40%. At an assumed ₹400/hr:

- 80% utilisation: IRR ~21.5%
- 60% utilisation: ~10.5%
- 40% utilisation: negative

The AI neocloud is a leveraged bet on utilisation and on the next-generation price curve.

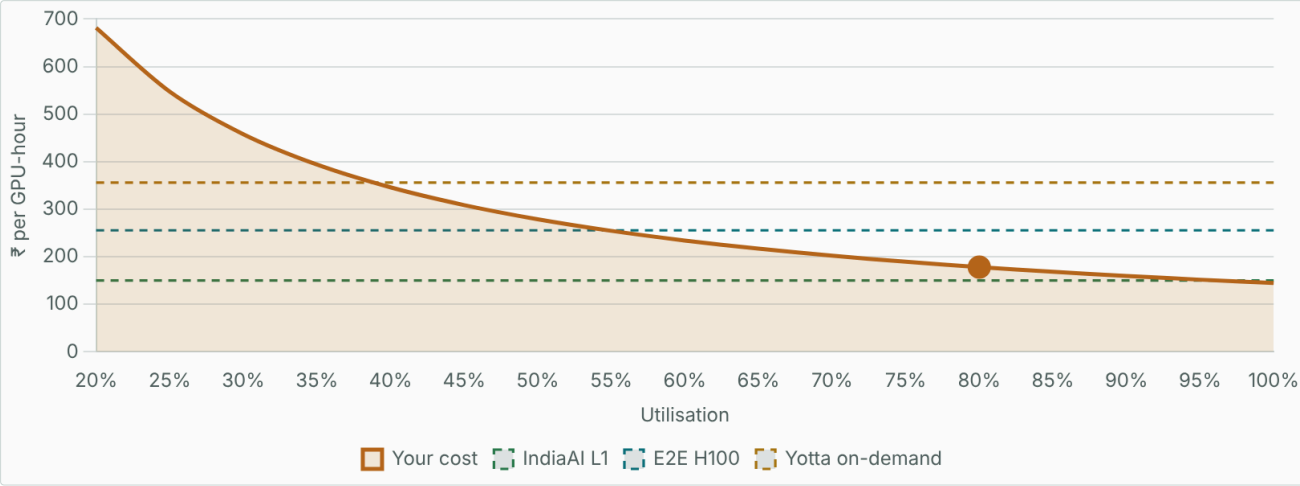


Figure 9.5 GPU-hour cost in India. GPU + network capex, colo hosting at ₹7,000/kW/month, power at 80% draw, ₹70k/yr ops. Dashed lines are market prices. *Source: India Data Centre Atlas, chapter built on this chapter's data.*

9.4.5 Sensitivity analysis

Table 9.22 Sensitivity analysis

Driver	Change	Case A (colo)	Case C (GB300 GPU-hour at 80%; base ₹308)
Tariff	±₹1/kWh	LCOD ±₹1.5/IT-kWh. EBITDA ±₹16.6 crore if power is bundled; only ±₹1.7 crore if passed through at cost +10%	±₹2.9/hr
Utilisation	A billed 70% vs 90%; C 60% / 40%	EBITDA −₹34 crore (yield 7.6%); LCOD ≈ ₹35	₹404 / ₹598
PUE	A 1.5→1.35; C 1.25→1.15	Power −10% (≈₹12.9 crore/yr); LCOD −₹1.2	−₹1.3/hr
Cost of capital	WACC 10% / 12%	LCOD ₹28.8 / ₹30.6 (annualised capex ₹143.6 / ₹163.6 crore)	₹301 / ₹314
FX	₹90 → ₹95/$	Facility capex +~2% (≈₹1.1 crore/MW; imported share ~35–40%, estimate)	+₹13.5/hr
Import duties	Hypothetical 10% basic duty + surcharge on servers; or IGST not creditable	—	+₹26.6/hr; or +₹43.6/hr
GPU life	4 / 6 years (H100 at 80%)	—	H100 ₹204 / ₹159 (base ₹177)

Ranking the drivers. For colo and hyperscale, the order of importance is utilisation and rent, then capex per MW and WACC, then PUE and tariff. Tariff ranks last only when power is passed through. For AI it is utilisation, then rack price and GPU life, then WACC and FX, then tariff.

9.4.6 Comparing models and markets

India vs US vs Singapore (Houlihan Lokey using Cushman & Wakefield/CBRE; Turner & Townsend; CBRE)

Table 9.23 Comparing models and markets

Metric	India	US	Singapore
Build cost, Turner & Townsend (US$/W)	6.64 (Mumbai)	9.5–13.3	14.5
Development cost (US$M/MW)	5.5–8.5	9–12 (Cushman & Wakefield 2026 US average US$17.6M all-in)	9–14.5
Rent (US$/kW/month)	80–100	160–225 (CBRE wholesale US$196)	300–475 (retail-skewed)
Electricity (US$M/MW/yr)	0.7–0.9	~1.22	~1.7
Total opex (US$M/MW/yr)	1.0–1.2	5.0–5.5 as reported by Houlihan Lokey (looks inconsistent; treat with caution)	2.0–2.4
Cap rate	7.5–8.5%	4.5–6.5%	5.0–6.5%
Gross rent ÷ capex	~15%	~22%	very high (retail mix)

India builds at about half the US cost. It rents at about half the US price and is valued at higher cap rates. Its unit margins are therefore thinner, not fatter. India wins on growth, data localisation under the DPDP Act, and cheaper land and labour, not on spread.

Colo vs hyperscale self-build vs AI neocloud

Table 9.24 Comparing models and markets

	Colocation (A)	Hyperscale build-to-suit / self-build (B)	AI neocloud (C)
Capex intensity	₹61 crore/MW	₹42 crore/MW	₹340 crore/MW
Revenue driver	₹/kW rent + cross-connects	Long-term ₹/kW lease (or avoided rent)	₹/GPU-hour × utilisation
Margin / return	42% EBITDA; IRR ~9.5–13.5% depending on rent	84% of rent; IRR ~14–15%; self-build 13–27% cheaper than leasing	90% EBITDA but 5-year asset; IRR −ve to ~21%
Power share of cost	~39% of LCOD	~48%	~6%
Key risk	Lease-up and rate pressure	Tenant concentration, pricing power	Utilisation, GPU price deflation, obsolescence

9.5 Recommendations

1. **Developers of colo (Case A-type).** - Underwrite to **₹7,700–8,200/kW/month or better**, or a facility cost of **≤₹50 crore/MW**. - Phase the M&E fit-out to match leasing rather than building all 20 MW on day one. - Push cross-connect and interconnection density. That is where retail colo earns its premium over the 9.5% base-case IRR.
2. **Hyperscale build-to-suit (Case B-type)** is the best risk-adjusted Indian data centre return available in this model, at about 14–15% unlevered on pre-lease. Lock the PUE caps and power pass-through clauses in contracts. Each 0.1 of PUE overrun costs the operator about ₹22 crore a year at 50 MW if it absorbs the overrun.
3. **Power procurement.** - In Maharashtra and Tamil Nadu, pursue **group captive RE or RTC** to avoid ₹2+/kWh of cross-subsidy and additional surcharge. Claim MERC's 10% wheeling discount and the state subsidies (₹1/unit in Maharashtra; electricity tax relief in Tamil Nadu). - In Telangana or Karnataka, buy RE for decarbonisation and price certainty, not for savings.
4. **AI neoclouds and GPU buyers.** - Do not buy GPUs without **contracted utilisation of at least 60–70%**. Below about 55%, renting from the Indian market at ₹218–256/hr beats owning an H100. - Model with capital recovery at your real WACC, not straight-line depreciation. - For IndiaAI empanelment, bid only with fully depreciated or cheaply acquired fleets. ₹150/hr is a cost-recovery price. - Budget the IGST working-capital drag (about 14% of IT capex carried for 2+ years in our case).
5. **Investors.** Platform deals near **US$7M per operating MW** (Nxtra pre-money) roughly equal replacement cost. Pay up only for secured power, land banks and hyperscale pre-leases. Those are the scarce inputs, not concrete and chillers.

9.6 Caveats

- **The case models are illustrative bottom-up estimates.** The three cases are built from the cited benchmarks plus our own assumptions. Line items marked "estimate" were not verified against Indian tenders or filings in this research. These include DG ₹/kVA, sq-ft costs, CPU server prices, staffing ratios, H200 server price, cross-connect fees, and land in Noida, Bengaluru and Pune.
- **Contested figures.**
- GB300 NVL72 rack price: US$3.7–6.5M, with analysts disagreeing.
- GPU economic life: 4–6 years.
- The Maharashtra tariff, given MERC's subsequent review order.
- Houlihan Lokey's US opex figure, which looks internally inconsistent.
- Chennai land prices, which come from a low-quality blog.
- The Cushman & Wakefield (2026) India per-MW range, taken via a secondary summary.
- **Not covered here.** The following were not reviewed, and their figures should be checked before investment use:
- Uttar Pradesh, Andhra Pradesh and Gujarat tariff orders
- MIDC, CMWSSB and HMWSSB water tariffs
- Lumina/Blackstone, STT GDC, Yotta and CtrlS valuations
- ICRA, CRISIL and CareEdge reports
- Morgan Stanley and Jefferies notes
- Techno Electric, RailTel and Netweb filings
- **Model choices affect the results.**
- All IRRs are pre-tax and unlevered.
- Tax holidays under the draft national data centre policy (reported at up to 20 years, via an unverified secondary source) would raise post-tax returns materially.
- FX is set at ₹90/$. Some 2026 sources use about ₹95.6, which would raise USD-denominated capex by about 6%.
- GPU market prices move weekly; the H100 index moved 4.5% in a single day in early October 2026.

Sources cited in this chapter

Organisations and publications named in this chapter's text, in order of how often they are cited. Full details are in the bibliography.

- Cushman & Wakefield, India data centre market updates (H1 2026)
- Houlihan Lokey India data centre edition (December 2025)
- Turner & Townsend, Data Centre Cost Index 2025
- State electricity regulatory commissions (KERC, MERC, TNERC, TGERC) tariff and open-access orders
- CBRE, India data centre market reports
- Nxtra by Airtel disclosures
- Solar Energy Corporation of India tenders
- Amazon Web Services sustainability reporting
- Yotta Data Services disclosures
- Ministry of Electronics and Information Technology (MeitY), Data Centre Policy draft and IndiaAI
- STT GDC India ESG report
- Sify Infinit Spaces sustainability report
- State Bank of India sector report on data centres (April 2026)
- Ministry of Power, orders on ISTS waiver and green open access
- Google environmental reports, efficiency data and Gemini inference paper
- Microsoft sustainability reports and datacenter design posts
- CtrlS disclosures
- Bangalore Water Supply and Sewerage Board
- Central Electricity Regulatory Commission
- NVIDIA product specifications and roadmap
- Equinix sustainability reports

PART III · ECONOMICS AND SUSTAINABILITY

Chapter 10
Green data centres: a decision-grade synthesis for India and Karnataka

WHAT THIS CHAPTER COVERS

This chapter defines what a green data centre is and how to tell it apart from greenwashing. It sets out twelve dimensions of environmental performance, the metrics and standards used to measure them, the regulations and certifications that apply around the world and in India, and the design levers that deliver quantified savings. It explains the options for buying renewable and carbon-free power in India, compares global benchmarks, works through the economics of a green 20 megawatt colocation hall in Bengaluru, and proposes a maturity model and policy recommendations.

KEY TAKEAWAYS

1. A data centre is green only if it is measured, disclosed and verified on at least four dimensions: energy efficiency, hourly carbon-free power, freshwater impact and embodied or life-cycle carbon.
2. For a Bengaluru operator the cheapest and most bankable route is a design PUE of 1.20 or lower, at least 80% round-the-clock renewable supply through open access or group captive arrangements, and cooling that uses dry coolers or treated water.
3. On the worked numbers, that package costs about 5% more to build, pays back in about two to three years on energy savings alone, and reduces carbon at a negative cost, which means it saves money while cutting emissions.
4. Annual renewable certificates are weaker evidence than hourly matching, and claims that rest only on offsets or unbundled certificates are a warning sign of greenwashing.
5. A five-level maturity model, calibrated to Indian practice, lets an operator see where it stands on each dimension and what a leading facility looks like.

How this chapter connects to the others. The carbon accounting behind this chapter is explained in Chapter 5, and Chapter 11 shows which clean-power mix also minimises water.

Bottom line: A data centre is "green" only if it is measured, disclosed and verified on at least four dimensions (efficiency, hourly carbon-free power, freshwater impact, and embodied/lifecycle carbon), and for a Bengaluru operator the cheapest, most bankable route is a design-PUE of ≤1.20, ≥80% round-the-clock renewable supply through open access or group captive, and dry/treated-water cooling. On our worked numbers that package costs roughly 5% more capex, pays back in about 2–3 years on energy savings alone, and abates carbon at a *negative* cost.

10.1 What is a "green data centre"?

10.1.1 A working definition

A green data centre delivers digital services with the lowest feasible **lifecycle** environmental impact per unit of useful computing, and proves it with **independently verifiable, site-level, time-resolved** data. That definition has three implications:

1. **"Per unit of useful compute"**, not per MW. A very efficient building full of idle servers is not green.
2. **"Lifecycle"**: operational energy, plus construction materials, IT hardware manufacturing (Scope 3), refrigerants, water and end-of-life.
3. **"Time-resolved"**: carbon depends on *when* and *where* electricity is consumed. As the earlier report "Every TWh is not the same" argued, an annual REC-matched TWh in a coal-heavy grid at 8 pm is not equivalent to a physically delivered solar-plus-storage TWh.

10.1.2 The twelve dimensions

Table 10.1 The twelve dimensions

#	Dimension	Core metric(s)	What "green" looks like (India-tailored)	Why it matters
1	Energy efficiency	PUE (ISO/IEC 30134-2)	≤1.30 annualised (air); ≤1.20 (liquid-cooled AI)	Overhead energy is pure waste
2	Carbon-free energy	REF, hourly CFE score, CUE	≥70–80% hourly-matched CFE; CUE ≤0.3 $kgCO_2$/IT-kWh	India's grid factor 0.710 tCO_2/MWh makes power the dominant footprint
3	Water	WUE (site), source WUE	≤0.2 L/kWh freshwater; treated wastewater for any evaporative trim	Bengaluru water stress
4	Embodied carbon (building)	$kgCO_2e/m^2$ or /MW (EN 15978-style LCA)	30–40% below baseline via low-clinker concrete, recycled steel, hybrid timber	Grows in share as grids decarbonise
5	IT hardware Scope 3	$kgCO_2e$ per server/GPU, lifecycle	Supplier EPDs/PCFs; extended refresh cycles	Now the largest share for hyperscalers
6	Circularity	% IT assets reused/refurbished; e-waste diversion	≥90% assets reused or recycled through authorised E-Waste (Management) Rules 2022 channels	Avoids manufacturing emissions and hazardous waste
7	Heat reuse	ERF (ISO/IEC 30134-6), ERE	"Heat-reuse ready" design; absorption cooling or process heat where offtake exists	Turns waste into a product
8	Refrigerants	GWP of charge, leakage rate	Low-GWP (HFO, natural refrigerants); chiller-less where possible	HFC leakage is a hidden Scope 1 source
9	Land and biodiversity	Footprint, green cover, heat-island	Brownfield/industrial-park siting; native landscaping	Social licence, local heat
10	Backup power	Diesel litres/yr, run-hours	HVO-ready gensets, BESS for short events, fuel cells long term	Diesel is local pollution and Scope 1
11	Grid-friendliness	Demand response, flexibility MW	BESS/UPS grid services; schedulable AI training	Helps integrate renewables
12	Social licence	Disclosure, local water/power impact, jobs	Public site-level dashboards; community water commitments	Without it, moratoria follow (Amsterdam, Dublin, Singapore 2019)

10.1.3 Green vs greenwashing: a quick test

Table 10.2 Green vs greenwashing: a quick test

Claim	Credible if...	Greenwashing red flag
"100% renewable"	Hourly or at least monthly matching, same grid region, additional (new) capacity, location-based Scope 2 also disclosed	Annual matching with unbundled RECs/I-RECs from old hydro elsewhere
"Carbon neutral"	Deep reductions first; removals (not avoidance offsets) for residuals	Cheap avoidance offsets netted against growing emissions
"PUE 1.2"	Annualised, measured at the utility meter boundary (ISO/IEC 30134-2 Category 2/3), at real load	Design PUE at 100% load, or a single cool-season month
"Water positive"	Site-level WUE disclosed; replenishment in the *same* basin	Global replenishment credits while a Bengaluru site draws borewells
"Green certified"	Operational certification with measured data	Design-stage label only, never re-verified

The industry is moving in this direction. Microsoft's FY25 reported emissions rose partly because it **stopped counting non-additional unbundled RECs**. The EU's 2026 rating scheme gives guarantees of origin less credit unless they are from the same bidding zone, time-matched, and from plants commissioned or substantially refurbished within the last 10 years.

10.2 Metrics and standards

10.2.1 Core metrics and formulas

Table 10.3 Core metrics and formulas

Metric	Formula	Standard	Notes
PUE	Total facility energy ÷ IT equipment energy	ISO/IEC 30134-2 (The Green Grid, 2007)	Ideal 1.0; global weighted average 1.54 (Uptime 2025)
WUE	Annual site water (L) ÷ IT energy (kWh)	ISO/IEC 30134-9	Karnataka policy uses this definition; the EU uses freshwater input
CUE	Total CO_2 from facility energy ÷ IT energy = grid EF × PUE (for grid-only supply)	ISO/IEC 30134-8	Bengaluru grid-only at PUE 1.5: 0.710 × 1.5 = **1.07 kgCO_2/IT-kWh**
ERF	Reused energy ÷ total facility energy	ISO/IEC 30134-6	German EnEfG requires 10→20% for new sites
ERE	(Total energy – reused energy) ÷ IT energy	The Green Grid	Can be <1.0 with heat reuse
REF	Renewable energy ÷ total facility energy	ISO/IEC 30134-3	Annual; does not capture timing
24/7 CFE score	Σ_h min(load_h, CFE supply_h) ÷ Σ_h load_h	Google/UN 24/7 CFE Compact methodology	The honest metric; Google fleet ≈65% (2025)
ITEEsv / ITEUsv	IT equipment energy efficiency / utilisation for servers	ISO/IEC 30134-4/-5	Captures the compute side that PUE ignores
GEC	Green energy share of facility energy	ISO/IEC 30134 family (Japan-originated)	Similar in spirit to REF

Worked CUE illustration (Karnataka): at PUE 1.25 and 80% hourly-matched CFE, CUE = 1.25 × 0.20 × 0.710 ≈ **0.18 kgCO_2/IT-kWh**, an 83% reduction from 1.07. Using the lower Karnataka-specific factor (~0.5), the baseline CUE is 0.75 and the green case 0.125.

10.2.2 Global regulatory landscape

European Union. The recast Energy Efficiency Directive (EU 2023/1791) and Delegated Regulation (EU) 2024/1364 created mandatory reporting for sites with ≥500 kW of installed IT power. Reports go to a European database by 15 September 2024 and every 15 May thereafter. On **21 September 2026** the Commission adopted the second step, a Delegated Regulation establishing a **common Union rating scheme**, as part of the Data Centre Energy Efficiency Package: - Two separate A–G grades. **Energy:** A = PUE ≤1.15, G = PUE >1.90. **Water:** A = WUE ≤0.10 L/kWh, G = >1.0. - The label also shows energy sourcing (on-site, PPAs, guarantees of origin; renewables and, newly, nuclear as "low-emission"), grid flexibility, "waste heat reuse ready" status, and cooling degree days. The grades themselves are not climate-adjusted. - Labels are generated automatically, first by **15 August 2027**. The 2-month Parliament/Council scrutiny period can be extended by two months. - Step three is a **consultation on binding minimum performance standards**, closing 14 December 2026, with a legislative proposal targeted for Q2 2027. - Context: EU data centres used **68 TWh in 2024**, projected to reach **114 TWh by 2030** (3.2% of EU demand). - The **EU Code of Conduct for Energy Efficiency in Data Centres** (voluntary, since 2007) remains the best-practice catalogue underpinning these rules. - Spain's draft Royal Decree goes further. It would make grid access conditional on Class-A PUE/WUE (≤1.15 / ≤0.1 in the transition) and, in one option, on **80% hourly renewable matching**. This is a preview of where minimum standards may land.

Climate Neutral Data Centre Pact (industry self-regulation, launched January 2021): - PUE 1.3 (cool climates) / 1.4 (warm) for new sites at full capacity by 1 January 2025; existing sites by 1 January 2030. Applies to sites >50 kW. - **75% renewable or hourly CFE by 31 December 2025, 100% by 31 December 2030.** The Pact marks the 2025 milestone "achieved". - WUE: maximum 0.4 L/kWh for new cool-climate sites using potable water in water-stressed areas from 2025. This is a design maximum adjustable by climate, stress and water type, not a universal cap.

Germany: Energy Efficiency Act (EnEfG). The strictest binding regime: - New sites commissioned from **1 July 2026**: **PUE ≤1.2** (to be reached within two years of operation). - Existing sites: PUE ≤1.5 from July 2027 and ≤1.3 from July 2030. A 2026 draft amendment proposes relaxing these to **1.6 and 1.4**, a sign that the targets proved hard for legacy stock. - **ERF** for new sites: 10% (from July 2026), 15% (July 2027), 20% (July 2028). The draft amendment adds flexibility where heat networks cannot absorb the heat. - **Renewable electricity:** 50% from 2024, **100% from 1 January 2027**. - Certified energy/environmental management systems are mandatory for larger sites.

Singapore: the closest climate analogue for coastal and hot-humid India. - After the 2019 moratorium, Singapore moved to selective allocation. The **Green Data Centre Roadmap** (IMDA, May 2024) aims to bring all data centres to **PUE ≤1.3 at 100% IT load** within about a decade, with at least 300 MW of new capacity. - **DC-CFA2** (opened 1 December 2025, closed 31 March 2026) required **PUE 1.25 at full IT load**, **≥50% green energy** (biomethane, low-carbon ammonia/hydrogen, novel fuel cells with carbon capture, on-site solar), **BCA-IMDA Green Mark for Data Centres 2024 Platinum**, and IT equipment beyond **SS 715:2025** efficiency. - On **21 August 2026**, 200 MW was provisionally allocated (50 MW each to Digital Realty, Equinix, Keppel DC and STT GDC) at a Jurong Island low-carbon park. The 2023 pilot allocated 80 MW. - Lesson for India: tropical sites can hit 1.25, but in practice only with liquid cooling and very high-efficiency plant.

China. - New large and mega data centres must achieve **PUE ≤1.25**; sites in national computing hubs face **≤1.2**. - A 2025 green data centre action plan requires new projects in the eight national hubs to source **≥80% renewable electricity**, tracked through Green Electricity Certificates. - "East Data, West Computing" steers compute-heavy, latency-tolerant loads to Guizhou, Inner Mongolia, Gansu and Ningxia. - RMI puts the national average PUE at around 1.48 in 2023, so the caps bind.

United States. No federal performance mandate. **ENERGY STAR** certification is limited to the top 25% of facilities by energy performance, and **LEED v4/v4.1 BD+C: Data Centers** (with LEED v5 rolling out) is the dominant certification. State action is mostly about grid cost allocation and disclosure rather than PUE caps. *(State-specific rules were not re-verified in this research cycle.)*

Netherlands/Amsterdam, Ireland, Japan *(background, not re-verified this cycle)*: - Amsterdam paused new builds in 2019 and now zones hyperscale growth with efficiency and heat-reuse conditions. - Ireland's grid operator has effectively restricted new Dublin connections, and connection policy increasingly requires on-site dispatchable generation or storage. - Japan uses energy-conservation-law benchmarking for data centres, with a PUE-type benchmark target for 2030.

10.2.3 Certifications

Table 10.4 Certifications

Certification	Scope	Strength	Weakness
LEED (v4/v4.1 BD+C Data Centers; v5)	Building lifecycle; energy modelled against ASHRAE 90.4	Globally recognised by tenants	Design-stage heavy; weak on hourly carbon
BREEAM	Building sustainability (UK/EU)	Strong on materials, ecology	Less used in India
IGBC Green Data Center Rating System (CII)	Design and Operational certification; enterprise, colo, hyperscale, edge, campuses	India-specific; aims to improve national average PUE by 20–25%	Voluntary; few disclosed operational ratings
GRIHA	Indian green building (TERI/MNRE)	Government projects	No DC-specific variant of note
BCA-IMDA Green Mark for DCs 2024	Singapore	Platinum is now a licence condition	Singapore-specific
Uptime Tier + Sustainability assessments	Resilience; operational sustainability	Credible with enterprise buyers	Tier is resilience, not green
ISO 50001 / 14001 / 14064	Energy management, environmental management, GHG verification	Process discipline; EnEfG requires EMS	Not a performance threshold
ENERGY STAR (US)	Top-quartile operational energy	Performance-based	US only

India's first IGBC Platinum data centre, ICICI Bank's Hyderabad facility, illustrates both the value and the limits of labels. It reported a **PUE of 1.36 at only 33% load**, with **15 MW of wheeled off-site renewable energy** and façade PV. That was good practice for its time but would rate only around "C" on the EU scale.

10.2.4 Indian standards and policy instruments

Table 10.5 Indian standards and policy instruments

Instrument	Status (Oct 2026)	Relevance
BEE / ECBC & Super ECBC	PUE benchmarks aligned with ECBC classes	Building-level; no binding DC PUE cap
MeitY tenders	AI cloud infrastructure tenders (since 2024) require **PUE <1.35**	De facto public-procurement standard
TRAI	Recommendations on a regulatory framework for data centres (incl. green measures)	Advisory
IGBC Green DC	Voluntary design/operational certification	Market signalling
SEBI BRSR / BRSR Core	Listed companies disclose energy, water, emissions; assurance phased in	Covers listed operators and parents (Bharti, Adani, Reliance, Sify via US listing etc.)
Carbon Credit Trading Scheme	Compliance market: ~490 obligated entities in 7 industrial sectors; first CCC trading expected ~October 2026	Data centres are *not* obligated entities; the offset mechanism and price signal matter
Draft National Data Centre Policy	Announced 2020; MeitY revived consultations (economic zones, single window, RE use)	No binding green thresholds yet
State DC policies	Karnataka 2026–31 (PUE/WUE reporting + incentives); UP 2026 (50% RE target for new DCs by 2030, green "booster" interest subsidy); others	States are leading on green conditions
Draft Climate Finance Taxonomy	DEA draft framework, May 2025 (comments to 25 June 2025); sectoral annexes pending	Inclusion of DCs with PUE/CFE criteria would unlock cheaper capital

10.3 Design levers with quantified savings

The table below ranks levers for an Indian air-and-liquid hybrid facility. Savings are indicative ranges drawn from engineering practice and the series' prior reports. **These are estimates, not measured Indian fleet data.**

Table 10.6 Design levers with quantified savings

Lever	Typical impact	Cost/complexity	India note
Site selection (climate, RE access, water, grid)	0.05–0.15 PUE; large water and carbon effects	Low if early	Bengaluru's mild, elevated climate beats Chennai/Mumbai for free cooling; Hoskote park pairs Pavagada solar + 60 MLD treated water
Hot/cold-aisle containment	10–20% cooling energy	Low	Table stakes
Raised supply air temperature (ASHRAE recommended 18–27 °C)	2–4% chiller energy per °C	Low	Germany will require 27 °C inlet from 2028
Free cooling / economisers, dry coolers	0.05–0.15 PUE in Bengaluru	Medium	Many hours below 24 °C wet-bulb
Direct-to-chip liquid cooling (DLC)	Removes ~70–80% of heat with warm water (30–45 °C); enables PUE 1.1–1.2	Medium-high	Essential for 50–130 kW AI racks
Immersion cooling	PUE ~1.03–1.1; no fans	High; ecosystem immature	Niche/edge

Lever	Typical impact	Cost/complexity	India note
High-efficiency UPS / eco-mode	2–4% of facility energy	Low-medium	Eco-mode at 99% efficiency vs ~94–96% double-conversion
Low-loss transformers, 415 V / higher-voltage distribution; 800 VDC for AI racks	1–3%	Medium	800 VDC is emerging with next-generation GPU racks
Lithium-ion UPS batteries	Smaller footprint, longer life	Medium	Fire-safety design essential
Server utilisation, virtualisation, consolidation	Can halve IT energy per workload	Low (software)	The biggest lever that PUE does not see
Efficient chips (ARM, accelerators), power capping	10–30% per workload	Medium	Procurement criterion: SS 715-type standards
AI model efficiency (distillation, quantisation, batching)	Large; Google reports a 33× fall in median Gemini prompt energy over one year (self-reported, unverified)	Software	Demand-side efficiency matters most for AI
AI-driven cooling optimisation, DCIM, digital twins	5–15% cooling energy	Low-medium	Quick win on existing Indian stock
Carbon-aware scheduling	Shifts flexible loads to solar hours	Low	India's solar-heavy midday supply favours daytime batch/training
Low-carbon materials (low-clinker concrete, recycled steel, hybrid timber)	Microsoft: hybrid mass timber cut embodied carbon an estimated **35% vs steel and 65% vs precast concrete**; Google cites up to 40% from low-carbon materials	Medium	Fly-ash/slag cements are abundant in India
Cool roofs, passive design	Small energy effect; reduces heat island	Low	Supports social licence
Treated wastewater (TTW) for cooling	Cuts freshwater to near zero	Medium (treatment plant)	Karnataka reimburses treated-water charges

Heat reuse: global examples and Indian possibilities. - **Microsoft–Fortum (Espoo/Kirkkonummi, Finland):** waste heat is expected to cover **about 40% of district-heating demand** (around 2 TWh/yr) for some 250,000 users in Espoo, Kauniainen and Kirkkonummi. CO_2 savings are expected at **about 400,000 t/yr**. The heat-pump plants are running on ambient air for now; data-centre heat is due "step by step" from about 2027. *These are projections, not achieved results.* - **Meta Odense (Denmark):** the 2020 goal was 100,000 MWh/yr (6,900 homes); the expansion target was 165,000 MWh (11,000 homes); Ramboll, which designed the plant, cites 215,000 MWh and >12,000 homes. - **Stockholm Data Parks:** roughly 30 data centres are connected to district heating/cooling (Space Daily). Stockholm Exergi CEO Anders Egelrud set the objective that "10% of Stockholm's heating needs will be provided by recovered data centre waste heat"; H&M's 1 MW Stockholm data centre can heat about 2,500 apartments. - **India has no district heating**, so the reuse logic flips to *cooling and process heat*: - Warm-water DLC loops (45–60 °C) can drive **absorption or adsorption chillers** for offices or adjacent buildings. - Low-grade process heat can go to food processing, laundries, pharma, or **greenhouse/agri-drying**. The Hoskote and Mysuru industrial belts are plausible offtakers. - Desalination pre-heating is an option for coastal Mangaluru. - Realistic Indian ERF in the near term is **2–10%**. The policy goal should be "heat-reuse ready" design (EU definition) plus zoning that co-locates offtakers.

Backup power. - Diesel gensets are the default in India and a growing Scope 1 source; Google's Scope 1 rose 20% in 2025 partly from backup generator use. - The pathway runs: 1. **HVO** (hydrotreated vegetable oil) drop-in, subject to Indian supply and certification; 2. **BESS** to ride through short outages and provide grid services, cutting genset run-hours; 3. **Gas gensets or fuel cells** for longer outages; 4. **Hydrogen fuel cells** (Microsoft has tested multi-MW hydrogen backup) once green hydrogen is available.

10.4 Renewable and carbon-free energy procurement

10.4.1 Procurement options in India

Table 10.7 Procurement options in India

Route	How it works	Typical landed cost (est.)	Pros	Cons
On-site rooftop solar	Behind-the-meter	₹3–4/kWh	No OA charges	Meets <2–5% of a DC's load (density)
Group captive	Consumer holds ≥26% equity and consumes ≥51%	₹5.5–7/kWh	Exempt from cross-subsidy and additional surcharges	Equity lock-in; state rules vary
Third-party open access PPA	Wheeled RE under state OA rules	₹6.5–8/kWh	Flexible	Surcharges, banking limits, regulatory risk
Green Energy Open Access (2022 Rules)	≥100 kW consumers; faster approvals, capped charges	Varies	National framework	Karnataka High Court struck down the state application; KERC issued new OA Regulations 2025
ISTS hybrid PPAs	Inter-state solar-wind(-storage)	₹5–6.5/kWh at bus + charges	Better hourly profile	Transmission waiver phase-down
SECI RTC/FDRE auctions	Firm, dispatchable RE (prior report: **₹5.25/kWh**)	~₹5.25 + delivery	High CFE share	Mainly for discoms; replicable bilaterally
State green tariffs	Discom supplies RE at a premium	Tariff + premium	Simple	Additionality unclear
RECs / I-RECs	Unbundled certificates	₹0.1–0.3/kWh	Cheap	Weakest claim; Microsoft has paused non-additional unbundled RECs
BESS / pumped storage	Firming	Adds ₹1.5–3/kWh to firmed supply	Lifts hourly CFE	Capex
Nuclear	Future PPAs	n/a in India today	24/7 CFE	India's SHANTI Act (December 2025) opened private participation; ₹20,000 crore Nuclear Energy Mission targets ≥5 indigenous SMRs by 2033; draft rules August 2026

Karnataka specifics: - KERC's **Open Access Regulations 2025** introduced monthly banking at **8% of banked energy in kind** and eight categories of consumer charges (transmission, wheeling, CSS, additional surcharge, banking, standby at 125% of energy charges, losses, scheduling). - In October 2025 BESCOM petitioned for an **additional surcharge of ₹1.65/kWh** on all OA consumers, a material headwind. For comparison, green OA consumers had their additional surcharge halved to ₹0.70 in FY24. - KERC's benchmark solar tariff is **₹3.07/kWh** for MW-scale plants (FY26). - **Implication:** in Karnataka, *group captive* (which avoids CSS and additional surcharge) is the structurally cheaper route for data centres. Stand-alone OA economics can swing by more than ₹1.5/kWh on regulatory decisions.

10.4.2 Hourly matching and additionality

Annual 100% matching remains the hyperscaler norm (Google for the ninth year, Microsoft in FY25). Yet Google's hourly CFE stayed **~65% in 2025** (66% in 2024) despite **12 GW of new clean-energy contracts**, because its electricity use grew **37%**. In India, solar-heavy portfolios meet daytime load but leave nights on coal. Moving from 50% to 80% hourly CFE typically requires wind, storage or firm RTC/FDRE contracts. **Recommendation:** Indian buyers should contract on *hourly* profiles, report a 24/7 CFE score alongside REF, and prefer ISTS hybrid and FDRE products over solar-only.

10.4.3 Nuclear and gas

- Global hyperscalers are signing firm clean power deals: Microsoft–Constellation (Crane/Three Mile Island restart), Amazon–Talen and X-energy, and Google–Kairos (500 MW of SMRs, first corporate SMR PPA, ~2030–2035).
- In India, SHANTI makes private nuclear legally possible, with a **100 GW-by-2047** national target and BSMR-200/220 designs. Realistically, nuclear will not supply Indian data centres before the early-to-mid 2030s. It belongs in a long-dated procurement roadmap, not current green claims.
- **Gas as a bridge:** Stand.earth reports that Microsoft is pursuing dedicated gas plants in the US that would emit >15 Mt CO_2/yr, which it says are undisclosed in the 2026 report. That is a cautionary example. In India, gas backup is defensible; gas baseload is not "green".

10.4.4 Indian operator snapshot

Table 10.8 Indian operator snapshot

Operator	Green credentials (latest disclosed)	Assessment
Nxtra by Airtel	49% RE in core DCs (FY25); 482,800 MWh RE contracted (3× FY21); PUE down 10% vs FY21; first Indian DC in RE100 (June 2024); 125.65 MW ISTS hybrid with AMPIN (205,167 MWh/yr); FY26 report targets 70% RE	Best-disclosed Indian operator; still annual matching
AdaniConneX	1 GW platform; **US$875m sustainability-linked loan (up to US$1.44bn)** with KPIs on energy efficiency, RE adoption and safety (first SLL in Indian DC sector, ING-led); partner in Google Vizag	Strong financing signal; site PUE/CFE data limited
Google–AdaniConneX–Nxtra, Vizag	US$15bn (2026–30), ~1 GW across three campuses; ground broken 28 April 2026; new RE + storage by Adani	Reports conflict on whether power comes from new Andhra Pradesh generation or Khavda (Gujarat); the hourly-matching design is undisclosed
Sify	>231 MW RE contracted; AI racks up to 200 kW	Good procurement; limited PUE disclosure
CtrlS	Captive solar (Nagpur) for Mumbai; NTPC Green partnership	Procurement-led
Yotta	Green tariff via Hiranandani group utility TUCO (hydro, biogas)	Novel structure; additionality unclear
Reliance–Meta, Jamnagar	168 MW AI DC announced June 2026, "supported by renewable energy"	Too early to assess
STT GDC India, NTT India, AWS/Microsoft India	RE PPAs announced in several states	*Site-level figures not re-verified this cycle*

The leading Indian operators report **PUE ~1.4–1.5** (SBI sector report, April 2026: "industry leaders achieving PUE values below 1.5"). That is better than the global average but well short of Singapore's 1.25 or Germany's 1.2.

10.5 Global benchmarks and the AI problem

Table 10.9 Global benchmarks and the AI problem

Company	Efficiency	Clean energy	Emissions trend	Water / other
Google	Fleet PUE **1.09** (2025); 83% less overhead than the 1.54 industry average	~65% hourly CFE; 12 GW new contracts; 100% annual match (9th year)	Ambition-based footprint **+18%** to ~14.5 Mt; Scope 3 **+25%**; DC construction ~2.3 Mt; electricity **+37%**	Target 24/7 CFE and 50% cut by 2030
Microsoft	n/a fleet PUE	40 GW contracted (19 GW online); 100% annual match	FY25 **20.29 Mt** vs 12.88 Mt FY20 (+~58% market-based; Stand.earth estimates ~80% gross); +25% YoY; Scope 2 now 13% of total after dropping unbundled RECs	Owned-DC WUE **0.27 L/kWh**; zero-water-for-cooling design (from August 2024) avoids **>125 million L/yr per DC**; mass-timber DCs
Meta, AWS, Equinix, Digital Realty, NTT, Iron Mountain, Apple	100% annual RE matching widely claimed; AWS "water positive by 2030"; Iron Mountain an early 24/7 CFE adopter	—	Mostly rising absolute emissions with AI	*Figures not re-verified this cycle*

Exemplars for cold climates (Verne Global Iceland, EcoDataCenter Sweden, Green Mountain Norway) run on hydro/geothermal at PUE ~1.1–1.2 with heat reuse. **They are not replicable in India.** Singapore's Jurong Island cohort is the relevant model: liquid cooling, low-carbon molecules and imported power, with PUE 1.25 in the tropics.

What this means: efficiency gains (PUE 1.09) and record clean-energy buying have not stopped absolute emissions rising, because AI load growth and hardware/construction Scope 3 dominate. Any Indian green standard focused only on PUE and annual RE will miss the main driver: lifecycle carbon of fast-growing IT fleets.

10.6 Economics of going green: a 20 MW Bengaluru colocation worked example

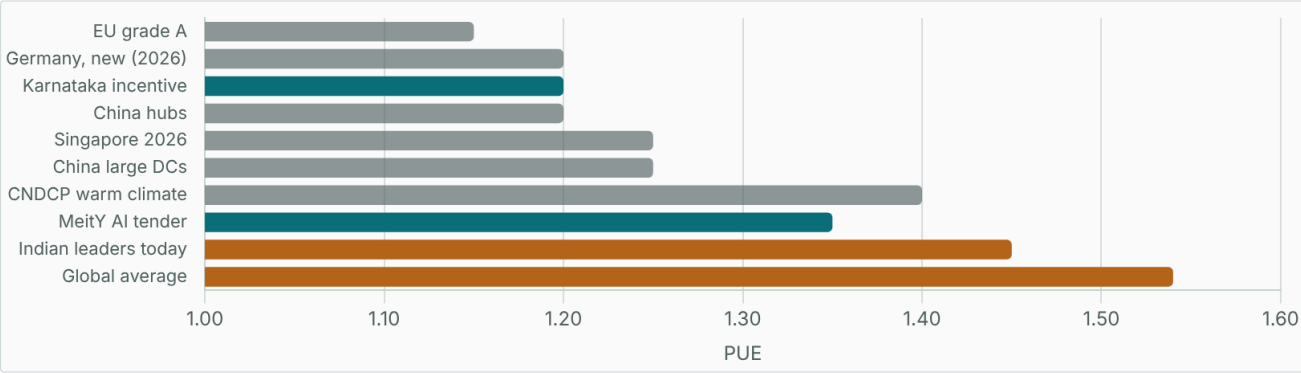


Figure 10.1 PUE thresholds around the world. Lower is stricter. *Source: India Data Centre Atlas, chapter built on this chapter's data.*

10.6.1 Assumptions (flagged estimates)

Table 10.10 Assumptions (flagged estimates)

Parameter	Baseline	Green design	Source/basis
IT capacity	20 MW	20 MW	—
Average IT utilisation	70% → 14 MW avg	Same	Estimate
Annual IT energy	14 × 8,760 = **122.64 GWh**	Same	Calc
PUE	1.50	1.25	Indian average vs green target
Facility energy	**183.96 GWh**	**153.30 GWh**	Calc
Power source	100% grid	80% RTC RE (group captive/OA), 20% grid	—
Grid all-in tariff	₹8.0/kWh	₹8.0/kWh	Estimate for HT commercial incl. demand charges and duty
RE landed cost	—	₹7.25/kWh (central); range ₹6.5–8.5	SECI RTC ₹5.25 + OA charges/banking (estimate)
Grid emission factor	0.710 tCO_2/MWh	0.710	CEA (prior report)
Cooling / WUE	Evaporative towers, 1.8 L/kWh freshwater	Dry coolers + DLC-ready, TTW adiabatic trim; 0.2 L/kWh freshwater	Estimate
Base capex	₹56 crore/MW → **₹1,120 crore**	+ green premium	Prior unit-economics report
Green capex premium	—	**₹60 crore (~5.4%)**	Estimate: efficient chillers/dry coolers, containment, eco-mode UPS, TTW plant, DCIM/digital twin, IGBC, low-carbon concrete, rooftop PV

10.6.2 Results

Energy saved by PUE: 183.96 – 153.30 = **30.66 GWh/yr** → × ₹8.0 = **₹24.5 crore/yr**.

RE procurement effect: 80% × 153.30 = 122.64 GWh of RE. At ₹0.75/kWh below grid, that saves **₹9.2 crore/yr**. *Contested:* if BESCOM's ₹1.65/kWh additional surcharge applies to third-party OA, the RE could cost ₹0.5/kWh *more* than grid (–₹6.1 crore/yr). Group captive structures avoid most of this.

Water: baseline 1.8 × 122.64 GWh = **220.8 million L/yr (~605 kL/day)**; green 0.2 × 122.64 = **24.5 million L/yr (~67 kL/day)**. Saving **~196 million L/yr** ≈ ₹1.8 crore at an assumed ₹90/kL. For scale, Karnataka's IT department, in an estimate "based on informal inputs from operators", puts the state's 32 data centres at roughly 4,000 kL/day (The News Minute), so this one baseline facility alone would be ~15% of that; a competing estimate from WELL Labs hydrologist Shashank Palur puts Bengaluru data centres at ~20 million litres/day.

CO_2: - Baseline: 183.96 GWh × 0.710 = **130,600 t/yr**. - Green: 153.30 × 20% × 0.710 = **21,800 t/yr**. - Reduction **~108,800 t/yr (–83%)**: 21,800 t from efficiency and 87,100 t from RE. - At Karnataka's ~0.5 factor, the reduction is ~76,700 t/yr.

Unit cost (links to the LCOD in the prior report): - Energy cost per IT-kWh: baseline 1.5 × ₹8.0 = ₹12.0; green 1.25 × (0.8 × 7.25 + 0.2 × 8.0) = **₹9.25**. - Saving **₹2.75/IT-kWh**, about **9% of the ~₹29.7/IT-kWh LCOD**. This is a material competitive edge in a price-sensitive colo market.

Simple payback: ₹60 crore ÷ (₹24.5 + ₹1.8) crore ≈ **2.3 years** on efficiency and water alone (excluding the RE effect).

10.6.3 Marginal abatement cost (MAC)

Annualised capex: ₹60 crore × CRF (10%, 15 yrs = 0.1315) = **₹7.9 crore/yr**.

MAC = (annualised capex + Δopex) ÷ tCO_2 abated

Table 10.11 Marginal abatement cost (MAC)

Package	Net annual cost (₹ crore)	tCO_2 abated/yr	MAC (₹/tCO_2)
Efficiency + water (PUE 1.5→1.25, dry/TTW)	7.9 – 24.5 – 1.8 = **–18.4**	21,800	≈ **–₹8,450**
RE at ₹0.75/kWh discount	–9.2	87,100	≈ **–₹1,060**
RE at ₹0.50/kWh premium (adverse OA case)	+6.1	87,100	≈ **+₹700**
Full package (central case)	**–27.6**	**108,800**	≈ **–₹2,540**

Interpretation: - Efficiency is a no-regret investment worth doing at any carbon price. - RE abatement costs between –₹1,060 and +₹700/t, below or within the ₹250–1,500/t range early estimates suggest for CCTS credits (Saur Energy), around Climate Risk Horizons' estimated initial price of about ₹955/t, and far below EU ETS levels (around €84/t). - **The carbon price is not the binding constraint in India; open-access regulation is.**

Karnataka incentive kicker: the ₹20 crore PUE incentive applies only to PUE **1.00–1.20**, sustained for three years and paid in year four. A design at 1.25 *misses* it. Pushing to 1.20 saves another ~6.1 GWh (≈₹4.9 crore/yr) and captures ₹20 crore, so **design to 1.20**. The WUE incentive (100% of water charges up to ₹30 lakh/yr for two years) needs WUE of 0.7–1.3 L/kWh. Our green case at 0.2 is better than that band, so check eligibility wording in the operational guidelines. Treated-water reimbursement is up to ₹8 lakh/yr for five years.

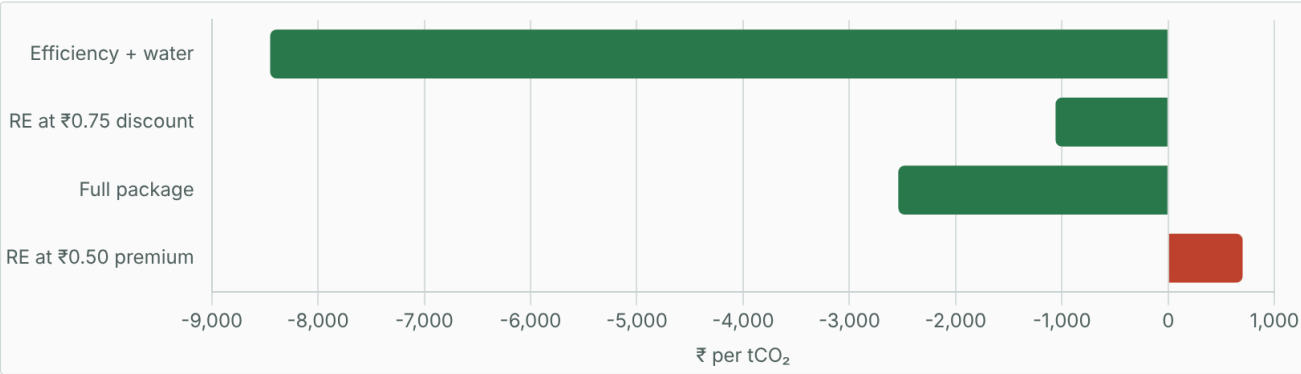


Figure 10.2 Abatement cost by measure. ₹ per tonne of CO_2, 20 MW Bengaluru colo. Negative means it saves money. *Source: India Data Centre Atlas, chapter built on this chapter's data.*

10.6.4 Green finance and valuation

- **Sustainability-linked loans:** AdaniConneX's US$875m SLL (expandable to US$1.44bn; eight international lenders) ties margins to energy efficiency, RE adoption and safety KPIs. Typical SLL margin step-downs of 5–15 bps on ₹800 crore of debt are worth ₹0.4–1.2 crore/yr. That is modest, but access to international lenders matters more than the discount.
- **Green bonds:** the draft climate finance taxonomy could make data centres eligible if PUE/CFE thresholds are written into a sectoral annex. Without that, Indian DC green bonds rely on international frameworks (ICMA, Climate Bonds).
- **Valuation:** hyperscaler RFPs increasingly require PUE, CFE and WUE commitments. Singapore and the EU are making ratings visible to buyers and procurement, so green performance is becoming a *pre-qualification*, not a premium. Treat the capex premium as the cost of staying in the market.

10.7 India and Karnataka

10.7.1 Landscape

- **Capacity:** CBRE's India Alternate Sectors Outlook 2026 says capacity "surpassed 1,700 MW in 2025, driven by a record 440 MW of new supply", with ~500 MW of new supply expected in 2026 (+30%). Karnataka has 32 data centres and targets **1 GW cumulative IT load by 2031** and a 10–12% share of India's AI-ready capacity.
- **Efficiency:** leaders report PUE ~1.4–1.5; the wider fleet is likely higher. **There is no public national dataset; this is a key data gap.**
- **Certified stock:** IGBC lists a small number of rated data centres. The first Platinum was ICICI Bank Hyderabad, and early certified projects include TCS, Wipro, HCL, HSBC and Netmagic. Operational certifications remain rare.

10.7.2 Challenges

1. **Hot-humid coastal hubs** (Mumbai, Chennai) limit free cooling. Bengaluru and the Deccan plateau are advantaged.
2. **Coal-heavy grid** (0.710 tCO_2/MWh): every unit of PUE overhead is carbon-intensive.
3. **Open-access friction:** surcharges, banking limits, the High Court annulment of Karnataka's GEOA application, and pending additional surcharge petitions.
4. **Water stress:** Bengaluru data centres sit largely in water-stressed areas, and until recently neither BWSSB nor KSPCB tracked their water use separately.
5. **Diesel dependence** for backup.
6. **E-waste:** the E-Waste (Management) Rules 2022 impose EPR, but server refurbishment and secondary markets are thin.

10.7.3 Opportunities

- **RE abundance:** Karnataka has one of India's highest RE shares, including Pavagada solar and strong wind, plus very low solar costs (KERC benchmark ₹3.07/kWh).
- **Hoskote park:** a ~500 MW concept with Pavagada solar and **60 MLD secondary-treated water**. It is a ready-made "green zone" if tertiary treatment and dual-grid supply are delivered.
- **Karnataka Sustainable Data Centre Policy 2026–31** (approved September 2026; ₹1,300.45 crore outlay):
- Mandatory plant-wise **PUE and WUE reporting**.
- **₹20 crore** one-time PUE incentive for PUE 1.00–1.20 sustained for three years.

- WUE incentive of up to ₹30 lakh/yr for two years at WUE 0.7–1.3 L/kWh.
- 100% treated-water reimbursement up to ₹8 lakh/yr for five years.
- Data Centre Resource Heat Mapping of water and power.
- Sustainable DC zones and parks (Hoskote, Mangaluru, Mysuru; ~1,000 MW combined).
- RE incentives and wheeling-charge reimbursement.
- A Centre of Excellence with R&D support up to 50% (capped at ₹15 crore).
- Customised packages for >₹3,000 crore projects, including subsea data centres and DC-linked desalination.
- **Assessment:** this is India's most advanced state green-DC framework, but it has gaps:
- No *binding* PUE ceiling.
- No carbon/CFE metric.
- The WUE band (0.7–1.3) is lax by EU standards (where 1.0 is grade G), so it rewards mediocre performance.
- No embodied-carbon, e-waste or diesel provisions.
- **National:** the draft National Data Centre Policy, the Green Credit Programme (mostly afforestation/water credits; marginal for DCs), CCTS (DCs not covered) and the draft taxonomy are all pending or peripheral. **States are setting the de facto standard.**

10.8 Green Data Centre maturity model (India-calibrated)

Table 10.12 Green Data Centre maturity model (India-calibrated)

Dimension	L1: Basic	L2: Managed	L3: Efficient	L4: Leading	L5: Regenerative
Energy (annual PUE, measured)	>1.7 or unmeasured	1.5–1.7	1.35–1.5	1.2–1.35	≤1.20 (air) / ≤1.15 (liquid)
Carbon (hourly CFE / CUE)	Grid only; CUE >1.0	Annual RECs; REF <50%	REF 50–80% via PPAs	≥70% hourly CFE; CUE ≤0.3	≥90% hourly CFE, additional; location-based disclosed
Water (freshwater WUE, L/kWh)	>2.0 or unmeasured	1.3–2.0	0.7–1.3	0.2–0.7; TTW	<0.2; zero freshwater for cooling; basin replenishment
Materials (embodied)	No LCA	LCA done	10–20% below baseline	30–40% below; EPDs required	>40% below; hybrid timber; low-clinker
Circularity	Ad hoc disposal	Authorised e-waste recycler	≥50% assets reused/refurbished	≥90% reuse/recycle; extended lifecycles	Closed-loop component recovery; supplier take-back
Heat	Rejected	Assessed	Heat-reuse ready (EU definition)	ERF 2–10% (absorption cooling/process heat)	ERF >10% with contracted offtaker
Backup	Diesel, unmonitored	Diesel run-hours tracked	HVO-ready, BESS for short events	BESS + HVO; diesel <20 h/yr	Fuel cells/zero-emission backup
Transparency	None	Annual report (BRSR)	Site-level PUE/WUE published	Third-party assured PUE/WUE/CFE; Scope 3	Real-time public dashboards; hourly CFE

Where Indian leaders sit today: mostly **L2-L3** (PUE ~1.4–1.5, ~50% annual RE, limited site disclosure). Google's global fleet sits at **L4-L5 on energy and L4 on carbon**. Singapore's DC-CFA2 conditions roughly equal **L4**.

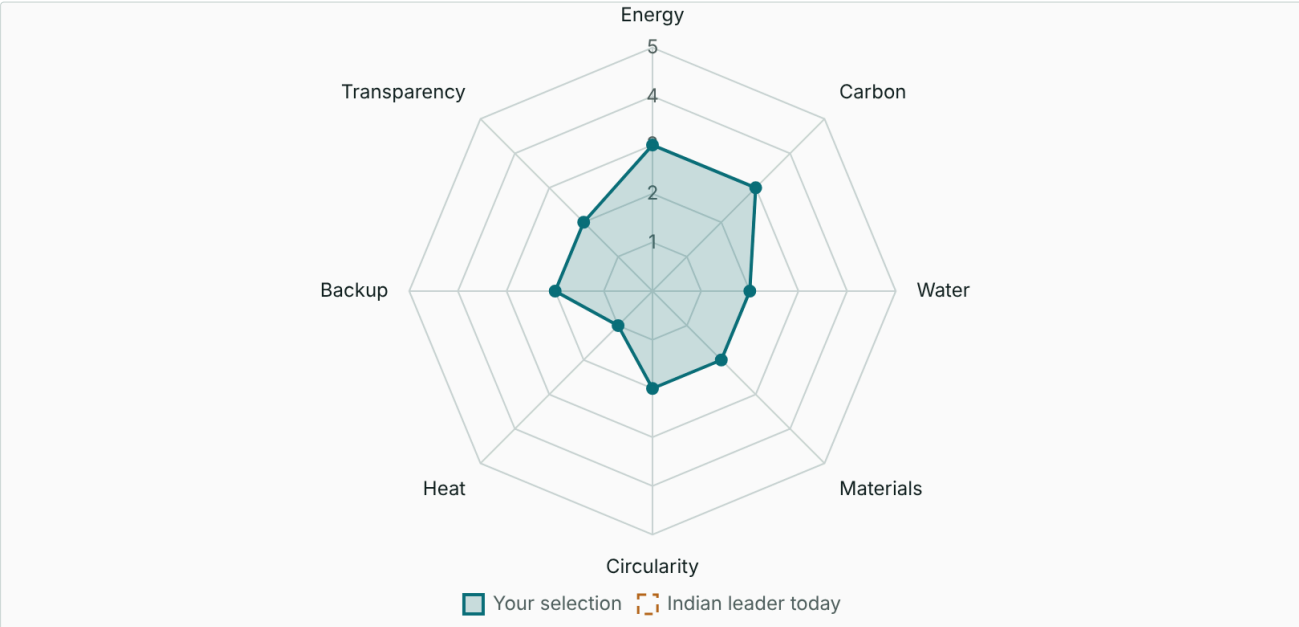


Figure 10.3 Green data-centre maturity: a self-assessment against today's Indian leaders. Eight dimensions scored from level 1 to level 5. *Source: India Data Centre Atlas, chapter built on this chapter's data.*

10.8.1 Blueprint: 20 MW Bengaluru colo ("L4 by design")

- **Site:** Hoskote park or an equivalent plug-and-play zone, chosen for dual-grid supply, TTW and Pavagada RE.
- **Cooling:** air-cooled halls with hot-aisle containment at 24–27 °C supply. Chillers with dry coolers and economiser; no open towers. Two DLC-ready halls (rear-door/CDU capable) for 40–80 kW racks.

- **Power:** eco-mode UPS, Li-ion, low-loss transformers; BESS of ~10 MWh for peak-shaving and outage ride-through.
- **Energy:** 80% hourly-profile RE via group captive (wind-solar hybrid plus storage), rooftop PV, and a target hourly CFE score reported.
- **Water:** TTW for all non-potable use; freshwater WUE ≤0.2.
- **Targets:** design PUE ≤1.20 (captures the ₹20 crore incentive); IGBC Platinum *operational*; ISO 50001/14001/14064.
- **Materials:** fly-ash/slag concrete; EPD-backed steel; 30% embodied-carbon reduction target.
- **Backup:** HVO-ready gensets; contractual diesel run-hour cap.
- **Disclosure:** quarterly site PUE/WUE/CFE to tenants and KITS.

10.8.2 Blueprint: 100 MW Bengaluru-region AI campus ("L4–L5")

- **Load:** 100 MW IT at ~80% utilisation = **700.8 GWh/yr IT**. At PUE 1.18 that is **~827 GWh/yr** of facility energy, about 587,000 tCO_2/yr if grid-supplied at 0.710, which is why procurement is the design.
- **Cooling:** ≥75% of heat to DLC warm-water loops (inlet 30–40 °C), dry coolers for the rest; WUE <0.1 freshwater (Microsoft-style closed-loop, zero-water-for-cooling).
- **Power architecture:** medium-voltage to the rack; 800 VDC readiness for next-generation GPU racks; grid-forming BESS (≥200 MWh) providing frequency services.
- **Energy:** a ≥600 MW hybrid portfolio (wind-solar-storage via ISTS and group captive), FDRE contracts and pumped-storage capacity rights. Target **≥80% hourly CFE by year three** and a nuclear option reserved for the 2030s under SHANTI.
- **Heat:** 45–60 °C return loops feeding absorption chillers for campus offices and an adjacent agri-processing or greenhouse cluster (target ERF 5%+).
- **Workload:** carbon-aware scheduling of training jobs to solar/wind-rich hours; power capping; model-efficiency KPIs (energy per inference).
- **Lifecycle:** hardware PCF requirements in GPU/server procurement; refurbishment/resale programme; e-waste EPR compliance.
- **Indicative capex:** ₹72.5 crore/MW (prior report) × 100 = ₹7,250 crore, with the green premium largely embedded because DLC is mandatory for AI densities anyway. *Estimate.*

10.9 Policy recommendations

10.9.1 For Government of India (MeitY, MoP, BEE, MoEFCC, DEA)

1. **Mandatory disclosure:** all DCs ≥500 kW IT report annual PUE, WUE, REF, *hourly CFE*, location- and market-based emissions, and refrigerant and diesel use to a national registry. This is a BEE-administered EU-style database, phased in from FY2027-28.
2. **Minimum performance standards (phased):** new DCs at PUE ≤1.35 from 2027 and ≤1.25 from 2030 (≤1.20 for liquid-cooled AI halls); existing DCs ≤1.6 by 2030. Use a climate-zone adjustment rather than a single number.
3. **Public procurement:** extend MeitY's PUE <1.35 tender rule to all NIC/government cloud and colocation contracts, adding WUE and ≥50% CFE criteria.
4. **Taxonomy inclusion:** add a data-centre annex to the Climate Finance Taxonomy. Eligibility should require PUE ≤1.3, ≥70% hourly CFE and freshwater WUE ≤0.4, with a transition category for retrofits.
5. **Open-access reform:** a national "digital infrastructure green OA" window with capped additional surcharge for hourly-matched RE, standardised banking, and FDRE access for large consumers.
6. **Diesel phase-down:** emissions norms plus run-hour reporting for DC gensets; tax parity for HVO; BESS-for-backup recognised in reliability codes.
7. **E-waste circularity:** EPR credits for certified server refurbishment and reuse; data-sanitisation standards that permit reuse rather than shredding.
8. **Nuclear readiness:** under SHANTI rules, allow long-term nuclear PPAs and co-location for data-centre clusters, with clear AERB siting rules.

10.9.2 For Government of Karnataka (DEIT/KITS, KERC, BESCOM, BWSSB)

1. **Convert reporting into thresholds:** a binding PUE ≤1.35 for new DCs in sustainable DC zones from 2028.
2. **Fix the WUE incentive band:** reward ≤0.4 L/kWh freshwater, not 0.7–1.3.
3. **Add a CFE incentive:** pay for hourly-matched RE, not annual REC matching.
4. **KERC:** exempt hourly-matched, additional RE for DCs in notified zones from the proposed additional surcharge, or cap it at ₹0.70. Approve a **"24/7 green tariff"** product from BESCOM backed by FDRE/storage contracts.
5. **Heat-reuse zoning:** co-locate DC parks with agri-processing, cold chain and pharma clusters; require "heat-reuse ready" design.
6. **Water:** make TTW use the default in Hoskote; publish the Resource Heat Map; meter all DC borewells.
7. **Publish** site-level PUE/WUE annually, as the EU now does.

10.9.3 For operators

1. **Design to PUE 1.20** in Bengaluru (and to 1.25–1.30 in coastal hubs with liquid cooling).
2. **Contract for hourly CFE** with hybrid plus storage; report the 24/7 score; stop relying on unbundled RECs.
3. **Eliminate freshwater cooling** via dry coolers, DLC and TTW.
4. **Embed lifecycle carbon:** EPDs, low-clinker concrete, hardware PCFs.
5. **Publish site-level metrics** with third-party assurance under BRSR Core.
6. **Use SLLs** tied to PUE, CFE and WUE KPIs to tap international lenders.

7. **Plan a backup transition** (HVO → BESS → fuel cells).

10.10 Caveats

- **Greenwashing risks:** unbundled RECs, annual matching and avoidance offsets can make a coal-powered site appear "100% renewable". Hyperscalers themselves show the gap: Google's location-based Scope 2 is reported at ~15 Mt (secondary source) versus far lower market-based figures. Treat any claim without hourly and location-based data as unverified.
- **Rebound (Jevons paradox):** efficiency lowers cost per computation and expands demand. Google's per-prompt energy fell sharply, yet its electricity use rose 37% and total emissions 18%. A green data centre cannot by itself offset unbounded AI demand growth. Policy must also address demand (efficiency standards for models and IT).
- **PUE limits:** PUE rewards inefficient IT (higher IT load lowers PUE) and penalises good practices such as raising temperatures, which shifts load to server fans. Pair it with utilisation (ITEUsv) and energy-per-workload metrics.
- **Water–energy trade-off:** dry coolers cut water but can raise PUE in hot hours. The Bengaluru climate makes the trade-off favourable; Chennai and Mumbai less so.
- **Data gaps:** there is no public national dataset of Indian DC PUE, WUE or CFE. Karnataka's water figure (roughly 4,000 kL/day, per the IT department via The News Minute) is an informal operator estimate, and WELL Labs' ~20 million litres/day estimate for Bengaluru data centres is about five times higher. Operator figures (e.g. Nxtra's 49% RE) are self-reported and use annual matching.
- **Estimates and contested figures:**
- The 20 MW worked example uses assumed tariffs (₹8.0 grid, ₹7.25 RE landed), utilisation (70%), WUE values and a capex premium (₹60 crore). Results are sensitive to KERC's surcharge decisions.
- CCTS prices are projections (₹250–1,500/t in early estimates cited by Saur Energy; about ₹955/t initially per Climate Risk Horizons) because trading had not begun at the time of writing.
- Heat-reuse figures for Espoo are projections.
- Reports conflict on the Google–Vizag RE source.
- Germany's existing-site PUE thresholds may be relaxed (draft amendment).
- The EU rating regulation is still in its scrutiny period.
- **Items not re-verified in this research cycle:** US state rules, Netherlands/Ireland/Japan specifics, Andhra Pradesh's reported RE condition for data centres, TRAI recommendation details, and individual figures for Meta, AWS, Equinix, Digital Realty, NTT, Iron Mountain and Apple. Treat these as background, not citations.

Sources cited in this chapter

Organisations and publications named in this chapter's text, in order of how often they are cited. Full details are in the bibliography.

- Google environmental reports, efficiency data and Gemini inference paper
- Microsoft sustainability reports and datacenter design posts
- ISO/IEC 30134 data-centre KPI standards
- State electricity regulatory commissions (KERC, MERC, TNERC, TGERC) tariff and open-access orders
- Ministry of Power, orders on ISTS waiver and green open access
- Amazon Web Services sustainability reporting
- AdaniConneX disclosures
- Singapore IMDA Green Data Centre Roadmap and DC-CFA
- Ministry of Electronics and Information Technology (MeitY), Data Centre Policy draft and IndiaAI
- Meta sustainability reports and data index
- Bureau of Energy Efficiency
- Equinix sustainability reports
- NTT DATA ESG data
- Nxtra by Airtel disclosures
- Germany, Energy Efficiency Act (EnEfG)
- Uptime Institute, Global Data Center Surveys and Tier Standard
- Bangalore Water Supply and Sewerage Board
- Solar Energy Corporation of India tenders
- ASHRAE TC 9.9 thermal guidelines
- STT GDC India ESG report
- Sify Infinit Spaces sustainability report
- Reliance Industries announcements
- WELL Labs, Bengaluru water research
- The Green Grid metric definitions
- European Union Energy Efficiency Directive and Delegated Regulation 2024/1364
- CBRE, India data centre market reports
- Central Electricity Authority (CEA), CO_2 Baseline Database and grid reports
- Ministry of Environment, Forest and Climate Change, thermal plant water norms and EIA
- Ministry of New and Renewable Energy statistics
- CtrlS disclosures
- Yotta Data Services disclosures
- State Bank of India sector report on data centres (April 2026)

PART III · ECONOMICS AND SUSTAINABILITY

Chapter 11
Least-water power: an hourly model for a Karnataka data centre

WHAT THIS CHAPTER COVERS

This chapter presents a model built for this report. It asks which combination of solar power, wind power, hydropower, batteries and grid electricity can supply a 50 megawatt-IT data centre in Karnataka every hour of the year at the lowest cost, and how that answer changes when the amount of water the supply may use is capped. It explains the model, presents every option it found, tests the result against the uncertain water use of reservoir hydropower, and sets out the reasons for the recommended option.

KEY TAKEAWAYS

1. The cheapest clean-power mix for the data centre costs ₹5.20 per kilowatt-hour, well below the ₹7.00 grid tariff, but it uses about 1,451 million litres of water a year, more than the 1,139 million litres used by buying all power from the grid, because reservoir hydropower evaporates water.
2. The recommended mix is 161 megawatts of solar, 86 megawatts of wind and 379 megawatt-hours of four-hour batteries with grid top-up and no hydro. It costs ₹5.39 per kilowatt-hour, uses about 218 million litres a year and supplies 83% of the hours from clean power.
3. Each kilolitre of water that the recommended mix saves, compared with the cheapest mix, costs about ₹85, which is less than the ₹99 per kilolitre that Bengaluru charges for municipal water. Saving more water beyond this point costs more than the water is worth.
4. Because the recommended mix contracts no reservoir hydro, its water use does not depend on how much water Indian reservoirs evaporate, which nobody measures, while the cheapest mix could use anywhere from 255 to 2,647 million litres depending on that unknown.
5. Against buying all power from the grid, the recommended mix is about 23% cheaper, uses about 81% less water and emits about 83% less carbon dioxide.

How this chapter connects to the others. The water factors come from Chapters 5 to 7, and the recommendation is carried into the procurement guidance of Chapter 15.

11.1 The question the model answers

A data centre that wants to cut its water footprint in India faces a problem that the earlier chapters made clear. Most of the water behind a data centre is not used on site. It is evaporated at the coal power stations that generate the electricity the data centre buys. Buying clean electricity therefore reduces water as well as carbon. But not every clean source is free of water. Solar panels need occasional washing, and reservoir hydropower loses water to evaporation from the surface of its reservoirs. Wind and batteries use almost none. A buyer who wants the least water at a reasonable cost needs to know which combination of sources to contract, and how much it will cost.

The model in this chapter answers that question for one specific case: a data centre with 50 megawatts of IT load in Karnataka, running at a power usage effectiveness of 1.30, so that it draws a steady 65 megawatts from the grid in every hour of the year. The model chooses how much solar capacity, wind capacity, hydropower contract and battery storage to buy, and how to use them hour by hour, so that the data centre is always supplied. Any hour that cannot be met from these sources is met from the grid.

11.2 How the model works

The model is a linear programme, which is a standard mathematical method for finding the best solution to a problem that can be written as a set of linear equations and limits. It was solved with the HiGHS solver in the SciPy library. Each solution involves 52,564 decision variables: four capacities (solar megawatts, wind megawatts, hydro contract megawatts and battery megawatt-hours) and six quantities for each of the 8,760 hours of the year (hydro used, grid electricity bought, battery charging, battery discharging, battery state of charge and surplus energy curtailed).

Three groups of rules bind every solution. First, energy must balance in every hour: the electricity from solar, wind, hydro, battery discharge and the grid, minus what goes into the battery and what is curtailed, must equal the 65 megawatt load. Second, the battery must behave like a real battery. It loses 12% of the energy that passes through it (an 88% round-trip efficiency), it cannot hold more than its capacity, it cannot charge or discharge faster than a four-hour rating allows, and its state of charge at the end of the year must equal its state at the start. Third, the hydro contract cannot deliver more power than its contracted megawatts in any hour, cannot deliver more energy in a month than the reservoir's monthly availability allows, and cannot exceed 40 megawatts in total, which is roughly what a buyer could realistically contract in Karnataka.

The solar and wind output in each hour comes from profiles shaped like Karnataka's resource. The solar profile resembles the Pavagada and Tumakuru region, with an annual capacity factor of about 20.5% and lower output in the monsoon months from June to September. The wind profile resembles the Chitradurga and Gadag region, with an annual capacity factor of about 33%, strong winds from May to August, and a tendency for the wind to blow harder in the evening. Both profiles are synthetic. They were generated from a seeded random process so that they have the right seasonal and daily shape, but they are not measurements from a particular site. This is the main limitation of the model, and it is discussed again at the end of the chapter.

Table 11.1 How the model works

Input	Value used	Why this value
Load	50 MW IT × PUE 1.30 = 65 MW, flat all year	An efficient mid-size facility; a flat load is the simplest case to optimise.

Input	Value used	Why this value
Solar profile	Capacity factor 20.0%, monsoon dip	Shaped like the Pavagada and Tumakuru solar region.
Wind profile	Capacity factor 33.0%, strong May–August, evening bias	Shaped like the Chitradurga and Gadag wind region.
Solar cost	₹3.8 crore/MW, 25 years, 10%, O&M 1.5% → ₹0.480 crore/MW/yr	2026 Indian utility-scale benchmark.
Wind cost	₹7.0 crore/MW, same terms → ₹0.880 crore/MW/yr	2026 Indian benchmark.
Battery cost	₹3.5 lakh per MW-month for a 4-hour unit → ₹0.1 crore/MWh/yr	Recent Indian battery storage tender levels.
Hydro contract	₹4.5/kWh take-or-pay, up to 40 MW, monthly availability 25–85%	Indicative reservoir hydro power purchase terms.
Grid power	₹7.0/kWh including fixed charges	Approximate high-tension industrial tariff in Karnataka.
Open-access charge	₹1.0/kWh on renewable energy used	Wheeling, transmission and loss charges on delivered renewables.
Water factors	Solar 0.1, wind 0, battery 0, grid 2.0, hydro 0 / 10 / 20 L/kWh	Panel washing; Karnataka grid coal share; unmeasured reservoir evaporation tested in three cases.
Carbon	Grid 0.71 t CO_2/MWh; renewables zero	Central Electricity Authority grid factor.

11.3 How the trade-off between cost and water is traced

The model first finds the cheapest possible mix, with no limit on water. It then finds the mix that uses the least possible water, with no limit on cost. These two mixes mark the two ends of a range. The model then re-solves for the cheapest mix seven more times, each time with a tighter cap on water. Each cap allows a set fraction of the gap between the minimum water and the water of the cheapest mix: 75%, 50%, 35%, 25%, 15%, 10% and 5%. Each solution is a point on a curve of cost against water, which economists call a frontier: a point on the frontier cannot be made cheaper without using more water, or drier without costing more.

The water used by reservoir hydropower is uncertain, because no one measures the evaporation from Indian reservoirs and allocates it to the electricity they generate. The model is therefore run three times. In the base case, hydro uses 10 litres of water per kilowatt-hour. In the run-of-river case, hydro uses no water, which would be true of a plant without a storage reservoir. In the high case, hydro uses 20 litres per kilowatt-hour. Grid electricity is assumed to use 2.0 litres per kilowatt-hour, reflecting the share of coal in Karnataka's supply, and solar power 0.1 litres per kilowatt-hour for panel washing.

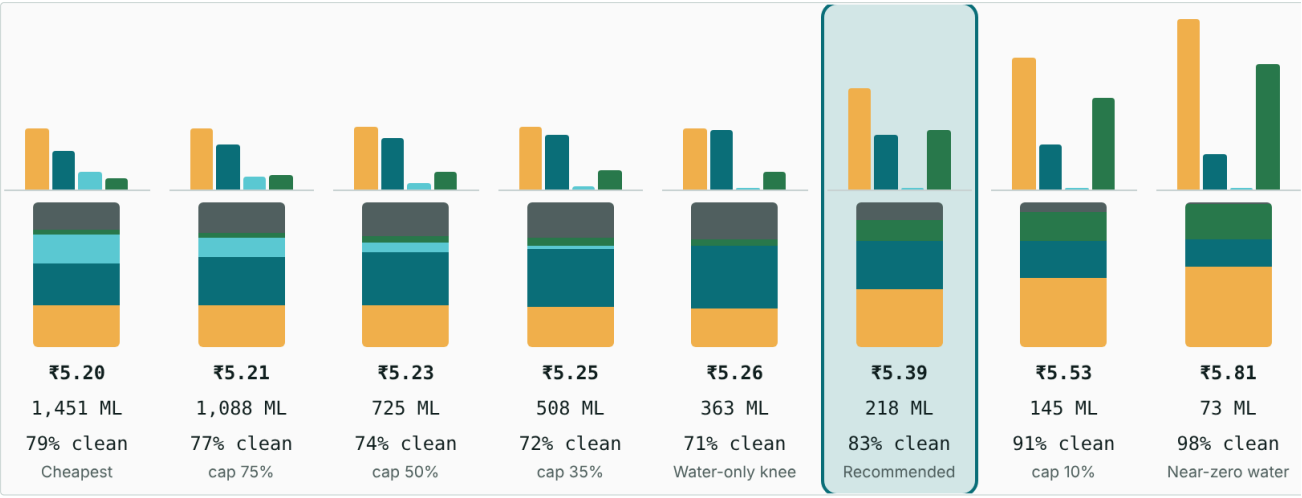


Figure 11.1 What gets built as the water cap tightens. Base case. Left to right: from the cheapest mix to the lowest-water mix. Top: capacity. Bottom: where each hour's energy comes from over a year. *Source: India Data Centre Atlas, chapter built on this chapter's data.*

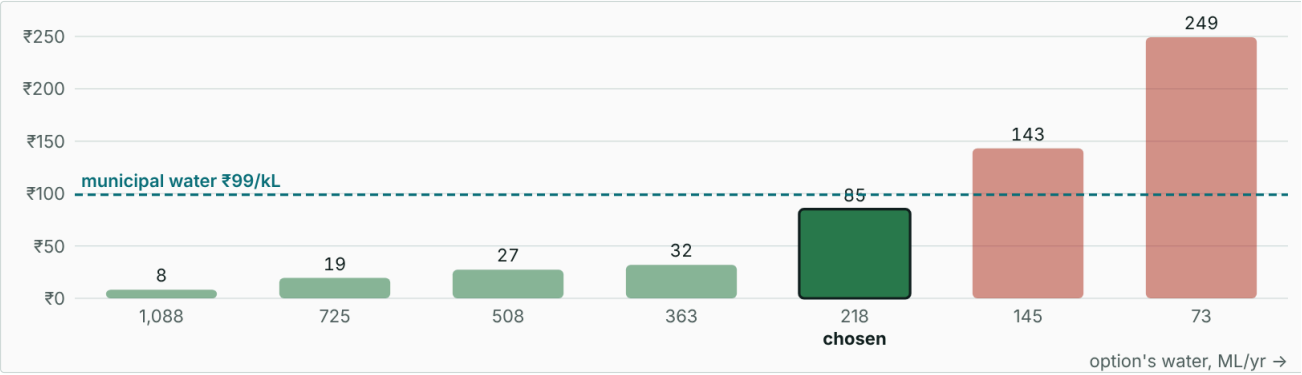


Figure 11.2 Rule 1 · Save water only while it is cheaper than buying it. Extra cost of each option per kilolitre of water saved, compared with the cheapest mix (base case). *Source: India Data Centre Atlas, chapter built on this chapter's data.*

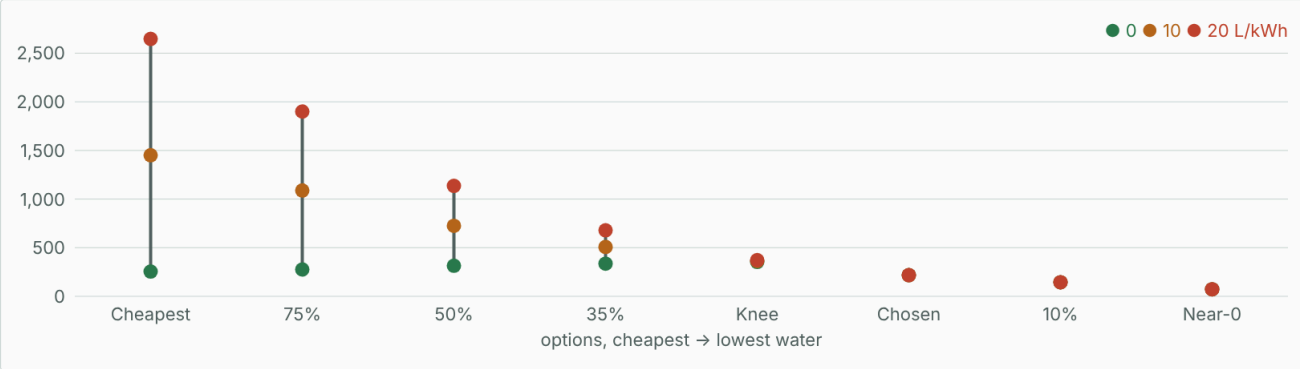


Figure 11.3 Rule 2 · Pick a mix that survives the hydro uncertainty. Annual water of each base-case mix if reservoir hydro really evaporates 0, 10 or 20 L/kWh. Nobody measures this for Indian reservoirs. *Source: India Data Centre Atlas, chapter built on this chapter's data.*

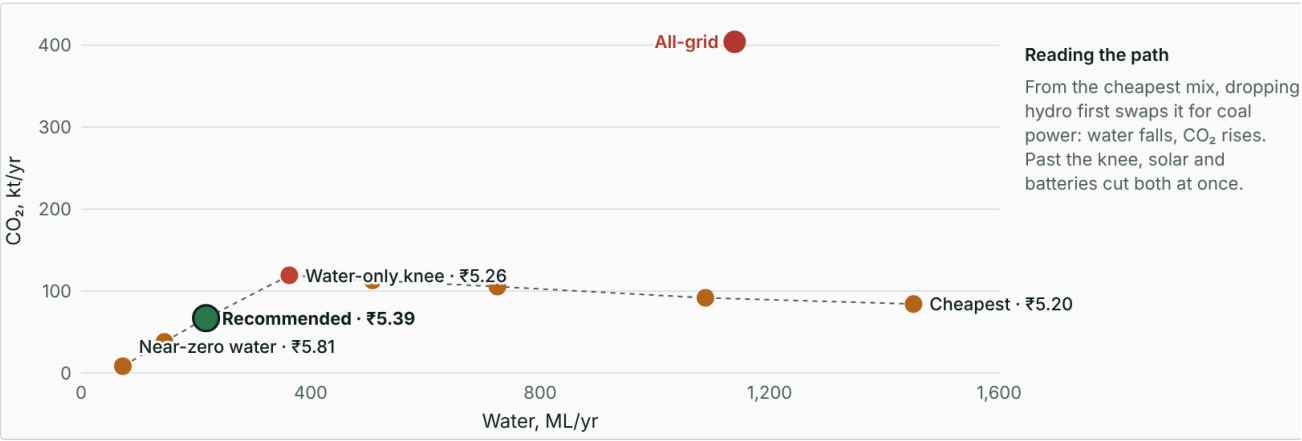


Figure 11.4 Rule 3 · Don't trade water for carbon. Water against CO_2 for each base-case option. The "water-only" knee cuts water by buying coal power; the recommended option cuts both. *Source: India Data Centre Atlas, chapter built on this chapter's data.*

11.4 Results: every option the model found

The table below lists every option the model found in the base case. Reading down the table, the water cap tightens. The first row is the cheapest mix and the last row is the mix that uses the least possible water. The column for clean energy shows the share of the year's consumption that is not supplied by the grid. The carbon dioxide column counts only grid electricity, at the Central Electricity Authority's factor of 0.71 tonnes per megawatt-hour, and treats solar, wind, hydro and battery output as zero-emission.

Table 11.2 Results: every option the model found

Option	Solar (MW)	Wind (MW)	Hydro (MW)	Battery (MWh)	Cost (₹/kWh)	Water (ML/yr)	Water (L/kWh)	Clean energy	CO_2 (kt/yr)
Cheapest mix	97	62	28	72	5.20	1,451	2.55	79%	84
Water cap 75%	98	72	20	91	5.21	1,088	1.91	77%	92
Water cap 50%	99	81	10	107	5.23	725	1.27	74%	105
Water cap 35%	99	88	4	117	5.25	508	0.89	72%	113
Water-only knee	97	95	0	112	5.26	363	0.64	71%	119
Recommended	161	86	0	379	5.39	218	0.38	83%	67
Water cap 10%	210	71	0	583	5.53	145	0.25	91%	38
Near-zero water	271	57	0	803	5.81	73	0.13	98%	8
Strict minimum water	0	593	0	4,356	18.16	0	0.00	100%	0

Three phases are visible in the table. In the first phase, from the cheapest mix down to the option labelled the water-only knee, the model removes the hydro contract and replaces it with grid electricity. Water falls by three-quarters, from 1,451 to 363 million litres a year, for an increase in cost of only about six paise per kilowatt-hour. But the share of clean energy falls from 79% to about 70%, because the grid supplies more of the load. In the second phase, solar capacity and batteries grow to replace the grid. Water continues to fall, and the clean share climbs back to 83% and then beyond 90%. In the third phase, only very large batteries can close the last night-time hours, and the cost rises steeply. The strict minimum-water solution uses no solar at all, because even panel washing uses some water, and instead builds 593 megawatts of wind and 4.4 gigawatt-hours of batteries, at a cost of ₹18.16 per kilowatt-hour. That option is shown for completeness but is not a practical choice.

A striking result appears in the first row. The cheapest clean-power mix uses more water, 1,451 million litres a year, than buying all power from the grid would use, 1,139 million litres a year. This is because the cheapest mix relies on a 28 megawatt hydro contract, and at 10 litres per kilowatt-hour reservoir hydro is five times as thirsty as Karnataka's grid. Clean power is not automatically water-light. The source matters.



Figure 11.5 Cost against water: every option the model found. Each dot is a least-cost mix for a given water cap. Moving left saves water; moving up costs more. Points below the dashed line save water for less than the city charges for it. Tap a dot. *Source: India Data Centre Atlas, chapter built on this chapter's data.*

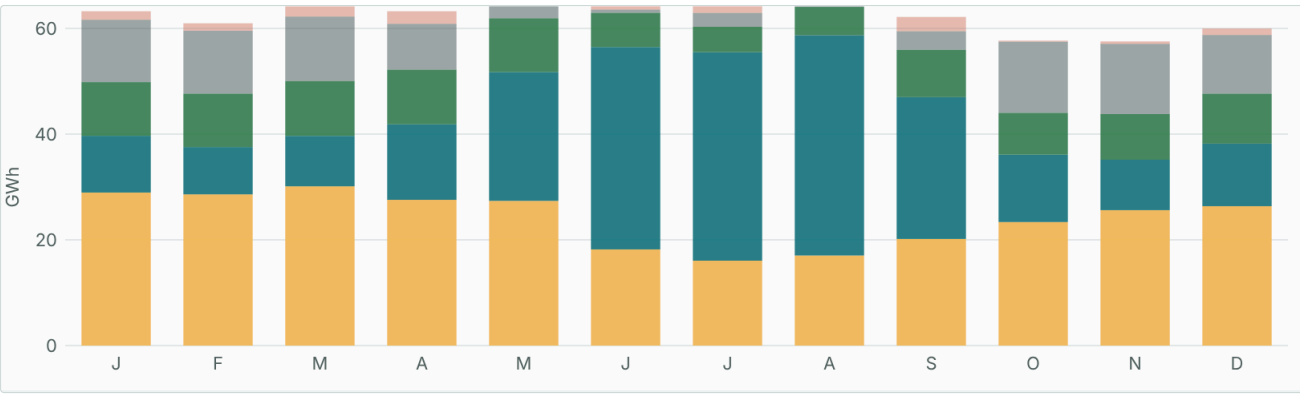


Figure 11.6 A year, month by month. Energy supplied to the load each month for the selected option (GWh). Curtailed surplus is shown as a thin cap. *Source: India Data Centre Atlas, chapter built on this chapter's data.*

11.5 The hydropower uncertainty

The next table shows how the cheapest mix and the lowest-cost options change when the water use of hydropower changes. In the run-of-river case, where hydro uses no water, the model keeps and even expands the hydro contract, because hydro is then both cheap and dry. In the high case, where hydro uses 20 litres per kilowatt-hour, the model drops hydro even sooner.

Table 11.3 The hydropower uncertainty

Hydro water case	Option	Solar (MW)	Wind (MW)	Hydro (MW)	Battery (MWh)	Cost (₹/kWh)	Water (ML/yr)	Clean energy
Hydro 0 L/kWh (run-of-river)	Cheapest mix	97	62	28	72	5.20	255	79%
Hydro 0 L/kWh (run-of-river)	Water cap 50%	145	43	38	252	5.30	127	91%
Hydro 0 L/kWh (run-of-river)	Water cap 15%	201	68	23	634	5.81	38	100%
Hydro 0 L/kWh (run-of-river)	Water cap 5%	40	290	40	132	7.57	13	100%
Hydro 10 L/kWh (base)	Cheapest mix	97	62	28	72	5.20	1,451	79%
Hydro 10 L/kWh (base)	Water cap 50%	99	81	10	107	5.23	725	74%
Hydro 10 L/kWh (base)	Recommended	161	86	0	379	5.39	218	83%
Hydro 10 L/kWh (base)	Near-zero water	271	57	0	803	5.81	73	98%
Hydro 20 L/kWh (high)	Cheapest mix	97	62	28	72	5.20	2,647	79%
Hydro 20 L/kWh (high)	Water cap 50%	99	79	13	102	5.22	1,324	75%
Hydro 20 L/kWh (high)	Water cap 15%	96	95	1	106	5.26	397	70%
Hydro 20 L/kWh (high)	Water cap 5%	221	66	0	628	5.56	132	92%

Because the true water use of reservoir hydro is not known, it is useful to ask how much water each base-case mix would use under each assumption. The table below does this. The cheapest mix could use anywhere from 255 to 2,647 million litres a year, depending on a number that no one measures. The recommended mix contracts no hydro, so its water use is the same, 218 million litres a year, whatever the true figure is.

Table 11.4 The hydropower uncertainty

Base-case option	Water if hydro uses 0 L/kWh	Water at 10 L/kWh	Water at 20 L/kWh	Spread
Cheapest mix	255	1,451	2,647	2,392
Water cap 75%	276	1,088	1,900	1,624
Water cap 50%	315	725	1,136	821
Water cap 35%	336	508	679	343
Water-only knee	353	363	372	19
Recommended	218	218	218	0
Water cap 10%	145	145	145	0
Near-zero water	73	73	73	0

11.6 Three rules for choosing an option

The model does not choose an option by itself. It shows the trade-off. To choose, this report applies three rules, each of which has a clear reason.

The first rule is to save water only while saving it is cheaper than buying it. For each option, the extra cost compared with the cheapest mix is divided by the water it saves compared with the cheapest mix. The result is the cost of each kilolitre of water saved. If that cost is lower than the price of water, saving the water is worth it on cost alone. The benchmark used here is ₹99 per kilolitre, the municipal water rate charged by Bengaluru's water board and used in the atlas simulator. Tanker water costs more, so this test is conservative.

Table 11.5 Three rules for choosing an option

Option	Extra cost vs cheapest (₹ crore/yr)	Water saved vs cheapest (ML/yr)	Cost per kL saved (₹)	Below ₹99 municipal price?
Water cap 75%	0.3	363	8	Yes
Water cap 50%	1.4	725	19	Yes
Water cap 35%	2.6	943	27	Yes
Water-only knee	3.5	1,088	32	Yes
Recommended	10.5	1,233	85	Yes
Water cap 10%	18.7	1,306	143	No
Near-zero water	34.4	1,378	249	No

Up to the recommended option, every kilolitre saved costs less than municipal water, at about ₹85 on average. The next option costs about ₹143 per kilolitre saved, and the one after about ₹249. At the margin, each additional kilolitre saved beyond the recommended option costs more than ₹1,100. Beyond the recommended option, the buyer would be paying more to save water than the water is worth.

The second rule is to choose a mix that does not depend on an unmeasured number. The recommended option has no hydro, so its water use is the same under all three hydro assumptions, as the robustness table shows. When an input is as uncertain as reservoir evaporation, a choice that does not depend on it is worth paying a little for.

The third rule is not to trade water for carbon. The water-only knee, at 363 million litres and ₹5.26 per kilowatt-hour, reduces water by buying more coal power, which raises carbon dioxide to about 119 thousand tonnes a year, compared with 84 thousand tonnes for the cheapest mix. The recommended option reduces both water and carbon, to 218 million litres and about 67 thousand tonnes.

11.7 The recommended mix

The recommended option is 161 megawatts of solar, 86 megawatts of wind and 379 megawatt-hours of four-hour batteries (about 95 megawatts of battery power), with grid electricity covering the remaining hours and no hydro contract. It costs ₹5.39 per kilowatt-hour, which is about ₹10.5 crore a year more than the cheapest mix (an increase of about 3.6%) but about ₹92 crore a year less than buying all power from the grid. It uses about 218 million litres of water a year, or 0.38 litres per kilowatt-hour, and supplies 83% of the year's consumption from clean power. Compared with buying all power from the grid, it is about 23% cheaper, uses about 81% less water and emits about 83% less carbon dioxide.

The hour-by-hour dispatch shows how this mix meets the load. On a typical April day, solar output rises to well above the 65 megawatt load at midday. The surplus charges the batteries, and a small amount is curtailed. In the evening the batteries discharge to carry the load past sunset, wind contributes through the night, and the grid fills the remaining gap in the early morning. In the monsoon months, from June to August, strong winds carry most of the load and the grid is almost unnecessary. On these synthetic profiles, the hardest month is October, when the monsoon winds have faded but cloud still reduces solar output. Grid use peaks in that month.

11.8 What the results mean for buyers and for policy

Several lessons follow from the model. A data centre that buys renewable power to reduce water should look at the water use of each source, not only at whether it is renewable, because reservoir hydro can use more water per kilowatt-hour than coal. Hourly-optimised solar, wind and storage can supply most of a data centre's hours at a cost well below the grid tariff in Karnataka, which means that water and carbon reductions do not require a cost premium compared with grid power. Policies that encourage round-the-clock renewable contracts, fair banking rules and co-located storage, all of which are recommended in Chapter 15, make this kind of portfolio easier to build. If a genuine run-of-river hydro contract were available, without reservoir evaporation, a mix including about 38 megawatts of it could reach about 127 million litres a year at ₹5.30 per kilowatt-hour, which would beat the recommended option. That is the one case in the model that does better.

The power supply is only part of a data centre's water footprint. The recommended mix addresses the water used at power stations. The water used on site depends on cooling, and Chapters 3, 7 and 8 show that dry or liquid cooling is needed to keep on-site water low as well.

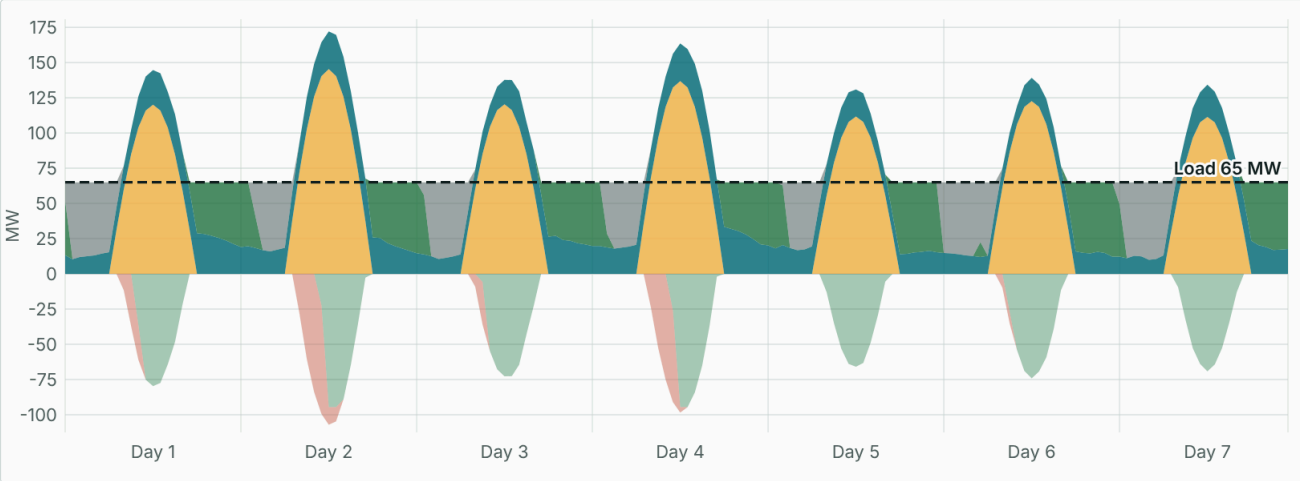


Figure 11.7 A week, hour by hour. How the chosen mix meets a flat 65 MW load. Above the line: supply. Below: energy going into the battery and surplus that is curtailed. *Source: India Data Centre Atlas, chapter built on this chapter's data.*

11.9 Limitations of the model

The model is illustrative. The solar and wind profiles are synthetic, with the right seasonal and daily shape for Karnataka but not measured at any site, and real weather includes multi-day lulls that the profiles may not capture. Costs are 2026 Indian benchmarks rather than quotes. The load is flat, whereas real data centres vary, and AI training loads can swing sharply. Open-access charges, banking rules and transmission charges differ by state and change often. Before any procurement decision, the model should be re-run with measured site data from the National Institute of Wind Energy and from satellite solar datasets, and with the actual open-access charges of the Karnataka Power Transmission Corporation and the relevant distribution company.

Sources for this chapter

The model file karnataka_dc_model.py, an hourly linear programme solved with SciPy HiGHS, with synthetic Karnataka-shaped solar and wind profiles and 2026 Indian cost benchmarks. Water factors and grid carbon follow Chapters 5 to 7 (CEA CO_2 Baseline Database; MoEFCC thermal plant water norms). The municipal water benchmark is the Bangalore Water Supply and Sewerage Board industrial tariff used in the atlas simulator.

PART IV · POLICY, PLACE AND THE FUTURE

Chapter 12
State data-centre policies: a state-by-state deep dive

WHAT THIS CHAPTER COVERS

This chapter reviews the national framework for data centres and then examines the policies of sixteen Indian states in turn: their instruments, power incentives, green conditions, land benefits, targets and projects. It compares the states in a master table, ranks them with an attractiveness index across seven factors, and discusses whether competition between states is producing a race to the bottom on standards or a race to attach sustainability conditions to incentives.

KEY TAKEAWAYS

1. As of October 2026 no national data-centre law is in force in India, and the real rulebook is about sixteen state regimes.
2. For a large project the deciding factors are the price and reliability of power, access to renewable energy, connectivity and policy stability, not headline subsidies. On those factors Maharashtra, Telangana, Andhra Pradesh, Tamil Nadu and Uttar Pradesh lead.
3. Karnataka's Sustainable Data Centre Policy 2026–31 is the only state policy that pays for measured efficiency, including a ₹20 crore incentive for a PUE of 1.20 or lower and reimbursement linked to water usage effectiveness and treated water.
4. A wave of policies in 2026, in Uttar Pradesh, Gujarat, Karnataka and Maharashtra, raised the ceiling on incentives and began to attach green conditions, although these conditions are usually soft. Maharashtra cut its green-power requirement from 100% to 51%.
5. Tamil Nadu's policy lapsed on 31 March 2026 and West Bengal's is about to lapse, so policy continuity is itself a factor that investors weigh.

How this chapter connects to the others. Chapter 13 uses these policy scores as one factor in ranking cities, and Chapter 15 recommends how national and state policy should change.

As of October 2026, no national data-centre law is in force. The real rulebook is about 16 state regimes, and for a large project the deciding factors are power price, renewable access and reliability, connectivity and policy stability, not headline subsidies. On those factors Maharashtra, Telangana, Andhra Pradesh, Tamil Nadu and Uttar Pradesh lead. Karnataka's new 2026–31 policy is the only one that pays for measured efficiency (PUE/WUE). A wave of 2026 policies — Uttar Pradesh, Gujarat, Karnataka and Maharashtra's amended parks policy — has raised the incentive ceiling. The same wave has started to attach green conditions to incentives, though usually soft ones.

12.1 Key Findings

1. **Count and coverage.** CEEW and SYSTEMIQ's February 2026 study, "Scaling India's Data Centre Ecosystem" (based on 24 stakeholder engagements), listed 15 states that support data centres through dedicated policies or IT/industrial frameworks: AP, Bihar, Chhattisgarh, Gujarat, Haryana, Karnataka, Maharashtra, MP, Odisha, Punjab, Rajasthan, Telangana, Tamil Nadu, UP and West Bengal. Since then Gujarat has notified a dedicated policy, UP and Karnataka have replaced theirs, and Maharashtra has overhauled its parks regime.
2. **CEEW's "5 of 15" finding is already out of date, but its diagnosis holds.** In February 2026 only Rajasthan, Odisha, Karnataka, Tamil Nadu and Andhra Pradesh embedded sustainability provisions. Since then Karnataka (PUE/WUE incentives and plant-wise reporting), UP (a PUE-linked "Green and Sustainable Operations Booster" and a 50% renewable goal) and Gujarat and Maharashtra (51% renewable floors) have added green conditions. The number has risen to roughly 8 of about 16. CEEW's three systemic gaps — weak water governance, no PUE/WUE/carbon performance standards, and weak climate-resilience framing — remain largely unaddressed.
3. **Power is what decides location.** CEEW's stakeholder consultations found electricity cost and reliability outweigh land, fiscal incentives and labour; citing DCD (2025), CEEW puts energy at typically 60–70% of operating cost. Land is a small share of build cost: about 1–3% according to Swarajya's analysis, though IMARC Engineering's 2026 guide puts it at 5–12%. That is why UP's 50% regional land subsidy drew zero data centres to Bundelkhand or Purvanchal in four years, while every operational UP facility sits in Noida or Greater Noida.
4. **Implementation is uneven and opaque.** - UP is the only state publishing a running tally: 644 MW proposed, ₹21,342.90 crore, 77.08 acres allotted or under consideration, about 101 MW commercialised. - Most states publish no data on disbursements. - CEEW's stakeholders report that despite single-window clearance provisions, delays persist in practice, particularly in land acquisition, building approvals, grid connectivity and fire safety clearances.
5. **Controversy has arrived.** The Google–Adani AI campus in Visakhapatnam faces: - a PIL by Jal Biradari in the AP High Court on water stress near the Kambalakonda Wildlife Sanctuary; - a PIL over 160 acres of temple endowment land; - protests over acquisition of assigned land held by Dalit farmers at Tarluvada.

It is India's first major test of whether data centres need environmental scrutiny. AP's IT minister maintains that it is "a state subject" and that "due process" was followed.

12.2 Part A — Central Framework

Table 12.1 Part A — Central Framework

Instrument	Status (Oct 2026)	What it does / implication
Draft National Data Centre Policy (MeitY)	Released 5 Nov 2020 for comments (deadline extended to 30 Nov 2020); **never notified**. MeitY has resumed industry consultations; one market-research release describes a "finalised draft" with up to 20-year conditional tax exemptions, but this is unverified.	Proposed four Data Centre Economic Zones (DCEZs), single-window clearance, infrastructure status, and support for indigenous servers, storage, networking and cooling. Lacked enforceable mandates. Treat "imminent" claims sceptically: it has been pending for nearly six years.
Infrastructure status	In the Harmonised Master List of Infrastructure sub-sectors since 11 Oct 2022	Applies to data centres of at least 5 MW IT load. Eases long-tenor bank and ECB financing. The Finance Act 2026 tax-holiday definition does not use the 5 MW threshold, so a facility can qualify under one and not the other.
Union Budget 2026–27 (1 Feb 2026)	Enacted in the Finance Act; **subordinate notifications on qualifying criteria still pending** (Norton Rose Fulbright)	Tax holiday until 31 March 2047 for foreign companies that provide cloud services globally using India-based data-centre services. Indian customers must be served through an Indian reseller. A 15% cost-plus safe harbour applies to related Indian data-centre service providers, and IT services generally move to a 15.5% safe harbour. This is the most important central incentive, but it benefits foreign cloud firms and Indian colocation operators indirectly and has no green conditions.
DPDP Act 2023 / DPDP Rules 2025	Rules notified 14 Nov 2025	"Soft localisation": cross-border transfers are allowed unless the government blacklists a country. Demand for local hosting comes more from sector rules — RBI payment-data localisation (all payment data stored in India) and the SEBI Cloud Framework 2023 (data and logs on Indian servers) — than from the DPDP regime itself.
MeitY cloud empanelment (GI Cloud / MeghRaj)	Ongoing	Government workloads must use MeitY-empanelled cloud service providers hosted in audited Indian data centres. This creates steady sovereign demand, especially for Tier III facilities. *(Not re-verified in this research.)*
IndiaAI Mission	Ongoing	Subsidised GPU compute procured from private data-centre operators. It shapes demand for high-density, liquid-cooled capacity. *(Not re-verified in this research.)*
Green Energy Open Access Rules 2022 (MoP)	In force	Lowered the open-access threshold to 100 kW and capped additional surcharges, enabling corporate renewable procurement. CEEW stakeholders still name high cross-subsidy surcharge (CSS) and additional surcharge, inconsistent banking rules and sudden withdrawal of waivers as the main barriers.
ISTS charges waiver	Being phased down for projects commissioned after mid-2025 *(general knowledge, not re-verified)*	Makes inter-state renewable PPAs dearer over time, so state-level wheeling and banking concessions matter more.
TRAI recommendations on data centres, CDNs and interconnect exchanges	Issued Nov 2022 (often cited as 2023); not implemented by statute	Recommended a legal framework, a single-window system, a data-centre-specific building-code category and incentives. Industry still asks for a National Building Code category for data centres.
CEA/MoP draft guidance on power for data centres	*Not verified in this research.*	Flagged as a gap: no publicly verified central power-supply standard for data centres was found.
SEZ route	Available	Data centres can be set up in IT/ITeS SEZs, for example the 2.7 ha SEZ portion of RackBank's Nava Raipur park. The Budget's IFSC tax-holiday extension (20 of 25 years) helps GIFT City-type locations.
GST	Standard rate on data-centre/colocation services *(not re-verified)*	States compete through SGST reimbursement, which is the reason "net SGST reimbursement" appears in almost every policy.

Assessment: the Centre has paid for demand (tax holiday, IndiaAI) and for financing (infrastructure status). It has not set siting, efficiency or water rules. That leaves states to compete on subsidies.

12.3 Part B — State-by-State Deep Dive

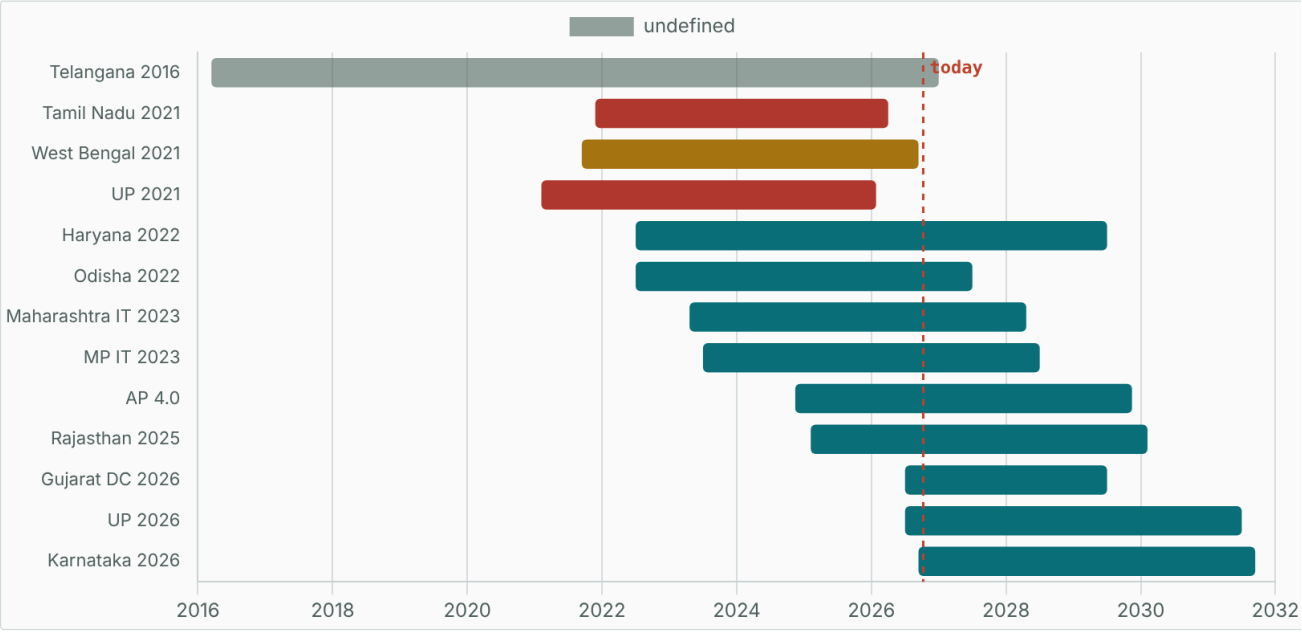


Figure 12.1 Policy validity. When each state's main data-centre instrument runs. The line marks today (Oct 2026). *Source: India Data Centre Atlas, chapter built on this chapter's data.*

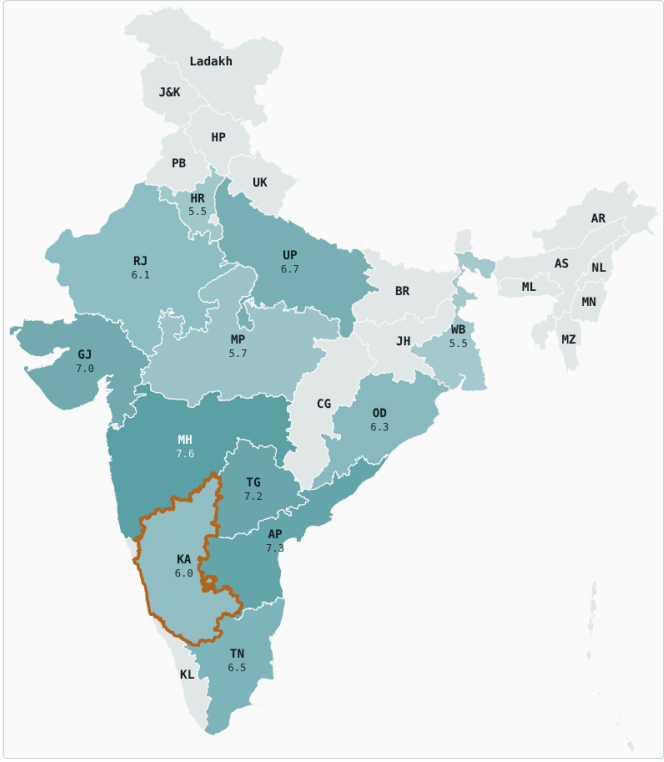


Figure 12.2 India at a glance. Colour the map by a measure, then tap or hover a state for details. *Source: India Data Centre Atlas, chapter built on this chapter's data.*

12.3.1 Maharashtra — the incumbent hub

- **Instruments:**
- IT/ITeS Policy 2023 (data centres get IT-unit benefits and essential-service status).
- Integrated Green Data Centre Parks (IGDCP) policy, Government Resolution (GR) of 9 Oct 2024, amended by GR of 9 June 2026 (Cabinet Sub-Committee approval 22 April 2026).
- Electricity-duty notification of 22 June 2026 under the Maharashtra Electricity Duty Act, 2016.
- **Nodal agencies:** Industries, Energy & Labour Department; MIDC for land. **Status:** active.
- **Targets:** the 2024 GR allowed 3 parks of at least 500 MW and at least ₹30,000 crore each, limited to the Mumbai Metropolitan Region (about 1.5 GW, about ₹1.6 lakh crore). The 2026 amendment allows **20 parks statewide**, tied to a "Developed Maharashtra 2047" ambition of 30–40 GW of green data-centre capacity. Officials have said state capacity will grow from about 500 MW to 800 MW by 2028 and want Mumbai's share to rise towards 70%.
- **Fiscal incentives:**
- **Electricity duty:** 20-year exemption for new and existing data-centre units (June 2026 notification).
- **Stamp duty:** 50–100% exemption by location under the IT policy; 100% on conveyance or lease for IGDCP projects.
- **Power tariff (2026 amendment, trade-press reporting):** ₹1/unit for 10 years for ₹30,000 crore projects, and for 20 years for ₹60,000 crore or more.
- **Investment subsidy:** Industrial Promotion Subsidy of 75% of eligible fixed capital investment for projects of ₹60,000 crore or more.
- **Interest:** 4% subsidy on term loans for up to 10 years, capped at ₹25 crore a year.
- **Non-fiscal:** essential service; data centres allowed in any zone, including residential, no-development and green zones; additional FSI; free MIDC sub-letting. Eligibility now counts combined investment by owner, developer and park units. The 2% incubation-land rule was dropped, and the benefits apply retrospectively to projects that already hold Letters of Approval.
- **Green provisions:** the mandatory green-power share for core data-centre activity was **cut from 100% to 51%** in June 2026. This is the clearest case of sustainability conditions being diluted to win projects. CEEW notes Maharashtra informally prefers coastal and peri-urban clusters (Rabale) over water-stressed inland Nagpur, but this is not codified.
- **Implementation:** CEEW counts about 271 data centres in India as of January 2026, with Mumbai hosting nearly a quarter of them. A Lodha Palava park MoU (about 370 acres, about ₹1.3 lakh crore) is reported in secondary sources only.
- **Assessment:** strongest ecosystem (subsea cables, internet exchanges, tenants). Policy stability is good, but the green floor has been weakened. Water governance is not codified. **Rank: 1.**

12.3.2 Tamil Nadu — connectivity leader in a policy vacuum

- **Instrument:** Tamil Nadu Data Centre Policy 2021. Government Order dated 23.11.2021 (Finance concurrence U.O. No. 52074/Fin(Ind.)/2021), launched 26 Nov 2021 at CII Connect.
- **Nodal agencies:** IT Department/ELCOT; Guidance TN for MoUs; SIPCOT disburses fiscal incentives.
- **Validity:** data centres incorporated from 1 April 2021 to 31 March 2026. **Status: lapsed.** No extension had been notified as of April 2026 (Ahlawat & Associates). A new TVK government under CM C. Joseph Vijay took office on 10 May 2026, and no successor policy was found.

- **Eligibility:** investment above ₹500 crore and at least **30% renewable energy**.
- **Incentives:**
- Industrial tariff.
- 100% Electricity Tax subsidy for 5 years on TANGEDCO power, refundable if data-centre use stops within 10 years.
- Stamp duty: 50% concession in Category A/B districts and 100% in Category C, for land from ELCOT/SIPCOT/SIDCO.
- 50% land-cost subsidy in Category C districts.
- Training incentive for 6 months.
- 10% reimbursement (up to ₹10 lakh) for using Tamil Nadu MSMEs.
- **Non-fiscal:**
- Dual power for loads of 50 MW or more.
- Multi-level generator stacking.
- Rooftop chillers excluded from FSI.
- No zoning or floor-to-ceiling restrictions.
- Lower parking and window requirements.
- Uninterrupted water supply.
- Sub-lease without charges.
- Single window covering CMDA, DTCP, Fire, Police, TNPCB, CMWSSB and TWAD.
- **Implementation:** about ₹18,000 crore of data-centre MoUs in FY2020-21, plus six MoUs (Adani, L&T, STT GDC, CtrlS, Nxtra and others) around the launch. Chennai ranks second nationally by data-centre count.
- **Assessment:** it was early to set a renewable-linked eligibility rule (30% RE), and its building-code relaxations became a template for other states. The 2026 lapse is now the main risk. Former senior official Sharmila Chavaly (in an interview with The Wire) cites a Tamil Nadu minister describing a deliberate "opting out of the rat race to the bottom" in favour of a distributed model. This is secondary and unconfirmed. **Rank: 4 (would be 2 with a renewed policy).**

12.3.3 Karnataka — the efficiency-linked model

- **Instruments:** Karnataka Data Centre Policy 2022–27 (superseded), and the **Karnataka Sustainable Data Centre Policy 2026–31**, approved by the Cabinet in Mangaluru in September 2026.
- **Nodal agencies:** Department of Electronics, IT, BT and S&T. Entities register with KITS; Karnataka Udyoga Mitra is the single window. *G.O. number not yet located.*
- **Targets and outlay:** 1 GW cumulative IT load by 2031; 10–12% of India's AI-ready capacity; ₹1,300.45 crore over five years.
- **Coverage:** data-centre developers, park developers, operators, cloud service providers, edge data centres, submarine cables and cable landing stations. Registered office must be in Karnataka.
- **Fiscal incentives:**
- Capital subsidies, building-fee reimbursement, land concessions, power-tariff benefits, connectivity support, electricity-duty exemption, renewable incentives and wheeling-charge reimbursement, in Bengaluru and beyond.
- R&D support of 50% of expenditure, up to ₹15 crore per project.
- Customised packages for strategic projects above **₹3,000 crore**, explicitly naming cable landing stations, subsea/underwater data centres and data-centre-linked desalination plants.
- **Sustainability provisions (India's first performance-linked design):**
- **Mandatory plant-wise PUE and WUE reporting.**
- **₹20 crore one-time support for PUE of 1.00–1.20**, sustained for three consecutive years and paid in year four.
- **WUE of 0.7–1.3 L/kWh** sustained for three years earns 100% water-charge reimbursement up to ₹30 lakh a year for two years.
- **Treated-water reimbursement** of up to ₹8 lakh a year for five years.
- A state water-and-energy "resource heat map" to guide siting.
- **Parks:** Hoskote, Mangaluru and Mysuru, about 1,000 MW combined. Land on 30+30-year leases, with dual-grid connectivity and assured water. The Hoskote park is planned at about 500 MW with Pavagada solar and 60 MLD of secondary-treated water. About 350 acres have been identified at Baikampady, Mangaluru.
- **Context:** the IT department estimates (from operators' informal inputs) that Karnataka's 32 data centres use about 4,000 kL of water a day. Neither BWSSB nor KSPCB tracks data-centre water use separately.
- **Old 2022–27 policy (for legacy claims):**
- 7% capital subsidy up to ₹10 crore outside Bengaluru Urban.
- Industrial tariff if at least 30% of power is green.
- Surcharge reimbursement for renewable share above 50%.
- Special packages above ₹4,000 crore.
- **Assessment:** the best sustainability design in India. The weaknesses are that the incentive amounts are small relative to hyperscale capex (₹20 crore against multi-thousand-crore projects), that Bengaluru has grid and water constraints, and that there is no subsea landing near Bengaluru (Mangaluru is the hedge). The minister called AP's Google package "an economic disaster", which shows Karnataka is choosing not to outbid. **Rank: 7 overall; 1 on sustainability.**

12.3.4 Telangana — the inland AI challenger

- **Instrument:** Telangana Data Centres Policy 2016 (ITE&C Department), still the base framework and re-hosted on the state IT portal in 2025. **No new notified data-centre policy for 2026 was found**; the state relies on tailor-made packages.

- **Incentives:**
- Data centres declared an **essential service** (24×7 operation, exemption from power cuts and certain labour provisions).
- Exemption from the purview of the Telangana Pollution Control Act.
- 50% building-fee rebate.
- Land-cost subsidy on government land.
- 25% lease-rental subsidy up to ₹5 lakh a year for 3 years.
- Startup internet reimbursement of 25%, up to ₹2.5 lakh a year.
- R&D grants; patent reimbursement of ₹2 lakh (Indian) or ₹10 lakh (foreign).
- Plans for dual power grids. IT/ITeS incentives also apply.
- **Targets:** officials cite about 5 GW by 2029–30.
- **Implementation:** According to Knight Frank India's "India Data Centre Market Update 2025", Hyderabad live capacity rose from 60.9 MW (2022) to 151.4 MW (end-2025), with a 1.9 GW committed pipeline, second only to Mumbai. AWS, which runs three availability zones in the city, holds 46% of live IT capacity. Named projects:

Table 12.2 Telangana — the inland AI challenger

Project	Size / investment	Details
AWS	₹60,000 crore	Foundation stone July 2026
TCS HyperVault	Up to 1 GW, up to ₹70,000 crore	264 acres, Future City; announced September 2026
CtrlS	400 MW, ₹10,000 crore	MoU 2025
NTT DATA–Neysa	400 MW, ₹10,500 crore	MoU 2025
Microsoft India South Central region	—	Launched 2026

- **Assessment:** inland and disaster-safe, with strong hyperscaler anchoring. But it runs on a ten-year-old policy and opaque bespoke deals, with no water or efficiency provisions. **Rank: 2.**

12.3.5 Andhra Pradesh — the mega-deal state

- **Instrument: AP Data Center Policy (4.0) 2024-29, G.O.Ms.No.6, ITE&C (Promotion Wing) Dept., dated 14.11.2024.** It runs alongside Industrial Development Policy 4.0 and Green Energy Policy 4.0.
- **Validity:** 5 years from notification or until a new policy, whichever is later. Amendments apply prospectively only.
- **Target:** inconsistent within the order itself. The G.O. text says 200 MW; the annexed policy says "few hundred MWs (up to 1 GW)". The state now claims agreements for **6.5 GW**.
- **Incentives:**
- 100% net SGST reimbursement on capital goods **or** 10% capital subsidy on plant and machinery.
- 100% stamp duty exemption on the first sale.
- Industrial tariff.
- Parking relaxation.
- Data-centre parks under the Private Industrial Parks (Plug & Play) policy, G.O. 67 dated 26.10.2024.
- Case-by-case: generator stacking, relaxed window and common-area norms, floor-height relaxations, rooftop chillers, dual grid via PPP with viability-gap funding, transmission and wheeling concessions.
- **Mega projects:** more than **50 MW or more than ₹5,000 crore** fixed capital investment (plant and equipment only) qualify for tailor-made benefits, including land-cost discounts and PPP water and road infrastructure. These are approved by the SIPB (CM-chaired, G.O. 73 of 5.11.2024) on SIPC advice (G.O. 72). The Finance Department required land discounts to follow APLMA guidelines.
- **Green provisions:** the policy promotes seawater cooling and renewable integration. **The "70% renewable energy" rule is a ministerial statement** ("the first thumb rule…", Nara Lokesh, 2026). No notified G.O. for it was found. Water is to come from the Polavaram Left Bank Canal and desalination.
- **Implementation:**
- **Google** ($15 bn, 2026–30, about 1 GW; SIPB earlier approved about $10 bn), with Adani.
- Reliance (1 GW).
- Meta–Sify (500 MW, about $1.5 bn).
- Adani's earlier 130-acre Madhurawada allotment (₹14,634 crore claimed).
- Karnataka's minister alleged AP's Google package is worth ₹22,000 crore, with a 25% land subsidy, a 25% water-tariff subsidy, "free electricity" and 100% SGST. This is a rival state's claim; AP has not published the package.
- **Controversies:**
- Land: 601.4 acres across three sites, including 266.6 acres at Tarluvada, held through three SPVs (Vizag Hyperscale, Vizag Mega, Vizag Rambilli Data Center Park Ltd).
- Acquisition of assigned land from Dalit farmers.
- A PIL on 160 acres of endowment land.
- Jal Biradari's water/wildlife PIL in the AP High Court (hearing listed 24 Aug 2026).
- Matters before the NGT Southern Bench.
- **Assessment:** the most generous and fastest state for anchors, with coast, cables and renewables. Its key weakness is the opacity of bespoke packages and the litigation risk they now carry. **Rank: 3.**

12.3.6 Uttar Pradesh — northern hub, data-transparent

- **2021 policy** (Invest UP/UPLC; amended 2022; expired 27 Jan 2026).
- **Targets:** 250 MW, ₹20,000 crore, 3 private parks.
- **Eligibility:** units above 2 MW and below 40 MW.
- **Incentives:**
  - 7% capital subsidy up to ₹20 crore (excluding land and building), paid over 10 years at up to ₹2 crore a year.
  - Park interest subsidy of 60% for 7 years, up to ₹10 crore a year and ₹50 crore per park.
  - Land subsidy of 25% (Madhyanchal/Paschimanchal) or 50% (Bundelkhand/Purvanchal), capped at 7.5% of project cost or ₹75 crore.
  - Stamp duty: 100% on the first transaction, 50% on the second.
  - 100% electricity duty exemption for 10 years.
  - Transmission and wheeling relief; dual grid.
  - CSS to be cut progressively to 20% of its opening level.
  - Sub-leasing without charges.
  - Edge incentives if at least 25 edge data centres form one proposal.
  - Proposed essential-service status under ESMA.
- **Outcome (June 2025):** 644 MW proposed, ₹21,342.90 crore, 77.08 acres. About 101 MW commercialised. Seven operational facilities (Yotta, NTT, STT GDC, AdaniConneX, CtrlS), all in Noida/Greater Noida. The original 250 MW target was beaten on proposals, though not on commissioning.
- **2026 policy:** Cabinet approved in early July 2026 (unveiled 6 July by IT Minister Sunil Sharma). Department of IT & Electronics with Invest UP.
- **Targets:** 2 GW additional capacity and more than ₹2 lakh crore.
- **Incentives:** Tier III/IV focus. "AI Compute Booster": parks installing at least 700 FP64-grade AI compute units get +₹10 crore on the interest-subsidy ceiling and +₹2 crore on the annual cap; units with 350 units get +₹5 crore capital subsidy. Core incentives continue (7% capex up to ₹20 crore; 60% interest; 25%/50% land; 100% stamp duty; 10-year ED waiver).
- **Green provisions:** "Green and Sustainable Operations Booster" linking extra interest subsidy to green certification, PUE and renewable use; at least 50% renewable sourcing for new data centres by 2030 (per the draft).
- **Edge:** internet subsidy for edge data centres.
- **Assessment:** the most transparent state, and it first introduced AI-specific incentives. The regional land subsidy is a proven non-driver (Bundelkhand groundwater over-extraction; 14–18-hour rural supply in Purvanchal). **Rank: 5.**

12.3.7 Gujarat — the newest, biggest bidder

- **Instruments:** Gujarat IT/ITeS Policy 2022–27 (DST), plus a **dedicated Data Centre Policy (2026–29)** unveiled in July 2026, billed as "first-of-its-kind". *G.O. number not located.*
- **IT/ITeS 2022–27:**
- Capex support of 25%, up to ₹150 crore per data centre.
- ₹1/unit for 5 years.
- Opex support of ₹20–40 crore a year.
- Cable landing station capex of 25%, up to ₹20 crore.
- Data centres treated as "strategic institutions".
- **Data Centre Policy 2026–29:**
- **Targets:** 7.5 GW and ₹6 lakh crore. Officials expect 7–8 GW at Dholera alone and claim requests for about twice the target (these are projections).
- **Incentives:**
  - 2.5% capital subsidy on eligible fixed capital investment in Dholera.
  - Interest subsidy up to 4% for 10 years, capped at ₹25 crore a year.
  - **₹1/unit for 20 years.**
  - 100% stamp duty and registration exemption.
  - **Electricity-duty reimbursement for 20 years.**
  - SGST reimbursement on plant, building and services for up to 20 years.
  - Total capped at 75% of eligible fixed capital investment over 20 years.
  - FSI and building relaxations; assured water and power.
- **Green provision:** at least **51% renewable** for all incentive-takers.
- **Implementation:** L&T Vyoma, 250 MW and ₹25,000 crore at Dholera by 2028 (MoU February 2026). GIFT City benefits from the IFSC tax-holiday extension. *Reliance's Jamnagar AI data-centre plans were not verified in this research.*
- **Assessment:** has renewable land, cheap solar and port logistics. It lacks subsea landings at scale and a water plan for drought-prone areas, and has no measurable outcome indicators. **Rank: 6.**

12.3.8 Haryana — Gurugram's policy

- **Instrument:** Haryana State Data Centre Policy 2022, approved by the Cabinet on 27 June 2022 (HEPC/DITECH). The draft specified 7-year validity; *the final term should be verified.*
- **Targets:** 115–120 data centres, ₹7,500 crore.

- **Eligibility:** at least 1 MW. Parks need at least 20 MW and at least 20 acres.
- **Incentives:**
- 100% stamp-duty reimbursement on the first transaction.
- **100% electricity-duty exemption for 20 years.**
- Power-bill reimbursement up to 25% of net SGST for 3 years, in exchange for equivalent government hosting through DITECH.
- Net SGST reimbursement of 50% (A/B blocks) or 75% (C/D) for 10 years.
- Employment subsidy of ₹48,000 per employee a year for 10 years.
- Property tax at industrial rates.
- **Non-fiscal:** essential service under the **Haryana ESMA 1974**; separate infrastructure and energy-intensive industry status; separate building-code category with FAR relaxation; the draft earmarked about 10-acre parcels in Tier-III cities (Yamunanagar, Hisar, Panipat).
- **Assessment:** solid incentives near Delhi demand, but no green provisions and expensive Gurugram land. **Rank: 8.**

12.3.9 West Bengal — the eastern hub

- **Instrument:** West Bengal Data Centre Policy 2021, **Notification No. 490-ITE dated 06.09.2021**; nodal agency WEBEL.
- **Validity:** 5 years from notification, which **lapses about September 2026**. *No renewal was found; treat as expiring.*
- **Targets:** 400 MW by 2025 and 3 parks.
- **Incentives:**
- 100% stamp duty and registration exemption.
- Electricity-duty waiver for 5 years.
- Industrial power status; dual grid.
- 60% ground coverage.
- Right of way under the WB Broadband Policy 2020.
- Essential services.
- Quality power and internet for **edge data centres in IT/industrial parks**.
- 24×7 water and captive water-treatment plants.
- **Implementation:** Kolkata about 67 MW (NTT 40 MW). Adani holds 51.8 acres (₹10,000 crore) and Reliance Jio 40 acres at Bengal Silicon Valley, New Town. A Tajpur/Digha cable landing is planned.
- **Assessment:** the only eastern hub with a dedicated policy, but it is about to lapse and the 400 MW target was missed by a wide margin. **Rank: 9.**

12.3.10 Odisha

- **Instruments:** Odisha State Data Centre Policy 2022 (E&IT, Invest Odisha); the IT Policy 2025 has replaced the IT Policy 2022.
- **Incentives:**
- 20% capital subsidy on fixed capital investment (excluding land), up to ₹25 crore.
- Interest subsidy of 5%, up to ₹25 crore, for 7 years.
- 30% power-bill subsidy for 5 years.
- 100% electricity-duty exemption for 10 years.
- 100% net SGST for 7 years.
- Stamp duty: 100% on the first transaction, 50% on the second.
- Land at IDCO concessional rates; 100% conversion-charge reimbursement.
- **100% transmission and wheeling reimbursement for captive renewable power.**
- Clean-and-green subsidy of 25%, up to ₹1 crore.
- Anchor-unit incentives for the first two park units.
- Essential service; exemption from sections 9, 10, 11 and 23 of the Odisha Shops & Commercial Establishments Act.
- **Assessment:** among the five states CEEW credits with sustainability framing (explicit carbon-footprint link). Few projects so far. **Rank: 10.**

12.3.11 Rajasthan

- **Instrument:** Rajasthan Data Centre Policy 2025 (DoIT&C/Rising Rajasthan).
- **Targets:** ₹20,000 crore over 5 years, 300 MW, 5 parks.
- **Incentives:**
- Annual asset-creation incentive of ₹10–20 crore for 10 years.
- 25% extra "sunrise" incentive for the first 3 data centres investing more than ₹100 crore.
- 5% interest subsidy for 5 years.
- **100% waiver of banking, transmission and wheeling charges.**
- Exemptions on stamp duty, land conversion and electricity duty (7 years).
- External development charge exemption up to ₹10 crore.
- 50% skilling reimbursement.
- **Green Solution Incentive of 50%, up to ₹12.5 crore.**
- 50% of IPR costs, up to ₹1 crore.

- Building by-law exemptions.
- *(One secondary summary cites a 25% capital subsidy and ₹1–1.5/unit tariff support; these conflict with official summaries and are unverified.)*
- **Green provisions:** CEEW calls it **India's most comprehensive sustainability-oriented framework**: mandatory wastewater recycling, ZLD, rainwater harvesting, groundwater recharge, smart water management and green-building standards.
- **Assessment:** excellent renewable resources and a model water framework, but distant from connectivity and demand. **Rank: 11.**

12.3.12 Madhya Pradesh

- **Instrument:** MP IT/ITeS & ESDM Investment Promotion Policy 2023 (MPSEDC). No standalone data-centre policy.
- **Incentives:** the first 5 anchor data centres get 25% capex subsidy on gross fixed capital investment (excluding land) for minimum investment of ₹500 crore, capped at ₹125 crore and paid over 5 years, plus a power-tariff rebate, ED waiver and stamp-duty reimbursement. *These figures come from consultancy summaries and need checking against the PDF.*
- **Projects:** CtrlS Bhopal (₹500 crore); RackBank in Indore.
- **Rank: 12.**

12.3.13 Bihar

- **Instrument:** Bihar IT Policy 2024. Data centres are covered as an ancillary sector.
- **Incentives:**
- **Either** 30% capital subsidy on fixed capital investment (including land), up to ₹30 crore (minimum ₹5 crore), **or** 10% interest subvention up to ₹40 crore.
- 50% lease-rental subsidy for 5 years.
- 25% power-tariff subsidy for 5 years.
- 100% net SGST.
- +10% for units outside Patna.
- *Validity end-date in secondary summaries (Dec 2029) needs verification.* No data-centre projects identified.

12.3.14 Chhattisgarh

- **Instrument:** Industrial Development Policy 2024–30. CEEW lists it as covering data centres; *the specific data-centre clauses were not verified.*
- **Project:** RackBank AI Data Centre Park, Nava Raipur — 13.5 acres including a 2.7 ha SEZ, 5 MW scalable to 150 MW, about ₹1,000 crore initial. A further ₹2,000 crore is a projection. Foundation stone laid 3 May 2025.

12.3.15 Punjab

- **Instruments:** Industrial & Business Development Policy 2022 (data-centre services treated as IT) and IT Policy 2022.
- **Incentives (secondary):** net SGST up to 200% of fixed capital investment for 15 years; ED and property-tax exemption; 100% stamp duty and CLU/EDC exemption; employment subsidy. No projects identified.

12.3.16 Long tail

- **Jharkhand:** draft IT, Data Center & BPO Investment Promotion Policy 2021 (25% land subsidy for parks). *Final notification not confirmed.*
- **Himachal Pradesh:** IT-policy incentives (100% stamp duty, 5% interest subvention up to ₹30 lakh a year, 100% net SGST for 7 years). *Policy name and G.O. not verified.*
- **Kerala:** no data-centre policy found. The IT Policy 2023 and GCC Policy 2025–30 apply. Kochi and Thiruvananthapuram subsea landings are an underused asset.
- **Assam/Northeast:** no data-centre policy found (the Assam ESDM Policy 2025 covers electronics only). Tripura reportedly positions itself on data-centre incentives (unverified).
- **Delhi, Goa, Uttarakhand, Jammu & Kashmir: no data-centre-specific policy found.** Delhi demand is served from Noida and Gurugram.

12.4 Part C — Master Comparison Table

Table 12.3 Part C — Master Comparison Table

State	Instrument / status	Target	Capex / interest	Power	Land / stamp	Green conditions	Validity
Maharashtra	IT/ITeS 2023 + IGDCP GRs (Oct 2024; Jun 2026) — active	20 parks; 30–40 GW by 2047	IPS 75% of fixed capital investment (≥₹60k cr); 4% interest, ₹25 cr/yr	ED 20 yrs; ₹1/unit 10–20 yrs	Stamp duty 100% (IGDCP); any-zone siting	51% RE (cut from 100%)	IT policy to 2028; GR open-ended
Tamil Nadu	DC Policy 2021 — **lapsed 31.3.2026**	—	—	Electricity tax 100%, 5 yrs	Stamp duty 50–100%; 50% land (Cat C)	≥30% RE eligibility	Expired
Karnataka	Sustainable DC Policy 2026–31 — approved Sept 2026	1 GW; ₹1,300.45 cr outlay	Capital subsidy; R&D 50% up to ₹15 cr	ED exemption; wheeling reimbursement	Parks on 30+30-yr lease	**PUE/WUE reporting; ₹20 cr PUE; WUE & treated-water reimbursement**	2031
Telangana	DC Policy 2016 + bespoke deals	~5 GW by 2029–30 (stated)	Lease rental 25%; R&D	Power incentives; dual grid	Land subsidy (govt land); 50% building fee	None	Legacy
Andhra Pradesh	DC Policy 4.0, G.O.Ms.6 (14.11.2024)	200 MW–1 GW (policy); 6.5 GW (claimed)	100% SGST on capital goods **or** 10% P&M; mega >50 MW/>₹5k cr bespoke	Industrial tariff; T&W case-by-case	Stamp duty 100% first sale; land discount (mega)	Seawater cooling; "70% RE" (stated, not notified)	2029 or new policy

State	Instrument / status	Target	Capex / interest	Power	Land / stamp	Green conditions	Validity
Uttar Pradesh	DC Policy 2026 (replaced 2021)	+2 GW; ₹2 lakh cr	7% up to ₹20 cr (+₹5 cr AI); 60% interest (+₹10 cr AI)	ED 10 yrs; T&W relief; dual grid	25–50% land; stamp duty 100%	PUE/green-cert booster; 50% RE by 2030	~2031 (assumed)
Gujarat	DC Policy 2026–29 + IT/ITeS 2022–27	7.5 GW; ₹6 lakh cr	2.5% capex (Dholera); 4% interest, ₹25 cr/yr; cap 75% of fixed capital investment	**₹1/unit 20 yrs; ED 20 yrs**	Stamp duty & registration 100%	51% RE	2029
Haryana	DC Policy 2022	115–120 DCs; ₹7,500 cr	Net SGST 50–75%, 10 yrs	ED 20 yrs; power bill ≤25% of SGST	Stamp duty 100%	None	~2029 (verify)
West Bengal	DC Policy 2021 (No. 490-ITE)	400 MW by 2025	—	ED 5 yrs; dual grid	Stamp duty & registration 100%	Captive water treatment	**Lapsing ~Sept 2026**
Odisha	DC Policy 2022	—	20% up to ₹25 cr; 5% interest	30% bill 5 yrs; ED 10 yrs; captive-RE T&W 100%	Stamp duty 100/50%	Clean-green subsidy; carbon framing	Verify
Rajasthan	DC Policy 2025	300 MW; ₹20,000 cr	Asset-creation ₹10–20 cr/yr; 5% interest	Banking/T&W 100% waiver; ED 7 yrs	Stamp duty, conversion exemptions	**ZLD, recycling, green-building; green incentive ₹12.5 cr**	~2030
Madhya Pradesh	IT/ITeS & ESDM 2023	—	25% up to ₹125 cr (anchors)	Tariff rebate; ED	Stamp reimbursement	Green DC support (vague)	~2028
Bihar	IT Policy 2024	—	30% up to ₹30 cr or interest	25% tariff 5 yrs	Lease subsidy	None	~2029

12.5 Part D — Attractiveness Index

Methodology (author's weighted judgment, 0–10 per factor). Weights: power cost and reliability 25%; renewable access 15%; connectivity (subsea/IX/fibre) 15%; incentives 15%; land and park readiness 10%; water security 10%; regulatory ease and policy stability 10%. Scores reflect the evidence above, not a published index.

Table 12.4 Part D — Attractiveness Index

Rank	State	Power	RE	Connectivity	Incentives	Land	Water	Regulatory	Weighted
1	Maharashtra	8	6	10	8	6	6	8	**7.8**
2	Telangana	8	7	6	7	9	6	7	**7.2**
3	Andhra Pradesh	7	9	7	9	7	5	6	**7.3**
4	Tamil Nadu	7	8	10	4	6	4	5	**6.8**
5	Uttar Pradesh	7	5	6	8	7	6	8	**6.7**
6	Gujarat	7	9	4	9	8	4	7	**6.8**
7	Karnataka	6	8	5	6	6	4	7	**6.0**
8	Haryana	6	4	6	7	4	4	7	**5.5**
9	West Bengal	6	4	6	5	6	7	4	**5.4**
10	Odisha	7	6	4	7	7	7	6	**6.2**
11	Rajasthan	6	10	2	8	8	2	6	**5.9**
12	Madhya Pradesh	6	6	3	7	7	5	6	**5.6**

Interpretation: AP's raw score edges Telangana, but litigation risk and opaque packages put it third. Gujarat and Tamil Nadu score similarly for opposite reasons: Gujarat has incentives but little connectivity; Tamil Nadu has connectivity but no current policy. Odisha scores above its rank on paper but has little track record. The ordering reflects demonstrated execution as well as policy.

Sustainability sub-ranking: 1 Karnataka; 2 Rajasthan; 3 UP (2026); 4 Andhra Pradesh (stated 70% RE, seawater cooling, desalination); 5 Gujarat (51% RE); 6 Odisha; 7 Tamil Nadu (30% RE, lapsed); 8 Maharashtra (diluted to 51%). Haryana, Telangana, West Bengal, MP, Bihar and Punjab have essentially none.

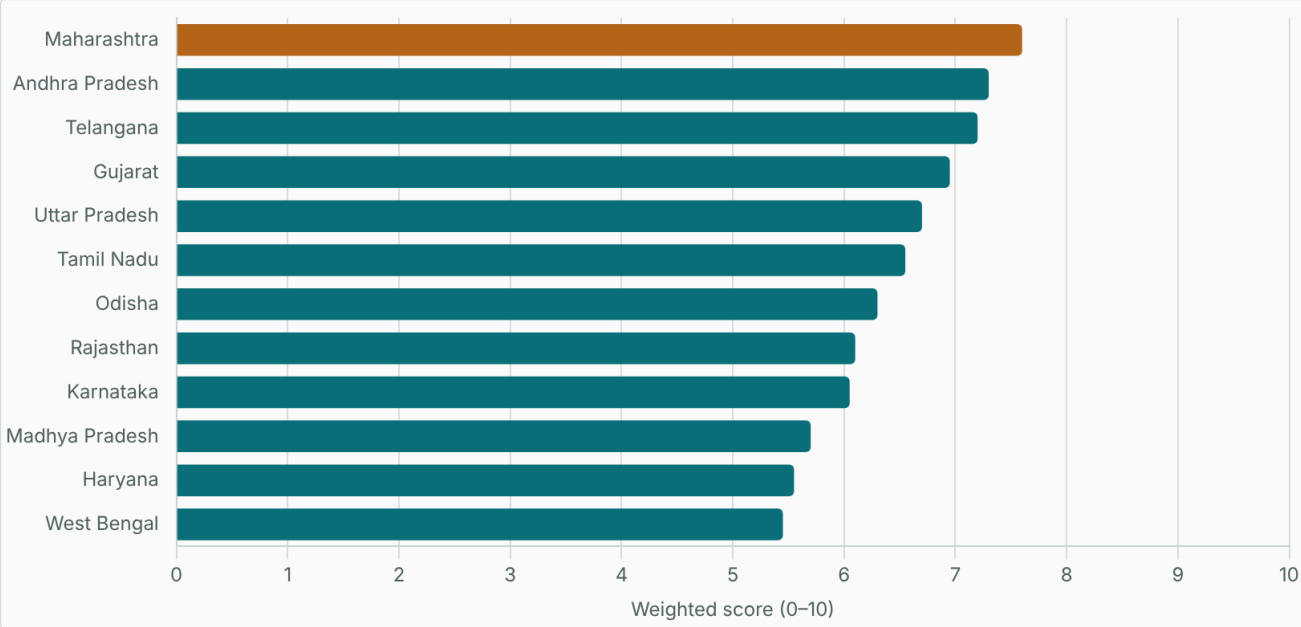


Figure 12.3 State attractiveness index at the report weights. Weighted score out of 10 across power, renewables, connectivity, incentives, land, water and regulation. *Source: India Data Centre Atlas, chapter built on this chapter's data.*

12.6 Part E — Competition Dynamics: Race to the Bottom vs Sustainability Conditions

- **Escalation is real.** Electricity-duty holidays have stretched from 5 years (Tamil Nadu, West Bengal) to 10 (UP, Odisha) to 20 (Haryana, Maharashtra, Gujarat). ₹1/unit subsidies have moved from 5 years (Gujarat IT 2022) to 20 years (Gujarat 2026, Maharashtra ≥₹60k crore). Bespoke packages for anchors — AP's Google deal, reportedly about ₹22,000 crore by a rival's account — set expectations that every hyperscaler will ask for.
- **Green conditions bend under competition.** Maharashtra cut its renewable mandate from 100% to 51% within 20 months. Gujarat launched at 51%, establishing a de facto national floor set by competition rather than by science or grid planning.
- **Incentives are often misdirected.** Land subsidies and regional top-ups barely move siting decisions. Power price, reliability and fibre do. Subsidy money would work better spent on substations, dual-grid connections, treated-water pipelines and cable landing stations, which is the logic behind Karnataka's and UP's park models.
- **Fiscal exposure is unmeasured.** No state publishes disbursements or tax expenditure. Twenty-year SGST and ED waivers stacked on a central tax holiday to 2047 mean the public gets data-centre growth with little direct revenue. That is defensible only if it brings jobs, a local supply chain or grid investment, and the policies rarely require any of these.
- **Water is the next flashpoint.** CEEW estimates a 100 MW hyperscale facility can use about 2 million litres a day. Visakhapatnam's PILs and Bengaluru's lack of tracking show water governance trailing capacity. Only Karnataka (WUE), Rajasthan (ZLD/recycling) and partly UP 2026 address it.

12.7 Recommendations

For the Centre (MeitY, MoP, MoEFCC): 1. **Notify a slim National Data Centre Framework.** It should be a set of standards, not another subsidy scheme: - mandatory annual PUE, WUE, renewable-share and diesel-hours disclosure for facilities of 5 MW or more (matching the infrastructure-status threshold); - phased PUE/WUE benchmarks by climate zone; - a BIS/NBC building-code category for data centres. 2. **Make central benefits depend on disclosure.** Make the 2047 tax holiday and infrastructure-status lending conditional on certified reporting. This is the one lever states cannot undercut. 3. **Set up a siting decision-support tool.** It would combine CEA grid headroom, CGWB groundwater stage and climate-hazard layers, as CEEW recommends, so that states stop subsidising unbuildable regions. 4. **Fix open-access friction.** Standardise banking, cap CSS and additional surcharge for renewable-linked data-centre load, and enable revenue stacking for battery storage so diesel backup can shrink. 5. **Model state policy and limit stacking.** Publish a model state data-centre policy and consider a soft cap (for example, total state incentives at or below 50–75% of eligible fixed capital investment, as Gujarat already applies) through a GST Council or NITI Aayog forum.

For states: 1. **Tamil Nadu and West Bengal:** renew urgently. A lapsed policy costs more in investor uncertainty than any subsidy saves. 2. **Copy Karnataka's metrics.** Move from renewable-share floors alone to PUE/WUE-linked payouts, and raise the amounts so they matter at hyperscale (₹20 crore is symbolic for a ₹10,000 crore campus). 3. **Publish deals.** Bespoke packages above a threshold (AP's 50 MW / ₹5,000 crore) should be disclosed. This is the best defence against PILs like Visakhapatnam's. 4. **Spend on infrastructure, not land.** Fund dual-grid substations, treated-water mains, cable landing stations and parks; drop regional land top-ups for data centres. 5. **Edge in Tier-2 and Tier-3 cities:** use UP's aggregation model (25 or more edge sites per proposal) and West Bengal's IT-park edge provisions, linked to 5G and state data-centre needs, rather than hyperscale subsidies in water-stressed districts.

For investors: in Telangana and AP, negotiate explicit, contract-backed packages. Price in Tamil Nadu and West Bengal renewal risk. Treat Karnataka's PUE/WUE reporting as a preview of national rules.

12.8 Caveats

- Several 2026 policies (Karnataka, UP, Gujarat, the Maharashtra amendment) are known from cabinet notes, press and trade reports. **Gazette G.O. numbers were not located** for Karnataka 2026, UP 2026, Gujarat 2026, Haryana 2022, Rajasthan 2025 or Odisha 2022. Verified references include AP G.O.Ms.No.6 (14.11.2024), WB No. 490-ITE (06.09.2021), the TN G.O. of 23.11.2021, the Maharashtra GRs of 9 Oct 2024 and 9 June 2026, and the Maharashtra ED notification of 22 June 2026.
- AP's 70% renewable rule, the 6.5 GW figure and the Google package value are statements or rival claims, not published instruments.
- MP, Bihar, Punjab, Himachal and Rajasthan details rely partly on consultancy and secondary summaries. Rajasthan figures conflict across sources.
- Central items marked "not re-verified" (MeghRaj, IndiaAI, ISTS, GST, CEA guidance) are from general knowledge and should be checked before citing.
- Capacity and investment targets (Gujarat 7.5 GW, UP 2 GW, Telangana 5 GW, Maharashtra 30–40 GW) are aspirations. Their sum far exceeds national demand projections — 4.5–6.5 GW by 2030 per Colliers and S&P Global (cited by CEEW), and even the higher estimates of nearly 8 GW (an August 2026 official factsheet) or 12 GW (Wood Mackenzie) — so many will not be met.
- The attractiveness index is the author's weighted judgment, not a published dataset.

Sources cited in this chapter

Organisations and publications named in this chapter's text, in order of how often they are cited. Full details are in the bibliography.

- Council on Energy, Environment and Water (CEEW) studies
- Ministry of Electronics and Information Technology (MeitY), Data Centre Policy draft and IndiaAI
- Google environmental reports, efficiency data and Gemini inference paper
- AdaniConneX disclosures
- Ministry of Power, orders on ISTS waiver and green open access
- CtrlS disclosures
- Central Electricity Authority (CEA), CO_2 Baseline Database and grid reports
- NTT DATA ESG data
- Reliance Industries announcements
- Amazon Web Services sustainability reporting
- STT GDC India ESG report
- Central Ground Water Board and state groundwater assessments
- Bangalore Water Supply and Sewerage Board
- Ministry of Environment, Forest and Climate Change, thermal plant water norms and EIA
- Bureau of Indian Standards (IS 1893, ISO/IEC 30134 adoption)
- Microsoft sustainability reports and datacenter design posts
- Meta sustainability reports and data index
- Sify Infinit Spaces sustainability report
- Nxtra by Airtel disclosures
- Yotta Data Services disclosures
- Wood Mackenzie India data centre outlook
- S&P Global (451 Research) India data centre analysis
- DatacenterDynamics reporting

PART IV · POLICY, PLACE AND THE FUTURE

Chapter 13
Where to build: a city-by-city shortlist for India

WHAT THIS CHAPTER COVERS

This chapter ranks fifty candidate locations across the Indian states for new data centres. It first removes locations that fail hard limits for seismic risk, distance to a 220 kilovolt substation, cyclone or flood exposure and over-exploited groundwater, and then scores the remaining locations on power, connectivity, latency, water, climate, natural hazards, land, policy and skills. It presents shortlists for every state, a national master table, tests of how stable the ranking is when the weights change, and top picks for each use case.

KEY TAKEAWAYS

1. The periphery of Hyderabad (Chandanvelly, Meerkhanpet and Shadnagar) is the strongest place in India to build a new data centre and ranks first or second under every weighting tested.
2. Hyderabad wins because it is the only candidate that combines a low-seismic, low-cyclone inland site with a strong high-voltage grid, a tariff of about ₹6.65 per kilovolt-ampere-hour at 132 kilovolts, latency of about 10 milliseconds to the southern and western hubs, and a 710 megawatt pipeline.
3. Visakhapatnam is the clear second choice for AI-scale campuses, now that new subsea cable landings and Google's one gigawatt hub are under way.
4. For colocation that depends on interconnection with many networks, nothing replaces Navi Mumbai and Chennai.
5. Latency is estimated from distance using the speed of light in fibre and a route factor of 1.4, which gives about 0.014 milliseconds of round-trip time per kilometre.

How this chapter connects to the others. Chapter 12 supplies the policy scores used here, and Chapter 15 turns the ranking into hub-level design guidance.

Hyderabad's periphery (Chandanvelly–Meerkhanpet–Shadnagar) is the strongest place in India to build a new data centre. It ranks first or second under every weighting tested, because it is the only candidate that combines a low-seismic, low-cyclone inland site with a strong EHV grid, a ₹6.65/kVAh tariff at 132 kV, 10 ms-class latency to the southern and western hubs, and a 710 MW pipeline. Visakhapatnam is the clear #2 for AI-scale campuses, now that new subsea landings and Google's 1 GW hub are under way. For interconnection-heavy colocation, nothing replaces Navi Mumbai and Chennai.

13.1 Key Findings

1. Market context (reported). Cushman & Wakefield's H1 2026 update puts operational capacity at 1,789 MW across seven markets, with 3,860 MW more due by 2030 (509 MW under construction and 3,351 MW planned). By city:

Table 13.1 Key Findings

Market	Operational (MW)	Upcoming (MW)
Mumbai	890	1,726
Chennai	214	368
Delhi NCR	183	387
Bengaluru	162	100
Hyderabad	151	710
Pune	133	258
Kolkata	22	84

Colocation vacancy stood at 14.9%. JLL's India Data Centre 2026 Mid-Year Report (press release, 2 September 2026) puts capacity at 1,637 MW in June 2026 and expects 6 GW by 2029, a build-out it says "will require USD 110 billion in capital investments". It reports H1 2026 absorption of 101 MW (82% pre-committed hyperscale) and vacancy at "a record low of 2.8%". That conflicts with C&W's 14.9%, probably because the two firms define vacancy differently. CBRE South Asia's October 2026 report, "Beyond Growth: India's Rise Among APAC's Leading Data Centre Markets", counts about 130 MW added in H1 2026: 94% of it in Mumbai and 48% from domestic developers. It expects more than 200 MW in H2, putting India "on track to reach 2 GW by the end of the year". These counts differ by methodology (city coverage, IT vs gross MW). Treat about 1.6–1.8 GW as the range.

2. Hyderabad is under-priced relative to its fundamentals. It has the second-largest pipeline in India (710 MW), against Bengaluru's 100 MW. That pipeline reflects what the model shows: a TGERC HT-I(A) tariff of ₹6.65/kVAh plus ₹500/kVA/month at 132 kV and above, held flat for FY2026-27. It also has seismic zone II, no cyclone exposure, and better groundwater than Bengaluru or Chennai.

3. Visakhapatnam has moved from "emerging" to "anchor". Google broke ground on 28 April 2026 on a 1 GW AI hub, part of a US$15 bn commitment, with AdaniConneX and Airtel Nxtra. The state allocated 601 acres across Tarluvada, Adavivaram and Rambilli. According to The Ken, "Andhra Pradesh has promised Google, Meta, Reliance, Tata Consultancy Services, and half a dozen others a combined 5 GW of data-centre capacity in Vizag alone." DatacenterDynamics reports that Google's subsidiary Raiden "has applied for almost 2.1GW of power allowance from local distribution companies." Google announced America-India Connect at the India AI Impact Summit in February 2026. It adds three new subsea paths (to Singapore, South Africa and Australia), which Google says "will establish Vizag as a major international subsea gateway". Meta's Waterworth also lists Vizag as a planned landing. The main weakness is cyclone exposure (Hudhud, 2014); design for it, don't walk away from it.

4. Water is the factor that most separates hubs from greenfield sites. - **Bengaluru Urban:** all six assessment units are over-exploited in IN-GRES 2025-26 (Bangalore-East 395%, North 257.9%, Yelahanka 251.5%, City 223.4%, South 121.9%, Anekal 106.4%). These figures come from a secondary compilation of IN-GRES; verify them on the portal. Hosakote and Devanahalli are on Karnataka's list of over-exploited taluks. - **Chennai district:** stage of extraction was 127.47% in CGWB 2023, with 13 of 16 taluks over-exploited. - **Haryana:** the state is at 136.75%, with 91 of 143 units over-exploited in 2025 and 5 over-exploited blocks in Gurugram district. - **National picture:** per the Ministry of Jal Shakti/CGWB Dynamic Groundwater Resources Assessment 2025, India-wide stage of extraction is 60.63%. Of 6,762 units, 730 (10.8%) are over-exploited, 201 critical, 758 semi-critical and 4,946 (73.14%) safe. Most of the country is safe; the established hubs mostly are not.

5. Seismic. The BIS IS 1893:2025 map, which added a Zone VI covering the Himalayan arc, was withdrawn by a gazette notification of 3 March 2026, reverting to IS 1893:2016. This report therefore uses the 2016 zones. Treat the withdrawn Zone VI (Dehradun, Shimla and others) as a design-risk signal for Himalayan sites.

6. Tariffs vary by about ₹2.7/kWh across states. Over 20 years at 100 MW, that gap is worth more than any capex incentive. Effective cost is computed at 80% load factor, as energy charge + demand charge ÷ 584 h:

Table 13.2 Key Findings

State	Tariff basis	Effective ₹/kWh
Odisha	EHT, ₹250/kVA, 580/470 paise slabs	≈ ₹5.96 (cheapest)
Kerala	EHT 220 kV ₹5.55, demand charge estimated	≈ ₹6.23
Haryana	132 kV	≈ ₹7.1
Uttar Pradesh	NIDP HV-2	≈ ₹7.41
Telangana	132 kV	≈ ₹7.51
Karnataka	HT-2(a)	≈ ₹7.61
Madhya Pradesh	HV-3 132 kV	≈ ₹8.2
Tamil Nadu	HT-I	≈ ₹8.54
Maharashtra	HT-I(A)	≈ ₹8.7

Open access to RE changes this materially for anyone above about 10 MW.

13.2 Method

Hard filters. - **Excluded:** seismic zone V; or no 220 kV+ node within about 25 km with no credible plan for one; or high flood/cyclone exposure with no elevated land available. - **Conditional:** over-exploited groundwater. The site passes only if it has a contracted treated-wastewater (STP/TTRO) or desalinated supply.

Normalisation (0–10). - **Latency:** anchored min–max on mean RTT to Mumbai, Delhi, Chennai and Bengaluru, using score = 10 × (26 − RTT)/16, clipped. - **Tariff:** min–max on effective ₹/kWh between ₹5.96 (score 10) and ₹8.70 (score 0). - **Power** = 50% tariff score + 50% grid/RE-access judgement. - **Water, hazards, climate, land, policy, demand:** rubric scores. Lower-is-better factors are flipped. Policy scores: AP, UP 9; Karnataka, Maharashtra, Gujarat 8; Rajasthan 7; Telangana, Odisha, MP 6; Haryana 5; Tamil Nadu, West Bengal, Kerala and secondary states 3–4; Delhi 3.

Weights.

Table 13.3 Method

Factor	Default	AI campus	Interconnect	Sustainability
Power	25	40	15	20 (RE-access score replaces tariff)
Connectivity	15	5	30	5
Latency	10	5	20	5
Water	15	15	5	30
Climate	5	5	0	15
Hazards	10	10	10	10
Land	5	10	2	5
Policy	10	7	5	5
Demand/talent	5	3	13	5

A weighted geometric mean, using the default weights, is also shown to penalise fatal weaknesses.

Latency formula. RTT (ms) ≈ 2 × d × 1.4 ÷ 200 = 0.014 × d (km). Inter-hub baselines:

Table 13.4 Method

Route	Distance	RTT
Mumbai–Delhi	1,150 km	16.1 ms
Mumbai–Chennai	1,030 km	14.4 ms
Mumbai–Bengaluru	845 km	11.8 ms
Delhi–Chennai	1,760 km	24.6 ms
Delhi–Bengaluru	1,740 km	24.4 ms
Chennai–Bengaluru	290 km	4.1 ms

Data labels. - **[R]** means reported, with a named source. - **[E]** means analyst estimate. - **[U]** means unpublished.

These rules apply throughout: - **EHV node names and distances** are [E]. They come from public grid knowledge and must be verified against STU single-line diagrams and CEA/CTUIL maps. - **Spare MVA** is [U] everywhere. - **Design temperatures** are [E], approximations of ASHRAE 0.4% dry-bulb / mean-coincident wet-bulb values. - **Free-cooling hours** are [E]: order-of-magnitude hours per year with dry-bulb ≤ 20 °C, not computed from TMY files.

13.3 Latency table (great-circle km to Mumbai / Delhi / Chennai / Bengaluru → mean RTT)

Table 13.5 Latency table (great-circle km to Mumbai / Delhi / Chennai / Bengaluru → mean RTT)

City	M	D	C	B	Mean RTT (ms)	Score
Anantapur	705	1,547	338	190	9.7	10.0
Bengaluru / Hoskote	845	1,740	290	0–25	10.1–10.2	10.0–9.9
Hyderabad	620	1,250	520	500	10.1	10.0
Hosur	850	1,760	260	35	10.2	9.9
Pune	120	1,170	910	735	10.3	9.9
Sri City	980	1,700	55	250	10.4	9.8
Navi Mumbai	15	1,150	1,020	835	10.6	9.7
Chennai	0–1,030	1,760	0	290	10.8	9.6
Nashik	140	1,040	1,040	900	10.9	9.6
Amaravati	856	1,385	381	504	10.9	9.6
Mysuru	820	1,810	385	125	11.0	9.5
Goa (Verna)	428	1,508	735	479	11.0	9.4
Warangal	760	1,180	560	640	11.0	9.5
Mangaluru	710	1,760	580	300	11.7	9.0
Nagpur	685	860	925	900	11.8	9.0
Indore	510	667	1,167	1,098	12.1	8.8
Bhopal	663	594	1,170	1,142	12.5	8.5
Coimbatore	1,000	1,960	430	225	12.7	8.4
Dholera	360	850	1,330	1,170	13.0	8.2
Ahmedabad / GIFT	440	775	1,380	1,230	13.4	8.0
Nava Raipur	957	953	913	1,013	13.4	8.0
Visakhapatnam	1,100	1,340	610	800	13.5	7.9
Kochi	1,080	2,060	545	355	14.1	7.5
Jamnagar / Jaipur	480 / 920	950 / 240	1,530 / 1,600	1,350 / 1,560	15.1	6.9
Jharsuguda	1,201	1,022	1,051	1,197	15.6	6.5
Thoothukudi	1,270	2,200	530	470	15.6	6.6
Thiruvananthapuram	1,200	2,210	620	490	15.8	6.5
Jodhpur	795	540	1,700	1,570	16.1	6.3
Delhi / Noida / Gurugram / Greater Noida	1,130–1,150	0–60	1,720–1,760	1,700–1,740	16.2–16.3	6.1
Lucknow / Kanpur	1,190	420	1,550	1,580	16.6	6.0
Bhubaneswar	1,365	1,274	996	1,195	16.9	5.8
Ranchi	1,380	1,006	1,257	1,410	17.7	5.3
Patna / Dehradun	1,457 / 1,350	860 / 205	1,479 / 1,930	1,607 / 1,926	18.9	4.5
Digha / Kharagpur	1,552	1,297	1,213	1,412	19.2	4.3
Mohali / Baddi	1,380	240	1,990	1,960	19.5	4.2
Kolkata	1,650	1,300	1,360	1,560	20.5	3.5
Guwahati	2,079	1,461	1,880	2,078	26.2	0.0

The mean-to-four-hubs metric penalises NCR and Kolkata. For local users, NCR sites are under 1 ms from Delhi, so read their latency score as "national reach", not "local quality".

13.4 Connectivity backbone (applies across states)

- **Subsea cables [R].** India has 17–18 international cables landing at 14–15 stations in five cities: Mumbai (Versova/Chimbai), Chennai, Kochi, Thoothukudi and Thiruvananthapuram. JLL counts 8 cables at Chennai. 2Africa (Airtel) lands at Mumbai.
- **Under development:**
- **Digha (West Bengal):** led by Reliance Jio, a ₹1,000 cr project targeted for "Q1 2026". Commissioning is unconfirmed as of October 2026.
- **Visakhapatnam:** Google's America-India Connect gateway and three new cable paths; Meta's Waterworth is also planned.
- **Others:** Puducherry and Bhubaneswar/Puri.
- **Internet exchanges [R].** NIXI reports 77 nodes. The core nodes are Delhi (Noida), Mumbai (Vashi), Chennai, Kolkata, Bengaluru, Hyderabad, Ahmedabad and Guwahati. Regional nodes listed include Lucknow, Patna, Ranchi, Bhubaneswar, Cuttack, Nagpur, Nashik, Indore, Bhopal, Raipur, Jabalpur, Panjim, Rajkot, Surat, Vadodara, Kochi, Coimbatore, Thiruvananthapuram, Visakhapatnam, Chandigarh, Ludhiana, Bareilly, Siliguri, Salem, Tuticorin, Durgapur and Bardhaman.
- **Private exchanges [E].** DE-CIX and Extreme IX are concentrated in Mumbai, Delhi NCR, Chennai, Kolkata, Bengaluru and Hyderabad. Confirm current locations on PeeringDB.

13.5 State-by-state shortlists

Score order in each table: P / C / L / W / Cl / H / La / Po / D. "Eff. tariff" is ₹/kWh at 80% load factor.

13.5.1 Maharashtra

- **Policy:** active; 51% RE floor cut in June 2026 (series Part 9).
- **Tariff [R]:** MSEDCL HT-I(A) FY2026-27 is about ₹7.11–7.27/kVAh plus about ₹0.60 wheeling, with a ₹560–565/kVA demand charge. MERC documents conflict slightly, giving an effective ≈ ₹8.7. MERC projects an average HT-industry tariff of about ₹9.00 for FY2026-27.
- **Land [R]:** MIDC TTC base rate is ₹31,390/m² (≈ ₹12.7 cr/acre). The market rate is ₹45,000–60,000/m² or more, and STT paid ₹282 cr for 14,093 m² including a building in May 2026. Taloja is ₹12,100/m² (≈ ₹4.9 cr/acre).

Table 13.6 Maharashtra

Rank	City (lat, long; district)	EHV node [E]	Water	Hazard (zone/other)	Design DB/WB °C; FC h [E]	Demand	Scores	Total	Tier
1	Navi Mumbai – TTC/Mahape/Rabale (19.03, 73.03; Thane/Raigad)	400 kV Kharghar / Kalwa, 5–15 km	Dam-fed municipal supply plus STP; Thane district not over-exploited [E]	III; monsoon flooding	35/26; ~50	Mumbai 890 MW op / 1,726 MW upcoming [R]	4.5/10/9.7/7/2/5/2/8/10	6.6	A
2	Pune – Chakan/Talegaon/Hinjewadi (18.52, 73.86; Pune)	400 kV Chakan / Lonikand, 10–20 km	Haveli semi-critical [E]	III	38/25; ~1,500	133 MW op / 258 MW upcoming [R]	4.0/7/9.9/6/5/7/5/8/8	6.3	A
3	Nagpur – MIHAN/Butibori (21.15, 79.09; Nagpur)	400 kV Koradi ~25 km; 220 kV Butibori <10 km	Safe [E]	II; heat (44 °C)	44/27; ~1,500	NIXI node; small DC base	4.0/5/9.0/7/5/9/8/8/4	6.3	A
4	Nashik (20.00, 73.79; Nashik)	400 kV Eklahare, 10–20 km	Safe [E]	III	39/25; ~1,800	NIXI node	3.0/5/9.6/7/6/7/7/8/3	5.8	B

13.5.2 Tamil Nadu

- **Policy:** DC policy lapsed 31 March 2026.
- **Tariff [R]:** TNERC T.O. No. 6 of 2025 sets HT-I at ₹7.50/kWh and ₹608/kVA/month, with no revision announced for 2026. Effective ≈ ₹8.54.

Table 13.7 Tamil Nadu

Rank	City	EHV node [E]	Water	Hazard	DB/WB; FC h [E]	Demand	Scores	Total	Tier
1	Hosur (12.74, 77.83; Krishnagiri)	400 kV Hosur ~10 km	Semi-critical to over-exploited [E]	II	35/24; ~2,500	Bengaluru spill-over	3.8/6/9.9/4/6/9/7/4/6	5.7	B
2	Chennai – Ambattur/Sriperumbudur/Siruseri (13.08, 80.27; Chennai/Tiruvallur/Kancheepuram/Chengalpattu)	400 kV Alamathy / Sriperumbudur, 5–20 km	District 127.47%, 13/16 taluks over-exploited [R]; TTRO/desal available → conditional	III; floods 2015/2023; cyclone	38.5/28.5; ~0	214 MW op / 368 MW upcoming; 8 cables [R]	4.3/10/9.6/3/1/4/4/4/9	5.5	B (A for colo)
3	Coimbatore (11.02, 76.96; Coimbatore)	400 kV Arasur ~20 km	Mixed [E]	III	37/25; ~400	NIXI node; wind belt	4.3/5/8.4/5/5/7/7/4/6	5.4	B
4	Thoothukudi (8.76, 78.13; Thoothukudi)	765/400 kV Tuticorin PS, 15–25 km	Seawater/desal option	II; coastal cyclone	38/28; ~0	BSNL CLS (Sri Lanka) + NIXI	4.8/7/6.6/5/1/6/8/4/2	5.2	C

13.5.3 Karnataka

- **Policy:** Sustainable DC Policy 2026–31, with a ₹20 cr incentive for PUE ≤ 1.20.
- **Tariff [R/E]:** KERC Order of 27 March 2025 sets HT-2(a) at about ₹700 paise/kWh for FY27 and about ₹355/kVA. The subagent could not confirm the exact row. Data centres are billed as industrial with a DoIT certificate. Effective ≈ ₹7.61.

Table 13.8 Karnataka

Rank	City	EHV node [E]	Water	Hazard	DB/WB; FC h [E]	Demand	Scores	Total	Tier
1	Mysuru (12.30, 76.64; Mysuru)	400 kV Mysuru (Kadakola), ~15 km	Cauvery-fed; safe to semi-critical [E]	II	35/24; ~2,000	Nascent	5.0/4/9.5/7/8/9/7/8/3	6.5	A
2	Hoskote (13.07, 77.80; Bengaluru Rural)	400 kV Hoody ~20 km	Hosakote over-exploited [R] → conditional	II	34.5/23.5; ~2,200	Hyperscale land bank	5.5/6/9.9/2/8/9/6/8/8	6.4	A (conditional)
3	Bengaluru core – Whitefield/EC (12.97, 77.59; Bengaluru Urban)	400 kV Hoody / Somanahalli, 5–25 km	All 6 units over-exploited; East 395% [R] → conditional	II; urban flooding	34.5/23.5; ~2,000	162 MW op / 100 MW upcoming [R]	5.0/7/10/1/8/8/3/8/10	6.1	B (conditional)
4	Mangaluru (12.91, 74.86; Dakshina Kannada)	220 kV local <15 km; 400 kV ~50 km	High rainfall; safe [E]	III; monsoon	34/27; ~0	Coastal edge	5.0/6/9.0/7/2/6/6/8/3	6.1	B

13.5.4 Telangana

- **Policy:** 2016 DC policy plus bespoke deals.
- **Tariff [R]:** TGERC FY2026-27 RST order, unchanged from FY26, sets HT-I(A) at ₹6.65/kVAh (132 kV and above), ₹7.15 (33 kV) and ₹7.65 (11 kV), with ₹500/kVA. Effective ≈ ₹7.51. FY27 adds a grid support charge of ₹19.77/kW/month and a green tariff of +₹0.66/kWh.

Table 13.9 Telangana

Rank	City	EHV node [E]	Water	Hazard	DB/WB; FC h [E]	Demand	Scores	Total	Tier
1	Hyderabad periphery – Chandanvelly/Meerkhanpet/Shadnagar (17.10, 78.20; Rangareddy)	400 kV Shankarpally / 765 kV Maheshwaram, 10–25 km	Mixed; Krishna/Godavari municipal supply [E]	II	40.5/25.5; ~900	151 MW op / 710 MW upcoming [R]	6.7/7/10/6/5/9/7/6/9	7.2	A

Rank	City	EHV node [E]	Water	Hazard	DB/WB; FC h [E]	Demand	Scores	Total	Tier
2	Hyderabad core – Gachibowli/Nanakramguda (17.44, 78.35; Rangareddy)	400 kV Shankarpally, ~15 km	Urban over-draft pockets [E]	II	40.5/25.5; ~900	as above	6.5/7/10/4/5/9/3/6/10	6.7	A
3	Warangal (17.97, 79.59; Hanamkonda)	400 kV Warangal, ~10 km	Safe [E]	II	42/26; ~800	Low	5.65/4/9.5/7/5/9/9/6/3	6.4	A

13.5.5 Andhra Pradesh

- **Policy:** DC Policy 4.0.
- **Tariff:** FY2026-27 APERC tariffs are unchanged from FY26 [R]. HT-I demand charge is about ₹475/kVA [R, older order]; the energy charge is not confirmed. Effective ≈ ₹7.3 [E].

Table 13.10 Andhra Pradesh

Rank	City	EHV node [E]	Water	Hazard	DB/WB; FC h [E]	Demand	Scores	Total	Tier
1	Visakhapatnam – Tarluvada/Adavivaram/Rambilli (17.69, 83.22; Visakhapatnam/Anakapalli)	400 kV Kalpaka, 10–25 km; new dedicated EHV planned	Safe [E]	II; very high cyclone	37/29; ~100	Google 1 GW (601 acres) [R]; 5 GW promised [R]	6.55/8/7.9/7/1/4/9/9/6	6.8	A
2	Sri City / Tirupati (13.53, 79.99; Tirupati)	220 kV in park; 400 kV unverified	Mixed [E]	III; moderate cyclone	39/28; ~0	55 km from Chennai cable landings	6.05/7/9.8/5/1/6/8/9/4	6.4	A
3	Anantapur (14.68, 77.60; Anantapur)	400 kV RE pooling (Uravakonda/Hindupur), 20–40 km	Drought-prone; over-exploited blocks [E] → conditional	II	40/25; ~600	Nil; RE-rich	7.05/3/10/3/4/9/10/9/2	6.3	A (conditional)
4	Amaravati / Vijayawada (16.51, 80.52; Guntur/NTR)	400 kV Vijayawada area, ~20 km	Safe [E]	III; Krishna floods	41/28; ~150	Capital build-out	6.05/4/9.6/7/2/6/7/9/3	6.2	B

13.5.6 Uttar Pradesh

- **Policy:** UP DC Policy 2026.
- **Tariff [R]:** UPERC's FY2026-27 order sets a standalone tariff for the New Integrated Data Centre Park (NIDP), Greater Noida: HV-2 at ₹270/kVA and ₹6.95/kVAh (effective ≈ ₹7.41), with optional green power at +₹0.34. The general HV-2 average billing rate is ₹7.97.
- **Land [R]:** YEIDA's 2026 residential allotment rate is ₹36,260/m². Industrial/DC rates are lower [U].

Table 13.11 Uttar Pradesh

Rank	City	EHV node [E]	Water	Hazard	DB/WB; FC h [E]	Demand	Scores	Total	Tier
1	Greater Noida – NIDP/Knowledge Park (28.47, 77.50; Gautam Buddh Nagar)	765/400 kV Greater Noida, 5–15 km	Semi-critical to over-exploited [E]; Ganga canal + STP	IV	43.5/29; ~2,800	NCR 183 MW op / 387 MW upcoming [R]	6.35/8/6.1/4/4/6/7/9/8	6.4	A
2	Noida – Sec 62/132/143 (28.54, 77.39; Gautam Buddh Nagar)	400 kV Noida, 5–10 km	Over-exploited [E] → conditional	IV	43.5/29; ~2,800	NIXI Delhi node in Noida	5.35/9/6.1/3/4/6/4/9/9	6.1	B
3	YEIDA / Jewar (28.17, 77.60; Gautam Buddh Nagar)	400 kV Jahangirpur ~15 km	Semi-critical [E]	IV	43.5/29; ~2,800	Airport-led	5.35/6/6.2/5/4/6/9/9/5	6.0	B
4	Lucknow (26.85, 80.95; Lucknow)	765 kV Unnao / 400 kV Lucknow, 20–40 km	Urban over-exploited [E]	III	43/29.5; ~2,500	NIXI node	4.85/5/6.0/3/4/7/8/9/5	5.5	B
5	Kanpur (26.45, 80.33; Kanpur Nagar)	400 kV Panki ~10 km	Semi-critical [E]	III	44/29.5; ~2,400	Low	4.85/3/6.0/4/4/7/8/9/3	5.2	C

13.5.7 Haryana

- **Policy:** Haryana 2022.
- **Tariff [R]:** HERC kept FY27 tariffs equal to FY26. HT is ₹290/kVA plus ₹6.95/kVAh at 11 kV, ₹6.85 at 33 kV, and about ₹6.6 at 66–132 kV (OCR uncertain). Effective ≈ ₹7.1.

Table 13.12 Haryana

Rank	City	EHV node [E]	Water	Hazard	DB/WB; FC h [E]	Demand	Scores	Total	Tier
1	Gurugram – Manesar/Sohna (28.46, 77.03; Gurugram)	PGCIL 400 kV Gurugram / Daultabad, 5–15 km	State 136.75%; 5 over-exploited blocks [R] → conditional	IV	44/28.5; ~2,800	NCR enterprise/cloud	6.4/8/6.1/1/4/6/3/5/9	5.5	B (conditional)
2	Sonipat – Kundli/Kharkhoda (28.99, 77.02; Sonipat)	400 kV Deepalpur ~15 km	Over-exploited [R, district list] → conditional	IV	44/29; ~3,000	Low	6.4/5/6.1/1/4/6/7/5/5	5.0	C

13.5.8 Delhi

- **Policy:** no DC policy.
- **Tariff:** DERC figures not retrieved [U]; effective ≈ ₹8.5 [E].

Table 13.13 Delhi

Rank	City	EHV node [E]	Water	Hazard	DB/WB; FC h [E]	Demand	Scores	Total	Tier
1	Delhi – Okhla/Narela/Bawana (28.61, 77.21; various)	400 kV Bawana / Mundka / Tughlakabad, 5–15 km	Largely over-exploited [E]	IV; Yamuna floodplain	43.5/29; ~2,800	Core NIXI; edge demand	3.35/10/6.1/1/4/6/1/3/10	4.7	C (edge only)

13.5.9 Gujarat

- **Policy:** Gujarat DC Policy 2026–29.
- **Tariff:** GERC FY27 tariffs are unchanged. The HTP-I energy charge is ₹4.30/unit plus FPPPA (base about ₹2.15), and the demand-charge slabs were not verified. The all-in figure is ₹7.5–8.5 (secondary source), so ≈ ₹8.0 is used. Open-access additional surcharge is ₹0.99/kWh for October 2026 – March 2027 [R].

Table 13.14 Gujarat

Rank	City	EHV node [E]	Water	Hazard	DB/WB; FC h [E]	Demand	Scores	Total	Tier
1	GIFT City / Gandhinagar (23.16, 72.68; Gandhinagar)	220 kV in city; 400 kV Dehgam ~25 km	Narmada supply; semi-critical [E]	III	43/28; ~900	IFSC, BFSI DR	5.3/6/8.0/5/3/7/6/8/6	6.0	B
2	Ahmedabad (23.02, 72.57; Ahmedabad)	400 kV Dehgam / Vadavi, 15–25 km	Critical [E]	III	43/28; ~900	NIXI Ahmedabad	5.3/6/8.0/4/3/7/6/8/7	5.9	B
3	Jamnagar (22.47, 70.06; Jamnagar)	400 kV Hadala/Jamnagar ~20 km	Desal	IV [E]; cyclone	40/28; ~600	Reliance AI campus [R, announced]	5.8/4/6.9/4/3/5/8/8/3	5.3	B
4	Dholera SIR (22.25, 72.19; Ahmedabad)	New 400 kV in SIR [E]	Saline; canal water	III; low-lying	42/28; ~800	Greenfield	5.3/3/8.2/4/3/5/9/8/2	5.2	C

13.5.10 West Bengal

- **Policy:** expiring around September 2026, so likely in renewal limbo now.
- **Tariff:** WBERC FY26 sets a ₹320/kVA demand charge [R]; the energy charge is unverified, and ≈ ₹7.5 effective is used [E].

Table 13.15 West Bengal

Rank	City	EHV node [E]	Water	Hazard	DB/WB; FC h [E]	Demand	Scores	Total	Tier
1	Kharagpur (22.35, 87.32; Paschim Medinipur)	400 kV Kharagpur ~10 km	Safe to semi-critical [E]	III; cyclone moderate	40/29.5; ~800	Backhaul for Digha CLS	4.7/6/4.3/6/2/5/8/4/1	4.9	C
2	Kolkata – New Town / Bengal Silicon Valley (22.57, 88.36; North 24 Parganas)	400 kV Subhasgram ~25 km	Hooghly supply; mixed [E]	III; flood; cyclone (Amphan)	37.5/29.5; ~1,000	22 MW op / 84 MW upcoming [R]	5.2/6/3.5/5/2/4/6/4/5	4.8	C
3	Digha (21.63, 87.51; Purba Medinipur)	Nearest 400 kV ~90 km → fails 25 km filter	Safe [E]	Very high cyclone	36/29.5; ~300	Jio CLS (pending)	4.7/6/4.3/6/2/2/8/4/1	4.6	C (CLS only)

13.5.11 Kerala

- **Policy:** no dedicated DC policy.
- **Tariff [R]:** KSERC Order of 5 December 2024, valid to 31 March 2027, sets energy charges of ₹6.75 (HT-I), ₹6.15 (110 kV) and ₹5.55 (220 kV). Demand charges are about ₹390–400/kVA [E]. Effective ≈ ₹6.23 at 220 kV.

Table 13.16 Kerala

Rank	City	EHV node [E]	Water	Hazard	DB/WB; FC h [E]	Demand	Scores	Total	Tier
1	Kochi – Kakkanad/Infopark (9.93, 76.27; Ernakulam)	400 kV Kochi East ~10 km	Safe; ~3,000 mm rainfall	III; 2018 floods	33.5/27.5; 0	Cable landings (Tata, others) + NIXI	7.5/8/7.5/7/1/5/4/4/6	6.3	A
2	Thiruvananthapuram – Technopark (8.52, 76.94; Thiruvananthapuram)	400 kV Pothencode ~15 km	Safe	III	33/27; 0	CLS + NIXI	7.0/7/6.5/7/1/6/4/4/5	6.0	B

13.5.12 Odisha

- **Policy:** Odisha 2022.
- **Tariff [R]:** OERC RST order of 24 March 2026 (FY27) sets EHT large industry at ₹250/kVA, with 580 paise/kVAh up to 60% load factor and 470 paise above. It also gives a 30 p rebate above 80% load factor and a 10% discount during solar hours from July 2026. Effective ≈ ₹5.96, the lowest found.

Table 13.17 Odisha

Rank	City	EHV node [E]	Water	Hazard	DB/WB; FC h [E]	Demand	Scores	Total	Tier
1	Jharsuguda (21.86, 84.01; Jharsuguda)	765/400 kV Jharsuguda (PGCIL) ~10 km	Hirakud; safe [E]	II–III; inland	45/27; ~1,000	Nil; power-intensive industry hub	9.0/2/6.5/7/3/8/9/6/1	6.3	A
2	Bhubaneswar – Infovalley (20.30, 85.82; Khordha)	400 kV Mendhasal ~15 km	Safe [E]	III; very high cyclone (Fani)	41/29; ~400	NIXI Bhubaneswar/Cuttack; planned CLS	8.5/5/5.8/7/2/4/8/6/4	6.2	B

13.5.13 Rajasthan

- **Policy:** Rajasthan 2025.
- **Tariff:** about ₹380/kVA + ₹6.50/kWh (aggregator, unconfirmed); effective ≈ ₹7.7 [E].

Table 13.18 Rajasthan

Rank	City	EHV node [E]	Water	Hazard	DB/WB; FC h [E]	Demand	Scores	Total	Tier
1	Jaipur – Mahindra World City/Sitapura (26.91, 75.79; Jaipur)	400 kV Heerapura, 10–20 km	Over-exploited [E] → conditional	II	43/27.5; ~2,300	Moderate	5.8/4/6.9/1/4/9/8/7/5	5.3	B (conditional)
2	Jodhpur (26.24, 73.02; Jodhpur)	400 kV Surpura ~10 km	Over-exploited [E] → conditional	II; heat	44/26; ~1,700	Bhadla solar	6.3/2/6.3/1/3/9/9/7/2	5.0	C

13.5.14 Madhya Pradesh

- **Policy:** MP IT Policy 2023.
- **Tariff:** MPERC FY27 HV-3 at 132 kV is ₹729/kVA, with 734/635 paise split by load factor (snippets; column mapping uncertain). Effective ≈ ₹8.2.

Table 13.19 Madhya Pradesh

Rank	City	EHV node [E]	Water	Hazard	DB/WB; FC h [E]	Demand	Scores	Total	Tier
1	Bhopal (23.26, 77.41; Bhopal)	400 kV Bhopal ~15 km	Semi-critical [E]	II	41.5/26; ~1,800	NIXI node	4.4/4/8.5/5/5/9/8/6/4	5.7	B
2	Indore – Super Corridor/Pithampur (22.72, 75.86; Indore/Dhar)	400 kV Indore / Pithampur ~15 km	Over-exploited [E] → conditional	III	41/26; ~1,800	NIXI node	4.4/5/8.8/2/5/7/8/6/5	5.2	C

13.5.15 Secondary states

Table 13.20 Secondary states

State	City (lat, long; district)	Key raw facts	Scores	Total	Tier
Chhattisgarh	Nava Raipur (21.16, 81.79; Raipur)	Surplus coal-based state grid; 400 kV Raita ~25 km [E]; zone II; NIXI Raipur	7.45/4/8.0/7/4/9/9/4/2	6.4	A
Jharkhand	Ranchi (23.34, 85.31; Ranchi)	400 kV Ranchi ~15 km [E]; zone II; mild plateau climate	5.75/4/5.3/5/6/9/8/3/2	5.3	B
Goa	Verna (15.36, 73.93; South Goa)	NIXI Panjim; ~430 km from Mumbai cable landings; zone III	5.1/4/9.4/7/2/6/4/4/2	5.3	B
Uttarakhand	Dehradun (30.32, 78.03; Dehradun)	Zone IV (Zone VI in the withdrawn 2025 draft); cool climate; hydro	6.5/3/4.5/7/8/4/4/4/3	5.1	C
Himachal Pradesh	Baddi–Nalagarh (30.95, 76.79; Solan)	Low hydro tariff [E]; 400 kV Nalagarh ~10 km; zone IV	8.0/2/4.2/5/7/4/6/3/1	4.9	C
Punjab	Mohali (30.70, 76.72; SAS Nagar)	PSERC FY26 industrial ₹5.82–7.04/kVAh [R]; over-exploited [E]; zone IV; NIXI Chandigarh	6.1/5/4.2/2/5/6/6/4/5	4.8	C
Bihar	Patna (25.59, 85.14; Patna)	Zone IV; Ganga floods; NIXI node	3.8/4/4.5/6/3/3/7/4/2	4.2	C
Assam / NE	Guwahati (26.14, 91.74; Kamrup Metro)	Zone V; Brahmaputra floods; NIXI core node	4.7/4/0/7/4/1/7/4/2	4.0	**Excluded (zone V)** – regional edge only, with seismic isolation

13.6 National master table (default weights)

Table 13.21 National master table (default weights)

#	City	State	District	Lat	Long	Score	Tier	Key strength	Key weakness
1	Hyderabad periphery	Telangana	Rangareddy	17.10	78.20	7.2	A	Grid + ₹6.65/kVAh + zone II	No cable landing (Vizag/Chennai backhaul)
2	Visakhapatnam	AP	Visakhapatnam/Anakapalli	17.69	83.22	6.8	A	Gigawatt anchor + new cable landings + policy	Cyclone
3	Hyderabad core	Telangana	Rangareddy	17.44	78.35	6.7	A	Ecosystem, talent	Land cost, water
4	Navi Mumbai	Maharashtra	Thane/Raigad	19.03	73.03	6.6	A	Cables, interconnection	Tariff ≈ ₹8.7, land
5	Mysuru	Karnataka	Mysuru	12.30	76.64	6.5	A	Climate, water, policy	Thin ecosystem
6	Greater Noida (NIDP)	UP	Gautam Buddh Nagar	28.47	77.50	6.4	A	DC tariff ₹6.95, 765 kV	Zone IV, heat
7	Hoskote	Karnataka	Bengaluru Rural	13.07	77.80	6.4	A*	Bengaluru latency, cooler climate	Over-exploited water
8	Sri City	AP	Tirupati	13.53	79.99	6.4	A	Near Chennai cable landings, AP policy	Humid, water
9	Nava Raipur	Chhattisgarh	Raipur	21.16	81.79	6.4	A	Cheap surplus power, zone II	Coal grid, no ecosystem
10	Warangal	Telangana	Hanamkonda	17.97	79.59	6.4	A	Land, grid	Talent
11	Pune	Maharashtra	Pune	18.52	73.86	6.3	A	120 km to Mumbai, land	Tariff
12	Nagpur	Maharashtra	Nagpur	21.15	79.09	6.3	A	Geographic centre, zone II	Tariff, heat
13	Kochi	Kerala	Ernakulam	9.93	76.27	6.3	A	Cable landings + cheap EHT	Policy, land
14	Anantapur	AP	Anantapur	14.68	77.60	6.3	A*	RE pooling, land	Water, connectivity
15	Jharsuguda	Odisha	Jharsuguda	21.86	84.01	6.3	A	Cheapest tariff, 765 kV	Connectivity, talent
16	Bhubaneswar	Odisha	Khordha	20.30	85.82	6.2	B	Tariff	Cyclone
17	Amaravati	AP	Guntur/NTR	16.51	80.52	6.2	B	Policy, land	Connectivity
18	Bengaluru core	Karnataka	Bengaluru Urban	12.97	77.59	6.1	B*	Demand, talent	Water (395% East)
19	Mangaluru	Karnataka	Dakshina Kannada	12.91	74.86	6.1	B	Water, coast	EHV distance
20	Noida	UP	Gautam Buddh Nagar	28.54	77.39	6.1	B	NIXI, demand	Water, land
21	GIFT City	Gujarat	Gandhinagar	23.16	72.68	6.0	B	IFSC, RE	Heat
22	YEIDA/Jewar	UP	Gautam Buddh Nagar	28.17	77.60	6.0	B	Land	Ecosystem
23	Thiruvananthapuram	Kerala	Thiruvananthapuram	8.52	76.94	6.0	B	Cable landing, tariff	Latency to north
24	Ahmedabad	Gujarat	Ahmedabad	23.02	72.57	5.9	B	NIXI, policy	Water
25	Nashik	Maharashtra	Nashik	20.00	73.79	5.8	B	Near Mumbai, safe water	Tariff
26	Hosur	Tamil Nadu	Krishnagiri	12.74	77.83	5.7	B	Bengaluru spill-over	TN policy lapsed
27	Bhopal	MP	Bhopal	23.26	77.41	5.7	B	Zone II, land	Tariff ≈ ₹8.2
28	Chennai	Tamil Nadu	Chennai and neighbours	13.08	80.27	5.5	B*	8 cables, 214 MW	Water, flood, policy lapse
29	Gurugram	Haryana	Gurugram	28.46	77.03	5.5	B*	Demand	Water (136.75% state)
30	Lucknow	UP	Lucknow	26.85	80.95	5.5	B	Policy	Water, connectivity

#	City	State	District	Lat	Long	Score	Tier	Key strength	Key weakness
31	Coimbatore	Tamil Nadu	Coimbatore	11.02	76.96	5.4	B	Wind RE	Policy
32	Jamnagar	Gujarat	Jamnagar	22.47	70.06	5.3	B	Captive RE	Connectivity
33	Jaipur	Rajasthan	Jaipur	26.91	75.79	5.3	B*	Zone II, RE	Water
34	Ranchi	Jharkhand	Ranchi	23.34	85.31	5.3	B	Climate, zone II	Policy
35	Goa (Verna)	Goa	South Goa	15.36	73.93	5.3	B	Latency, water	Scale
36	Thoothukudi	Tamil Nadu	Thoothukudi	8.76	78.13	5.2	C	Cable landing + RE	Latency, talent
37	Dholera	Gujarat	Ahmedabad	22.25	72.19	5.2	C	Land	Connectivity
38	Indore	MP	Indore	22.72	75.86	5.2	C*	Latency	Water, tariff
39	Kanpur	UP	Kanpur Nagar	26.45	80.33	5.2	C	Policy	Connectivity
40	Dehradun	Uttarakhand	Dehradun	30.32	78.03	5.1	C	Free cooling	Seismic
41	Sonipat	Haryana	Sonipat	28.99	77.02	5.0	C*	Grid	Water
42	Jodhpur	Rajasthan	Jodhpur	26.24	73.02	5.0	C*	Solar	Water, connectivity
43	Kharagpur	West Bengal	Paschim Medinipur	22.35	87.32	4.9	C	Digha backhaul	Policy, demand
44	Baddi	HP	Solan	30.95	76.79	4.9	C	Hydro tariff	Seismic, connectivity
45	Kolkata	West Bengal	North 24 Parganas	22.57	88.36	4.8	C	Eastern demand	Hazard, latency, policy
46	Mohali	Punjab	SAS Nagar	30.70	76.72	4.8	C*	Tariff	Water, latency
47	Delhi	Delhi	various	28.61	77.21	4.7	C	Interconnection	Water, land, tariff
48	Digha	West Bengal	Purba Medinipur	21.63	87.51	4.6	C	Cable landing	Cyclone, no EHV within 25 km
49	Patna	Bihar	Patna	25.59	85.14	4.2	C	—	Flood, zone IV
—	Guwahati	Assam	Kamrup Metro	26.14	91.74	4.0	Excluded	NE edge	Zone V

* = conditional on a contracted treated-water source (groundwater over-exploited).

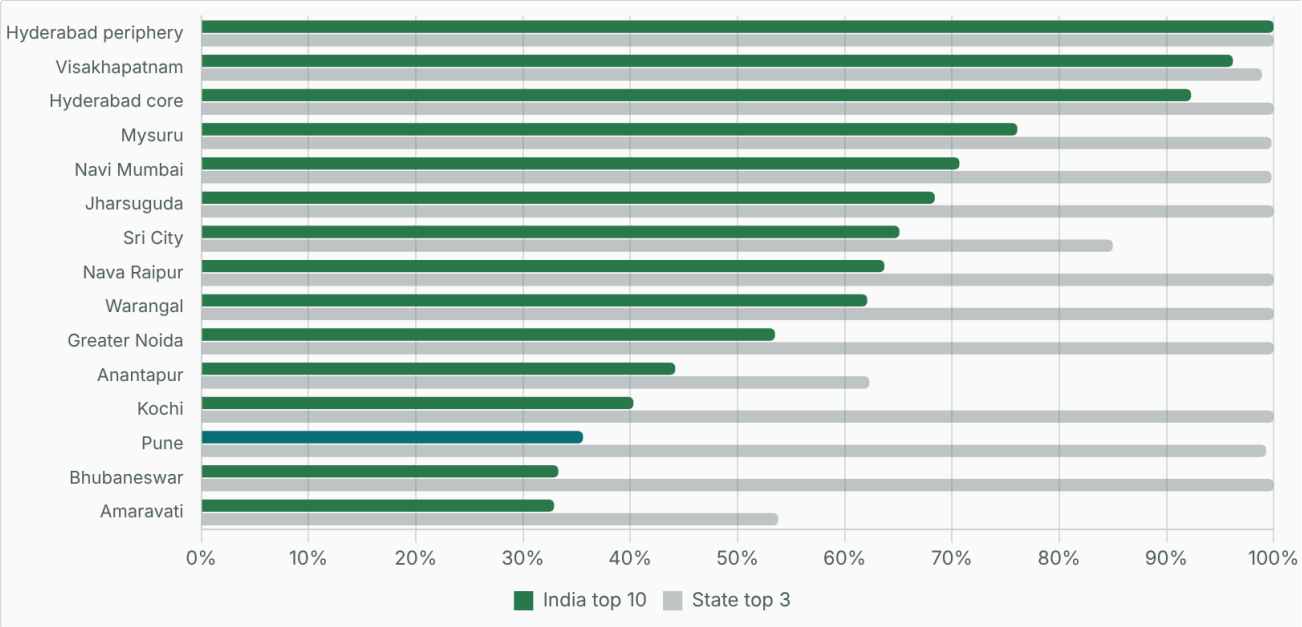


Figure 13.1 How stable is the ranking?. Share of 1,000 randomised runs (weights and scores each varied ±20%) in which a city lands in the national top 10. *Source: India Data Centre Atlas, chapter built on this chapter's data.*

13.7 Alternate weightings and rank stability

Table 13.22 Alternate weightings and rank stability

City	Default	AI campus	Interconnect	Sustainability	Geo-mean
Hyderabad periphery	7.2	7.0	7.9	6.6	7.0
Visakhapatnam	6.8	6.6	7.1	6.2	6.3
Hyderabad core	6.7	6.2	7.8	5.9	6.3
Navi Mumbai	6.6	5.5	**8.2**	6.1	6.0
Mysuru	6.5	6.4	6.0	**7.2**	6.1
Greater Noida	6.4	6.2	7.0	5.3	6.2
Hoskote	6.4	6.0	7.2	6.0	5.8
Sri City	6.4	6.2	6.9	5.5	—
Nava Raipur	6.4	7.0	5.8	5.6	5.9
Pune	6.3	5.6	7.2	6.3	6.1
Nagpur	6.3	6.0	6.2	6.5	—

City	Default	AI campus	Interconnect	Sustainability	Geo-mean
Kochi	6.3	6.2	6.9	5.6	—
Anantapur	6.3	6.7	5.9	6.1	—
Jharsuguda	6.3	**7.4**	5.0	5.2	—
Bhubaneswar	6.2	6.8	5.7	5.2	—
Bengaluru core	6.1	5.4	7.5	5.6	5.0
Noida	6.1	5.4	7.2	5.0	—
GIFT City	6.0	5.8	6.4	6.2	—
Chennai	5.5	4.6	7.6	4.7	4.7
Gurugram	5.5	5.1	6.7	4.1	4.6
Jamnagar	5.3	5.6	5.1	5.6	—
Kolkata	4.8	4.8	4.9	4.0	—
Delhi	4.7	3.7	6.8	4.0	3.7

What the sensitivity shows. - **Robust:** Hyderabad periphery is first or second in every lens. Visakhapatnam, Hyderabad core and Greater Noida stay at 6.2 or above in four of five lenses. - **Lens-dependent:** - Navi Mumbai is #1 for interconnection but only mid-table for AI campuses, because of tariff and land. - Odisha and Chhattisgarh lead for AI campuses but trail for interconnection. - Mysuru leads for sustainability; Pune and Nagpur also rise there. - **Fragile:** Bengaluru core, Gurugram, Chennai and Delhi lose 0.8–1.0 points under the geometric mean, because water is a near-fatal weakness that an arithmetic average hides. The geometric mean also penalises every humid coastal site (Vizag, Chennai, Kochi), since a climate score of 1 is treated as fatal. That is an artefact: humid sites can use closed-loop liquid cooling. Read their geo-mean scores with that in mind.

13.8 Recommendations — top picks by use case

- **AI training campus (100 MW–1 GW, power/RE-led):** 1. **Visakhapatnam:** best combination of power, policy, subsea access and anchor tenants. Design for cyclones, with an elevated plinth, wind-rated envelopes and N+2 grid feeds. 2. **Hyderabad periphery.** 3. **Odisha (Jharsuguda/Bhubaneswar):** cheapest tariff at ≈ ₹5.96. Pair it with ISTS RE open access, because the base grid is coal-heavy. 4. **Anantapur:** RE-pooling land, but only with a treated-water or air-cooled design. 5. **Nava Raipur.**
- **Interconnection / colocation:** Navi Mumbai (8.2), Hyderabad (7.8–7.9), Chennai (7.6), Bengaluru (7.5), and Noida/Greater Noida for NCR. In Chennai and Bengaluru, buy into TTRO/recycled-water-served parks; do not rely on borewells.
- **Edge / regional:** Lucknow, Bhubaneswar, Kochi, Ahmedabad, Indore/Bhopal, Ranchi, Patna, Guwahati, Nagpur and Goa. Size them at 1–10 MW and co-locate near NIXI regional nodes. The NIXI node map shows where ISP peering density already exists.
- **Sustainability-first:**
- **Mysuru:** best combination of climate and water, with Karnataka's ₹20 cr incentive for PUE ≤ 1.20 achievable at about 2,000 free-cooling hours.
- **Hyderabad periphery, Nagpur, Pune.**
- **GIFT City:** Gujarat RE.
- **Kochi:** water plus hydro, but humid.
- **Avoid for new greenfield builds:** Delhi proper, Gurugram and Bengaluru core, unless the site comes with contracted recycled water. Also avoid Digha (cable landing only, no EHV within 25 km), Guwahati (zone V) and Patna (flood plus zone IV).

13.9 Caveats and data gaps (and how to close them)

1. **Substation spare capacity [U].** No STU publishes per-substation MVA headroom. - **Request:** a feasibility/load-flow study from MSETCL, TANTRANSCO, KPTCL, TGTRANSCO, APTRANSCO, UPPTCL, HVPNL, DTL, GETCO, WBSETCL, KSEB-T, OPTCL, RVPN, MPPTCL and CSPTCL. - **Central sources:** CTUIL's ISTS connectivity/GNA margins, and CEA's transmission plans. Grid-India publishes some regional ATC/TTC data. - **Unverified:** all substation names and distances above. Confirm them on STU single-line diagrams.
2. **Tariff conflicts.** - Maharashtra's HT-I(A) FY27 figure appears as ₹7.11, ₹7.27 or ₹7.48/kVAh across MERC documents, with ₹555–565/kVA. Rely on the MERC Case 217/2024 schedule. - Karnataka HT-2(a), Gujarat demand-charge slabs, West Bengal energy charges, Andhra Pradesh HT-I energy charges, Rajasthan, MP column mapping, Kerala demand charges, and Delhi are unverified or estimated. - Effective ₹/kWh excludes electricity duty, which varies by state (for example, 15% for HT in Gujarat and 9.3% in Maharashtra [secondary]), and excludes fuel surcharges.
3. **Water.** Only Bengaluru Urban, Chennai, Haryana/Gurugram and the national figures are reported. Everything else is [E]. - **Pull from IN-GRES** (ingres.iith.ac.in): the block-level 2025 categories. - **Pull from state STP inventories:** BWSSB, CMWSSB, HMWSSB, NMMC.
4. **Climate.** Design values and free-cooling hours are approximations. Compute exact hours from ISHRAE/ASHRAE TMY files, counting hours where wet-bulb + 4 K approach ≤ 27 °C for evaporative designs, or dry-bulb ≤ 20–25 °C for air-side economisers.
5. **Hazards.** Flood and cyclone ratings are qualitative. Check the NRSC Bhuvan flood hazard atlas and NDMA cyclone hazard zones at plot level. Seismic zones use IS 1893:2016, because the 2025 revision was withdrawn in March 2026.
6. **Forward-looking claims.** These are announcements or plans, not commissioned assets: - Visakhapatnam's "5 GW", the 2.1 GW Raiden power application, and the Vizag cable landings. - The Digha cable landing ("Q1 2026" target). - Jamnagar. - Pipeline MW figures (C&W's 3,351 MW is still at "planning" stage).
7. **Talent.** AISHE graduate counts were not retrieved. The demand/talent scores are [E], based on IT-employment scale.

Sources cited in this chapter

Organisations and publications named in this chapter's text, in order of how often they are cited. Full details are in the bibliography.

- State electricity regulatory commissions (KERC, MERC, TNERC, TGERC) tariff and open-access orders
- Google environmental reports, efficiency data and Gemini inference paper
- Bureau of Indian Standards (IS 1893, ISO/IEC 30134 adoption)
- Meta sustainability reports and data index
- Reliance Industries announcements
- JLL, Global and India Data Center Outlooks
- Central Electricity Authority (CEA), CO_2 Baseline Database and grid reports
- Central Ground Water Board and state groundwater assessments
- Ministry of Power, orders on ISTS waiver and green open access
- ASHRAE TC 9.9 thermal guidelines
- CBRE, India data centre market reports
- Cushman & Wakefield, India data centre market updates (H1 2026)
- Bangalore Water Supply and Sewerage Board
- STT GDC India ESG report
- Nxtra by Airtel disclosures
- AdaniConneX disclosures
- DatacenterDynamics reporting
- ISHRAE climate design data
- National Disaster Management Authority guidelines

PART IV · POLICY, PLACE AND THE FUTURE

Chapter 14
India data centres to 2035: a forecast

WHAT THIS CHAPTER COVERS

This chapter forecasts India's operational data-centre capacity, electricity use, carbon emissions, capital cost, GPU-hour prices and water demand from 2026 to 2035. It builds three capacity models (constant growth, an S-shaped adoption curve and a bottom-up pipeline model), combines them into an integrated path, compares the result with every published forecast, and corrects for the historical tendency of forecasts to overshoot. It also distributes new capacity by city, tests demand drivers by regression, provides a parameter table with uncertainty ranges, and ranks the risks.

KEY TAKEAWAYS

1. In the base case, India's operational capacity reaches about 4.7 gigawatts of IT load by the end of 2030 and about 10.8 gigawatts by the end of 2035.
2. This is about 28% below the median published forecast for 2030 of about 6.5 gigawatts. The reduction is deliberate, because consultants have historically delivered only 58 to 78% of the capacity additions they forecast.
3. By 2035 data centres would use about 83 terawatt-hours of electricity a year, about 2.5% of India's electricity, and would emit about 18 million tonnes of carbon dioxide after renewable procurement.
4. Cumulative facility investment would reach about ₹8 lakh crore, about US$90 billion, by 2035.
5. On-site water use plateaus at about 60 million litres a day nationally, provided that liquid and dry cooling spread as modelled.

How this chapter connects to the others. Chapter 9 supplies the cost assumptions, and Chapter 15 notes that the Central Electricity Authority's projection of 26 gigawatts of new demand by 2031–32 is far higher than this base case.

Bottom line: In our base case, India's operational data-centre capacity reaches about 4.7 GW (IT) by end-2030 and about 10.8 GW by end-2035. That is roughly 28% below the median published 2030 forecast (about 6.5 GW). The haircut is deliberate: consultants have historically delivered only 58–78% of the capacity additions they forecast. By 2035 data centres would use about 83 TWh a year (about 2.5% of national electricity), emit about 18 $MtCO_2$ after renewable procurement, and require about ₹8 lakh crore (about US$90 bn) of cumulative facility capex. On-site water use plateaus at about 60 MLD nationally, provided liquid and dry cooling spread as modelled.

Labels used throughout: [R] = reported by a named source; **[E]** = analyst estimate or assumption; **[M]** = model output from the formulas in this report. All US$ figures are converted at ₹89/US$ [E] unless the source gives its own conversion. Capacity means operational IT MW in the major colocation and hyperscale markets unless stated otherwise.

14.1 Key Findings

1. **History, harmonised [E from R].** Operational IT capacity grew from about 375 MW (2019) to 1,611 MW (end-2025). A log-linear fit gives g = 27.9%/yr with $R^2 \approx 0.99$ [M]. The 1,611 MW figure is derived from Cushman & Wakefield's (C&W) H1 2026 total of 1,789 MW minus 178 MW added in H1 2026.
2. **Sources disagree by about ±7% on today's stock:** - C&W: 1,789 MW (H1 2026) [R]. - JLL: 1,637 MW (about 1.64 GW) at June 2026, per its India Data Centre 2026 Mid-Year Report [R]. JLL projects 6 GW by 2029, needing US$110 bn. - PIB factsheet (14 Sep 2026): 1.57 GW "as of August 2026", up from about 375 MW in 2020 [R]. - The calculator should use C&W as the anchor because its city split is the most complete.
3. **Track record on forecasts:** - Measured as delivered additions ÷ forecast additions, past consultant forecasts delivered 0.58–0.74. - Measured as actual level ÷ forecast level, they delivered 0.78–0.87. - Projects already under construction deliver close to 100% within 4–5 years. - This supports a 25–35% haircut on headline forecasts of additions.
4. **Published 2030 forecasts range from 4 GW (CareEdge) to 12 GW (Wood Mackenzie); the median is about 6.5 GW** [R]. Our base of 4.7 GW sits close to the delivered pipeline.
5. **Demand-driver regressions point lower still (2.5–4.9 GW in 2030)** [M]. Consumer and fintech drivers (UPI, mobile data) are decelerating. The gap between those drivers and the supply-side forecasts is AI, hyperscale and sovereign compute, which these regressions cannot see.
6. **Electricity is the binding constraint at the high end.** - A PIB factsheet (14 Sep 2026) states: "According to the Central Electricity Authority (CEA), Ministry of Power, demand could reach 17 GW by 2031–32" [R]. The Ministry of Power's March 2026 estimate was 13.56 GW [R]. In a 29 July 2026 written reply to Parliament, MoS Power Shripad Naik cited "an additional load of 26.3 GW from artificial intelligence data centres" by 2031–32 (Business Standard) [R], which is above even our high case. - Those figures match our *high* case for connected facility load (about 14 GW in 2032) [M], not our base (about 9.6 GW).
7. **Cost per MW is rising.** - Turner & Townsend reports a 5.5% global inflation in data-centre construction costs in 2025 (9.0% in 2024) [R]. - CareEdge reports 50–70% cumulative cost increases in Indian data-centre construction [R]. - Like-for-like learning ($b \approx 0.10$, i.e. about 7% cheaper per doubling) [E] is swamped by inflation and the premium for AI density.

14.2 Capacity (MW IT): data, three models, comparison, backtest

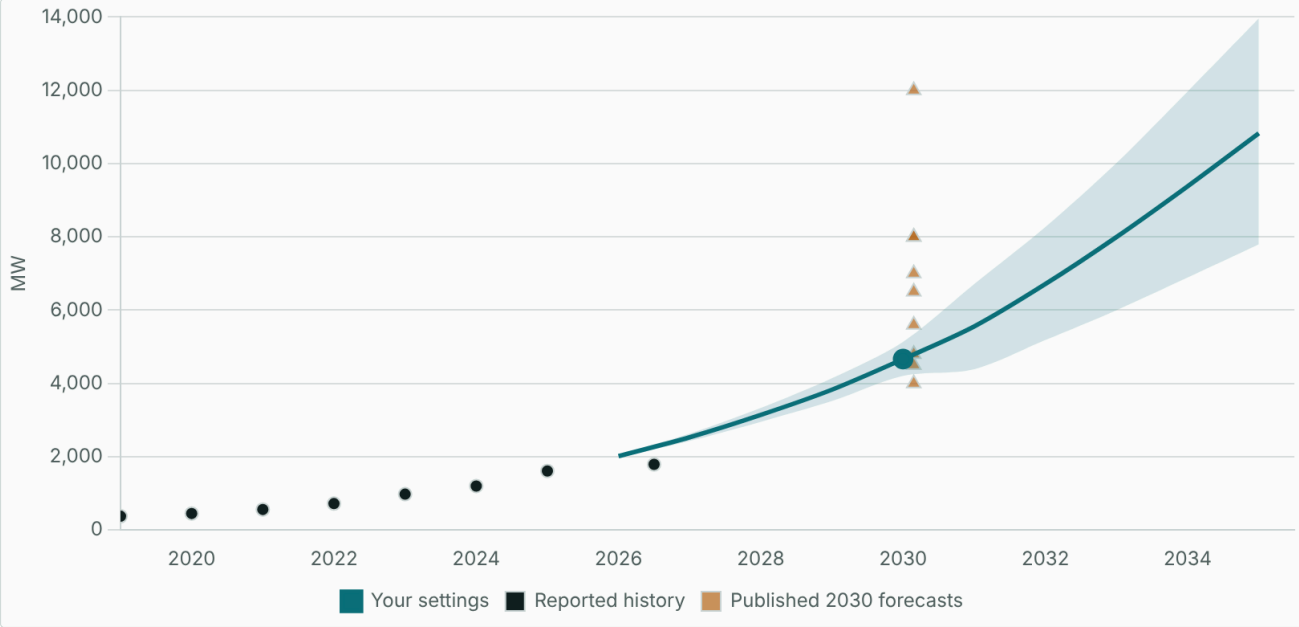


Figure 14.1 Capacity. MW of IT load *Source: India Data Centre Atlas, chapter built on this chapter's data.*

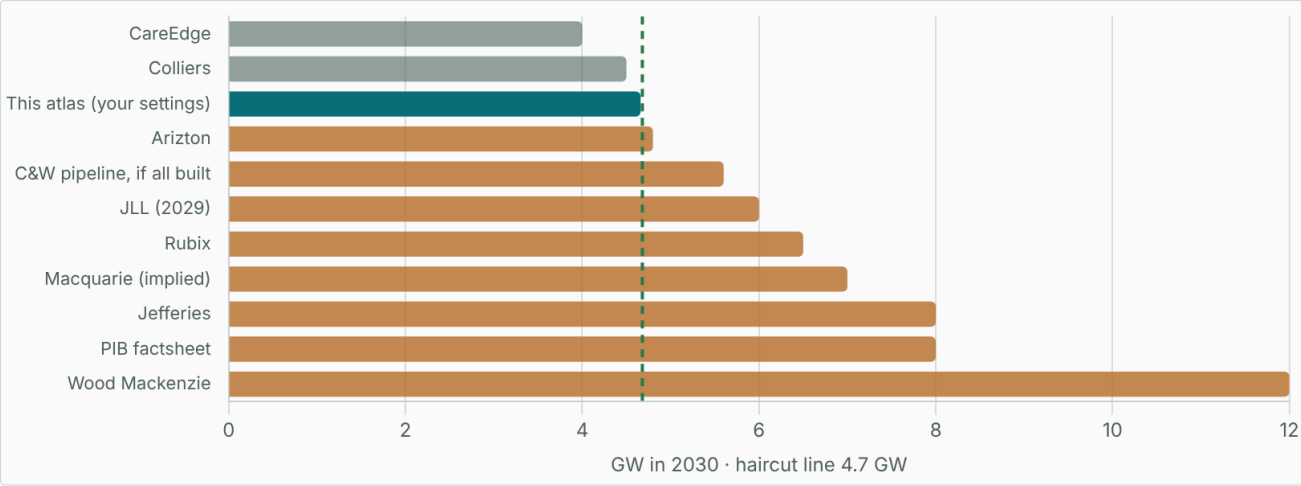


Figure 14.2 Our forecast against everyone else's. 2030 capacity, GW. Line = median published forecast × your backtest haircut. *Source: India Data Centre Atlas, chapter built on this chapter's data.*

14.2.1 Historical series

Table 14.1 Historical series

Year-end	Harmonised base series used for fitting (MW IT)	Reported points (source)	Label
2019	375	350 (JLL); about 400 (C&W)	E from R
2020	450	375 (MeitY, Parliament reply, Mar 2026); 520 (JM Financial via CEEW, broader definition)	E from R
2021	560	499 at H1 2021 (JLL); 824 at end-2021 (C&W, colocation stock, inconsistent with C&W's later series)	E from R
2022	720	722 (JLL)	R
2023	977	977 at H2 2023 (C&W; 258 MW built in 2023, +105% YoY); 950 in FY24 (ICRA)	R
2024	1,200	1,030 (JLL); about 1,150 in Dec 2024 (ICRA); 1,255 (CBRE)	E from R
H1 2025	–	1,280 (C&W); 1,263 in Apr 2025 (Colliers); 1,123 (JLL)	R
2025	1,611	1,789 − 178 (C&W, derived); about 1,500 (MeitY); about 1.7 GW (CBRE); 387 MW added in 2025 (Rubix)	E from R
H1 2026	1,789	C&W: 1,789 across 7 markets, 147 facilities, 33 providers, 14.9% vacancy; JLL: 1,637 MW (2026 Mid-Year Report, June 2026); PIB factsheet (14 Sep 2026): 1.57 GW as of Aug 2026	R
2026E	2,000	CBRE: about 2 GW by end-2026; C&W: Mumbai to exceed 1 GW by end-2026	E

Definitional cautions: - JLL counts "IT design power load" for colocation. - C&W covers the top 7 markets and appears to include hyperscale self-builds more fully. - MeitY's figure is national. - Wood Mackenzie's 2.2 GW for 2025 uses a broader base. - Some sources quote gross (facility) MW. - JLL's claim of a "60% CAGR since 2021" does not reconcile with its own 350 MW (2019) → 1,030 MW (2024) series, which is about 24% a year. We treat it as a CAGR of *supply additions*.

14.2.2 Model (a): CAGR, C(t) = $C_o(1+g)^t$

Fit [M]: OLS of ln C on year, 2019–2025, n = 7. - Slope = 0.2461, so g = e^0.2461 − 1 = **27.9%**. $R^2 \approx 0.99$. - Arithmetic check: the end-point CAGR over 2019–2025 is (1,611/375)^(1/6) − 1 = 27.5%.

Forward scenarios from C_o = 1,611 MW (end-2025) [M]:

Table 14.2 Model (a): CAGR, C(t) = $C_o(1+g)^t$

g	2027	2028	2030	2033	2035
Low 18%	2,243	2,647	3,686	6,055	8,431
Base 25%	2,517	3,146	4,916	9,602	15,003
High 32%	2,807	3,705	6,456	14,849	25,872

Constant CAGR is a poor tool beyond 2030: the base case would reach 15 GW by 2035 with no deceleration. We use it only as a cross-check up to 2030.

14.2.3 Model (b): logistic S-curve, C(t) = K / (1 + e^(−r(t − t_o)))

Choosing the ceiling K [E]: - India today has about 1.2 W of data-centre capacity per person: 1,789 MW across about 1.46 bn people. - CareEdge gives 1.2 MW per million internet users [R]. - The global average is about 12 W per person: the IEA puts global data-centre capacity at "almost 100 GW" [R], spread over about 8.2 bn people. - One secondary source cites about 51 MW per million users for the US and about 4 for China [R, secondary; the China figure looks too low and is not used]. - **Base K = 25 GW** is about 16 W per person on a 2040 population of about 1.6 bn, which is roughly the global average in the mid-2020s. It also sits just above the CEA's 17 GW data-centre demand for 2031-32, before PUE. - **Low K = 12 GW**: the constraints are power and connection queues. - **High K = 45 GW**: India becomes a regional AI export hub.

Solving for t_o: t_o is fixed so that the curve passes through C(2025) = 1,611:

$t_o = 2025 + \ln(K/1{,}611 - 1) / r$

Table 14.3 Model (b): logistic S-curve, C(t) = K / (1 + e^(−r(t − t_o)))

Parameter	Low	Base	High
K (MW)	12,000	25,000	45,000
r (/yr)	0.24	0.26	0.30
t_o	2032.8	2035.3	2036.0

r is anchored on the observed log-growth of 0.246 (because r ≈ g_log when C ≪ K).

Logistic output (MW) [M]:

Table 14.4 Model (b): logistic S-curve, C(t) = K / (1 + e^(−r(t − t_o)))

Year	Low	Base	High
2026	1,975	2,051	2,146
2027	2,403	2,596	2,850
2028	2,897	3,266	3,763
2029	3,457	4,077	4,936
2030	4,076	5,044	6,417
2031	4,745	6,171	8,249
2032	5,447	7,458	10,465
2033	6,166	8,884	13,062
2034	6,879	10,425	16,008
2035	7,568	12,030	19,215

Caveat: with seven points all in the early exponential phase, K and t_o are *not* statistically identified by the history. K is a judgement constrained by power, land and demand. This is why it gets a wide prior in the Monte Carlo (section 9).

14.2.4 Model (c): pipeline conversion, C(t+1) = C(t) + Σ P_stage(t − L) × c_stage

Current pipeline (C&W India Data Centre H1 2026 Update, press release 16 Sep 2026) [R]: 3,860 MW to 2030, of which 509 MW is under construction (UC) and 3,351 MW is planned. The pipeline "includes both colocation facilities and self-built infrastructure" by hyperscalers and needs US$29.9 bn of development capital.

Conversion evidence:

Table 14.5 Model (c): pipeline conversion, C(t+1) = C(t) + Σ P_stage(t − L) × c_stage

Source and date	Forecast	Outcome	Ratio	Label
JLL H1 2021 update	499 → 1,008 MW by 2023 (+509)	JLL's 2023 figure was not published in our sources. Interpolated between 722 (2022) and 1,030 (2024) it is about 875; C&W reports 977	Additions about 0.74; level about 0.87	E
JLL (about 2023)	+681 MW to 1,318 MW by end-2024	1,030 MW (JLL 2024)	Additions 0.58; level 0.78	R/M
C&W, Feb 2022	"over 1.0 GW incremental ... in various stages of construction, expected to start operations over the next 4–5 years"	About 1,070 MW added from end-2022 to H1 2026	About 1.0 within 4.5 years	R/M

Source and date	Forecast	Outcome	Ratio	Label
ICRA, Oct 2024	950 MW (FY24) → 2,000–2,100 MW by Mar 2027	On track on C&W's basis (1,789 at H1 2026); marginal on JLL's basis (1,637 MW at June 2026)	About 0.9–1.0 (provisional)	M
Arizton / ResearchAndMarkets	3.5 GW by 2030 (Feb 2025), revised to 4.8 GW (Jul 2025)	Forecast inflated by 37% in 5 months	n/a	R
CEEW / Systemiq	"Only about 50% of the USD ~95 bn worth of capital committed ... between 2019 and 2025 has a ..." (text truncated in source)	Treated as a conversion ceiling for announcements of about 50%	about 0.5	R (partial)

Parameters [E]:

Table 14.6 Model (c): pipeline conversion, C(t+1) = C(t) + Σ P_stage(t − L) × c_stage

Stage	Conversion c	Lag L
Under construction	Low 0.85 / Base 0.95 / High 1.0	1–2 years
Planned	Low 0.40 / Base 0.60 / High 0.80	2–4 years (base 3)
MoU / announced only	0.2–0.5	3–5 years

Typical construction lead times are 18–30 months per 20–50 MW block [E]. JLL notes that blocks have grown from 20 MW to 50+ MW and that prefabricated steel is shortening build times [R].

Output to 2030 [M]: - Base: 1,789 + 509 × 0.95 + 3,351 × 0.60 = **4,284 MW**. - Low: 1,789 + 433 + 1,340 = **3,562 MW**. - High: 1,789 + 509 + 2,681 = 4,979 MW, plus about 1,000 MW of gigawatt-campus phases outside C&W's 7 markets (Visakhapatnam, Jamnagar) [E] = **about 6,000 MW**.

Table 14.7 Model (c): pipeline conversion, C(t+1) = C(t) + Σ P_stage(t − L) × c_stage

Path (MW)	2026	2027	2028	2029	2030
Low	1,950	2,250	2,650	3,100	3,560
Base	1,990	2,450	3,050	3,650	4,280
High	2,050	2,800	3,800	4,900	6,000

14.2.5 Integrated capacity path

Construction of the path [M]: - 2026–2030: the mean of the pipeline model and the logistic model. - 2031–2035: the logistic model, with the base case trimmed by about 10% to stay consistent with the conversion haircut.

Table 14.8 Integrated capacity path

MW IT	2026	2027	2028	2029	2030	2031	2032	2033	2034	2035
Low	1,950	2,250	2,650	3,100	3,600	4,200	4,800	5,450	6,100	6,800
Base	**2,000**	**2,500**	**3,100**	**3,850**	**4,700**	**5,700**	**6,850**	**8,100**	**9,450**	**10,800**
High	2,100	2,850	3,800	4,950	6,400	8,200	10,300	12,800	15,600	18,800

Implied base CAGR: 2025–30 = 23.9%; 2030–35 = 18.1% [M].

14.2.6 Published forecasts compared

Table 14.9 Published forecasts compared

Forecaster	2026–27	2028–29	2030	2031–35	Label
ICRA	2,000–2,100 MW by Mar 2027; edge capacity 200–210 MW by 2027	–	–	3.0–3.5 GW pipeline over 7–10 years, ₹2.0–2.3 lakh crore	R
JLL	about 2.1 GW (2027); earlier 1,825 MW (2027)	6 GW (2029)	–	–	R
CBRE	about 2 GW (end-2026); about 500 MW of supply in 2026	–	–	–	R
C&W	–	–	1,789 + 3,860 pipeline = about 5.6 GW if fully delivered; US$29.9 bn	–	R
Colliers	–	–	more than 4,500 MW (top 7 cities)	–	R
CareEdge	–	–	about 4 GW (FY30), about ₹1.5 lakh crore	–	R
Arizton (via ResearchAndMarkets)	–	–	4.8 GW	–	R
Rubix	–	–	6.5 GW	–	R
Macquarie	2× by 2027	–	5× by 2030 (about 6.5–7.5 GW implied)	–	R/E
Jefferies	–	–	8 GW; US$30 bn capex	–	R
PIB factsheet (14 Sep 2026)	–	–	"nearly 8 GW"; US$70 bn ongoing plus US$90 bn announced	–	R
Wood Mackenzie	–	–	12 GW (from 2.2 GW in 2025); AI capacity 275 → 6,546 MW	–	R
CEA / Ministry of Power (demand, GW of power)	–	–	–	17 GW (CEA) / 13.56 GW (MoP, Mar 2026) by 2031-32; 26.3 GW additional AI data-centre load (MoS Power, Jul 2026)	R
This report (base)	2.0 / 2.5 GW	3.1 / 3.85 GW	**4.7 GW**	6.85 GW (2032); **10.8 GW (2035)**	M

Coverage gaps: we found no public, source-verifiable India MW forecasts from Morgan Stanley, Bernstein, Nomura, Goldman, CRISIL, NASSCOM or the IEA (country-level). They are omitted rather than guessed. The IEA's critical review does cite Nomura as estimating that data centres "currently consume 8.4 TWh" in India [R].

Haircut test [M]: the median of the nine 2030 point forecasts is about 6.5 GW. Our base of 4.7 GW is 0.72× that median, which sits inside the historical delivered-to-forecast range of 0.58–0.87.

14.3 Electricity (TWh) and share of the grid

Formulas: - IT energy (TWh) = MW × u × 8,760 ÷ 10^6 - Facility energy (TWh) = IT energy × PUE

Utilisation u [E]: - Colocation vacancy is 14.9% (C&W) [R], so about 85% of capacity is leased. - Leased load typically draws about 65% of contracted power while it ramps. - So u ≈ 0.55 in 2026 [E]. It rises to 0.65 by 2035 as AI training and inference, which run at high load factors, grow to more than 60% of new capacity.

PUE [E/R]: - tele.net reports that Indian data centres "typically run at a PUE of around 1.45–1.55" [R]. - CtrlS reports a company-wide design PUE of 1.35 [R]. - Takshashila assumes 1.6 for 2024 [R]. - SBI's sector report says "Industry leaders are achieving PUE values below 1.5" [R]. - Karnataka's 2026–31 policy pays incentives for PUE of 1.00–1.20 [R]. - Base path: 1.50 (2026) → 1.42 (2030) → 1.35 (2035).

Calibration check [M]: for 2024, 1,200 MW × 0.55 × 1.52 × 8.76 ≈ 8.8 TWh. That matches Nomura's 8.4 TWh [R]. For 2025 the model gives about 10 TWh, about 0.57% of demand, which agrees with CEEW's "about 0.5% of national electricity consumption" [R].

National denominator [R]: - Electrical energy requirement was 1,694 BU in FY2024-25 (CEA LT-NRAP). - The mid-term review of the 20th Electric Power Survey projects 3,365 BU and a 459 GW peak in 2035-36, a CAGR of 6.41%. - Model path: 1,694 × 1.0641^n [M]. - The NEP 2022-32 figure was 2,473.8 BU for 2031-32 [R]. - Downside risk: peak demand was flat in FY26 (about 245 GW against a projected 270 GW) [R].

Base case [M] (calendar-year data-centre energy against the matching fiscal year's requirement; year-end MW basis, so use the mid-year average to get figures about 8–10% lower):

Table 14.10 Electricity (TWh) and share of the grid

Year	MW	u	PUE	IT TWh	Facility TWh	India requirement (BU)	Data-centre share
2026	2,000	0.55	1.50	9.6	14.5	1,918	0.75%
2027	2,500	0.56	1.48	12.3	18.2	2,041	0.89%
2028	3,100	0.57	1.46	15.5	22.6	2,172	**1.04%**
2029	3,850	0.58	1.44	19.6	28.2	2,311	1.22%
2030	4,700	0.60	1.42	24.7	35.1	2,459	1.43%
2031	5,700	0.61	1.41	30.5	43.0	2,617	1.64%
2032	6,850	0.62	1.40	37.2	52.1	2,785	1.87%
2033	8,100	0.63	1.38	44.7	61.7	2,963	**2.08%**
2034	9,450	0.64	1.37	53.0	72.6	3,153	2.30%
2035	10,800	0.65	1.35	61.5	83.0	3,355	2.47%

Scenarios [M]:

Table 14.11 Electricity (TWh) and share of the grid

Scenario	2030 TWh	2030 share	2035 TWh	2035 share	Year share reaches 1% / 2% / 3%
Low (u 0.50→0.58; PUE 1.52→1.42)	24.7	1.0%	49.1	1.5%	2030 / not reached / not reached
Base	35.1	1.4%	83.0	2.5%	2028 / 2033 / after 2035
High (u 0.60→0.72; PUE 1.50→1.30)	51.0	2.1%	154.2	4.6%	2027 / 2030 / about 2032

Context on other estimates: - The IEA puts global data-centre use at 415 TWh (2024) → about 945 TWh (2030) [R]. Its 2026 update gives 485 → 950 TWh [R]. - India's base case would be about 3.7% of global data-centre electricity in 2030 [M]. - A trade-press figure of 191 TWh for Indian AI data centres (GreentechLead) has no stated year or method. Treat it as an unverified outlier.

State-level shares, base case [M; state denominators are E]. These are approximate state requirements, extrapolated about 6%/yr from FY25. Replace them with the state tables of the 20th EPS mid-term review before publishing.

Table 14.12 Electricity (TWh) and share of the grid

State (data-centre cities)	2030 DC TWh	≈ State BU 2030 [E]	Share	2035 DC TWh	≈ State BU 2035 [E]	Share
Maharashtra (Mumbai/Navi Mumbai, Pune)	17.9	250	7.2%	39.9	330	12.1%
Telangana (Hyderabad)	4.8	120	4.0%	12.2	160	7.6%
Tamil Nadu (Chennai)	3.5	175	2.0%	11.0	230	4.8%
Andhra Pradesh (Visakhapatnam)	1.9	105	1.8%	4.9	140	3.5%
Karnataka (Bengaluru, Mysuru)	1.7	120	1.4%	3.9	160	2.4%
Uttar Pradesh (Noida/Greater Noida, about 70% of NCR)	2.4	210	1.1%	5.5	280	2.0%

Maharashtra is the clear grid-stress state. Data centres would take about 1 in 8 units there by 2035 in the base case. This is consistent with the CEA's Maharashtra resource-adequacy study, which already shows 4.1% unserved energy in 2031-32 without additional capacity [R].

14.4 Carbon trajectory

Formula: CO_2 (Mt) = facility TWh × EF(t) [tCO_2/MWh] × (1 − clean(t))

Grid emission factor history (CEA CO_2 Baseline Database, weighted average including renewables) [R]:

Table 14.13 Carbon trajectory

Fiscal year	EF (tCO_2/MWh)
FY14	0.774
FY21	0.703 (COVID year)
FY23	0.716
FY24	0.727
FY25	**0.710** (v21: 1,234.19 Mt ÷ 1,739.06 BU)

Projections: - NEP 2022-32 [R]: 0.548 (2026-27) and 0.430 (2031-32). - The NEP numbers are kg/kWh_net. The draft plan gave 0.524 / 0.441. - **The NEP path is already about 0.15 off track.** FY25 came in at 0.710, so we do not use NEP as the base. - CEA LT-NRAP [R]: non-fossil sources reach 70% of installed capacity by 2035-36, against 52% in January 2026. Renewables supplied about 27% of generation in FY26.

EF paths (tCO_2/MWh) [E]:

Table 14.14 Carbon trajectory

Year	Slow	Base	Fast
2026	0.71	0.70	0.69
2030	0.69	0.62	0.55
2035	0.65	0.535	0.45

Between these years the paths are linear. Base runs at about −2.7%/yr; fast is NEP delayed by about 4 years.

Operators' clean-power position [R]: - Nxtra (Airtel): about 70% renewables for core data centres, targeted for 2027; net zero by 2031. Its 2024 wheeling deal for 140,208 MWh/yr cut about 99,547 tCO_2e. - CtrlS: 100% renewables and carbon neutral by 2030; renewables tie-up of up to 2 GW with NTPC Green. - AdaniConneX: "up to 100% renewable energy"; US$1.44 bn of sustainability-linked financing. - Yotta: exploring wind, solar and pumped storage for round-the-clock power. - Meta: about 1 GW of renewables procurement in India. - CB Insights notes that data centres make up 13.6% of India's open-access demand.

Fleet clean share [E]: - About 30% in 2026. - Low: 25% → 40%. Base: 30% → 43% (2030) → 60% (2035). High: 35% → 85%.

Emissions, base case [M]:

Table 14.15 Carbon trajectory

Year	Facility TWh	EF	Clean share	Net $MtCO_2$	Location-based Mt (clean = 0)
2026	14.5	0.70	0.30	7.1	10.1
2027	18.2	0.68	0.33	8.3	12.3
2028	22.6	0.66	0.37	9.4	14.9
2029	28.2	0.64	0.40	10.8	18.0
2030	35.1	0.62	0.43	12.4	21.7
2031	43.0	0.60	0.47	13.7	25.8
2032	52.1	0.58	0.50	15.1	30.2
2033	61.7	0.565	0.53	16.4	34.9
2034	72.6	0.55	0.57	17.2	39.9
2035	83.0	0.535	0.60	17.8	44.4

Range in 2035 [M]: - Low: about 3 Mt (low capacity, fast EF, 85% clean). - High: about 60 Mt (high capacity, slow EF, 40% clean).

The "clean share" is market-based (PPAs, open access, RECs) and rarely matched hour by hour. Location-based emissions are the honest upper bound.

14.5 Costs: build cost per MW and GPU-hour prices

14.5.1 Build cost

Data points:

Table 14.16 Build cost

Source	Figure	Implied ₹ crore/MW	Label
Turner & Townsend 2025 index	Mumbai US$6.64/W (₹589/W per IBEF); 51st of 52 markets. Tokyo US$15.2, Singapore US$14.5. Global +5.5% in 2025, +9.0% in 2024	about 58.4	R
ICRA (Oct 2024)	₹50,000–55,000 crore for about 1,100 MW (FY25–27)	about 47	R/M
ICRA (Mar 2025)	₹2.0–2.3 lakh crore for a 3.0–3.5 GW pipeline	about 62–70	R/M
C&W (Sep 2026)	US$29.9 bn for 3,860 MW	about 69 (US$7.75m/MW)	R/M
Jefferies	US$4–5m/MW	36–45	R
C&W 2026 APAC guide (via secondary source)	US$5.9–9.0m/MW across India's five major cities	52–80	R (secondary)
CareEdge	4 GW by FY30 needs about ₹1.5 lakh crore; costs up 50–70% recently	about 54	R/M
US AI vs standard (Quartz, citing industry)	US$20m+/MW vs US$10–12m/MW	AI premium ×1.7–2.0	R

Wright's-law diagnosis [M]: - ICRA's figure of about ₹47 crore/MW at about 1,000 MW cumulative, against C&W's about ₹69 crore/MW at 1,789 MW, implies b = −ln(69/47)/ln(1.789) ≈ −0.66. - That is *negative* learning: unit cost rose with cumulative volume. - The aggregate data do not support a cost-decline curve. We therefore split the model into parts:

cost(t) = C_{2025} × (X_t/X_{2025})^(−b) × (1+π)^(t−2025) × (1 + a × AI_share(t))

Parameters [E]: - C_{2025} = ₹58.4 crore/MW (Turner & Townsend shell and core plus M&E; excludes land and IT). - b = 0.10, so the learning rate is 1 − 2^(−0.10) = 6.7% per doubling. - π = 4%/yr inflation in electrical and cooling equipment. - a = 0.5 (AI facility = 1.5× air-cooled). - AI share of new builds: 40% (2026) → 60% (2030) → 70% (2035).

Cost path [M]:

Table 14.17 Build cost

₹ crore/MW	2026	2027	2028	2029	2030	2032	2035
Air-cooled facility	59.4	60.4	61.5	62.6	63.8	66.5	71.5
AI / liquid-cooled facility (×1.5)	89	91	92	94	96	100	107
Blended new build	71.3	74.0	76.9	79.8	82.9	87.8	96.5

Ranges [M]: - Low (b = 0.15, π = 2%, a = 0.3): air-cooled about ₹58 crore/MW in 2030. - High (b = 0.05, π = 7%, a = 0.8): about ₹80 crore/MW air-cooled and about ₹145 crore/MW AI in 2030.

Cumulative facility capex, base case (2026 onward; excludes land and IT/GPU hardware) [M]:

Table 14.18 Build cost

Year	ΔMW	Annual ₹ crore	Cumulative ₹ lakh crore	Cumulative US$ bn
2026	389	27,700	0.28	3.1
2027	500	37,000	0.65	7.3
2028	600	46,100	1.11	12.5
2029	750	59,900	1.71	19.2
2030	850	70,500	2.42	27.2
2031	1,000	85,300	3.27	36.7
2032	1,150	101,000	4.28	48.1
2033	1,250	113,000	5.41	60.8
2034	1,350	126,000	6.67	75.0
2035	1,350	130,300	7.97	89.6

Cross-check: our 2026–30 total of US$27 bn for 3.1 GW is close to C&W's US$29.9 bn for 3.86 GW [R].

14.5.2 GPU-hour prices

Data [R]: - Global H100 on-demand peaked at about US$8/hr in early 2024 (Neysa, citing Value Add VC and Thunder Compute). The median is now about US$2.50–3.50. - SemiAnalysis reports a rebound from US$1.70 (October 2025) to US$2.35 (early 2026) as supply tightened. - India commercial rates:

Table 14.19 GPU-hour prices

Provider	H100 rate
JarvisLabs	₹217.89/hr (US$2.69)
Cyfuture	₹219/hr (SXM)
Neysa	₹395/hr on demand; about ₹219/hr on a 36-month commitment
Hyperscalers in India (8-GPU instances)	about ₹850–1,050 per GPU-hour

- IndiaAI tender L1 rates: ₹115.85 (standard) and ₹150 (high-end) per GPU-hour.
- IndiaAI subsidised rates: about ₹65–67 average and about ₹92 for H100.
- IndiaAI has 38,231 GPUs onboarded across 14 providers (PIB, March 2026). At about 1.0–1.4 kW of IT per GPU [E], that is only about 40–55 MW, or about 3% of national capacity.

Fitted decline [M]: - From US$8 to about US$2.4 over about 2.25 years: ln(0.30)/2.25 = −0.535, i.e. **−41%/yr** for a single generation after launch. - The 2025–26 rebound shows this is not a stable trend. - We use −20%/yr as the base for an H100-class hour from 2026.

Table 14.20 GPU-hour prices

₹ per GPU-hour (India commercial on-demand)	2026	2027	2028	2029	2030
H100-class, base (−20%/yr)	220	176	141	113	90
H100-class, low decline (−10%/yr)	220	198	178	160	144
H100-class, high decline (−30%/yr)	220	154	108	75	53
Frontier class (B200/GB200 → successors) [E]	about 450–550	about 450–550	about 450–550	about 450–550	about 450–550

Frontier class: we found no verified Indian B200 list price. The estimate assumes a launch price of about 2× H100. The price per *frontier* hour stays roughly flat because each new generation resets it. The price per unit of compute falls about 30–40%/yr [E]. IndiaAI subsidies will keep a floor of about ₹65–92 for eligible users and will pull down commercial prices.

14.6 Water demand by city (MLD)

Formulas: - Water (L) = IT kWh × WUE_mix(t) - WUE_mix = s_evap × 1.8 + s_adiab × 0.5 + s_dry/liquid × 0.1 (L/kWh) - MLD = IT TWh × WUE × 2.7397

WUE values: - tele.net estimates Indian facilities at 1.5–2.5 L/kWh, against a global average of about 1.8–1.9 [R, estimate]. - Adiabatic cooling uses 0.2–0.8 L/kWh [R, trade guide]. - Karnataka's policy incentivises WUE of 0.7–1.3 L/kWh [R]. - Operator disclosure is thin. Down To Earth found no published WUE for Nxtra, AdaniConneX or CtrlS. STT GDC India reports only a 29% WUE reduction in FY24 [R]. - Calibration [M]: the IEA's "about 2 million litres per day for a 100 MW hyperscale facility" [R] works out to 0.83–1.4 L/kWh at u = 0.6–1.0. This is consistent with our WUE_mix of 1.14 in 2025.

Cooling mix in 2025 [E]: 55% evaporative, 25% adiabatic, 20% dry/closed-loop or direct liquid cooling.

Bass diffusion of the dry/liquid share:

$L(t) = L_0 + (m - L_0) \times F(t)$, with $F(t) = (1 - e^{-(p+q)t}) / (1 + (q/p)e^{-(p+q)t})$

- t = year − 2025; L_0 = 0.20; m = 0.85; **p = 0.03; q = 0.40** [E].
- **Analogue evidence [R]:**
- Dell'Oro: direct liquid cooling grows from US$1.1 bn (2024) to US$5.8 bn (2029).
- Dell'Oro: liquid-cooling revenue roughly doubles in 2025 to about US$3 bn and approaches US$7 bn by 2029.
- Dell'Oro: direct liquid cooling grew 156% YoY in Q2 2025.
- Turner & Townsend: 26–50% of projects use liquid-cooled environments for AI.
- That is a fast imitation-driven curve, so q ≫ p.
- The evaporative share of the non-dry remainder falls from 69% to 50% by 2035 [E].

Table 14.21 Water demand by city (MLD)

Year	Dry/liquid share	WUE_mix (L/kWh)	National on-site water, base (MLD)
2026	22%	1.09	28.7
2027	26%	1.03	34.5
2028	30%	0.95	40.4
2029	36%	0.87	46.5
2030	43%	0.77	52.4
2031	50%	0.68	56.3
2032	57%	0.58	59.1
2033	64%	0.50	60.6
2034	70%	0.43	61.7
2035	74%	0.37	62.3

Scenarios [M]: - Low: low capacity with fast diffusion. About 30 MLD (2030) and about 24 MLD (2035). - High: high capacity with slow diffusion (q = 0.25). About 95 MLD (2030) and about 210 MLD (2035).

Discrepancy flag: CEEW, citing Mordor Intelligence, gives about 150 bn litres/yr in 2025 (about 410 MLD) [R]. That is about 20 L per IT kWh, roughly 15× any plausible on-site WUE. It must include water embedded in power generation and possibly semiconductors, which CEEW lists as separate components. It is not comparable with on-site WUE. Mongabay reports CEEW finding data centres at about 0.02–0.03% of national water demand [R].

City water, base case [M]: - Intensity: 11.1 MLD per GW in 2030 (u = 0.60, WUE = 0.774); 5.8 MLD per GW in 2035 (u = 0.65, WUE = 0.370). - City supply figures are [R] via the subagent's sources.

Table 14.22 Water demand by city (MLD)

City	MW 2030	MLD 2030	MW 2035	MLD 2035	City supply (MLD)	Data-centre share 2030 / 2035	High-case 2035 MLD
Mumbai MMR	2,083	23.2	4,499	26.0	BMC about 4,100, against 4,600 demand (Oct 2026)	0.57% / 0.63%	about 85
Hyderabad	641	7.1	1,587	9.2	HMWSSB 545 MGD ≈ 2,478	0.29% / 0.37%	about 30
Chennai	468	5.2	1,426	8.2	Metrowater 1,245 (Apr 2025, incl. tankers)	0.42% / 0.66%	about 27
Delhi NCR	452	5.0	1,025	5.9	DJB about 1,002 MGD target ≈ 4,555 (Delhi only)	0.11% / 0.13%	about 19
Pune	311	3.5	692	4.0	PMC about 1,222 (older figure)	0.29% / 0.33%	about 13
Bengaluru	232	2.6	510	2.9	BWSSB about 1,935 supplied (2,225 capacity)	0.13% / 0.15%	about 10
Kolkata	80	0.9	147	0.85	KMC 1,654–2,045 (sources conflict)	about 0.05%	about 3
Visakhapatnam	250	2.8	637	3.7	GVMC 84.44 MGD = 383.86	0.73% / 0.96%	about 17 (4.4%)

Stress points:

1. **Navi Mumbai/Thane.** Most "Mumbai" capacity sits in Navi Mumbai and on the Thane–Belapur belt, which are served by NMMC, CIDCO and MIDC rather than the BMC. So the 0.6% share of BMC supply understates the local share.
2. **Chennai.** The city already runs a structural deficit, and data-centre water use rises about 60% between 2030 and 2035 in the base case.
3. **Visakhapatnam.** A single 1 GW AI campus on an evaporative design would need about 17 MLD, about 4–5% of the city's supply. Yovizag reports that the Google AI data centre is part of the reason demand is projected to reach 726 MLD by 2040 [R].
4. **Policy lever.** Mandating dry, adiabatic or recycled-water cooling in these three locations removes most of the high-case risk.

14.7 Where new capacity goes, city by city

Allocation formula:

$share_i = \alpha \times s_i(t-1) + (1 - \alpha) \times score_i^k / \Sigma_j score_j^k$

- $s_i(t-1)$ is the incumbency share.

- score is the chapter-10 composite.
- For 2026–2030, the allocation uses the *revealed* pipeline (C&W upcoming MW by city) with stage conversion. The softmax-plus-incumbency blend is used for 2030–2035.

Calibration evidence [R]: - Mumbai's share of operational capacity was 45% (JLL, H1 2021) and is 49.7% now (C&W: 890/1,789). Incumbency has *strengthened*. - The pipeline shares (C&W, on the 3,633 MW city-sum basis) are below.

Table 14.23 Where new capacity goes, city by city

City	Pipeline MW	Share of pipeline
Mumbai	1,726	47.5%
Hyderabad	710	19.5%
NCR	387	10.7%
Chennai	368	10.1%
Pune	258	7.1%
Bengaluru	100	2.8%
Kolkata	84	2.3%

- The city figures sum to 3,633 MW, against C&W's 3,860 MW total. The 227 MW gap is unexplained in the press materials.
- Hyderabad's pipeline share (19.5%) is about 2.3× its stock share (8.4%), so it is the main share-gainer.
- **Fitting k:** if scores alone had to reproduce Mumbai's 4.7× pipeline lead over Chennai, k would have to be about 25. That is implausible, and it shows that incumbency (α) does most of the work. Matching the pipeline with α ≈ 0.5 needs k ≈ 6–10. We use **k = 8, α = 0.5** [E].
- Illustrative scores used [E; substitute the chapter-10 values]:

Table 14.24 Where new capacity goes, city by city

City	Score
Mumbai	85
Chennai	80
Hyderabad	78
NCR	72
Visakhapatnam	70
Pune	68
Bengaluru	65
Kolkata	50

- The resulting softmax shares are: Mumbai 34.5%, Chennai 21.4%, Hyderabad 17.3%, NCR 9.1%, Visakhapatnam 7.3%, Pune 5.8%, Bengaluru 4.1%, Kolkata 0.5% [M].

City forecast, base case [M]: - 2030: 400 MW goes to sites outside the 7 markets (Visakhapatnam about 250, Jamnagar/other about 150) [E]. The remaining 2,511 MW is allocated by pipeline share. - 2035: the 6,100 MW increment is allocated with α = 0.5, k = 8.

Table 14.25 Where new capacity goes, city by city

City	H1 2026 [R]	2030 base	2035 base	2035 high (indicative)	Share of national, 2035 base
Mumbai MMR (incl. Navi Mumbai)	890	2,083	4,499	about 7,700	41.7%
Hyderabad (incl. periphery)	151	641	1,587	about 2,900	14.7%
Chennai	214	468	1,426	about 2,600	13.2%
Delhi NCR (incl. Noida/Greater Noida)	183	452	1,025	about 1,800	9.5%
Pune	133	311	692	about 1,200	6.4%
Visakhapatnam	0	250	637	about 1,500	5.9%
Bengaluru (incl. Mysuru spill-over)	162	232	510	about 850	4.7%
Kolkata	22	80	147	about 250	1.4%
Other (Jamnagar etc.)	–	150	248	about 0 (inside the figures above)	2.3%
Total	**1,789**	**about 4,670**	**about 10,770**	**about 18,800**	100%

Mumbai's share drifts from 50% to about 42%. Hyderabad and Chennai together approach 28%. Visakhapatnam becomes a top-6 market if its announced gigawatt campus delivers even one-third of its capacity.

14.8 Demand drivers (regression)

Driver series [R]: - UPI annual volume (NPCI via PIB): 24,162 crore transactions in FY26, +30% YoY, so FY25 = 18,586 crore [M]. - Earlier fiscal years used: FY20 1,250; FY21 2,233; FY22 4,597; FY23 8,375; FY24 13,116 crore. These are NPCI figures as widely reported, not re-verified in this round [E]. - Ericsson data per smartphone in India: 13 GB (2019), 14.6 GB (2020), 26 GB (2022, after a methodology revision), 32–36 GB (2024, revised), 37 GB (2025). Forecast: 70 GB by 2031 (11% CAGR). - Gartner India public cloud: US$13.7 bn (2025) → US$17.5 bn (2026), +28.1%. IaaS grows +40%. - Policy step-changes: RBI payment-data localisation (2018) and the DPDP Act (2023) with its 2025 Rules.

Regression 1 [M]: ln MW_t on ln UPI_FY(t+1), 2019–2025, n = 7. - ln MW = 2.462 + **0.471** × ln UPI. - **R^2 = 0.958**. - Elasticity: 1% more UPI volume goes with 0.47% more data-centre MW.

Regression 2 (cloud intensity) [M]: - 2025 intensity is 117.6 MW per US$ bn of public cloud spend. - With cloud spend growing 25%/yr to US$41.8 bn in 2030 [E], elasticity 1.0 gives 4.9 GW and elasticity 0.8 gives 3.9 GW.

Regression 3 (traffic) [E/M]: - Historical capacity growth (about 28%/yr) against mobile-traffic growth (about 20%/yr) implies an elasticity of about 1.3. - Ericsson's path implies traffic growth of about 14%/yr. - That gives about 18%/yr, or about 3.7 GW in 2030.

Driver-based forecasts for 2030 [M]:

Table 14.26 Demand drivers (regression)

Driver model	2030 MW
UPI (UPI +20%/yr)	2.5 GW
Traffic	3.7 GW
Cloud	3.9–4.9 GW
Mean	**about 3.7 GW**

Interpretation: consumer and enterprise drivers support about 3.7 GW by 2030. The supply-side base of 4.7 GW and the consultant median of 6.5 GW therefore assume roughly 1–3 GW of capacity driven by AI and hyperscalers' export demand. This is consistent with Wood Mackenzie's AI estimate (275 → 6,546 MW) and with JLL's 2026 Mid-Year Report, which found that pre-committed hyperscale capacity accounted for 82% of the 101 MW absorbed in H1 2026, "primarily driven by high-density AI computing workloads". That AI tranche is the most cyclical part of the forecast.

Caveats: - n = 7 with five degrees of freedom. - Every regressor trends upward, so multicollinearity makes multivariate coefficients meaningless. - There is spurious-regression risk with trending series, and first differences leave too few points. - The definitions of the dependent variable vary across sources. - Treat the elasticities as descriptive, not causal.

14.9 Integrated base-case table (2026–2035) [M]

Table 14.27 Integrated base-case table (2026–2035) [M]

Year	MW low / base / high	IT TWh	Facility TWh	% of India	Net $MtCO_2$	Water, national MLD	Water, Mumbai / Chennai / Hyderabad MLD	Cumulative capex ₹ lakh crore (US$ bn)	₹ crore/MW (air / AI)	₹/GPU-hr H100-class
2026	1,950 / 2,000 / 2,100	9.6	14.5	0.75%	7.1	28.7	14 / 3.5 / 2.0	0.28 (3.1)	59 / 89	220
2027	2,250 / 2,500 / 2,850	12.3	18.2	0.89%	8.3	34.5	16 / 4.0 / 3.0	0.65 (7.3)	60 / 91	176
2028	2,650 / 3,100 / 3,800	15.5	22.6	1.04%	9.4	40.4	19 / 4.4 / 4.3	1.11 (12.5)	62 / 92	141
2029	3,100 / 3,850 / 4,950	19.6	28.2	1.22%	10.8	46.5	21 / 4.8 / 5.7	1.71 (19.2)	63 / 94	113
2030	3,600 / 4,700 / 6,400	24.7	35.1	1.43%	12.4	52.4	23 / 5.2 / 7.1	2.42 (27.2)	64 / 96	90
2031	4,200 / 5,700 / 8,200	30.5	43.0	1.64%	13.7	56.3	24 / 6.0 / 7.6	3.27 (36.7)	65 / 98	72
2032	4,800 / 6,850 / 10,300	37.2	52.1	1.87%	15.1	59.1	25 / 6.8 / 8.1	4.28 (48.1)	67 / 100	58
2033	5,450 / 8,100 / 12,800	44.7	61.7	2.08%	16.4	60.6	25 / 7.4 / 8.5	5.41 (60.8)	68 / 102	46
2034	6,100 / 9,450 / 15,600	53.0	72.6	2.30%	17.2	61.7	26 / 7.8 / 8.9	6.67 (75.0)	70 / 104	37
2035	6,800 / 10,800 / 18,800	61.5	83.0	2.47%	17.8	62.3	26 / 8.2 / 9.2	7.97 (89.6)	72 / 107	30

Notes: - City water for intermediate years is linearly interpolated between the modelled years 2026, 2030 and 2035. - GPU prices after 2030 extrapolate the −20%/yr trend and are increasingly speculative. - Capex excludes land and IT/GPU hardware. Including servers would multiply AI capex several-fold.

14.10 Parameter table for the calculator, with Monte Carlo distributions

Table 14.28 Parameter table for the calculator, with Monte Carlo distributions

Parameter	Symbol	Low	Base	High	Suggested distribution	Basis
Starting capacity (end-2025)	C_o	1,500	1,611	1,700	Uniform	R (MeitY / C&W / CBRE)
CAGR	g	0.18	0.25	0.32	Triangular(0.15, 0.25, 0.33)	M (fitted 0.279)
Logistic ceiling	K (MW)	12,000	25,000	45,000	Lognormal, median 25 GW, 90% CI 12–45 GW	E
Logistic rate	r	0.24	0.26	0.30	Uniform(0.22, 0.32)	M
Logistic midpoint	t_o	2032.8	2035.3	2036.0	Derived: $t_o = 2025 + \ln(K/C_o - 1)/r$	M
Conversion, under construction	c_UC	0.85	0.95	1.00	Beta(19, 1.5)	R/M
Conversion, planned	c_P	0.40	0.60	0.80	Beta(6, 4)	R/M
Conversion, MoU / announced	c_A	0.2	0.35	0.5	Uniform	R (CEEW) / E
Lag	L (years)	2	3	4	Discrete {2: 0.4, 3: 0.45, 4: 0.15}	E
Forecast haircut (delivered/forecast)	h	0.58	0.70	0.87	Triangular	M (backtest)
Utilisation	u(2026→2035)	0.50→0.58	0.55→0.65	0.60→0.72	Triangular on the 2030 value (0.52, 0.60, 0.68)	E
PUE	PUE(2026→2035)	1.52→1.42	1.50→1.35	1.50→1.30	Triangular on 2030 (1.32, 1.42, 1.50)	R/E
Grid emission factor	EF (2026 / 2030 / 2035)	0.71 / 0.69 / 0.65	0.70 / 0.62 / 0.535	0.69 / 0.55 / 0.45	Triangular on 2030 (0.55, 0.62, 0.69)	R (CEA) / E

Parameter	Symbol	Low	Base	High	Suggested distribution	Basis
Clean share	c_clean (2026 / 2030 / 2035)	0.25 / 0.32 / 0.40	0.30 / 0.43 / 0.60	0.35 / 0.60 / 0.85	Triangular on 2030	R (operators) / E
Learning exponent	b	0.15	0.10	0.05	Uniform(0.05, 0.15); LR = 1 − 2^(−b)	E
Equipment inflation	π	0.02	0.04	0.07	Triangular	R (T&T 5.5–9%) / E
Base build cost	C_{2025} (₹ crore/MW)	50	58.4	69	Triangular	R (T&T, C&W)
AI premium	a	0.3	0.5	0.8	Uniform	R (US) / E
GPU price decline	d (%/yr)	10%	20%	30%	Triangular(0.10, 0.20, 0.30)	M
H100-class price, 2026	P_0 (₹/hr)	150	220	395	Triangular	R
Bass innovation	p	0.01	0.03	0.05	Uniform	E (Dell'Oro analogue)
Bass imitation	q	0.25	0.40	0.55	Uniform	E
Dry/liquid ceiling	m	0.70	0.85	0.95	Uniform	E
WUE, evaporative	L/kWh	1.5	1.8	2.5	Triangular	R
WUE, adiabatic	L/kWh	0.2	0.5	0.8	Uniform	R
WUE, dry / liquid	L/kWh	0.0	0.1	0.2	Uniform	E
Allocation exponent	k	4	8	12	Uniform	M
Incumbency weight	α	0.3	0.5	0.7	Uniform	M
National demand growth	g_grid	4.5%	6.41%	7.3%	Triangular	R (CEA EPS; 7.35% FY21–25)
FX rate	₹/US$	88	89	92	Uniform	E

Monte Carlo guidance: - Correlate K with u (ρ ≈ +0.5): both rise with AI. - Correlate EF with c_clean (ρ ≈ −0.3). - Correlate p/q with the AI share (ρ ≈ +0.6). - Draw capacity from a 50/50 mixture of the logistic and pipeline models to 2030, and the logistic model thereafter.

14.11 Risks and swing factors, ranked by impact on 2035 MW

1. **Power and grid connection (largest swing, ±5 GW).** - The CEA has set up a committee, which includes the Central Transmission Utility and Nasscom, because data-centre load could reach 17 GW by 2031-32 (PIB factsheet, 14 Sep 2026) [R]. - Wood Mackenzie's VP Research Rashika Gupta says "Land and capital are no longer the limiting factors for data centre developers in India" (DataCenterNews Asia, 29 Jul 2026) [R]. The binding constraint is access to electricity. - Maharashtra already shows unserved energy in the CEA's adequacy study. - Transformer lead times (128–144 weeks, per Wood Mackenzie Q2 2025) [R] put a floor on how fast projects can be delivered.
2. **The AI demand cycle (±3-6 GW).** - JLL's 2026 Mid-Year Report says pre-committed hyperscale capacity "accounted for 82% of total absorption" of 101 MW in H1 2026, "primarily driven by high-density AI computing workloads" [R], so demand depends on a few buyers. - The 2025–26 rebound in GPU rental prices points to tightness today, not a guarantee of future demand.
3. **GPU export controls and supply.** Restrictions under any US AI-diffusion framework, or limits on Blackwell allocation, would slow AI-density builds. They would also raise the GPU-hour price path to the low-decline case.
4. **Policy.** - Positive levers: infrastructure status, state incentives (for example Karnataka's PUE and WUE-linked incentives), tax holidays, and the localisation requirements of the DPDP Rules 2025. - Negative: tariff or open-access reversals.
5. **Water and local licence to operate.** Protests over a facility said to need 12 million litres a day (The Better India) [R] are a sign of what may follow. Evaporative designs in Chennai, Navi Mumbai and Visakhapatnam carry permitting risk.
6. **Forecast inflation in the source data.** Arizton raised its 2030 forecast by 37% in five months, and JLL's "60% CAGR" cannot be reconciled with its own series. Rebase the model every six months to C&W's or JLL's actuals.

14.12 Recommendations

- **For the calculator:**
- Default to the integrated base path.
- Expose K, c_P, u and EF as the four headline sliders; together they explain most of the output variance.
- Show the backtested haircut h explicitly next to any consultant forecast.
- **For planners and utilities:**
- Plan for the **high** case for connected load (about 14 GW in 2032), not the base case.
- Over-planning transmission is cheap; stranding a 1 GW campus is not.
- **For investors:**
- Underwrite at no more than 0.7× of announced pipeline.
- Model ₹ crore/MW as *rising* 2–4%/yr in nominal terms, not falling.
- **For state governments:**
- Make dry or recycled-water cooling a condition of approval in Chennai, Navi Mumbai and Visakhapatnam.
- Tie incentives to measured, published WUE and PUE. Most operators publish neither today.

14.13 Caveats

- Historical capacity series mix definitions (colocation vs total, IT vs gross, 7 cities vs national). The harmonised series is an estimate.
- Several inputs come from trade press, aggregators or vendor blogs and are labelled accordingly:

- GPU prices.
- C&W's 2026 APAC cost guide (seen via a secondary source).
- Some city water figures.
- State electricity denominators and the earlier UPI years were not re-verified in this round.
- We did not find public India MW forecasts from several banks named in the brief; we did not invent them.
- The NEP emission-factor path is already off track, so carbon outcomes depend heavily on how fast the grid actually decarbonises.

Sources cited in this chapter

Organisations and publications named in this chapter's text, in order of how often they are cited. Full details are in the bibliography.

- JLL, Global and India Data Center Outlooks
- Central Electricity Authority (CEA), CO_2 Baseline Database and grid reports
- Ministry of Electronics and Information Technology (MeitY), Data Centre Policy draft and IndiaAI
- Council on Energy, Environment and Water (CEEW) studies
- Press Information Bureau releases
- Wood Mackenzie India data centre outlook
- International Energy Agency (IEA), Energy and AI (2025) and updates
- CBRE, India data centre market reports
- Turner & Townsend, Data Centre Cost Index 2025
- Ministry of Power, orders on ISTS waiver and green open access
- Dell'Oro Group, liquid-cooling market data
- CtrlS disclosures
- Nxtra by Airtel disclosures
- AdaniConneX disclosures
- Cushman & Wakefield, India data centre market updates (H1 2026)
- Bangalore Water Supply and Sewerage Board
- Google environmental reports, efficiency data and Gemini inference paper
- Meta sustainability reports and data index
- STT GDC India ESG report
- Yotta Data Services disclosures
- State Bank of India sector report on data centres (April 2026)
- Down To Earth reporting
- Mordor Intelligence market estimates

PART IV · POLICY, PLACE AND THE FUTURE

Chapter 15
Recommendations: design, siting and policy for India's next data centres

WHAT THIS CHAPTER COVERS

This chapter brings the research together into recommendations. The first part covers design and siting: a framework for choosing sites, siting rules for each kind of data centre, a rating of eleven Indian hubs, and recommendations for electrical design, cooling by climate, heat reuse, building resilience, water and clean-power procurement, together with performance targets. The second part covers policy: a review of India's current framework, a comparison with other countries, eighteen numbered recommendations with lead agencies and timelines, and guidance on how policy should differ between states.

KEY TAKEAWAYS

1. The Central Electricity Authority projects about 26.3 gigawatts of additional data-centre demand by 2031–32 across eleven states, and about 17 gigawatts of this has already applied for connection to state grids.
2. About 80% of India's operating capacity is in Mumbai, Chennai and Delhi-NCR, and no national data-centre policy is in force; the 2020 draft was never notified.
3. Cloud and colocation facilities should stay near users and cable landings and achieve sustainability through design, while AI training campuses should go where renewable power, land, water and low hazard coincide.
4. AI campuses should be designed for an annual PUE of about 1.15 to 1.22, an on-site water usage effectiveness of 0.2 litres per kilowatt-hour or less, at least 80% hourly clean power by 2030, and direct-to-chip liquid cooling for racks of 200 kilowatts and more.
5. The eighteen policy recommendations start with a national policy with mandatory, assured disclosure, water and clean-power criteria in public tenders, grid-connection rules for large loads, a statutory water hierarchy, and outcome-based incentives with sunset clauses.

How this chapter connects to the others. This chapter draws on every other chapter of the report.

India should build its next wave of data centres in places where power, water and connectivity have been planned together. Grid capacity and water should be allocated against verified performance: hourly clean power, low freshwater use and grid flexibility. Today's patchwork of state capex subsidies should give way to one notified national framework. Without that change, the roughly 26 GW of new demand the Central Electricity Authority projects will land on the grid and aquifers that are already most stressed.

15.1 Key Findings

1. **There is still no national policy in force.** MeitY's Draft Data Centre Policy 2020 was never notified. MeitY reopened consultations in August 2025. Computer Weekly, citing The Economic Times, reported a plan to replace "more than 30 permissions" with a single window and to create four Data Centre Economic Zones, because Mumbai, Chennai and Delhi-NCR hold "nearly 80%" of capacity. CBRE's October 2026 report puts their share higher, at "more than 80% of operational capacity". [Reported] The Union Budget 2026–27 added a tax holiday to 2047 for foreign companies selling global cloud services from MeitY-notified "specified data centres" owned by Indian companies, plus a 15% safe-harbour margin (PIB). [Reported] As of mid-2026, secondary trackers reported that the specified-data-centre scheme itself had not been issued. [Reported, secondary]
2. **Grid demand is the binding constraint.** The Ministry of Power told the Rajya Sabha that, as of 14 May 2026, CEA projects 26.3 GW of additional data-centre demand by 2031–32 across 11 states. About 17 GW of that has already filed for connectivity with State Transmission Utilities and 9.3 GW has not (Telborg briefing citing PIB, August 2026). [Reported, via secondary briefing] That is many times today's operational base of nearly 1.8 GW (Cushman & Wakefield H1 2026). CBRE puts the base at ~1.75 GW, and MeitY told the Rajya Sabha about 1,575 MW. CEA's draft Technical Standards for Connectivity to the Grid Regulations, 2026 cover bulk consumers at 33 kV and above. They would impose ride-through and reactive-power duties on inverter-interfaced bulk consumers of 50 MW or more. [Reported] They do not yet impose flexibility or cost-allocation obligations specific to data centres.
3. **The seismic baseline moved and then moved back.** BIS published IS 1893:2025 in November 2025. It created a new Zone VI across the Himalayan arc (zone factor 0.75 per a 2026 comparative study) and placed 61% of land in moderate-to-high hazard. BIS then withdrew it by a 3 March 2026 gazette notification and reverted to the 2016 code, after MoHUA objected over consultation and cost increases of 10–15%. [Reported] Designers face regulatory uncertainty, so the prudent course is to design critical facilities to the more conservative 2025 hazard values even where the 2016 code governs legally.
4. **ISTS-waiver economics for clean power are now graded, not free.** For projects commissioned after June 2025, 25% of ISTS charges applied in the first year, stepping up toward full charges by 2028. The Ministry of Power extended a 100% waiver for co-located BESS commissioned by June 2028 and for pumped storage awarded by June 2028 (Mercom, JMK). Later orders offered graded 75/50/25% waivers for projects with firm PPAs as on 31 March 2026, plus relief for transmission-delayed projects (Saur Energy). [Reported] Interstate RE now costs an estimated ₹1–2/kWh more for transmission than it did under the waiver (JMK estimate). [Reported estimate] That tilts the economics toward in-state group captive and co-located storage.
5. **Green open access rules diverge by state, and the divergence is getting worse.** The central GEOA Rules 2022 set a 100 kW eligibility threshold and minimum monthly banking of 30% (IEEFA). [Reported] Tamil Nadu's GEOA Regulations 2025, effective 1 October 2025, imposed an 8% in-kind banking charge, barred banking for third-party open access and tightened time-of-day banking, which cut group-captive savings by an estimated ~15% (Neufin). [Reported] Gujarat's GERC has charged ₹1.50/unit for banking and proposed ₹1/unit in a 2026 draft (Saur Energy). [Reported] For a 24×7 load, these rules matter more than headline subsidies.

6. **Water regulation exists but is not data-centre-specific.** CGWA's 2020 guidelines bar groundwater NOCs for new industries in over-exploited units, except MSMEs, staff drinking water and infrastructure. The 2023 amendment requires commercial users to file online water audits (Haryana WRA copy of the guidelines; TeamLease). [Reported] The 2025 Dynamic Ground Water Assessment classes 730 of 6,762 assessment units (10.8%) as over-exploited (PIB, Dec 2025). [Reported] No national rule requires data centres to disclose WUE, and the IndiaAI tender sets PUE <1.35 with no WUE criterion. Per our earlier part on AI vs conventional data centres, only STT GDC India publishes audited WUE (0.58).
7. **States are starting to tie incentives to performance, unevenly.** Karnataka's Sustainable DC Policy 2026–31 (₹1,300.45 crore; 1 GW by 2031) requires PUE and WUE reporting for each plant. It reimburses 100% of treated-water charges (capped at ₹8 lakh a year) and plans resource heat-mapping and a 500 MW Hoskote park with 60 MLD of secondary-treated water (The News Minute, Medianama). [Reported] Maharashtra went the other way. Its 2026 revision cut the green-power requirement from 100% to 51%, widened eligibility to 20 projects statewide and added ₹1/unit for 10–20 years and a 4% interest subvention (Whalesbook). [Reported] That is a live example of the race to the bottom on incentives.
8. **The market is already moving toward the siting logic recommended here.** Google broke ground on 28 April 2026 on a ~1 GW, US$15 billion (2026–2030) AI hub at Vizag with AdaniConneX and Airtel, including a new cable landing station and three subsea cables (Google Cloud press release). [Reported] Meta will lease a 168 MW Reliance facility in Jamnagar "powered by renewable energy and cooled with desalinated seawater", with Meta covering "the full cost of the energy and water" (Meta newsroom, June 2026). [Reported] Reliance targets more than 120 MW of AI capacity online in H2 2026 (TechCrunch). [Reported]
9. **International peers have converged on four tools: disclosure, hard efficiency floors, grid conditions and water pricing.** Germany's EnEfG requires PUE ≤1.2 for data centres opening from 1 July 2026, energy-reuse factors of 10/15/20% and renewable power. The EU database (Delegated Regulation 2024/1364) collects PUE, WUE, ERF and REF from sites of 500 kW or more. Singapore's DC-CFA2 allocates ≥200 MW only to designs with PUE ≤1.25 and ≥50% green pathways. Ireland requires dispatchable on-site or nearby generation or storage equal to import capacity, plus 80% additional renewables. Texas SB 6 mandates curtailability for loads of 75 MW or more. Johor charges RM5.33/m³ for water and paused water-cooled expansions. China requires PUE ≤1.25 for new large sites (≤1.2 at hub nodes). [All Reported]

15.2 Part 1 — Design and Site Recommendations

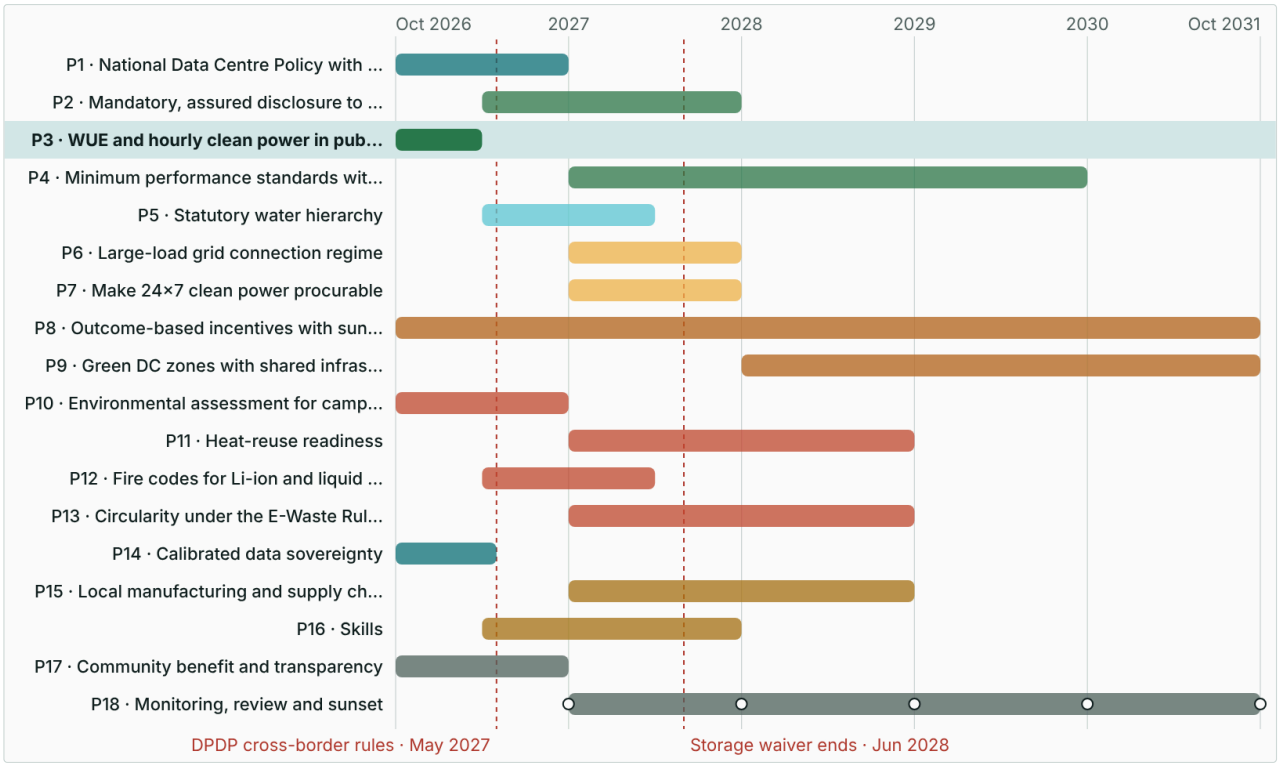


Figure 15.1 Eighteen recommendations, on a five-year roadmap. Each bar runs from start to target date (Oct 2026 → Oct 2031). Filter by theme or lead agency; tap a bar for the detail. *Source: India Data Centre Atlas, chapter built on this chapter's data.*

15.2.1 Site-selection framework (recommended weighting, estimated)

Table 15.1 Site-selection framework (recommended weighting, estimated)

Criterion	Weight: Cloud/Colo	Weight: AI training	What "good" looks like in India	Data to check
Firm power and grid headroom	25%	30%	A 220/400 kV substation within ~10–20 km with confirmed STU/CTU bay allocation; a dual independent feed; the queue position known	STU connectivity queue, CEA transmission plans, GNA applications
Clean-power access	10%	20%	In-state RE-rich DISCOM with workable group-captive/GEOA banking; co-located BESS eligible for the ISTS waiver to June 2028	SERC GEOA regs, banking charges, ToD rules
Water security	15%	15%	Not in a CGWB over-exploited or critical unit; an STP/TSE contract or seawater/desal option	CGWB 2025 assessment, municipal TSE availability
Hazard (seismic, cyclone, flood, heat)	10%	10%	Zone II/III; outside 1-in-100-year flood plain + freeboard; outside cyclone storm-surge zone	IS 1893 (2016 legal; 2025 values as stress test), NDMA, IMD cyclone atlas

Criterion	Weight: Cloud/Colo	Weight: AI training	What "good" looks like in India	Data to check
Latency and fibre	20%	5%	<5 ms to a top-8 metro for cloud; ≥3 diverse long-haul routes; CLS within ~200 km	Carrier maps, CLS list
Land and zoning	10%	10%	Industrial/IT zoned land, ≥20–50 acres for campuses, DC-park status	State IT/industrial land banks
Policy and incentives	5%	5%	Notified, time-bound policy with performance conditions	State DC policies
Skills and supply chain	5%	5%	Within 2–3 hours of an engineering talent pool and logistics hub	

Rationale: AI training is latency-tolerant: model training traffic is bulk and asynchronous. So it should trade latency for power, clean energy and water. Cloud availability zones and colocation serve interactive users and interconnection. They must stay near demand and cable landings, so their sustainability has to come from design rather than location.

15.2.2 Siting logic by archetype

Table 15.2 Siting logic by archetype

Archetype	Typical size	Recommended locations	Siting rule
Hyperscale cloud region/AZ	50–300 MW per campus	Navi Mumbai, Pune, Chennai, Hyderabad, NCR (Noida/Greater Noida), Bengaluru periphery	AZs 30–100 km apart on separate substations and flood basins; within latency of a cable landing station (CLS)
Retail/wholesale colocation	5–100 MW	Mumbai, Chennai, NCR, Hyderabad, Kolkata, Bengaluru	Carrier density and interconnection come first; choose existing industrial land with TSE
AI training campus	100 MW–1 GW+	Vizag, Jamnagar/Kutch periphery, Kurnool/Anantapur (AP), Tumakuru/Chitradurga corridor (Karnataka), Telangana interior, western Rajasthan with transmission, Odisha (Bhubaneswar/Paradip)	Co-locate with RE + BESS at a 400 kV pooling station; desal or TSE water; Zone II/III; hourly CFE
AI inference / edge	0.5–20 MW	Tier-2 cities: Lucknow, Jaipur, Ahmedabad, Indore, Nagpur, Coimbatore, Kochi, Bhubaneswar, Guwahati, Patna	Near users and 5G aggregation; air-cooled/zero-water; standardised prefab
Sovereign/government cloud	10–100 MW	NIC/State DCs; Naya Raipur AI DC park; Bhubaneswar; Tier III+ sites in two seismic/flood-independent regions	Geo-redundant pairs at least 500 km apart; Indian-owned operator (specified DC); highest disclosure standard

15.2.3 Hub-by-hub suitability (scores are our estimates; 5 = best)

Table 15.3 Hub-by-hub suitability (scores are our estimates; 5 = best)

Hub	Power headroom	Clean-power access	Water	Hazard	Latency/CLS	Land	Best-fit archetype	Key design response
Mumbai/Navi Mumbai	3	3	3	2 (monsoon flood, Zone III)	5	1	Colo, cloud AZ	Multi-storey; plinth above design flood level; closed-loop liquid + dry coolers; TSE only; 51%+ RE via MSEDCL GEOA
Pune	3	3	3	4	4	3	Cloud AZ, colo overflow	Pair with Navi Mumbai as a second AZ; adiabatic-assisted dry coolers
Chennai	3	4 (TN wind/solar)	1 (2019 crisis)	2 (cyclone, flood)	5	3	Colo, cloud, CLS-adjacent	Desal/TSE only; raised plinth; cyclone-rated envelope; note TN policy lapsed 31 Mar 2026 and TN banking restrictions
Hyderabad	4	3	3	5 (Zone II, inland)	3	4	Cloud AZ, AI inference, mid-size training	Dry coolers with trim chillers; Telangana needs a policy update beyond 2016
Bengaluru	2 (grid-constrained)	4 (Pavagada solar, wind)	1	5	3	2	Colo, enterprise, inference	Hoskote TSE park; best free-cooling climate of the metros; zero groundwater; PUE ≤1.20 bonus under state policy
NCR/Noida	3	2	1 (over-exploited aquifers)	3 (Zone IV Delhi; 45 °C+ heat)	4	3	Cloud AZ, colo, government	Air-cooled/hybrid chillers sized for 47 °C+ dry bulb; no groundwater; UP DC Policy 2026
Kolkata	3	2	3	2 (cyclone, flood)	3 (limited CLS)	3	Eastern colo, edge	Raised plinth; demand-side diversity for the east
Vizag	4	4 (AP RE + ISTS)	3 (sea/desal)	2 (cyclone, e.g., Hudhud)	4 (new CLS, three cables)	4	AI training, hyperscale	Cyclone-rated structure; desal or closed-loop; seawater heat rejection only with marine ecology assessment
Jamnagar/Gujarat	4	5	3 (desal)	3 (cyclone; Kutch is Zone V)	3	5	AI training	Desalinated seawater; dry coolers in a hot-arid climate; avoid Kutch Zone V for critical halls
GIFT City	3	4	3	4	3	3	Fintech colo, IFSC workloads	Tie into GIFT's district-cooling network for absorption/heat reuse; SEZ/IFSC benefits
Tier-2 (Lucknow, Bhubaneswar, Nagpur, Coimbatore, Indore, Naya Raipur)	3–4	3	3–4	3–5	2–3	5	Edge, inference, state/sovereign	Prefab modules; air-cooled zero-water; anchor on government demand

Interpretation: No metro scores well on water. Water performance in Mumbai, Chennai, Bengaluru and NCR therefore has to be designed in through closed-loop cooling and TSE, because site selection cannot deliver it. Vizag and Jamnagar score best for AI because RE, land and seawater coincide, but both carry cyclone risk, which must be engineered for.

15.2.4 Recommendations by archetype

R1. Hyperscale cloud. Build paired AZ campuses on separate substations and flood basins. Use 33 kV medium-voltage distribution into the campus with N+1 transformers and block-redundant (distributed-redundant or catcher) UPS rather than full 2N. That captures most of 2N's availability at lower capex and loss. Use Li-ion UPS, closed-loop air plus liquid hybrid halls (30–50% liquid-ready) and dry coolers with adiabatic assist only above ~35 °C wet-bulb-equivalent hours. Procure in-state group-captive RE plus co-located BESS.

R2. Colocation. Design for mixed tenancy: 10–20 kW air racks today, with capacity reserved for 50–130 kW liquid pods. Provide facility-water (FWS) take-off points at W32–W40 on every floor. In land-scarce metros, go multi-storey (4–8 floors) with dedicated heavy-load slabs (≥20–25 kN/m² in liquid-cooled rooms) and vertical riser space for coolant distribution. Offer tenants hourly clean-power and water reporting as a product.

R3. AI training campus (100 MW–1 GW+). Treat the campus as a power-plant-plus-load project: - Connect at 220/400 kV with an on-site substation. - Size co-located BESS (≥2–4 h at 30–50% of IT load) to absorb training-job power swings, provide grid-forming support and firm RE. Our Part 14 LP found that 161 MW solar + 86 MW wind + 379 MWh BESS served a 50 MW-IT Karnataka DC at ₹5.39/kWh vs ₹7.00/kWh grid. - Use 100% direct-to-chip liquid cooling for GPU racks (GB200/GB300-class at 120–142 kW, scalable to 200–600 kW) with W40–W45 facility water, so dry coolers can reject heat without evaporation for almost all hours in most Indian climates. - Keep a small evaporative trim for peak days only, sized from peak-day modelling (April–May).

R4. AI inference/edge. Standardise prefabricated 1–5 MW modules with air-cooled DX or pumped-refrigerant economisers and rear-door heat exchangers (RDHx) for 30–60 kW inference racks. Run zero water. Use HVO-ready gensets or BESS for backup, and site at telecom aggregation nodes.

R5. Sovereign/government cloud. Require Tier III (or Uptime/TIA-942 Rated-3) minimum, geo-redundancy across seismic and flood regions, and Indian ownership/operation, consistent with the Budget's "specified data centre" construct. Set the strictest disclosure: audited PUE, WUE and CUE published quarterly. Government should lead on transparency rather than lag.

15.2.5 Electrical design

- **Redundancy:** Use 2N for colocation where SLAs demand it. For hyperscale and AI, prefer N+1/block-redundant with software resilience. AI training clusters can tolerate partial-hall loss with checkpointing, so 2N everywhere is over-engineering. [Estimated judgement]
- **Medium voltage:** 33 kV internal rings for campuses above 50 MW; 11 kV for smaller sites. Plan from day one for 800 V DC / ±400 V DC busways for future 200–600 kW racks.
- **UPS and BESS:** Li-ion (LFP preferred for thermal stability) UPS; a separate grid-scale BESS that can deliver frequency response, peak shaving and RE firming. The 100% ISTS waiver for co-located BESS commissioned by June 2028 is a time-limited window to capture. [Reported basis]
- **Grid-forming:** Specify grid-forming inverters on BESS. CEA's draft 2026 connectivity regulations introduce grid-forming provisions and ride-through for inverter-interfaced bulk consumers of 50 MW or more. Large AI loads should meet these voluntarily now. [Reported basis]
- **Backup generation:** Diesel gensets remain the default. Specify HVO-compatible engines and test on HVO where supply exists. For campuses above 200 MW, evaluate gas engines on the city-gas grid only as backup or flexibility, never as baseload. Ireland's CRU explicitly states that "islanded", mainly fossil-powered data centres are not in line with national policy.
- **Power-swing management:** AI training produces sharp, synchronised load steps. Require power-smoothing (rack-level capacitors/BESS, GPU power capping) so the site's ramp rate stays within DISCOM/STU limits.

15.2.6 Cooling design by climate

Table 15.4 Cooling design by climate

Climate (examples)	Design dry-bulb / wet-bulb (approx., estimated, 0.4% design)	Recommended primary heat rejection	Water strategy
Mild plateau (Bengaluru, Pune, Hyderabad)	~35–40 °C DB / ~23–25 °C WB	Liquid-to-dry coolers at W32–W40; air halls with indirect evaporative only on peak hours	TSE for any evaporation; target WUE <0.2
Hot-humid coastal (Mumbai, Chennai, Kolkata, Vizag)	~35–38 °C DB / ~28–29 °C WB	High-temperature liquid (W40–W45) + dry coolers; magnetic-bearing chillers as trim; free-cooling hours are limited at low supply temperatures and much greater at W40+	Desal/TSE; seawater once-through only after a marine EIA; avoid towers on potable water
Hot-arid (NCR, Jamnagar, Rajasthan, Ahmedabad/GIFT)	~44–47 °C DB / ~26–28 °C WB	Dry coolers oversized for 47 °C+; adiabatic assist only for short peaks; dust filtration	Zero groundwater; desal in Gujarat; size storage for April–June peaks

Key points: - **Raise the supply temperature.** The ASHRAE TC 9.9 liquid classes W32–W45 let dry coolers reject heat at ambient temperatures in the high 30s °C, which is what makes zero-evaporation feasible in India's climate. Air halls should run at the upper end of the ASHRAE recommended envelope (27 °C inlet) with containment. - **Zero-water designs are proven at scale.** Microsoft says its design, used for all new data centres since August 2024, "will avoid the need for more than 125 million liters of water per year per datacenter", taking WUE to "near zero". Its FY2024 global average withdrawal WUE was 0.30 L/kWh, a 39% improvement on 2021. [Reported] The trade-off is a modest energy penalty, which Microsoft calls a "nominal increase". That penalty matters in India because MoEFCC norms allow coal plants 3.0–3.5 L/kWh, and CSE found half of surveyed freshwater-based plants exceeding the caps. A dry design that adds 0.03–0.05 to PUE can add 0.1–0.15 L/kWh of off-site water on a coal-heavy grid. [Estimated] Zero on-site water is therefore only net-positive when paired with clean power. - **Seawater cooling** works for Vizag, Chennai and Jamnagar, preferably indirectly through desalinated make-up or a closed loop with seawater heat exchangers. Discharge temperature and intake ecology need EIA-grade study. - **Immersion** suits niche high-density and edge sites. Direct-to-chip has the stronger vendor ecosystem for GB-class systems.

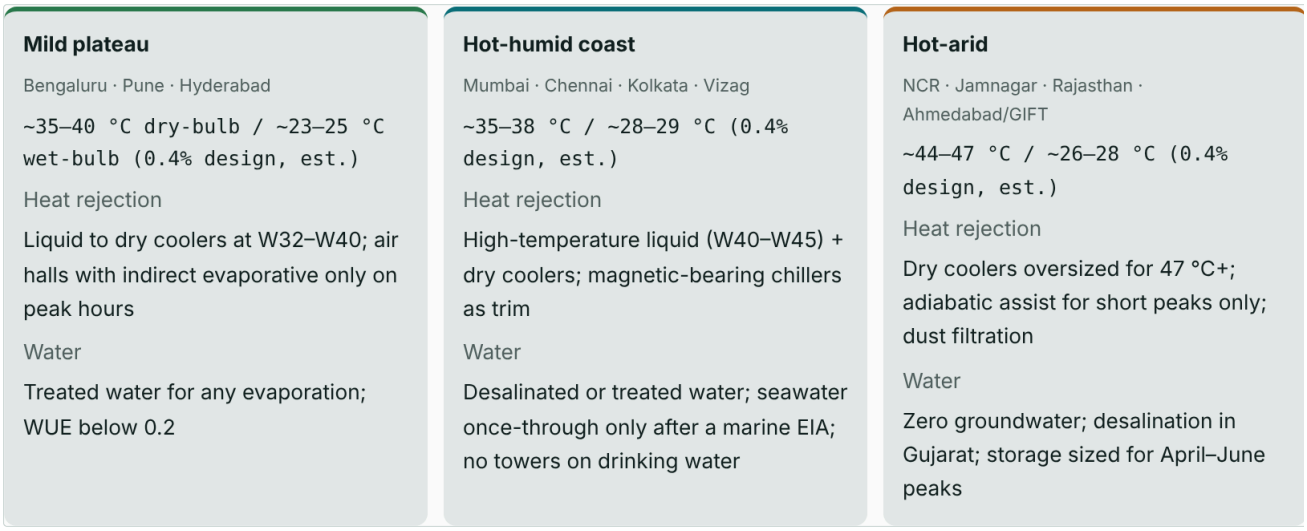


Figure 15.2 Cooling by climate. Raise the water temperature so dry coolers can reject heat without evaporation; keep evaporation for peak hours, on treated or desalinated water.
Source: India Data Centre Atlas, chapter built on this chapter's data.

15.2.7 Heat reuse in India

Space heating is mostly irrelevant in India, so heat reuse must target cooling and process heat: 1. **Absorption chilling.** At 45–60 °C return temperatures, single-effect absorption output is marginal. Pairing it with heat pumps to lift heat to 80–90 °C enables chilled water for adjacent offices or cold chains. [Estimated feasibility] 2. **District cooling at GIFT City**, which already operates a district-cooling network. It is the natural pilot for feeding DC heat-pump output into absorption chillers. 3. **Industrial process heat:** food processing, dairy pasteurisation, laundries, pharmaceutical drying and desalination pre-heat (thermal-assisted desalination at Jamnagar/Vizag). 4. **Agriculture:** grain and spice drying and aquaculture in tier-2 parks.

Germany's ERF quotas (10% rising to 20%) are not transferable to India. A practical Indian target is an ERF ≥5% for campuses co-located in industrial parks with an identified offtaker, plus a heat-reuse feasibility study for all sites of 10 MW or more. [Estimated recommendation]

15.2.8 Building, hazard and resilience design

- **Seismic:** Design critical facilities for the more conservative of IS 1893:2016 and the withdrawn IS 1893:2025 hazard values, with importance factor 1.5. Give non-structural elements, including raised floors, overhead busways, liquid manifolds and battery racks, seismic restraint. Avoid Zone V (Kutch, the Northeast, the Himalaya) for primary campuses.
- **Flood:** Set the finished floor level and all MV/UPS/battery rooms ≥1 m above the higher of the 1-in-100-year flood level and the highest recorded flood (Mumbai 2005; Chennai 2015/2023). [Estimated recommendation] Do not put critical plant in basements, and put fuel and water tanks on elevated plinths. NDMA urban flood guidelines and the local DCR are the floor, not the target.
- **Cyclone (east coast, Gujarat):** Design the envelope, louvres, rooftop dry coolers and solar to the IS 875 (Part 3) wind speeds for the site, plus an uplift margin. Provide storm-surge setbacks and 72–96 h on-site fuel to cover post-landfall grid outages.
- **Heat:** Size all heat-rejection plant for current extremes plus a climate allowance (+2 °C to the 0.4% design dry-bulb). [Estimated]
- **Floors:** Use slab-on-grade with overhead distribution for liquid-cooled AI halls (heavier racks, easier leak management). Keep raised floors only where tenants require them.
- **Fire safety:** Segregate Li-ion UPS/BESS rooms with 2-hour rated construction, gas detection (off-gas sensing), deflagration venting and NFPA 855 / IS-aligned spacing. For liquid cooling, use leak detection at manifolds and propylene-glycol or dielectric fluids with known flash points. NBC 2016 Part 4 is the legal baseline, but it does not cover these risks adequately (see policy recommendation P12).
- **Modular/prefab:** Use prefabricated power skids, e-houses and coolant distribution units (CDUs) to compress schedules and labour risk.
- **Embodied carbon:** Specify ≥30–50% GGBS/fly-ash cement replacement, recycled-content steel and an LCA at design stage. Report embodied carbon per MW. [Estimated recommendation]

15.2.9 Water design and peak-day sizing

- Apply a hierarchy: (1) zero-evaporation closed loop; (2) TSE from municipal STPs (e.g., Hoskote's 60 MLD); (3) desalinated seawater on the coast; (4) surface allocation with a drought-curtailment clause; (5) never fresh groundwater in over-exploited, critical or semi-critical units.
- Size storage and treatment for the April–May peak, when adiabatic systems use the most water (Part 12), with ≥7 days' storage for TSE-dependent sites. Report peak-day withdrawal (m^3/day) as well as the annual average.
- Report withdrawal, consumption and scarcity-weighted water separately (Part 6), and include the off-site grid water embedded in electricity.

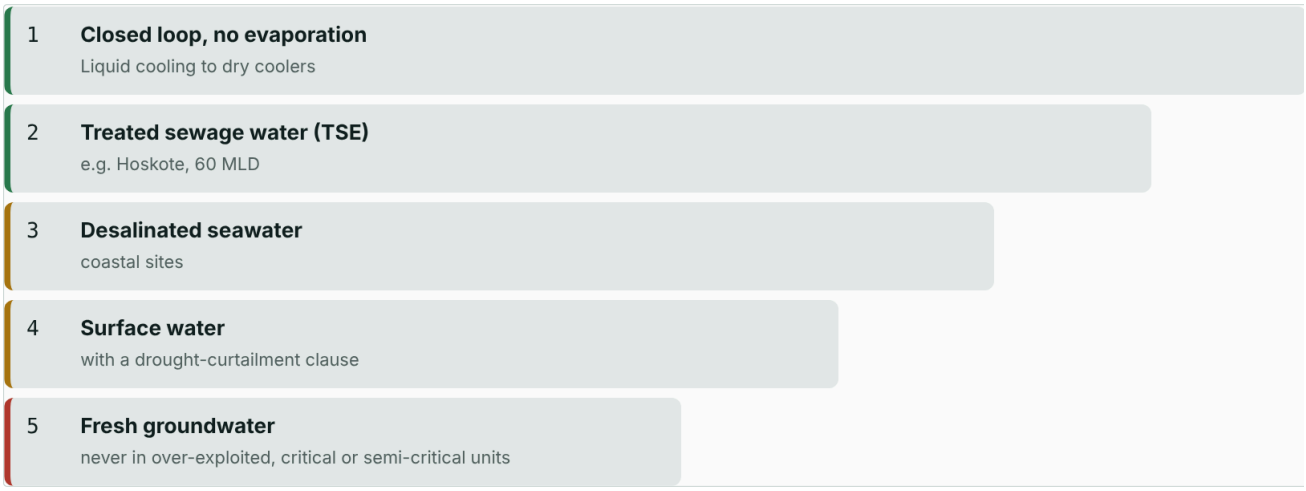


Figure 15.3 Water: a hierarchy, sized to the peak day. Use the first source that works; never the last. *Source: India Data Centre Atlas, chapter built on this chapter's data.*

15.2.10 Clean-power procurement design

- Anchor on in-state group captive (26% equity) solar+wind+BESS, because the ISTS waiver is now graded. Use interstate RE where co-located BESS still earns the waiver.
- Contract for hourly matching (24×7 CFE), not annual RECs. The Part 14 LP shows hourly-optimised solar+wind+BESS beats grid tariff on both cost and water.
- Use banking strategically, but model state rules: Tamil Nadu's 8% banking charge in kind and its third-party banking ban, and Gujarat's ₹1–1.50/unit.
- Use IEX price spreads (Part 12) to run BESS arbitrage and flexible AI scheduling.

15.2.11 Performance targets by archetype (recommended design targets; estimated)

Table 15.5 Performance targets by archetype (recommended design targets; estimated)

Archetype	Annualised PUE (mild / hot-humid / hot-arid)	Site WUE (L/kWh IT, consumption)	Freshwater share	Hourly CFE by 2030	CUE ($kgCO_2$/kWh IT)	ERF
Hyperscale cloud	≤1.20 / ≤1.28 / ≤1.25	≤0.3	≤20%	≥70%	≤0.25	Feasibility study
Colocation	≤1.30 / ≤1.38 / ≤1.35	≤0.5	≤30%	≥60%	≤0.35	Feasibility study
AI training campus	≤1.15 / ≤1.22 / ≤1.20	≤0.2	0% (TSE/desal only)	≥80%	≤0.15	≥5% where offtaker exists
AI inference/edge	≤1.35 / ≤1.45 / ≤1.40	~0 (air-cooled)	n/a	≥50%	≤0.45	—
Sovereign/government	≤1.30 (tighten to 1.25 by 2030) / ≤1.35 / ≤1.35	≤0.4, disclosed quarterly	≤20%	≥60%	≤0.35	Feasibility study

Context: The MeitY IndiaAI tender floor is PUE <1.35. Leading Indian operators report 1.4–1.5 (SBI sector report, April 2026). [Reported] The best published audited WUE is STT GDC India's 0.58. These targets are ambitious but within reach with W40+ liquid cooling: Germany's legal new-build ceiling is 1.2, and Singapore's DC-CFA2 requires 1.25 in a tropical climate.

15.3 Part 2 — Policy Recommendations

15.3.1 The current national framework at a glance

Table 15.6 The current national framework at a glance

Instrument	Status (Oct 2026)	Gap for data centres
MeitY Draft DC Policy 2020	Never notified; consultations revived Aug 2025 (single window, 4 DCEZs)	No binding national framework, efficiency standard or water rule
Infrastructure status (harmonised master list)	DCs included (2022)	Financing benefit, but not conditioned on performance
Budget 2026–27 tax holiday to 2047 + 15% safe harbour	Enacted; "specified DC" scheme reported as not yet issued	The scheme could carry PUE/WUE/CFE conditions, but none are announced
DPDP Act 2023 + Rules 2025	Notified 13/14 Nov 2025; consent, rights and cross-border Rule 15 commence 13 May 2027	Blacklist model, not blanket localisation; sectoral rules (RBI payments) still localise
IndiaAI Mission	₹10,372 crore; 38,000+ GPUs empanelled; around ₹65/GPU-hr (Business Standard, Jun 2026; the IT minister cites ₹67)	PUE <1.35 only; no WUE, CFE or disclosure conditions
BEE / ECBC	General building codes; no DC star label or designated-consumer status	No mandatory DC benchmark or reporting
CEA grid connectivity draft 2026	Draft; comments closed 2 Sep 2026	Technical standards only; no large-load flexibility or cost-causation rules
GEOA Rules 2022 (amended)	In force; state implementation diverges	Banking/ToD restrictions vary; no hourly-matching certificate
ISTS waiver	Graded phase-out; 100% for co-located BESS to June 2028	Data centres are not a distinct category
CGWA guidelines 2020 (amended 2023)	In force	No WUE disclosure; MSME exemption could be gamed by small edge sites
EIA Notification 2006	DCs generally not a listed category (building/construction thresholds may apply)	100 MW–1 GW campuses escape cumulative water/power impact assessment

Instrument	Status (Oct 2026)	Gap for data centres
NBC 2016 / fire norms	In force	Inadequate for Li-ion BESS and liquid cooling at scale
E-Waste Rules 2022	In force (EPR)	No server reuse/refurbishment targets
PLI IT Hardware 2.0, ISM	In force	Weak linkage to DC procurement/localisation

15.3.2 International comparison

Table 15.7 International comparison

Jurisdiction	Instrument	Efficiency floor	Water	Grid/connection	Heat reuse	Lesson for India
EU	EED recast + Delegated Reg. 2024/1364; draft rating scheme	Reporting now; draft label PUE A ≤1.15, G >1.90	WUE reported; label A ≤0.10, G >1.00	—	ERF reported; Art. 26 waste-heat CBA for sites above 1 MW	Mandatory KPI disclosure to a public database, from 500 kW
Germany	EnEfG 2023	New DCs from 1 Jul 2026: PUE ≤1.2; existing ≤1.5 (2027), ≤1.3 (2030)	—	Renewable electricity requirement	ERF 10% (2026), 15% (2027), 20% (2028)	Hard floors with a glide path for existing stock
Singapore	Green DC Roadmap (May 2024); DC-CFA2 (Dec 2025)	≤1.3 aspiration (10 yrs); ≤1.25 at 100% load to win capacity	—	Capacity rationed (≥200 MW in CFA2)	—	Allocate scarce grid capacity competitively on performance
Ireland	CRU LEU Connection Policy (12 Dec 2025)	—	—	≥10 MVA: dispatchable on-site/nearby generation/storage = MIC; SEM participation; 80% additional Irish RE	—	Connection is conditional on flexibility and additionality
Netherlands/Amsterdam	2019–20 Amsterdam pause; national hyperscale siting limits (2022)	—	—	Zoning restricts hyperscale to designated areas	—	Zone-based siting (not independently re-verified in this research)
Malaysia (Johor)	DC water tariff (Aug 2025), approval vetting, pause on water-cooled expansion (Nov 2025)	PUE thresholds in vetting	RM5.33/m³; reclaimed water for new DCs; WUE thresholds (1.4–1.8 reported, sources conflict)	National task force checks power/water capacity	—	Price water at cost and mandate reclaimed water
US – Texas	SB 6 (signed 20 Jun 2025); PUCT rules (Sep 2026)	—	—	≥75 MW: ≥$100k study fee, disclosure of duplicate requests and backup generation, curtailment in load shed	—	Large-load rules protect other consumers
US – Virginia	Large-load tariff class debates (Dominion)	—	—	Minimum-demand charges for large loads (not independently re-verified in this research)	—	Cost-causation tariffs
China	2024 Special Action Plan (NDRC/MIIT/NEA/NDA)	New large DCs ≤1.25 by end-2025; hub nodes ≤1.2; national average <1.5	—	>80% renewable at hub nodes	—	National hubs + floors steer siting
Japan	Watt-bit collaboration council (METI/MIC, Mar 2025; Report 1.0 Jun 2025)	—	—	Guide DCs to locations near decarbonised power	—	Joint power–telecom planning of DC zones

15.3.3 Numbered policy recommendations

P1. Notify a National Data Centre Policy with a statutory backbone (MeitY lead; MoP, MoEFCC, Jal Shakti, DoT; within 6–12 months). Rationale: The 2020 draft's six-year limbo has produced 16 divergent state regimes. The policy should: - define DC categories (edge, colo, hyperscale, AI campus, sovereign); - create the single-window portal already consulted on (replacing 30+ permissions) and designate DCEZs; - make the Budget's "specified data centre" notification conditional on the disclosure and performance standards in P2–P4. Trade-off: a binding national framework constrains state autonomy. Mitigate by setting floors, not ceilings.

P2. Mandatory disclosure with third-party assurance (BEE under the Energy Conservation Act, with MeitY; rules within 12 months; first reports FY2027–28). - Bring DCs of 500 kW IT or more under the Energy Conservation Act as designated consumers or a notified category. - Require annual public reporting to a national DC registry of: PUE; WUE split by withdrawal and consumption and by source (potable, ground, TSE, desal); peak-day water; CUE; hourly CFE share; REF; ERF; installed vs. utilised IT capacity. - Require BEE-accredited energy-auditor assurance, and align with ISO/IEC 30134 (adopted by BIS) and the EU 2024/1364 KPI set so multinationals report once. Evidence: only one Indian operator publishes audited WUE, while the EU now collects 24 data points from every site of 500 kW or more.

P3. Add WUE and hourly-clean-power criteria to all public compute procurement (MeitY/IndiaAI, NIC, GeM; next tender round, within 6 months). - Keep PUE <1.35 as the floor and tighten it to ≤1.30 by 2028. - Add WUE ≤0.5 L/kWh (site), zero groundwater in over-exploited, critical or semi-critical units, and disclosure of hourly CFE. - Score bids on ₹/GPU-hr *and* a sustainability index, so the lowest-price bidder does not automatically win. Evidence: the IndiaAI rate of around ₹65 per GPU-hour (Business Standard, June 2026) rewards cost alone, and subsidised public compute should set the national standard.

P4. Minimum performance standards with a glide path (BEE; notify 2027, effective for new builds from 2028). New DCs commissioned from 2028: annualised PUE ≤1.30 in hot-humid and ≤1.25 elsewhere; ≤1.20 for AI campuses above 50 MW by 2030. Existing DCs: ≤1.50 by 2030, ≤1.40 by 2033. Make climate-zone adjustments explicit. Model the approach on Germany's EnEfG, but with an extra 0.05–0.10 PUE allowance for tropical climates. [Estimated recommendation] Risk: operators may game PUE through IT fan power. Require rack-level IT metering, as the EU now does.

P5. Statutory water hierarchy for data centres (Jal Shakti/CGWA, MoEFCC, state water boards; 6–18 months). - Make "no fresh groundwater for cooling" absolute for DCs in over-exploited, critical and semi-critical units, and close the MSME loophole for DC use. - Require TSE or desal first for evaporative cooling. Set DC-specific water tariffs at full cost (cf. Johor's RM5.33/m³). Require peak-day

drought-curtailment plans. - Extend the 2023 online water-audit duty to all DCs. - States should replicate Karnataka's treated-water reimbursement, but cap it as a share of savings rather than a flat ₹8 lakh, which is immaterial for a 50 MW site.

P6. A large-load grid connection regime (MoP/CEA/CERC + SERCs; 12–24 months). - Define "large load" as 50 MW or more (aligning with CEA's draft 50 MW threshold). - Require: a study fee and financial security to deter speculative queue entries (Texas SB 6: ≥$100,000); disclosure of duplicate applications (17 GW is already in STU queues against a 26.3 GW projection); curtailability or ride-through and power-smoothing; co-located storage or generation sized to a share of import capacity in constrained nodes (a softer version of Ireland's 100%-of-MIC rule); and dedicated tariff categories with minimum-demand charges so that grid upgrades do not burden other consumers. - Publish a nodal "headroom map" jointly with the CTU and STUs, the Indian equivalent of Japan's watt-bit planning.

P7. Make 24×7 clean power procurable (MoP, CERC/SERCs, SECI/NTPC; 12–24 months). - Issue RTC/FDRE tenders open to data-centre offtakers. - Introduce time-stamped (hourly) green certificates on top of the existing GEOA framework. - Harmonise state banking: a ceiling on banking charges, no outright ban for third-party OA for loads above 10 MW, and ToD-consistent settlement in 15-minute blocks. - Extend the co-located BESS ISTS waiver logic to DC-linked storage beyond June 2028, with a sunset. Evidence: Tamil Nadu's 2025 banking rules cut group-captive value by ~15%. Without hourly products, annual "100% RE" claims mask coal-heavy night-time supply.

P8. Outcome-based incentives with sunset clauses (state IT departments + DISCOMs; with each policy revision). - Replace flat capex subsidies and blanket ₹/unit discounts with payments tied to verified outcomes: PUE ≤1.20 (as in Karnataka's ₹20 crore bonus), WUE ≤0.2, hourly CFE ≥70%, ERF > 0. - Claw back incentives if targets are missed for two consecutive years. - Adopt a national "incentive code of conduct" through NITI Aayog/MeitY to curb bidding wars. Maharashtra's move from 100% to 51% green power is the cautionary example. - Every incentive should sunset within 5–7 years and face a published cost-benefit review.

P9. Green DC zones and parks with shared infrastructure (states with MeitY DCEZ scheme, CTU/STUs, municipal STPs; 2–5 years). Plan zones where a 400/220 kV substation, RE pooling, TSE/desal pipelines and diverse fibre are built ahead of demand and costs are recovered through park charges. Karnataka's resource heat-mapping and the Hoskote park are templates. Priority zones: Vizag–Anakapalli, Jamnagar, Hoskote/Tumakuru, Navi Mumbai–Panvel (ReNew's 750 MW MoU), Noida/YEIDA, Hyderabad–Shamshabad, Bhubaneswar–Paradip.

P10. Bring large campuses into environmental assessment (MoEFCC; 12 months). Add DC campuses above 100 MW (or with water demand above a set threshold) to the EIA Notification schedule as a category with cumulative power-water assessment and public consultation. Otherwise 1 GW sites will escape scrutiny that a small thermal plant would face. Trade-off: possible delays. Mitigate with standard ToRs and a fixed 120-day timeline within the single window.

P11. Heat-reuse readiness (BEE, MoHUA, state industrial development corporations; medium term). Require a heat-reuse feasibility study for sites of 10 MW or more, and ERF disclosure. Offer capital support or concessional finance for absorption-chiller, process-heat and district-cooling links, starting with GIFT City and food-processing parks. Do not copy Germany's quotas until offtake infrastructure exists.

P12. Fire and life-safety codes for Li-ion and liquid cooling (BIS/NBC committee, state fire services; 12–18 months). Issue an NBC Part 4 amendment or a BIS standard for BESS/UPS rooms (aligned with NFPA 855), liquid-cooling fluid classification, leak detection and firefighter access for multi-storey DCs. Train state fire services in DC incidents.

P13. Circularity under the E-Waste Rules 2022 (MoEFCC, CPCB, MeitY; medium term). Set targets for server and GPU refurbishment and reuse. Require ITAD data-sanitisation certification that is DPDP-compliant. Include lithium battery second-life provisions, and require disclosure of tonnage retired and reused.

P14. Calibrated data sovereignty (MeitY, sector regulators; before 13 May 2027). Keep the DPDP blacklist approach rather than blanket localisation. Localisation mandates raise capacity demand without regard to resource efficiency. Pair sovereign workload requirements (government, critical sectors) with the strictest efficiency and disclosure rules, and publish the Section 16 negative list well before the May 2027 commencement to reduce investment uncertainty.

P15. Local manufacturing and supply chain (MeitY PLI IT Hardware 2.0, ISM, DPIIT; medium term). Extend PLI support to DC electrical and cooling equipment: transformers, switchgear, CDUs, dry coolers, LFP battery packs. Rationalise customs duty on items not made domestically. Link IndiaAI empanelment scoring to domestic-value-add disclosure, but not to mandates that would raise GPU-hour costs.

P16. Skills (MSDE, Electronics Sector Skills Council, AICTE; short to medium term). Create a national DC technician and critical-facility-engineer curriculum covering liquid cooling, MV switching and BESS safety, with tier-2 training centres co-located with DC parks.

P17. Community benefit and transparency (states, ULBs; with every allocation). - Publish water and power allocations per project and require public hearings for sites above 50 MW. - Ring-fence a share of DC water and electricity duty revenue for local water infrastructure, such as STP upgrades. - Johor's experience, where residents protested over water in February 2026, shows that legitimacy depends on visible fairness.

P18. Monitoring, review and sunset (MeitY + NITI Aayog; annual). Publish an annual "State of Indian Data Centres" report from registry data. Review incentives every three years and auto-sunset any that cannot show verified outcomes.

15.3.4 Policy recommendation matrix

Table 15.8 Policy recommendation matrix

#	Recommendation	Lead agency	Supporting	Timeline	Instrument	Key risk
P1	National DC policy + single window + specified-DC conditions	MeitY	MoP, MoEFCC, Jal Shakti, DoT, MoF	Short (≤12 mo)	Cabinet-approved policy + notification	Delay; state pushback
P2	Mandatory assured disclosure + registry	BEE	MeitY, CEA	Short–medium	EC Act notification	Reporting burden on small sites; use thresholds
P3	WUE/CFE in public tenders	MeitY (IndiaAI), NIC	GeM	Short (next round)	Tender RFP	Higher GPU-hour bids

#	Recommendation	Lead agency	Supporting	Timeline	Instrument	Key risk
P4	PUE floors with glide path	BEE	MeitY	Medium (2027–2030)	EC Act standards	Climate fairness; gaming
P5	Water hierarchy + groundwater ban	CGWA/Jal Shakti	MoEFCC, state water boards, ULBs	Short–medium	CGWA guideline amendment; state rules	Enforcement capacity
P6	Large-load connection rules	CEA, CERC	SERCs, CTU, STUs, DISCOMs	Medium (12–24 mo)	Regulations + tariff orders	Slower connections; investor flight
P7	24×7 clean power products, banking harmonisation	MoP	CERC, SERCs, SECI, Grid-India	Medium	Rules amendment, tenders	DISCOM revenue loss
P8	Outcome-based, sunset incentives	State IT depts	DISCOMs, NITI Aayog	With each revision	State policies	Interstate competition
P9	Green DC zones/parks	States, MeitY	CTU/STUs, ULBs	Medium–long (2–5 yrs)	DCEZ scheme	Stranded park infrastructure
P10	EIA coverage for >100 MW campuses	MoEFCC	State SEIAAs	Short–medium	EIA notification amendment	Delays
P11	Heat-reuse readiness	BEE	MoHUA, state IDCs, GIFT City	Medium	Guidelines + finance	Few offtakers
P12	Li-ion/liquid fire codes	BIS (NBC)	State fire services	Short–medium	NBC amendment	Code lag
P13	Circularity targets	MoEFCC/CPCB	MeitY	Medium	E-Waste Rules amendment	Data-security concerns
P14	Calibrated sovereignty	MeitY	RBI, SEBI, IRDAI	Before May 2027	DPDP S.16 notification	Over-localisation
P15	Local manufacturing	MeitY, DPIIT	ISM, MoF (customs)	Medium	PLI guidelines	Cost inflation
P16	Skills	MSDE	ESSC, AICTE	Short–medium	Curriculum	Uptake
P17	Community transparency	States, ULBs	—	Short	Allocation rules	Local opposition
P18	Monitoring and sunset	MeitY, NITI Aayog	BEE	Annual	Report	Political will

15.3.5 State-level differentiation

- **Water-stressed metros (Karnataka/Bengaluru, Tamil Nadu/Chennai, Delhi-NCR/UP, Telangana):** Pay incentives only for zero-freshwater designs, and allocate TSE first. Tamil Nadu should notify a successor to its lapsed policy with Karnataka-style PUE/WUE conditions and fix its 2025 GEOA banking restrictions for large DC loads. Telangana should replace its 2016 policy.
- **RE-rich states (Gujarat, Rajasthan, AP, Karnataka interior, Odisha):** Market "RE + land + water" campuses with pre-built 400 kV substations and desal or TSE, and compete on infrastructure readiness, not tariff discounts.
- **Maharashtra:** Restore a rising green-power trajectory (51% now, 75% by 2030, hourly) rather than a fixed 51% floor, and attach WUE conditions to the ₹1/unit concession.
- **UP and the eastern states:** Use DC policies to anchor edge and sovereign capacity for underserved populations, prioritising reliability and fibre.

15.4 Caveats

- **Reported vs. estimated:** Items marked [Reported] come from named public sources. Some, including the CEA 26.3 GW figure, the specified-DC status, the Texas PUCT final rules, Singapore DC-CFA2 awardees and the Johor WUE thresholds, rest on secondary trackers or press summaries rather than primary gazettes. Hub scores, climate design values, performance targets, plinth and BESS sizing, and the tropical PUE allowance are our engineering estimates and should be validated with site-specific ASHRAE/IMD data and studies.
- **Conflicts noted:** Johor's WUE threshold appears as 1.8 (ArusGrid) and 1.4 (S&P via BigGo). Secondary sources disagree on whether China's 2024 plan sets 1.25 or 1.3. The NDRC primary text says ≤1.25 for new large data centres and ≤1.2 at hub nodes, and we use that. Netherlands and Virginia entries were not re-verified in this research cycle.
- **Regulatory flux:** IS 1893:2025 has been withdrawn, CEA's connectivity regulations are in draft, the DPDP cross-border provisions start in May 2027 and the national DC policy is unfinalised. Recommendations are written to stay robust to these outcomes.
- **Forward-looking statements:** Capacity figures for Google Vizag (~1 GW; some reports cite a 5 GW long-term target) and Reliance Jamnagar (multi-GW) are company plans, not operational capacity.

Sources cited in this chapter

Organisations and publications named in this chapter's text, in order of how often they are cited. Full details are in the bibliography.

- Ministry of Electronics and Information Technology (MeitY), Data Centre Policy draft and IndiaAI
- Ministry of Power, orders on ISTS waiver and green open access
- Central Electricity Authority (CEA), CO_2 Baseline Database and grid reports
- Bureau of Indian Standards (IS 1893, ISO/IEC 30134 adoption)
- Ministry of Environment, Forest and Climate Change, thermal plant water norms and EIA
- Bureau of Energy Efficiency
- Central Ground Water Authority guidelines
- Texas Senate Bill 6 and PUCT rules
- Central Electricity Regulatory Commission
- European Union Energy Efficiency Directive and Delegated Regulation 2024/1364
- Singapore IMDA Green Data Centre Roadmap and DC-CFA
- Central Ground Water Board and state groundwater assessments
- ASHRAE TC 9.9 thermal guidelines
- Google environmental reports, efficiency data and Gemini inference paper
- Meta sustainability reports and data index
- Reliance Industries announcements
- Germany, Energy Efficiency Act (EnEfG)
- Press Information Bureau releases
- CBRE, India data centre market reports
- State electricity regulatory commissions (KERC, MERC, TNERC, TGERC) tariff and open-access orders
- Solar Energy Corporation of India tenders
- Microsoft sustainability reports and datacenter design posts
- STT GDC India ESG report
- India Meteorological Department climate normals
- Ireland Commission for Regulation of Utilities, large-energy-user connection policy
- China National Development and Reform Commission action plan
- Japan METI watt-bit collaboration council
- National Disaster Management Authority guidelines
- Uptime Institute, Global Data Center Surveys and Tier Standard

- Cushman & Wakefield, India data centre market updates (H1 2026)
- Indian Energy Exchange price data
- AdaniConneX disclosures
- State Bank of India sector report on data centres (April 2026)
- Centre for Science and Environment, thermal plant water survey
- ISO/IEC 30134 data-centre KPI standards
- IEEFA analysis of green open access
- Telborg data-centre briefings

Web sources: the 66 web pages consulted for this chapter are listed with their full addresses in Appendix B.2.

APPENDICES

Appendix A · Simulator assumptions and key numbers

The atlas includes a simulator that lets a reader build a data centre by choosing a city, a workload, a size, a cooling system, a power supply and a water source. The first table lists the inputs the simulator uses and the evidence behind each. The second table lists the figures the research series relies on most often, with their sources.

Table A.1 Inputs behind the atlas simulator

Input	Value used	Evidence
Grid tariffs (energy + demand)	KA ₹6.70 + ₹365/kVA; MH ₹7.27 + ₹560; TN ₹7.50 + ₹608; TG ₹6.65 + ₹500 (FY2025–26 orders)	Reported
UP and AP tariffs	₹7.4 and ₹7.2/kWh (typical ₹7–8)	Estimate
Renewable landed cost	₹5.5–6.2/kWh open access; ≥₹6.4 for 90% hourly clean	Estimate
Facility capex	Colo ₹56.1, hyperscale ₹39.2, AI ₹72.5 cr/MW before land; cooling adds ₹0–3 cr/MW	Model
GPU capex	₹266 cr per MW of IT (~425 GB300-class GPUs/MW)	Model
Land	₹3–20 cr per acre by city; 0.25–0.4 acre/MW	Estimate
Cooling overhead and WUE	Function of design dry and wet bulb; towers ~1.8 L/kWh, adiabatic ~0.4, liquid + dry ~0.03	Model
Grid carbon	0.710 tCO_2/MWh for all states	Reported
Power-station water	Karnataka 1.5 L/kWh; other states 2.2 L/kWh	Estimate
Water price	BWSSB ₹99/kL; ₹80 elsewhere; groundwater ₹20; treated + RO ₹40	Estimate
Electricity duty	7.5% of the energy charge, where waived	Estimate
Finance	WACC 11%; facility 20 years, GPUs 5 years	Model
Scarcity weights	Groundwater 1.0, municipal 0.7, treated 0.1, power stations 0.4, × city stress	Model

Table A.2 Key numbers used across the series

Measure	Value	Source	Basis
CEA projected extra DC demand by 2031–32	26.3 GW (11 states)	MoP reply to Rajya Sabha, May 2026 (secondary)	Reported
AI campus design target (mild climate)	PUE ≤1.15 · WUE ≤0.2 · CFE ≥80%	Report 15 estimate	Estimate
Germany new-DC PUE limit	≤1.2 from 1 Jul 2026	EnEfG	Reported
Least-water mix, 50 MW-IT Karnataka DC	161 MW solar · 86 MW wind · 379 MWh	Chapter 14 hourly LP	Model
Recommended mix cost · water	₹5.39/kWh · 218 ML/yr	Chapter 14 model	Model
Cheapest clean mix vs all-grid water	1,451 vs 1,139 ML/yr	Chapter 14 (hydro at 10 L/kWh)	Model
AI rack power (GB200 / GB300 NVL72)	120–142 kW	NVIDIA, HPE (Report 13)	Reported
AI vs colo energy per MW of IT	≈1.0× (per site ~5×)	Report 13 model	Model
Grid water, India (at plant)	≈2.6 L/kWh	Report 13 assumption from MoEFCC norms	Model
Gemini median prompt	0.24 Wh · 0.26 mL	Google, Aug 2025	Reported
Meta direct vs electricity water	2,974 vs 72,207 ML	Meta 2024 (Report 13)	Reported
India AI-dedicated capacity 2025 → 2030	275 → 6,546 MW	Wood Mackenzie (Report 13)	Estimate
India DC electricity 2024 → 2030	13 → 57 TWh	S&P Global (Report 13)	Estimate
STT GDC India WUE FY25	0.58 L/kWh (assured)	STT GDC India ESG report	Reported
Global DC electricity 2024 → 2030	415 → 945 TWh	IEA Energy and AI, Apr 2025	Estimate
Global capacity 2025 → 2030	~103 → ~200 GW	JLL, Jan 2026	Estimate
Hyperscale share of capacity	~48% (→ 67% by 2031)	Synergy, Apr 2026	Estimate
India operational capacity	1,789 MW	Cushman & Wakefield, H1 2026	Reported
India pipeline to 2030	3,860 MW · US$29.9 bn	Cushman & Wakefield	Estimate
Mumbai share of India capacity	~50% (890 MW)	Cushman & Wakefield	Reported
Global average PUE	1.54	Uptime 2025	Reported
Google fleet PUE	1.09	Google, 2025	Reported
MeitY AI-cloud tender PUE	<1.35	MeitY, since 2024	Reported
Indian leaders' PUE	1.4–1.5	SBI, Apr 2026	Estimate
India grid emission factor	0.710 tCO_2/MWh	CEA v21, FY25	Reported
Karnataka consumption mix	~0.50 tCO_2/MWh	Series estimate	Model
Evening fossil peak	~174 GW at 7 pm	Ember, H1 2026	Reported
Coal lifecycle carbon	820 g/kWh (India actual 900–1,100)	IPCC AR5	Reported
Indian coal water use	2.8–4.0+ m^3/MWh	CSE / CEA research	Reported
Karnataka grid water	~1.5 L/kWh	Series estimate	Model
Evaporation per kWh of heat	~1.5 L	Latent heat 2,400 kJ/kg	Reported
50 MW evaporative withdrawal	1.77 MLD (peak ~2.4)	Report 6 model	Model
BWSSB delivered supply	~1,935 MLD	Legislative Council, Aug 2026	Reported
Bangalore East groundwater stage	~305%	2024 assessment	Reported
Bengaluru DC water today	~2–20 MLD (estimates)	DTE, IT dept, WELL Labs	Estimate
100 MW AI campus energy	~841 MU/yr · ~140 MVA	Report 4	Model

Measure	Value	Source	Basis
Karnataka 1 GW target, evaporative	~32 MLD	Report 4	Model
Tier III build cost	₹50–55 cr/MW (series model ₹56.1)	SBI / Report 7	Estimate
AI / Tier IV build cost	up to ₹75 cr/MW	SBI, Apr 2026	Estimate
Mumbai construction cost	US$6.64/W (51st of 52)	Turner & Townsend 2025	Reported
Colo LCOD (20 MW)	₹29.7 per IT-kWh	Report 7	Model
Mumbai colo rent, 1–5 MW	₹6,500–8,000/kW/month	Cushman & Wakefield	Reported
H100 cost at 80% use	~₹177/GPU-hour	Report 7	Model
IndiaAI L1 rate (H100-class)	₹150/GPU-hour	IndiaAI	Reported
IndiaAI GPUs	38,231 across 14 providers	PIB, Mar 2026	Reported
SECI round-the-clock RE	₹5.25–5.26/kWh	SECI tranche V, Aug 2026	Reported
Green 20 MW colo payback	~2.3 years	Report 8	Model
Green package abatement cost	≈ −₹2,540/tCO_2	Report 8	Model
Karnataka PUE incentive	₹20 cr for PUE 1.00–1.20	Karnataka policy 2026–31	Reported
Karnataka policy outlay	₹1,300.45 cr · 1 GW by 2031	Karnataka, Sept 2026	Reported
HFC freeze in India	1 Jan 2028	Kigali Amendment	Reported
Liquid-cooling revenue	~$3 bn (2025) → ~$7 bn (2029)	Dell'Oro, Jan 2026	Estimate
GB200 / GB300 NVL72 rack	120–150 kW	Nvidia / OEMs	Reported
Kyber rack (2027)	~600 kW, 100% liquid, 800 VDC	Nvidia GTC	Estimate
Budget tax holiday	to 31 Mar 2047	Finance Act 2026	Reported
India subsea bandwidth via Mumbai	~95%	Series sources	Estimate

APPENDICES

Appendix B · Bibliography

The research series cites its sources by name in the text. The first list below gives every organisation and publication named across the chapters, with the chapters that cite it. The second list gives the web addresses recorded by the research for Chapters 7, 8 and 15. Readers should consult the original documents before relying on any figure for an investment or policy decision.

B.1 Organisations and publications cited

1. **AdaniConneX disclosures.** Cited in chapters 1, 3, 6, 7, 8, 10, 12, 13, 14, 15.
2. **Amazon Web Services sustainability reporting.** Cited in chapters 1, 2, 3, 5, 6, 8, 9, 10, 12.
3. **ASHRAE TC 9.9 thermal guidelines.** Cited in chapters 3, 8, 10, 13, 15.
4. **Bangalore Water Supply and Sewerage Board.** Cited in chapters 4, 6, 7, 8, 9, 10, 12, 13, 14.
5. **Bureau of Energy Efficiency.** Cited in chapters 1, 3, 4, 7, 10, 15.
6. **Bureau of Indian Standards (IS 1893, ISO/IEC 30134 adoption).** Cited in chapters 1, 7, 12, 13, 15.
7. **CBRE, India data centre market reports.** Cited in chapters 1, 2, 9, 10, 13, 14, 15.
8. **Central Electricity Authority (CEA), CO_2 Baseline Database and grid reports.** Cited in chapters 4, 5, 7, 8, 10, 11, 12, 13, 14, 15.
9. **Central Electricity Regulatory Commission.** Cited in chapters 9, 15.
10. **Central Ground Water Authority guidelines.** Cited in chapters 4, 15.
11. **Central Ground Water Board and state groundwater assessments.** Cited in chapters 4, 6, 7, 12, 13, 15.
12. **Centre for Science and Environment, thermal plant water survey.** Cited in chapters 5, 7, 15.
13. **China National Development and Reform Commission action plan.** Cited in chapter 15.
14. **Council on Energy, Environment and Water (CEEW) studies.** Cited in chapters 1, 2, 6, 7, 12, 14.
15. **CPHEEO urban water supply norms.** Cited in chapters 4, 7.
16. **CtrlS disclosures.** Cited in chapters 1, 3, 6, 7, 8, 9, 10, 12, 14.
17. **Cushman & Wakefield, India data centre market updates (H1 2026).** Cited in chapters 1, 2, 4, 9, 13, 14, 15.
18. **DatacenterDynamics reporting.** Cited in chapters 1, 8, 12, 13.
19. **De Vries-Gao, Patterns (2025).** Cited in chapter 8.
20. **Dell'Oro Group, liquid-cooling market data.** Cited in chapters 3, 14.
21. **Down To Earth reporting.** Cited in chapters 4, 7, 8, 14.
22. **Electric Power Research Institute.** Cited in chapter 8.
23. **Ember, Global Electricity Review and India data.** Cited in chapter 5.
24. **Epoch AI energy estimates.** Cited in chapters 2, 8.
25. **Equinix sustainability reports.** Cited in chapters 1, 6, 7, 8, 9, 10.
26. **European Union Energy Efficiency Directive and Delegated Regulation 2024/1364.** Cited in chapters 4, 6, 7, 10, 15.
27. **Germany, Energy Efficiency Act (EnEfG).** Cited in chapters 10, 15.
28. **Google environmental reports, efficiency data and Gemini inference paper.** Cited in chapters 1, 2, 3, 5, 6, 7, 8, 9, 10, 12, 13, 14, 15.
29. **Hewlett Packard Enterprise product data.** Cited in chapters 2, 7, 8.
30. **Houlihan Lokey India data centre edition (December 2025).** Cited in chapters 4, 9.
31. **IEEFA analysis of green open access.** Cited in chapters 5, 15.
32. **India Meteorological Department climate normals.** Cited in chapter 15.
33. **Indian Energy Exchange price data.** Cited in chapters 7, 15.
34. **International Energy Agency (IEA), Energy and AI (2025) and updates.** Cited in chapters 1, 2, 5, 7, 8, 14.
35. **IPCC AR5 Annex III life-cycle emission factors.** Cited in chapter 5.
36. **Ireland Commission for Regulation of Utilities, large-energy-user connection policy.** Cited in chapters 1, 15.
37. **ISHRAE climate design data.** Cited in chapters 3, 7, 13.
38. **ISO/IEC 30134 data-centre KPI standards.** Cited in chapters 6, 7, 10, 15.
39. **Japan METI watt-bit collaboration council.** Cited in chapter 15.
40. **JLL, Global and India Data Center Outlooks.** Cited in chapters 1, 2, 13, 14.
41. **Kigali Amendment to the Montreal Protocol.** Cited in chapters 2, 3.
42. **Lawrence Berkeley National Laboratory, 2024 US Data Center Energy Usage Report.** Cited in chapters 2, 3, 5, 6, 7, 8.
43. **Li, Yang, Islam and Ren, "Making AI Less Thirsty".** Cited in chapters 2, 6, 7, 8.
44. **MarketsandMarkets market estimates.** Cited in chapter 3.
45. **Meta sustainability reports and data index.** Cited in chapters 1, 2, 3, 5, 6, 8, 10, 12, 13, 14, 15.
46. **Microsoft sustainability reports and datacenter design posts.** Cited in chapters 1, 2, 3, 5, 6, 7, 8, 9, 10, 12, 15.
47. **Ministry of Electronics and Information Technology (MeitY), Data Centre Policy draft and IndiaAI.** Cited in chapters 1, 2, 3, 4, 6, 8, 9, 10, 12, 14, 15.
48. **Ministry of Environment, Forest and Climate Change, thermal plant water norms and EIA.** Cited in chapters 3, 4, 5, 7, 8, 10, 12, 15.
49. **Ministry of New and Renewable Energy statistics.** Cited in chapters 7, 10.
50. **Ministry of Power, orders on ISTS waiver and green open access.** Cited in chapters 1, 5, 7, 8, 9, 10, 12, 13, 14, 15.
51. **Mistral AI life-cycle assessment (ADEME / Carbone 4).** Cited in chapters 2, 6, 8.
52. **Mordor Intelligence market estimates.** Cited in chapters 1, 2, 14.
53. **National Disaster Management Authority guidelines.** Cited in chapters 13, 15.
54. **National Renewable Energy Laboratory water factors.** Cited in chapters 3, 5, 7.
55. **North American Electric Reliability Corporation, Large Loads alert (May 2026).** Cited in chapters 2, 7.
56. **NTT DATA ESG data.** Cited in chapters 1, 3, 6, 7, 10, 12.

57. **NVIDIA product specifications and roadmap.** Cited in chapters 1, 2, 3, 7, 8, 9.
58. **Nxtra by Airtel disclosures.** Cited in chapters 1, 3, 6, 7, 8, 9, 10, 12, 13, 14.
59. **Omdia server market data.** Cited in chapter 8.
60. **Press Information Bureau releases.** Cited in chapters 1, 14, 15.
61. **Reliance Industries announcements.** Cited in chapters 1, 2, 8, 10, 12, 13, 15.
62. **S&P Global (451 Research) India data centre analysis.** Cited in chapters 1, 6, 12.
63. **Sify Infinit Spaces sustainability report.** Cited in chapters 1, 3, 6, 7, 8, 9, 10, 12.
64. **Sightline Climate (Currence) delay tracking.** Cited in chapter 1.
65. **Singapore IMDA Green Data Centre Roadmap and DC-CFA.** Cited in chapters 10, 15.
66. **Solar Energy Corporation of India tenders.** Cited in chapters 7, 9, 10, 15.
67. **State Bank of India sector report on data centres (April 2026).** Cited in chapters 1, 2, 3, 4, 8, 9, 10, 14, 15.
68. **State electricity regulatory commissions (KERC, MERC, TNERC, TGERC) tariff and open-access orders.** Cited in chapters 4, 5, 7, 9, 10, 13, 15.
69. **STT GDC India ESG report.** Cited in chapters 1, 3, 6, 7, 8, 9, 10, 12, 13, 14, 15.
70. **Synergy Research Group, capacity share data.** Cited in chapter 1.
71. **Telborg data-centre briefings.** Cited in chapter 15.
72. **Texas Senate Bill 6 and PUCT rules.** Cited in chapter 15.
73. **The Green Grid metric definitions.** Cited in chapters 3, 5, 6, 10.
74. **Turner & Townsend, Data Centre Cost Index 2025.** Cited in chapters 2, 9, 14.
75. **Uptime Institute, Global Data Center Surveys and Tier Standard.** Cited in chapters 1, 2, 3, 6, 7, 8, 10, 15.
76. **WELL Labs, Bengaluru water research.** Cited in chapters 4, 6, 10.
77. **Wood Mackenzie India data centre outlook.** Cited in chapters 8, 12, 14.
78. **World Resources Institute (Aqueduct, thermal water studies).** Cited in chapters 2, 5, 6, 7.
79. **Yotta Data Services disclosures.** Cited in chapters 1, 2, 3, 4, 6, 7, 8, 9, 10, 12, 14.

B.2 Web sources recorded by the research

Chapter 7 · Water and energy calculated

1. LBNL24 — Shehabi et al., 2024 US Data Center Energy Usage Report: https://eta-publications.lbl.gov/sites/default/files/2024-12/lbnl-2024-united-states-data-center-energy-usage-report_1.pdf
2. UPT25 — Uptime Institute Global Data Center Survey 2025: https://datacenter.uptimeinstitute.com/rs/711-RIA-145/images/2025.Annual.Survey.Report.pdf?version=0
3. GOOG25 — Elsworth et al., Measuring the environmental impact of delivering AI at Google Scale: https://arxiv.org/pdf/2508.15734 ; Google PUE: https://datacenters.google/efficiency/
4. MSFT — Microsoft datacenter efficiency: https://datacenters.microsoft.com/sustainability/efficiency/
5. LI — Li, Yang, Islam & Ren, Making AI Less "Thirsty": https://dl.acm.org/doi/10.1145/3724499
6. NTT — NTT DATA ESG environment data: https://www.nttdata.com/global/en/about-us/sustainability/esg-data/environment-data
7. DTE — Down To Earth, India's digital thirst: https://www.downtoearth.org.in/science-technology/indias-digital-thirst-what-data-centre-giants-arent-saying-about-their-water-use
8. CEEW — Scaling India's Data Centre Ecosystem: https://www.ceew.in/sites/default/files/ceew-data-centre-study-web-ready-final.pdf
9. KADC — Karnataka Sustainable Data Centre Policy 2026–31 (TNM): https://www.thenewsminute.com/karnataka/karnatakas-new-data-centre-policy-puts-water-energy-in-focus-what-does-it-say
10. MOEF15 — MoEFCC S.O. 3305(E), 7 Dec 2015 (CPCB): https://cpcb.nic.in/uploads/Thermal_Power_Plant_overview.pdf ; amendments: https://powerline.net.in/2026/09/04/water-stress-thermal-power-plants-take-steps-to-optimise-consumption/
11. SD23 — Vulnerability assessment of thermal power plants in India under water stress: https://www.sciencedirect.com/science/article/abs/pii/S0360544223009477
12. WRI — WRI, India's thermal plants in water-scarce areas: https://www.wri.org/insights/40-indias-thermal-power-plants-are-water-scarce-areas-threatening-shutdowns
13. NREL — Macknick et al., operational water factors: https://docs.nrel.gov/docs/fy11osti/50900.pdf
14. MNRE25 — MNRE Renewable Energy Statistics 2024-25: https://cdnbbsr.s3waas.gov.in/s3716e1b8c6cd17b771da77391355749f3/uploads/2025/11/2025110616276787 82.pdf
15. CREA — FY25 generation: https://www.deccanherald.com/amp/story/india%2Fthermal-generation-growth-slowest-in-fy-2024-25-since-covid-3497645
16. EU24 — Delegated Regulation (EU) 2024/1364: https://www.eudca.org/documents/content/_wU9NIdhRn2O7FEPP-YZ2Qo33?download=0 ; rating scheme: https://www.philiplee.ie/eu-publishes-draft-mandatory-sustainability-rating-scheme-for-data-centres/
17. NERC — NERC/FERC large loads: https://www.ferc.gov/sites/default/files/2025-04/Presentation%20NERC%20Seeks%20to%20Address%20Reliability%20Impacts%20from%C2%A0Large%20Load%20Integration_1.pdf
18. IEX — https://www.saurenergy.com/solar-energy-news/iex-dam-prices-drop-nearly-18-yoy-despite-rising-demand-11176360 ; FY25: https://www.business-standard.com/amp/markets/capital-market-news/iex-records-highest-ever-electricity-traded-volumes-in-fy25-125040400233_1.html ; intraday: https://earthsync.io/resources/iex-dam-gdam-rtm-explained
19. SECI — FDRE-RTC: https://www.pv-magazine.com/2026/08/10/secis-1-gw-fdre-rtc-power-tender-discovers-inr-5-25-kwh-tariff/ ; FDRE-IX: https://www.pv-magazine.com/2026/08/24/seci-discovers-inr-5-99-kwh-tariff-for-4-hour-peak-renewable-power/
20. ISTS — MoP ISTS waiver phase-out: https://www.tndindia.com/waiver-of-ists-charges-on-renewable-energy-to-be-phased-out/
21. MERC — MSEDCL MYT press note: https://merc.gov.in/wp-content/uploads/2025/03/Press-Note_MSEDCL-MYT-Order_English.pdf ; TPC-D: https://merc.gov.in/wp-content/uploads/2025/03/Press-Note_MYT-Order_TPC_English.pdf
22. TGERC — TGSPDCL FY26 tariff schedule: https://tgsouthernpower.org/resources/PDF/Tariffs/63tarifffile.pdf
23. KERC — KERC Tariff Order 2025 schedule: https://hrecs.co.in/app/documents/TariffRates2026-27-28.pdf
24. TNERC — TN Tariff Order 6/2025 summary: https://www.ecloudsenergy.com/news-details.php?nid=52
25. GEOA — GEOA Rules 2022 gazette: https://www.acma.in/uploads/ciculer-attachement/Electricity%20(Promoting%20Renewable%20Energy%20through%20Green%20Energy%20Open%20Access)%20Rules,2022.pdf ; FOR model: https://forumofregulators.gov.in/Data/Reports/Revised%20Model%20regulation%20GEOA.pdf
26. NVDA — NVIDIA DGX B200 guide: https://docs.nvidia.com/dgx/dgxb200-user-guide/introduction-to-dgxb200.html ; GB200 NVL72: https://pantheon.run/learn/nvidia-gb200-nvl72-power-and-cooling

Chapter 8 · Conventional and AI data centres compared

1. datacenterdynamics — https://www.datacenterdynamics.com/en/news/roughly-one-third-of-data-center-owners-and-operators-doing-ai-training-or-inference-uptime-institute/
2. amazon — https://aws.amazon.com/sustainability/
3. datacenterdynamics — https://www.datacenterdynamics.com/en/news/meta-data-center-electricity-consumption-hits-14975gwh-leased-data-center-use-nearly-doubles/
4. datacenters — https://datacenters.google/efficiency/
5. dcpostmea — https://www.dcpostmea.com/2025/08/uptime-institute-release-15th-annual-global-data-center-survey-2025/
6. datacenterdynamics — https://www.datacenterdynamics.com/en/news/microsoft-launches-atlanta-fairwater-data-center-two-stories-no-ups-or-gen-sets/
7. pantheon — https://pantheon.run/learn/nvidia-gb200-nvl72-power-and-cooling
8. hpe — https://buy.hpe.com/my/en/options/server-accelerators/nvidia-ai-accelerated-computing/nvidia-ai-accelerated-computing/nvidia-gb200-nvl72-by-hpe/p/1014890104
9. crbonfree — https://crbonfree.com/blog/how-much-energy-does-ai-use
10. technologyreview — https://www.technologyreview.com/2025/08/21/1122288/google-gemini-ai-energy/
11. google — https://cloud.google.com/blog/products/infrastructure/measuring-the-environmental-impact-of-ai-inference?hl=en
12. towardsdatascience — https://towardsdatascience.com/is-googles-reveal-of-geminis-impact-progress-or-greenwashing/
13. epoch — https://epoch.ai/gradient-updates/how-much-energy-does-chatgpt-use
14. greenly — https://greenly.earth/en-us/leaf-media/data-stories/ai-efficiency-a-gemini-reality-check-with-greenly

15. arxiv — https://arxiv.org/pdf/2508.15734
16. arxiv — https://arxiv.org/pdf/2304.03271
17. felloai — https://felloai.com/how-much-water-does-ai-use/
18. cpcb — https://cpcb.nic.in/uploads/Industry-Specific-Standards/Effluent/TPP_Effluent_Norms.pdf
19. downtoearth — https://www.downtoearth.org.in/water/widespread-non-compliance-as-thermal-power-plants-don-t-meet-water-use-norms-65391
20. thehawk — https://www.thehawk.in/news/economy-and-business/indias-data-centre-capacity-likely-to-surge-40-pc-annually-to-12-gw-by-2030-report
21. spglobal — https://www.spglobal.com/en/research-insights/special-reports/india-forward/shifting-horizons/will-data-center-growth-india-propel-country-global-hub-status
22. sttelemediagdc — https://assets.sttelemediagdc.com/sttgdc/global_en/public/2026-03/STT-GDC-India-ESG-Report-FY2025.pdf
23. lbl — https://eta-publications.lbl.gov/sites/default/files/2024-12/lbnl-2024-united-states-data-center-energy-usage-report_1.pdf
24. uptimeinstitute — https://journal.uptimeinstitute.com/large-data-centers-are-mostly-more-efficient-analysis-confirms/
25. ashb — https://www.ashb.com/public_research/is-2025-107-uptime-institute-global-data-center-survey-2024/
26. marketscreener — https://www.marketscreener.com/quote/stock/AMAZON-COM-INC-12864605/news/Amazon-com-2024-Amazon-Sustainability-Report-AWS-Summary-2024-amazon-sustainability-report-aws-50528004/
27. yotta — https://colocation.yotta.com/data-center/
28. yotta — https://yotta.com/data-center/
29. communicationstoday — https://www.communicationstoday.co.in/indiaai-missions-sustainability-push-sparks-data-center-cost-concerns/
30. mlq — https://mlq.ai/news/v2/microsoft-deploys-closed-loop-cooling-at-fairwater-campus-claims-restaurant-level-water-use/
31. iea — https://www.iea.org/reports/energy-and-ai/energy-demand-from-ai
32. iea — https://www.iea.org/reports/energy-and-ai/executive-summary
33. Wikipedia — https://en.wikipedia.org/wiki/Data_center
34. windows — https://iea.blob.core.windows.net/assets/3179f7f8-01f6-4dd6-bffa-c9f7b73f1dc9/KeyQuestionsonEnergyandAI.pdf
35. energy — https://www.energy.gov/articles/doe-releases-new-report-evaluating-increase-electricity-demand-data-centers
36. energynews — https://energynews.pro/en/india-to-reach-12-gw-of-data-centre-capacity-by-2030
37. euronews — https://www.euronews.com/next/2025/08/23/using-googles-ai-prompt-akin-to-five-drops-of-water-or-the-energy-needed-to-watch-9-second
38. techbytehaven — https://techbytehaven.com/how-much-electricity-does-chatgpt-actually-use/
39. hhc — https://www.hhc.earth/knowledge-base/articles/french-mistral-ai-sets-new-environmental-standard-with-comprehensive-lca-report
40. the-decoder — https://the-decoder.com/mistral-ai-publishes-the-first-comprehensive-life-cycle-assessment-of-a-large-language-model/
41. acm — https://cacm.acm.org/sustainability-and-computing/making-ai-less-thirsty/
42. arxiv — https://arxiv.org/html/2603.02705v2
43. aboutchromebooks — https://www.aboutchromebooks.com/ai-data-center-water-usage-statistics/
44. microsoft — https://www.microsoft.com/en-us/microsoft-cloud/blog/2024/12/09/sustainable-by-design-next-generation-datacenters-consume-zero-water-for-cooling/
45. aboutamazon — https://sustainability.aboutamazon.com/products-services/aws-cloud
46. atmeta — https://sustainability.atmeta.com/wp-content/uploads/2025/10/Meta_2025-Environmental-Data-Index.pdf
47. Wikipedia — https://en.wikipedia.org/wiki/Water_usage_effectiveness
48. redmondmag — https://redmondmag.com/articles/2024/12/09/microsoft-rolls-out--water-saving-datacenter-design.aspx
49. stanford — https://ojs.stanford.edu/ojs/index.php/surj/article/download/4407/1933/13349
50. techtarget — https://www.techtarget.com/it-infrastructure/news/366650693/A-shadow-water-problem-AIs-power-demand-shifts-consumption-upstream
51. sifyinfinitspaces — https://sifyinfinitspaces.com/wp-content/uploads/2025/09/Sify-Infinit-Spaces-Limited_Sustainability-Report-2024-25.pdf
52. earthjournalism — https://earthjournalism.net/stories/what-data-center-giants-arent-saying-about-their-water-use-in-india
53. downtoearth — https://www.downtoearth.org.in/science-technology/indias-digital-thirst-what-data-centre-giants-arent-saying-about-their-water-use
54. esgdive — https://www.esgdive.com/news/google-commits-to-replenish-more-water-than-it-uses-by-2030/822021/
55. smartwatermagazine — https://smartwatermagazine.com/news/smart-water-magazine/microsoft-unveils-next-generation-datacenters-zero-water-cooling
56. techradar — https://www.techradar.com/pro/did-extreme-overclockers-inspire-microsoft-to-develop-closed-loop-liquid-cooling-for-data-centers-id-like-to-think-so
57. networkworld — https://www.networkworld.com/article/3631084/microsoft-raises-bar-with-water-saving-data-center-design.html
58. datacentremagazine — https://datacentremagazine.com/news/microsoft-inside-the-worlds-first-ai-superfactory
59. microsoft — https://blogs.microsoft.com/blog/2025/09/18/inside-the-worlds-most-powerful-ai-datacenter/
60. Substack — https://measuredai.substack.com/p/microsoft-fairwater-atlanta-data-center
61. texastribune — https://www.texastribune.org/2025/09/25/texas-data-center-water-use/
62. governing — https://www.governing.com/resilience/wastewater-will-cool-this-memphis-data-center
63. wreg — https://wreg.com/news/local/xai-to-resume-construction-on-wastewater-recycling-facility-memphis-mayor-says/
64. The Cool Down — https://www.thecooldown.com/green-tech/musk-ai-water-recycling-project-memphis/
65. yahoo — https://www.yahoo.com/news/articles/xai-puts-water-recycling-project-163519747.html
66. wwdmag — https://www.wwdmag.com/wastewater-treatment/news/55386236/xai-commits-to-restarting-memphis-wastewater-recycling-project-in-2027
67. powerline — https://powerline.net.in/2026/09/04/water-stress-thermal-power-plants-take-steps-to-optimise-consumption/
68. downtoearth — https://www.downtoearth.org.in/pollution/it-is-imperative-to-improve-water-efficiency-in-india-s-coal-fired-thermal-power-plants-89484
69. scribd — https://www.scribd.com/document/989264277/CEA-Indian-Grid-Emission-Factor
70. sciencedirect — https://www.sciencedirect.com/science/article/pii/S2666389925002788
71. arxiv — https://arxiv.org/abs/2304.03271
72. researchgate — https://www.researchgate.net/publication/369855581_Making_AI_Less_Thirsty_Uncovering_and_Addressing_the_Secret_Water_Footprint_of_AI_Models
73. sinkproject — https://www.sinkproject.com/company/mistral-ai
74. countercurrents — https://countercurrents.org/2026/10/people-need-decentralized-distributed-data-centres-not-hyperscale-centralized-ones
75. latestly — https://www.latestly.com/technology/meta-reliance-partnership-tech-giants-unveil-indias-1st-ai-enabled-data-centre-in-jamnagar-7467321.html/amp
76. psuconnect — https://www.psuconnect.in/corporate-news/reliance-meta-to-build-india-first-ai-data-centre-in-jamnagar-168-mw-powered-by-renewable-energy
77. vaidicslucknow — https://www.vaidicslucknow.com/current-affair/data-centres-in-karnataka-sustainability-concerns-sustainable-data-centre-policy-2026-31-t/
78. reclimatize — https://reclimatize.in/india-grid-emission-factor-cea-ccts/
79. thebatterymagazine — https://www.thebatterymagazine.com/indias-data-centre-capacity-set-to-surge-nearly-sixfold-to-12-gw-by-2030-wood-mackenzie/
80. woodmac — https://www.woodmac.com/press-releases/india-data-centre-capacity-to-reach-12-gw-by-2030/
81. re24 — https://re24.energy/india-data-centre-boom-digital-infrastructure/
82. enterpriseam — https://enterpriseam.com/mena-india/2025/12/24/ai-to-drive-20-of-indias-incremental-power-demand-by-2030/
83. constructionworld — https://www.constructionworld.in/energy-infrastructure/power-and-renewable-energy/india-data-centres-to-consume-191-twh-by-2040-driving-renewables/95229
84. thenextweb — https://thenextweb.com/news/google-india-data-centre-water-wildlife-visakhapatnam
85. datacenterdynamics — https://www.datacenterdynamics.com/en/news/google-confirms-15bn-data-center-project-in-andhra-pradesh-india/
86. adani — https://www.adani.com/newsroom/media-releases/google-breaks-ground-on-india-ai-hub-launching-a-national-industrial-ecosystem
87. TechCrunch — https://techcrunch.com/2025/10/14/google-to-invest-15b-in-indian-ai-infrastructure-hub
88. voicendata — https://www.voicendata.com/data-centre/reliances-sovereign-ai-push-exposes-indias-model-compute-gap-12544068
89. inc42 — https://inc42.com/buzz/coreweave-enters-india-partners-adaniconnex-for-240-mw-data-centre-capacity/
90. tele — https://tele.net.in/smart-energy-management-role-of-technology-and-green-policy-mandates-in-building-sustainable-data-centres/
91. bank — https://sbi.bank.in/documents/57189647/58231607/Sector+Report+DataCentre.pdf/f1919592-ee66-b50b-ea20-75fbb27c9bb5?t=1779885206866
92. cbinsights — https://www.cbinsights.com/company/cyfuture
93. cbinsights — https://www.cbinsights.com/investor/netmagic-solutions
94. indiaai — https://www.indiaai.gov.in/article/union-minister-of-electronics-it-railways-and-i-b-announces-the-availability-of-18-000-affordable-ai-compute-units
95. medianama — https://www.medianama.com/2026/09/223-karnataka-sustainable-data-centre-policy/
96. newslaundry — https://www.newslaundry.com/2026/10/01/karnatakas-new-data-centre-policy-puts-water-energy-in-focus-what-does-it-say</untrusted_external_data>

Chapter 15 · Recommendations

1. telborg — https://telborg.com/datacentre-briefings/256
2. ecorpit — https://ecorpit.com/karnataka-data-centre-policy-gcc-ai-infrastructure-2026/
3. computerweekly — https://www.computerweekly.com/news/366629299/India-revives-national-datacentre-policy-amid-AI-push
4. treelife — https://treelife.in/reports/india-budget-2026-data-centres-it-tech-global-ai/
5. powerpeakdigest — https://powerpeakdigest.com/cea-draft-2026-grid-connectivity-regulations/
6. raceias — https://www.raceias.com/current-affairs/india-revises-earthquake-design-code-2025-is-18932025
7. ijcope — https://ijcope.org/wp-content/uploads/2026/05/Comparative-Study-of-Seismic-Design-Provisions-Impact-of-Is-1893-2025-Part-1-And-Part-5-on-Multi-Storey-RC-Buildings.pdf
8. visionias — https://visionias.in/current-affairs/news-today/2025-11-29/environment/indias-new-earthquake-zonation-map-by-bureau-of-indian-standards
9. visionias — https://visionias.in/current-affairs/upsc-daily-news-summary/article/2026-03-07/the-economic-times/environment/bis-revokes-new-seismic-map-with-high-risk-zone-vi
10. mercomindia — https://www.mercomindia.com/power-ministry-announces-100-ists-charges-waiver-for-psps-co-located-bess
11. saurenergy — https://www.saurenergy.com/solar-energy-news/mop-extends-ists-charges-waiver-for-solar-wind-projects-facing-transmission-delays-11756500
12. jmkresearch — https://jmkresearch.com/inter-state-transmission-waiver-for-energy-storage-projects-extended-by-three-years/
13. ieefa — https://ieefa.org/sites/default/files/2024-12/Impact%20of%20Green%20Energy%20Open%20Acess_Dec2024.pdf
14. etpi — https://etpi.in/sites/default/files/publications/ETPI_Green%20Energy%20Open%20Access%20Regulations%20in%20India.pdf
15. neufin — https://neufin.co/blog/tamil-nadu-open-access-regulation-2025-for-business/
16. saurenergy — https://www.saurenergy.com/solar-energy-news/gerc-proposes-lower-banking-charges-for-green-energy-open-access-12062017
17. hwra — https://hwra.org.in/Content/Images/Revised_Guidelines_Section_2(3).pdf
18. teamleaseregtech — https://teamleaseregtech.com/updates/article/22624/central-ground-water-authority-amended-the-guidelines-to-regulate-and/
19. enviliance — https://enviliance.com/regions/south-asia/in/report_1185
20. pib — https://www.pib.gov.in/PressReleasePage.aspx?PRID=2206073&reg=3&lang=2
21. bank — https://sbi.bank.in/documents/57189647/58231607/Sector+Report+DataCentre.pdf/f1919592-ee66-b50b-ea20-75fbb27c9bb5?t=1779885206866
22. medianama — https://www.medianama.com/2026/09/223-karnataka-sustainable-data-centre-policy/
23. thenewsminute — https://www.thenewsminute.com/karnataka/karnatakas-new-data-centre-policy-puts-water-energy-in-focus-what-does-it-say
24. whalesbook — https://www.whalesbook.com/news/English/real-estate/Maharashtra-Expands-Green-Data-Centre-Policy-Investor-Impact/6a30b10ad017fdb509977cc9
25. googlecloudpresscorner — https://www.googlecloudpresscorner.com/2026-04-28-Google-Breaks-Ground-on-India-AI-Hub,-Launching-a-National-Industrial-Ecosystem-Alongside-Indias-Digital-Infrastructure-Milestone
26. startupfeed — https://startupfeed.in/google-breaks-ground-vizag-ai-hub-adani-airtel-15-billion/
27. subseacables — https://www.subseacables.net/infrastructure-news/google-unveils-three-new-subsea-cables-to-strengthen-india-us-ai-connectivity/
28. fb — https://about.fb.com/news/2026/06/meta-partners-with-reliance-on-ai-enabled-data-center-in-india/
29. TechCrunch — https://techcrunch.com/2026/02/19/reliance-unveils-110b-ai-investment-plan-as-india-ramps-up-tech-ambitions/
30. wfw — https://www.wfw.com/articles/data-centres-an-international-legal-and-regulatory-perspective-spotlight-on-germany/
31. mayerbrown — https://www.mayerbrown.com/en/insights/publications/2024/02/sustainable-data-centers-the-german-energy-efficiency-act-what-data-center-operators-need-to-consider-now-and-in-the-future
32. mondaq — https://www.mondaq.com/climate-change/1851256/european-commission-advances-data-center-sustainability-ratings-and-consults-on-minimum-performance-standards
33. eudca — https://www.eudca.org/energy-efficiency-directive
34. lexology — https://www.lexology.com/library/detail.aspx?g=46b6cc4e-011c-45ac-9c4c-9ac58df0b76b
35. dcbyte — https://www.dcbyte.com/news-blogs/ireland-reopens-the-grid-for-data-centres/
36. arthurcox — https://www.arthurcox.com/knowledge/new-connection-policy-for-data-centres-in-ireland/
37. relae — https://relae.co/blog/data-center-load-flexibility
38. waterwastewaterasia — https://waterwastewaterasia.com/higher-costs-tighter-rules-johors-data-centres-turn-to-reclaimed-water/
39. malaysia4u — https://malaysia4u.com/data-centres-guide
40. arusgrid — https://arusgrid.com/policy/malaysia
41. asiadaily — https://asiadaily.org/news/10940/
42. cls — https://www.cls.cn/detail/1743451
43. solarquarter — https://solarquarter.com/2026/07/16/cea-releases-draft-2026-grid-connectivity-regulations-to-strengthen-renewable-energy-integration-and-grid-stability/
44. itpro — https://www.itpro.com/infrastructure/data-centres/data-center-water-consumption-is-skyrocketing-but-microsoft-thinks-it-has-a-solution-the-companys-new-closed-loop-cooling-system-consumes-zero-water-and-could-save-millions-of-liters-per-year
45. saurenergy — https://www.saurenergy.com/solar-energy-news/tnerc-proposes-draft-banking-rules-for-green-power-open-access-10511916
46. whitecase — https://www.whitecase.com/insight-alert/data-center-requirements-under-new-german-energy-efficiency-act
47. introl — https://introl.com/blog/singapore-green-data-center-mandate-dc-cfa2-2026
48. dlapiperdataprotection — https://www.dlapiperdataprotection.com/?t=law&c=IN
49. pib — https://static.pib.gov.in/WriteReadData/specificdocs/documents/2025/nov/doc20251117695301.pdf
50. veritect — https://veritect.ai/digital-data-ai-law/dpdp-rules-2025-phased-rollout
51. biggo — https://finance.biggo.com/news/63b93a0b-f368-4c8f-94fb-56b7351b6c30
52. wuhan — https://home.wuhan.gov.cn/zcfg/202408/t20240808_2439562.shtml
53. news — http://www.news.cn/energy/20240725/2256950d364e4b7c8579a9466c0b1811/c.html
54. meti — https://www.meti.go.jp/english/press/2025/0612_001.html
55. meti — https://www.enecho.meti.go.jp/category/others/basic_plan/pdf/2025_strategic_energy_plan.pdf
56. india-briefing — https://www.india-briefing.com/news/india-bis-standards-data-centres-46991.html
57. moduledge — https://www.moduledge.com/blog/eu-data-center-regulations-2026
58. mayerbrown — https://www.mayerbrown.com/en/insights/publications/2025/07/important-texas-regulatory-updates-for-data-centers
59. mri — https://www.mri.co.jp/en/knowledge/insight/20251219.html
60. businessworld — https://www.businessworld.in/article/karnataka-approves-sustainable-data-centre-policy-targets-1-gw-it-load-by-2031-624457
61. eqmagpro — https://www.eqmagpro.com/renew-subsidiary-secures-%E2%82%B970000-crore-maharashtra-investment-pact-for-750-mw-green-data-centre-park-eq/
62. legal500 — https://www.legal500.com/intelligence/india/privacy/cross-border-data-transfers-under-indias-dpdp-act-rules-restrictions-and-compliance-roadmap-for-2027
63. thediplomat — https://thediplomat.com/2026/04/whose-water-powers-the-cloud-data-centers-and-the-right-to-water-in-johor/
64. ndrc — https://www.ndrc.gov.cn/xwdt/tzgg/202407/P020240723625616053849.pdf
65. indianinfrastructure — https://indianinfrastructure.com/2026/07/14/cea-issues-draft-cea-technical-standards-for-connectivity-to-the-grid-regulations-2026/
66. indianweb2 — https://www.indianweb2.com/2026/04/google-breaks-ground-on-15-billion-ai.html</untrusted_external_data>